



21 世纪高等院校经典教材同步辅导
ERSHIYISHIJIGAODENGYUANXIAOJINGDIANJIAOCAITONGBUFUDAO

数学物理方法

第四版

全程导学及习题全解

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内 容 提 要

本书是为高等教育出版社出版、梁昆森先生编著的《数学物理方法》(第四版 刘法、缪国庆老师修订)一书所编写的配套学习参考书。《数学物理方法习题指导》对《数学物理方法》(第四版)的习题按顺序进行了详细的解答,以方便读者较好地掌握各章知识要点和基本解题方法。每章均分为本章知识要点、习题全解两部分。本章知识要点部分将《数学物理方法》中主要的概念、理论、公式加以归纳总结。习题全解是本书的重点,按原教科书所列习题进行详细的求解和证明。对其中部分习题还给出了多种解法或是不同的解题思路。本书可作为理工科院校学生或相关专业人员的参考书,亦可帮助报考研究生的同学复习备考,或作为相关教学人员的教学参考书。

前 言

数学物理方法是物理学中一门重要的基础理论课程。是物理本科各专业以及部分工科专业学生必修的重要基础课，它将为学习物理专业课程提供基础的数学处理工具。数学物理方法是为大学本科的理论物理学准备数学工具的。由于理论物理学的结构使用的是像欧几里得几何学那样严密的公理化体系，从公理（原理）演绎推导出一系列可以用实验验证的具体推论，步步紧扣，逻辑体系严密，简洁而优美。在它的演绎推导中大量地使用了数学，鉴于该课程的重要性，国内高校的物理类专业及某些工科类专业均开设了数理方法课程。本书与梁昆森先生编著的《数学物理方法》（第四版）相配套，全书内容包括（一）复变函数论（复变函数、复变函数的积分、幂级数展开、留数定理、傅立叶变换、拉普拉斯变换等）；（二）数学物理方程（数学物理定解问题、分离变量法、二阶常微分方程级数解法、本征值问题、球函数、柱函数、格林函数法、积分变换法、保角变换法、非线性数学物理问题简介等）。本辅导教材根据《数学物理方法》，致力于以下几方面内容的编写：一、知识点归纳：概括了各章中的主要知识点，理清各知识点之间的脉络联系，囊括了主要定理及相关推论和重要公式等，帮助读者迅速了解本章的重要知识点，系统理解各章的体系结构，奠定扎实的理论基础。二、习题全解：依据教材各章节的习题，进行详尽的解答。在解题中既注重物理上的分析，也注意数学的演算过程，使书中每道题都得到了详尽的解答。我们希望通过编写这本《数学物理方法解题指导》来帮助同学们更好地掌握、理解数理方法基本理论并掌握相关问题的分析和处理方法，以便更好地解决同学们在解题过程中所遇到的问题。

本书由苗明川、龚云、任正等编写。邓胜华教授主审全书，并对书中内容和编写提出了许多宝贵的意见和建议。邓胜华教授高深的学术造诣，严谨的治学态度，使编者受益匪浅，在此深表感谢。本书编写过程中还得到黄江、杨蕤、林一强等给予的大力支持和帮助，同时得到中国时代经济出版社的领导和有关编辑的大力支持，编者深表谢意。

对《数学物理方法》教材的作者梁昆森教授，编者在此表示崇高的敬意，对刘法、缪国庆教授，表示衷心的感谢！

由于编者水平有限，加之时间仓促，本书难免有缺点和疏漏，存在不妥之处，敬请各位专家及广大读者批评指正，以便再版时修正。

目 录

第一篇 复变函数论

第一章 复变函数	3
本章知识要点	3
习题全解	5
§ 1.1 复数与复数运算	5
§ 1.2 复变函数	11
§ 1.3 导数	15
§ 1.4 解析函数	15
§ 1.5 平面标量场	21
§ 1.6 多值函数	26
第二章 复变函数的积分	29
本章知识要点	29
习题全解	30
§ 2.4 柯西公式	30
第三章 幂级数展开	32
本章知识要点	32
习题全解	33
§ 3.2 幂级数	33
§ 3.3 泰勒级数展开	36
§ 3.5 洛朗级数展开	39
§ 3.6 孤立奇点的分类	46
第四章 留数定理	48
本章知识要点	48
习题全解	49
§ 4.1 留数定理	49

§ 4.2 应用留数定理计算实变函数定积分	53
第五章 傅里叶变换	70
本章知识要点	70
习题全解	73
§ 5.1 傅里叶级数	73
§ 5.2 傅里叶积分与傅里叶变换	85
§ 5.3 δ 函数	89
第六章 拉普拉斯变换	91
本章知识要点	91
习题全解	92
§ 6.1 拉普拉斯变换	92
§ 6.2 拉普拉斯变换的反演	94
§ 6.3 应用例	100

第二篇 数学物理方程

第七章 数学物理定解问题	115
本章知识要点	115
习题全解	120
§ 7.1 数学物理方程的导出	120
§ 7.2 定解条件	125
§ 7.3 数学物理方程的分类	129
§ 7.4 达朗贝尔公式 定解问题	135
第八章 分离变数法	143
本章知识要点	143
习题全解	144
§ 8.1 齐次方程的分离变数法	144
§ 8.2 非齐次振动方程和输运方程	172
§ 8.3 非齐次边界条件的处理	182
§ 8.4 泊松方程	188

第九章 二阶常微分方程级数解法 本征值问题	194
本章知识要点	194
习题全解	200
§ 9.1 特殊函数常微分方程	200
§ 9.2 常点邻域上的级数解法	202
§ 9.3 正则奇点邻域上的级数解法	206
§ 9.4 施图母-刘维尔本征值问题	216
第十章 球函数	221
本章知识要点	221
习题全解	226
§ 10.1 轴对称球函数	226
§ 10.3 一般的球函数	237
第十一章 柱函数	244
本章知识要点	244
习题全解	247
§ 11.2 贝塞尔方程	247
§ 11.4 虚宗量贝塞尔方程	259
§ 11.5 球贝塞尔函数	268
第十二章 格林函数法	277
本章知识要点	277
习题全解	279
§ 12.2 用电像法求格林函数	279
§ 12.4 用冲量定理法求格林函数	281
§ 12.5 推广的格林公式及其应用	288
第十三章 积分变换法	291
本章知识要点	291
习题全解	292
§ 13.1 傅里叶变换法	292
§ 13.2 拉普拉斯变换法	307
第十四章 保角变换法	311
本章知识要点	311
习题全解	313

第十五章 非线性数学物理问题简介	321
本章知识要点	321
孤立子	321
混沌	321

第一篇

复变函数论

复变函数论是数学的一个分支,主要研究以复数作为自变量的函数,以及相关的理论。复变函数涉及复数域上的解析函数。本篇介绍解析函数,复变函数的积分,幂级数展开,留数定理和傅氏及变换等。

第一章 复变函数

本章知识要点

复数与复数运算

(一) 复数的基本概念

一个复数总可以表为实数 x 与纯虚数 iy 之和,

$$z = x + iy$$

此为复数的代数式. x 和 y 分别为该复数的实部和虚部, 记为 $\operatorname{Re} z$ 和 $\operatorname{Im} z$. 复数 z 与平面上的点相对应, 该平面称为复数平面, x 轴为实轴, y 轴为虚轴. 复数可用复数平面上的矢量来表示.

极坐标

$$\begin{cases} \rho = \sqrt{x^2 + y^2}, \\ \varphi = \arctan(y/x). \end{cases} \quad \begin{cases} x = \rho \cos \varphi, \\ y = \rho \sin \varphi \end{cases}$$

三角式

$$z = \rho(\cos \varphi + i \sin \varphi)$$

指数式

$$z = \rho e^{i\varphi}$$

ρ 称为该复数的模, 记作 $|z|$. φ 为该复数的辐角, 记作 $\operatorname{Arg} z$.

表示为 $\varphi = \operatorname{Arg} z = \arg z + 2k\pi \quad (k = 0, \pm 1, \pm \dots)$ 其中 $0 \leq \arg z \leq 2\pi$

z 共轭复数表示为 z^* . 记为

$$z^* = x - iy = \rho(\cos \varphi - i \sin \varphi) = \rho e^{-i\varphi}$$

(二) 无限远点

模为无限大的复数与复平面上的一点相对应, 称为无限远点.

(三) 复数的运算

复数的运算满足加法的交换律和结合律.

复数的运算满足乘法的交换律、结合律与分配律.

复变函数

(一) 复变函数的定义

若复数平面上存在一个点集 E , 对于 E 的每个点, 有一个或多个复数值 ω 与之对应, 称 ω 为 z 的函数 —— 复变函数. 记作

$$\omega = f(z) \quad z \in E$$

(二) 区域的概念

邻域、内点、境界点、境界线、闭区域.

(三) 复变函数例

$$e^z = e^{x+iy} = e^x(\cos y + i\sin y)$$

$$\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$$

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$$

$$\operatorname{sh} z = \frac{1}{2}(e^z - e^{-z})$$

$$\operatorname{ch} z = \frac{1}{2}(e^z + e^{-z})$$

$$\ln z = \ln(|z| e^{i\operatorname{Arg} z}) = \ln |z| + i\operatorname{Arg} z$$

$$z^j = e^{j \ln z} \quad (z \text{ 为复数})$$

$$|\sin z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\sin^2 x - \cos^2 x)}$$

$$|\cos z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)}$$

$|\sin z|$ 和 $|\cos z|$ 可以大于 1.

$$f(z) = u(x, y) + iv(x, y)$$

复变函数可归结为一对二元实变函数, 实变函数的许多定义、公式、定理可直接移植到复变函数中.

导 数

柯西 - 黎曼方程.

若 $f(z)$ 在点 z 可导, 则 $f(z) = u(x, y) + iv(x, y)$

满足
$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y},$$

即
$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

称为柯西 - 黎曼方程, 或柯西 - 黎曼条件 (C-R 条件).

它是复变函数可导的必要条件. 极坐标系中的柯西 - 黎曼方程

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases}$$

直角坐标系与极坐标系的柯西 - 黎曼方程是等价的, 可相互推导得出. 复变函数的实部与虚部通过柯西 - 黎曼方程联系起来.

解析函数

(一) 若函数 $f(z) = u + iv$ 在区域 B 上解析, 则

$$u(x, y) = C_1, v(x, y) = C_2$$

(C_1, C_2 为常数) 是 B 上的两组正交曲线族.

(二) 若函数 $f(z) = u + iv$ 在区域 B 上解析, 则 u, v 均为 B 上的调和函数.

平面标量场

如电磁场、声场、温度场等, 这些场均随时间、空间变化. 当场与时间无关, 则称为恒定场. 如静电场, 流体中的定常流速场等.

若所研究的场在空间某方向上是均匀的, 只需在垂直于该方向的平面上进行研究, 这样的场称为平面场.

平面静电场:

$$f(z) = u(x, y) + iv(x, y)$$

设 $u(x, y)$ 为电势, 曲线 $u(x, y) = \text{常数}$, 为等势线族. 设 $v(x, y)$ 为电场线族, $v(x, y)$ 垂直于等势线族.

习题全解

§ 1.1 复数与复数运算

1. 下列式子在复数平面上各具有怎样的意义?

(1) $|z| \leq 2$

解: 在复数平面上表示为以 $(0, 0)$ 为圆心, 2 为半径的圆周及圆内部. 如图 1-1-1 所示.

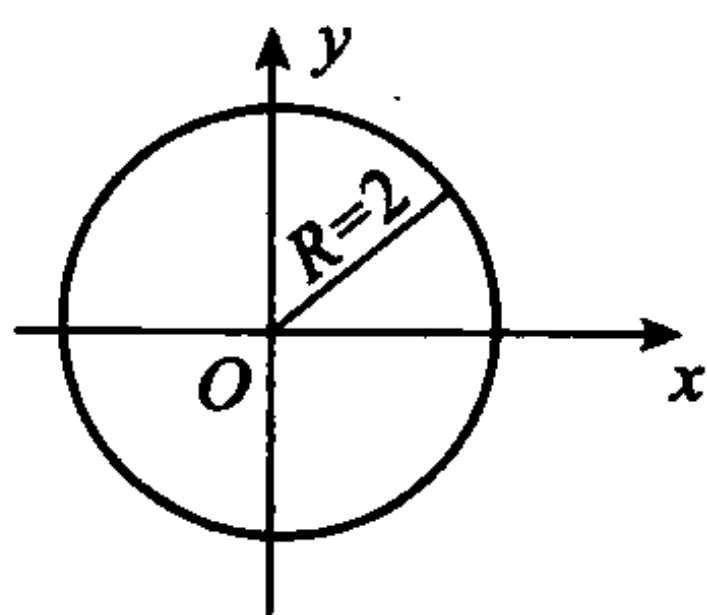


图 1-1-1

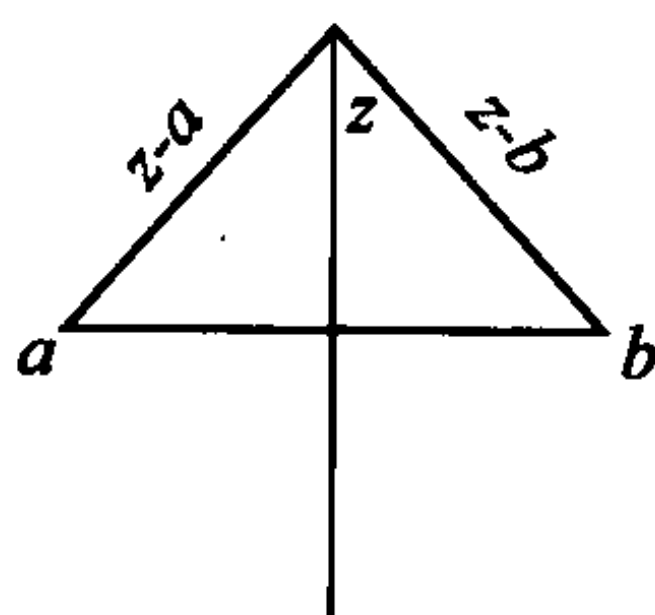


图 1-1-2

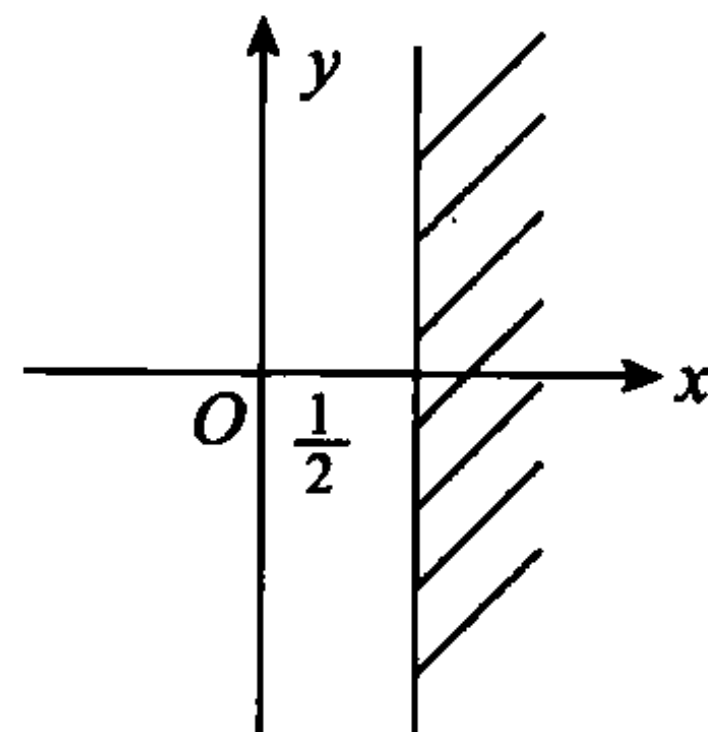


图 1-1-3

(2) $|z - a| = |z - b|$ (a, b 为复常数)

解: 在复数平面上表示为满足到 a 点与 b 点的距离相等的点的集合. 在复数平面上作点 a 与 b 的连线 ab , 线段 ab 的垂直平分线上点 z 均满足 $|z - a| = |z - b|$. 如图 1-1-2 所示.

$$(3) \operatorname{Re} z > \frac{1}{2}$$

解:在复数平面上表示为实部大于 $1/2$ 的部分,即 $x > 1/2$ 的平面区域.如图 1-1-3 所示.

$$(4) |z| + \operatorname{Re} z \leq 1$$

解:原式可写成 $|x + iy| + x \leq 1$

即

$$|x + iy| \leq 1 - x$$

由于

$$|x + iy| = \sqrt{x^2 + y^2}$$

所以有

$$0 \leq \sqrt{x^2 + y^2} \leq 1 - x.$$

即

$$x^2 + y^2 \leq 1 - 2x + x^2$$

$$y^2 \leq 1 - 2x.$$

(如图 1-1-4 所示)在复数平面上,为抛物线 $y^2 \leq 1 - 2x$ 所围区域(含抛物线上点).

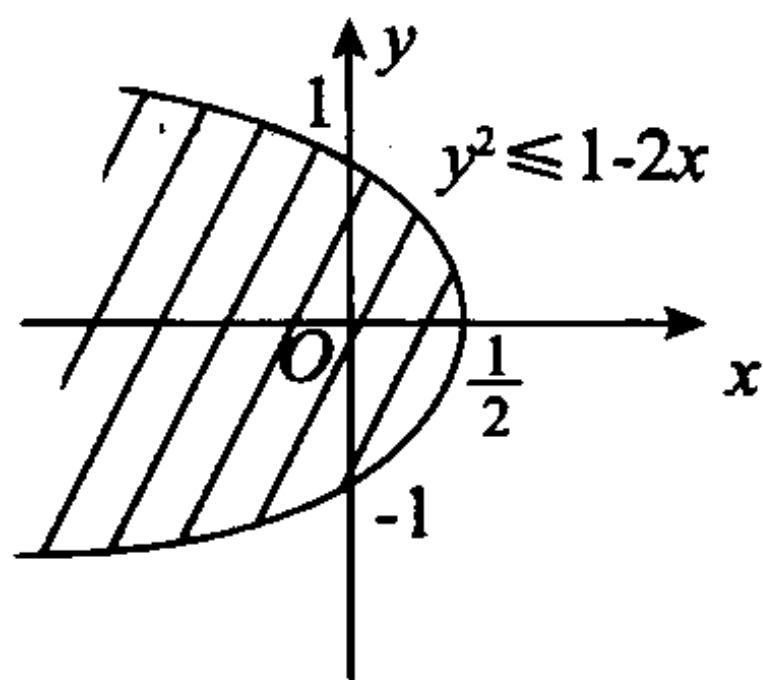


图 1-1-4

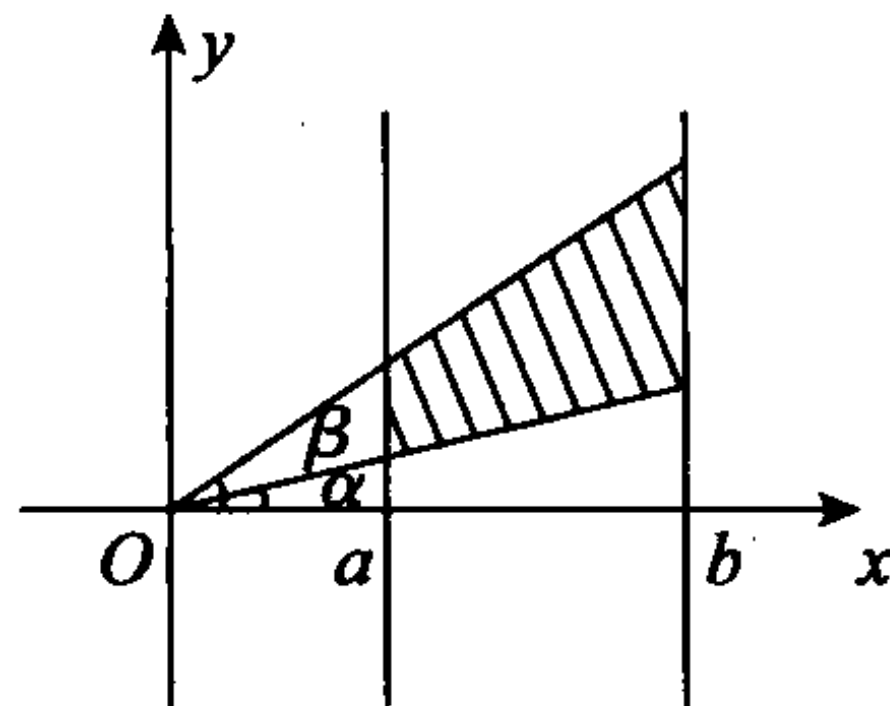


图 1-1-5

$$(5) \alpha < \arg z < \beta, \quad a < \operatorname{Re} z < b \quad (\alpha, \beta, a \text{ 和 } b \text{ 为实常数})$$

解:在复数平面上为主辐角 $\varphi > \alpha$ 和 $\varphi < \beta$ 与直线 $x > a$ 和 $x < b$ 所围成的梯形区域(不含边界上点).如图 1-1-5 所示.

$$(6) 0 < \arg \frac{z-i}{z+i} < \frac{\pi}{4}$$

解:设 $z = x + iy$, 则 $\frac{z-i}{z+i} = \frac{x+i(y-1)}{x+i(y+1)}$, 即 $\frac{z-i}{z+i} = \frac{x+i(y-1)}{x+i(y+1)}$

分子、分母同乘以 $x - i(y+1)$,

$$\text{分母为 } [x + i(y+1)][x - i(y+1)] = x^2 + (y+1)^2$$

$$\text{分子为 } [x + i(y-1)][x - i(y+1)] = x^2 + y^2 - 1 - i2x$$

设分母 $x^2 + (y+1)^2 \neq 0$ 即 $x = 0$ 时 $y \neq -1$.

$$\text{则分子有 } \operatorname{Re}(x^2 + y^2 - 1 - i2x) = x^2 + y^2 - 1 > 0$$

$$\operatorname{Im}(x^2 + y^2 - 1 - i2x) = -2x > 0$$

$$\text{且 } 0 < \arctan \frac{-2x}{x^2 + y^2 - 1} < \frac{\pi}{4}$$

$$\text{即 } \begin{cases} 0 < \frac{-2x}{x^2 + y^2 - 1} < 1 \\ x < 0 \end{cases}$$

$$-2x < x^2 + y^2 - 1$$

$$0 < x^2 + 2x + y^2 - 1 = (x+1)^2 + y^2 - 2$$

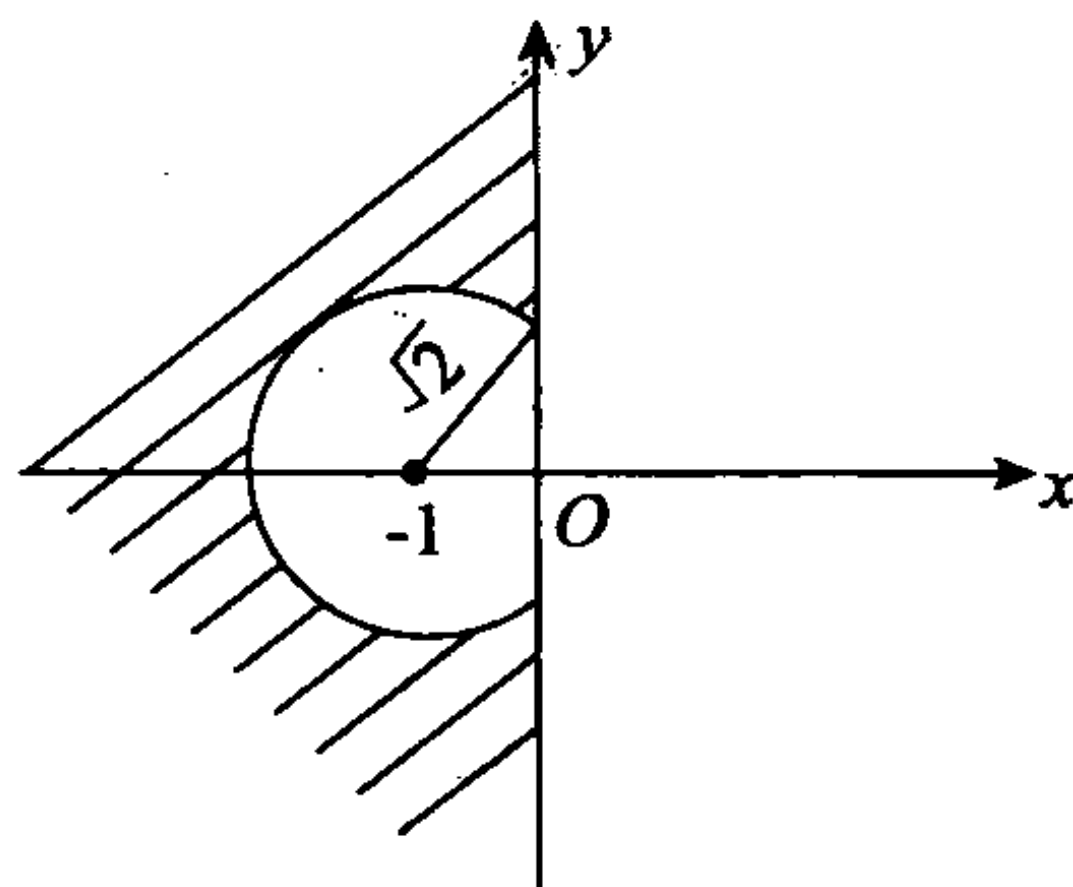


图 1-1-6

即

$$\begin{cases} x < 0 \\ (x+1)^2 + y^2 > 2 \end{cases}$$

将 $x=0, y=-1$ 代入, x, y 仍能满足条件.

所以, 在复数平面上表示为 $x < 0$ 的左半平面, 且在 $(x+1)^2 + y^2 > 2$ 的区域. 如图 1-1-6 所示.

$$(7) \left| \frac{z-1}{z+1} \right| \leq 1$$

解: 将 $z = x + iy$ 代入

$$\text{即 } \left| \frac{x+iy-1}{x+iy+1} \right| = \left| \frac{(x-1)+iy}{(x+1)+iy} \right| = \frac{|(x-1)+iy|}{|(x+1)+iy|} \leq 1$$

$$\text{有: } \frac{x^2 - 2x + 1 + y^2}{x^2 + 2x + 1 + y^2} \leq 1$$

$$\text{设 } x^2 + 2x + 1 + y^2 \neq 0$$

$$\text{则 } x^2 - 2x + 1 + y^2 \leq x^2 + 2x + 1 + y^2$$

$$\text{有 } -2x \leq 2x, \text{ 即 } 0 \leq 4x, 0 \leq x.$$

将 $x \geq 0$ 代入, $x^2 + 2x + 1 + y^2$ 恒大于 0. 故 $x \geq 0$ 满足条件. 即在复数平面上 $x \geq 0$ 的整个区域(含 y 轴). 如图 1-1-7 所示.

$$(8) \operatorname{Re}\left(\frac{1}{z}\right) = 2$$

$$\text{解: 因为 } \frac{1}{z} = \frac{1}{x+iy} = \frac{(x-iy)}{(x+iy)(x-iy)} = \frac{x-iy}{x^2+y^2}, \text{ 由原题 } \operatorname{Re}\left(\frac{1}{z}\right) = 2$$

$$\text{所以 } \operatorname{Re}\left(\frac{1}{z}\right) = \frac{x}{x^2+y^2} = 2$$

$$\text{即 } x^2 + y^2 - \frac{1}{2}x = 0.$$

$$\text{有 } \left(x - \frac{1}{4}\right)^2 + y^2 = \frac{1}{16} = \left(\frac{1}{4}\right)^2.$$

如图 1-1-8 所示.

$$\operatorname{Re}(1/z) = 2 \text{ 为复数平面上的圆 } \left(x - \frac{1}{4}\right)^2 + y^2 = \frac{1}{16}.$$

$$(9) \operatorname{Re} z^2 = a^2 \quad (a \text{ 是实常数}),$$

$$\text{解: 因为 } z^2 = (x+iy)^2 = x^2 - y^2 + i2xy, \text{ 且 } \operatorname{Re} z^2 = a^2, \\ \text{则 } \operatorname{Re} z^2 = x^2 - y^2 = a^2$$

$$\text{所以 } \operatorname{Re} z^2 = a^2 \text{ 为复数平面上的双曲线 } x^2 - y^2 = a^2.$$

如图 1-1-9 所示.

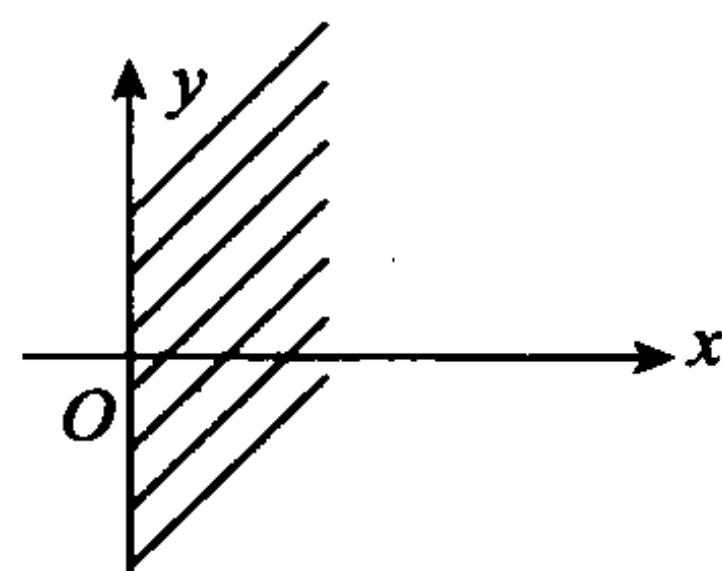


图 1-1-7

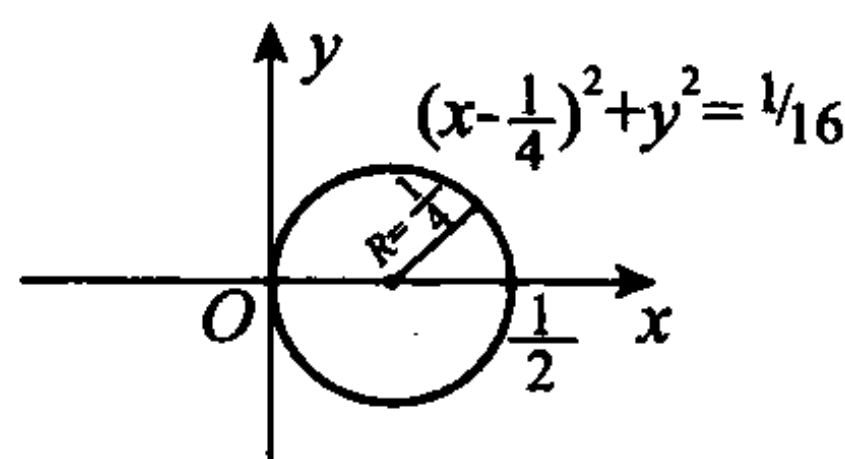


图 1-1-8

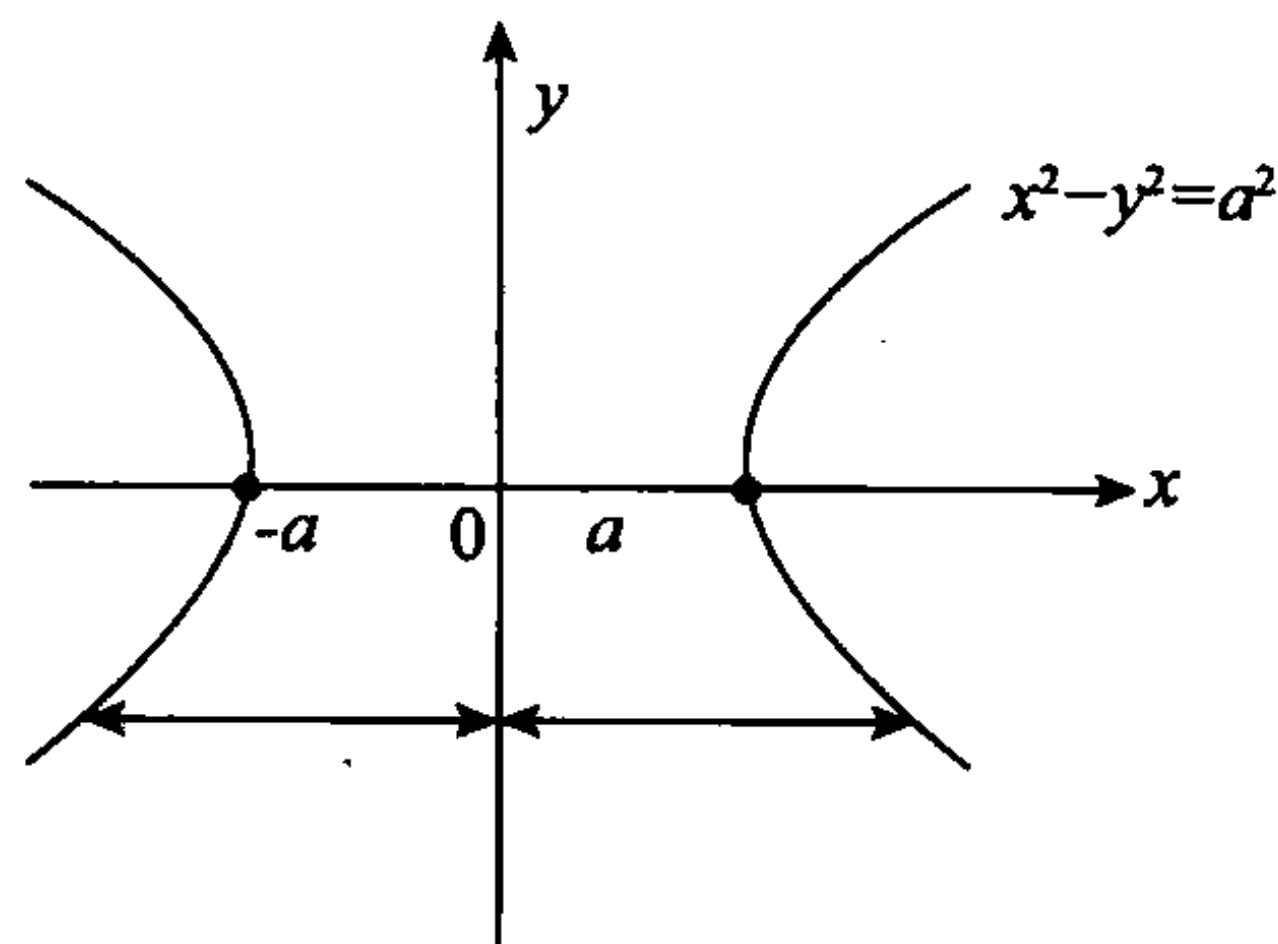


图 1-1-9

$$(10) |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2.$$

解: 设 $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$

则

$$|z_1 + z_2|^2 = (x_1 + x_2)^2 + (y_1 + y_2)^2$$

$$|z_1 - z_2|^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$2|z_1|^2 = 2(x_1^2 + y_1^2)$$

$$2|z_2|^2 = 2(x_2^2 + y_2^2)$$

$$\text{左端} = x_1^2 + 2x_1x_2 + x_2^2 + y_1^2 + 2y_1y_2 + y_2^2 + x_1^2 - 2x_1x_2 + x_2^2$$

$$+ y_1^2 - 2y_1y_2 + y_2^2 = 2(x_1^2 + y_1^2) + 2(x_2^2 + y_2^2) = \text{右端}$$

在复数平面上表示为平行四边形两对角线平方之和等于两邻边平方和的两倍. 如图 1-1-10 所示.

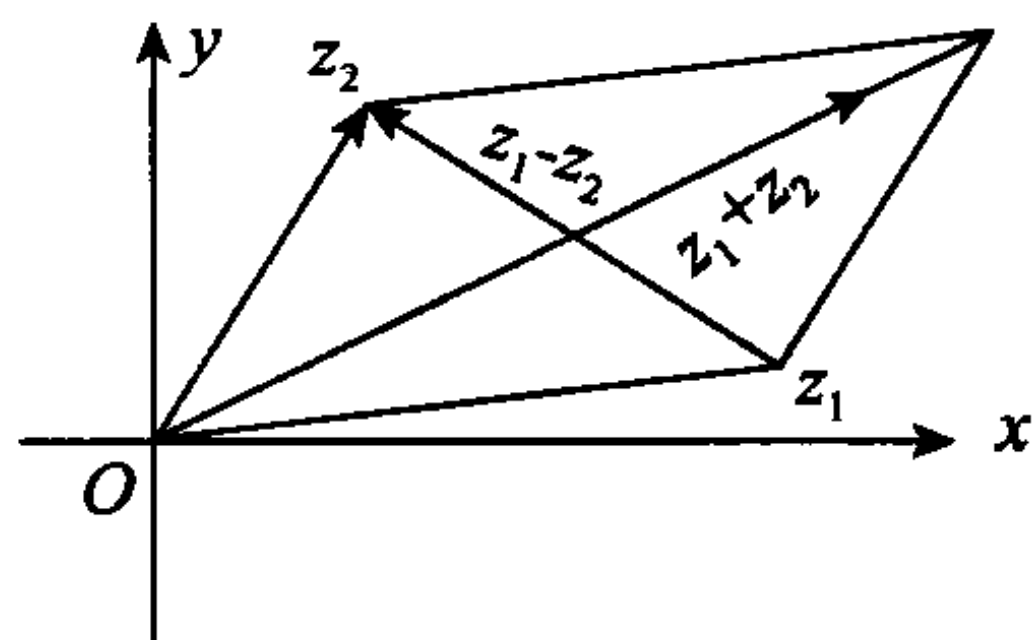


图 1-1-10

2. 将下列复数用代数式、三角式和指数式几种形式表示出来.

(1) i

解: 代数式

$$z_0 = i$$

三角式

$$|z_0| = \sqrt{1^2} = 1. \quad \arg z_0 = \arctan\left(\frac{1}{0}\right) = \frac{\pi}{2}$$

所以

$$z_0 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

指数式

$$z_0 = e^{i\frac{\pi}{2}}$$

(2) -1

解: 代数式

$$z_0 = -1$$

三角式

$$|z_0| = 1. \quad \arg(z_0) = \pi + \arctan\left(\frac{0}{-1}\right) = \pi$$

所以

$$z_0 = \cos \pi + i \sin \pi$$

指数式

$$z = e^{i\pi}.$$

(3) $1 + i\sqrt{3}$

解: 代数式

$$z_0 = 1 + i\sqrt{3}$$

三角式

$$|z_0| = \sqrt{1+3} = 2. \quad \arg z_0 = \arctan \frac{\sqrt{3}}{1} = \frac{\pi}{3}$$

所以

$$z_0 = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

指数式

$$z_0 = 2e^{i\frac{\pi}{3}}$$

(4) $1 - \cos \alpha + i \sin \alpha$ (α 是实常数),

解: 代数式 $z_0 = 1 - \cos \alpha + i \sin \alpha$

$$\text{三角式} \quad |z_0| = \sqrt{(1 - \cos \alpha)^2 + \sin^2 \alpha} = \sqrt{2(1 - \cos \alpha)} = 2 \sin \frac{\alpha}{2}$$

$$\arg z_0 = \arctan \frac{\sin \alpha}{1 - \cos \alpha} = \arctan \left(\cotan \frac{\alpha}{2} \right)$$

$$\text{所以} \quad z_0 = 2 \sin \frac{\alpha}{2} \cdot \cos \left[\arctan \left(\cotan \frac{\alpha}{2} \right) \right] + i \cdot 2 \sin \frac{\alpha}{2} \cdot \sin \left[\arctan \left(\cotan \frac{\alpha}{2} \right) \right]$$

指数式 $z_0 = 2\sin \frac{\alpha}{2} e^{i \arctan(\tan \frac{\alpha}{2})}$

(5) z^3

解: 设 $z = x + iy$

代数式 $z_0 = z^3 = (x + iy)^2(x + iy) = (x^2 - y^2 + i2xy)(x + iy)$
 $= x^3 - 3xy^2 + i(3x^2y - y^3)$

三角式 $z_0 = \rho^3(\cos 3\varphi + i\sin 3\varphi)$, 其中 $\rho = \sqrt{x^2 + y^2}$, $\varphi = \arctan\left(\frac{y}{x}\right)$

指数式 $z_0 = \rho^3 e^{i3\varphi}$, 其中 $\rho = \sqrt{x^2 + y^2}$, $\varphi = \arctan\left(\frac{y}{x}\right)$

(6) e^{1+i}

解: 代数式 $z_0 = e \cdot e^i = e \cos 1 + i e \sin 1 = 1.469 + i2.287$

三角式 $z_0 = e(\cos 1 + i \sin 1)$

指数式 $z_0 = ee^i$

(7) $\frac{1-i}{1+i}$

解: 原式 $= \frac{(1-i)(1-i)}{(1+i)(1-i)} = \frac{1}{2}(1 - 2i + i^2) = -i$

代数式 $z_0 = -i$

三角式 $z_0 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$

指数式 $z_0 = e^{i\frac{3\pi}{2}}$

3. 计算下列数值. (a, b 和 φ 为实常数)

(1) $\sqrt{a+ib}$

解: 设 $\sqrt{a+ib} = x + iy$, 则 $a + ib = (x + iy)^2$

$$a + ib = x^2 - y^2 + i2xy$$

$$\begin{cases} x^2 - y^2 = a \\ 2xy = b \end{cases} \begin{cases} x^2 - y^2 = a \\ x = \frac{b}{2y} \end{cases} \quad (\text{将 } x \text{ 或 } y \text{ 约去}) \text{ 得}$$

$$\begin{cases} y^4 + ay^2 - \frac{b^2}{4} = 0 \\ x^4 - ax^2 - \frac{b^2}{4} = 0 \end{cases} \begin{cases} y = \sqrt{\frac{-a + \sqrt{a^2 + b^2}}{2}} \\ x = \sqrt{\frac{a + \sqrt{a^2 + b^2}}{2}} \end{cases} \quad (\text{其它根舍去})$$

$$\text{所以 } z_0 = \sqrt{a+ib} = \frac{\sqrt{2}}{2} [(\sqrt{a + \sqrt{a^2 + b^2}}) + i(\sqrt{-a + \sqrt{a^2 + b^2}})]$$

(2) $\sqrt[3]{i}$

解: 因为 $i = e^{i(\frac{\pi}{2} + 2k\pi)}$

$$\text{所以 } z_0 = \sqrt[3]{i} = \sqrt[3]{e^{i(\frac{\pi}{2} + 2k\pi)}} = e^{i(\pi/2 + 2k\pi)/3} = e^{i(\pi/6 + 2k\pi/3)} \quad (k = 0, 1, 2, \dots)$$

(3) i^i 解: 因为 $i = e^{i(\pi/2+2k\pi)}$

$$\text{所以 } z_0 = i^i = e^{i \cdot i(\pi/2+2k\pi)} = e^{-(\pi/2+2k\pi)} \quad (k=0,1,2,\dots)$$

(4) $\sqrt{i}$ 解: 因为 $i = e^{i(\pi/2+2k\pi)}$

$$\text{所以 } z_0 = \sqrt{i} = e^{i(\pi/2+2k\pi)/i} = e^{(\pi/2+2k\pi)} \quad (k=0,1,2,\dots)$$

(5) $\cos 5\varphi$

解法一:

$$\begin{aligned} \cos 5\varphi &= \cos \varphi \cdot \cos 4\varphi - \sin \varphi \sin 4\varphi \\ &= \cos \varphi (\cos^2 2\varphi - \sin^2 2\varphi) - \sin \varphi \times 2\sin 2\varphi (\cos 2\varphi) \\ &= \cos \varphi [(\cos^2 \varphi - \sin^2 \varphi)^2 - (2\sin \varphi \cos \varphi)^2] - 4\sin^2 \varphi \cos \varphi (\cos^2 \varphi - \sin^2 \varphi) \\ &= \cos \varphi [\cos^4 \varphi - 2\sin^2 \varphi \cos^2 \varphi + \sin^4 \varphi - 4\sin^2 \varphi \cos^2 \varphi] - 4\sin^2 \varphi \cos^3 \varphi + 4\sin^4 \varphi \cos \varphi \\ &= \cos^5 \varphi + 5\sin^4 \varphi \cos \varphi - 10\sin^2 \varphi \cos^3 \varphi \end{aligned}$$

解法二:

$$\begin{aligned} \text{因为 } (\cos \varphi + i\sin \varphi)^5 &= (\cos \varphi + i\sin \varphi)^4 (\cos \varphi + i\sin \varphi) \\ &= \cos^5 \varphi + 5\sin^4 \varphi \cos \varphi - 10\sin^2 \varphi \cos^3 \varphi + i(\sin^5 \varphi + 5\sin \varphi \cos^4 \varphi - 10\sin^3 \varphi \cos^2 \varphi) \end{aligned}$$

$$\text{且 } (\cos \varphi + i\sin \varphi)^5 = \cos 5\varphi + i\sin 5\varphi$$

$$\text{所以 } \cos 5\varphi = \cos^5 \varphi + 5\sin^4 \varphi \cos \varphi - 10\sin^2 \varphi \cos^3 \varphi$$

$$\sin 5\varphi = \sin^5 \varphi + 5\sin \varphi \cos^4 \varphi - 10\sin^3 \varphi \cos^2 \varphi \text{ 得证.}$$

(6) $\sin 5\varphi$

解法一:

$$\begin{aligned} \sin 5\varphi &= \sin 4\varphi \cos \varphi + \sin \varphi \cdot \cos 4\varphi \\ &= 2\sin 2\varphi \cos 2\varphi \cos \varphi + \sin \varphi (\cos^2 2\varphi - \sin^2 2\varphi) \\ &= 4\sin^2 \varphi \cos^2 \varphi (\cos^2 \varphi - \sin^2 \varphi) + \sin \varphi [(\cos^2 \varphi - \sin^2 \varphi)^2 - (2\sin \varphi \cos \varphi)^2] \\ &= 4\sin \varphi \cos^4 \varphi - 4\sin^3 \varphi \cos^2 \varphi + \sin \varphi [\cos^4 \varphi - 2\sin^2 \varphi \cos^2 \varphi + \sin^4 \varphi - 4\sin^2 \varphi \cos^2 \varphi] \\ &= \sin^5 \varphi + 5\sin \varphi \cos^4 \varphi - 10\sin^3 \varphi \cos^2 \varphi \end{aligned}$$

解法二: 见题(5) 解法二

(7) $\cos \varphi + \cos 2\varphi + \cos 3\varphi + \dots + \cos n\varphi$ (8) $\sin \varphi + \sin 2\varphi + \sin 3\varphi + \dots + \sin n\varphi$ 解: (7)、(8) 因为 $e^{i\varphi} = (\cos \varphi + i\sin \varphi)$, 又 $e^{in\varphi} = (\cos n\varphi + i\sin n\varphi)$

$$\text{所以 } e^{i\varphi} + e^{i2\varphi} + e^{i3\varphi} + \dots + e^{in\varphi} = (\cos \varphi + \cos 2\varphi + \cos 3\varphi + \dots + \cos n\varphi) + i(\sin \varphi + \sin 2\varphi + \sin 3\varphi + \dots + \sin n\varphi)$$

$$\begin{aligned} \text{又 } e^{i\varphi} + e^{i2\varphi} + e^{i3\varphi} + \dots + e^{in\varphi} &= \frac{e^{i\varphi} - e^{i(n+1)\varphi}}{1 - e^{i\varphi}} = e^{i\varphi} \frac{1 - e^{in\varphi}}{1 - e^{i\varphi}} = \frac{e^{i\varphi}(1 - e^{in\varphi}) \cdot e^{-i\frac{\varphi}{2}}}{(1 - e^{i\varphi}) \cdot e^{-i\frac{\varphi}{2}}} = \frac{e^{i\varphi/2}(1 - e^{in\varphi})}{e^{-i\frac{\varphi}{2}} - e^{i\frac{\varphi}{2}}} \\ &= \frac{(\cos \varphi/2 + i\sin \varphi/2)(1 - \cos n\varphi - i\sin n\varphi)}{-i \cdot 2\sin \varphi/2} \end{aligned}$$

$$= \frac{(\cos \varphi/2 + i \sin \varphi/2)(1 - \cos n\varphi - i \sin \varphi)}{2 \sin \varphi/2} \quad (i)$$

$$= \frac{i \cos \varphi/2 - \sin \varphi/2 - i \cos \varphi \cos \varphi/2 + i \sin n\varphi \sin \varphi/2 + \sin \varphi/2 \cos n\varphi + \cos \frac{\varphi}{2} \sin n\varphi}{2 \sin \varphi/2}$$

$$= \frac{\left[\sin \left(n + \frac{1}{2} \right) \varphi - \sin \frac{\varphi}{2} \right] + i \left[\cos \frac{\varphi}{2} - \cos \left(n + \frac{1}{2} \right) \varphi \right]}{2 \sin \varphi/2}$$

复数的实部和虚部分别相等,有

$$\cos \varphi + \cos 2\varphi + \cos 3\varphi + \cdots + \cos n\varphi = \frac{\sin \left(n + \frac{1}{2} \right) \varphi - \sin \frac{\varphi}{2}}{2 \sin \varphi/2} \text{ 为(7) 的解.}$$

$$\sin \varphi + \sin 2\varphi + \sin 3\varphi + \cdots + \sin n\varphi = \frac{\cos \frac{\varphi}{2} - \cos \left(n + \frac{1}{2} \right) \varphi}{2 \sin \varphi/2} \text{ 为(8) 的解.}$$

§ 1.2 复变函数

1. 试验证(1.2.11) ~ (1.2.14) 几个式子.

证明:(1) $\sin(z+2\pi) = \sin z$, $\cos(z+2\pi) = \cos z$ (1.2.11)

(a) $\sin(z+2\pi) = \sin z$

$$\begin{aligned} \text{证明: 因为 } \sin(z+2\pi) &= \frac{1}{2i} (e^{i(z+2\pi)} - e^{-i(z+2\pi)}) = \frac{1}{2i} (e^{iz} \cdot e^{i2\pi} - e^{-iz} \cdot e^{-i2\pi}) \\ &= \frac{1}{2i} [e^{iz} (\cos 2\pi + i \sin 2\pi) - e^{-iz} (\cos(-2\pi) + i \sin 2\pi)] \\ &= \frac{1}{2i} (e^{iz} - e^{-iz}) = \sin z \end{aligned}$$

所以 $\sin(z+2\pi) = \sin z$ 得证.

(b) $\cos(z+2\pi) = \cos z$

$$\begin{aligned} \text{证明: 因为 } \cos(z+2\pi) &= \frac{1}{2} (e^{i(z+2\pi)} + e^{-i(z+2\pi)}) \\ &= \frac{1}{2} [e^{iz} (\cos 2\pi + i \sin 2\pi) + e^{-iz} (\cos(-2\pi) + i \sin 2\pi)] \\ &= \frac{1}{2} (e^{iz} + e^{-iz}) = \cos z. \end{aligned}$$

所以 $\cos(z+2\pi) = \cos z$ 得证.

$$(2) |\sin z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\sin^2 x - \cos^2 x)} \quad (1.2.12)$$

证明: 因为 $e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y)$

所以 $e^{iz} = e^{-y+ix} = e^{-y} e^{ix} = e^{-y} (\cos x + i \sin x)$

$$\text{又 } \sin z = \frac{1}{2i} (e^{iz} - e^{-iz}) = \frac{1}{2i} (e^{-y+ix} - e^{y-ix})$$

$$\begin{aligned}
&= \frac{1}{2i} \{ e^{-y}(\cos x + i\sin x) - e^y[\cos(-x) + i\sin(-x)] \} \\
&= \frac{1}{2i} [e^{-y}\cos x - e^y\cos x + i(e^{-y}\sin x + e^y\sin x)] \\
&= \frac{1}{2i} [(e^{-y} - e^y)\cos x + (e^{-y} + e^y)i\sin x] \\
&= \frac{1}{2} [(e^y + e^{-y})\sin x + (e^y - e^{-y})i\cos x] \\
&= \operatorname{ch} y \sin x + \operatorname{sh} y \cos x
\end{aligned}$$

所以 $|\sin z| = |\sin x \operatorname{ch} y + i \cos x \operatorname{sh} y|$

$$\begin{aligned}
\text{即 } |\sin z| &= \sqrt{\sin^2 x \operatorname{ch}^2 y + \cos^2 x \operatorname{sh}^2 y} \\
&= \sqrt{\sin^2 x \left(\frac{e^y + e^{-y}}{2} \right)^2 + \cos^2 x \left(\frac{e^y - e^{-y}}{2} \right)^2} \\
&= \frac{1}{2} \sqrt{\sin^2 x (e^{2y} + 2 + e^{-2y}) + \cos^2 x (e^{2y} - 2 + e^{-2y})} \\
&= \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\sin^2 x - \cos^2 x)}
\end{aligned}$$

所以 $|\sin z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\sin^2 x - \cos^2 x)}$ 得证.

$$(3) |\cos z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)} \quad (1.2.13)$$

$$\begin{aligned}
\text{证明: 因为 } \cos z &= \frac{1}{2}(e^{iz} + e^{-iz}) = \frac{1}{2}(e^{-y+ix} + e^{y-ix}) = \frac{1}{2}[e^{-y}(\cos x + i\sin x) + e^y(\cos x - i\sin x)] \\
&= \frac{1}{2}[(e^y + e^{-y})\cos x - i\sin x(e^y - e^{-y})] \\
&= \cos x \operatorname{ch} y - i \sin x \operatorname{sh} y
\end{aligned}$$

$$\begin{aligned}
\text{所以 } |\cos z| &= \sqrt{\cos^2 x \operatorname{ch}^2 y + \sin^2 x \operatorname{sh}^2 y} \\
&= \sqrt{\cos^2 x \left(\frac{e^{2y} + 2 + e^{-2y}}{4} \right) + \sin^2 x \left(\frac{e^{2y} - 2 + e^{-2y}}{4} \right)} \\
&= \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)}
\end{aligned}$$

即 $|\cos z| = \frac{1}{2} \sqrt{(e^{2y} + e^{-2y}) + 2(\cos^2 x - \sin^2 x)}$ 得证.

$$(4) e^{z+2\pi i} = e^z, \operatorname{sh}(z+2\pi i) = \operatorname{sh} z, \operatorname{ch}(z+2\pi i) = \operatorname{ch} z. \quad (1.2.14)$$

$$(a) e^{z+2\pi i} = e^z$$

$$\begin{aligned}
\text{证明: 因为 } e^{z+2\pi i} &= e^{i2\pi} \cdot e^z = e^{x+i(y+2\pi)} = e^x [\cos^2(y+2\pi) + i\sin(y+2\pi)] = e^x (\cos y + i\sin y) \\
&= e^x e^{iy} = e^{x+iy} = e^z.
\end{aligned}$$

所以 $e^{z+2\pi i} = e^z$. 得证.

$$(b) \operatorname{sh}(z+2\pi i) = \operatorname{sh} z.$$

$$\text{证明: 因为 } \operatorname{sh}(z+2\pi i) = \frac{(e^{z+2\pi i}) - e^{-z-2\pi i}}{2} \quad \text{又 } e^{z+2\pi i} = e^z \quad \text{且 } e^{-z-2\pi i} = \frac{1}{e^{z+2\pi i}} = \frac{1}{e^z} = e^{-z}$$

所以 $\operatorname{sh}(z+2\pi i) = \frac{e^{z+2\pi i} - e^{-(z+2\pi i)}}{2} = \frac{e^z - e^{-z}}{2} = \operatorname{sh} z$ 得证.

(c) $\operatorname{ch}(z+2\pi i) = \operatorname{ch} z$

因为 $\operatorname{ch}(z+2\pi i) = \frac{e^{z+2\pi i} + e^{-(z+2\pi i)}}{2} = \frac{e^z + e^{-z}}{2}$

理同(b) 得 $\operatorname{ch}(z+2\pi i) = \frac{e^z + e^{-z}}{2} = \operatorname{ch} z$ 得证.

2. 计算下列数值. (a 和 b 为实常数, x 为实变数)

(1) $\sin(a+ib)$

$$\begin{aligned}\text{解: } \sin(a+ib) &= \frac{e^{i(a+ib)} - e^{-i(a+ib)}}{2i} = \frac{e^{-b}e^{ia} - e^b e^{-ia}}{2i} = \frac{1}{2i} [e^{-b}(\cos a + i\sin a) - e^b(\cos a - i\sin a)] \\ &= \frac{i}{2} [e^b(\cos a - i\sin a) - e^{-b}(\cos a + i\sin a)] \\ &= \frac{1}{2}(e^b + e^{-b})\sin a + \frac{i}{2}(e^b - e^{-b})\cos a\end{aligned}$$

(2) $\cos(a+ib)$

$$\begin{aligned}\text{解: } \cos(a+ib) &= \frac{e^{i(a+ib)} + e^{-i(a+ib)}}{2} = \frac{1}{2}(e^{-b}e^{ia} + e^b e^{-ia}) \\ &= \frac{1}{2} [e^{-b}(\cos a + i\sin a) + e^b(\cos a - i\sin a)] \\ &= \frac{1}{2} [(e^b + e^{-b})\cos a + i(e^{-b} - e^b)\sin a] \\ &= \frac{1}{2}(e^b + e^{-b})\cos a + \frac{i}{2}(e^{-b} - e^b)\sin a\end{aligned}$$

(3) $\ln(-1)$

解: 取 $z = x + iy$

设 $z = \ln(-1)$ 得 $e^z = -1$ 又 $e^z = e^{x+iy} = e^x(\cos y + i\sin y) = -1$

所以
$$x = 0, \begin{cases} \sin y = 0 \\ \cos y = -1 \end{cases} \quad y = (2n+1)\pi$$

即 $z = iy = i(2n+1)\pi$.

所以 $\ln(-1) = i(2n+1)\pi. \quad (n = 0, 1, 2, \dots)$

(4) $\operatorname{ch}^2 z - \operatorname{sh}^2 z$

$$\begin{aligned}\text{解: } \operatorname{ch}^2 z - \operatorname{sh}^2 z &= \frac{1}{4} [(e^z + e^{-z})^2 - (e^z - e^{-z})^2] \\ &= \frac{1}{4} (e^z + e^{-z} + e^z - e^{-z})(e^z + e^{-z} - e^z + e^{-z}) \\ &= \frac{1}{4} (2e^z)(2e^{-z}) = \frac{4}{4} = 1.\end{aligned}$$

(5) $\cos ix$

$$\text{解: } \cos ix = \frac{1}{2}(e^{i(ix)} + e^{-i(ix)}) = \frac{1}{2}(e^{-x} + e^x) = \operatorname{ch} x$$

(6) $\sin ix$

$$\text{解: } \sin ix = \frac{1}{2i}(e^{i(ix)} - e^{-i(ix)}) = \frac{1}{2i}(e^{-x} - e^x) = \frac{i}{2}(e^x - e^{-x}) = i \operatorname{sh} x$$

(7) $\operatorname{ch} ix$

$$\begin{aligned} \text{解: } \operatorname{ch} ix &= \frac{1}{2}(e^{ix} + e^{-ix}) = \frac{1}{2}(\cos x + i \sin x + \cos x - i \sin x) \\ &= \frac{1}{2}(2 \cos x) = \cos x \end{aligned}$$

(8) $\operatorname{sh} ix$

$$\text{解: } \operatorname{sh} ix = \frac{1}{2}(e^{ix} - e^{-ix}) = \frac{1}{2}(\cos x + i \sin x - \cos x + i \sin x) = i \sin x.$$

(9) $|e^{iaz - ib \sin z}|$

$$\text{解: } |e^{iaz - ib \sin z}| = |e^{iaz} e^{-ib \sin z}| = |e^{-ay} \cdot e^{iax} e^{-ib \sin z}| = |e^{-ay} e^{-ib \sin z}|$$

$$\begin{aligned} ib \sin z &= \frac{ib}{2i}(e^{iz} - e^{-iz}) = \frac{b}{2}(e^{iz} - e^{-iz}) \\ &= \frac{b}{2}(e^{ix} e^{-y} - e^{-ix} e^y) = \frac{b}{2}[(e^{-y} - e^y) \cos x + (e^y + e^{-y}) i \sin x] \end{aligned}$$

$$\text{所以 } |e^{iaz - ib \sin z}| = |e^{-ay} e^{-\frac{b}{2}(e^y - e^{-y}) \cos x}| = |e^{-ay - \frac{b}{2}(e^y - e^{-y}) \cos x}|$$

3. 求解方程 $\sin z = 2$.解: 设 $z = x + iy$

$$\begin{aligned} \sin z &= \frac{e^{iz} - e^{-iz}}{2i} = \frac{-i}{2}(e^{-y} e^{ix} - e^y e^{-ix}) \\ &= \frac{-i}{2}[e^{-y}(\cos x + i \sin x) - e^y(\cos x - i \sin x)] \\ &= \frac{1}{2}(e^y + e^{-y}) \sin x - \frac{i}{2}(e^{-y} - e^y) \cos x \end{aligned}$$

又 $\sin z = 2$.

$$\text{有 } \begin{cases} \frac{1}{2}(e^y + e^{-y}) \sin x = 2 \\ \frac{1}{2}(e^y - e^{-y}) \cos x = 0 \end{cases}$$

当 $\cos x = 0$ 有 $x = (2n + \frac{1}{2})\pi$ 则

$$e^{2y} - 4e^y + 1 = 0$$

$$e^y = 2 \pm \sqrt{3}, y = \ln(2 \pm \sqrt{3})$$

当 $e^y - e^{-y} = 0$, 有 $y = 0$, 则 $\sin x = 2$ (x 为实数, 无解)

$$\text{所以 } \begin{cases} x = (2n + \frac{1}{2})\pi \\ y = \ln(2 \pm \sqrt{3}) \end{cases}$$

$$\text{即 } z = (2n + \frac{1}{2})\pi + i \ln(2 \pm \sqrt{3}) \quad (n = 0, 1, 2, \dots)$$

§ 1.3 导数

试推导极坐标系中的柯西 - 黎曼方程(1.3.4).

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases} \quad (1.3.4)$$

证明:在极坐标下 $f(z) = u(\rho, \varphi) + iv(\rho, \varphi)$

设 Δz 沿径向趋于零,有 $\Delta z = \Delta \rho e^{i\varphi} \rightarrow 0, (\Delta \varphi = 0)$ 则

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} &= \lim_{\Delta \rho e^{i\varphi} \rightarrow 0} \frac{u(\rho + \Delta \rho, \varphi) + iv(\rho + \Delta \rho, \varphi) - u(\rho, \varphi) - iv(\rho, \varphi)}{\Delta \rho e^{i\varphi}} \\ &= \lim_{\Delta \rho \rightarrow 0} \frac{u(\rho + \Delta \rho, \varphi) - u(\rho, \varphi) + i[v(\rho + \Delta \rho, \varphi) - v(\rho, \varphi)]}{\Delta \rho e^{i\varphi}} \\ &= \left[\frac{\partial u}{\partial \rho} + i \frac{\partial v}{\partial \rho} \right] e^{-i\varphi} \end{aligned}$$

设 Δz 沿切向趋于零,有 $\Delta z = i\rho e^{i\varphi} \Delta \varphi \rightarrow 0 (\Delta \rho = 0)$ 则

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} &= \lim_{i\rho e^{i\varphi} \Delta \varphi \rightarrow 0} \frac{u(\rho, \varphi + \Delta \varphi) + iv(\rho, \varphi + \Delta \varphi) - u(\rho, \varphi) - iv(\rho, \varphi)}{i\rho e^{i\varphi} \Delta \varphi} \\ &= \lim_{\Delta \varphi \rightarrow 0} \frac{u(\rho, \varphi + \Delta \varphi) - u(\rho, \varphi) + i[v(\rho, \varphi + \Delta \varphi) - v(\rho, \varphi)]}{i\rho e^{i\varphi} \Delta \varphi} \\ &= \left[\frac{\partial v}{\partial \varphi} - i \frac{\partial u}{\partial \varphi} \right] e^{-i\varphi} \cdot \frac{1}{\rho}. \end{aligned}$$

由柯西 - 黎曼条件,上面两式极限值相等.

所以

$$\frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi}, \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi}.$$

即

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases} \quad \text{得证.}$$

§ 1.4 解析函数

1. 某个区域上的解析函数如为实函数,试证它必为常数.

证明:由题意知 $f(z) = u(x, y) + iv(x, y)$ 在区域上解析,且 $v = 0$. 则由柯西 - 黎曼条件

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad \text{将 } v = 0 \text{ 代入得} \quad \begin{cases} \frac{\partial u}{\partial x} = 0 \\ \frac{\partial u}{\partial y} = 0 \end{cases}$$

即

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0$$

所以

$$u = \int du = C(\text{常数}) \quad \text{得证.}$$

2. 已知解析函数 $f(z)$ 的实部 $u(x, y)$ 或虚部 $v(x, y)$, 求该解析函数.

(1) $u = e^x \sin y$

解:
$$\begin{cases} \frac{\partial u}{\partial x} = e^x \sin y \\ \frac{\partial u}{\partial y} = e^x \cos y \end{cases} \quad \text{即} \quad \begin{cases} \frac{\partial v}{\partial y} = e^x \sin y \\ \frac{\partial v}{\partial x} = -e^x \cos y \end{cases}$$

因为 $dv = \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial x} dx$

$$dv = e^x \sin y dy - e^x \cos y dx = d(-e^x \cos y + C)$$

所以 $v = -e^x \cos y + C$ (C 为常数)

$$\begin{aligned} f(z) &= e^x \sin y - ie^x \cos y + iC = e^x (\sin y - i \cos y) + iC \\ &= -ie^x (\cos y + i \sin y) + iC = -ie^{x+iy} + iC \\ &= -ie^z + iC \end{aligned}$$

(2) $u = e^x (x \cos y - y \sin y), f(0) = 0$

解: $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = e^x (x \cos y - y \sin y) + e^x \cos y,$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -e^x (-x \sin y - \sin y - y \cos y) = e^x [(x+1) \sin y + y \cos y]$$

$$\begin{aligned} dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = e^x (x \cos y - y \sin y + \cos y) dy + e^x [(x+1) \sin y + y \cos y] dx \\ &= d(e^x x \sin y + e^x y \cos y) \end{aligned}$$

$$v = \int dv = e^x x \sin y + e^x y \cos y + C \quad (C \text{ 为常数})$$

$$\begin{aligned} f(z) &= u + iv = e^x (x \cos y - y \sin y) + ie^x (x \sin y + y \cos y) + iC \\ &= e^x (x + iy) (\cos y + i \sin y) + iC = e^x z e^{iy} + iC \\ &= z e^{x+iy} + iC = z e^z + iC \end{aligned}$$

又 $f(0) = 0$, 得 $C = 0$, 所以 $f(z) = z e^z$

(3) $u = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}, f\left(\frac{\pi}{2}\right) = 0,$

解: $u = \frac{2 \sin 2x}{2(\operatorname{ch} 2y - \cos 2x)} = \frac{\sin 2x}{\operatorname{ch} 2y - \cos 2x}$

$$\begin{aligned} \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = \frac{2 \cos 2x}{\operatorname{ch} 2y - \cos 2x} - \frac{2 \sin^2 2x}{(\operatorname{ch} 2y - \cos 2x)^2} \\ &= \frac{2(\operatorname{ch} 2y \cos 2x - 1)}{(\operatorname{ch} 2y - \cos 2x)^2} \end{aligned}$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = \frac{2 \sin 2x \operatorname{sh} 2y}{(\operatorname{ch} 2y - \cos 2x)^2}$$

$$v = \int \frac{2 \sin 2x \operatorname{sh} y}{(\operatorname{ch} 2y - \cos 2x)^2} dx = \int \frac{d \operatorname{sh} 2y (-\cos 2y)}{(\operatorname{ch} 2y - \cos 2x)^2} = \frac{-\operatorname{sh} 2y}{\operatorname{ch} 2y - \cos 2x} + C(y)$$

将所求 v 对 y 求导有

$$\frac{\partial v}{\partial y} = \frac{2(\operatorname{ch} 2y \cos 2x - 1)}{(\operatorname{ch} 2y - \cos 2x)^2} + C'(y).$$

与 $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}$ 比较 $C'(y) = 0$ $C(y) = C(\text{常数})$

所以
$$f(z) = \frac{\sin 2x}{\operatorname{ch} 2y - \cos 2x} - i \frac{\operatorname{sh} 2y}{\operatorname{ch} 2y - \cos 2x} + iC$$

又 $\sin z = \frac{1}{2}[(e^y + e^{-y})\sin x + i(e^y - e^{-y})\cos x] = \operatorname{ch} y \sin x + i \operatorname{sh} y \cos x$

同理 $\cos z = \cos x \operatorname{ch} y - i \sin x \operatorname{sh} y$

$\operatorname{sh} z = -i \sin iz, \operatorname{ch} z = \cos iz$

所以
$$\begin{aligned} \operatorname{ctan} z &= \frac{\cos z}{\sin z} = \frac{\cos x \operatorname{ch} y - i \sin x \operatorname{sh} y}{\sin x \operatorname{ch} y + i \cos x \operatorname{sh} y} \\ &= \frac{\sin 2x - i \operatorname{sh} 2y}{(2\sin^2 x - 1) + (1 + 2\operatorname{sh}^2 y)} = \frac{\sin 2x}{\operatorname{ch} 2y - \cos 2x} - i \frac{\operatorname{sh} 2y}{\operatorname{ch} 2y - \cos 2x} \end{aligned}$$

所以 $f(z) = \frac{\sin 2x}{\operatorname{ch} 2y - \cos 2x} - i \frac{\operatorname{sh} 2y}{\operatorname{ch} 2y - \cos 2x} + iC = \operatorname{ctan} z + iC$

又 $f\left(\frac{\pi}{2}\right) = 0$, 有 $iC = 0$ 即 $C = 0$, 所以 $f(z) = \operatorname{ctan} z$

(4) $v = \frac{y}{x^2 + y^2}, f(2) = 0$

解: 取极坐标 $y = \rho \sin \varphi, x^2 + y^2 = \rho^2$, 所以 $v = \frac{\sin \varphi}{\rho}$

极坐标下的柯西 - 黎曼方程

$$\begin{cases} \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{\partial u}{\partial \rho} \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} \end{cases}$$

$$\begin{aligned} \frac{\partial u}{\partial \rho} &= \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{1}{\rho^2} \cos \varphi \begin{cases} du = \frac{\cos \varphi}{\rho^2} d\rho + \frac{\sin \varphi}{\rho} d\varphi \\ u = \int du = \int d\left(-\frac{\cos \varphi}{\rho}\right) \end{cases} \\ \frac{\partial u}{\partial \varphi} &= -\rho \frac{\partial v}{\partial \rho} = \frac{\sin \varphi}{\rho} \end{aligned}$$

所以 $u = -\frac{\cos \varphi}{\rho} + C$

即 $f(z) = -\frac{\cos \varphi}{\rho} + C + i \frac{\sin \varphi}{\rho} = -\frac{1}{\rho} e^{-i\varphi} + C = \frac{-1}{\rho e^{i\varphi}} + C = \frac{-1}{z} + C$

$f(2) = -\frac{1}{2} + C = 0, C = \frac{1}{2}$. 所以 $f(z) = -\frac{1}{z} + \frac{1}{2}$.

(5) $u = \frac{x^2 - y^2}{(x^2 + y^2)^2}, f(\infty) = 0$

解: 化成极坐标形式

$$u = \frac{\rho^2 \cos^2 \varphi - \rho^2 \sin^2 \varphi}{\rho^4} = \frac{\cos 2\varphi}{\rho^2}$$

$$\frac{\partial v}{\partial \varphi} = \rho \frac{\partial u}{\partial \rho} = -\frac{2 \cos 2\varphi}{\rho^2}$$

$$\frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = +\frac{2 \sin 2\varphi}{\rho^3}$$

$$dv = \frac{\partial v}{\partial \rho} d\rho + \frac{\partial v}{\partial \varphi} d\varphi = +\frac{2 \sin 2\varphi}{\rho^3} d\rho - \frac{2 \cos 2\varphi}{\rho^2} d\varphi$$

$$v = \int dv = + \int d\left(-\frac{\sin 2\varphi}{\rho^2}\right) = -\frac{\sin 2\varphi}{\rho^2} + C$$

$$f(z) = u + iv = \frac{\cos 2\varphi}{\rho^2} - i \frac{\sin 2\varphi}{\rho^2} + iC = \frac{1}{\rho^2} e^{-i2\varphi} + iC = \frac{1}{\rho^2 e^{i2\varphi}} + iC = \frac{1}{z^2} + iC$$

因为当 $z \rightarrow \infty$ 时 $f(z) = 0$, 所以 $C = 0$. $f(z) = \frac{1}{z^2}$

$$(6) u = x^2 - y^2 + xy, f(0) = 0$$

解:
$$\left. \begin{aligned} \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = 2x + y \\ \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = 2y - x \end{aligned} \right\} dv = d\left(2x - y + \frac{1}{2}(y^2 - x^2)\right)$$

$$v = \int dv = 2xy + \frac{1}{2}(y^2 - x^2) + C$$

$$\begin{aligned} f(z) &= x^2 - y^2 + xy + i\left[2xy + \frac{1}{2}(y^2 - x^2) + C\right] \\ &= (x^2 - y^2 + i2xy) + i\left[\frac{1}{2}(y^2 - x^2) - ixy + C\right] \\ &= (x + iy)^2 - i\frac{1}{2}(x + iy)^2 + iC \\ &= z^2 - i\frac{1}{2}z^2 + iC \end{aligned}$$

因为当 $z = 0$, $f(0) = 0$, 所以 $C = 0$.

即
$$f(z) = z^2 - i\frac{1}{2}z^2 = z^2\left(1 - \frac{i}{2}\right)$$

$$(7) u = x^3 - 3xy^2, f(0) = 0$$

解: 因为
$$\left\{ \begin{aligned} \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = 3x^2 - 3y^2 \\ \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = 6xy \end{aligned} \right.$$

所以 $dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = 6xy dx = (3x^2 - 3y^2) dy = d(3x_y^2 - y^3)$

$$v = \int dv$$

所以 $v = 3x^2 y - y^3 + C$

$$f(z) = u + iv = x^3 - 3xy^2 + i(3x_y^2 - y^3) + iC = z^3 + iC$$

因为 $z^3 = (x + iy)^3 = (x^2 + 2xyi - y^2)(x + iy) = x^3 + i3x_y^2 - 3xy^2 - iy^3$

又 $f(0) = 0$, 所以 $C = 0$.

即 $f(z) = z^3$

$$(8) u = x^3 + 6x^2y - 3xy^2 - 2y^3, f(0) = 0$$

解:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 3x^2 + 12xy - 3y^2$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -6x^2 + 6xy + 6y^2$$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = (-6x^2 + 6xy + 6y^2)dx + (3x^2 + 12xy - 3y^2)dy$$

$$= d(-2x^3 + 3x^2y + 6xy^2 - y^3), v = \int dv$$

$$v = -2x^3 + 3x^2y + 6xy^2 - y^3 + c.$$

$$f(z) = u + iv = x^3 + 6x^2y - 3xy^2 - 2y^3 + i(-2x^3 + 3x^2y + 6xy^2 - y^3) + iC$$

$$= x^3 + i3x^2y - 3xy^2 - iy^3 + [6x^2y - 2y^3 + i(-2x^3 + 6xy^2)] + iC$$

$$= z^3 - 2i(x^3 + i3x^2y - 3xy^2 - iy^3) + iC = z^3 - 2iz^3 + iC$$

$$= z^3(1 - 2i) + iC$$

又 $f(0) = 0$, 所以 $C = 0$, 即 $f(z) = z^3(1 - 2i)$

$$(9) u = x^4 - 6x^2y^2 + y^4, f(0) = 0$$

解:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 4x^3 - 12xy^2$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = +12x^2y - 4y^3$$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = (12x^2y - 4y^3)dx + (4x^3 - 12xy^2)dy$$

$$= d(4x^3y - 4y^3x)$$

$$v = \int dv = 4x^3y - 4y^3x + C$$

$$\text{所以 } f(z) = u + iv = x^4 - 6x^2y^2 + y^4 + i(4x^3y - 4y^3x) + iC$$

$$\text{又 } z^4 = (x + iy)^4 = x^4 + i4x^3y - 6x^2y^2 - i4xy^3 + y^4$$

所以 $f(z) = z^4 + iC$, 又 $f(0) = 0$, 所以 $C = 0$. 即 $f(z) = z^4$.

$$(10) u = \ln \rho, f(1) = 0$$

解: 因为

$$\begin{cases} \frac{\partial v}{\partial \varphi} = \rho \frac{\partial u}{\partial \rho} = 1 \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = 0 \end{cases}$$

$$dv = \frac{\partial v}{\partial \rho}d\rho + \frac{\partial v}{\partial \varphi}d\varphi = d\varphi$$

$$\text{所以 } v = \int dv = \varphi + C$$

$$\text{即 } f(z) = u + iv = \ln \rho + i\varphi + iC = \ln \rho + i\varphi \ln e + iC = \ln(\rho e^{i\varphi}) + iC = \ln z + iC$$

因为 $f(1) = 0$, 所以 $C = 0$, 即 $f(z) = \ln z$

$$(11) u = \varphi, f(1) = 0$$

解: 因为

$$\begin{cases} \frac{\partial v}{\partial \varphi} = \rho \frac{\partial u}{\partial \rho} = 0 \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{1}{\rho} \end{cases}$$

$$\text{所以 } dv = \frac{\partial v}{\partial \rho} d\rho + \frac{\partial v}{\partial \varphi} d\varphi = -\frac{1}{\rho} d\rho = -d(\ln \rho)$$

$$v = \int dv = -\ln \rho + C.$$

$$\text{所以 } f(z) = u + iv = \varphi + i(-\ln \rho) + iC = -i(\ln \rho + i\varphi) + iC = -i \ln(\rho e^{i\varphi}) + iC \\ = -i \ln z + iC$$

因为 $f(1) = 0$, 所以 $C = 0$, 即 $f(z) = -i \ln z$

3. 试从极坐标系中的柯西 - 黎曼方程(1.3.4) 消去 u 或 v .

本题答案就是拉普拉斯方程(1.4.2) 在极坐标系中的表示式.

解: 柯西 - 黎曼方程(1.3.4) 的极坐标表示为

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \end{cases} \quad (1)$$

$$\begin{cases} \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases} \quad (2)$$

首先消去 v , 将柯西 - 黎曼方程(1) 和(2) 式变形为

$$\begin{cases} \rho \frac{\partial u}{\partial \rho} = \frac{\partial v}{\partial \varphi} \end{cases} \quad (3)$$

$$\begin{cases} -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} = \frac{\partial v}{\partial \rho} \end{cases} \quad (4)$$

将(3) 式、(4) 式分别对 ρ 和 φ 求微分.

$$\text{对(3) 式} \quad \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) = \frac{\partial^2 v}{\partial \rho \partial \varphi} \quad (5)$$

$$\text{对(4) 式} \quad -\frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2} = \frac{\partial^2 v}{\partial \varphi \partial \rho} \quad (6)$$

由于 $\frac{\partial^2 v}{\partial \varphi \partial \rho} = \frac{\partial^2 v}{\partial \rho \partial \varphi}$, 有 $\frac{\partial}{\partial \rho} \rho \frac{\partial u}{\partial \rho} = -\frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2}$. 即 $\frac{\partial}{\partial \rho} \rho \frac{\partial u}{\partial \rho} + \frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2} = 0$ 为消去 v 的结果.

再将 v 从柯西 - 黎曼方程中消去. 将柯西 - 黎曼方程变形为

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \end{cases} \quad (a)$$

$$\begin{cases} \frac{\partial u}{\partial \varphi} = -\rho \frac{\partial v}{\partial \rho} \end{cases} \quad (b)$$

将(a)(b) 式分别对 φ 和 ρ 求微分.

$$\text{对(a) 式} \quad \frac{\partial^2 u}{\partial \varphi \partial \rho} = \frac{\partial}{\partial \varphi} \left(\frac{1}{\rho} \frac{\partial v}{\partial \varphi} \right) = \frac{1}{\rho} \frac{\partial^2 v}{\partial \varphi^2} \quad (c)$$

$$\text{对(b) 式} \quad \frac{\partial^2 u}{\partial \rho \partial \varphi} = \frac{\partial}{\partial \rho} \left(-\rho \frac{\partial v}{\partial \rho} \right) = -\frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) \quad (d)$$

由于 $\frac{\partial^2 u}{\partial \varphi \partial \rho} = \frac{\partial^2 u}{\partial \rho \partial \varphi}$, 所以 $\frac{1}{\rho} \frac{\partial^2 v}{\partial \varphi^2} = -\frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right)$

即 $\frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial^2 v}{\partial \varphi^2} = 0$ 为消去 u 的结果.

$$\begin{cases} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial^2 u}{\partial \varphi^2} = 0 \\ \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial^2 v}{\partial \varphi^2} = 0 \end{cases}$$

为极坐标系中拉普拉斯方程的表示式.

§ 1.5 平面标量场

1. 已知复势 $f(z) = 1/(z-2+i)$, 试描画等温网.

$$\begin{aligned} \text{解: } f(z) &= 1/(z-2+i) = \frac{1}{(x-2)+i(y+1)} \\ &= \frac{(x-2)-i(y+1)}{(x-2)^2+(y+1)^2} = \frac{x-2}{(x-2)^2+(y+1)^2} - i \frac{y+1}{(x-2)^2+(y+1)^2} \end{aligned}$$

等温网曲线方程

$$\begin{cases} \frac{x-2}{(x-2)^2+(y+1)^2} = \frac{1}{C_1} \\ \frac{y+1}{(x-2)^2+(y+1)^2} = \frac{1}{C_2} \end{cases}$$

$$\text{即 } \begin{cases} \left(x-2-\frac{C_1}{2}\right)^2 + (y+1)^2 = \left(\frac{C_1}{2}\right)^2 \\ (x-2)^2 + \left(y+1-\frac{C_2}{2}\right)^2 = \left(\frac{C_2}{2}\right)^2 \end{cases} \quad \text{取 } \begin{cases} C_1 = 2C_3 \\ C_2 = 2C_4 \end{cases}$$

$$\text{得 } \begin{cases} (x-2-C_3)^2 + (y+1)^2 = C_3^2 \\ (x-2)^2 + (y+1-C_4)^2 = C_4^2 \end{cases}$$

等温网为在 $(2, -1)$ 点上与直线 $x=2, y=-1$ 相切的圆族. 如图 1-5-1 所示.

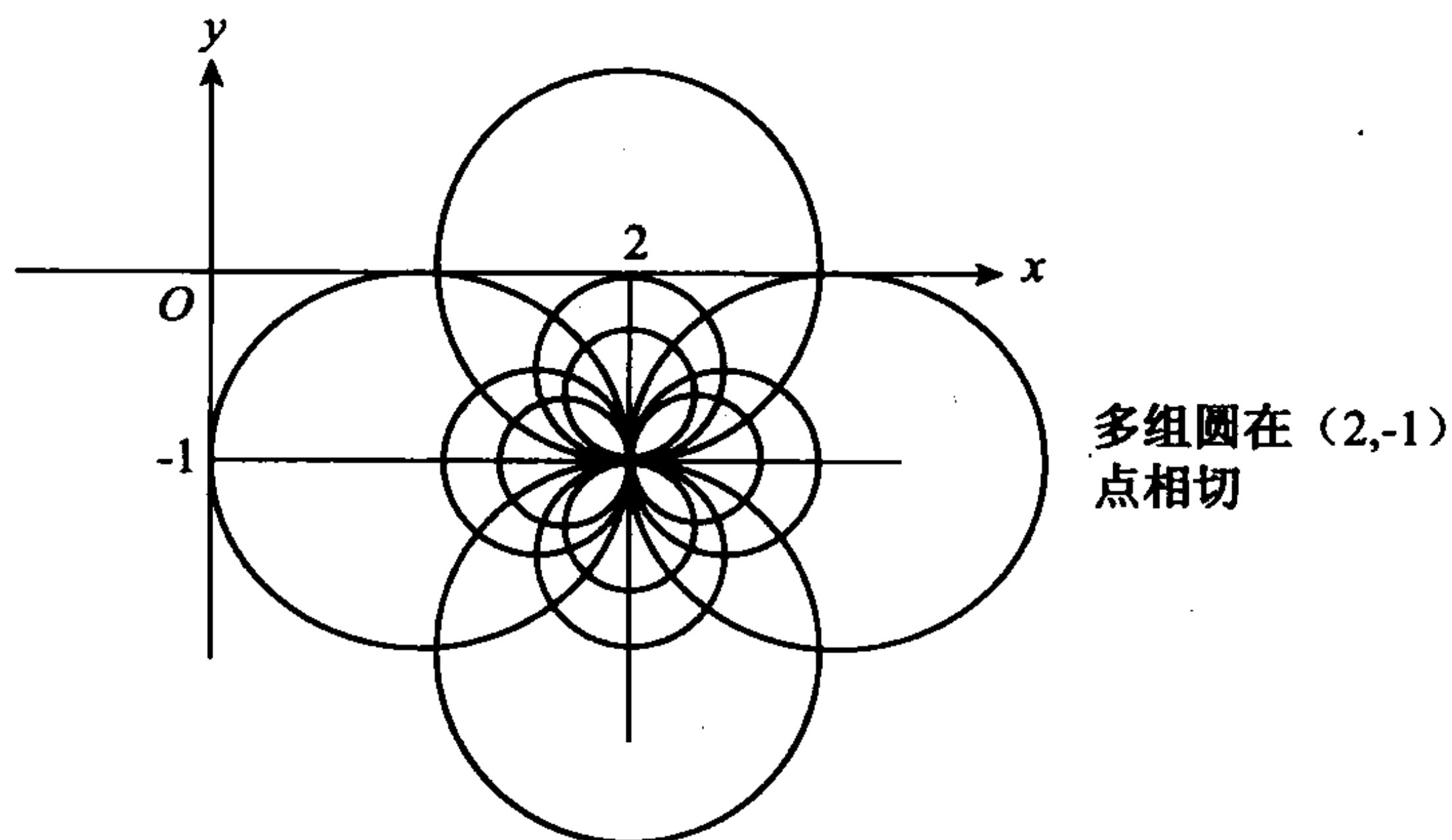


图 1-5-11

2. 已知流线族的方程为“ $y/x = \text{常数}$ ”, 求复势.

解: $\frac{y}{x} = C$ (C 为常数)

令 $v = F(t)$ ($t = y/x$) F 为尚待确定的函数关系

$$\begin{cases} \frac{\partial v}{\partial x} = F'(t) \left(\frac{-y}{x^2} \right) & \begin{cases} \frac{\partial^2 v}{\partial x^2} = F''(t) \left(\frac{y^2}{x^4} \right) + F' \left(\frac{2y}{x^3} \right) \\ \frac{\partial^2 v}{\partial y^2} = F''(t) \left(\frac{1}{x^2} \right) \end{cases} \\ \frac{\partial v}{\partial y} = F'(t) \left(\frac{1}{x} \right) \end{cases}$$

代入拉普拉斯方程

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

得

$$F''(t) \left(\frac{y^2}{x^4} \right) + F''(t) \left(\frac{1}{x^2} \right) + F' \left(\frac{2y}{x^3} \right) = 0$$

$$\frac{F''(t)}{F'(t)} = - \frac{\left[\frac{y^2}{x^4} + \frac{1}{x^2} \right]^{-1}}{\left[\frac{2y}{x^3} \right]^{-1}} = - \frac{2yx^4}{x^3(x^2 + y^2)}$$

$$\frac{F''(t)}{F'(t)} = \frac{-2yx}{x^2 + y^2} = - \frac{2}{\frac{x}{y} + \frac{y}{x}} = \frac{-2}{t + \frac{1}{t}} = - \frac{2t}{1 + t^2}$$

$$[\ln F'(t)]' = \frac{-2t}{1 + t^2},$$

$$\ln F'(t) = \int \frac{-2t}{1 + t^2} dt = - \int \frac{1}{1 + t^2} d(t^2 + 1)$$

$$\ln F'(t) = - \ln(1 + t^2) + C.$$

$$F'(t) = \frac{C_1}{1 + t^2}$$

$$F(t) = C_1 \arctan(t) + C_2$$

所以

$$v = C_1 \arctan \left(\frac{y}{x} \right) + C_2$$

根据柯西 - 黎曼方程

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left(C_1 \arctan \left(\frac{y}{x} \right) + C_2 \right) = \frac{C_1 x}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = - \frac{\partial v}{\partial x} = \frac{C_1 y}{x^2 + y^2}$$

$$\begin{aligned} du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = C_1 \left(\frac{x dx + y dy}{x^2 + y^2} \right) \\ &= \frac{1}{2} C_1 \frac{d(x^2 + y^2)}{x^2 + y^2} = \frac{1}{2} C_1 d[\ln(x^2 + y^2) + C_3] \end{aligned}$$

$$\text{复势 } f(z) = u + iv = \frac{1}{2} C_1 \ln(x^2 + y^2) + C_3 + i(C_1 \arctan \frac{y}{x} + C_2)$$

$$= \frac{1}{2} C_1 \ln \rho^2 + C_3 + i[C_1 \arctan(\tan \varphi) + C_2]$$

$$\begin{aligned}
 &= \frac{1}{2}C_1 \ln \rho^2 + iC_1 \varphi + C_3 + iC_2 = C_1 \ln \rho e^{i\varphi} + C_3 + iC_2 \\
 &= C_1 \ln z + C_3 + iC_2
 \end{aligned}$$

3. 已知等势线族的方程为“ $x^2 + y^2 = \text{常数}$ ”, 求复势.

解: $x^2 + y^2 = C$ (常数)

$$\begin{aligned}
 u &= F(t), & t &= x^2 + y^2, \\
 \frac{\partial u}{\partial x} &= F'(2x) = 2xF' & \frac{\partial^2 u}{\partial x^2} &= 2F' + (2x)^2 F'' \\
 \frac{\partial u}{\partial y} &= F'(2y) = 2yF' & \frac{\partial^2 u}{\partial y^2} &= 2F' + (2y)^2 F'' \\
 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 4F' + 4(x^2 + y^2)F'' = 0
 \end{aligned}$$

有

$$F' + (x^2 + y^2)F'' = 0.$$

$$\frac{F''}{F'} = -\frac{1}{x^2 + y^2} = -\frac{1}{t}$$

$$\ln F' = -\ln t + C_1, F' = \frac{C_1}{t}$$

$$F = C_1 \ln t + C_2 = C_1 \ln(x^2 + y^2) + C_2$$

得

$$u = C_1 \ln(x^2 + y^2) + C_2$$

由柯西 - 黎曼方程

$$\begin{aligned}
 \frac{\partial v}{\partial y} &= \frac{\partial u}{\partial x} = \frac{2xC_1}{x^2 + y^2}, & \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y} = \frac{-2yC_1}{x^2 + y^2} \\
 dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy = \frac{2C_1}{x^2 + y^2} (xdy - ydx) \\
 &= 2C_1 \left[\frac{d\left(\frac{y}{x}\right)_y}{1 + \left(\frac{y}{x}\right)^2} - \frac{d\left(\frac{y}{x}\right)_x}{1 + \left(\frac{y}{x}\right)^2} \right] = 2C_1 \arctan \frac{y}{x} + C_3
 \end{aligned}$$

$$\text{复势 } f(z) = u + iv = C_1 \ln(x^2 + y^2) + C_2 + i2C_1 \arctan \frac{y}{x} + iC_3$$

$$= C_1 (\ln \rho^2 + i2\varphi) + C_2 + iC_3$$

$$= C_1 \ln z^2 + C_2 + iC_3 = 2C_1 \ln z + C_2 + iC_3$$

$$= C'_1 \ln z + C'_2 \quad (\text{其中 } x^2 + y^2 = \rho^2, \arctan \frac{y}{x} = \varphi).$$

4. 已知电场线为跟实轴相切于原点的圆族, 求复势.

解: 由题意知该圆族的方程为 (如图 1-5-4 所示)

$$x^2 + (y - R)^2 = R^2 \quad (R \text{ 为实数})$$

$$x^2 + y^2 - 2Ry = 0$$

$$\frac{1}{2R} = \frac{y}{x^2 + y^2}$$

令 $v = \frac{y}{x^2 + y^2} + C_1$

有 $\frac{\partial v}{\partial x} = \frac{-2xy}{(x^2 + y^2)^2},$

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} &= \frac{-2y}{(x^2 + y^2)^2} + \frac{2xy \cdot 2x \cdot 2(x^2 + y^2)}{(x^2 + y^2)^4} \\ &= \frac{-2y(x^2 + y^2) + 8x^2y}{(x^2 + y^2)^3} = \frac{-6yx^2 + 2y^3}{(x^2 + y^2)^3}\end{aligned}$$

$$\frac{\partial v}{\partial y} = \frac{1}{x^2 + y^2} - \frac{2y^2}{(x^2 + y^2)^2}$$

$$\begin{aligned}\frac{\partial^2 v}{\partial y^2} &= \frac{-2y}{(x^2 + y^2)^2} - \frac{4y}{(x^2 + y^2)^2} + \frac{8y^3(x^2 + y^2)}{(x^2 + y^2)^4} \\ &= \frac{-2y(x^2 + y^2) + 8x^2y}{(x^2 + y^2)^3} = \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3}\end{aligned}$$

所以 $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{-6yx^2 + 2y^3}{(x^2 + y^2)^3} + \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3} = 0$

即 $v = \frac{y}{x^2 + y^2}$ 是调和函数. 极坐标下 $v = \frac{y}{x^2 + y^2} = \frac{\rho \sin \rho}{\rho^2} = \frac{\sin \rho}{\rho}$

极坐标下的柯西 - 黎曼方程

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases} \quad \begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{\cos \rho}{\rho^2} \\ \frac{\partial u}{\partial \varphi} = -\rho \frac{\partial v}{\partial \rho} = \frac{\sin \rho}{\rho} \end{cases}$$

有 $du = \frac{\partial u}{\partial \varphi} d\varphi + \frac{\partial u}{\partial \rho} d\rho = d\left(-\frac{\cos \varphi}{\rho}\right)$

有 $u = -\frac{\cos \varphi}{\rho} + c$

$$\begin{aligned}\text{所以复势 } f(z) = u + iv &= -\frac{\cos \rho}{\rho} + C + i\left(\frac{\sin \varphi}{\rho}\right) = -\left[\frac{\cos \varphi}{\rho} - i\frac{\sin \varphi}{\rho}\right] + C \\ &= -\frac{1}{\rho}(\cos \varphi - i \sin \varphi) + C = -\frac{1}{\rho} e^{i\varphi} + C = -\frac{1}{\rho e^{i\varphi}} + C \\ &= -\frac{1}{z} + C\end{aligned}$$

由于差常数项, 所以 $f(z) = \frac{-C_1}{z} + C_2 + iC_3 = \frac{C'i}{z} + C_2 + iC_3$ (取 $C'i = -C_1$)

5. 在圆柱 $|z| = R$ 的外部的平面静电场的复势为 $f(z) = i2\sigma \ln(R/z)$, 求柱面上的电荷面密度.

解: 静电场的复势 $f(z) = i2\sigma \ln \frac{R}{z} = i2\sigma \left(\ln \frac{R}{\rho e^{i\varphi}}\right) = i2\sigma \left(\ln \frac{R}{\rho} - i\varphi\right) = 2\sigma\varphi + i2\sigma \ln \frac{R}{\rho}$

电势 $u = 2\sigma \ln \frac{R}{\rho}$

则 $E = -\frac{\partial u}{\partial \rho} = -2\sigma \frac{\rho}{R} \left(-\frac{R}{\rho^2}\right) = \frac{2\sigma}{\rho}$

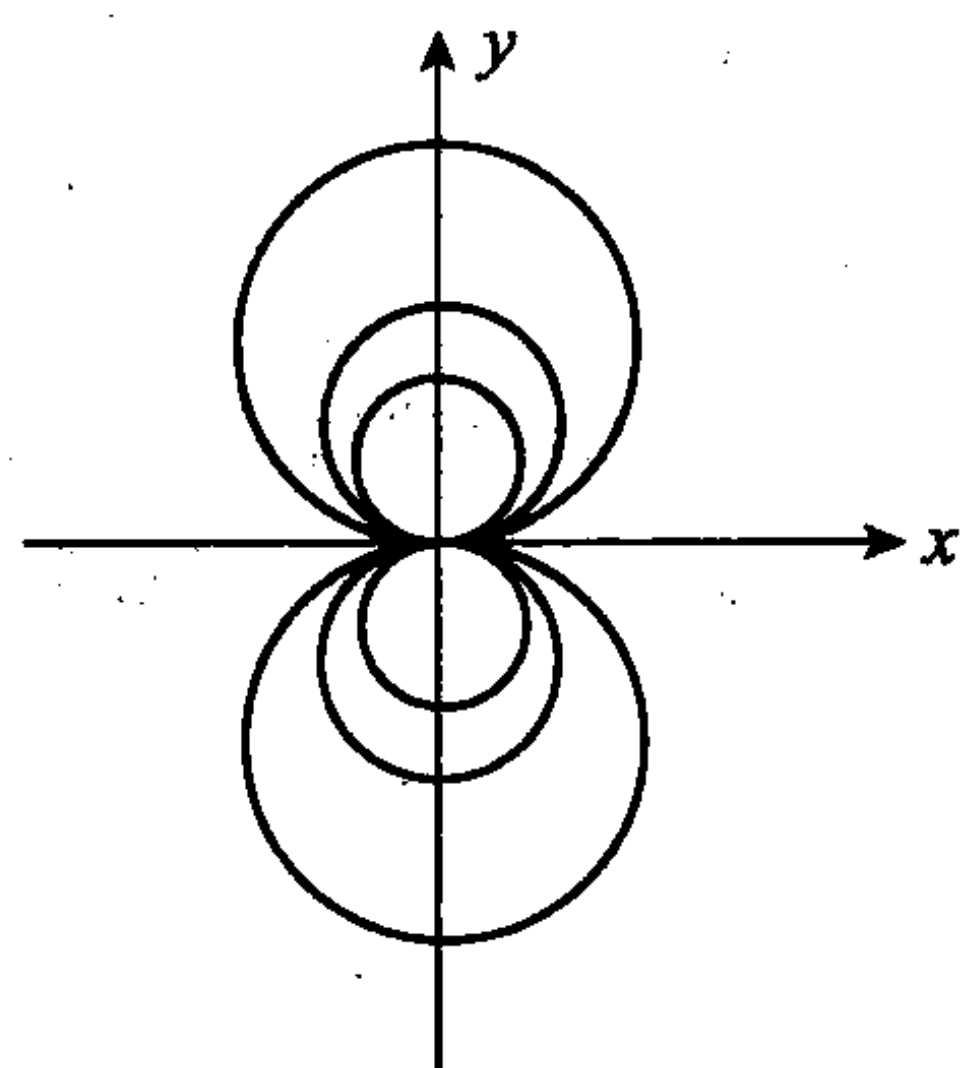


图 1-5-4

当 $\rho = R$ 时, $E = \frac{2\sigma}{R}$

由高斯定理有 $\oint E \cdot dS = 4\pi q, ES = 4\pi q, q = \sigma_e S$

$$\frac{2\sigma}{R} \cdot S = 4\pi q = 4\pi\sigma_e S$$

所以 $\sigma_e = \frac{\sigma}{2\pi R}$ (σ_e 为电荷面密度)

6. 有两个平行而均匀带电的线电荷, 每单位长度所带电量分别是 $+q$ 和 $-q$, 两线相距 $2a$. 求这个平面静电场的复势、电场线和等势线.

解: 设一线电荷在原点, 单位长度所带电量为 Q , 取通量函数为 $v = 2Q\varphi$, 其中 u 为电势, 代入极坐标系中的柯西 - 黎曼方程

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{1}{\rho} \frac{\partial u}{\partial \varphi} = -\frac{\partial v}{\partial \rho} \end{cases}$$

$$\begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} = \frac{2Q}{\rho} \\ \frac{\partial u}{\partial \varphi} = -\rho \frac{\partial v}{\partial \rho} = 0 \end{cases}$$

得

所以 $u = 2Q \ln \rho + C$

$$f(z) = u + iv = 2Q \ln \rho + i2Q\varphi + C$$

$$\begin{aligned} \text{复势} \quad &= 2Q(\ln \rho + i\varphi) + C = 2Q(\ln \rho e^{i\varphi}) + C \\ &= 2Q \ln z + C \end{aligned}$$

(1) 取 $Q = +q$, 线电荷点坐标为 $(a, 0)$

$$\text{复势} \quad f_1(z) = 2q \ln(z - a) + C_1$$

(2) 取 $Q = -q$, 线电荷点坐标为 $(-a, 0)$

$$\text{复势} \quad f_2(z) = -2q \ln(z + a) + C_2$$

由叠加原理, 所求复势为 $f(z) = f_1(z) + f_2(z)$

$$\text{即} \quad f(z) = f_1(z) + f_2(z) = 2q \ln(z - a) - 2q \ln(z + a) + C \quad (\text{其中 } C = C_1 + C_2)$$

$$f(z) = 2q \ln \left(\frac{z - a}{z + a} \right) + C$$

电力线族为虚部

$$\operatorname{Im} f(z) = \text{常数}$$

等势线族为实部

$$\operatorname{Re} f(z) = \text{常数}$$

又

$$\begin{aligned} \ln \frac{z - a}{z + a} &= \ln \frac{x + iy - a}{x + iy + a} = \ln \frac{x^2 + y^2 - a^2 + 2iay}{(x + a)^2 + y^2} \\ &= \ln \frac{\sqrt{(x^2 + y^2 - a^2)^2 + 4a^2 y^2}}{(x + a)^2 + y^2} e^{i \arctan \frac{2ay}{x^2 + y^2 - a^2}} \\ &= \frac{1}{2} \ln \frac{(x^2 + y^2 - a^2)^2 - 4a^2 y^2}{[(x + a)^2 + y^2]^2} + i \arctan \frac{2ay}{x^2 + y^2 - a^2} \end{aligned}$$

$$= \frac{1}{2} \ln \frac{(x-a)^2 + y^2}{(x+a)^2 + y^2} + i \arctan \frac{2ay}{x^2 + y^2 - a^2}$$

虚部(Im) 为电力线族

$$\arctan \frac{2ay}{x^2 + y^2 - a^2} = C'_1 \Rightarrow \frac{2ay}{x^2 + y^2 - a^2} = C''$$

即

$$x^2 + y^2 - 2ayC''_1 - a^2 = 0 \quad (C''_1 = 1/C'_1)$$

$$x^2 + y^2 - 2ayC''_1 + a^2 C''_1{}^2 = a^2 + a^2 C''_1{}^2$$

$$x^2 + (y - aC''_1)^2 = a^2(1 + C''_1{}^2)$$

实部(Re) 为等势线族

$$\frac{(x-a)^2 + y^2}{(x+a)^2 + y^2} = C'_2 \Rightarrow [(x-a)^2 + y^2] = C'_2 [(x+a)^2 + y^2]$$

取($C'_2 = 1/C'_2$)

有

$$C''_2 [(x-a)^2 + y^2] = (x+a)^2 + y^2$$

化简后

$$\left(x - \frac{C''_2 + 1}{C''_2 - 1}a\right)^2 + y^2 = \frac{4C''_2}{(C''_2 - 1)}a^2$$

§ 1.6 多值函数

指出下列多值函数的支点及其阶,并作出黎曼面.

(1) $\sqrt{z-a}$

解: 设 $w = \sqrt{z-a}$, $w^2 = z-a$

指数表达式

$$w = \rho e^{i\varphi_1} = |z-a| e^{i\varphi/2}$$

由对应关系

$$\rho = |z-a| \text{ 是唯一的}$$

$$e^{i\varphi_1} = e^{i\varphi/2}, \text{ 得 } 2\varphi = \varphi + 2n\pi$$

$\varphi_1 = \frac{\varphi}{2} + n\pi$, 对于不同的 n , w 只有两个值

$w_1 = \rho e^{i\frac{\varphi}{2}}$ 和 $w_2 = -\rho e^{i\frac{\varphi}{2}}$, 其它值为这两个值的重复.

当 $z=a$ 时具有这样的特征: 当 z 绕 a 点一周回到原处时, 对应的函数值 w 不还原, 当 z 绕 a 点 2 周时 ($n=2$), 函数值复原, 由此知 a 为多值函数的一阶支点 ($n-1$ 阶). 又 $z=\infty$ 亦是 $w = \sqrt{z-a}$ 的一阶支点. 设 $z = \frac{1}{t}$, 则有

$w = \frac{\sqrt{1-at}}{\sqrt{t}}$, 当 z 绕 $t=0$ 一周, 函数 w 值不还原, 绕 2 周后 w 值复原.

$z=\infty$ ($t=0$) 为 $\sqrt{z-a}$ 的一阶支点.

黎曼面如图 1-6-1 所示.

(2) $\sqrt{(z-a)(z-b)}$

解: 设 $w = \sqrt{(z-a)(z-b)} = \rho e^{i\varphi}$, $z-a = \rho_1 e^{i\theta_1}$, $z-b = \rho_2 e^{i\theta_2}$

$$\text{得 } w = \sqrt{(z-a)(z-b)} = \rho e^{i\varphi} = \sqrt{\rho_1 \rho_2} e^{i\frac{\theta_1 + \theta_2}{2}}$$

所以 $\rho = \sqrt{\rho_1 \rho_2}$, $2\theta = \theta_1 + \theta_2 + 2n\pi$ ($n=0, \pm 1, \dots$)

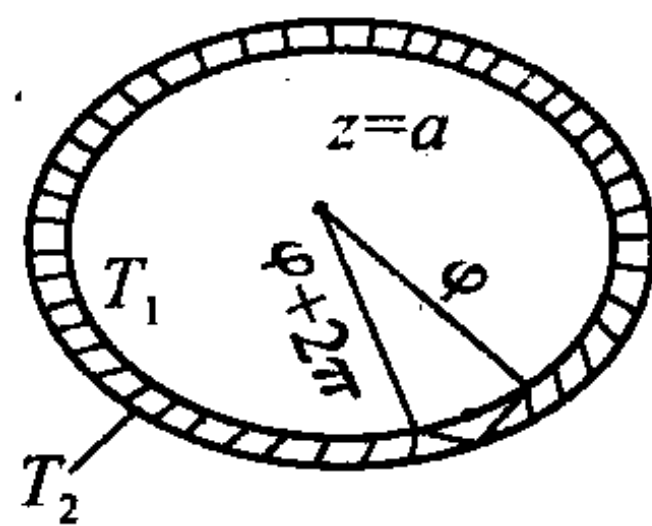


图 1-6-1

$z = a, z = b$, 是多值函数的一阶支点.

令 $z = \frac{1}{t}$, 则 $w = \frac{\sqrt{(1-at)(1-bt)}}{t}$, 当 t 绕 $t = 0$, 转一圈回到原处时, w 值不能还原, 绕两圈回到原处时, w 值复原, 所以 $z = \infty$ 也是一阶支点.

黎曼面如图 1-6-2 所示.

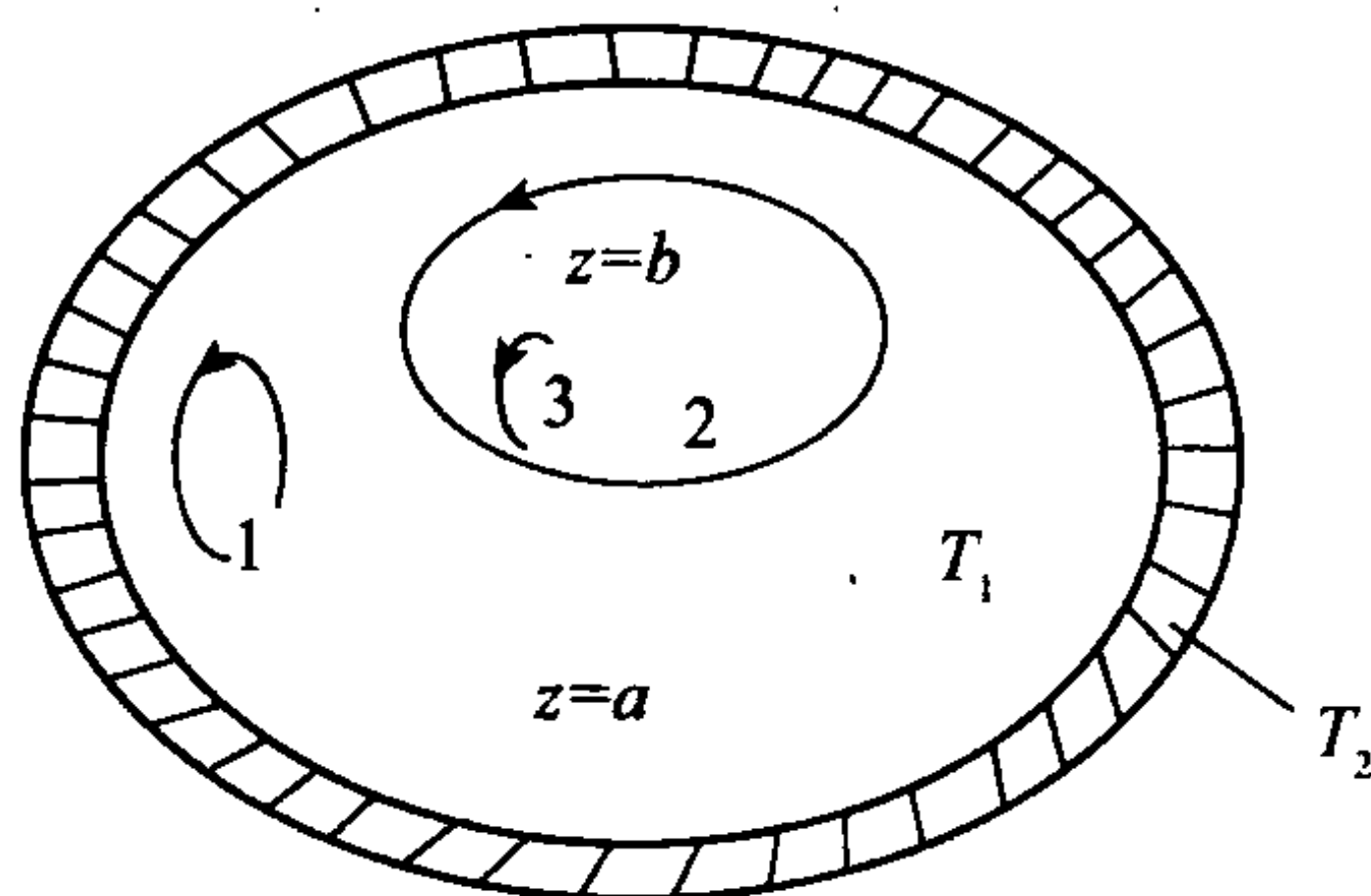


图 1-6-2

沿不包围 a, b 的闭合回路 1 环行一周, 辐角 θ_1 和 θ_2 返回原值. 沿包围两个点的 a 和 b 闭合回路 2 环行一周, 此二辐角各增加 2π , 而 $\frac{1}{2}(\theta_1 + \theta_2)$ 也增加 2π , 函数值 w 还原. 若只包围 a 点 (或 b 点) 的闭合回路 3 环行一周, 函数值 w 不还原.

(3) $\ln z$

解: 设 $w = \ln z$, 则 $e^w = z$. 若取 $w = u + iv$.

$$z = \rho e^{iQ}, \text{ 则: } e^w = e^u e^{iv} = z = \rho e^{iQ}$$

$$e^u = \rho, \quad u = \ln \rho = \ln |z| \quad (\rho = |z|)$$

$$e^{iv} = e^{iQ}, \quad v = Q + 2\pi n = \text{Arg } z \quad (n = 0, \pm 1, \pm 2, \dots)$$

所以 $w = \ln |z| + i \text{Arg}(z) = \ln \rho + i(Q + 2\pi n)$

对于每一个 z , 有无穷多个 w 值, 其支点为 $z = 0$ 和 $z = \infty$, 为无限阶支点.

黎曼面如图 1-6-3 所示.

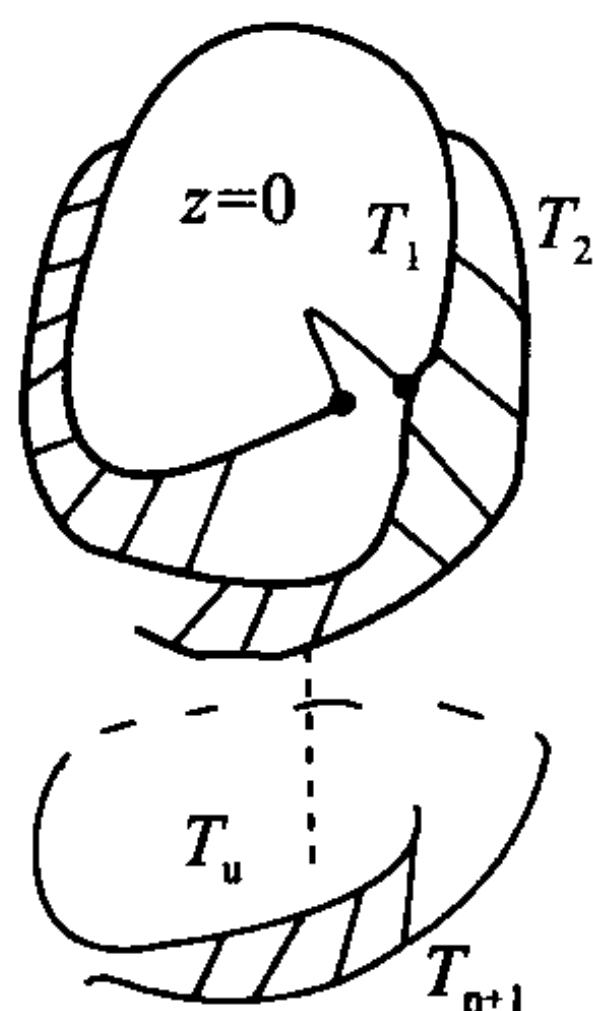


图 1-6-3

(4) $\ln(z-a)$

解: 设 $w = \ln(z-a)$, 则 $e^w = z-a$ 取 $w = u + iv$,

$$(z-a) = \rho e^{iQ}, e^w = e^u e^{iv} = (z-a) = \rho e^{iQ}$$

$$e^u = \rho = |z-a|, \text{ 所以 } u = \ln \rho = \ln |z-a|.$$

$$e^{iQ} = e^{iv}, v = Q + 2\pi n = \text{Arg}(z-a) \quad (n = 0, \pm 1, \pm 2, \dots)$$

所以

$$w = \ln(z-a) + i\text{Arg}(z-a) = \ln \rho + i(Q + 2\pi n)$$

对于 $z-a$, 有无穷多个 w 值. 其支点为 $z=a$ 和 $z=\infty$. 且为无限阶支点.

黎曼面如图 1-6-4 所示.

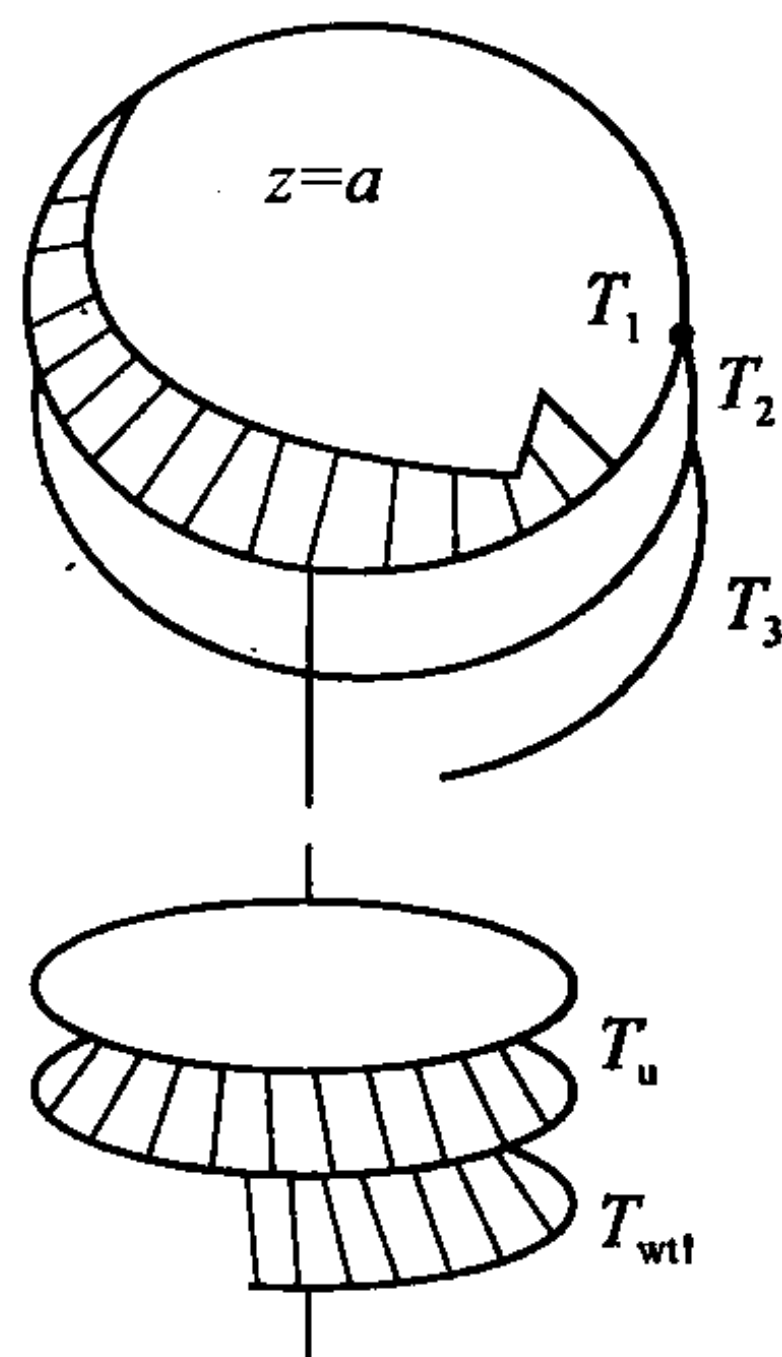


图 1-6-4

第二章 复变函数的积分

本章知识要点

复变函数的积分

$$\int_l f(z) dz = \int_l u(x, y) dx - v(x, y) dy + i \int_l v(x, y) dx + u(x, y) dy$$

柯西定理

单通区域柯西定理

$$\oint_l f(z) dz = 0$$

复通区域柯西定理

$$\oint_l f(z) dz + \sum_{i=1}^n \oint_{l_i} f(z) dz = 0$$

不定积分

若函数 $f(z)$ 在单通区域上解析, 则沿 B 上任一路径 l 的积分 $\int_l f(z) dz$ 的值只与起点和终点有关, 与路径无关. 若起点 z_0 固定, 则不定积分就是定义了一个单值函数

$$F(z) = \int_{z_0}^z f(\zeta) d\zeta$$

柯西公式

若 $f(z)$ 在闭单通域 $\bar{B}$ 上解析, l 为 $\bar{B}$ 的境界线, α 为 $\bar{B}$ 内的任一点.

则有柯西公式

$$f(\alpha) = \frac{1}{2\pi i} \oint_l \frac{f(z)}{z - \alpha} dz$$

习题全解

§ 2.4 柯西公式

1. 已知函数 $\psi(t, x) = e^{2tx - t^2}$. 将 x 作为参数, t 为复变数, 试应用柯西公式将 $\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0}$ 表示为回路积分. 对回路积分进行积分变数的代换 $\zeta = x - z$, 并借以证明

$$\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

(本题的 $\psi(t, x)$ 是埃尔米特多项式的母函数, 见教材附录十).

证明: (1) 应用柯西公式, 把 $\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0}$ 表示为回路积分

由柯西公式
$$f^n(z) = \frac{n!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

取 $f(\zeta) = e^{2\zeta x - \zeta^2}$

有
$$\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{2\zeta x - \zeta^2}}{(\zeta - t)^{n+1}} d\zeta \Big|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{2\zeta x - \zeta^2}}{(\zeta)^{n+1}} d\zeta$$

(2) 对回路积分进行积分变数代换. 将 $\zeta = x - z$ 代入上式, 当 $t = 0$ 时有

$$\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{2(x-z)x - (x-z)^2}}{(x-z)^{n+1}} dz$$

其中 $2(x-z)x - (x-z)^2 = 2x^2 - 2xz - x^2 + 2xz - z^2 = x^2 - z^2$

有
$$\begin{aligned} \left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} &= \frac{n!}{2\pi i} \oint \frac{e^{x^2 - z^2}}{(x-z)^{n+1}} dz = \frac{n!}{2\pi i} \oint \frac{e^{x^2 - z^2}}{(x-z)^{n+1}} d(-z) \\ &= e^{x^2} \cdot \frac{n!}{2\pi i} \oint \frac{(-1)^n e^{-z^2}}{(z-x)^{n+1}} dz \\ &= (-1)^n e^{x^2} \cdot \frac{n!}{2\pi i} \oint \frac{e^{-z^2}}{(z-x)^{n+1}} dz \end{aligned}$$

又
$$f^n(z) = \frac{n!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

所以
$$\frac{n!}{2\pi i} \oint \frac{e^{-z^2}}{(z-x)^{n+1}} dz = (e^{-x^2})^{(n)} = \frac{d^n}{dx^n} e^{-x^2}$$

即
$$\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \quad \text{证毕.}$$

2. 已知函数 $\psi(t, x) = e^{-x/(1-t)}/(1-t)$. 将 x 作为参数, t 为复变数, 试应用柯西公式将 $\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0}$ 表为回路积分.

对回路积分进行积分变数的代换 $\zeta = (z-x)/z$, 并借以证明

$$\left. \frac{\partial^n \psi}{\partial t^n} \right|_{t=0} = e^x \frac{d^n}{dx^n} (x^n e^{-x}).$$

(本题的 $\phi(t, x)$ 是拉盖尔多项式的母函数, 见教材附录十一).

证明: (1) 应用柯西公式把 $\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0}$ 表示为回路积分.

由柯西公式
$$f^n(z) = \frac{n!}{2\pi i} \oint \frac{f(\zeta) d\zeta}{(\zeta - z)^{n+1}}$$

取 $f(\zeta) = e^{-x/(1-\zeta)}/(1-\zeta)$

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{-x/(1-\zeta)}/(1-\zeta)}{(\zeta - t)^{n+1}} d\zeta \Big|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{-x/(1-\zeta)}}{\zeta^{n+1}(1-\zeta)} d\zeta$$

(2) 对回路积分进行积分变数代换. 将 $\zeta = (z-x)/z$ 代入上式.

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = \frac{n!}{2\pi i} \oint \frac{e^{-x/(1-\frac{z-x}{z})}/(1-\frac{z-x}{z})}{\left(\frac{z-x}{z}\right)^{n+1} \left(1-\frac{z-x}{z}\right)} d\left(\frac{z-x}{z}\right)$$

由于

$$\frac{d\left(\frac{z-x}{z}\right)}{dz} = \frac{x}{z^2}$$

有

$$\begin{aligned} \frac{\partial^n \psi}{\partial t^n} &= \frac{n!}{2\pi i} \oint \frac{z^{n+1} e^{-(z-x)}}{(z-x)^{n+1} \cdot \left(\frac{x}{z}\right)} \cdot \frac{x}{z^2} dz \\ &= \frac{n!}{2\pi i} \oint \frac{z^n e^{-(z-x)}}{(z-x)^{n+1}} dz \\ &= e^x \frac{n!}{2\pi i} \oint \frac{z^n e^{-z}}{(z-x)^{n+1}} dz \end{aligned}$$

又

$$f^n(z) = \frac{n!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

所以

$$\frac{n!}{2\pi i} \oint \frac{e^{-z} dz}{(z-x)^{n+1}} = [(e^{-x} x^n)]^{(n)} = \frac{d^n (e^{-x} x^n)}{dx^n}$$

即

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = e^x \frac{d^n}{dx^n} (x^n e^{-x}), \text{证毕.}$$

第三章 幂级数展开

本章知识要点

复数项级数

$$w_k = u_k + i v_k$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n w_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n u_k + i \lim_{n \rightarrow \infty} \sum_{k=1}^n v_k \quad (1)$$

对复数项级数的收敛性问题,柯西收敛判据成立.

若 $\sum_{k=1}^{\infty} w_k = \sum_{k=1}^{\infty} \sqrt{u_k^2 + v_k^2}$ 收敛,则复数项级数(1) 绝对收敛.

若 $\left| \sum_{k=n+1}^{n+p} \dots \right| < \epsilon$, (p 为任意正整数)

ϵ 为任一给定的小正数时,当 $n > N$ 时,复变项级数叫作在 B 上一致收敛.

幂级数

幂级数

$$\sum_{k=0}^{\infty} a_k (z - z_0)^k = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \dots$$

比值判别法(达朗贝尔判别法)

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| \left| \frac{(z - z_0)^{k+1}}{(z - z_0)^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| |z - z_0|$$

收敛半径

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|,$$

根值判别法

$$\begin{aligned} \lim_{k \rightarrow \infty} \sqrt[k]{|a_k|} |z - z_0| < 1 & \quad \text{收敛} \\ \lim_{k \rightarrow \infty} \sqrt[k]{|a_k|} |z - z_0| > 1 & \quad \text{发散} \end{aligned}$$

收敛半径

$$R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}}$$

泰勒级数展开

$f(z)$ 展为幂级数

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$$

其中

$$a_k = \frac{1}{2\pi i} \oint_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta = \frac{f^{(k)}(z_0)}{k!}$$

解析延拓

解析延拓: 解析函数定义域的扩大, 可利用泰勒级数进行解析延拓.

洛朗级数展开

洛朗展开

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k \quad (\text{洛朗级数})$$

其中

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta$$

孤立奇点的分类

若函数 $f(z)$ 在某点 z_0 不可导, 而在 z_0 的任意小邻域内除 z_0 外 Q 上处处可导, 称 z_0 为 $f(z)$ 的孤立奇点. 若在 z_0 的无论多么小的邻域内总可以找到除 z_0 以外的不可导的点, 则称 z_0 为 $f(z)$ 的非孤立奇点.

习题全解

§ 3.2 幂级数

1. 将幂级数(3.2.1)逐项求导, 求所得级数的收敛半径, 以此验证逐项求导不改变收敛半径.

解: (3.2.1) 为 $\sum_{k=1}^{\infty} a_k (z - z_0)^k = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \dots$

原幂级数的收敛半径为

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right|$$

对该级数逐项求导后的新级数为

$$\left[\sum_{k=0}^{\infty} \frac{d}{dz} a_k (z - z_0)^k \right] = a_1 + 2a_2 (z - z_0) + \dots + k a_k (z - z_0)^{k-1} + \dots$$

新级数的收敛半径为

$$R_1 = \lim_{k \rightarrow \infty} \left| \frac{ka_k}{(k+1)a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{k}{k+1} \right| \cdot \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = R.$$

所以 $R_1 = R$, 由此验证幂级数逐项求导, 并不改变其收敛半径.

2. 将幂级数(3.2.1)逐项积分, 求所得级数的收敛半径, 以此验证逐项积分不改变收敛半径.

解: (3.2.1) 为 $\sum a_k(z-z_0)^k = a_0 + a_1(z-z_0) + \cdots + a_k(z-z_0)^k + \cdots$

对原级数逐项积分为:

$$\sum_{k=0}^{\infty} \int a_k(z-z_0)^k d(z-z_0) = a_0(z-z_0) + \frac{1}{2}(z-z_0)^2 + \cdots + \frac{a_k}{k+1}(z-z_0)^{k+1} + \cdots$$

积分后新的级数的收敛半径为 R_1

$$\begin{aligned} R_1 &= \lim_{k \rightarrow \infty} \left| \frac{\frac{1}{k+1}a_k}{\frac{1}{k+2}a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{(k+2)a_k}{(k+1)a_{k+1}} \right| \\ &= \lim_{k \rightarrow \infty} \left| \frac{k+2}{k+1} \right| \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = R \end{aligned}$$

R 为积分前级数的收敛半径. 所以 $R_1 = R$. 由此验证幂级的逐项积分, 并不改变收敛半径.

(本题实际上与上一题为逆运算, 也可以将题 1 的结论倒推, 以验证本题.)

3. 求下列幂级数的收敛圆.

$$(1) \sum_{k=1}^{\infty} \frac{1}{k} (z-i)^k$$

解: 用比值判别法(达朗贝尔判别法)计算

$$\text{收敛半径} \quad R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{\frac{1}{k}}{\frac{1}{k+1}} \right| = 1$$

所以收敛圆为

$$|z-i|=1.$$

$$(2) \sum_{k=1}^{\infty} k^{\ln k} (z-2)^k$$

解: 用根值判别法计算

$$\text{收敛半径} \quad R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}}$$

因为

$$\sqrt[k]{|a_k|} = \sqrt[k]{k^{\ln k}} = k^{\frac{\ln k}{k}}$$

$$\ln(k^{\frac{1}{k}}) \ln k = \frac{1}{k} (\ln k)^2 = \frac{(\ln k)^2}{k}$$

由罗比达法则

$$\lim_{k \rightarrow \infty} \frac{(\ln k)^2}{k} = \lim_{k \rightarrow \infty} \frac{2 \ln k}{k} = \lim_{k \rightarrow \infty} \frac{2}{k} = 0$$

所以

$$\lim_{k \rightarrow \infty} \ln(k^{\frac{1}{k}}) \ln k = 0, \text{ 即 } \lim_{k \rightarrow \infty} k^{\frac{1}{k}} \ln k = 1$$

收敛半径

$$R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} \frac{1}{k^{\frac{\ln k}{k}}} = 1.$$

所以收敛圆为

$$|z-2|=1.$$

$$(3) \sum_{k=1}^{\infty} \left(\frac{z}{k}\right)^k$$

解:用根值判别法计算

收敛半径

$$R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{\left(\frac{1}{k}\right)^k}} = \lim_{k \rightarrow \infty} \frac{1}{\frac{1}{k}} = \lim_{k \rightarrow \infty} k = \infty.$$

只要 z 是有限的, $\sum_{k=1}^{\infty} \left(\frac{z}{k}\right)^k$ 就收敛. 且收敛圆 $|z| = R < \infty$.

$$(4) \sum_{k=1}^{\infty} k! \left(\frac{z}{k}\right)^k$$

解:用比值判别法计算.

$$\begin{aligned} R &= \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{k!}{k^k} \cdot \frac{(k+1)^{k+1}}{(k+1)!} \right| \\ \text{收敛半径} \quad &= \lim_{k \rightarrow \infty} \frac{(k+1)^k}{k^k} = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = e \end{aligned}$$

所以收敛圆是 $|z| = e$

$$(5) \sum_{k=1}^{\infty} k^k (z-3)^k$$

解:用根值判别法计算

$$\text{收敛半径} \quad R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{k^k}} = \lim_{k \rightarrow \infty} \frac{1}{k} = 0$$

所以收敛圆是 $|z-3| = 0$. 只要 $z \neq 3$, $\sum k^k (z-3)^k$ 就发散.

4. 已知幂级数 $\sum_{k=0}^{\infty} a_k z^k$ 及 $\sum_{k=0}^{\infty} b_k z^k$ 的收敛半径分别是 R_1 及 R_2 . 求下列幂级数的收敛半径.

$$(1) \sum_{k=0}^{\infty} (a_k + b_k) z^k$$

$$\text{解:原式} = \sum_{k=0}^{\infty} (a_k) z^k + \sum_{k=0}^{\infty} b_k z^k$$

已知 $\sum_{k=0}^{\infty} a_k z^k$ 的收敛半径为 R_1 , $\sum_{k=0}^{\infty} b_k z^k$ 的收敛半径为 R_2 . 所以 $\sum_{k=0}^{\infty} (a_k + b_k) z^k$ 的收敛半径为 R_1 与 R_2 的交集 $(R_1 \cap R_2)$, $|z|$ 的收敛圆即存在于 R_1 和 R_2 . 因此, 其 R_1 和 R_2 中较小的即满足其条件.

$$(2) \sum_{k=0}^{\infty} (a_k - b_k) z^k$$

$$\text{解:原式} = \sum_{k=0}^{\infty} (a_k) z^k - \sum_{k=0}^{\infty} b_k z^k.$$

$\sum_{k=0}^{\infty} a_k z^k$ 的收敛半径为 R_1 , $\sum_{k=0}^{\infty} b_k z^k$ 的收敛半径为 R_2 , 所以 $\sum_{k=0}^{\infty} (a_k - b_k) z^k$ 的收敛半径为其共有

部分, 即 R_1 与 R_2 的交集. 因此, R_1 和 R_2 中较小的收敛半径即为 $\sum_{k=0}^{\infty} (a_k - b_k) z^k$ 的收敛半径.

$$(3) \sum_{k=0}^{\infty} a_k b_k z^k$$

解法一:用比值法计算

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k b_k}{a_{k+1} b_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| \lim_{k \rightarrow \infty} \left| \frac{b_k}{b_{k+1}} \right| = R_1 R_2$$

所以收敛半径 $R = R_1 R_2$

解法二:用根值法计算

$$\frac{1}{R} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{a_k b_k}} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{a_k}} \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{b_k}} = \frac{1}{R_1 R_2}$$

所以收敛半径 $R = R_1 R_2$

$$(4) \sum_{k=0}^{\infty} \frac{a_k}{b_k} z^k \quad (b_k \neq 0)$$

解法一:用比值法计算

$$\frac{1}{R} = \lim_{k \rightarrow \infty} \left| \frac{a_k}{b_k} \cdot \frac{b_{k+1}}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| \lim_{k \rightarrow \infty} \left| \frac{b_{k+1}}{b_k} \right| = \frac{1}{R_1 R_2}$$

所以收敛半径 R 为 R_1/R_2 .

解法二:用根值法计算

$$\frac{1}{R} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{a_k/b_k}} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{a_k}} \lim_{k \rightarrow \infty} \sqrt[k]{b_k} = \frac{1}{R_1 R_2}$$

所以收敛半径 R 为 R_1/R_2 .

§ 3.3 泰勒级数展开

在指定的点 z_0 的邻域上将下列函数展开为泰勒级数.

(1) $\arctan z$ 在 $z_0 = 0$

解:由泰勒级数 $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k$

取 $z_0 = 0$, 则 $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k$

$$f(z) = \arctg z \quad f(0) = 0$$

$$f'(z) = \frac{1}{1+z^2} \quad f'(0) = 1$$

$$f''(z) = \frac{-2z}{(1+z^2)^2} \quad f''(0) = 0$$

$$f'''(z) = \frac{-2+6z^2}{(1+z^2)^3} \quad f'''(0) = -2$$

$$f^{(4)}(z) = \frac{24z(1-z^2)}{(1+z^2)^4} \quad f^{(4)}(0) = 0$$

$$f^{(5)}(z) = \frac{24(1-10z^2+5z^4)}{(1+z^2)^5} \quad f^{(5)}(0) = 24$$

.....,

.....,

$$\text{所以 } f(z) = \sum_{k=1}^{\infty} \frac{f^{(k)}(z_0)(z-z_0)^k}{k!} = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \dots \quad (|z| < 1)$$

通项 $a_k = \frac{(-1)^{k+1}}{2k-1}$, 且 z 仅有奇次方项.

(2) $\sqrt[3]{z}$ 在 $z_0 = i$

解: 泰勒级数展开

$$f(z) = z^{1/3}$$

$$f(i) = i^{1/3}$$

$$f'(z) = \frac{1}{3}z^{-\frac{2}{3}}$$

$$f'(i) = \frac{1}{3}i^{-\frac{2}{3}}$$

$$f''(z) = \frac{-2}{9}z^{-\frac{5}{3}}$$

$$f''(i) = \frac{-2}{9}i^{-\frac{5}{3}}$$

$$f'''(z) = \frac{5 \times 2}{27}z^{-\frac{8}{3}}$$

$$f'''(i) = \frac{10}{27}i^{-\frac{8}{3}}$$

$$f^{(4)}(z) = \frac{-80}{81}z^{-\frac{11}{3}}$$

$$f^{(4)}(i) = \frac{-80}{81}i^{-\frac{11}{3}}$$

$$\text{所以 } f(z) = \sqrt[3]{i} \left[1 + \frac{1}{3i}(z-i) - \frac{1}{9i^2}(z-i)^2 + \frac{5}{81i^3}(z-i)^3 - \frac{10}{243i^4}(z-i)^4 + \dots \right]$$

$$\text{通项 } a_n = \sum_{n=2}^{\infty} \frac{(-1)^{n-1}(3n-4)!!!}{3^n i^n n!} \quad (|z-i| < 1).$$

(3) $\ln z$ 在 $z_0 = i$

解: 泰勒级数展开

$$f(z) = \ln z$$

$$f(i) = \ln i$$

$$f'(z) = \frac{1}{z}$$

$$f'(i) = \frac{1}{i}$$

$$f''(z) = -\frac{1}{z^2}$$

$$f''(i) = -\frac{1}{i^2}$$

$$f'''(z) = \frac{2}{z^3}$$

$$f'''(i) = \frac{2}{i^3}$$

$$f^{(4)}(z) = \frac{-6}{z^4}$$

$$f^{(4)}(i) = \frac{-6}{i^4}$$

$$\text{所以 } f(z) = \ln i + \frac{1}{i}(z-i) - \frac{1}{2i^2}(z-i)^2 + \frac{1}{3i^3}(z-i)^3 - \frac{1}{4i^4}(z-i)^4 + \dots$$

$$\text{通项 } a_k = \frac{(-1)^{k+1}}{ki^k} \quad (k=1, 2, \dots) \quad |z-i| < 1$$

(4) $\sqrt[m]{z}$ 在 $z_0 = 1$

解: 泰勒级数展开

$$\begin{aligned}
f(z) &= \sqrt[m]{z} & f(1) &= 1 \\
f'(z) &= \frac{1}{m} z^{\frac{1}{m}-1} & f'(1) &= \frac{1}{m} \\
f''(z) &= \frac{1-m}{m^2} z^{\frac{1}{m}-2} & f''(1) &= \frac{1-m}{m^2} \\
f'''(z) &= \frac{(1-m)(1-2m)}{m^3} z^{\frac{1}{m}-3} & f'''(1) &= \frac{(1-m)(1-2m)}{m^3} \\
f^{(4)}(z) &= \frac{(1-m)(1-2m)(1-3m)}{m^4} z^{\frac{1}{m}-4} & f^{(4)}(1) &= \frac{(1-m)(1-2m)(1-3m)}{m^4} \\
&\dots\dots, & & \dots\dots \\
f^{(n)}(z) &= \frac{(1-m)(1-2m)\cdots(1-nm)}{m^n} z^{\frac{1}{m}-n}, & f^{(n)}(1) &= \frac{(1-m)(1-2m)\cdots[1-(n-1)m]}{m^n}
\end{aligned}$$

所以 $f(z) = 1 + \frac{1}{m}(z-1) + \frac{1-m}{2!m^2}(z-1)^2 + \frac{(1-m)(1-2m)}{3!m^3}(z-1)^3 + \cdots \quad |z-1| < m^{-1}$

通项 $a_k = \frac{(1-m)(1-2m)\cdots[1-(k-1)m]}{k!m^k}$

(5) $e^{1/(1-z)}$ 在 $z_0 = 0$

解: 泰勒级数展开

$$\begin{aligned}
f(z) &= e^{1/(1-z)} & f(0) &= e \\
f'(z) &= \frac{1}{(1-z)^2} e^{1/(1-z)} & f'(0) &= e \\
f''(z) &= \left[\frac{2}{(1-z)^3} + \frac{1}{(1-z)^4} \right] e^{1/(1-z)} & f''(0) &= 3e \\
f'''(z) &= \left[\frac{2}{(1-z)^5} + \frac{1}{(1-z)^6} + \frac{6}{(1-z)^4} + \frac{4}{(1-z)^5} \right] e^{1/(1-z)} \\
&= \left[\frac{1}{(1-z)^6} + \frac{6}{(1-z)^5} + \frac{6}{(1-z)^4} \right] e^{1/(1-z)} & f'''(0) &= 13e
\end{aligned}$$

所以 $f(z) = e \left(1 + z + \frac{3}{2!}z^2 + \frac{13}{3!}z^3 + \cdots \right)$

(6) $\ln(1+e^z)$ 在 $z_0 = 0$

解: 泰勒级数展开

$$\begin{aligned}
f(z) &= \ln(1+e^z) & f(0) &= \ln 2 \\
f'(z) &= \frac{e^z}{1+e^z} & f'(0) &= 1/2 \\
f''(z) &= \frac{e^z}{(1+e^z)^2} & f''(0) &= 1/4 \\
f'''(z) &= \frac{2e^z}{(1+e^z)^3} - \frac{e^z}{(1+e^z)^2} & f'''(0) &= 0 \\
f^{(4)}(z) &= \frac{e^4}{(1+e^z)^2} - \frac{6e^z}{(1+e^z)^3} + \frac{6e^z}{(1+e^z)^4} & f^{(4)}(0) &= -1/8
\end{aligned}$$

所以 $f(z) = \ln 2 + \frac{1}{2}z + \frac{1}{2!4}z^2 - \frac{1}{4!8}z^4 + \dots$

(7) $(1+z)^{1/z}$ 在 $z_0 = 0$

解: 泰勒级数展开

$$f(z) = (1+z)^{1/z} = e^{\frac{1}{z}\ln(1+z)} \quad (1)$$

$$\text{又 } \ln(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots$$

代入(1)得

$$\begin{aligned} f(z) &= (1+z)^{1/z} = e^{\frac{1}{z}\ln(1+z)} = e^{\frac{1}{z}(z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots)} \\ &= e^{1 - \frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots} = e \cdot e^{-\frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots} \end{aligned}$$

$$\text{又 } e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3, \text{ 令 } -\frac{1}{2}z + \frac{1}{3}z^2 - \frac{1}{4}z^3 + \dots = A$$

$$\text{则 } f(z) = e(e^A) = e\left[1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots\right]$$

将 A 代入并化简, 得

$$f(z) = e\left[1 - \frac{z}{2} + \frac{11}{24}z^2 + \dots\right]$$

本题也可对 $f(z)$ 求导, 代入得到

$$\begin{aligned} f(z) &= f(z_0) + \sum_{k=1}^n \frac{f^{(k)}(z_0)}{k!}(z-z_0)^k = f(0) + \sum_{k=1}^n \frac{f^{(k)}(0)}{k!}z^k \\ &= e + \sum_{k=1}^n \frac{f^{(k)}(0)}{k!}z^k \end{aligned}$$

(8) $\sin^2 z$ 和 $\cos^2 z$ 在 $z_0 = 0$

解: 已知泰勒级数

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

$$\cos 2z = 1 - \frac{(2z)^2}{2!} + \frac{(2z)^4}{4!} - \frac{(2z)^6}{6!} + \dots$$

$$2\sin^2 z = 1 - \cos 2\theta = \frac{(2z)^2}{2!} - \frac{(2z)^4}{4!} + \frac{(2z)^6}{6!}$$

$$\sin^2 z = z^2 - \frac{2^3}{4!}z^4 + \frac{2^5}{6!}z^6 - \dots = \frac{1}{2} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(2z)^{2k}}{(2k)!}$$

$$\text{所以 } \cos^2 z = 1 - \sin^2 z = 1 - z^2 + \frac{2^3}{4!}z^4 - \frac{2^5}{6!}z^6 + \dots$$

$$= 1 + \frac{1}{2} \sum_{k=1}^{\infty} (-1)^k \frac{(2z)^{2k}}{(2k)!}$$

§ 3.5 洛朗级数展开

在挖去奇点 z_0 的环域上或指定的环域上将下列函数展开为洛朗级数.

(1) $z^5 e^{1/z}$ 在 $z_0 = 0$

解: 由 $e^x = 1 + x + \frac{1}{2!}x^2 + \cdots + \frac{1}{k!}x^k + \cdots$ ($|x| < \infty$)

有 $e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!} \cdot \frac{1}{z^2} + \cdots + \frac{1}{k!} \frac{1}{z^k} + \cdots$ ($|\frac{1}{z}| < \infty$)

$$z^5 e^{\frac{1}{z}} = z^5 + z^4 + \frac{1}{2!}z^3 + \cdots + \frac{1}{k!}z^{5-k} + \cdots \quad (|z| < 0)$$

展为洛朗级数

$$f(z) = z^5 e^{\frac{1}{z}} = z^5 + z^4 + \frac{1}{2!}z^3 + \cdots + \frac{1}{k!}z^{5-k} + \cdots \quad (|z| > 0).$$

(2) $1/z^2(z-1)$ 在 $z_0 = 1$

解: 由于 $z^2 = [1 - (1-z)]^2$, 所以 $f(z) = \frac{1}{z^2(z-1)} = \frac{1}{[1 - (z-1)]^2(z-1)}$

$$\text{又 } \left(\frac{1}{1-x}\right)' = \frac{1}{(1-x)^2}$$

且 $\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^k + \cdots$ ($|x| < 1$)

所以 $\frac{1}{1 - (1-z)} = 1 + (-z) + (1-z)^2 + \cdots + (1-z)^k + \cdots$ ($|z-1| < 1$)

$$\begin{aligned} \frac{1}{[1 - (1-z)]^2} &= \left[\frac{1}{1 - (1-z)} \right]'_{(1-z)} \\ &= +1 + 2(1-z) + \cdots + n(1-z)^{-1} \\ &= \sum_{k=1}^{\infty} (-1)^{k+1} k (z-1)^{k-1} \quad (0 < |1-z| < 1) \end{aligned}$$

洛朗级数为

$$\begin{aligned} \frac{1}{z^2(z-1)} &= \frac{1}{(z-1)} \frac{1}{[1 - (1-z)]^2} = \frac{1}{z-1} \sum_{k=1}^{\infty} (-1)^{k+1} k (z-1)^{k-1} \\ &= \sum_{k=1}^{\infty} (-1)^{k+1} k (z-1)^{k-2} \\ &= \sum_{k=-1}^{\infty} (-1)^{k+1} (k+2) (z-1)^{k-2} \quad (0 < |z-1| < 1) \end{aligned}$$

(3) $1/z(z-1)$ 在 $z_0 = 0$, 在 $z_0 = 1$

解: 在 $z_0 = 0$ 处展开

由于 $\frac{1}{z-1} = \frac{-1}{1-z} = -[1 + z + \cdots + z^n + \cdots] = -\sum_{k=0}^{\infty} z^k$

所以 $\frac{1}{z} \frac{1}{z-1} = \frac{-1}{z} \sum_{k=0}^{\infty} z^{k+1} = (-1) \sum_{k=-1}^{\infty} z^k \quad (0 < |z| < 1)$

在 $z_0 = 1$ 处展开:

$$\frac{1}{z} = \frac{1}{1 - (1-z)} = \sum_{k=0}^{\infty} (1-z)^k = \sum_{k=0}^{\infty} (-1)^k (z-1)^k$$

$$\begin{aligned}\frac{1}{z-1} \cdot \frac{1}{z} &= \frac{1}{z-1} \sum_{k=0}^{\infty} (-1)^k (z-1)^k = \sum_{k=0}^{\infty} (-1)^k (z-1)^{k-1} \\ &= \sum_{k=-1}^{\infty} (-1)^{k+1} (z-1)^k, \quad (0 < |z-1| < 1)\end{aligned}$$

(4) $e^{1/(1-z)}$ 在 $|z| > 1$

解: $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

$$e^{1/(1-z)} = e^{\frac{-1}{z-1}} = \sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{z-1}\right)^k / k! \quad (1)$$

又 $|z| > 1$

且 $\frac{1}{1-t} = \sum_{k=0}^{\infty} t^k, (|t| < 1)$, 取 $\frac{1}{z}$ 的形式

$$\text{所以 } \frac{1}{z-1} = \frac{1}{z\left(1-\frac{1}{z}\right)} = \frac{1}{z} \left(1 + \frac{1}{z} + \cdots + \frac{1}{z^k} + \cdots\right) = \left(\frac{1}{z} + \frac{1}{z^2} + \cdots + \frac{1}{z^{k+1}} + \cdots\right), \quad (2)$$

$|z| < 1$ 即 $|z| > 1$. 将(2)式代入(1)式, 有

$$\begin{aligned}e^{\frac{1}{1-z}} &= \sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{z-1}\right)^k / k! \\ &= 1 - \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right) + \frac{1}{2!} \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^2 - \frac{1}{3!} \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)^3 + \cdots \\ &= 1 - \frac{1}{z} - \frac{1}{2z^2} - \frac{1}{6z^3} + \frac{1}{24z^4} - \frac{19}{120z^5} + \cdots \quad (|z| > 1).\end{aligned}$$

(5) $1/(z-2)(z-3)$ 在 $|z| > 3$

解: $\frac{1}{3} - \frac{1}{z-2} = \frac{1}{(z-2)(z-3)}$ 且 $\frac{1}{1-t} = \sum_{k=0}^{\infty} t^k \quad (|t| < 1)$ 而本题中 $|z| > 1$.

$$\text{取 } \frac{1}{z-3} = \frac{1}{z} \left[\frac{1}{1-\frac{3}{z}} \right] = \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{3}{z}\right)^k \quad (|z| > 3)$$

$$\text{同理: } \frac{1}{z-2} = \frac{1}{z} \left[\frac{1}{1-\frac{2}{z}} \right] = \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^k \quad (|z| > 2)$$

$$\begin{aligned}\frac{1}{(z-2)(z-3)} &= \frac{1}{z-3} - \frac{1}{z-2} = \frac{1}{z} \sum_{k=0}^{\infty} \left[\left(\frac{3}{z}\right)^k - \left(\frac{2}{z}\right)^k \right] \\ &= \sum_{k=0}^{\infty} (3^k - 2^k) \left(\frac{1}{z}\right)^{k+1} \\ &= \sum_{k=-\infty}^{k=-1} (3^{-(k+1)} - 2^{-(k+1)}) z^k \quad |z| > 3\end{aligned}$$

(6) $(z-1)(z-2)(z-3)(z-4)$ 在 $R < |z| < \infty$ (R 很大)

$$\text{解: } \frac{(z-1)(z-2)}{(z-3)(z-4)} = \frac{\left(1-\frac{1}{z}\right)\left(1-\frac{2}{z}\right)}{\left(1-\frac{3}{z}\right)\left(1-\frac{4}{z}\right)}$$

设 $t = 1/z$,

$$\begin{aligned}\text{则原式} &= \frac{(1-t)(1-2t)}{(1-3t)(1-4t)} = \frac{(1-4t+3t)(1-2t)}{(1-3t)(1-4t)} \\ &= \frac{1-2t}{1-3t} + \frac{3t(1-2t)}{(1-3t)(1-4t)} = 1 + \frac{t}{1-3t} + \frac{3t}{1-4t} + \frac{3t^2}{(1-3t)(1-4t)} \\ &= 1 + \frac{t}{1-3t} + \frac{3t}{1-4t} + 3t \left(\frac{t}{(1-3t)(1-4t)} \right)\end{aligned}$$

$$\text{又 } \frac{t}{(1-3t)(1-4t)} = \frac{1}{1-4t} - \frac{1}{1-3t}$$

$$\text{原式} = 1 + \frac{t}{1-3t} + \frac{3t}{1-4t} + \frac{3t}{1-4t} - \frac{3t}{1-3t} = 1 + \frac{6t}{1-4t} - \frac{2t}{1-3t}$$

$$\text{又 } \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

$$\text{即 } \frac{6t}{1-4t} = 6t \sum_{k=0}^{\infty} (4t)^k \quad |4t| < 1$$

$$\text{同时 } \frac{2t}{1-3t} = 2t \sum_{k=0}^{\infty} (3t)^k \quad (|3t| < 1)$$

将 $t = \frac{1}{z}$ 代回

$$\begin{aligned}\text{原式} &= 1 + \frac{6}{z} \sum_{k=0}^{\infty} \left(\frac{4}{z}\right)^k - \frac{2}{z} \sum_{k=0}^{\infty} \left(\frac{3}{z}\right)^k = 1 + \sum_{k=0}^{\infty} 6 \cdot 4^k \cdot \left(\frac{1}{z}\right)^{k+1} - \sum_{k=0}^{\infty} 2 \cdot 3^k \left(\frac{1}{z}\right)^{k+1} \\ &= 1 + \sum_{k=-\infty}^{-1} (6 \cdot 4^{-(k+1)} - 2 \cdot 3^{-(k+1)}) z^k \quad \left(\left|\frac{4}{z}\right| < 1, \text{即 } |z| > 4\right)\end{aligned}$$

(7) $1/(z^2 - 3z + 2)$ 在 $1 < |z| < 2$

$$\text{解: } \frac{1}{(z^2 - 3z + 2)} = \frac{1}{(z-2)(z-1)} = \frac{1}{z-2} - \frac{1}{z-1}$$

$$\text{又 } \frac{1}{z-2} = -\frac{\frac{1}{2}}{1-\frac{z}{2}} = -\frac{1}{2} \left[\frac{1}{1-\frac{z}{2}} \right] = -\frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \quad \left(\left|\frac{z}{2}\right| < 1\right)$$

$$\text{且 } \frac{-1}{z-1} = -\left[\frac{\frac{1}{z}}{1-\frac{1}{z}} \right] = -\frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^k \quad (1 < |z|, |z| < 2)$$

$$\begin{aligned}\text{原式} &= \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^k \\ &= -\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{k+1} z^k - \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^{k+1} \\ &= -\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{k+1} z^k - \sum_{k=-\infty}^{-1} z^k, (1 < |z| < 2)\end{aligned}$$

(8) $1/(z^2 - 3z + 2)$ 在 $1 < |z| < \infty$

$$\text{解: } 1/(z^2 - 3z + 2) = \frac{1}{z-2} - \frac{1}{z-1} = \left[\frac{1}{1 - \frac{2}{z}} - \frac{1}{1 - \frac{1}{z}} \right] z^{-1}$$

$$\text{又 } \frac{1}{1 - \frac{2}{z}} = \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^k, (|z| > 2), \text{ 且 } \frac{1}{1 - \frac{1}{z}} = \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^k \quad (|z| > 1)$$

$$\begin{aligned} \text{原式} &= \frac{1}{z} \sum_{k=0}^{\infty} \left[\left(\frac{2}{z}\right)^k + \left(\frac{1}{z}\right)^k \right] = \sum_{k=0}^{\infty} (2^k - 1) \left(\frac{1}{z}\right)^{k+1} \\ &= \sum_{k=-\infty}^{-1} (2^{-(k+1)} - 1) z^k \quad (2 < |z| < \infty) \end{aligned}$$

(9) e^z/z 在奇点

解: e^z/z 的奇点为 $z=0$, e^z/z 在 $z=0$ 展开为洛朗级数.

$$\text{又 } e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!},$$

$$\frac{e^z}{z} = \sum_{k=0}^{\infty} \frac{z^{k-1}}{k!} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{z^{k-1}}{k!} \quad (0 < |z| < \infty).$$

(10) $(1 - \cos z)/z$ 在奇点

解: $(1 - \cos z)/z$ 的奇点为 $z=0$,

$(1 - \cos z)/z$ 在 $z=0$ 展开为洛朗级数.

$$\text{又 } \frac{1 - \cos z}{z} = \frac{1}{z} - \frac{\cos z}{z}$$

$$\text{且 } \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}$$

$$\begin{aligned} (1 - \cos z)/z &= \frac{1}{z} - \frac{\cos z}{z} = \frac{1}{z} - \frac{1}{z} \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{(2k)!} \\ &= - \sum_{k=1}^{\infty} (-1)^k \frac{z^{2k-1}}{(2k)!} = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{z^{2k-1}}{(2k)!} \quad (|z| < \infty) \end{aligned}$$

(11) $\sin(1/z)$ 在奇点

解: $\sin(1/z)$ 的奇点为 $z=0$, $\sin(1/z)$ 在 $z=0$ 展开为洛朗级数.

$$\text{又 } \sin t = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} t^{2k+1}$$

$$\begin{aligned} \text{取 } t = 1/z, \text{ 则 } \sin(1/z) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (z)^{-(2k+1)} \\ &= \sum_{k=-\infty}^0 \frac{(-1)^k}{(2k+1)!} z^{2k-1} \quad (0 < |z| < \infty) \end{aligned}$$

(12) $\cot z$ 在奇点

解: $\cot z = \frac{\cos z}{\sin z}$ 的奇点为 $z=0$ 和 $z=\pi$

$$\operatorname{ctan} z = \frac{\cos z}{\sin z}, \quad \cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$$

$$\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$$

设 $\operatorname{ctan} z = \frac{1}{z} \sum_{i=0}^{\infty} a_i z^{2i}$, 且 $\cos z = \operatorname{ctan} z \sin z$

即
$$\sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} \cdot \frac{1}{z} \sum_{i=0}^{\infty} a_i z^{2i}$$

根据展开的唯一性, 得系数的递推关系

$$k=0, a_0=1, k=1, a_1=1/3, \dots$$

$$\operatorname{ctan} z = \frac{1}{z} \sum a_i z^{2i}$$

(13) $z/(z-1)(z-2)^2$ 在 $|z|<1$, 在 $1<|z|<2$, 在 $2<|z|$

解:
$$\frac{z}{(z-1)(z-2)^2} = \frac{2(z-1)-(z-2)}{(z-2)^2(z-1)} = \frac{2}{(z-2)^2} - \frac{1}{(z-1)(z-2)}$$

$$= \frac{2}{(z-2)^2} - \frac{1}{z-2} + \frac{1}{z-1}$$

又
$$\frac{2}{(z-2)^2} = \left(\frac{1}{1-\frac{z}{2}} \right)', \text{ 且 } \frac{1}{1-\frac{z}{2}} = \sum_{k=0}^{\infty} (z/2)^k, (|z|<2)$$

(a) 在 $|z|<1$, 有

$$\begin{aligned} \frac{2}{(z-2)^2} &= \left(\frac{1}{1-z/2} \right)' = \left[\sum_{k=0}^{\infty} (z/2)^k \right]' = \frac{1}{2} \sum_{k=1}^{\infty} k \left(\frac{z}{2} \right)^{k-1} \\ &= \frac{1}{2} \sum_{k=0}^{\infty} (k+1) \left(\frac{z}{2} \right)^k \quad (|z|<2) \end{aligned} \quad (1)$$

$$\frac{-1}{z-2} = \frac{1/2}{1-z/2} = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2} \right)^k \quad (|z|<2) \quad (2)$$

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{k=0}^{\infty} z^k \quad (|z|<1) \quad (3)$$

由式(1)(2)(3)

$$\begin{aligned} \frac{1}{(z-1)(z-2)^2} &= \frac{1}{2} \sum_{k=0}^{\infty} (k+1) \left(\frac{z}{2} \right)^k + \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2} \right)^k - \sum_{k=0}^{\infty} z^k \\ &= \sum_{k=0}^{\infty} \left[\frac{k+1+1}{2} \left(\frac{z}{2} \right)^k - z^k \right] \\ &= \sum_{k=0}^{\infty} \left[\left(\frac{k+2}{2^{k+1}} \right) - 1 \right] z^k \quad (|z|<1) \end{aligned}$$

(b) 在 $1<|z|<2$, 式(1), (2) 不变, 式(3) 按 $1/z$ 展开为(4)

$$\frac{1}{z-1} = \frac{1}{z} \left(\frac{1}{1-1/z} \right) = \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z} \right)^k = \sum_{k=0}^{\infty} \left(\frac{1}{z} \right)^{k+1} = \sum_{k=-\infty}^{-1} z^k \quad (|z|>1) \quad (4)$$

由式(1)(2)(4)

$$\begin{aligned}\frac{z}{(z-1)(z-2)^2} &= \frac{1}{2} \sum_{k=0}^{\infty} (k+1) \left(\frac{z}{2}\right)^k + \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k + \sum_{k=-\infty}^{-1} z^k \\ &= \sum_{k=0}^{\infty} \left(\frac{k+2}{2^{k+1}}\right) z^k + \sum_{k=-\infty}^{-1} z^k \quad (1 < |z| < 2)\end{aligned}$$

(c) 当 $|z| > 2$ 时, 式(4) 不变, (1)(2) 式按 $\frac{2}{z}$ 展开为(5)(6)

$$\frac{2}{(z-2)^2} = \frac{2}{z^2} \left[\frac{1}{\left(1 - \frac{2}{z}\right)^2} \right] = \frac{2}{z^2} \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^k \cdot (k+1) = - \sum_{k=-\infty}^{-2} (k+1) z^k / 2^{k+1} \quad (|z| > 2) \quad (5)$$

$$\frac{-1}{z-2} = \frac{-1}{z} \left[\frac{1}{1 - \frac{2}{z}} \right] = - \sum_{k=0}^{\infty} \frac{1}{z} \left(\frac{2}{z}\right)^k = - \sum_{k=-\infty}^{-1} 2^{-(k+1)} z^k \quad (|z| > 2) \quad (6)$$

由式(4)(5)(6)

$$\frac{z}{(z-1)(z-2)^2} = - \sum_{k=-\infty}^{-2} \frac{(k+1)}{2^{k+1}} z^k - \sum_{k=-\infty}^{-1} \frac{z^k}{2^{k+1}} + \sum_{k=-\infty}^{-1} z^k$$

$$\text{又 } \frac{-z^k}{2^{k+1}} \Big|_{k=-1} + z^k \Big|_{k=-1} = 0$$

$$\text{原式} = \sum_{k=-\infty}^{-2} \left(1 - \frac{k+2}{2^{k+1}}\right) z^k, \quad (|z| > 2)$$

(14) $z/(z-1)(z-2)$ 在 $|z| < 1$, 在 $1 < |z| < 2$, 在 $2 < |z|$

$$\begin{aligned}\text{解: } \frac{z}{(z-1)(z-2)} &= \frac{z-1+1}{(z-1)(z-2)} = \frac{1}{z-2} + \frac{1}{(z-1)(z-2)} \\ &= \frac{1}{z-2} + \left[\frac{1}{z-2} - \frac{1}{z-1} \right] = \frac{2}{z-2} - \frac{1}{z-1}\end{aligned}$$

(a) 在 $|z| < 1$ 同上题(13题) 式(2), (3) 有

$$\frac{2}{z-2} = \frac{-1}{1-z/2} = - \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \quad (|z| < 2)$$

$$\frac{-1}{z-1} = \frac{1}{1-z} = \sum_{k=0}^{\infty} z^k \quad (|z| < 1)$$

$$\frac{z}{(z-1)(z-2)} = - \sum_{k=0}^{\infty} \left[\left(\frac{z}{2}\right)^k - z^k \right] = \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^k}\right) z^k \quad (|z| < 1)$$

(b) 在 $1 < |z| < 2$, 同上题式(2)(4), 有

$$\frac{2}{z-2} = \frac{-1}{1-z/2} = - \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \quad (|z| < 2)$$

$$\text{及 } \frac{1}{1-z} = - \frac{1}{z\left(1 - \frac{1}{z}\right)} = \frac{-1}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^k = - \sum_{k=-\infty}^{-1} z^k \quad (|z| > 1)$$

$$\frac{z}{(z-1)(z-2)} = - \sum_{k=0}^{\infty} \frac{1}{2^k} z^k - \sum_{k=-\infty}^{-1} z^k$$

(c) 在 $2 < |z|$, 同上题式(4), (6) 有

$$\frac{1}{1-z} = - \sum_{k=-\infty}^{-1} z^k \quad (|z| > 1)$$

$$\begin{aligned} \frac{2}{z-2} &= \frac{2}{z} \left[\frac{1}{1-\frac{2}{z}} \right] = \frac{2}{z} \sum_{k=0}^{\infty} \left(\frac{2}{z} \right)^k = \sum_{k=0}^{\infty} \left(\frac{2}{z} \right)^{k+1} = \sum_{k=1}^{\infty} \left(\frac{2}{z} \right)^k \\ &= \sum_{k=-\infty}^{-1} \left(\frac{z}{2} \right)^k \quad (|z| > 2) \end{aligned}$$

所以 $\frac{z}{(z-1)(z-2)} = \sum_{k=-\infty}^{-1} \left[\left(\frac{z}{2} \right)^k - z^k \right] = \sum_{k=-\infty}^{-1} \left(\frac{1}{2^k} - 1 \right) z^k \quad (2 < |z|)$

(15) $1/z^2(z^2-1)^2$ 在 $0 < |z| < 1$, 在 $1 < |z| < \infty$

解: $\frac{1}{z^2(z^2-1)^2} = \frac{1}{z^2} \frac{1}{(1-z^2)^2}$

又 $\frac{1}{1-t^2} = \sum_{k=0}^{\infty} t^{2k}$, 且 $\left[\frac{1}{(1-t^2)} \right]' = \frac{2t}{(1-t^2)^2}$

(a) 在 $0 < |z| < 1$, 有

$$\begin{aligned} \frac{1}{z^2(1-z^2)^2} &= \frac{1}{z^2} \cdot \frac{1}{2z} \left[\frac{1}{1-z^2} \right]' = \frac{1}{z^3} \left(\sum_{k=0}^{\infty} z^{2k} \right)' = \frac{1}{2z^3} \sum_{k=1}^{\infty} 2kz^{2k-1} = \sum_{k=1}^{\infty} kz^{2(k-2)} \\ &= \sum_{k=-1}^{\infty} (k+2)z^{2k}, \quad (0 < |z| < 1) \end{aligned}$$

(b) 在 $1 < |z| < \infty$, 有

$$\begin{aligned} \frac{1}{z^2(1-z^2)^2} &= \frac{1}{z^6} \frac{1}{\left(1-\frac{1}{z^2}\right)^2} = \frac{1}{z^6} \left(\frac{-z^3}{2} \right) \left[\frac{1}{1-\frac{1}{z^2}} \right]'_{1/z^2} \\ &= -\frac{1}{2z^3} \left[\sum_{k=0}^{\infty} z^{-2k} \right]' = \frac{1}{2z^3} \sum_{k=0}^{\infty} 2kz^{-2k-1} \\ &= \sum_{k=0}^{\infty} kz^{-2k-4} = \sum_{k=1}^{\infty} kz^{-(2k+2)} = \sum_{k=-\infty}^{-2} z^{2k} \\ &= \sum_{k=-\infty}^{-3} (k+2)z^{2k}, \quad (z < |z| < \infty). \end{aligned}$$

§ 3.6 孤立奇点的分类

设函数 $f(z)$ 和 $g(z)$ 分别以点 z_0 为 m 阶和 n 阶极点, 问对于下列函数而言, z_0 是何种性质的点?

(1) $f(z)g(z)$ (2) $f(z)/g(z)$ (3) $f(z)+g(z)$.

解: 由于 $f(z)$ 和 $g(z)$ 分别以点 z_0 为 m 阶和 n 阶极点,

设 $f(z) = \frac{\varphi(z)}{(z-z_0)^m}$, 且 $\varphi(z_0) \neq 0$, 在 $z = z_0$ 邻域解析

$g(z) = \frac{\psi(z)}{(z-z_0)^n}$, 且 $\psi(z_0) \neq 0$, 在 $z = z_0$ 邻域解析.

(1) $f(z)g(z) = \frac{\varphi(z)\psi(z)}{(z-z_0)^m(z-z_0)^n} = \frac{\varphi(z)\psi(z)}{(z-z_0)^{m+n}}$

函数 $f(z) \cdot g(z)$ 是以 z_0 为 $(m+n)$ 阶极点.

$$(2) f(z)/g(z) = \frac{\phi(z)/\psi(z)}{(z-z_0)^m(z-z_0)^n} = \frac{\phi(z)/\psi(z)}{(z-z_0)^{m+n}}$$

当 $m > n$ 时, 则 $f(z)/g(z)$ 是以 z_0 为 $(m-n)$ 阶极点.

当 $m < n$ 时, $\frac{1}{(z-z_0)^{m-n}} = (z-z_0)^{n-m}, (n-m > 0)$ 不是奇点.

$$(3) f(z) + g(z) = \frac{\phi(z)}{(z-z_0)^m} + \frac{\varphi(z)}{(z-z_0)^n}$$

函数 $f(z) + g(z)$, z_0 是以 m 或 n 中较大的为阶数的极点. 若 $m = n$, 则极点的阶数 $\leq m$ 或 n .

若设 $f(z) + g(z)$ 在 z_0 点的阶数为 k , 则 $k = \max(m, n)$, 当 $m \neq n$ 时.

当 $m = n$ 时, 阶数 $k \leq m$ 或 n .

(例: 当 $m = n$ 时, 如 $f(z) = z, g(z) = z_0$, 则 $f(z) + g(z) = \frac{z-z_0}{(z-z_0)^m} = \frac{1}{(z-z_0)^{m-1}},$

$k = m-1 < m$)

第四章 留数定理

本章知识要点

留数定理

函数 $f(z)$ 在 z_0 点的留数记为 $\text{Res } f(z_0)$

$$\oint_l f(z) dz = 2\pi i \text{Res } f(z_0)$$

留数定理: 设函数 $f(z)$ 在回路 l 所围区域 B 上除有限个孤立奇点 $b_1, b_2, \dots, b_n$ 外解析, 在闭区域 $\bar{B}$ 上除 $b_1, b_2, \dots, b_n$ 处连续, 则

$$\oint_l f(z) dz = 2\pi i \sum_{j=1}^n \text{Res } f(b_j)$$

留数定理将回路积分归结为被积函数在回路所围区域上各奇点的留数之和.

$$\text{Res } f(z_0) = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \left\{ \frac{d^{m-1}}{dz^{m-1}} [(z-z_0)^m f(z)] \right\}, (m \text{ 阶极点}, m \geq 1)$$

用上式可算出函数 $f(z)$ 在极点的留数.

应用留数定理计算实变函数定积分

几种类型的实变定积分

类型一 $\int_0^{2\pi} R(\cos x, \sin x) dx$

代换 $z = e^{ix}, \cos x = \frac{1}{2}(z + z^{-1}), \sin x = \frac{1}{2i}(z - z^{-1}) \quad dx = \frac{1}{iz} dz$

原积分化为

$$\int_0^{2\pi} R(\cos x, \sin x) dx = \oint_{|z|=1} R\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right) \frac{dz}{iz}$$

类型二 $\int_{-\infty}^{\infty} f(x) dx$

原积分化为

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \{f(z) \text{ 在上半平面所有奇点的留数之和}\}$$

类型三 $\int_0^{\infty} F(x) \cos mx dx, \int_0^{\infty} G(x) \sin mx dx$

原积分化为

$$\int_0^{\infty} F(x) \cos mx dx = \pi i \{F(z) e^{imz} \text{ 在上半平面所有奇点的留数之和} \}$$

$$\int_0^{\infty} G(x) \sin mx dx = \pi i \{G(z) e^{imz} \text{ 在上半平面所有奇点的留数之和} \}$$

习题全解

§ 4.1 留数定理

1. 确定下列函数的奇点, 求出函数在各奇点的留数.

(1) $e^z/(1+z)$

解: $e^z/(1+z)$ 的极点 $z_0 = -1$

$$\operatorname{Res} f(-1) = \lim_{z \rightarrow -1} \left[(z - (-1)) \frac{e^z}{1+z} \right] = \lim_{z \rightarrow -1} e^z = \frac{1}{e}$$

$$\operatorname{Res} f(-1) = 1/e$$

又 $\lim_{z \rightarrow \infty} \left(\frac{e^z}{1+z} \right) \rightarrow \infty$, 所以 $z = \infty$ 是本性奇点.

因全平面各留数之和为 0, 有 $\operatorname{Res} f(\infty) = -\operatorname{Res} f(-1) = -1/e$

(2) $z/(z-1)(z-2)^2$

解: $z_0 = 1, z_0 = 2$ 分别为单极点和二阶极点

$$\operatorname{Res} f(1) = \lim_{z \rightarrow 1} (z-1) \frac{z}{(z-1)(z-2)^2} = \lim_{z \rightarrow 1} \frac{z}{(z-2)^2} = 1$$

$$\begin{aligned} \operatorname{Res} f(2) &= \lim_{z \rightarrow 2} \frac{1}{1!} \frac{d}{dz} \left[(z-2)^2 \frac{z}{(z-1)(z-2)^2} \right] \\ &= \lim_{z \rightarrow 2} \frac{-1}{(z-1)^2} = -1 \end{aligned}$$

(3) $e^z(z^2 + a^2)$

解: 单极点 $z_0 = ia, z_0 = -ia$

本性奇点 $z_0 = \infty$

$$\operatorname{Res} f(ia) = \lim_{z \rightarrow ia} (z-ia) \frac{e^z}{(z-ia)(z+ia)} = \lim_{z \rightarrow ia} \frac{e^z}{z+ia} = \frac{e^{ia}}{2ia}$$

$$\operatorname{Res} f(-ia) = \lim_{z \rightarrow -ia} (z+ia) \frac{e^z}{(z-ia)(z+ia)} = \lim_{z \rightarrow -ia} \frac{e^z}{z-ia} = \frac{e^{-ia}}{-2ia}$$

$$\operatorname{Res} f(\infty) = -\sum \operatorname{Res} f(z_i) = -[\operatorname{Res} f(-ia) + \operatorname{Res} f(ia)]$$

$$= \frac{e^{-ia} - e^{ia}}{2ia} = \frac{-i \sin a - i \sin a}{2ia} = \frac{-\sin a}{a}$$

$$(4) e^z(z^2 + a^2)$$

解:单极点 $z = ia$, 单极点 $z_0 = -ia$

本性奇点 $z_0 = \infty$

$$\text{Res } f(ia) = \lim_{z \rightarrow ia} (z - ia) \frac{e^z}{(z - ia)(z + ia)} = \frac{e^{-a}}{2ia}$$

$$\text{Res } f(-ia) = \lim_{z \rightarrow -ia} (z + ia) \frac{e^z}{(z - ia)(z + ia)} = \frac{-e^a}{2ia}$$

$$\begin{aligned} \text{Res } f(\infty) &= -[\text{Res } f(-ia) + \text{Res } f(ia)] \\ &= -\left(\frac{-e^a}{2ia} + \frac{e^{-a}}{2ia}\right) = -(e^{-a} - e^a)/2ia = \frac{sha}{ia} \end{aligned}$$

$$(5) ze^z/(z-a)^3$$

解: $z_0 = a$ 为三阶极点, $z = \infty$ 为本性奇点

$$\begin{aligned} \text{Res } f(a) &= \lim_{z \rightarrow a} \frac{1}{2!} \frac{d^2}{dz^2} \left[(z-a)^3 \frac{ze^z}{(z-a)^3} \right] \\ &= \frac{1}{2} \frac{d^2}{dz^2} (ze^z) \Big|_{z=a} = \left(e^z + \frac{z}{2} e^z \right) \Big|_{z=a} \\ &= e^a + \frac{a}{2} e^a = \left(1 + \frac{a}{2} \right) e^a \end{aligned}$$

本性奇点 $\text{Res } f(\infty) = -\text{Res } f(a) = -\left(1 + \frac{a}{2} \right) e^a$

$$(6) 1/(z^3 - z^5)$$

解: $1/(z^3 - z^5) = -1/z^3(z^2 - 1)$

$z_0 = 0$ 为三阶极点, $z_0 = 1, z_0 = -1$ 为单极点

$$\begin{aligned} \text{Res } f(0) &= \lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \left(z^3 \cdot \frac{-1}{z^3(z^2 - 1)} \right) = \frac{-1}{2} \frac{d^2}{dz^2} \left(\frac{-1}{z^2 - 1} \right) \Big|_{z=0} \\ &= \frac{-1}{2} \left[\frac{-2}{(z^2 - 1)^2} + \frac{4z^2}{(z^2 - 1)^3} \right] \Big|_{z=0} \\ &= \frac{-1}{2} \times (-2) = 1 \end{aligned}$$

$$\text{Res } f(1) = \lim_{z \rightarrow 1} (z - 1) \frac{-1}{(z - 1)(z + 1)z^3} = -\frac{1}{2}$$

$$\text{Res } f(-1) = \lim_{z \rightarrow -1} (z + 1) \frac{-1}{(z + 1)(z - 1)z^3} = -\frac{1}{2}$$

$$(7) z^2/(z^2 + 1)^2$$

解: $z_0 = i, z_0 = -i$ 分别为二阶极点

$$\begin{aligned} \text{Res } f(i) &= \lim_{z \rightarrow i} \frac{d}{dz} \left(\frac{(z-i)^2 z^2}{(z+i)^2 (z-i)^2} \right) = \lim_{z \rightarrow i} \frac{d}{dz} \left(\frac{z}{z+i} \right)^2 \\ &= 2 \cdot \left(1 - \frac{i}{z+i} \right) \cdot \left(\frac{i}{(z+i)^2} \right) \Big|_{z=i} = 2 \times \frac{1}{2} \times \frac{i}{(2i)^2} = -\frac{i}{4} \end{aligned}$$

$$\text{Res } f(-i) = \lim_{z \rightarrow -i} \frac{d}{dz} \frac{(z+i)^2 z^2}{(z+i)^2 (z-i)^2} = \lim_{z \rightarrow -i} \frac{d}{dz} \left(\frac{z}{z-i} \right)^2$$

$$= 2\left(1 + \frac{i}{z-i}\right)\left(\frac{i}{(z-i)^2}\right)\Big|_{z=-i} = 2\left(\frac{1}{2}\right)\frac{i}{(-2i)^2} = \frac{i}{4}$$

(8) $z^{2n}/(z+1)^n$

解: $z_0 = -1, z_0 = \infty$, 分别为 n 阶极点

$$\begin{aligned}\operatorname{Res} f(-1) &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left[(z+1)^n \frac{z^{2n}}{(z+1)^n} \right] = \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1} z^{2n}}{dz^{n-1}} \\ &= \frac{1}{(n-1)!} 2n(2n-1)\cdots(n+2) z^{2n-n+1} \Big|_{z=-1} \\ &= \frac{(2n)!}{(n+1)!} \frac{(-1)^{n+1}}{(n-1)!} = (-1)^{n+1} \frac{(2n)!}{(n-1)!(n+1)!}\end{aligned}$$

$$\operatorname{Res} f(\infty) = -\operatorname{Res} f(-1) = \frac{(2n)!}{(n+1)!} \frac{(-1)^n}{(n-1)!} = (-1)^n \frac{(2n)!}{(n-1)!(n+1)!}$$

(9) $e^{1/(1-z)}$

解: $e^{1/(1-z)} = 1 - \frac{1}{z} - \frac{1}{2z^2} - \frac{1}{6z^3} + \frac{1}{24z^4} + \cdots$ [见 § 3.5 习题(4), 本书 P 41]

且本性奇点 $z_0 = 1$, 对应 $\frac{1}{z}$ 项的系数 $a_1 = -1$ 即为留数.

$\operatorname{Res} f(1) = -1$

(10) $1/(1+z^{2n})$

解: 令 $z_0^{2n} + 1 = 0$ 得 $z_0^{2n} = -1, z_0^n = \pm i = e^{i(2k+1)\pi/2}$

$z_0 = e^{i(2k+1)\pi/2n}, (k = 1, 2, \cdots, 2n-1)$ 为单极点.

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} \left[(z - z_0) \frac{1}{(1 + z^{2n})} \right], \text{ 由罗毕达法则得}$$

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} \frac{1}{2nz^{2n-1}} = \frac{1}{2nz_0^{2n-1}} = \frac{1}{2n} z_0^{-2n+1}, \text{ 将 } z_0 = e^{i(2k+1)\pi/2n} \text{ 代入得}$$

$$\begin{aligned}\operatorname{Res} f(z_0) &= \frac{1}{2n} e^{-i(2k+1)\pi(2n-1)/2n} = \frac{1}{2n} \frac{e^{i(2k+1)\pi/2n}}{e^{i(2k+1)\pi}} = \frac{1}{2n} (-1) e^{i(2k+1)\pi/2n} \\ &= -\frac{1}{2n} e^{i(2k+1)\pi/2n}\end{aligned}$$

2. 计算下列回路积分.

$$(1) \oint_l \frac{dz}{(z^2+1)(z-1)^2} \quad (l \text{ 的方程是 } x^2 + y^2 x - 2y = 0)$$

解: (1) $z_0 = \pm i$ 是两个单极点, $z = 1$ 是一个二阶极点

由于 l 的方程为 $x^2 + y^2 - 2x - 2y = 0$

即

$$(x^2 - 2x + 1) + (y^2 - 2y + 1) = 2$$

$$(x-1)^2 + (y-1)^2 = \sqrt{2}$$

在复数平面上, 圆心为 $(1, i)$, 半径 $R = \sqrt{2}$ 的圆.

由此可知 $z_0 = -i$ 不在该区域内. 故 $z_0 = i$ 和 $z_0 = 1$ 是回路 l 内极点.

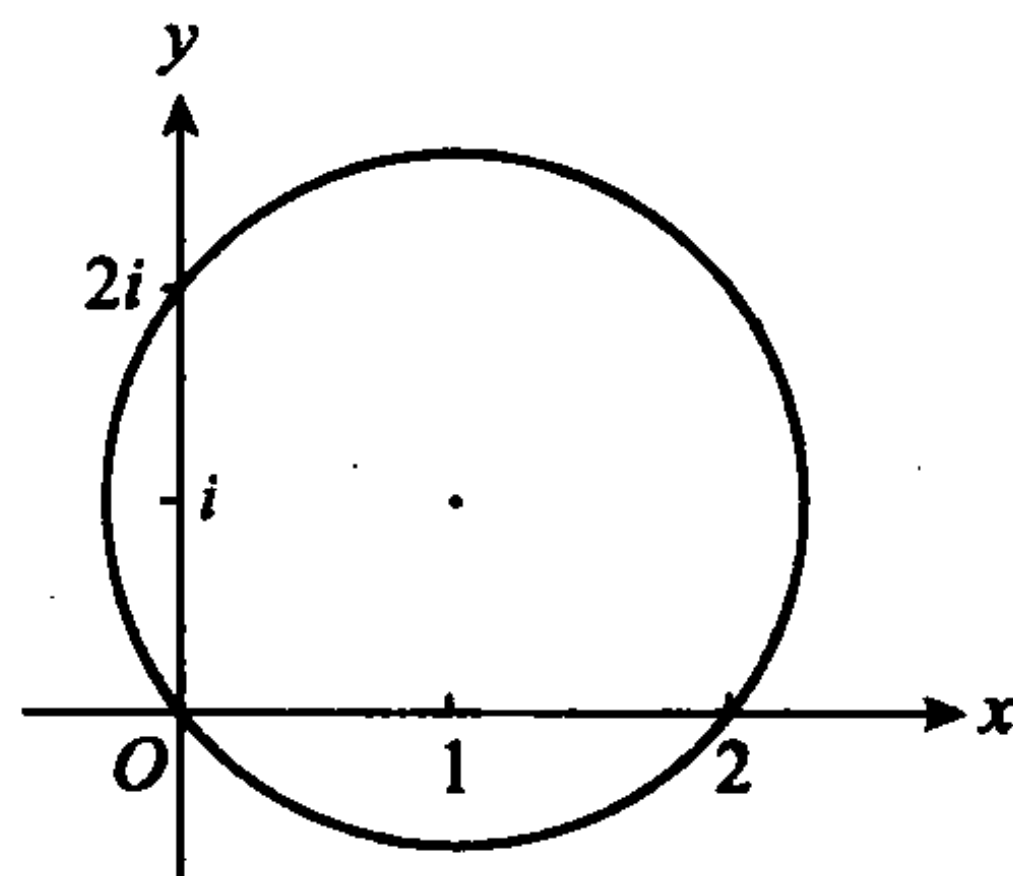
$$\operatorname{Res} f(i) = \lim_{z \rightarrow i} \left[(z-i) \frac{1}{(z-i)(z+i)(1-z)^2} \right]$$

$$= \lim_{z \rightarrow i} \frac{1}{(z+i)(1-z)^2}$$

$$= \frac{1}{2i} - \frac{1}{(1-i)^2} = \frac{-1}{4(i)^2} = \frac{1}{4}$$

$$\text{Res } f(1) = \lim_{z \rightarrow 1} \frac{d}{dz} (z-1)^2 \frac{1}{(z^2+1)(z-1)^2} = \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{1}{z^2+1} \right)$$

$$= \lim_{z \rightarrow 1} \frac{-2z}{(z^2+1)^2} = -\frac{1}{2}$$



由留数定理

$$\oint_C \frac{dz}{(z^2+1)(z-1)^2} = 2\pi i [\text{Res } f(i) + \text{Res } f(1)]$$

$$= 2\pi i \left(\frac{1}{4} - \frac{1}{2} \right) = -\frac{\pi i}{2}$$

$$(2) \oint_{|z|=1} \frac{\cos z}{z^3} dz$$

解: $z_0 = 0$ 是三阶极点, 且在 $|z|=1$ 的圆内

故

$$\text{Res } f(0) = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(z^3 \frac{\cos z}{z^3} \right) / dz^2 = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2 \cos z}{dz^2}$$

$$= \frac{1}{2} (-\cos z) \Big|_{z=0} = -\frac{1}{2}$$

所以

$$\oint_{|z|=1} \frac{\cos z}{z^3} dz = 2\pi i \text{Res } f(0) = -\pi i$$

$$(3) \oint_{|z|=2} e^{1/z^2} dz$$

解: $z=0$ 是本性奇点, 且在 $|z|=2$ 的圆内

$$\text{Res } f(0) = 0, \oint_{|z|=2} e^{1/z^2} dz = 0$$

$$(4) \oint_{|z|=2} \frac{z dz}{1/2 - \sin^2 z}$$

$$f(z) = \frac{z}{\frac{1}{2} - \sin^2 z} = \frac{2z}{1 - 2\sin^2 z} = \frac{2z}{\cos^2 z - \sin^2 z} = \frac{2z}{\cos 2z}$$

当 $\cos 2z = 0, z = \frac{1}{2} \left[\frac{\pi}{2} + k\pi \right] = \frac{\pi}{4} + \frac{k}{2}\pi, (k=0, \pm 1, \pm 2, \dots)$ 为单极点,

在 $|z|=2$ 的区域内, $z_0 = \pm \frac{\pi}{4}$,

$$\text{Res } f\left(\frac{\pi}{4}\right) = \lim_{z \rightarrow \frac{\pi}{4}} \left(z - \frac{\pi}{4} \right) \frac{2z}{\cos 2z} = \lim_{z \rightarrow \frac{\pi}{4}} \frac{2/z^2 - \frac{\pi}{2}z}{\cos 2z}$$

由罗比达法则

$$\text{Res } f\left(\frac{\pi}{4}\right) = \lim_{z \rightarrow \frac{\pi}{4}} \frac{4z - \frac{\pi}{2}}{-2\sin 2z} = \frac{\frac{\pi}{2}}{-2} = -\frac{\pi}{4}$$

$$\operatorname{Res} f\left(-\frac{\pi}{4}\right) = \lim_{z \rightarrow -\frac{\pi}{4}} \frac{\left(z + \frac{\pi}{4}\right) \cdot 2z}{\cos 2z} = \lim_{z \rightarrow -\frac{\pi}{4}} \frac{2z^2 + \frac{\pi}{2}z}{\cos 2z}$$

再由罗比达法则

$$\operatorname{Res} f\left(-\frac{\pi}{4}\right) = \lim_{z \rightarrow -\frac{\pi}{4}} \frac{4z + \frac{\pi}{2}}{-2\sin^2 z} = \frac{-\frac{\pi}{2}}{2} = -\frac{\pi}{4}$$

$$\begin{aligned} \text{所以 } \oint_{|z|=2} \frac{3dz}{\frac{1}{2} - \sin^2 z} &= 2\pi i \left[\operatorname{Res} f\left(\frac{\pi}{4}\right) + \operatorname{Res} f\left(-\frac{\pi}{4}\right) \right] = \frac{-\pi \times 2\pi i}{2} \\ &= -\pi^2 i \end{aligned}$$

3. 应用留数定理计算回路积分 $\frac{1}{2\pi i} \oint_l \frac{f(z)}{z-a} dz$, 函数 $f(z)$ 在 l 所围区域上是解析的, a 是这区域的一个内点.

解: 因为 $f(z)$ 在 l 所围区域内解析, 所以 $z_0 = a$ 点为 $f(z)/(z-a)$ 惟一的单极点.

$$\operatorname{Res} f(a) = \lim_{z \rightarrow a} (z-a) \frac{f(z)}{(z-a)} = \lim_{z \rightarrow a} f(z) = f(a)$$

$$\oint_l \frac{f(z)}{z-a} dz = 2\pi i \operatorname{Res} f(a) = 2\pi i f(a)$$

$$\frac{1}{2\pi i} \oint_l \frac{f(z)}{z-a} dz = f(a), \text{ 此即为科希公式.}$$

§ 4.2 应用留数定理计算实变函数定积分

1. 计算下列实变函数定积分.

$$(1) \int_0^{2\pi} \frac{dx}{2 + \cos x}$$

解: 属类型一积分: $I = \int_0^{2\pi} R(\cos x, \sin x) dx$, 取 $z = e^{ix}$, 将积分化为单位圆的回路积分, 有

$$z = e^{ix}, dz = e^{ix} i dx = z i dx, dx = \frac{1}{iz} dz$$

$$\begin{aligned} z = e^{ix} &= \cos x + i \sin x & \cos x &= \frac{1}{2}(z + z^{-1}) \\ z^{-1} = e^{-ix} &= \cos x - i \sin x & \Rightarrow \sin x &= \frac{1}{2i}(z - z^{-1}) \end{aligned}$$

$$\begin{aligned} \text{所以原积分 } I &= \oint_{|z|=1} \frac{\frac{dz}{iz}}{2 + \frac{1}{2}(z + z^{-1})} = \oint_{|z|=1} \frac{-idz}{2z + \frac{1}{2}z^2 + \frac{1}{2}} = -2i \oint_{|z|=1} \frac{dz}{z^2 + 4z + 1} \\ &= -2i \oint_{|z|=1} \frac{dz}{(z + 2 + \sqrt{3})(z + 2 - \sqrt{3})} \end{aligned}$$

设 $f(z) = \frac{1}{(z + 2 + \sqrt{3})(z + 2 - \sqrt{3})}$, $f(z)$ 有两个单极点, $z_1 = -2 - \sqrt{3}$ 和 $z_2 = -2 + \sqrt{3}$, 其中

$z_2 = -2 + \sqrt{3}$ 在积分回路内.

$$\begin{aligned}\operatorname{Res} f(-2 + \sqrt{3}) &= \lim_{z \rightarrow -2 + \sqrt{3}} \left[(z + 2 - \sqrt{3}) \frac{1}{(z + 2 + \sqrt{3})(z + 2 - \sqrt{3})} \right] \\ &= \frac{1}{(-2 + \sqrt{3} + 2 + \sqrt{3})} = \frac{1}{2\sqrt{3}}\end{aligned}$$

$$I = 2\pi i \times \frac{(-2i)}{2\sqrt{3}} = 2\pi/\sqrt{3}.$$

$$(2) \int_0^{2\pi} \frac{dx}{(1 + \epsilon \cos x)^2} \quad (0 < \epsilon < 1)$$

解: 取 $z = e^{ix}$, $dx = dz/iz \cos x = \frac{1}{2}(z + z^{-1})$

$$\begin{aligned}I &= \int_0^{2\pi} \frac{dx}{(1 + \epsilon \cos x)^2} = \oint_{|z|=1} \frac{dz/iz}{\left[1 + \epsilon \left(\frac{z + z^{-1}}{2}\right)\right]^2} = \oint_{|z|=1} \frac{-4izdz}{(2z + \epsilon z^2 + \epsilon)^2} \\ &= \oint_{|z|=1} \frac{-4i}{\epsilon^2} \left[\frac{zdz}{\left(z^2 + \frac{2z}{\epsilon} + 1\right)^2} \right] = \frac{-4i}{\epsilon^2} \oint_{|z|=1} \frac{zdz}{\left(z^2 + \frac{2z}{\epsilon} + 1\right)}\end{aligned}$$

$$\text{设 } f(z) = \frac{z}{\left(z^2 + \frac{2z}{\epsilon} + 1\right)^2}, \text{ 当 } z_0 = \frac{-\frac{2}{\epsilon} \pm \sqrt{\frac{4}{\epsilon^2} - 4}}{2} = \frac{1}{\epsilon}(-1 \pm \sqrt{1 - \epsilon^2})$$

$f(z)$ 的分母为零. $z_0 = \frac{1}{\epsilon}(-1 \pm \sqrt{1 - \epsilon^2})$ 为 $f(z)$ 的两个二阶极点, 且 $|z_0| = \left| \frac{1}{\epsilon}(-1 - \sqrt{1 - \epsilon^2}) \right| > \frac{1}{\epsilon} > 1$, 在 $|z| = 1$ 的回路外. $|z| = \left| \frac{1}{\epsilon}(-1 + \sqrt{1 - \epsilon^2}) \right| = \frac{\epsilon}{1 + \sqrt{1 + \epsilon^2}} < \epsilon/2$, 在 $|z| = 1$ 的回路内.

当 $z_0 = \frac{1}{\epsilon}(-1 + \sqrt{1 - \epsilon^2})$ 时

$$\begin{aligned}\operatorname{Res}(z_0) &= \lim_{z \rightarrow z_0} \frac{d}{dz} \left[(z - z_0)^2 \frac{z}{(z - z_0) \left(z + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^2} \right] \\ &= \lim_{z \rightarrow z_0} \frac{d}{dz} \left[\frac{z}{\left(z + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^2} \right] \\ &= \frac{1}{\left(z_0 + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^2} - \frac{2z_0}{\left(z_0 + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^3} \\ &= \frac{z_0 + 1 + \sqrt{1 - \epsilon^2}/\epsilon - 2z_0}{\left(z_0 + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^3} = \frac{\frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon} - z_0}{\left(z_0 + \frac{1 + \sqrt{1 - \epsilon^2}}{\epsilon}\right)^3}\end{aligned}$$

将 $z_0 = (-1 + \sqrt{1-\epsilon^2})/\epsilon$ 代入, 得

$$\operatorname{Res}(z_0) = \frac{2/\epsilon}{(\sqrt{1-\epsilon^2}/\epsilon)^3} = \frac{2\epsilon^3}{8\epsilon} \frac{1}{(\sqrt{1-\epsilon^2})^3} = \frac{\epsilon^2}{4} \frac{1}{(1-\epsilon^2)^{3/2}}$$

$$\text{所以} \oint_{|z|=1} \frac{zdz}{z^2 + \frac{2z}{\epsilon} + 1} = 2\pi i \left[\frac{\epsilon^2}{4(1-\epsilon^2)^{3/2}} \right] = \frac{\pi i \epsilon^2}{2(1-\epsilon^2)^{3/2}}$$

$$I = -\frac{4i}{\epsilon^2} \oint_{|z|=1} \frac{zdz}{\left(z^2 + \frac{2z}{\epsilon} + 1\right)} = \frac{-4i}{\epsilon^2} \times \frac{\pi i \epsilon^2}{2(1-\epsilon^2)^{3/2}} = \frac{2\pi}{(1-\epsilon^2)^{3/2}}$$

$$(3) \int_0^{2\pi} \frac{\cos^2 x dx}{1 - 2\epsilon \cos x + \epsilon^2} \quad (0 < \epsilon < 1)$$

解: 取 $z = e^{ix}$, $dx = \frac{dz}{iz}$, $\cos x = \frac{1}{2}(z + z^{-1})$

$$\cos 2x = 2\cos^2 x - 1 = \frac{1}{2}(z^2 + 2 - z^{-2}) - 1 = \frac{z^2 + z^{-2}}{2} = \frac{1}{2}(z^2 + z^{-2})$$

$$\begin{aligned} I &= \int_0^{2\pi} \frac{\cos^2 2x dx}{1 - 2\epsilon \cos x + \epsilon^2} = \oint_{|z|=1} \frac{\left[\frac{1}{2}(z^2 + z^{-2})\right]^2 \frac{dz}{iz}}{1 - 2\epsilon(z + z^{-1})/2 + \epsilon^2} \\ &= \oint_{|z|=1} \frac{(z^4 + 1)^2 dz}{i4z^4(z - \epsilon z^2 - \epsilon + \epsilon z^2)} \\ &= \frac{1}{4i} \oint_{|z|=1} \frac{(z^4 + 1)^2 dz}{z^4[-\epsilon z^2 + (1 + \epsilon^2)z - \epsilon]} = \frac{1}{4i} \oint_{|z|=1} \frac{(z^4 + 1)^2 dz}{z^4(1 - \epsilon z)(z - \epsilon)} \end{aligned}$$

设 $f(z) = \frac{(z^4 + 1)^2}{z^4(1 - \epsilon z)(z - \epsilon)}$, $f(z)$ 在 $z_0 = 0$ 为四阶极点, $z_0 = \epsilon, z = \frac{1}{\epsilon}$ 为两个单极点.

$z_0 = \epsilon, z_0 = 0$, 在 $|z| = 1$ 的区域内, $z_0 = 1/\epsilon, (|\epsilon| < 1)$ 在 $|z| < 1$ 的回路外

$$\text{又 } \operatorname{Res} f(\epsilon) = \lim_{z \rightarrow \epsilon} \left[(z - \epsilon) \frac{(z^4 + 1)^2}{z^4(z - \epsilon)(1 - \epsilon z)} \right] = \lim_{z \rightarrow \epsilon} \frac{(z^4 + 1)^2}{(1 - \epsilon z)z^4} = \frac{(\epsilon^4 + 1)^2}{(1 - \epsilon^2)\epsilon^4}$$

$$\begin{aligned} \operatorname{Res} f(0) &= \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left(z^4 \cdot \frac{(z^4 + 1)^2}{z^4(1 - \epsilon z)(z - \epsilon)} \right) = \frac{1}{6} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left[\frac{(z^4 + 1)^2}{(1 - \epsilon z)(z - \epsilon)} \right] \\ &= \frac{1}{6} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(\frac{2(1 + z^4)4z^3}{(1 - \epsilon z)(z - \epsilon)} + \frac{(1 + z^4)^2 [2\epsilon z - (1 + \epsilon^2)]}{[(1 - \epsilon z)(z - \epsilon)]^2} \right) \\ &= \frac{1}{6} \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{8(3z^2 + 7z^3)}{(1 - \epsilon z)(z - \epsilon)} + \frac{8z^3(1 + z^4)[2\epsilon z - 1 - \epsilon^2]}{[(1 - \epsilon z)(z - \epsilon)]^2} \right) \\ &\quad + \frac{2(1 + z^4)4z^3(2\epsilon z - 1 - \epsilon^2)}{[(1 - \epsilon z)(z - \epsilon)]^2} + \frac{2\epsilon(1 + z^4)}{[(1 - \epsilon z)(z - \epsilon)]^2} + \frac{2(1 + z^4)(2\epsilon z - 1 - \epsilon^2)^2}{[(1 - \epsilon z)(z - \epsilon)]^3} \\ &= \frac{1}{6} \lim_{z \rightarrow 0} \left(\frac{48z + 168z^3}{(1 - \epsilon z)(z - \epsilon)} + \frac{8(3z^2 + 7z^3)(2\epsilon z - 1 - \epsilon^2)}{[(1 - \epsilon z)(z - \epsilon)]^2} + \frac{2\epsilon(8z^3)(1 + z^4)2}{[(1 - \epsilon z)(z - \epsilon)]^2} \right. \\ &\quad + \frac{(24z^2 + 32z^3)(2\epsilon z - 1 - \epsilon^2)2}{[(1 - \epsilon z)(z - \epsilon)]^2} + \frac{2 \times 8z^3(1 + z^4)[2\epsilon z - 1 - \epsilon^2]^2}{[(1 - \epsilon z)(z - \epsilon)]^3} \\ &\quad \left. + \frac{8\epsilon z^3}{[(1 - \epsilon z)(z - \epsilon)]^2} + \frac{4\epsilon(1 + z^4)(2\epsilon z - 1 - \epsilon^2)}{[(1 - \epsilon z)(1z - \epsilon)]^3} \right) \end{aligned}$$

$$\begin{aligned}
& + \frac{8z^3(2\epsilon z - 1 - \epsilon^2)^2 + 8(1+z^4)(2\epsilon z - 1 - \epsilon^2)\epsilon}{(1-\epsilon z)(z-\epsilon)^3} \\
& + \frac{2(1+z^4)(2\epsilon z - 1 - \epsilon^2)^2(2\epsilon z - 1 - \epsilon^2)3}{[(1-\epsilon z)(z-\epsilon)]^4} \\
& = \frac{1}{6} \left(\frac{-4\epsilon(\epsilon^2+1)}{-\epsilon^3} + \frac{-8\epsilon(\epsilon^2+1)}{-\epsilon^3} + \frac{-2(1+\epsilon^2)^3 3}{\epsilon^4} \right) \\
& = \frac{2\epsilon(\epsilon^2+1)}{\epsilon^3} - \frac{(1+\epsilon^2)^3}{\epsilon^4} = \frac{(1+\epsilon^2)}{\epsilon^4} [2\epsilon^2 - (1+\epsilon^2)] \\
& = \frac{1+\epsilon^2}{\epsilon^4} (-1-\epsilon^4) = -\frac{(1+\epsilon^2)(\epsilon^4+1)}{\epsilon^4}
\end{aligned}$$

$$\begin{aligned}
\text{所以 } I &= \frac{1}{4i} \oint_{|z|=1} f(z) dz = \frac{1}{4i} 2\pi i [\text{Res } f(\epsilon) + \text{Res } f(0)] = \frac{\pi}{2} \left(\frac{(\epsilon^4+1)^2}{(1-\epsilon^2)\epsilon^4} - \frac{(1+\epsilon^2)(\epsilon^4+1)}{\epsilon^4} \right) \\
&= \frac{\pi}{2} \left(\frac{\epsilon^4+1}{\epsilon^4} \right) \left(\frac{1+\epsilon^4}{1-\epsilon^2} - \frac{1-\epsilon^4}{1-\epsilon^2} \right) = \frac{\pi}{2} \frac{\epsilon^4+1}{\epsilon^4} \left(\frac{2\epsilon^4}{1-\epsilon^2} \right) \\
&= \pi \frac{\epsilon^4+1}{1-\epsilon^2} = \pi(\epsilon^4+1)/(1-\epsilon^2)
\end{aligned}$$

$$(4) \int_0^{2\pi} \frac{\sin^2 x dx}{a + b \cos x} \quad (a > b > 0)$$

解: 取 $z = e^{ix}$, $dx = dz/iz$, $\cos x = \frac{1}{2}(z + z^{-1})$, $\sin x = \frac{1}{2i}(z - z^{-1})$ 代入

$$\begin{aligned}
I &= \int_0^{2\pi} \frac{\sin^2 x dx}{a + b \cos x} = \oint_{|z|=1} \frac{[(z - z^{-1})/2i]^2 dz/iz}{a + b(z + z^{-1})/2} \\
&= \frac{-1}{2i} \oint_{|z|=1} \frac{(z^2 - 1)^2 dz}{2z^3 [a + b(z^2 + 1)/2z]} = \frac{-1}{2ib} \oint_{|z|=1} \frac{(z^2 - 1)^2 dz}{z^2 (z^2 + \frac{2az}{b} + 1)}
\end{aligned}$$

设 $f(z) = (z^2 - 1)^2 / z^2 (z^2 + 2az/b + 1)$

令其分母: $z^2 (z^2 + \frac{2az}{b} + 1) = 0$, 则 $z_0^2 = 0, z_0 = 0$

$$z_0 = \left(-\frac{2a}{b} \pm \sqrt{\frac{4a^2}{b^2} - 4} \right) / 2 = -\frac{a}{b} \pm \sqrt{\frac{a^2}{b^2} - 1} = -\frac{1}{b} (a \pm \sqrt{a^2 - b^2})$$

所以 $f(z)$ 有两个单极点 $z_0 = -\frac{1}{b} (a \pm \sqrt{a^2 - b^2})$

且 z_0 为二阶极点. 由于 $z_0 = -\frac{1}{b} (a + \sqrt{a^2 - b^2})$, $(a > b > 0)$ 的模

$$|z_0| = \left| \frac{a}{b} + \sqrt{\frac{a^2}{b^2} - 1} \right|, \text{ 且 } \left| \frac{a}{b} \right| > 1.$$

$|z_0| = \left| \frac{a}{b} + \sqrt{\frac{a^2}{b^2} - 1} \right| > \left| \frac{a}{b} \right| > 1$, 在积分回路外. 其留数分别为

$$\begin{aligned}
\text{Res } f(0) &= \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \frac{(z^2 - 1)^2}{z^2 (z^2 + \frac{2az}{b} + 1)} \right] = \lim_{z \rightarrow 0} \frac{d}{dz} \left[\frac{(z^2 - 1)^2}{z^2 + \frac{2az}{b} + 1} \right] \\
&= \lim_{z \rightarrow 0} \left[\frac{2(z^2 - 1) \cdot 2z}{z^2 + \frac{2az}{b} + 1} - \frac{(z^2 - 1)^2 (2z + \frac{2a}{b})}{(z^2 + \frac{2az}{b} + 1)^2} \right] = -\frac{2a}{b}
\end{aligned}$$

设 $z_0 = -\frac{1}{b}(a - \sqrt{a^2 - b^2})$

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} (z - z_0) \frac{(z^2 - 1)^2}{(z - z_0) \left[z + \frac{1}{b}(a + \sqrt{a^2 - b^2}) \right] z^2}$$

则

$$= \lim_{z \rightarrow z_0} \frac{(z^2 - 1)^2}{z^2 \left(z + \frac{1}{b}(a + \sqrt{a^2 - b^2}) \right)}$$

将 $z_0 = -\frac{1}{b}(a - \sqrt{a^2 - b^2})$ 代入得

$$\begin{aligned} \operatorname{Res} f(z_0) &= \frac{(z_0^2 - 1)^2}{2z_0^2 (\sqrt{a^2 - b^2})/b} = \frac{b^2 (a^2 - 2a\sqrt{a^2 - b^2} + a^2 - b^2 - b^2)^2}{2b^4 (a - \sqrt{a^2 - b^2})^2 \sqrt{a^2 - b^2}/b} \\ &= \frac{1}{2b} \left(\frac{(2a^2 - 2b^2 - 2a\sqrt{a^2 - b^2})^2}{a^2 - 2a\sqrt{a^2 - b^2} + a^2 - b^2} \right) \frac{1}{\sqrt{a^2 - b^2}} \\ &= \frac{4}{2b} \left(\frac{(a^2 - b^2 - a\sqrt{a^2 - b^2})^2}{(2a^2 - 2a\sqrt{a^2 - b^2} - b^2)} \right) \frac{1}{\sqrt{a^2 - b^2}} \left(\frac{\sqrt{a^2 - b^2}}{\sqrt{a^2 - b^2}} \right) \\ &= \frac{2}{b} \sqrt{a^2 - b^2} \left(\frac{(\sqrt{a^2 - b^2} - a)^2}{2a^2 - 2a\sqrt{a^2 - b^2} - b^2} \right) \\ &= \frac{2(2a^2 - b^2 - 2a\sqrt{a^2 - b^2}) \sqrt{a^2 - b^2}}{b(2a^2 - b^2 - 2a\sqrt{a^2 - b^2})} \\ &= \frac{2}{b} \sqrt{a^2 - b^2} \end{aligned}$$

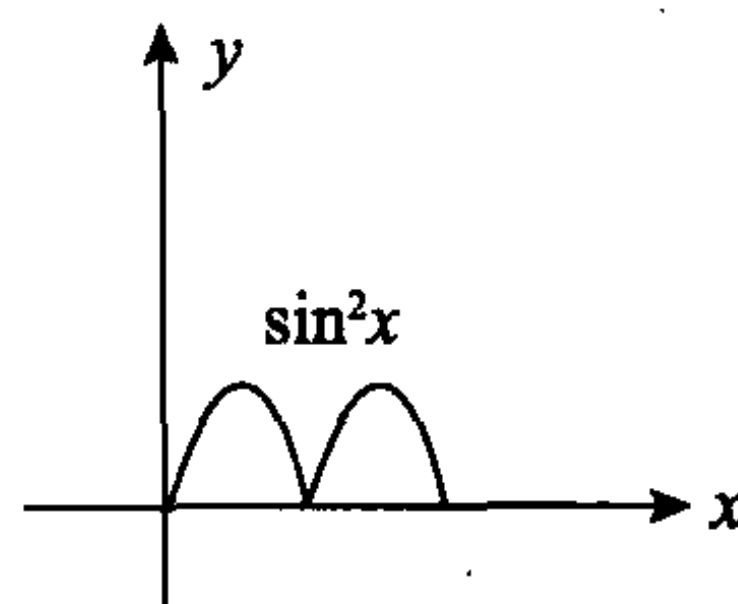
$$\begin{aligned} I &= \frac{-1}{2bi} \oint_{|z|=1} f(z) dz = \frac{-2\pi i}{2bi} (\operatorname{Res} f(0) + \operatorname{Res} f(z_0)) = \frac{-\pi}{b} \left(-\frac{2a}{b} + \frac{2}{b} \sqrt{a^2 - b^2} \right) \\ &= \frac{-2\pi}{b^2} (\sqrt{a^2 - b^2} - a) = \frac{2\pi}{b^2} (a - \sqrt{a^2 - b^2}) \end{aligned}$$

$$(5) \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} \quad (a > 0)$$

$$\text{解: } I = \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} = a \int_0^\pi \frac{dx}{a^2 + \sin^2 x} = \frac{a}{2} \int_0^{2\pi} \frac{dx}{a^2 + \sin^2 x}$$

将 $z = e^{ix}$, $dz/iz = dx$, $\sin x = \frac{1}{2i}(z - z^{-1})$ 代入

$$\begin{aligned} I &= \frac{a}{2} \int_0^{2\pi} \frac{dx}{a^2 + \sin^2 x} = \frac{a}{2} \oint_{|z|=1} \frac{dz/iz}{a^2 + [(z - z^{-1})/2i]^2} \\ &= \frac{a}{2} \oint_{|z|=1} \frac{dz/iz}{a^2 - [(z - z^{-1})/2]^2} \\ &= \frac{a}{2} \oint_{|z|=1} \frac{dz/iz}{\left(a - \frac{z - z^{-1}}{2} \right) \left(a + \frac{z - z^{-1}}{2} \right)} \\ &= \frac{a}{2i} \oint_{|z|=1} \frac{4z dz}{(2az - z^2 + 1)(2az + z^2 - 1)} \end{aligned}$$



$$\begin{aligned}
 &= \frac{-a}{2i} \oint_{|z|=1} \frac{4zdz}{(z^2 - 2az - 1)(z^2 + 2az - 1)} \\
 &= 2ai \oint_{|z|=1} \frac{4zdz}{(z^2 - 2az - 1)(z^2 + 2az - 1)}
 \end{aligned}$$

$$\text{取 } f(z) = \frac{z}{(z^2 - 2az - 1)(z^2 + 2az - 1)}$$

当分母为零时, $(z^2 - 2az - 1)(z^2 + 2az - 1) = 0$ 有 $\begin{cases} z^2 - 2az - 1 = 0 \\ z^2 + 2az - 1 = 0 \end{cases}$

分别解方程得 $z_0 = a \pm \sqrt{a^2 + 1}$ 和 $z_0 = -a \pm \sqrt{a^2 + 1}$

在 $z_0 = a + \sqrt{a^2 + 1}$ 和 $z_0 = -a - \sqrt{a^2 + 1}$ 时, $|z_0| > 1$,

故仅有 $z_0 = -a + \sqrt{a^2 + 1}$ 和 $z_0 = a - \sqrt{a^2 + 1}$, 当 a 一定范围内可满足在积分回路内的条件, 求留数: 设 $z_0 = -a + \sqrt{a^2 + 1}$.

$$\begin{aligned}
 \text{则 Res}(z_0) &= \lim_{z \rightarrow z_0} \frac{(z - z_0)z}{(z^2 - 2az - 1)(z^2 + 2az - 1)} \\
 &= \lim_{z \rightarrow z_0} \frac{(z + a - \sqrt{a^2 + 1})z}{(z^2 - 2az - 1)(z + a + \sqrt{a^2 + 1})(z + a - \sqrt{a^2 + 1})} \\
 &= \lim_{z \rightarrow z_0} \frac{z}{(z^2 - 2az - 1)(z + a + \sqrt{a^2 + 1})}, \quad (z_0 = -a + \sqrt{a^2 + 1}) \\
 &= \frac{-a + \sqrt{a^2 + 1}}{(1 - a + \sqrt{a^2 + 1} - a - \sqrt{a^2 + 1})(-a + \sqrt{a^2 + 1} - a + \sqrt{a^2 + 1})(-a + \sqrt{a^2 + 1} + a + \sqrt{a^2 + 1})} \\
 &= \frac{-a + \sqrt{a^2 + 1}}{(-2a)2(-a + \sqrt{a^2 + 1})2\sqrt{a^2 + 1}} = \frac{-1}{8a(a^2 + 1)}
 \end{aligned}$$

设 $z_0 = a - \sqrt{a^2 + 1}$

$$\begin{aligned}
 \text{则 Res}(z_0) &= \lim_{z \rightarrow z_0} \frac{(z - z_0)z}{(z^2 - 2az - 1)(z^2 + 2az - 1)} \\
 &= \lim_{z \rightarrow z_0} \frac{z}{(z - a - \sqrt{a^2 + 1})(z^2 + 2az - 1)} \\
 &= \frac{a - \sqrt{a^2 + 1}}{-2(\sqrt{a^2 + 1})[(a - \sqrt{a^2 + 1})^2 + 2a\sqrt{a^2 + 1} - 1]} \\
 &= \frac{a - \sqrt{a^2 + 1}}{-2\sqrt{a^2 + 1}[a^2 - 2a\sqrt{a^2 + 1} + a^2 + 1 + 2a(a - \sqrt{a^2 + 1}) - 1]} \\
 &= \frac{a - \sqrt{a^2 + 1}}{-2\sqrt{a^2 + 1}(4a^2 - 4a\sqrt{a^2 + 1})} \\
 &= \frac{1}{-8a\sqrt{a^2 + 1}}
 \end{aligned}$$

$$I = \int_0^\pi \frac{dx}{a^2 + \sin^2 x} = 2ai2\pi i \left[\frac{1}{-8a\sqrt{a^2 + 1}} + \frac{1}{-8a\sqrt{a^2 + 1}} \right]$$

$$= \frac{-2a\pi i}{2a\sqrt{a^2+1}} = \frac{\pi}{\sqrt{a^2+1}}$$

$$(6) \int_0^{2\pi} \frac{\cos x dx}{1-2\epsilon \cos x + \epsilon^2} \quad (|\epsilon| < 1), dx = dz/iz, \cos x = \frac{1}{2}(z+z^{-1})$$

解:

$$I = \int_0^{2\pi} \frac{\cos x dx}{1-2\epsilon \cos x + \epsilon^2} = \oint_{|z|=1} \frac{\frac{1}{2}(z+z^{-1})dz/iz}{1-2\epsilon\left(\frac{z+z^{-1}}{2}\right)+\epsilon^2}$$

$$= \frac{1}{2i} \oint_{|z|=1} \frac{(z^2+1)dz}{z^2+\epsilon z^2-\epsilon z^3-\epsilon z}$$

其中分母为:

$$z^2 + \epsilon^2 z^2 - \epsilon z^3 - \epsilon z = z[(1+\epsilon^2)z - \epsilon z^2 - \epsilon] = z(1-\epsilon z)(z-\epsilon) = 0$$

$f(z)$ 有三个单极点, 分别为 $z_0 = 0, z_0 = \epsilon, z_0 = \frac{1}{\epsilon}$.

又 $|\epsilon| < 1$, 则 $\left|\frac{1}{\epsilon}\right| > 1, z_0 = 1/\epsilon$ 不在积分回路内满足条件的为 $z_0 = 0, z_0 = \epsilon$.

$$\text{Res } f(0) = \lim_{z \rightarrow 0} z \cdot f(z) = \lim_{z \rightarrow 0} \frac{(z^2+1)}{(1-\epsilon z)(z-\epsilon)} = -\frac{1}{\epsilon}$$

$$\text{Res } f(\epsilon) = \lim_{z \rightarrow \epsilon} (z-\epsilon) f(z) = \lim_{z \rightarrow \epsilon} \frac{z^2+1}{z(1-\epsilon z)} = \frac{\epsilon^2+1}{\epsilon(1-\epsilon^2)}$$

$$I = 2\pi i \frac{1}{2i} [\text{Res } f(0) + \text{Res } f(\epsilon)]$$

$$= \pi \left(-\frac{1}{\epsilon} + \frac{1+\epsilon^2}{\epsilon(1-\epsilon^2)} \right) = \pi \left(\frac{1+\epsilon^2-1+\epsilon^2}{\epsilon(1-\epsilon^2)} \right) = \frac{2\pi\epsilon}{1-\epsilon^2}$$

$$(7) \int_0^{\pi/2} \frac{dx}{1+\cos^2 x}$$

$$\text{解: } dx = dz/iz, \cos x = \frac{1}{2}(z+z^{-1})$$

$$I = \int_0^{\pi/2} \frac{dx}{1+\cos^2 x} = \frac{1}{2} \int_0^{\pi} \frac{dx}{1+\cos^2 x} = \frac{1}{4} \int_0^{2\pi} \frac{dx}{1+\cos^2 x}$$

$$= \frac{1}{4} \oint_{|z|=1} \frac{dz/iz}{1+\left[\frac{1}{2}(z+z^{-1})\right]^2} = \frac{1}{i} \oint_{|z|=1} \frac{dz}{6z+z^3+z^{-1}}$$

$$= \frac{1}{i} \oint_{|z|=1} \frac{zdz}{z^4+6z+1} = \frac{1}{i} \oint_{|z|=1} \frac{zdz}{(z^2+3+2\sqrt{2})(z^2+3-2\sqrt{2})}$$

$$= \frac{1}{i} \oint_{|z|=1} f(z) dz$$

$f(z)$ 的分母 $(z^2+3+2\sqrt{2})=0$ 和 $(z^2+3-2\sqrt{2})=0$

即 $z_0 = \pm \sqrt{3+2\sqrt{2}}$ 和 $z_0 = \pm \sqrt{3-2\sqrt{2}}$ 为 4 个单极点.

由于 $|z_0| > 1$, 所以 $z_0 = \pm \sqrt{3+2\sqrt{2}}$ 在积分回路外,

$$\begin{aligned}
 \operatorname{Res} f(2\sqrt{2}-3) &= \lim_{z \rightarrow \sqrt{2\sqrt{2}-3}} (z + \sqrt{2\sqrt{2}-2}) f(z) \\
 &= \lim_{z \rightarrow \sqrt{2\sqrt{2}-3}} \frac{z}{(z^2 + 3 + 2\sqrt{2})(z + \sqrt{2\sqrt{2}-3})} \\
 &= \frac{1}{2} \cdot \frac{1}{4\sqrt{2}} = \frac{\sqrt{2}}{16}
 \end{aligned}$$

$$\begin{aligned}
 \operatorname{Res} f(-\sqrt{2\sqrt{2}-3}) &= \lim_{z \rightarrow -\sqrt{2\sqrt{2}-3}} (z + \sqrt{2\sqrt{2}-3}) f(z) \\
 &= \lim_{z \rightarrow -\sqrt{2\sqrt{2}-3}} \frac{z}{(z^2 + 3 + 2\sqrt{2})(z - \sqrt{2\sqrt{2}-3})} \\
 &= \frac{1}{2} \cdot \frac{1}{4\sqrt{2}} = \frac{\sqrt{2}}{16}
 \end{aligned}$$

$$\text{所以 } I = \frac{1}{i} 2\pi i \left(\frac{\sqrt{2}}{16} + \frac{\sqrt{2}}{16} \right) = \frac{\sqrt{2}}{4} \pi$$

$$(8) \int_0^{2\pi} \cos^{2n} x dx$$

$$\text{解: 令 } dx = dz/iz, \cos x = \frac{1}{2}(z + z^{-1})$$

$$\begin{aligned}
 I &= \int_0^{2\pi} \cos^{2n} x dx = \oint_{|z|=1} \left[\frac{1}{2}(z + z^{-1}) \right]^{2n} dz/iz \\
 &= \frac{1}{2^{2n}i} \oint_{|z|=1} (z + z^{-1})^{2n} dz/z = \frac{1}{2^{2n}i} \oint_{|z|=1} \frac{(z^2 + 1)^{2n} dz}{z^{2n+1}} \\
 &= \frac{1}{2^{2n}i} \oint_{|z|=1} f(z) dz
 \end{aligned}$$

分母为 0, 有 $z_0 = 0$ 是 $2n+1$ 阶极点.

$$\text{留数为 } \operatorname{Res} f(0) \lim_{z \rightarrow 0} \frac{1}{(m-1)!} \left\{ \frac{d^{m-1}}{dz^{m-1}} [z^m f(z)] \right\} = \frac{1}{(2n)!} \frac{d^{2n}}{dz^{2n}} (z^2 + 1)^{2n}$$

由于 $(1+z^2)^{2n}$ 的 k 项为 $(2n)! z^{2k} / (2n-k)! k!$ 且对 z 微分 $2n$ 次, 故 $2k < 2n$ 项均为零, 又 $z_0 = 0$, 故 $2k > 2n$ 项均有 $z = 0$, 故也均为零. 所以仅有 $2k = 2n$ 项不为零.

$$\operatorname{Res} f(0) = \frac{1}{(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} \left[\frac{(2n)! z^{2n}}{(2n-n)! n!} \right] = \lim_{z \rightarrow 0} \frac{1}{(n!)^2} \frac{d^{2n}}{dz^{2n}} (z^{2n}) = \frac{(2n)!}{(n!)^2}$$

所以

$$\begin{aligned}
 I &= \frac{1}{2^{2n}i} \cdot 2\pi i \cdot \oint_{|z|=1} f(z) dz = \frac{2\pi i}{2^{2n}i} (\operatorname{Res} f(0)) \\
 &= \frac{2\pi}{2^{2n}} \frac{(2n)!}{(n!)^2} = 2\pi \frac{(2n)!}{(2^n n!)^2} = 2\pi \frac{1 \cdot 2 \cdot 3 \cdots 2n}{2^n n! 2^n n!} \\
 &= 2\pi \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)(2 \cdot 4 \cdot 6 \cdots 2n)}{2^n n! 2^n n!} \\
 &= 2\pi \frac{1 \cdot 3 \cdot 5 \cdots (2n-1) 2^n (1 \cdot 2 \cdot 3 \cdot 4 \cdots n)}{2^n n! 2^n n!} \\
 &= 2\pi \frac{1 \cdot 3 \cdot 5 \cdots (2n-1) 2^n n!}{2^n n! 2^n n!} = 2\pi \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n n!}
 \end{aligned}$$

$$= 2\pi \frac{1 \cdot 3 \cdot 5 \cdot 7 \cdot \dots (2n-1)}{2 \cdot 4 \cdot 6 \cdot 8 \cdot \dots 2n}$$

2. 计算下列实变函数定积分.

$$(1) \int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1} dx$$

解: 积分区间是 $(-\infty, +\infty)$; 复变函数 $f(z) = \frac{z^2+1}{z^4+1}$ 在实轴上没有奇点, 在上半平面除有限个奇点外是解析的; 且 z 在上半平面 $\rightarrow \infty$ 和在实轴上 $\rightarrow \infty$ 时, $zf(z)$ 一致地 $\rightarrow 0$, 所以本题属于类型二,

则 $\int_{-\infty}^{\infty} f(x) dx = 2\pi i \{f(z) \text{ 在上半平面所有奇点的留数之和}\}.$

$$\text{因为 } f(z) = \frac{z^2+1}{z^4+1} = \frac{z^2+1}{(z^2+i)(z^2-i)} = \frac{z^2+1}{(z+\sqrt{i})(z-\sqrt{i})(z-\sqrt{-i})(z+\sqrt{-i})}$$

共有四个单极点, 分别为 $z = \sqrt{i}, z_0 = -\sqrt{i}, z_0 = \sqrt{-i}, z_0 = -\sqrt{-i}$

在上半平面的有 $z_0 = \sqrt{i}$ 和 $z_0 = -\sqrt{-i}$

$$\begin{aligned} \text{其留数分别为} \quad \text{Res } f(\sqrt{i}) &= \lim_{z \rightarrow \sqrt{i}} \frac{(z^2+1)(z-\sqrt{i})}{(z^2+i)(z+\sqrt{i})(z-\sqrt{i})} \\ &= \lim_{z \rightarrow \sqrt{i}} \frac{z^2+1}{(z^2+i)(z+\sqrt{i})} = \frac{i+1}{2i2\sqrt{i}} = \frac{i+1}{2i \left[2 \frac{\sqrt{2}}{2} (1+i) \right]} = \frac{1}{2\sqrt{2}i} \end{aligned}$$

[注: $\sqrt[n]{z} = \sqrt[n]{\rho}(\cos \varphi/n + i \sin \varphi/n), n=2, \varphi=90^\circ, \sqrt{i} = i^{\frac{1}{2}} = \sqrt{1}(\cos 45^\circ + i \sin 45^\circ) = \frac{\sqrt{2}}{2}(1+i)$]

$$\begin{aligned} \text{Res } f(\sqrt{-i}) &= \lim_{z \rightarrow \sqrt{-i}} \frac{(z^2+1)(z+\sqrt{-i})}{(z^2-i)(z-\sqrt{-i})(z+\sqrt{-i})} = \frac{1-i}{+2i2\sqrt{-i}} \\ &= \frac{1-i}{2i(2 \times \cos 45^\circ - i \sin 45^\circ)} = \frac{1}{2\sqrt{2}i} \end{aligned}$$

$$\sqrt{-i} = \sqrt{1} \left(\cos \frac{90^\circ}{2} - i \sin \frac{90^\circ}{2} \right) = \frac{\sqrt{2}}{2}(1-i)$$

$$I = \int_{-\infty}^{\infty} f(z) dz = 2\pi i \left(\frac{1}{2\sqrt{2}i} + \frac{1}{2\sqrt{2}i} \right) = \sqrt{2}\pi$$

$$(2) \int_0^{\infty} \frac{x^2 dx}{(x^2+9)(x^2+4)^2}$$

解: 本题属于类型二, 且为偶函数

$$I = \int_0^{\infty} \frac{x^2 dx}{(x^2+9)(x^2+4)^2} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+9)(x^2+4)^2} \quad (\text{偶函数})$$

$$\text{取 } f(z) = \frac{z^2}{(z^2+9)(z^2+4)^2}$$

其中分母部分: $(z^2+9)(z^2+4)^2 = (z+3i)(z-3i)(z+2i)^2(z-2i)^2$

共有 4 个奇点, 分别为 $z_0 = -3i, z_0 = 3i, z_0 = -2i, z_0 = 2i$.

其中 $z_0 = \pm 2i$ 是二阶极点.

在上半面的有 $z_0 = 3i$ 为单极点, $z_0 = 2i$ 为二阶极点.

其留数分别为

$$\begin{aligned}\operatorname{Res} f(3i) &= \lim_{z \rightarrow 3i} \frac{(z-3i)z^2}{(z-3i)(z+3i)(z^2+4)^2} = \frac{-9}{6i \times (-5)^2} \\ &= \frac{-3}{2i \times 25} = \frac{3i}{50}\end{aligned}$$

$$\begin{aligned}\operatorname{Res} f(2i) &= \lim_{z \rightarrow 2i} \frac{d}{dz} \left(\frac{z^2(z-2i)^2}{(z^2+9)(z^2+4)^2} \right) = \lim_{z \rightarrow 2i} \frac{d}{dz} \left[\frac{z^2}{(z^2+9)(z+2i)^2} \right] \\ &= \lim_{z \rightarrow 2i} \left[\frac{2z}{(z^2+9)(z+2i)^2} + \frac{-2z^2(z^2+9)(z+2i) - 2z^3(z+2i)^2}{[(z^2+9)(z+2i)^2]^2} \right] \\ &= \frac{-i}{20} - \frac{96i}{640} = \frac{-416i}{6400} = \frac{-13i}{200}\end{aligned}$$

$$I = 2\pi i [\operatorname{Res} f(3i) + \operatorname{Res} f(2i)] \times \frac{1}{2} = \frac{2\pi i}{2} \left(\frac{3i}{50} - \frac{13i}{200} \right) = \frac{\pi}{200}$$

$$(3) \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2(x^2+b^2)}$$

解: 本题属于类型二.

$$I = \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2(x^2+b^2)}, \text{ 取 } f(z) = \frac{1}{(z^2+a^2)^2(z^2+b^2)}, (a>0, b>0)$$

分母为 $(z-ai)^2(z+ai)^2(z-bi)(z+bi)$

有 4 个奇点 $z_0 = ai, z_0 = -ai, z_0 = bi, z_0 = -bi$

其中两个二阶极点 $z_0 = \pm ai$, 两个单极点 $z_0 = \pm bi$

在上半面上有 $z_0 = ai$ (二阶极点) 和 $z_0 = bi$ (单极点)

其留数为

$$\operatorname{Res} f(bi) = \lim_{z \rightarrow bi} (z-bi) f(z) = \lim_{z \rightarrow bi} \frac{1}{(z+bi)(z^2+a^2)} = \frac{1}{2bi(a^2-b^2)^2}$$

$$\begin{aligned}\operatorname{Res} f(ai) &= \lim_{z \rightarrow ai} \frac{d}{dz} \left(\frac{(z-ai)^2}{(z^2+b^2)(z+ai)^2(z-ai)^2} \right) = \lim_{z \rightarrow ai} \frac{d}{dz} \left(\frac{1}{(z^2+b^2)(z+ai)^2} \right) \\ &= \lim_{z \rightarrow ai} \frac{-2z(z+ai)^2 - 2(z^2+b^2)(z+ai)}{[(z^2+b^2)(z+ai)^2]^2} = \frac{-4ai(b^2-3a^2)}{[(b^2-a^2)(2ai)^2]^2} \\ &= \frac{(3a^2-b^2)i}{4a^3(a^2-b^2)^2}\end{aligned}$$

$$\begin{aligned}I &= 2\pi i [\operatorname{Res} f(ai) + \operatorname{Res} f(bi)] = 2\pi i \left[\frac{(3a^2-b^2)i}{4a^3(a^2-b^2)^2} + \frac{1}{2bi(a^2-b^2)^2} \right] \\ &= \pi \left[\frac{b^2-3a^2}{2a^3(a^2-b^2)^2} + \frac{2}{2b(a^2-b^2)^2} \right] = \frac{b^3-3a^2b+2a^3}{2a^3b(a^2-b^2)^2} \pi \\ &= \frac{(a-b)^2(2a+b)}{2a^3b(a^2-b^2)^2} \pi = \frac{(2a+b)}{2a^3b(a+b)^2} \pi\end{aligned}$$

$$(4) \int_0^{\infty} \frac{dx}{x^4+a^4}$$

解: 本题属于类型二, 且为偶函数

$$I = \int_0^{\infty} \frac{dx}{x^4 + a^4} \text{ 取 } f(z) = \frac{1}{z^4 + a^4}, (a > 0)$$

$$f(z) \text{ 为偶函数, } I = \int_0^{\infty} \frac{dx}{x^4 + a^4} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{x^4 + a^4}$$

$$f(z) = \frac{1}{z^4 + a^4} \text{ 其分母为零时有四个单极点.}$$

$$z_0 = \sqrt[4]{-a^4} = ae^{i\frac{\pi}{2} \cdot \frac{1}{2} + \frac{2k\pi i}{4}} = ae^{i\frac{\pi}{4} + \frac{k\pi}{2}i}$$

$$z_0 = ae^{i\frac{\pi}{4}} (k=0), z_0 = ae^{i\frac{3\pi}{4}} (k=1)$$

$$z_0 = ae^{i\frac{5\pi}{4}} (k=3), z_0 = ae^{i\frac{7\pi}{4}} (k=3) \text{ 上半平面有 } \varphi = \frac{1}{4}\pi, \frac{3}{4}\pi.$$

$z_0 = ae^{i\frac{\pi}{4}}, z_0 = ae^{i\frac{3\pi}{4}}$ 分别为所求两个单极点. 求留数:

$$\text{Res } f(ae^{i\frac{\pi}{4}}) = \lim_{z \rightarrow ae^{i\frac{\pi}{4}}} \left[(z - ae^{i\frac{\pi}{4}}) \frac{1}{z^4 + a^4} \right] \quad (\text{用罗毕达法则对分子分母分别求导})$$

$$= \lim_{z \rightarrow ae^{i\frac{\pi}{4}}} \frac{1}{4z^3} = \frac{1}{4a^3 e^{i\frac{3\pi}{4}}} = \frac{1}{4a^3} e^{-i\frac{3}{4}\pi}$$

$$\begin{aligned} \text{Res } f(ae^{i\frac{3\pi}{4}}) &= \lim_{z \rightarrow ae^{i\frac{3\pi}{4}}} \left[(z - ae^{i\frac{3\pi}{4}}) \frac{1}{z^4 + a^4} \right] = \lim_{z \rightarrow ae^{i\frac{3\pi}{4}}} \left[\frac{1}{4z^3} \right] = \frac{1}{4a^3 e^{i\frac{9}{4}\pi}} \\ &= \frac{1}{4a^3} e^{-i\frac{9}{4}\pi} \end{aligned}$$

$$\begin{aligned} I &= 2\pi i \times \frac{1}{2} [\text{Res } f(ae^{i\frac{\pi}{4}}) + \text{Res } f(ae^{i\frac{3\pi}{4}})] \\ &= \pi i \left(\frac{1}{4a^3} \right) [e^{-i\frac{3}{4}\pi} + e^{-i\frac{9}{4}\pi}] = \pi i \frac{1}{4a^3} \left[i(-2\sin \frac{\pi}{4}) \right] \\ &= \frac{\pi}{2a^3} \cdot \sin \frac{\pi}{4} = \frac{\pi}{2a^3} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{4a^3} = \frac{\pi}{2\sqrt{2}a^3} \end{aligned}$$

$$(5) \int_0^{\infty} \frac{x^2 + 1}{x^6 + 1} dx$$

解: 本题属于类型二. (且为偶函数)

$$I = \int_0^{\infty} \frac{x^2 + 1}{x^6 + 1} dx \xrightarrow{\text{偶函数}} \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 + 1}{x^6 + 1} dx$$

$$\text{取 } f(z) = \frac{z^2 + 1}{z^6 + 1} = \frac{(z^2 + 1)}{(z^2 + 1)(z^4 - z^2 + 1)} = \frac{1}{z^4 - z^2 + 1}, \text{分母为}$$

$$\begin{aligned} z^4 - z^2 + 1 &= \left(z^2 - \frac{1 + \sqrt{3}i}{2} \right) \left(z^2 - \frac{1 - \sqrt{3}i}{2} \right) \\ &= \left(z - \sqrt{\frac{1 + \sqrt{3}i}{2}} \right) \left(z + \sqrt{\frac{1 + \sqrt{3}i}{2}} \right) \cdot \left(z - \sqrt{\frac{1 - \sqrt{3}i}{2}} \right) \left(z + \sqrt{\frac{1 - \sqrt{3}i}{2}} \right) \end{aligned}$$

共有 4 个单极点, 又因为,

$$\sqrt{1+\sqrt{3}i} = \sqrt{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{1}{2}(\sqrt{3} + i)$$

$$\sqrt{1-\sqrt{3}i} = \sqrt{\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}} = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = \frac{1}{2}(-\sqrt{3} + i)$$

所以只有两个单极点在上半平面. 求留数:

$$\begin{aligned} \operatorname{Res} f\left(\sqrt{\frac{1+\sqrt{3}i}{2}}\right) &= \operatorname{Res} f\left[\frac{1}{2}(\sqrt{3} + i)\right] \\ &= \lim_{z \rightarrow \frac{1}{2}(\sqrt{3}+i)} \frac{\left[z - \frac{1}{2}(\sqrt{3} + i)\right]}{\left(z^2 - \frac{1-\sqrt{3}i}{2}\right)} \cdot \frac{1}{\left(z - \sqrt{\frac{1+\sqrt{3}i}{2}}\right)\left(z + \sqrt{\frac{1+\sqrt{3}i}{2}}\right)} \\ &= \frac{1}{(\sqrt{3} + i)(\sqrt{3}i)} = \frac{1}{3i - \sqrt{3}} = \frac{1}{\sqrt{3}(\sqrt{3}i - 1)} \end{aligned}$$

$$\begin{aligned} \operatorname{Res} f\left(\sqrt{\frac{1-\sqrt{3}i}{2}}\right) &= \operatorname{Res} f\left[(-\sqrt{3} + i)/2\right] \\ &= \lim_{z \rightarrow \frac{-\sqrt{3}+i}{2}} \frac{\left(z - \frac{-\sqrt{3}+i}{2}\right)}{\left(z^2 - \frac{1+\sqrt{3}i}{2}\right)} \cdot \frac{1}{\left(z - \sqrt{\frac{1-\sqrt{3}i}{2}}\right)\left(z + \sqrt{\frac{1-\sqrt{3}i}{2}}\right)} \\ &= \frac{1}{(-\sqrt{3} + i)(1 - \sqrt{3}i)} = \frac{1}{3i + \sqrt{3}} \\ &= \frac{1}{\sqrt{3}(\sqrt{3}i + 1)} \end{aligned}$$

$$\begin{aligned} I &= 2\pi i \cdot \frac{1}{2} \left[\operatorname{Res}\left(\sqrt{\frac{1+\sqrt{3}i}{2}}\right) + \operatorname{Res}\left(\sqrt{\frac{1-\sqrt{3}i}{2}}\right) \right] = \frac{\pi i}{\sqrt{3}} \left(\frac{1}{(\sqrt{3}i + 1)} + \frac{1}{\sqrt{3}i - 1} \right) \\ &= \frac{\pi i \sqrt{3}i}{\sqrt{3}(-2)} = \frac{\pi}{2} \end{aligned}$$

$$(6) \int_0^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx$$

解: 本题属于类型二, 且为偶函数

$$I = \int_0^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx, \text{ 取 } f(z) = \frac{z^2}{(z^2 + a^2)^2}$$

分母为

$$(z^2 + a^2)^2 = (z + ai)^2 (z - ai)^2$$

有两个二阶极点

$$z_0 = ai, z_0 = -ai$$

其中在上半平面 $z_0 = ai$ 的留数为

$$\begin{aligned}\operatorname{Res} f(ai) &= \lim_{z \rightarrow ai} \frac{d}{dz} \frac{(z-ai)^2 z^2}{(z+ai)^2 (z-ai)^2} = \lim_{z \rightarrow ai} \frac{d}{dz} \left[\frac{z^2}{(z+ai)^2} \right] \\ &= \lim_{z \rightarrow ai} \left[\frac{2z}{(z+ai)^2} - \frac{z^2 \cdot 2(z+ai)}{(z+ai)^4} \right] = \frac{2ai}{-4a^2} + \frac{2a^2}{(2ai)^3} \\ &= -\frac{1}{2a}i + \frac{-1}{4ai} = \frac{-1}{4a}i\end{aligned}$$

$$\text{所以 } I = 2\pi i \cdot \frac{1}{2} \cdot \operatorname{Res}(ai) = \pi i \left(\frac{-1}{4a} \right) i = \frac{\pi}{4a}$$

$$(7) \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx \quad (m < n)$$

解: 本题属于类型二.

$$I = \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx, \text{ 则 } f(z) = \frac{z^{2m}}{1+z^{2n}}$$

若分母为 $1+z^{2n}=0$, 则 $z^{2n}=-1$, 又因为 $e^{i\varphi}=-1, \varphi=(2k+1)\pi$,

所以 $z^{2n}=e^{i(\pi+2k\pi)/2n} \quad (k=1,2,\dots,n-1)$

其留数为 $\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} \frac{(z-z_0)z^{2m}}{(z^{2n}+1)}$, 由罗比达法则

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} \frac{z^{2m} + 2m(z-z_0)z^{2m-1}}{2nz^{2n-1}} = \lim_{z \rightarrow z_0} \frac{z^{2m}}{2nz^{2n-1}} = \frac{e^{i\pi(2k+1) \cdot 2m/2n}}{2ne^{i\pi(2k+1)(2n-1)/2n}}$$

$$I = 2\pi i \sum_{i=0}^{n-1} \operatorname{Res} f(z_i)$$

$$\begin{aligned}\text{又 } \sum_{i=0}^{n-1} \operatorname{Res} f(z_i) &= \sum_{k=0}^{n-1} \frac{e^{i\pi(2k+1)2m/2n}}{2ne^{i\pi(2k+1)(2n-1)/2n}} = \sum_{k=0}^{n-1} \frac{1}{2ne^{(2k+1)(2n-2m-1)/2n}} \\ &= \frac{1}{2n} \frac{1}{e^{2n-2m-1} \pi i / 2n} \sum_{k=0}^{n-1} \left[\frac{1}{e^{i\pi(2k)(2n-2m-1)/2n}} \right] \\ &= \frac{1}{2n} \frac{1}{e^{(2n-2m-1)\pi i / 2n}} \sum_{k=0}^{n-1} \left[\frac{1}{e^{i\pi k(2n-2m-1)/n}} \right] \\ &= \frac{1}{2n} \frac{e^{(2m+1)\pi i / 2n}}{e^{\pi i}} \sum_{k=0}^{n-1} \frac{1}{e^{i\pi k(2n-2m-1)/n}}\end{aligned}$$

$$\text{由公式 } S = 1 + q + \dots + q^n = \frac{1-q^{n+1}}{1-q}$$

$$\text{则 } \sum_{k=0}^{n-1} e^{-i\pi k(2n-2m-1)/n} = \frac{1 - e^{-i\pi(2n-2m-1)}}{1 - e^{-i\pi(2n-2m-1)/n}} = \frac{2}{1 - e^{-i\pi(2n-2m-1)/n}}$$

$$\begin{aligned}\text{有: } \sum_{i=0}^{n-1} \operatorname{Res} f(z_i) &= \frac{1}{2n} \cdot \frac{2e^{i\pi(2m+1)/2n}}{(1 - e^{-i\pi(2n-2m-1)/n})e^{\pi i}} \\ &= -\frac{2}{2n} \left(\frac{1}{e^{-i\pi(2m+1)/2n} - e^{-i\pi(4n-2m-1)/2n}} \right), \quad (\text{其中 } e^{\pi i} = -1, e^{4\pi i/2n} = 1) \\ &= -\frac{2}{2n} \left(\frac{1}{e^{-i\pi(2m+1)/2n} - e^{i\pi(2m+1)/2n}} \right) \\ &= \frac{1}{2n} \left(\frac{2}{e^{i\pi(2m+1)/2n} - e^{-i\pi(2m+1)/2n}} \right) = \frac{1}{2n} \left(\frac{2}{2i \sin(2m+1)\pi/2n} \right)\end{aligned}$$

$$= \frac{1}{2ni\sin[(2m+1)\pi/2n]}$$

$$I = 2\pi i \sum_{i=0}^{n-1} \operatorname{Res} f(z_i) = \frac{2\pi i}{2ni\sin[(2m+1)\pi/2n]} = \frac{\pi}{n} \cdot \frac{1}{\sin[(2m+1)\pi/2n]}$$

3. 计算下列实变函数定积分.

$$(1) \int_0^{\infty} \frac{\cos mx}{1+x^4} dx \quad (m>0)$$

解: 本题属于 $\int_0^{\infty} F(x) \cdot \cos mx dx$ 或 $\int_0^{\infty} G(x) \cdot \sin mx dx$ 的类型即类型三. 积分区间是 $[0, \infty)$, $F(x)$ 是偶函数, $G(x)$ 是奇函数, 且在实轴上没有奇点, 在上半平面除有限个奇点外是解析的; 当 z 在上半平面或实轴上 $\rightarrow \infty$ 时, $F(x)$ 和 $G(x)$ 一致地 $\rightarrow 0$.

$$\text{则 } \int_0^{\infty} F(x) \cos mx dx = \pi i \{F(z)e^{imz} \text{ 在上半平面所有奇点的留数之和}\}$$

$$\int_0^{\infty} G(x) \sin mx dx = \pi \{G(z)e^{imz} \text{ 在上半平面所有奇点的留数之和}\}$$

$$\text{由 } F(z) = \frac{1}{1+z^4}, \text{ 则 } F(z)e^{imz} = \frac{e^{imz}}{1+z^4}$$

$$F(z) = \frac{1}{1+z^4}, \text{ 其分母为零时有四个单极点.}$$

$$\text{分别为 } z_0^4 = -1, z_0 = e^{i\frac{\pi}{4}}, z_0 = e^{i\frac{3}{4}\pi}, z_0 = e^{i\frac{5}{4}\pi}, z_0 = e^{i\frac{7}{4}\pi}$$

上半平面有 $z_0 = e^{i\frac{\pi}{4}}, z_0 = e^{i\frac{3}{4}\pi}$ 分别为两个单极点. (见本章习题 2(4) 中 $a=1$)

取 $f(z) = F(z)e^{imz}$, 留数为

$$\begin{aligned} \operatorname{Res} f(z_0) &= \operatorname{Res} f(e^{i\frac{\pi}{4}}) = \lim_{z \rightarrow e^{i\frac{\pi}{4}}} \left[(z - e^{i\frac{\pi}{4}}) \frac{e^{imz}}{1+z^4} \right] \\ &= \lim_{z \rightarrow e^{i\frac{\pi}{4}}} \left[\frac{e^{imz} + e^{imz}(z - e^{i\frac{\pi}{4}})}{4z^3} \right] = \lim_{z \rightarrow e^{i\frac{\pi}{4}}} \left[\frac{e^{imz}}{4z^3} \right] \\ &= \frac{e^{imz}}{4e^{i\frac{3}{4}\pi}} \Big|_{z=e^{i\frac{\pi}{4}}} \end{aligned}$$

$$\begin{aligned} \operatorname{Res} f(z_0) &= \operatorname{Res} f(e^{i\frac{3}{4}\pi}) = \lim_{z \rightarrow e^{i\frac{3}{4}\pi}} \left[(z - e^{i\frac{3}{4}\pi}) \left(\frac{e^{imz}}{1+z^4} \right) \right] \\ &= \lim_{z \rightarrow e^{i\frac{3}{4}\pi}} \left[\frac{e^{imz}}{4z^3} \right] = \frac{e^{imz}}{4e^{i\frac{5}{4}\pi}} \Big|_{z=e^{i\frac{3}{4}\pi}} \end{aligned}$$

上半平面留和之和为

$$\begin{aligned} \frac{e^{imz}}{4e^{i\frac{3}{4}\pi}} \Big|_{z=e^{i\frac{\pi}{4}}} + \frac{e^{imz}}{4e^{i\frac{5}{4}\pi}} \Big|_{z=e^{i\frac{3}{4}\pi}} &= \frac{e^{i\frac{5}{4}\pi} e^{imz_1} + e^{i\frac{3}{4}\pi} e^{imz_2}}{4} = \frac{(-1-i)e^{imz_1} + (1-i)e^{imz_2}}{4\sqrt{2}} \\ &= \frac{(-i-1)e^{im[\frac{\sqrt{2}}{2}(1+i)]} + (1-i)e^{im[\frac{\sqrt{2}}{2}(-1+i)]}}{4\sqrt{2}} \\ &= \frac{(-i-1)e^{-\frac{\sqrt{2}}{2}m} e^{\frac{\sqrt{2}}{2}im} + (1-i)e^{-\frac{\sqrt{2}}{2}m} e^{(-\frac{\sqrt{2}}{2}mi)}}{4\sqrt{2}} \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-\frac{\sqrt{2}}{2}m}}{4\sqrt{2}} \left[(-i-1) \left(\cos \frac{\sqrt{2}}{2}m + i \sin \frac{\sqrt{2}}{2}m \right) + (1-i) \right. \\
&\quad \left. \left(\cos \frac{\sqrt{2}}{2}m - i \sin \frac{\sqrt{2}}{2}m \right) \right] \\
&= \frac{-\sqrt{2}e^{-\frac{\sqrt{2}}{2}m}}{4} i \left(\cos \frac{\sqrt{2}}{2}m + \sin \frac{\sqrt{2}}{2}m \right)
\end{aligned}$$

$$I = \pi i \left[\frac{-\sqrt{2}}{4} e^{-\frac{\sqrt{2}}{2}m} \times i \left(\cos \frac{\sqrt{2}}{2}m + \sin \frac{\sqrt{2}}{2}m \right) \right] = \frac{\sqrt{2}\pi}{4} e^{-\frac{\sqrt{2}}{2}m} \left(\cos \frac{\sqrt{2}}{2}m + \sin \frac{\sqrt{2}}{2}m \right)$$

$$(2) \int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx \quad (m>0, a>0)$$

$$\text{解: } \frac{\sin mx}{x(x^2+a^2)} = \left[\frac{1}{a^2 x} - \frac{x}{a^2(x^2+a^2)} \right] \sin mx$$

$$\text{有 } I = \int_0^{\infty} \frac{\sin mx}{x(x^2+a^2)} dx = \int_0^{\infty} \frac{\sin mx}{a^2 x} dx - \int_0^{\infty} \frac{x \sin(mx) dx}{a^2(x^2+a^2)}$$

$$\text{又 } \int_0^{\infty} \frac{\sin mx}{a^2 x} dx = \frac{1}{a^2} \int_0^{\infty} \frac{\sin mx}{mx} dmx = \frac{1}{a^2} \int_0^{\infty} \frac{\sin u}{u} du = \frac{1}{a^2} \left(\frac{\pi}{2} \right) = I_0$$

$$\text{且 } \int_0^{\infty} \frac{x \sin mx}{a^2(a^2+x^2)} dx = \frac{1}{a^2} \int_0^{\infty} \frac{x \sin mx}{a^2+x^2} dx$$

$$\text{取: } f(z) = \int_0^{\infty} F(z) e^{imx} dz = \int_0^{\infty} \frac{ze^{imx}}{a^2+z^2} dz$$

$F(z) = \frac{z}{a^2+z^2}$ 为奇函数, 有 $z_0 = ia$ 和 $z_0 = -ia$ 两个单极点. 上半平面有 $z_0 = ia$, 属类型三.

$$\text{其留数为 } \text{Res}f(ia) = \lim_{z \rightarrow ia} \frac{(z-ia)ze^{imx}}{(z-ia)(z+ia)} = \lim_{z \rightarrow ia} \frac{iae^{-ma}}{(z+ia)} = \frac{1}{2} e^{-ma}$$

$$I_1 = \pi [\text{Res}f(ia)] = \pi \left(\frac{1}{2} e^{-ma} \right),$$

$$I = I_0 - I_1 = \frac{1}{a^2} \left(\frac{\pi}{2} \right) - \frac{1}{a^2} \left(\frac{\pi}{2} e^{-ma} \right) = \frac{\pi}{2a^2} (1 - e^{-ma})$$

$$(3) \int_{-\infty}^{\infty} \frac{x \sin x}{1+x^2} dx$$

$$\text{解: } \int_{-\infty}^{\infty} \frac{x \sin x}{1+x^2} dx = 2 \int_0^{\infty} \frac{x \sin x}{1+x^2} dx \quad (\text{偶函数})$$

$G(z) = \frac{z}{1+z^2}$, 属于类型三.

$$I = \int_0^{\infty} \frac{ze^{ix}}{1+z^2} dz$$

分母为 $1+z^2 = (z+i)(z-i)$ 有两个单极点, $z_0 = i, z_0 = -i$, 在上半平面 $z_0 = i$

$$\text{Res } f(i) = \lim_{z \rightarrow i} \frac{(z-i)ze^{ix}}{(z-i)(z+i)} = \frac{1}{2} e^{-1}$$

$$\text{所以 } I = \frac{\pi}{2} e^{-1}$$

原积分
$$\int_{-\infty}^{\infty} \frac{x \sin x}{1+x^2} dx = 2 \int_0^{\infty} \frac{x \sin x}{1+x^2} dx = 2I = \pi e^{-1}$$

(4)
$$\int_{-\infty}^{\infty} \frac{x \sin mx}{2x^2 + a^2} dx \quad (m > 0, a > 0)$$

解:
$$I = \int_{-\infty}^{\infty} \frac{x \sin mx}{2x^2 + a^2} dx = 2 \int_0^{\infty} \frac{x \sin mx}{2x^2 + a^2} dx, \quad (\text{偶函数})$$

$$G(z) = \frac{z \sin(mz)}{2z^2 + a^2}, I = \int_0^{\infty} G(z) e^{imz} dz, \text{属于类型三.}$$

分母为
$$2z^2 + a^2 = \frac{1}{2} \left[\left(z + \frac{a}{\sqrt{2}} i \right) \left(z - \frac{a}{\sqrt{2}} i \right) \right]$$

有两个单极点, 在上半平面有一个单极点 $z_0 = \frac{a}{\sqrt{2}} i$

$$\text{Res } f\left(\frac{a}{\sqrt{2}} i\right) = \lim_{z \rightarrow \frac{a}{\sqrt{2}} i} \frac{\left(z - \frac{a}{\sqrt{2}} i\right) z e^{imz}}{2 \left(z - \frac{a}{\sqrt{2}} i\right) \left(z + \frac{a}{\sqrt{2}} i\right)} = \frac{1}{4} e^{-\frac{a}{\sqrt{2}} m}$$

所以
$$I = 2\pi \left(\frac{1}{4}\right) e^{-\frac{a}{\sqrt{2}} m} = \frac{\pi}{2} e^{-\frac{a}{\sqrt{2}} m}$$

(5)
$$\int_0^{\infty} \frac{\cos mx}{(x^2 + a^2)^2} dx$$

解: 本题属于类型三, 其中
$$F(z) = \frac{1}{(a^2 + z^2)^2}$$

$$I = \int_0^{\infty} F(z) e^{imz} dz$$

$F(z) e^{imz}$ 分母为 $(a^2 + z^2)^2 = a(z - ai)^2 (z + ai)^2$ 只两个二阶极点 $z_0 = ai$ 和 $z_0 = -ai$, 且 $z_0 = ai$ 在上半平面, 其留数为

$$\begin{aligned} \text{Res } f(ai) &= \lim_{z \rightarrow ai} \frac{d}{dz} \left(\frac{(z - ai)^2 e^{imz}}{(z - ai)^2 (z + ai)^2} \right) = \lim_{z \rightarrow ai} \frac{d}{dz} \left(\frac{e^{imz}}{(z + ai)^2} \right) \\ &= \lim_{z \rightarrow ai} \left\{ \left[\frac{ime^{imz}}{(z + ai)^2} \right] - \left[\frac{2e^{imz}}{(z + ai)^3} \right] \right\} = \frac{-ime^{-ma}}{4a^2} + \frac{2e^{-ma}}{8a^3 i} \\ &= -\frac{(ima + i)e^{-ma}}{4a^3} \end{aligned}$$

$$I = \pi i [\text{Res } f(ia)] = \frac{\pi(ma + 1)}{4a^3} e^{-ma}$$

(6)
$$\int_0^{\infty} \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx$$

解: 本题属于类型三.
$$F(z) = \frac{1}{(z^2 + a^2)(z^2 + b^2)}$$

$$I = \int_0^{\infty} F(z) e^{imz} dz$$

分母为
$$(z^2 + a^2)(z^2 + b^2) = (z - ia)(z + ia)(z - ib)(z + ib)$$

有四个单极点, 且 $z_0 = ia$ 和 $z_0 = ib$ 在上半平面.

其留数为 $\text{Res } f(ia) = \lim_{z \rightarrow ia} \frac{(z-ia)e^{iz}}{(z-ia)(z+ia)(z^2+b^2)} = \frac{e^{-a}}{2ai(b^2-a^2)}$

$$\text{Res } f(ib) = \lim_{z \rightarrow ib} \frac{(z-ib)e^{iz}}{(z^2+a^2)(z-ib)(z+ib)} = \frac{e^{-b}}{(-b^2+a^2)2ib}$$

$$\begin{aligned} I &= \pi i [\text{Res } f(ia) + \text{Res } f(ib)] = \pi i \left[\frac{1}{2i} \left(\frac{e^{-a}}{(b^2-a^2)a} + \frac{-e^{-b}}{(b^2-a^2)b} \right) \right] \\ &= \frac{\pi}{2} \left(\frac{1}{b^2-a^2} \right) \left(\frac{e^{-a}}{a} - \frac{e^{-b}}{b} \right) \end{aligned}$$

$$(7) \int_0^\infty \frac{\sin^2 x}{x^2} dx$$

解: $\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{1}{2i} \int_{-\infty}^\infty \frac{e^{ix} \sin x}{x^2} dx = I$

因为 $\oint_l \frac{e^{iz} \sin z}{z^2} dz = \left[\int_{C_r} + \int_{C_\epsilon} \right] \frac{e^{iz} \sin z}{z^2} dz + \left(\int_{-\epsilon}^\epsilon + \int_\epsilon^R \right) \frac{e^{ix} \sin x}{x^2} dx$

如图, l 中无奇点, 所以上式左端为零.

令 $\epsilon \rightarrow 0, R \rightarrow \infty$,

右端第一项为 $\int_{C_r} \frac{e^{iz} (e^{iz} - e^{-iz})}{2iz^2} dz = \frac{1}{2i} \int_{C_r} \left[\frac{e^{imz}}{z^2} - \frac{1}{z^2} \right] dz$

上式中, 第一项按约当引理 $\rightarrow 0$, 第二项 $\frac{1}{z^2}$ 因 z 一致趋于 ∞ , 也 $\rightarrow 0$, $\therefore \lim_{R \rightarrow \infty} \int_{C_r} = 0$

$$2iI = \lim_{\epsilon \rightarrow 0} - \int_{C_\epsilon} \frac{e^{iz} \sin z}{z^2} dz = \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^\epsilon - \left[\frac{1}{z} + \text{解析部分 } p(z) \right] dz = \int_\pi^0 - \frac{i\epsilon e^{i\varphi}}{\epsilon e^{i\varphi}} d\varphi = i\pi,$$

即 $I = \pi/2$. 所以 $\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$

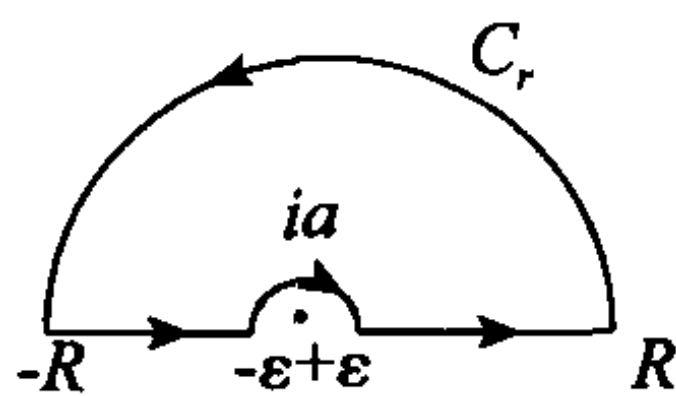
$$(8) \int_{-\infty}^\infty \frac{e^{imx}}{x-i\alpha} dx, \int_{-\infty}^\infty \frac{e^{imx}}{x+i\alpha} dx, (m>0, \text{Re } \alpha>0)$$

解: $I_1 = \int_{-\infty}^\infty \frac{e^{imz}}{z-i\alpha} dz$, 在上半平面 $\frac{e^{imz}}{z-i\alpha}$ 有单极点 $z_0 = i\alpha$

$I_2 = \int_{-\infty}^\infty \frac{e^{imz}}{z+i\alpha} dz$, 在上半平面 $\frac{e^{imz}}{z+i\alpha}$ 无奇点.

留数为 $\text{Res } f(z_0) = \lim_{z \rightarrow i\alpha} \frac{(z-i\alpha)e^{imz}}{(z-i\alpha)} = e^{-m\alpha}$

$I_1 = 2\pi i e^{-m\alpha}, I_2 = 0$ (无奇点)



第五章 傅里叶变换

本章知识要点

傅里叶级数

(一) 以 $2l$ 为周期的函数 $f(x)$ 展开为傅里叶级数

$$f(x) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos \frac{k\pi x}{l} + b_k \sin \frac{k\pi x}{l} \right)$$

$$a_k = \frac{1}{\delta_k l} \int_{-l}^l f(\xi) \cos \frac{k\pi \xi}{l} d\xi$$

$$b_k = \frac{1}{l} \int_{-l}^l f(\xi) \sin \frac{k\pi \xi}{l} d\xi$$

其中 $\delta_k = \begin{cases} 2 & (k=0) \\ 1 & (k \neq 0) \end{cases}$, a_k, b_k 称为傅里叶系数.

(二) 奇函数及偶函数的傅里叶展开

奇函数 $f(x) = \sum_{k=1}^{\infty} b_k \sin \frac{k\pi x}{l}$, 称傅里叶的正弦级数

展开系数为 $b_k = \frac{2}{l} \int_0^l f(\xi) \sin \frac{k\pi \xi}{l} d\xi$

偶函数 $f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi x}{l}$ 称傅里叶余弦级数

展开系数为 $a_k = \frac{2}{l} \int_0^l f(\xi) \cos \frac{k\pi \xi}{l} d\xi$

复数形式的傅里叶级数

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{i \frac{k\pi x}{l}}$$

傅里叶系数

$$c_k = \frac{1}{2l} \int_{-l}^l f(\xi) [e^{i \frac{k\pi \xi}{l}}]^* d\xi$$

傅里叶积分与傅里叶变换

(一) 实数形式的傅里叶积分

$$f(x) = \int_0^{\infty} A(\omega) \cos \omega x d\omega + \int_0^{\infty} B(\omega) \sin \omega x d\omega$$

其中

$$\begin{cases} A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \cos \omega \xi d\xi \\ B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \sin \omega \xi d\xi \end{cases}$$

称为 $f(x)$ 的傅里叶变换式

奇函数 $f(x) = \int_0^{\infty} B(\omega) \sin \omega x d\omega$ 傅里叶正弦积分

傅里叶正弦变换 $B(\omega) = \frac{2}{\pi} \int_0^{\infty} f(\xi) \sin \omega \xi d\xi$

偶函数 $f(x) = \int_0^{\infty} A(\omega) \cos \omega x d\omega$ 傅里叶余弦积分

傅里叶余弦变换 $A(\omega) = \frac{2}{\pi} \int_0^{\infty} f(\xi) \cos \omega \xi d\xi$

(二) 复数形式的傅里叶积分

$$f(x) = \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

傅里叶变换式

$$F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) [e^{i\omega x}]^* dx$$

用符号简写为

$$F(\omega) = \mathcal{F}[f(x)], f(x) = \mathcal{F}^{-1}[F(\omega)]$$

$f(x)$ 称为傅里叶变换的原函数.

$F(\omega)$ 称为傅里叶变换的像函数.

(三) 傅里叶变换的基本性质

假设 $f(x)$ 的傅里叶变换存在, 且 $\mathcal{F}[f(x)] = F(\omega)$

(1) 导数定理

$$\mathcal{F}[f'(x)] = i\omega F(\omega)$$

(2) 积分定理

$$\mathcal{F}\left[\int^{(x)} f(\xi) d\xi\right] = \frac{1}{i\omega F(\omega)}$$

(3) 相似性定理

$$\mathcal{F}[f(ax)] = \frac{1}{a} F\left(\frac{\omega}{a}\right)$$

(4) 延迟定理

$$\mathcal{F}[f(x-x_0)] = e^{-i\omega x_0} F(\omega)$$

(5) 位移定理

$$\mathcal{F}[e^{-i\omega_0 x} f(x)] = f(\omega - \omega_0)$$

(6) 卷积定理

$$\text{若 } \mathcal{F}[f_1(x)] = F_1(\omega), \mathcal{F}[f_2(x)] = F_2(\omega)$$

则

$$\mathcal{F}[f_1(x) * f_2(x)] = 2\pi F_1(\omega) F_2(\omega)$$

其中 $f_1(x) * f_2(x) = \int_{-\infty}^{\infty} f_1(\xi) f_2(x-\xi) d\xi$ 称为 $f_1(x)$ 与 $f_2(x)$ 的卷积.

(四) 多重傅里叶积分

三重傅里叶分积

$$f(\mathbf{r}) = \iiint F(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} dk_1 dk_2 dk_3$$

三重傅里叶变换

$$F(\mathbf{k}) = \frac{1}{(2\pi)^3} \iiint f(\mathbf{r}) [e^{i\mathbf{k}\cdot\mathbf{r}}]^* dx dy dz.$$

δ 函数

(一) δ 函数定义

$$\delta(x) = \begin{cases} 0, & (x \neq 0) \\ \infty, & (x = 0) \end{cases}$$

$$\int_a^b \delta(x) dx = \begin{cases} 0, & (a, b \text{ 都} < 0, \text{或} a, b \text{ 都} > 0) \\ 1 & (a < 0 < b) \end{cases}$$

(二) δ 函数的性质

$$1. \delta(-x) = \delta(x) \quad (\delta \text{ 是偶函数})$$

$$\delta'(-x) = -\delta'(x) \quad (\delta \text{ 的导数是奇函数})$$

$$2. H(x) = \int_{-\infty}^x \delta(t) dt = \begin{cases} 0, & (x < 0) \\ 1, & (x > 0) \end{cases}$$

$H(x)$ 为阶跃函数, 或亥维赛单位函数

$$\delta(x) = \frac{dH(x)}{dx}$$

$$3. \int_{-\infty}^{\infty} f(\tau) \delta(\tau - t_0) d\tau = f(t_0)$$

(三) δ 函数是一种广义函数

$$\delta(x) = \lim_{l \rightarrow 0} \frac{1}{l} \text{rect}\left(\frac{x}{l}\right)$$

$$\delta(x) = \lim_{K \rightarrow \infty} \frac{1}{\pi} \frac{\sin Kx}{x}$$

$$\delta(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon}{\epsilon^2 + x^2}$$

(四) δ 函数的傅里叶变换

$$\delta(x) = \int_{-\infty}^{\infty} C(\omega) e^{i\omega x} d\omega$$

其中

$$C(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(x) e^{-i\omega x} dx = \frac{1}{2\pi}$$

因此

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} d\omega$$

(五) 多维的 δ 函数

$$\delta(\mathbf{r}) = \begin{cases} 0, & (\mathbf{r} \neq 0) \\ 1, & (\mathbf{r} = 0) \end{cases}$$

$$\iiint_{-\infty}^{\infty} \delta(\mathbf{r}) dx dy dz = 1$$

$$\delta(\mathbf{r}) = \delta(x)\delta(y)\delta(z)$$

习题全解

§ 5.1 傅里叶级数

1. 交流电压 $E_0 \sin \omega t$ 经过全波整流, 成为 $E(t) = E_0 |\sin \omega t|$. 试将它展为傅里叶级数.

解: $E(t) = E_0 |\sin \omega t|$ 的波形如图 5-1-1 所示.

由图可知其周期为 π , 取 $x = \omega t$, 则 $E(x) = E_0 |\sin x|$, 由于 $E(x)$ 是偶函数, 将其展开为傅里叶级有

$$E(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx$$

其中 $x = \omega t$,

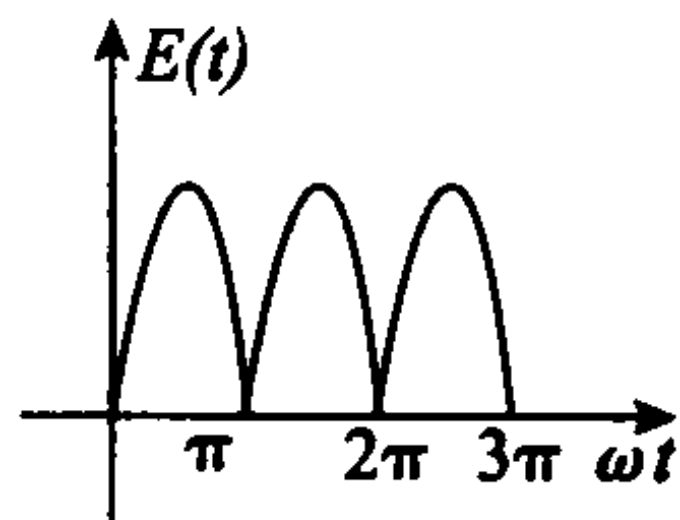


图 5-1-1

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{\pi} E(x) dx = \frac{E_0}{\pi} \int_0^{\pi} \sin x dx = \left. \frac{-E_0}{\pi} \cos x \right|_0^{\pi} \\ &= \frac{2E_0}{\pi} \end{aligned}$$

$$\begin{aligned} a_k &= \frac{2}{\pi} \int_0^{\pi} E_0 \sin x \cos kx dx \\ &= \frac{E_0}{\pi} \int_0^{\pi} \{ \sin [(k+1)x] - \sin [(k-1)x] \} dx \\ &= \frac{E_0}{\pi} \left[\frac{-\cos (k+1)x}{k+1} + \frac{\cos (k-1)x}{k-1} \right] \Big|_0^{\pi} \\ &= \frac{E_0}{\pi} \left[-\frac{\cos (k+1)\pi}{k+1} + \frac{\cos (k-1)\pi}{k-1} \right] + \frac{1}{k+1} - \frac{1}{k-1} \end{aligned}$$

$$= \begin{cases} \frac{E_0}{\pi} \times (0) = 0 & (k = 1, 3, 5 \cdots \text{奇数}) \\ \frac{E_0}{\pi} \left(\frac{2}{k+1} - \frac{2}{k-1} \right) = \frac{2E_0}{\pi} \left(\frac{k-1-k-1}{k^2-1} \right) = \frac{2E_0}{\pi} \frac{(-2)}{(k^2-1)} = \frac{4E_0}{\pi} \left(\frac{1}{1-k^2} \right), & (k = 2, 4, 6 \cdots \text{偶数}) \end{cases}$$

$$a_k = \frac{4E_0}{\pi} \left(\frac{1}{1-k^2} \right), k \text{ 为偶数, 取 } k = 2k_1$$

$$a_{k_1} = \frac{4E_0}{\pi} \left(\frac{1}{1-4k_1^2} \right), (k_1 = 1, 2 \cdots)$$

$$E(x) = a_0 + \sum_{k=1}^{\infty} a_{k_1} \cos 2k_1 x$$

$$E(t) = a_0 + \sum_{k=1}^{\infty} a_{k_1} \cos 2k_1 \omega t = \frac{2E_0}{\pi} + \frac{4E_0}{\pi} \sum_{k=1}^{\infty} \left(\frac{1}{1-4k_1^2} \right) \cos 2k_1 \omega t$$

整流的结果, 直流部分为 $2E_0/\pi$, 交流部分没有原来的基波 $\sin \omega t$ 部分, 基波消失.

2. 将锯齿波展为傅里叶级数. 在 $(0, T)$ 这个周期上, 该锯齿波

可表为 $f(x) = x/3$.

解: $f(x) = f(x+T)$, 且 $f(x) = x/3$, $f(x)$ 为奇函数.

$$\text{有: } f(x) = a_0 + \sum_{k=1}^{\infty} b_k \sin \frac{2k\pi}{T} x$$

$$a_0 = \frac{1}{T} \int_0^T f(x) dx = \frac{1}{T} \left. \frac{x^2}{6} \right|_0^T = \frac{T}{6}$$

$$\begin{aligned} b_k &= \frac{2}{T} \int_0^T f(x) \sin \left(\frac{2k\pi}{T} x \right) dx = \frac{2}{T} \int_0^T \frac{x}{3} \sin \left(\frac{2k\pi}{T} x \right) dx \\ &= \frac{2}{T} \left(-\frac{T}{2k\pi} \right) = \int_0^T \frac{x}{3} d \cos \left(\frac{2k\pi}{T} x \right) = -\frac{1}{k\pi} \left[\left(\frac{x}{3} \cos \frac{2k\pi}{T} x \right) \right]_0^T \\ &\quad - \frac{1}{3} \int_0^T \cos^2 \left(\frac{2k\pi}{T} x \right) dx \Bigg] = -\frac{1}{k\pi} \left[\frac{T}{3} - \frac{1}{3} \left(\frac{T}{2k\pi} \right) \sin \frac{2k\pi}{T} x \right]_0^T \\ &= -\frac{T}{3k\pi} \end{aligned}$$

所以 $f(x) = \frac{T}{6} - \sum_{k=1}^{\infty} \frac{T}{3k\pi} \sin \frac{2k\pi}{T} x$. 其波形如图 5-1-2 所示.

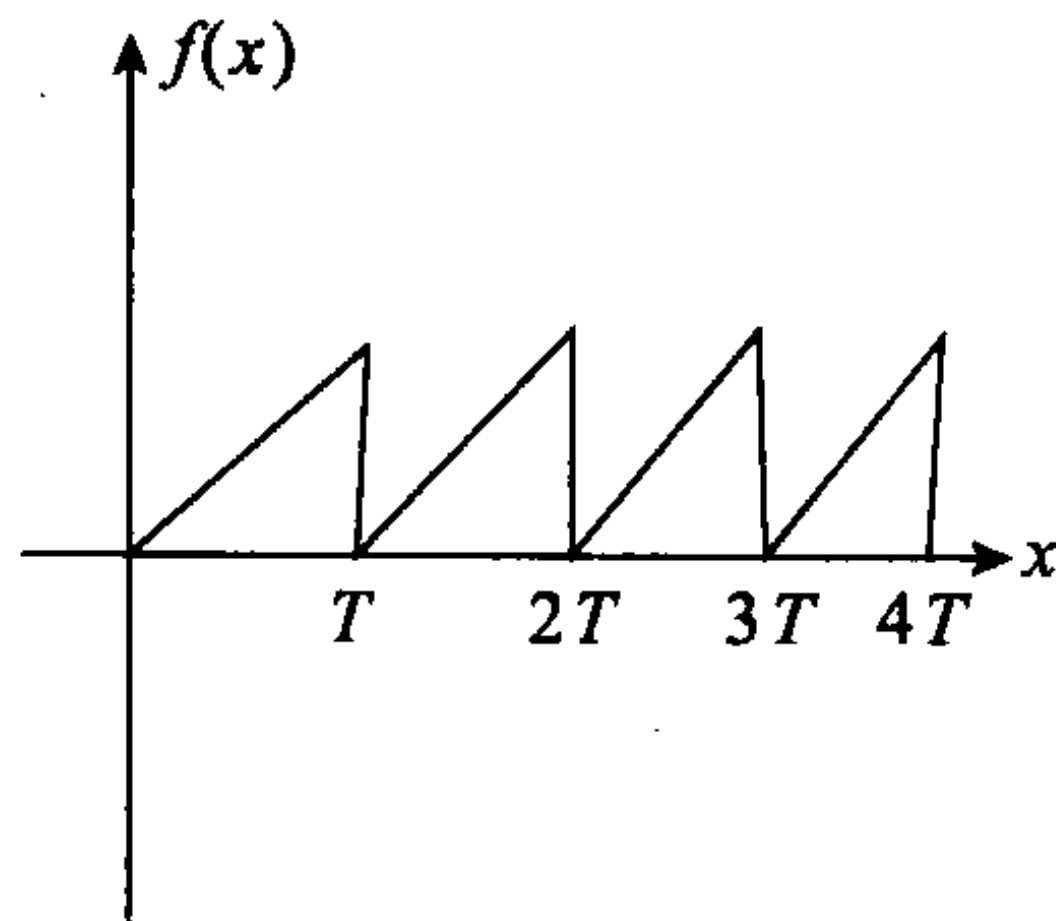


图 5-1-2

3. 在 $(-\pi, \pi)$ 这个周期上, $f(x) = x + x^2$. 试将它展为傅里叶级数, 又在本题所得展开式中置

$x = \pi$, 由此验证 $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}$.

解: 设 $f(x) = f_1(x) + f_2(x)$, $f_1(x) = x$, $f_2(x) = x^2$,

$$f_1(x) = x \text{ 是奇函数, } f_1(x) = \sum_{k=1}^{\infty} b_k \sin \left(\frac{k\pi x}{\pi} \right) = \sum_{k=1}^{\infty} b_k \sin kx,$$

$$f_2(x) = x^2 \text{ 是偶函数, } f_2(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \left(\frac{k\pi x}{\pi} \right) = \sum_{k=1}^{\infty} a_0 + \sum_{k=1}^{\infty} a_k \cos kx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f_2(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi} \left(\frac{x^3}{3} \right) \Big|_{-\pi}^{\pi} = \frac{\pi^2}{3}$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin kx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{x}{k} d(-\cos kx) = \frac{-1}{\pi k} (x \cos kx) \Big|_{-\pi}^{\pi} + \frac{1}{\pi k} \int_{-\pi}^{\pi} \cos kx dx$$

$$= \frac{\sin kx}{\pi k^2} \Big|_{-\pi}^{\pi} - \frac{2\pi}{\pi k} \cos n\pi = \frac{2}{k} (-1)^{k+1}$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos kx dx = \frac{1}{k\pi} \int_{-\pi}^{\pi} x^2 d\sin kx = \frac{1}{k\pi} x^2 \sin kx \Big|_{-\pi}^{\pi} - \frac{1}{k\pi} \int_{-\pi}^{\pi} \sin kx dx^2$$

$$= \frac{-1}{k\pi} \int_{-\pi}^{\pi} 2x \sin kx dx = \frac{-2}{k\pi} \frac{2\pi}{k} (-1)^{k+1} = \frac{4}{k^2} (-1)^k$$

$$f_1(x) = \sum_{k=1}^{\infty} \frac{2}{k} (-1)^{k+1} \sin kx, f_2(x) = \frac{\pi^2}{3} + \sum_{k=1}^{\infty} \frac{4}{k^2} (-1)^k \cos kx$$

$$f(x) = f_1(x) + f_2(x) = \frac{\pi^2}{3} + \sum_{k=1}^{\infty} \frac{2}{k} (-1)^{k+1} \sin kx + \sum_{k=1}^{\infty} \frac{4}{k^2} (-1)^k \cos kx \quad (1)$$

展式中置 $x = \pi$

此时函数为第一类间断点,由狄里希里定理得到

$$\text{等式左端} = f(\pi) = \frac{1}{2} [f(\pi) + f(-\pi)] = \pi^2$$

将 $x = \pi$ 代入(1),得

$$\text{等式右端} = \frac{\pi^2}{3} + \sum_{k=1}^{\infty} \frac{2}{k} (-1)^{k+1} \sin kx \Big|_{x=\pi} + \sum_{k=1}^{\infty} \frac{4}{k^2} (-1)^k \cos kx \Big|_{x=\pi}$$

$$= \frac{\pi^2}{3} + \sum_{k=1}^{\infty} \frac{4}{k^2} (-1)^k \cos k\pi = \frac{\pi^2}{3} + 4 \sum_{k=1}^{\infty} \frac{1}{k^2}$$

左端 = 右端,所以

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{k=1}^{\infty} \frac{1}{k^2}$$

$$\frac{\pi^2}{6} = \sum_{k=1}^{\infty} \frac{1}{k^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots, \text{得验证。}$$

4. 将下列函数展为傅里叶级数:

$$(1) f(x) = \cos^3 x$$

$$\text{解: } f(x) = \left(\frac{1 + \cos 2x}{2} \right) \cos x = \frac{1}{2} \cos x + \frac{1}{2} \cos 2x \cos x$$

$$= \frac{1}{2} \cos x + \frac{1}{2} \cos 3x + \sin^2 x \sin x = \frac{1}{2} \cos x + \frac{1}{2} \cos 3x + \sin^2 x \cos x$$

$$= \frac{1}{2} \cos x + \frac{1}{2} \cos 3x + (1 - \cos^2 x) \cos x = \frac{3}{2} \cos x + \frac{1}{2} \cos 3x - \cos^3 x$$

$$= \frac{3}{2} \cos x + \frac{1}{2} \cos 3x - f(x)$$

所以

$$2f(x) = \frac{3}{2} \cos x + \frac{1}{2} \cos 3x$$

$$f(x) = \cos^3 x = \frac{3}{4} \cos x + \frac{1}{4} \cos 3x$$

$$(2) f(x) = \frac{1 - a^2}{1 - 2a \cos x + a^2}, (|a| < 1)$$

解: $f(x) = \frac{1-\alpha^2}{1-2\alpha\cos x+\alpha^2}$ 为偶函数

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx$$

又 $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, 设 $u = e^{ix}$, 则 $\cos x = \frac{u+u^{-1}}{2}$

$$\begin{aligned} f(u) &= \frac{1-\alpha^2}{1-2\alpha\left(\frac{u+u^{-1}}{2}\right)+\alpha^2} = \frac{1-\alpha^2}{1-\alpha u-\alpha u^{-1}+\alpha^2} \\ &= \frac{1-\alpha^2}{(\alpha-u)(\alpha-u^{-1})} = \frac{(1-\alpha^2)u}{(\alpha-u)(\alpha u-1)} \\ &= \frac{u-\alpha^2 u+\alpha-\alpha}{(\alpha-u)(\alpha u-1)} = \frac{u-\alpha+\alpha-\alpha^2 u}{(\alpha-u)(\alpha u-1)} \\ &= \frac{1}{1-\alpha u} + \frac{\alpha(1-\alpha u)}{(\alpha-u)(\alpha u-1)} \\ &= \frac{1}{1-\alpha u} + \frac{\alpha}{-\alpha+u} = \frac{1}{1-\alpha u} + \frac{\alpha/u}{1-\alpha/u} \\ &= \sum_{k=1}^{\infty} \alpha^k u^k + \sum_{k=1}^{\infty} \left(\frac{\alpha}{u}\right)^{k+1} = 1 + \sum_{k=1}^{\infty} (\alpha^k u^k + \alpha^k u^{-k}) \\ &= 1 + \sum_{k=1}^{\infty} \alpha^k (u^k + u^{-k}) \end{aligned}$$

$$f(x) = 1 + \sum_{k=1}^{\infty} \alpha^k (e^{ikx} + e^{-ikx}) = 1 + \sum_{k=1}^{\infty} 2\alpha^k \cos kx = 1 + 2 \sum_{k=1}^{\infty} \alpha^k \cos kx$$

(3) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \cos \alpha x$ (α 非整数)

解: $f(x) = \cos \alpha x$ 是偶函数, $f(x) = f(-x)$.

所以 $f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx$

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \alpha x dx = \frac{1}{2\pi\alpha} \sin \alpha x \Big|_{-\pi}^{\pi} = \frac{2}{2\pi\alpha} \sin \alpha\pi = \frac{\sin \alpha\pi}{\pi\alpha} \\ a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos \alpha x \cos kx dx = \frac{1}{\pi} \int_0^{\pi} 2 \cos \alpha x \cos kx dx \\ &= \frac{1}{\pi} \int_0^{\pi} [\cos (\alpha+k)x + \cos (\alpha-k)x] dx \\ &= \frac{1}{\pi} \left[\frac{\sin (\alpha+k)x}{\alpha+k} + \frac{\sin (k-\alpha)x}{k-\alpha} \right] \Big|_0^{\pi} = \frac{1}{\pi} \left[\frac{\sin (\alpha+h)\pi}{\alpha+k} - \frac{\sin (k-\alpha)\pi}{k-\alpha} \right] \\ &= \frac{1}{\pi} \left[\frac{1}{k+\alpha} (\sin \alpha\pi \cos k\pi + \sin k\pi \cos \alpha\pi + \frac{\sin (k-\alpha)\pi}{k-\alpha}) \right] \\ &= \frac{1}{\pi} \left[\frac{\sin \alpha\pi}{k+\alpha} (-1)^k + \frac{1}{k-\alpha} (\sin k\pi \cos k\pi - \sin \alpha\pi \cos k\pi) \right] \\ &= \frac{1}{\pi} \left[\frac{\sin \alpha\pi}{k+\alpha} (-1)^k - \frac{\sin k\pi}{k-\alpha} (-1)^k \right] = \frac{\sin \alpha\pi}{\pi} (-1)^k \left[\frac{1}{k+\alpha} - \frac{1}{k-\alpha} \right] \\ &= (-1)^k \frac{\sin \alpha\pi}{\pi} \frac{k-\alpha-k-\alpha}{k^2-\alpha^2} = (-1)^k \frac{\sin \alpha\pi}{\pi} \frac{(-2\alpha)}{k^2-\alpha^2} \end{aligned}$$

$$= (-1)^{k+1} \frac{2(\sin \alpha \pi) \alpha}{(k^2 - \alpha^2) \pi}$$

$$\begin{aligned} f(x) &= a_0 + \sum_{k=1}^{\infty} a_k \cos kx = \frac{\sin \alpha \pi}{\pi \alpha} + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{2\alpha(\sin \alpha \pi) \cos kx}{(k^2 - \alpha^2) \pi} \\ &= \frac{2\sin \alpha \pi}{\pi} \left[\frac{1}{2\alpha} + \alpha \sum_{k=1}^{\infty} (-1)^{k+1} \frac{\cos kx}{k^2 - \alpha^2} \right] \end{aligned}$$

(4) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \operatorname{ch} \alpha x$ (α 非整数)

解: $f(x) = \operatorname{ch} \alpha x = \frac{e^{\alpha x} + e^{-\alpha x}}{2}$

因为 $f(-x) = f(x)$ 是偶函数, 所以 $f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx$

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} \operatorname{ch} \alpha x dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{e^{\alpha x} + e^{-\alpha x}}{2} \right) dx \\ &= \frac{1}{2\pi} \cdot \frac{1}{2} \cdot 2 \int_0^{\pi} (e^{\alpha x} + e^{-\alpha x}) dx = \frac{1}{2\pi \alpha} (e^{\alpha x} - e^{-\alpha x}) \Big|_0^{\pi} \\ &= \frac{1}{2\pi \alpha} (e^{\alpha \pi} - e^{-\alpha \pi}) = \frac{1}{\pi \alpha} \left(\frac{e^{\alpha \pi} - e^{-\alpha \pi}}{2} \right) = \frac{1}{\pi \alpha} \operatorname{sh} \alpha \pi \end{aligned}$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} \operatorname{ch} \alpha x \cos kx dx = \frac{2}{\pi} \int_0^{\pi} \operatorname{ch} \alpha x \cos kx dx$$

其中 $I = \int e^{\alpha x} \cdot \cos kx dx = \frac{1}{\alpha} \int \cos kx de^{\alpha x}$

$$\begin{aligned} &= \frac{1}{\alpha} (\cos kx) e^{\alpha x} - \frac{1}{\alpha} \int e^{\alpha x} \cdot d \cos kx = \frac{1}{\alpha} \cos(kx) e^{\alpha x} + \frac{k}{\alpha} \int e^{\alpha x} \sin kx dx \\ &= \frac{1}{\alpha} \cos kx e^{\alpha x} + \frac{k}{\alpha^2} \int \sin kx de^{\alpha x} = \frac{1}{\alpha} \cos kx e^{\alpha x} + \frac{k}{\alpha^2} \sin kx e^{\alpha x} - \frac{k}{\alpha^2} \int e^{\alpha x} d \sin kx \\ &= \frac{1}{\alpha} \cos kx e^{\alpha x} + \frac{k}{\alpha^2} \sin kx e^{\alpha x} - \frac{k^2}{\alpha^2} \int e^{\alpha x} \cos kx dx \end{aligned}$$

$$I + \frac{k^2}{\alpha^2} I = \frac{\alpha}{\alpha^2} \cos kx e^{\alpha x} + \frac{k}{\alpha^2} \sin kx e^{\alpha x}$$

$$I \left(1 + \frac{k^2}{\alpha^2} \right) = \frac{\alpha}{\alpha^2} \cos kx e^{\alpha x} + \frac{k}{\alpha^2} \sin kx e^{\alpha x}$$

$$I(\alpha^2 + k^2) = \alpha \cos kx e^{\alpha x} + k \sin kx e^{\alpha x}$$

$$I = \frac{\alpha \cos kx e^{\alpha x} + k \sin kx e^{\alpha x}}{\alpha^2 + k^2} = \frac{e^{\alpha x} (\alpha \cos kx + k \sin kx)}{\alpha^2 + k^2}$$

且 $\int -e^{-\alpha x} \cos kx dx = - \int e^{-\alpha x} \cos(-kx) d(-x)$

$$= - \frac{e^{-\alpha x} [\alpha \cos(-kx) + k \sin(-kx)]}{\alpha^2 + k^2}$$

$$= - \frac{e^{-\alpha x} (\alpha \cos kx - k \sin kx)}{\alpha^2 + k^2}$$

$$= \frac{e^{-\alpha x} (k \sin kx - \alpha \cos kx)}{\alpha^2 + k^2}$$

$$\begin{aligned}
 a_k &= \frac{2}{\pi} \int_0^\pi \operatorname{ch} \alpha x \cos kx \, dx = \frac{2}{\pi} \int_0^\pi \left(\frac{e^{\alpha x} + e^{-\alpha x}}{2} \right) \cos kx \, dx \\
 &= \frac{1}{\pi} \int_0^\pi (e^{\alpha x} + e^{-\alpha x}) \cos kx \, dx \\
 &= \frac{1}{\pi} \int_0^\pi e^{\alpha x} \cos kx \, dx + \frac{1}{\pi} \int_0^\pi e^{-\alpha x} \cos kx \, dx \\
 &= \frac{e^{\alpha x}}{\pi} \left[\frac{\alpha \cos kx + k \sin kx}{\alpha^2 + k^2} \right] \Big|_0^\pi + \frac{e^{-\alpha x}}{\pi} \left[\frac{k \sin kx - x \cos kx}{\alpha^2 + k^2} \right] \Big|_0^\pi \\
 &= \frac{e^{\alpha \pi}}{\pi} \frac{\alpha(-1)^k}{\alpha^2 + k^2} + \frac{e^{\alpha \pi}}{\pi} \left[\frac{-\alpha(-1)^k}{\alpha^2 + k^2} \right] \\
 &= \frac{2\alpha}{\pi} \frac{(-1)^k}{(\alpha^2 + k^2)} \left(\frac{e^{2\pi} - e^{-\alpha \pi}}{2} \right) = \frac{2\alpha(-1)^k}{\pi(\alpha^2 + k^2)} \operatorname{sh} \alpha \pi
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= a_0 + \sum_{k=1}^{\infty} a_k \cos kx \\
 &= \frac{1}{\pi \alpha} \operatorname{sh} \alpha \pi + \frac{2\alpha}{\pi} \operatorname{sh} \alpha \pi \sum_{k=1}^{\infty} (-1)^k \frac{1}{\alpha^2 + k^2} \cos kx \\
 &= \frac{2 \operatorname{sh} \alpha \pi}{\pi} \left[\frac{1}{2\alpha} + \sum_{k=1}^{\infty} (-1)^k \frac{\cos kx}{\alpha^2 + k^2} \right]
 \end{aligned}$$

(5) 在 $(-\pi, \pi)$ 这个周期上, $f(x) = \begin{cases} \cos x & (|x| < \frac{\pi}{2}) \\ 0 & (\frac{\pi}{2} < |x| < \pi) \end{cases}$

解: 由于 $f(x) = f(-x)$ 所以 $f(x)$ 是偶函数

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos kx$$

$$\begin{aligned}
 a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \cos x \, dx \\
 &= \frac{1}{\pi} \sin x \Big|_0^{\frac{\pi}{2}} = \frac{1}{\pi}
 \end{aligned}$$

$$\begin{aligned}
 a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos kx \, dx \\
 &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos x \cos kx \, dx = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} 2 \cos x \cos kx \, dx \\
 &= \frac{1}{\pi} \int_0^{\frac{\pi}{2}} [\cos(k+1)x + \cos(k-1)x] \, dx = \frac{1}{\pi} \left(\frac{\sin(k+1)x}{k+1} \Big|_0^{\frac{\pi}{2}} + \frac{\sin(k-1)x}{k-1} \Big|_0^{\frac{\pi}{2}} \right) \\
 &= \frac{1}{\pi} \left[\frac{\cos(k \frac{\pi}{2})}{k+1} + \frac{\sin(k-1) \frac{\pi}{2}}{k-1} \right]
 \end{aligned}$$

$$k=1 \text{ 时, } a_1 = \frac{1}{\pi} \lim_{u \rightarrow 0} \frac{\pi}{2} \left(\frac{\sin u}{u} \right) = \frac{1}{2}$$

$$k \neq 1 \text{ 时, } a_k = \frac{1}{\pi} \left[\frac{\cos \frac{\pi}{2} k}{k+1} + \frac{\sin(k-1) \frac{\pi}{2}}{k-1} \right]$$

当 k 为奇数时, 有 $\cos \frac{\pi}{2} k = 0, \sin(k-1) \frac{\pi}{2} = -\cos \frac{\pi}{2} k = 0$

所以 $a_k = 0$

当 k 为偶数时, 设 $k = 2n (n = 1, 2, 3 \cdots), \cos \frac{\pi}{2} k = \cos \left(\frac{\pi}{2} 2n \right) = \cos(n\pi) = (-1)^n$

$$\sin \frac{\pi}{2} (k-1) = -\cos \left(\frac{\pi}{2} k \right) = -\cos \left(\frac{\pi}{2} 2n \right) = -\cos(n\pi) = (-1)^{n+1}$$

$$\begin{aligned} a_{2n} &= \frac{1}{\pi} \left[\frac{(-1)^n}{2n+1} + \frac{(-1)^{n+1}}{2n-1} \right] = \frac{(-1)^n}{\pi} \left(\frac{1}{2n+1} - \frac{1}{2n-1} \right) \\ &= \frac{(-1)^n}{\pi} \left(\frac{-2}{4n^2-1} \right) = \frac{(-1)^{n+1}}{\pi} \left(\frac{2}{4n^2-1} \right) \end{aligned}$$

$$\begin{aligned} f(x) &= a_0 + \sum_{k=1}^{\infty} a_k \cos kx = \frac{1}{\pi} + \frac{\cos x}{2} + \sum_{k=2}^{\infty} a_k \cos kx \\ &= \frac{1}{\pi} + \frac{\cos x}{2} + \sum_{n=1}^{\infty} a_{2n} \cos(2nx) \\ &= \frac{1}{\pi} + \frac{\cos x}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{4n^2-1} \right) \cos(2nx) \end{aligned}$$

5. 要求下列函数 $f(x)$ 的值在它的定义区间的边界上为零, 试根据这一要求将 $f(x)$ 展为傅里叶级数.

(1) $f(x) = \cos ax$, 定义在 $(0, \pi)$ 上

解: 由于 $f(x)$ 在所定义区间的边界上为零, 即 $f(0) = f(\pi) = 0$, 则展开后的傅里叶级数为

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) \text{ 代入 } x=0, x=\pi, \text{ 得 } a_0 = 0, a_k = 0 (k=1, 2 \cdots)$$

$$\text{即 } f(x) = \sum_{k=1}^{\infty} b_k \cdot \sin kx$$

$$\begin{aligned} b_k &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin kx dx = \frac{2}{\pi} \int_0^{\pi} \cos ax \cdot \sin kx dx \\ &= \frac{1}{\pi} \int_0^{\pi} 2 \cos x \sin kx dx = \frac{1}{\pi} \int_0^{\pi} [\sin(k+x)x + \sin(k-x)x] dx \\ &= \frac{1}{\pi} \left(\frac{-\cos(k+\alpha)x}{k+\alpha} + \frac{-\cos(k-\alpha)x}{k-\alpha} \right) \Big|_0^{\pi} \\ &= \frac{1}{\pi} \left(\frac{-\cos(k+\alpha)\pi}{k+\alpha} + \frac{-\cos(k-\alpha)\pi}{k-\alpha} \right) + \frac{1}{\pi} \left(\frac{1}{k+\alpha} + \frac{1}{k-\alpha} \right) \\ &= \frac{-1}{\pi} \left(\frac{(k-\alpha)\cos(k+\alpha)\pi + (k+\alpha)\cos(k-\alpha)\pi}{k^2-\alpha^2} \right) + \frac{2k}{\pi} \cdot \frac{1}{k^2-\alpha^2} \\ &= \frac{2k}{\pi} \left(\frac{1}{k^2-\alpha^2} \right) + \frac{(-1)^{k+1}(k-\alpha)\cos \alpha\pi}{\pi(k^2-\alpha^2)} + \frac{(-1)^{k+1}(k+\alpha)\cos \alpha\pi}{\pi(k^2-\alpha^2)} \end{aligned}$$

$$= \frac{2k}{\pi(k^2 - a^2)} [1 + (-1)^{k+1} \cos a\pi]$$

所以
$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx = \sum_{k=1}^{\infty} \frac{2k}{\pi(k^2 - a^2)} [1 + (-1)^{k+1} \cos x\pi] \sin kx$$

$$= \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{k}{(k^2 - a^2)} [1 + (-1)^{k+1} \cos a\pi] \sin kx$$

(2) $f(x) = x^3$, 定义在 $(0, \pi)$ 上

解: 由于 $f(x)$ 在所定义区间的边界上为零, 即 $f(0) = f(\pi) = 0$,

则展开后的傅里叶级数为 $f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$

将 $x = 0, x = \pi$ 代入得 $a = 0, a_k = 0 \quad (k = 1, 2, \dots)$

即 $f(x) = \sum_{k=1}^{\infty} b_k \sin kx$

$$b_k = \frac{2}{\pi} \int_0^{\pi} f(x) \sin kx dx = \frac{2}{\pi} \int_0^{\pi} x^3 \sin kx dx$$

$$\begin{aligned} \int x^3 \sin kx dx &= \frac{1}{k} \int x^3 d\cos kx = -\frac{1}{k} (x^3 \cos kx) + \frac{1}{k} \int 3x^2 \cos kx dx \\ &= -\frac{1}{k} (x^3 \cos kx) + \frac{3}{k^2} \int x^2 d\sin kx = -\frac{1}{k} x^3 \cos kx + \frac{3x^2}{k^2} \sin kx \\ &\quad - \frac{3}{k^2} \int 2x \sin kx dx = -\frac{1}{k} x^3 \cos kx + \frac{3x^2}{k^2} \sin kx + \frac{6}{k^3} \int x d\cos kx \\ &= -\frac{x^3}{k} \cos kx + \frac{3x^2}{k^2} \sin kx + \frac{6x}{k^3} \cos kx - \frac{6}{k^3} \int \cos kx dx \\ &= -\frac{x^3}{k} \cos kx + \frac{3x^2}{k^2} \sin kx + \frac{6x}{k^3} \cos kx - \frac{6}{k^4} \sin kx + c \\ b_k &= \left[-\frac{\pi^3}{k} (-1)^k + \frac{3\pi^2}{k^2} \times 0 + \frac{6\pi}{k^3} (-1)^k + \frac{6}{k^4} \times 0 - 0 \right] \frac{2}{\pi} \\ &= \left[-\frac{\pi^3}{k} (-1)^k + \frac{6\pi}{k^3} (-1)^k \right] \frac{2}{\pi} = (-1)^k \left(\frac{12}{k^3} - \frac{2\pi^2}{k} \right) \end{aligned}$$

所以
$$f(x) = \sum_{k=1}^{\infty} b_k \sin kx = \sum_{k=1}^{\infty} (-1)^k \left(\frac{12}{k^3} - \frac{2\pi^2}{k} \right) \sin kx$$

(3) $f(x) = 1$, 定义在 $(0, \pi)$ 上

解: 由于 $f(x)$ 在所定义区间的边界上为零, 即 $f(0) = f(\pi) = 0$,

则展开后的傅里叶级数为 $f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$

将 $x = 0, x = \pi$ 代入得 $a_0 = 0, a_k = 0 (k = 1, 2, \dots)$

即 $f(x) = \sum_{k=1}^{\infty} b_k \sin kx$

$$b_k = \frac{2}{\pi} \int_0^{\pi} f(x) \sin kx dx = \frac{2}{\pi} \int_0^{\pi} \sin kx dx = \left. \frac{-2}{\pi k} \cos kx \right|_0^{\pi} = \frac{2}{\pi k} (1 - \cos k\pi)$$

$$= \frac{2}{\pi k} [1 - (-1)^k] = \begin{cases} 0, & (k = 2n) \\ \frac{4}{\pi k}, & (k = 2n+1), (n = 0, 1, 2, \dots) \end{cases}$$

$$\text{所以 } f(x) = \sum_{k=1}^{\infty} b_k \sin kx = \sum_{k=0}^{\infty} \frac{4}{\pi(2n+1)} \sin(2n+1)x = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{2n+1} \sin(2n+1)x$$

6. 要求下列函数 $f(x)$ 的导数 $f'(x)$ 在函数定义区间的边界上为零. 试根据这个要求将 $f(x)$ 展为傅里叶级数.

(1) 在 $(0, l/2)$ 上, $f(x) = \cos(\pi x/l)$, 而在 $(l/2, l)$ 上, $f(x) = 0$

解: 由题意, 要求 $f(x)$ 的导数 $f'(x)$ 在函数定义区间的边界上为零.

因此, 有 $f'(0) = f'(l) = 0$

$$\text{设 } f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos \frac{k\pi}{l}x + b_k \sin \frac{k\pi}{l}x)$$

$$f'(x) = \sum_{k=1}^{\infty} (-a_k \frac{k\pi}{l} \sin \frac{k\pi}{l}x + b_k \frac{k\pi}{l} \cos \frac{k\pi}{l}x)$$

$$f'(x)|_{x=0} = \sum_{k=1}^{\infty} b_k \frac{k\pi}{l}, \quad f'(x)|_{x=l} = \sum_{k=1}^{\infty} b_k \frac{k\pi}{l} (-1)^k$$

由题意有 $f'(x)|_{x=0} = 0$ 和 $f'(x)|_{x=l} = 0$, 则有 $b_k = 0$

$$\text{即 } f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi}{l}x$$

$$\text{其中 } a_0 = \frac{1}{l} \int_0^l f(x) dx = \frac{1}{l} \int_0^{l/2} \cos \left(\frac{\pi x}{l} \right) dx = \frac{1}{l} \left(\frac{l}{\pi} \right) \sin \frac{\pi x}{l} \Big|_0^{l/2} = \frac{1}{\pi} \sin \frac{\pi}{2} = \frac{1}{\pi}$$

$$\begin{aligned} a_k &= \frac{2}{l} \int_0^l f(x) \cos \left(\frac{k\pi}{l}x \right) dx = \frac{2}{l} \int_0^{l/2} \cos \frac{\pi x}{l} \cos \frac{k\pi}{l}x dx \\ &= \frac{1}{l} \int_0^{l/2} \cos \left(\frac{\pi x}{l} + \frac{k\pi}{l}x \right) dx + \frac{1}{l} \int_0^{l/2} \cos \left(\frac{\pi x}{l} - \frac{k\pi}{l}x \right) dx \\ &= \frac{1}{l} \frac{l}{(k+1)\pi} \sin \frac{\pi}{l} (1+k)x \Big|_0^{l/2} + \frac{1}{l} \frac{l}{(k-1)\pi} \sin \frac{\pi}{l} (k-1)x \Big|_0^{l/2} \\ &= \frac{1}{(k+1)\pi} \sin \frac{\pi}{2} (1+k) + \frac{1}{(k-1)\pi} \sin \frac{\pi}{2} (k-1) \end{aligned}$$

$$\text{当 } k=1 \text{ 时, } a_1 = \lim_{k \rightarrow 1} \frac{\sin \frac{\pi}{2} (k-1)}{2(k-1)\pi/2} = \frac{1}{2}$$

当 $k=2n+1$ 时, $a_k = 0$

$$\text{当 } k=2n \text{ 时, } a_k = \frac{1}{(k+1)\pi} (-1)^n + \frac{1}{(k-1)\pi} (-1)^{n-1} = \frac{-2(-1)^n}{\pi(k^2-1)} = \frac{-2(-1)^n}{\pi(4n^2-1)} = a_{2n}$$

$$\text{即 } a_k = a_{2n} = (-1)^{n+1} \frac{2}{\pi(4n^2-1)}$$

$$\text{所以 } f(x) = \frac{1}{\pi} + \frac{1}{2} \cos \frac{\pi x}{l} + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{\pi(4n^2-1)} \cos \left(\frac{2n\pi x}{l} \right)$$

$$= \frac{1}{\pi} + \frac{1}{2} \cos \frac{\pi x}{l} + \frac{2}{\pi} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{4n^2 - 1} \cos \frac{2n\pi x}{l}$$

(2) $f(x) = a(1 - x/l)$, 定义在 $(0, l)$ 上

解: 在 $(0, l)$ 上, $f(x) = a(1 - x/l)$ 且 $f'(0) = f'(l) = 0$, 即展成傅里叶级数为余弦级数.

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi}{l} x$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx = \frac{1}{l} \int_0^l a(1 - x/l) dx = a \left(\frac{1}{l} x - \frac{x^2}{2l^2} \right) \Big|_0^l$$

$$= a - \frac{a}{2} = \frac{a}{2}$$

$$a_k = \frac{2}{l} \int_0^l f(x) \cos \left(\frac{k\pi}{l} x \right) dx = \frac{2}{l} \int_0^l a(1 - x/l) \cos \left(\frac{k\pi}{l} x \right) dx$$

$$= \frac{2}{l} \left[a \sin \left(\frac{k\pi}{l} x \right) \right] \cdot \frac{l}{k\pi} \Big|_0^l - \frac{2a}{k\pi l^2} \int_0^l x d \left(\sin \frac{k\pi x}{l} \right)$$

$$= -\frac{2a}{k\pi l} x \sin \frac{k\pi x}{l} \Big|_0^l + \frac{2a}{k\pi l} \int_0^l \sin \frac{k\pi x}{l} dx$$

$$= \frac{-2a}{k^2 \pi^2} \cos \frac{k\pi x}{l} \Big|_0^l = (-\cos k\pi + 1) \frac{2a}{k^2 \pi^2}$$

$$= \frac{2a}{k^2 \pi^2} (1 + (-1)^{k+1})$$

$$a_k = \begin{cases} 0, & (k = 2n) \\ \frac{4a}{(2n+1)^2 \pi^2}, & (k = 2n+1) \end{cases}$$

所以
$$f(x) = \frac{a}{2} + \sum_{n=1}^{\infty} \frac{4a}{(2n+1)^2 \pi^2} \cos \left[\frac{(2n+1)\pi x}{l} \right]$$

(3) 在 $(0, l/2)$ 上, $f(x) = x$; 在 $(l/2, l)$ 上, $f(x) = l - x$

解: 由题意有 $f'(0) = 0, f'(l) = 0$, 展开成傅里叶级数为余弦级数

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{k\pi}{l} x$$

$$a_0 = \frac{1}{l} \int_0^l f(x) dx = \frac{1}{l} \int_0^{\frac{l}{2}} x dx + \frac{1}{l} \int_{\frac{l}{2}}^l (l - x) dx$$

$$= \frac{1}{2l} x^2 \Big|_0^{\frac{l}{2}} + x \Big|_{\frac{l}{2}}^l - \frac{1}{2l} x^2 \Big|_{\frac{l}{2}}^l$$

$$= \frac{1}{2l} \left(\frac{l}{2} \right)^2 + \frac{l}{2} - \frac{l^2}{2l} = \frac{l}{4}$$

$$a_k = \frac{2}{l} \int_0^l f(x) \cos \left(\frac{k\pi}{l} x \right) dx$$

$$= \frac{2}{l} \int_0^{\frac{l}{2}} x \cos \left(\frac{k\pi}{l} x \right) dx + \frac{2}{l} \int_{\frac{l}{2}}^l (l - x) \cos \left(\frac{k\pi}{l} x \right) dx$$

$$= \frac{2}{l} \left[\int_0^{\frac{l}{2}} x d \left(\sin \frac{k\pi x}{l} \right) \right] \frac{l}{k\pi} + \frac{2l}{l} \int_{\frac{l}{2}}^l \cos \frac{k\pi x}{l} dx - \frac{2}{l} \left[\int_{\frac{l}{2}}^l x d \left(\sin \frac{k\pi x}{l} \right) \right] \frac{l}{k\pi}$$

$$\begin{aligned}
&= \frac{2}{l} \frac{l}{k\pi} x \sin \frac{k\pi}{l} x \Big|_0^{\frac{l}{2}} + \frac{2l \sin \frac{k\pi}{l} x}{k\pi} \Big|_{l/2}^l - \frac{2l}{lk\pi} x \sin \frac{k\pi}{l} x \Big|_{\frac{l}{2}}^l + \frac{2}{k\pi} \int_0^{\frac{l}{2}} \sin \frac{k\pi}{l} x dx + \frac{2}{k\pi} \int_{\frac{l}{2}}^l \sin \frac{k\pi}{l} x dx \\
&= \frac{2}{k\pi} \frac{l}{2} \sin \frac{k\pi}{2} - \frac{2l}{k\pi} \sin \frac{k\pi}{2} + \frac{l}{k\pi} \sin \frac{k\pi}{2} + \frac{2l}{(k\pi)^2} \cos \frac{k\pi}{l} x \Big|_0^{\frac{l}{2}} - \frac{2l}{(k\pi)^2} \cos \frac{k\pi}{l} x \Big|_{\frac{l}{2}}^l \\
&= \frac{2l}{(k\pi)^2} \left(\cos \frac{k\pi}{2} - 1 - \cos k\pi + \cos \frac{k\pi}{2} \right) \\
&= \frac{2l}{(k\pi)^2} \left(2\cos \frac{k\pi}{2} - [1 + (-1)^k] \right) \\
&= \begin{cases} \frac{4l}{(k\pi)^2} \cos \frac{k\pi}{2} & (k = \text{奇数}) \\ \frac{4}{(k\pi)^2} \left(\cos \frac{k\pi}{2} - 1 \right) & (k = \text{偶数}) \end{cases} = \begin{cases} 0 & (k = \text{奇数}) \\ \frac{4l}{(k\pi)^2} [(-1)^{k/2} - 1] & (k = \text{偶数}) \end{cases}
\end{aligned}$$

当 $k = 2n$ 时

$$a_k = a_{2n} \frac{4l}{(k\pi)^2} [(-1)^{k/2} - 1] = \frac{4l}{(2n\pi)^2} [(-1)^n - 1] = \begin{cases} 0, & (n = \text{偶数}) \\ -\frac{8l}{(2n\pi)^2}, & (n = \text{奇数}) \end{cases}$$

$$\text{即 } a_n = \frac{-8l}{[2(2n+1)\pi]^2} = \frac{-2l}{(2n+1)^2 \pi^2}$$

$$\text{所以 } f(x) = \frac{l}{4} - \frac{8l}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(4n+2)^2} \cos \left[(4n+2) \frac{\pi x}{l} \right]$$

7. 在区间 $(0, l)$ 上定义了函数 $f(x) = x$. 试根据条件 $f'(0) = 0, f(l) = 0$ 将 $f(x)$ 展开为傅里叶级数.

解: 由题意 $f'(0) = 0$ 及 $f(l) = 0$

$$\text{设 } f(x) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos \frac{k\pi}{l} x + b_k \sin \frac{k\pi}{l} x \right)$$

$$\text{由 } f'(0) = 0 \text{ 有 } f'(x) = \sum_{k=1}^{\infty} \left(-a_k \frac{k\pi}{l} \sin \frac{k\pi}{l} x + b_k \frac{k\pi}{l} \cos \frac{k\pi}{l} x \right)$$

$$f'(0) = \sum_{k=1}^{\infty} b_k \frac{k\pi}{l} = 0 \quad \text{即 } b_k = 0$$

所以

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos \frac{k\pi}{l} x$$

由于在 $x = l$ 时, $f(l) = 0$ 时, 故将函数 $f(x)$ 在区间 $(0, l)$ 作奇延拓,

延拓后的函数周期为 $L = 4l$,

延拓后的函数图形如图所示:

拓展后的函数 $f(x)$ 为

$$f(x) = \begin{cases} x & (0, l) \\ x - 2l & (l, 2l) \end{cases}$$

展开后的傅里叶级数为

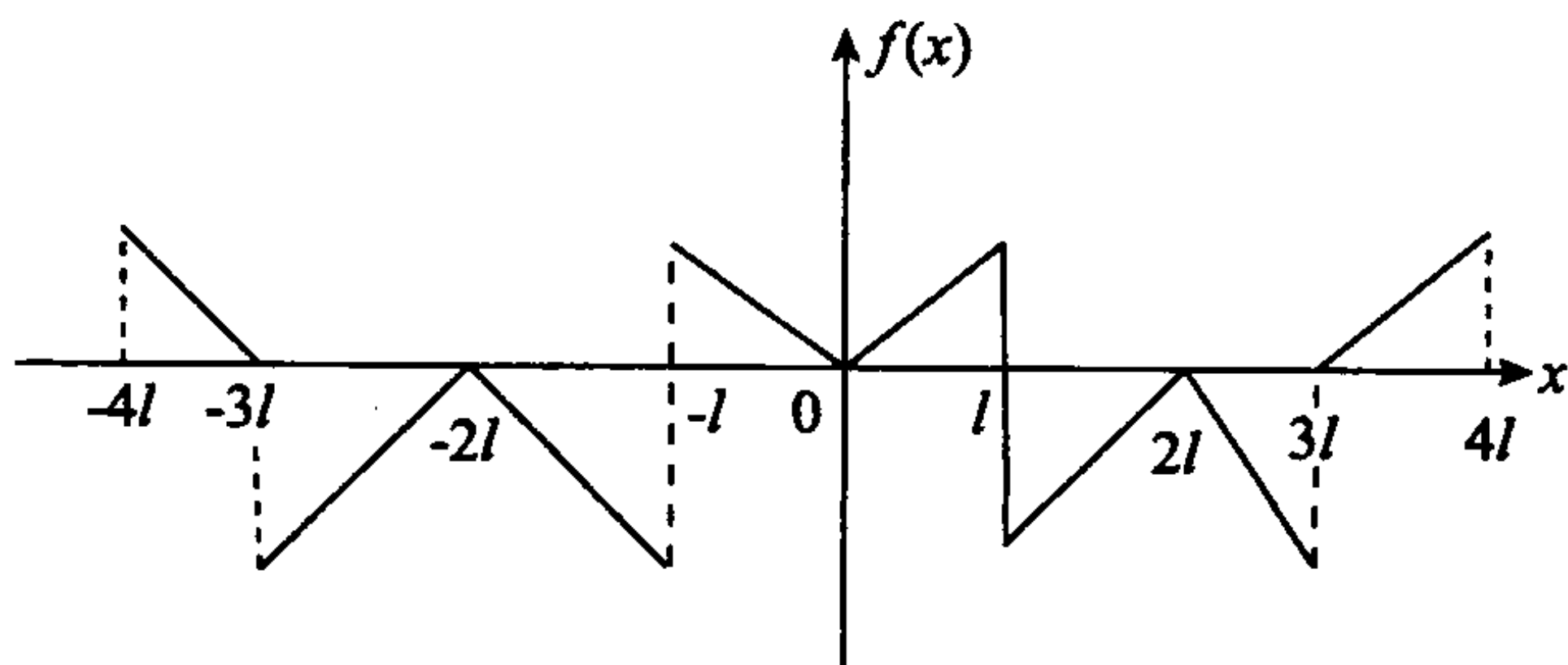


图 5-1-7

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos$$

$$a_0 = \frac{1}{2l} \int_0^{2l} f(x) dx = \frac{1}{2l} \int_0^l x dx + \frac{1}{2l} \int_l^{2l} (x-2l) dx$$

$$= \frac{1}{4l} x^2 \Big|_0^l + \frac{1}{4l} x^2 \Big|_l^{2l} - x \Big|_l^{2l} = \frac{l}{4} + \frac{l^2}{4l} - \frac{l}{4} - 2l + l = 0$$

$$a_k = \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{k\pi x}{2l}\right) dx$$

$$= \frac{1}{l} \int_0^l x \cos \frac{k\pi x}{2l} dx + \frac{1}{l} \int_l^{2l} (x-2l) \cos\left(\frac{k\pi x}{2l}\right) dx$$

其中 $\int x \cos \frac{k\pi x}{2l} dx = \int x \frac{2l}{k\pi} d\left(\sin \frac{k\pi x}{2l}\right) = \frac{2l}{k\pi} \left[x \sin \frac{k\pi x}{2l} - \int \sin \frac{k\pi x}{2l} dx \right]$

$$= \frac{2l}{k\pi} \left[x \sin \frac{k\pi x}{2l} + \frac{2l}{k\pi} \cos \frac{k\pi x}{2l} \right] + c$$

$$= \frac{2l}{k^2 \pi^2} (x \cdot \sin \frac{k\pi x}{2l}) + \frac{4l^2}{k^2 \pi^2} \cos \frac{k\pi x}{2l} + c$$

当定积分在 $(0, l)$ 范围内, 则

$$\int_0^l \cos \frac{k\pi x}{2l} dx = \frac{2l^2}{k^2 \pi^2} \sin \frac{k\pi}{2} + \frac{4l^2}{k^2 \pi^2} \left(\cos \frac{k\pi}{2} - 1 \right)$$

$$\int_l^{2l} x \cos \frac{k\pi x}{2l} dx = \frac{-2l^2}{k^2 \pi^2} \sin \frac{k\pi}{2} + \frac{4l^2}{k^2 \pi^2} (\cos k\pi - \cos \frac{k\pi}{2})$$

代入 $a_k = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{k\pi x}{2l} dx$

$$a_k = \frac{1}{l} \frac{4l^2}{k^2 \pi^2} (\cos k\pi - 1) - 2 \int_l^{2l} \cos \frac{k\pi x}{2l} dx$$

所以 $= \frac{4l}{k^2 \pi^2} (\cos k\pi - 1) - \frac{4l}{k\pi} \sin \frac{k\pi}{2}$

$$= \begin{cases} 0, & (k = \text{偶数}) \\ \frac{-8l}{(2n+1)^2 \pi^2} + \frac{4l}{(2n+1)\pi} (-1)^n, & (k = \text{奇数}, k = 2u+1) \end{cases}$$

所以 $f(x) = \sum_{k=1}^{\infty} a_k \cos \frac{k\pi x}{2l} = \sum_{k=1}^{\infty} \left[\frac{-8l}{(2n+1)^2 \pi^2} + \frac{4l(-1)^n}{(2n+1)\pi} \right] \cos \frac{(2n+1)\pi x}{2l}$

8. 矩形波 $f(x)$, 在 $(-T/2, T/2)$ 这个周期上可表为

$$f(x) = \begin{cases} 0, & \text{在 } (-T/2, -\tau/2) \text{ 上} \\ H, & \text{在 } (-\tau/2, \tau/2) \text{ 上} \\ 0, & \text{在 } (\tau/2, T/2) \text{ 上.} \end{cases}$$

试将它展为复数形式的傅里叶级数.

解: 矩形波函数 $f(x)$ 展开为复数形式的傅里叶级数.

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{\frac{ik\pi x}{l}} = \sum_{k=-\infty}^{\infty} e^{\frac{2ik\pi x}{T}} \quad (\text{其中 } l = T/2)$$

$$c_k = \frac{1}{2l} \int_{-l}^l f(\xi) [e^{\frac{ik\pi \xi}{l}}]^* d\xi = \frac{1}{T} \int_{-T/2}^{T/2} f(\xi) e^{-\frac{2ik\pi \xi}{T}} d\xi$$

$$c_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(\xi) d\xi = \frac{1}{T} \int_{-\tau/2}^{\tau/2} H d\xi = \frac{\tau}{T} H$$

$$\begin{aligned}
 c_k &= \frac{1}{T} \int_{-\tau/2}^{\tau/2} H e^{-i \frac{2k\pi}{T} \xi} d\xi = \frac{-H}{T} \cdot \frac{T}{2k\pi i} e^{-i \frac{2k\pi}{T} \xi} \Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \\
 &= \frac{-H}{2k\pi i} \left(e^{-i \frac{k\pi\tau}{T}} - e^{i \frac{k\pi\tau}{T}} \right) = \frac{Hi}{2k\pi} \left(-2i \sin \frac{k\pi\tau}{T} \right) \\
 &= \frac{H}{k\pi} \sin \left(\frac{k\pi\tau}{T} \right)
 \end{aligned}$$

当 $k=0$, $c_k = c_0 = \lim_{k \rightarrow 0} \frac{H\tau}{T k \pi \tau} \cdot \sin \frac{k\pi\tau}{T} = \frac{H\tau}{T}$ 与 c_0 相同.

$$\text{所以 } f(x) = \sum_{k=-\infty}^{\infty} c_k e^{i \frac{2k\pi}{T} x} = \sum_{k=-\infty}^{\infty} \frac{H}{k\pi} \sin \left(\frac{k\pi\tau}{T} \right) e^{i \frac{2k\pi}{T} x}$$

其中当 $k=0$ 时, $c_0 = \frac{\tau}{T} H$.

$$f(x) = \frac{\tau}{T} H + \left(\sum_{k=-\infty}^{-1} + \sum_{k=1}^{\infty} \right) \frac{H}{k\pi} \sin \left(\frac{k\pi\tau}{T} \right) e^{i \frac{2k\pi}{T} x}$$

§ 5.2 傅里叶积分与傅里叶变换

1. 求单个锯齿脉冲 $f(t) = kt \operatorname{rect}\left(\frac{t}{T} - \frac{1}{2}\right)$, 即

$$f(t) = \begin{cases} 0 & (t < 0), \\ kt & (0 < t < T), \\ 0 & (t > T) \end{cases}$$

的傅里叶变换.

解: $f(t)$ 的傅里叶积分为

$$f(t) = \int_0^{\infty} A(\omega) \cos \omega t d\omega + \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

$$\text{其中} \begin{cases} A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \cos \omega \xi d\xi \\ B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \sin \omega \xi d\xi \end{cases}$$

将 $f(t) = kt \operatorname{rect}\left(\frac{t}{T} - \frac{1}{2}\right)$ 代入即

$$\begin{aligned}
 A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \cos \omega \xi d\xi = \frac{1}{\pi} \int_0^T kt \cos \omega t dt \\
 &= \frac{k}{\pi\omega} \left(t \sin \omega t \Big|_0^T - \int_0^T \sin \omega t dt \right) = \frac{k}{\pi\omega} \left(t \sin \omega t + \frac{\cos \omega t}{\omega} \right) \Big|_0^T \\
 &= \frac{k}{\pi\omega} \left(T \sin \omega T + \frac{\cos \omega T}{\omega} - \frac{1}{\omega} \right) = \frac{k}{\pi\omega^2} (\cos \omega T + \omega T \sin \omega T - 1) \\
 B(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \sin \omega \xi d\xi = \frac{1}{\pi} \int_0^T kt \sin \omega t dx \\
 &= \frac{k}{\pi\omega} \left(-t \cos \omega t \Big|_0^T + \int_0^T \cos \omega t dx \right) = \frac{k}{\pi\omega} \left(-t \cos \omega t \Big|_0^T + \frac{\sin \omega t}{\omega} \Big|_0^T \right) \\
 &= \frac{k}{\pi\omega^2} (-\omega T \cos \omega T + \sin \omega T)
 \end{aligned}$$

$$\begin{aligned} \text{所以 } f(t) &= \int_0^\infty A(\omega) \cos \omega t d\omega + \int_0^\infty B(\omega) \sin \omega t d\omega \\ &= \frac{k}{\pi} \int_0^\infty \frac{1}{\omega^2} (\cos \omega T + \omega T \sin \omega T - 1) \cos \omega t d\omega + \frac{k}{\pi} \int_0^\infty \frac{1}{\omega^2} (\sin \omega T - \omega T \cos \omega T) \sin \omega t d\omega \end{aligned}$$

复数形式

$$\begin{aligned} F(\omega) &= \frac{1}{2\pi} \int_{-\infty}^\infty f(t) [e^{i\omega t}]^* dt = \frac{1}{2\pi} \int_0^T kt e^{-i\omega t} dt = \frac{ik}{2\pi\omega} \int_0^T t d e^{-i\omega t} = \frac{ik}{2\pi\omega} \left(t e^{-i\omega t} \Big|_0^T - \int_0^T e^{-i\omega t} dt \right) \\ &= \frac{ik}{2\pi\omega} \left(T e^{-i\omega T} - \frac{i}{\omega} e^{-i\omega t} \Big|_0^T \right) \\ &= \frac{ik}{2\pi\omega^2} \left(\omega T e^{-i\omega T} - \frac{i}{\omega} e^{-i\omega T} + \frac{i}{\omega} \right) = \frac{k}{2\pi\omega} \left[\frac{1}{\omega} (e^{-i\omega T} - 1) + iT e^{-i\omega T} \right] \end{aligned}$$

2. 求 $\text{sinc } t = \frac{\sin \pi t}{\pi t}$ 的傅里叶变换, 试以本题的傅里叶变换函数距教材图 5-1 比较, 又以本题的 $\text{sinc } t$ 跟教材图 5-2 比较, 比较的结果说明什么问题?

解: $f(t) = \text{sinc } t = \frac{\sin \pi t}{\pi t}$, $f(t)$ 为偶函数, $f(t) = f(-t)$, 所以傅里叶变换为

$$f(t) = \int_0^\infty A(\omega) \cos \omega t d\omega$$

$$A(\omega) = \frac{2}{\pi} \int_0^\infty f(\xi) \cos \omega \xi d\xi = \frac{2}{\pi} \int_0^\infty \frac{\sin \pi t \cos \omega t}{\pi t} dt$$

其中

$$\begin{aligned} &= \frac{2}{\pi^2} \int_0^\infty \frac{1}{2} [\sin(\omega + \pi)t - \sin(\omega - \pi)t] \frac{dt}{t} \\ &= \frac{1}{\pi^2} \left(\int_0^\infty \frac{\sin(\omega + \pi)t}{t} dt - \int_0^\infty \frac{\sin(\omega - \pi)t}{t} dt \right) \end{aligned}$$

$$\text{又 } \frac{1}{x} = \int_0^\infty e^{-\alpha x} d\alpha \quad (\alpha > 0)$$

所以

$$\begin{aligned} \int_0^\infty \frac{\sin x}{x} dx &= \int_0^\infty \sin x \left[\int_0^\infty e^{-\alpha x} d\alpha \right] dx = \int_0^\infty d\alpha \int_0^\infty e^{-\alpha x} \sin x dx \\ &= \int_0^\infty d\alpha \int_0^\infty e^{-\alpha x} \sin x dx = \int_0^\infty \frac{1}{1 + \alpha^2} d\alpha = \pi/2 \end{aligned}$$

因此有

$$I = \int_0^\infty \frac{\sin kx}{x} dx = \begin{cases} \frac{\pi}{2} & (k > 0) \\ 0 & (k = 0) \\ -\frac{\pi}{2} & (k < 0) \end{cases}$$

$$\text{即 } A(\omega) = \frac{1}{\pi^2} (I_1 - I_2)$$

$$\text{其中 } I_1 = \int_0^\infty \frac{\sin(\omega + \pi)t}{t} dt = \begin{cases} \frac{\pi}{2} & (\omega > -\pi) \\ 0 & (\omega = -\pi) \text{ (提示: 做 } \omega + \pi = u \text{ 变换)} \\ -\frac{\pi}{2} & (\omega < -\pi) \end{cases}$$

$$I_2 = \int_0^{\infty} \frac{\sin(\omega - \pi)}{t} dt = \begin{cases} \frac{\pi}{2} & (\omega > \pi) \\ 0 & (\omega = \pi) \text{ (提示: 做 } \omega - \pi = v \text{ 变换)} \\ -\frac{\pi}{2} & (\omega < \pi) \end{cases}$$

$$A(\omega) = \begin{cases} 0 & (\omega > \pi) \\ \frac{1}{\pi} & (-\pi < \omega < \pi) \\ 0 & (\omega < -\pi) \end{cases}$$

即:

$$A(\omega) = \begin{cases} 1/\pi & (|\omega| < \pi) \\ 0 & (|\omega| > \pi) \end{cases}$$

$$\text{所以 } A(\omega) = \frac{1}{\pi} \text{rect}\omega = \frac{1}{\pi} \text{rect}\left(\frac{x}{2\pi}\right), (x = 2\pi\omega) \text{ 其中 } \text{rect}x = \begin{cases} 1 & (|x| < \frac{1}{2}) \\ 0 & (|x| > \frac{1}{2}) \end{cases}$$

图 5-1 如下:

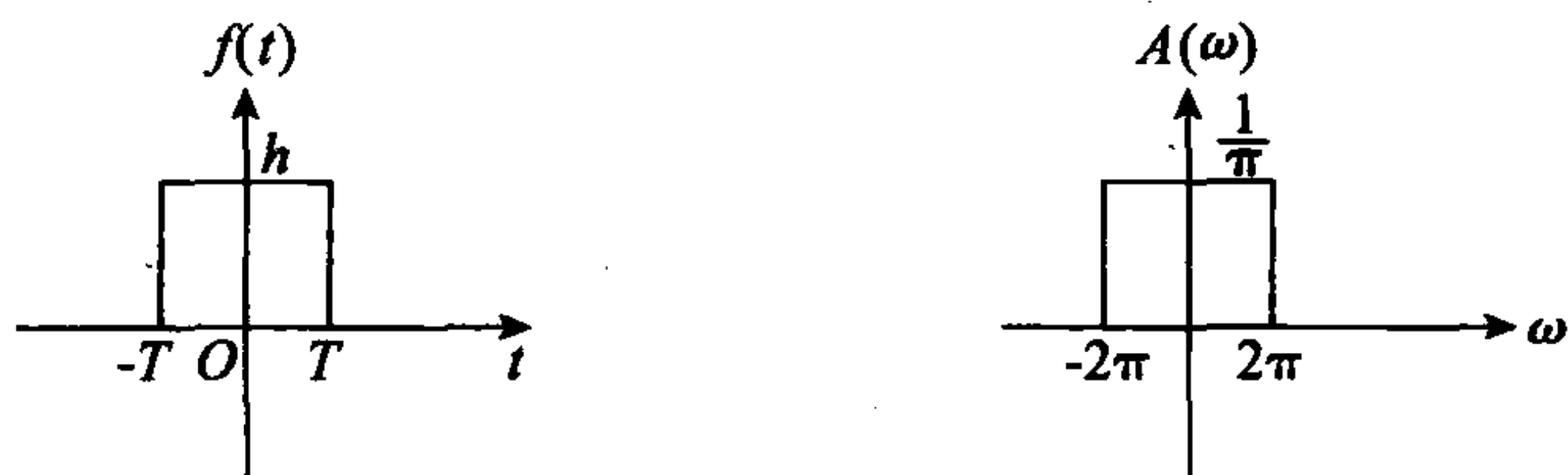


图 5-2-2(a)

图 5-2 如下:

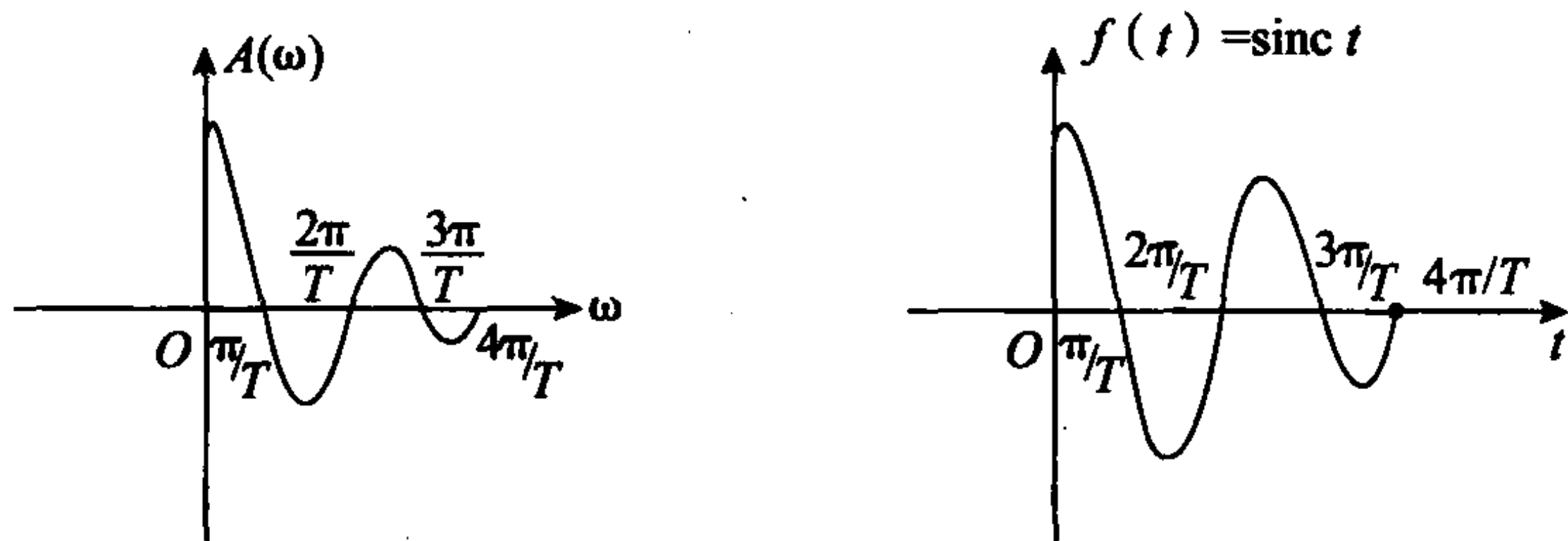


图 5-2-2(b)

由此可见, 本题与教材中例 1 题互为逆运算.

即 $f(t)$ 和 $A(\omega)$ 互为傅里叶变换式.

3. 将下列脉冲 $f(t)$ 展开为傅里叶积分.

$$f(t) = \begin{cases} 0 & (t < -T) \\ -h & (-T < t < 0), \\ h & (0 < t < T), \\ 0 & (T < t). \end{cases}$$

注意在半无界区间 $(0, \infty)$ 上, 本例的 $f(t)$ 与例 1 的 $f(t)$ 相同.

解: $f(t)$ 为奇函数, 所以傅里叶积分如下

$$f(t) = \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^{\infty} f(\xi) \sin \omega \xi d\xi = \frac{2}{\pi} \left(\int_{-T}^0 -h \sin \omega x dx + \int_0^T h \sin \omega x dx \right) \\ &= \frac{2h}{\pi} \int_0^T \sin \omega x dx = \frac{2h}{\pi \omega} (-\cos \omega x) \Big|_0^T \\ &= \frac{2h}{\pi \omega} (1 - \cos \omega T) \end{aligned}$$

$$\text{所以: } f(t) = \int_0^{\infty} \frac{2h}{\pi \omega} (1 - \cos \omega T) \sin \omega t d\omega = \frac{2h}{\pi} \int_0^{\infty} \frac{(1 - \cos \omega T) \sin \omega t}{\omega} d\omega$$

与教材中例 1 在半无界区间 $(0, \infty)$ 上相同, 但例 1 是偶函数而本题为奇函数.

4. $f(t)$ 是定义在半无界区间 $(0, \infty)$ 上的函数.

$$f(t) = \begin{cases} h & (0 < t < T). \\ 0 & (T < t). \end{cases}$$

(1) 在边界条件 $f'(0) = 0$ 下, 将 $f(t)$ 展为傅里叶积分;

(2) 在边界条件 $f(0) = 0$ 下, 将 $f(t)$ 展为傅里叶积分.

解: (1) 由于边界条件 $f'(0) = 0$, 则 $f(t)$ 须展开为傅里叶余弦积分

$$\text{即 } f(t) = \int_0^{\infty} A(\omega) \cos \omega t d\omega$$

$$A(\omega) = \frac{2}{\pi} \int_0^{\infty} f(\xi) \cos \omega \xi d\xi = \frac{2}{\pi} \int_0^T h \cos \omega t dt = \frac{2h}{\pi \omega} \sin \omega T$$

$$f(t) = \int_0^{\infty} \frac{2h}{\pi \omega} \sin \omega T \cos \omega t d\omega = \frac{2h}{\pi} \int_0^{\infty} \frac{\sin \omega T \cos \omega t}{\omega} d\omega$$

(2) 由于边界条件 $f(0) = 0$, 则 $f(t)$ 须展为傅里叶正弦积分

$$\text{即 } f(t) = \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^{\infty} f(\xi) \sin \omega \xi d\xi = \frac{2}{\pi} \int_0^T h \sin \omega t dt \\ &= \frac{2h}{\pi} \int_0^T \sin \omega t dt = \frac{-2h}{\pi \omega} \cos \omega t \Big|_0^T = \frac{2h}{\pi \omega} (1 - \cos \omega T) \end{aligned}$$

$$f(t) = \int_0^{\infty} \frac{2h}{\pi \omega} (1 - \cos \omega T) \sin \omega t d\omega = \frac{2h}{\pi} \int_0^{\infty} \frac{1 - \cos \omega T}{\omega} \sin \omega t d\omega$$

5. 在边界条件 $f(0) = 0$ 下, 将定义在 $(0, \infty)$ 上的函数 $f(x) = e^{-\lambda x}$ 展为傅里叶积分.

解: 由边界条件 $f(0) = 0$ 知 $f(t)$ 须展为傅里叶正弦积分

$$\text{即 } f(t) = \int_0^{\infty} B(\omega) \sin \omega t d\omega$$

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^{\infty} f(\xi) \sin \omega \xi d\xi = \frac{2}{\pi} \int_0^{\infty} e^{-\lambda x} \sin \omega t dt \\ &= \frac{-2}{\pi \omega} \int_0^{\infty} e^{-\lambda x} d(\cos \omega t) = \frac{-2}{\omega t} \left[e^{-\lambda x} \cos \omega t \Big|_0^{\infty} - \int_0^{\infty} \cos \omega t d(e^{-\lambda x}) \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{\pi\omega} - \frac{2\lambda}{\pi\omega} \int_0^\infty (\cos \omega t) e^{-\lambda t} dt = \frac{2}{\pi\omega} - \frac{2\lambda}{\pi\omega^2} \int_0^\infty e^{-\lambda t} d\sin \omega t \\
&= \frac{2}{\pi\omega} - \frac{2\lambda}{\pi\omega^2} e^{-\lambda t} \cdot \sin \omega t \Big|_0^\infty + \frac{2\lambda}{\pi\omega} \int_0^\infty \sin \omega t \cdot d e^{-\lambda t} \\
&= \frac{2}{\pi\omega} + \frac{2\lambda^2}{\pi\omega^2} \int_0^\infty (\sin \omega t) e^{-\lambda t} dt = \frac{2}{\pi\omega} - \frac{\lambda^2}{\omega^2} B(\omega)
\end{aligned}$$

$$B(\omega) + \frac{\lambda^2}{\omega^2} B(\omega) = \frac{2}{\pi\omega}$$

即

$$B(\omega) = \frac{2/\pi\omega}{1 + \frac{\lambda^2}{\omega^2}} = \frac{2\omega}{(\omega^2 + \lambda^2)\pi}$$

所以

$$f(t) = \frac{2}{\pi} \int_0^\infty \frac{\omega \sin \omega t}{(\omega^2 + \lambda^2)} d\omega$$

§ 5.3 δ 函数

1. 验证教材 § 5.2 中例 2 的频谱 $B(\omega)$ (图 5-4) 于 $N \rightarrow \infty$ 时成为 $A\delta(\omega - \omega_0) - A\delta(\omega + \omega_0)$, 并解释这结果的物理意义.

$$\text{解: } f(t) = \begin{cases} A \sin \omega_0 t & (|t| < \frac{2N\pi}{\omega_0}) \\ 0 & (|t| > \frac{2N\pi}{\omega_0}) \end{cases}$$

$$\text{当 } N \rightarrow \infty \text{ 时} \quad f(t) = A \sin \omega_0 t \quad (t < \infty)$$

由此可见, 将 § 5.2 例 2 转为 $f(t) = A \sin \omega_0 t$, 并求其傅里叶变换, 由于 $f(t) = A \sin \omega_0 t$ 是奇函数, 故展为正弦函数

$$\begin{aligned}
B(\omega) &= \frac{2}{\pi} \int_0^\infty f(t) \sin \omega t dt = \frac{2}{\pi} \int_0^\infty A \sin \omega_0 t \sin \omega t dt \\
&= \frac{-2A}{\pi} \int_0^\infty \frac{1}{2} [\cos(\omega_0 + \omega)t - \cos(\omega - \omega_0)t] dt \\
&= -\frac{A}{\pi} \left[\frac{\sin(\omega + \omega_0)t}{\omega + \omega_0} - \frac{\sin(\omega - \omega_0)t}{\omega - \omega_0} \right] \Big|_0^\infty
\end{aligned}$$

当 $N \rightarrow \infty$ 时, 即 $t \rightarrow \infty$

$$\begin{aligned}
B(\omega) &= A \lim_{t \rightarrow \infty} \frac{-1}{\pi} \left(\frac{\sin(\omega + \omega_0)t}{\omega + \omega_0} \right) + A \lim_{t \rightarrow \infty} \frac{1}{\pi} \left(\frac{\sin(\omega - \omega_0)t}{\omega - \omega_0} \right) \\
&= -A\delta(\omega + \omega_0) + A\delta(\omega - \omega_0)
\end{aligned}$$

其中

$$\delta(\omega + \omega_0) = \lim_{K \rightarrow \infty} \frac{1}{\pi} \frac{\sin K(\omega + \omega_0)}{\omega + \omega_0}$$

$$\delta(\omega - \omega_0) = \lim_{K \rightarrow \infty} \frac{1}{\pi} \frac{\sin K(\omega - \omega_0)}{\omega - \omega_0}$$

$$\lim_{K \rightarrow \infty} \int_{-\infty}^\infty \frac{1}{\pi} \frac{\sin Kx}{x} dx = \lim_{K \rightarrow \infty} \int_{-\infty}^\infty \frac{1}{\pi} \frac{\sin Kx}{Kx} d(Kx)$$

$$\begin{aligned}
 &= \frac{1}{\pi} \lim_{y \rightarrow \infty} \int_{-\infty}^{\infty} \frac{\sin y}{y} dy = \frac{2}{\pi} \lim_{y \rightarrow \infty} \int_0^{\infty} \frac{\sin y}{y} dy \\
 &= \frac{2}{\pi} \frac{\pi}{2} = 1
 \end{aligned}$$

所以 $\lim_{k \rightarrow \infty} \frac{1}{\pi} \frac{\sin kx}{x} = \delta(x)$

因此,正弦波的频谱线为图 5-3-1 所示.

该图表明:正弦波为单一频率的频谱线,这正是频谱线表明的单一正弦波的频率特性.

2. 将 $\delta(x)$ 展为实数形式的傅里叶积分.

解: $f(x) = \delta(x)$ 是偶函数,其傅里叶积分为余弦积分.

$$f(x) = \delta(x) = \int_0^{\infty} A(\omega) \cos \omega x d\omega$$

$$\begin{aligned}
 A(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(\xi) \cos \omega \xi d\xi = \frac{1}{\pi} \int_{-\infty}^{\infty} \delta(t) \cos \omega t dt \\
 &= \frac{1}{\pi} \cos \omega t \Big|_{t=0} = \frac{1}{\pi}
 \end{aligned}$$

所以

$$\delta(x) = \frac{1}{\pi} \int_0^{\infty} \cos \omega x d\omega$$

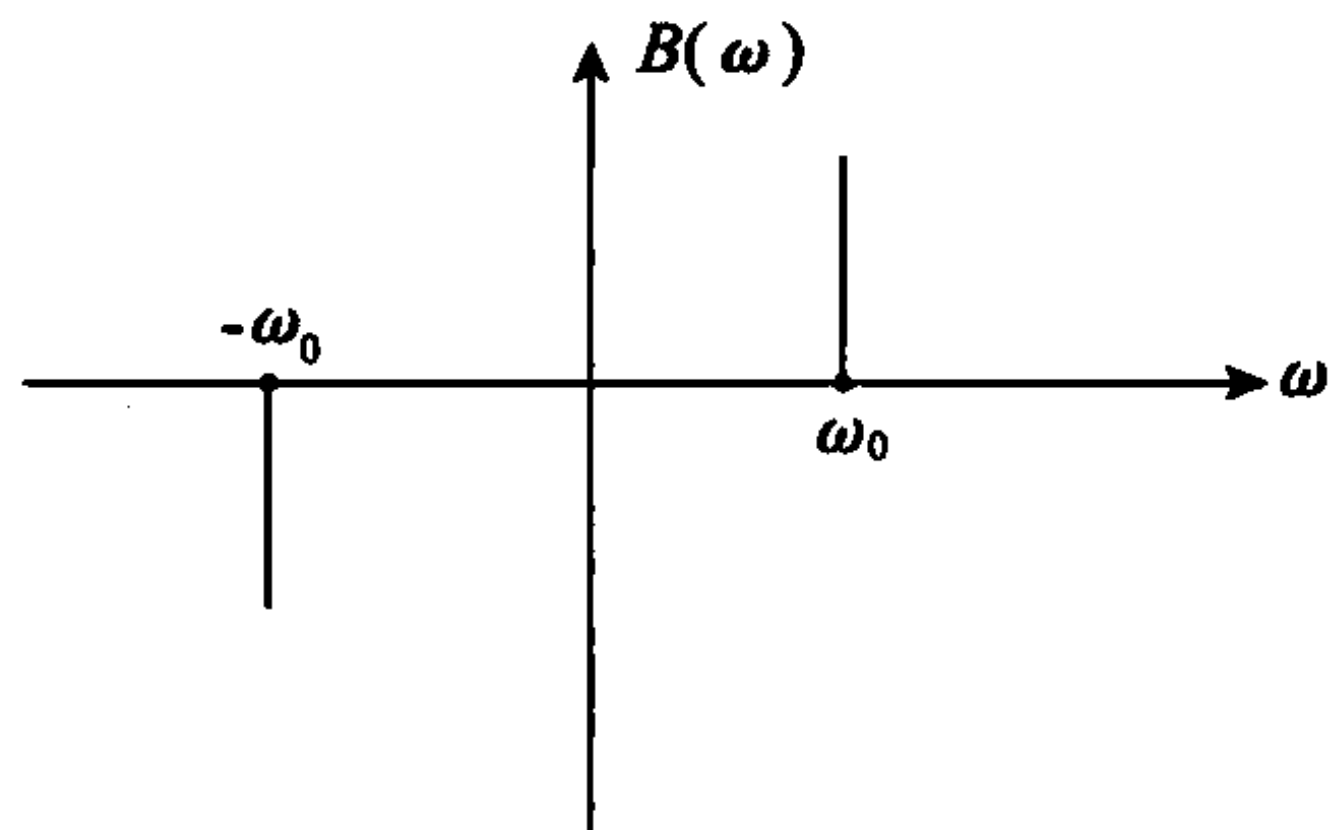


图 5-3-1

第六章 拉普拉斯变换

本章知识要点

拉普拉斯变换

(一) 拉普拉斯变换的定义

$$\bar{f}(p) = \int_0^{\infty} f(t)e^{-pt} dt \text{ (拉普拉斯积分)}$$

$\bar{f}(p)$ 为拉普拉斯变换函数, $f(t)$ 到 $\bar{f}(p)$ 的变换为拉普拉斯变换.

$\bar{f}(p)$ 为像函数, $f(t)$ 为原函数.

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \bar{f}(p)e^{pt} dp$$

$$\bar{f}(p) = \mathcal{L}[f(t)], \quad \bar{f}(p) \neq f(t)$$

$$f(t) = \mathcal{L}^{-1}[\bar{f}(p)], \quad f(t) \neq \bar{f}(p)$$

(二) 拉普拉斯变换的基本性质

(1) 线性定理

若

$$f_1(t) \rightleftharpoons \bar{f}_1(p), f_2(t) \rightleftharpoons \bar{f}_2(p)$$

则

$$c_1 f_1(t) + c_2 f_2(t) \rightleftharpoons c_1 \bar{f}_1(p) + c_2 \bar{f}_2(p)$$

(2) 导数定理

$$f'(t) \rightleftharpoons p\bar{f}(p) - f(0)$$

高阶导致 $f^n(t) \rightleftharpoons p^n \bar{f}(p) - p^{n-1} f(0) - p^{n-2} f'(0) \cdots - p f^{(n-2)}(p) - f^{(n-1)}(0)$

(3) 积分定理

$$\int_0^t \psi(\tau) d\tau \rightleftharpoons \frac{1}{p} \mathcal{L}[\psi(t)]$$

(4) 相似性定理

$$f(at) \rightleftharpoons \frac{1}{a} \bar{f}(p/a)$$

(5) 位移定理

$$e^{-\lambda t} f(t) \rightleftharpoons \bar{f}(p + \lambda)$$

(6) 延迟定理

$$f(t - t_0) \rightleftharpoons e^{-pt_0} \bar{f}(p)$$

(7) 卷积定理 若

$$f_1(t) \rightleftharpoons \bar{f}_1(p), f_2(t) \rightleftharpoons \bar{f}_2(p)$$

则 $f_1(t) * f_2(t) = \bar{f}_1(p) \bar{f}_2(p)$

其中 $f_1(t) * f_2(t) \equiv \int_0^t f_1(\tau) f_2(t-\tau) d\tau$ 称为 $f_1(t)$ 与 $f_2(t)$ 的卷积.

拉普拉斯变换的反演

由像函数求原函数的变换称为拉普拉斯变换的反演.

(一) 有理分式反演法

将有理分式的像函数分解成分项分式, 然后利用拉普拉斯变换的基本公式得到相应的原函数.

(二) 查表法

将拉普拉斯变换制成表(见教材附录二), 相应的拉普拉斯变换可直接查表得到. 有些像函数可以利用延迟定理、位移定理和卷积定理等拉普拉斯变换的基本性质, 再结合查表而解决其反演问题.

习题全解

§ 6.1 拉普拉斯变换

求下列函数的拉普拉斯变换函数.

(1) $\text{sh } \omega t, \quad \text{ch } \omega t$

解: (a) $f(t) = \text{sh } \omega t = \frac{e^{\omega t} - e^{-\omega t}}{2}$

$$\begin{aligned} \bar{f}(p) &= \int_0^\infty f(t) e^{-pt} dt = \int_0^\infty \text{sh } \omega t e^{-pt} dt \\ &= \frac{1}{2} \int_0^\infty (e^{\omega t} - e^{-\omega t}) e^{-pt} dt = \frac{1}{2} \int_0^\infty (e^{(\omega-p)t} - e^{-(\omega+p)t}) dt \\ &= \frac{1}{2} \left(\frac{-1}{\omega-p} - \frac{1}{\omega+p} \right) = \frac{1}{2} \frac{(-2\omega)}{\omega^2 - p^2} = \frac{\omega}{p^2 - \omega^2} \quad (\text{Re } p > \text{Re } \omega) \end{aligned}$$

(b) $f(t) = \text{ch } \omega t = \frac{e^{\omega t} + e^{-\omega t}}{2}$

$$\begin{aligned} \bar{f}(p) &= \int_0^\infty f(t) e^{-pt} dt = \int_0^\infty \text{ch } \omega t e^{-pt} dt \\ &= \frac{1}{2} \int_0^\infty (e^{\omega t} + e^{-\omega t}) e^{-pt} dt = \frac{1}{2} \int_0^\infty [e^{(\omega-p)t} + e^{-(\omega+p)t}] dt \\ &= \frac{1}{2} \left(\frac{1}{\omega-p} \int_0^\infty de^{(\omega-p)t} - \frac{1}{\omega+p} \int_0^\infty de^{-(\omega+p)t} \right) \\ &= \frac{1}{2} \left(\frac{-1}{\omega-p} + \frac{1}{\omega+p} \right) = \frac{-p}{\omega^2 - p^2} = \frac{p}{p^2 - \omega^2} \quad (\text{Re } p > \text{Re } \omega) \end{aligned}$$

$$(2) e^{-\lambda} \sin \omega t, \quad e^{-\lambda} \cos \omega t$$

$$\text{解: (a) } f(t) = e^{-\lambda} \sin \omega t = e^{-\lambda} \left[\frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \right]$$

$$\text{又} \quad \mathcal{L}[e^{st}] = \frac{1}{p-s}, \quad (\operatorname{Re} p > \operatorname{Re} s)$$

$$\begin{aligned} \text{所以 } \mathcal{L}[f(t)] &= \mathcal{L}\left\{e^{-\lambda} \left[\frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \right]\right\} = \frac{1}{2i} i\omega \mathcal{L}[e^{-(\lambda+i\omega)t}] - \frac{1}{2i} \mathcal{L}[e^{-(\lambda-i\omega)t}] \\ &= \frac{1}{2i} \left(\frac{1}{p+\lambda-i\omega} - \frac{1}{p+\lambda+i\omega} \right) = \frac{1}{2i} \left(\frac{2i\omega}{(p+\lambda)^2 + \omega^2} \right) \\ &= \frac{\omega}{(p+\lambda)^2 + \omega^2} \end{aligned}$$

$$(b) f(t) = e^{-\lambda} \cos \omega t = e^{-\lambda} \left[\frac{1}{2} (e^{i\omega t} + e^{-i\omega t}) \right]$$

$$\text{又} \quad \mathcal{L}[e^{st}] = \frac{1}{p-s}, \quad (\operatorname{Re} p > \operatorname{Re} s)$$

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[e^{-\lambda} (e^{i\omega t} + e^{-i\omega t})/2] = \frac{1}{2} \left(\frac{1}{p+\lambda-i\omega} + \frac{1}{p+\lambda+i\omega} \right) \\ &= \frac{p+\lambda}{(p+\lambda)^2 + \omega^2} \end{aligned}$$

$$(3) \frac{1}{\sqrt{\pi t}}$$

$$\text{解: } f(t) = \frac{1}{\sqrt{\pi t}}$$

$$\bar{f}(p) = \int_0^\infty f(t) e^{-pt} dt = \int_0^\infty \frac{1}{\sqrt{\pi t}} e^{-pt} dt = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{e^{-pt}}{\sqrt{t}} dt$$

取 $t = x^2$ 则

$$\bar{f}(p) = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{e^{-px^2}}{x} 2x dx = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-px^2} dx = \frac{2}{\sqrt{\pi} \sqrt{p}} \int_0^\infty e^{-u^2} d\sqrt{p}x = \frac{2}{\pi p} \int_0^\infty e^{-u^2} du$$

由于

$$\int_0^\infty e^{-u^2} du = \frac{\pi}{2}$$

所以

$$\bar{f}(p) = \frac{2}{\pi p} \frac{\sqrt{\pi}}{2} = \frac{1}{\sqrt{p}}.$$

$$(4) \delta(t-\tau)$$

$$\text{解: } f(t) = \delta(t-\tau)$$

$$\bar{f}(p) = \int_0^\infty f(t) e^{-pt} dt = \int_0^\infty \delta(t-\tau) e^{-pt} dt = \int_{\tau-\epsilon}^{\tau+\epsilon} \delta(t-\tau) e^{-pt} dt = e^{-p\tau}$$

所以

$$\bar{f}(p) = e^{-p\tau}.$$

§ 6.2 拉普拉斯变换的反演

1. 将下列像函数反演.

$$(1) \bar{y}(p) = \frac{6}{(p+1)^4}$$

解: 由 $\mathcal{L}[t^n e^s] = \frac{n!}{(p-s)^{n+1}}$

得 $\bar{y}(p) = \frac{6}{(p+1)^4} = \frac{3!}{(p+1)^{3+1}}, n=3, s=-1$

即 $\frac{6}{(p+1)^4} = \frac{3!}{(p+1)^{3+1}} \rightleftharpoons t^3 e^{-t}$

所以 $y(t) = t^3 e^{-t}$

$$(2) \bar{y}(p) = \frac{3p}{p^2-1}$$

解: $\frac{3p}{p^2-1} = \frac{3}{2} \left(\frac{1}{p-1} + \frac{1}{p+1} \right)$

又 $\mathcal{L}[e^t] = \frac{1}{p-1}, \mathcal{L}[e^{-t}] = \frac{1}{p+1}$

$$y(p) = \frac{3p}{p^2-1} = \frac{3}{2} \left(\frac{1}{p-1} + \frac{1}{p+1} \right) \rightleftharpoons \frac{3}{2} (e^t + e^{-t}) = 3 \operatorname{ch} t$$

所以 $y(t) = 3 \operatorname{ch} t$

$$(3) \bar{y}(p) = \frac{1}{p-2}, \bar{z}(p) = \frac{3}{p-2}$$

解: (a) $\bar{y}(p) = \frac{1}{p-2}$, 由于 $\frac{1}{p-2} \rightleftharpoons e^{2t}$ 所以 $\bar{y}(p) = \frac{1}{p-2} \rightleftharpoons e^{2t}$ 得: $y(t) = e^{2t}$

(b) $\bar{z}(p) = \frac{3}{p-2}$, 由于 $\frac{1}{p-2} \rightleftharpoons e^{2t}$ 所以 $\bar{z}(p) = \frac{3}{p-2} \rightleftharpoons 3e^{2t}$ 得: $z(t) = 3e^{2t}$

$$(4) \bar{y}(p) = \frac{2}{(p-1)^5}$$

解: $\mathcal{L}(t^n \cdot e^s) = \frac{n!}{(p-s)^{n+1}}$, (其中: $n=4, s=1, n!=4!=24$)

所以 $\bar{y}(p) = \frac{2}{(p-1)^5} = \frac{24}{12(p-1)^5} = \frac{24}{12} (p-1)^{-5}$

$$= \frac{4!}{12(p-1)^{4+1}} \rightleftharpoons t^4 e^{-t}/12 = 2t^4 e^{-t}/4!$$

2. 求 $\bar{j}(p) = \frac{E}{LP^2 + Rp + \frac{1}{C}}$ 的原函数.

解: $LP^2 + Rp + 1/C = L(p^2 + \frac{R}{L}p + \frac{1}{LC})$

$$= L \left(p + \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} \right) \left(p + \frac{R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} \right)$$

(a) 当 $R^2/4L^2 - 1/LC = 0$ 时, 则 $\bar{j}(p) = \frac{E}{L(p + \frac{R}{2L})} \rightleftharpoons \frac{E}{L} t e^{-\frac{R}{2L}t}$, $\therefore j(t) = \frac{E}{L} e^{-\frac{R}{2L}t}$

其中 $\mathcal{L}[t^n e^{st}] = \frac{n!}{(p-s)^{n+1}}$, $(s = -\frac{R}{2L}, n=1)$

(b) 当 $R^2/4L^2 - \frac{1}{LC} > 0$ 时

则
$$\bar{j} = \frac{E/L}{\left(\frac{p+R}{2L}\right)^2 - \left(\frac{R^2}{4L^2} - \frac{1}{LC}\right)}$$

将函数中 $p + \frac{R}{2L}$ 位移为 p , 且分子和分母同乘以 $\omega = \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$

得 $\bar{j}(p) = \left[\frac{E/L}{\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}} \right] \left(\frac{\omega}{p^2 - \omega^2} \right)$, (其中 $\omega = \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$)

从教材附录二拉普拉斯变换函数表查得第 7 项 $\frac{\omega}{p^2 - \omega^2} \rightleftharpoons \text{sh } \omega t$, 再应用位移定理

$$\bar{f}(p+\lambda) = e^{-\lambda t} f(t)$$

所以
$$\bar{j}(p) \rightleftharpoons \frac{E/L}{\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}} e^{-\frac{R}{2L}t} \text{sh} \left(\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} t \right)$$

所以
$$\bar{j}(p) = \frac{2E}{\sqrt{R^2 - 4L/C}} e^{-\frac{R}{2L}t} \text{sh} \frac{\sqrt{R^2 - 4L/C}}{2L} t$$

(c) 当 $R^2/4L^2 - \frac{1}{LC} < 0$ 时

则
$$\bar{j}(p) = \frac{E/L}{\left(\frac{p+R}{2L}\right)^2 - \left(\frac{R^2}{4L^2} - \frac{1}{LC}\right)} = \frac{E/l}{\left(p + \frac{R}{2L}\right)^2 + \left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

将函数中 $p + \frac{R}{2L}$ 位移为 p , 且分子和分母同乘 $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$

得
$$\bar{j}(p) = \frac{E/L}{\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} \cdot \frac{\omega}{p^2 + \omega^2}$$
, (其中 $\omega = \sqrt{\frac{1}{2C} - \frac{R^2}{4L^2}}$)

从教材附录二拉普拉斯变换函数表查得第 5 项 $\frac{\omega}{p^2 + \omega^2} \rightleftharpoons \sin \omega t$

再应用位移定理

$$\bar{f}(p+\lambda) = e^{-\lambda t} f(t)$$

所以
$$\bar{j}(p) \rightleftharpoons \frac{E/L}{\sqrt{-\frac{R^2}{4L^2} + \frac{1}{LC}}} \cdot e^{-\frac{R}{2L}t} \sin \left(\frac{1}{LC} - \frac{R^2}{4L^2} \right) t$$

$$= \frac{2E}{\sqrt{4L/C - R^2}} e^{-\frac{R}{2L}t} \sin\left(\frac{\sqrt{4L/C - R^2}}{2L}t\right)$$

所以
$$j(t) = \frac{2E}{\sqrt{4L/C - R^2}} e^{-\frac{R}{2L}t} \sin\left(\frac{\sqrt{4L/C - R^2}}{2L}t\right)$$

3. 求 $\bar{N}_4(p) = \frac{N_0 C_1 C_2 C_3}{p(p+C_1)(p+C_2)(p+C_3)}$ 的原函数.

解: 设 $\bar{N}_4 = \frac{N_0 C_1 C_2 C_3}{p(p+C_1)(p+C_2)(p+C_3)} = \frac{A_0}{p} + \frac{A_1}{p+C_1} + \frac{A_2}{p+C_2} + \frac{A_3}{p+C_3}$ (1)

则 $\frac{A_0}{p}$ 的原函数为 $\frac{A_0}{p} \Rightarrow A_0$

$\frac{A_1}{p+C_1}$ 的原函数为 $\frac{A_1}{p+C_1} \Rightarrow A_1 e^{-C_1 t}$

$\frac{A_2}{p+C_2}$ 的原函数为 $\frac{A_2}{p+C_2} \Rightarrow A_2 e^{-C_2 t}$

$\frac{A_3}{p+C_3}$ 的原函数为 $\frac{A_3}{p+C_3} \Rightarrow A_3 e^{-C_3 t}$

求解方程(1) 可得到 A_0, A_1, A_2, A_3 分别为

$$\begin{cases} A_0 = N_0 \\ A_1 = \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)} \\ A_2 = \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)} \\ A_3 = \frac{C_1 C_2 N_0}{(C_3 - C_1)(C_2 - C_3)} \end{cases}$$

由于 $\bar{N}_4 = \frac{A_0}{p} + \frac{A_1}{p+C_1} + \frac{A_2}{p+C_2} + \frac{A_3}{p+C_3}$

得 $N_4 = A_0 + A_1 e^{-C_1 t} + A_2 e^{-C_2 t} + A_3 e^{-C_3 t}$

将 A_1, A_2, A_3, A_4 分别代入, 有

$$N_4 = A_0 + A_1 e^{-C_1 t} + A_2 e^{-C_2 t} + A_3 e^{-C_3 t}$$

$$= N_0 + \frac{C_2 C_3 N_0 e^{-C_1 t}}{(C_1 - C_2)(C_3 - C_1)} + \frac{C_1 C_3 N_0 e^{-C_2 t}}{(C_1 - C_2)(C_2 - C_3)} + \frac{C_1 C_2 N_0 e^{-C_3 t}}{(C_3 - C_1)(C_2 - C_3)}$$

4. 求 $\bar{y}(p) = \lambda \mu \frac{p}{(p+C)^4}$ 的原函数.

解: $\bar{y}(p) = \lambda \mu \frac{p}{(p+C)^4} = \lambda \mu \left[\frac{p+C}{(p+C)^4} - \frac{C}{(p+C)^4} \right]$

设 $\bar{y}_1(p) = \frac{p+C}{(p+C)^4} = \frac{1}{(p+C)^3}, \bar{y}_2(p) = \frac{-C}{(p+C)^4}, \bar{y}(p) = \bar{y}_1(p) + \bar{y}_2(p)$

由 $\mathcal{L}[te^x] = \frac{n!}{(p-s)^{n+1}}$

所以 $y_1(t) = \frac{1}{2} t^2 e^{-\alpha}, y_2(t) = -\frac{C}{6} t^3 e^{-\alpha}$

$$y(t) = \lambda \mu (y_1 - y_2) = \lambda \mu \left(\frac{1}{2} t^2 e^{-\alpha} - \frac{C}{6} t^3 e^{-\alpha} \right) = \frac{1}{2} \lambda \mu t^2 \left(1 - \frac{C}{3} t \right) e^{-\alpha}$$

所以

$$y(t) = \frac{1}{2} \lambda \mu t^2 \left(1 - \frac{C}{3} t\right) e^{-\alpha}$$

5. 求 $\bar{j}(p) = \frac{E_0 \omega}{(R + 1/pc)(p^2 + \omega^2)}$ 的原函数.

$$\begin{aligned} \text{解: 设 } \bar{j}(p) &= \frac{E_0 \omega}{(R + 1/pc)(p^2 + \omega^2)} = \frac{E_0 \omega p}{R(p + 1/Rc)(p^2 + \omega^2)} \\ &= \frac{A_1 p}{p^2 + \omega^2} + \frac{A_2}{p^2 + \omega^2} + \frac{A_3 p}{p + 1/Rc} + \frac{A_4}{p + 1/Rc} \end{aligned}$$

求出 A_1, A_2, A_3, A_4

$$\begin{cases} A_1 = \frac{E_0}{R^2 \omega c + \frac{1}{c\omega}} & \frac{A_1 p}{p^2 + \omega^2} \text{ 的原函数为 } \frac{Ap}{p^2 + \omega^2} = A \cos \omega t \\ A_2 = \frac{E_0}{R} \left[\frac{\omega}{1 + \frac{1}{R^2 c^2 \omega^2}} \right] & \frac{A_2}{p^2 + \omega^2} \text{ 的原函数为 } \frac{B\omega}{(p^2 + \omega^2)} \frac{1}{\omega} = \frac{B}{\omega} \sin \omega t \\ A_3 = 0 \\ A_4 = -\frac{E_0}{R^2 \omega c + \frac{1}{c\omega}} & \frac{A_4}{p + 1/Rc} \text{ 的原函数为 } \frac{D}{p + 1/Rc} = D e^{-\frac{t}{Rc}} \end{cases}$$

$$\text{由于 } \bar{j}(p) = \frac{A_1 p}{p^2 + \omega^2} + \frac{A_2}{p^2 + \omega^2} + \frac{A_3 p}{p + 1/Rc} + \frac{A_4}{p + 1/Rc} \quad (\text{其中 } A_3 = 0)$$

$$\text{所以 } j(p) = A_1 \cos \omega t + \frac{A_2}{\omega} \sin \omega t + A_4 e^{-\frac{t}{Rc}}$$

将 A_1, A_2, A_4 代入

$$j = \frac{E_0 \cos \omega t}{R^2 \omega c + \frac{1}{c\omega}} + \frac{E_0 \sin \omega t}{R \left(\frac{1 + 1}{R^2 c^2 \omega} \right)} - \frac{E_0}{R^2 \omega c + \frac{1}{c\omega}} e^{-\frac{t}{Rc}}$$

得

$$= \frac{E_0}{R^2 + 1/c^2 \omega^2} \left[\left(R \sin \omega t + \frac{1}{c\omega} \cos \omega t \right) - \frac{1}{c\omega} e^{-\frac{t}{Rc}} \right]$$

6. 求 $\bar{T}(p) = A \frac{\omega}{p^2 + \omega^2} \frac{1}{p^2 + \pi^2 a^2 / l^2}$ 的原函数.

$$\text{解: 设 } \bar{T} = A \frac{\omega}{p^2 + \omega^2} \frac{1}{p^2 + \pi^2 a^2 / l^2} = \frac{A_1}{p^2 + \omega^2} + \frac{A_2}{p^2 + \pi^2 a^2 / l^2}$$

$$\text{则 } \frac{A_1}{p^2 + \omega^2} \text{ 的原函数为 } \frac{A_1}{p^2 + \omega^2} = \frac{A_1}{\omega^2} \sin \omega t$$

$$\frac{A_2}{p^2 + \pi^2 a^2 / l^2} \text{ 的原函数为 } \frac{A_2}{p^2 + \pi^2 a^2 / l^2} = \frac{A_2}{\pi a / l} \sin \frac{\pi a}{l} t$$

$$\text{所以 } T = \frac{A_1}{\omega} \sin \omega t + \frac{A_2}{\omega} \sin \frac{\pi a}{l} t$$

$$\text{由(1)解得 } A_1, A_2 \begin{cases} \text{展开(1)有 } A_1(p^2 + \pi^2 a^2/l^2) + A_2(p^2 + \omega^2) = A\omega \\ \text{得 } A_1 = -A_2, A_2 = \frac{A\omega}{\pi^2 a^2/l^2 - \omega^2} \\ A_1 = \frac{A\omega}{\pi^2 a^2/l^2 - \omega^2} \\ A_2 = \frac{-A\omega}{\pi^2 a^2/l^2 - \omega^2} \end{cases}$$

$$\begin{aligned} \text{代入 } T &= \frac{A_1}{\omega} \sin \omega t + \frac{A_2}{\pi a/l} \sin \frac{\pi a}{l} t = \frac{A \sin \omega t}{\pi^2 a^2/l^2 - \omega^2} - \frac{A\omega t/\pi a}{\pi^2 a^2/l^2 - \omega^2} \sin \frac{\pi a}{l} t \\ &= \frac{A}{\pi^2 a^2/l^2 - \omega^2} \left(\sin \omega t - \frac{\omega t}{\pi a} \sin \frac{\pi a}{l} t \right) = \frac{Al}{\pi a} \cdot \frac{1}{\omega^2 - \pi^2 a^2/l^2} (\omega \sin \frac{\pi a}{l} t - \frac{\pi a}{l} \sin \omega t) \end{aligned}$$

7. 求 $\bar{T}(p) = \frac{1}{p^2 + \omega^2 a^2} \bar{g}(p)$ 的原函数, $\bar{g}(p)$ 的某个已知的 $g(t)$ 的像函数.

$$\text{解: 设 } \bar{T}(p) = \frac{1}{p^2 + \omega^2 a^2} \bar{g}(p) = \bar{f}(p) \bar{g}(p)$$

$$\text{其中 } \bar{f}(p) = \frac{1}{p^2 + \omega^2 a^2}, \quad f(t) = \frac{1}{\omega a} \sin \omega a t$$

$$\begin{aligned} \text{由卷积定理 } \bar{f}(p) \rightleftharpoons f(t) &= \frac{1}{\omega a} \sin \omega t = \frac{1}{\omega a} \cdot \frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \\ \bar{g}(p) &\rightleftharpoons g(t) \end{aligned}$$

$$\begin{aligned} \text{所以 } \bar{T}(p) &= \bar{f}(p) \bar{g}(p) \rightleftharpoons \int_0^t g(\tau) f(t-\tau) d\tau \\ &= \frac{1}{\omega a 2i} \int_0^t g(\tau) [e^{i\omega(t-\tau)} - e^{-i\omega(t-\tau)}] d\tau = T(t) \end{aligned}$$

8. 求 $\bar{T}(p) = \frac{1}{p + \omega^2 a^2} \bar{g}(p)$ 的原函数, $\bar{g}(p)$ 的某个已知的 $g(t)$ 的像函数.

$$\text{解: 设 } \bar{T}(p) = \frac{1}{p + \omega^2 a^2} \bar{g}(p) = \bar{f}(p) \bar{g}(p)$$

$$\text{其中 } \bar{f}(p) = \frac{1}{p + \omega^2 a^2}, \quad f(t) = e^{-\omega^2 a^2 t}$$

$$\text{由卷积定理 } \bar{f}(p) \rightleftharpoons f(t) = e^{-\omega^2 a^2 t}, \quad \bar{g}(p) \rightleftharpoons g(t)$$

$$T(t) \rightleftharpoons \bar{T}(p) = \bar{f}(p) \cdot \bar{g}(p) \rightleftharpoons \int_0^t g(\tau) e^{-\omega^2 a^2 (t-\tau)} d\tau$$

9. 已知像函数 $\bar{y}(p) = e^{-p^2/4} p^{-(\lambda/2+1)} \int e^{p^2/4} p^{(\lambda/2+1)} (c_1 + \frac{c_2}{p}) dp$, 其中 C_1 和 C_2 是两个任意常数. 问 λ 应取怎样的数值才有可能选定 C_1 和 C_2 使原函数 $y(t)$ 为多项式?

解: 若 $\bar{y}(p)$ 的原函数 $y(t)$ 为多项式, 则 $\bar{y}(p)$ 为 $\frac{n!}{p^{n+1}}$ 的形式

$$\begin{aligned} \text{又 } \bar{y}(p) &= e^{-p^2/4} p^{-(\lambda/2+1)} \int e^{p^2/4} p^{(\lambda/2+1)} \left(c_1 + \frac{c_2}{p} \right) dp \\ &= e^{-p^2/4} p^{-(\lambda/2+1)} \left[\int c_1 e^{p^2/4} p^{(\lambda/2+1)} dp + \int c_2 e^{p^2/4} p^{\lambda/2} dp \right] \end{aligned} \quad (1)$$

$$\text{求} \quad \int e^{p^{1/4}} p^k dp = \int 2p^{k-1} de^{p^{1/4}} = 2e^{p^{1/4}} p^{k-1} - 2 \int e^{p^{1/4}} (k-1) p^{k-2} dp$$

多次分部积分后,可得到关于 $1/p$ 的多项式.

$$\text{对应(1) 取} \begin{cases} k_1 = \lambda/2 + 1 \\ k_2 = \lambda/2 \end{cases}, \text{则当 } k \text{ 为整数时上式成立,即} \begin{cases} \lambda = 2(k_1 - 1) \\ \lambda = 2k_2 \end{cases}$$

当 λ 为偶数时,即可使原函数为多项式,若 $\lambda \neq$ 偶数时(1) 式不能表示为 $\frac{1}{p}$ 的形式,故原函数不为多项式.

10. 已知 $\bar{y}(p) = \frac{(p-1)^\lambda}{p^{\lambda+1}}$. 问 λ 应取怎样的数值,原函数才是多项式?

解: λ 为正整数时, $\bar{y}(p) = \frac{(p-1)^\lambda}{p^{\lambda+1}}$ 可以展开成

$$\bar{y}(p) = \frac{1}{p} - \frac{\lambda^2}{p^2} + \frac{\lambda(\lambda-1)}{2!} - \frac{1}{p^3} + \dots + (-1)^\lambda \frac{\lambda(\lambda-1)\dots(\lambda-k+1)}{k!} p^{(-1-k)} + \frac{(-1)^\lambda}{p^{\lambda+1}}$$

$$\text{而 } \frac{1}{p^{n+1}} = \frac{1}{n!} t^n \quad (n \text{ 为整数})$$

因此,当 λ 为正整数时, $\bar{y}(p)$ 的原函数是多项式

$$y(t) = 1 - \lambda^2 t + \frac{\lambda(\lambda-1)}{(2!)^2} t^2 \dots + \frac{(-1)^\lambda \lambda(\lambda-1)\dots(\lambda-k+1)}{(k!)^2} t^k + \frac{(-1)^\lambda}{\lambda!} t^\lambda \quad (\lambda \text{ 为正整数})$$

此原函数即为多项式.

11. 已知 $\bar{X}(p) = F_0 \frac{\omega}{p^2 + \omega^2} \frac{mp^2 + k}{D(p)}$, 其中 $D(p) = (Mp^2 + Rp + K + k)(mp^2 + k) - k^2$, 而 $F_0, \omega, m, k, K, M, R$ 都是正的常数. 试论证 $D(p)$ 没有正的根, 也没有纯虚数根. 在什么条件下, 原函数 $X(t)$ 不含有稳定振荡的部分而只含有指数式衰减的部分或衰减振荡的部分?

解: ① 由于 F_0, ω, m, k, K, M 都是正的常数.

则 $D(p) = (mp^2 + Rp + K + k)(mp^2 + k) - k^2$

$$= Mmp^4 + Rmp^3 + (kM + Km + km)p^2 + Rkp + kK$$

若 $D(p)$ 有正根, 则 $D(p) = 0$ 成立, 此时的 p 即为其根, 由原题条件, $F_0, \omega, m, k, K, M, R$ 都是正常数, 即当 $p > 0$ 时, $D(p) > 0$, 故 $D(p)$ 没有正根.

② 若 $D(p)$ 有纯虚的根时, $p = iy$, 代入

$$D(p) = Mmy^4 + Rmy^3(-i) + y^2(kM + Km + km)(-1) + Rky(+i) + kK.$$

则 $D(p) = 0$, 即 $\operatorname{Re} D(p) = 0$ 且 $\operatorname{Im} D(p) = 0$

$$\text{有} \begin{cases} Mmy^4 - y^2(kM + Km + km) + kK = 0 \\ -Rmy^4 + Rky = 0 \end{cases}$$

结果 $my^2 - k = 0$, $y = \sqrt{k/m}$, $k = 0$, 与条件不一致, 故没有纯虚数根.

③ 虽然 $D(p)$ 没有纯虚数根, 但 $(p^2 + \omega^2)D(p)$ 有虚数根 $p = \pm \omega i$

$$\bar{X}(p) = F_0 \frac{\omega}{p^2 + \omega^2} \frac{mp^2 + k}{D(p)}$$

若要原函数 $x(t)$ 中不含有稳定振荡的部分, 则需要约去 $p^2 + \omega^2$ 与 $mp^2 + k$

即
$$p^2 + \omega^2 = p^2 + \frac{k}{m}$$

有 $\omega^2 = \frac{k}{m}$ 或 $\omega^2 = C \frac{k}{m} (C \neq 0)$

C 为任意常数,
$$\bar{X}(p) = \frac{F_0(mp^2 + k)}{D(p)}$$

所以当 $\omega^2 = \frac{k}{m}$ (或 $\omega^2 = C \frac{k}{m}$) 时, $x(t)$ 不含有稳定振荡的部分而只含有指数衰减的部分或衰减振荡部分.

$$\bar{X}(p) = F_0 / \left[(M^2 p + Rp + K + k) - \frac{k^2}{mp^2 + k} \right]$$

12. 求下列像函数的原函数.

(1) $\bar{I}(p) = \frac{\pi}{2a} \frac{1}{p+a}$

解: 可由拉普拉斯变换表查表得到.

$\bar{I}(p) = \frac{\pi}{2a} \frac{1}{p+a}$ (查教材附录二, P_{394} , 第 4 项得)

$$I(t) = \frac{\pi}{2a} e^{-at}$$

(2) $\bar{I}(p) = \frac{\pi}{2p}$

解: $\bar{I}(p) = \frac{\pi}{2p}$ (查教材附录二, P_{394} , 第 1 项得)

$$I(t) = \frac{\pi}{2}$$

(3) $\bar{I}(p) = \frac{\pi}{2} \frac{1}{p(p+1)}$

解: $\bar{I}(p) = \frac{\pi}{2} \frac{1}{p(p+1)} = \frac{\pi}{2} \left(\frac{1}{p} - \frac{1}{p+1} \right)$ (查教材附录二, P_{394} , 第 1, 4 项得)

$$I(t) = \frac{\pi}{2} (1 - e^{-t})$$

(4) $\bar{I}(p) = \frac{\pi}{2p^2}$

解: $\bar{I}(p) = \frac{\pi}{2p^2} = \frac{\pi}{2} t$ (查教材附录二, P_{394} , 第 2 项得)

$$I(t) = \frac{\pi}{2} t$$

(注: 教材附录二为拉普拉斯变换函数表)

§ 6.3 应用例

1. 求解下列常微分方程.

(1) $\frac{d^3 y}{dt^3} + 3 \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + y = 6e^{-t}, y(0) = \frac{dy}{dt} \Big|_{t=0} = \frac{d^2 y}{dt^2} \Big|_{t=0} = 0.$

解:对方程进行拉普拉斯变换,有

$$p^3 \bar{y}(p) + 3p^2 \bar{y}(p) + 3p \bar{y}(p) + \bar{y}(p) = 6 \frac{1}{p+1}$$

其中

$$6e^{-t} \doteq 6 \frac{1}{p+1}$$

$$(p+1)^3 \bar{y}(p) = \frac{6}{p+1} \Rightarrow (p+1)^4 \bar{y}(p) = 6$$

$$\bar{y}(p) = \frac{6}{(p+1)^4}$$

由拉普拉斯反演,再求 $\bar{y}(p)$ 的原函数 $y(t)$, 由于 $\bar{y}(p) = \frac{6}{(p+1)^4}$

且

$$\mathcal{L}[t^n e^s] = \frac{n!}{(p-s)^{n+1}}, \quad (\text{取 } n=3, s=-1.)$$

有

$$\frac{6}{(p+1)^4} = \frac{3!}{(p+1)^{3+1}} \doteq t^3 e^{-t}$$

常微分方程的解为 $y(t) = t^3 e^{-t}$, 满足 $y(0) = \frac{dy}{dt} \Big|_{t=0} = \frac{d^2 y}{dt^2} \Big|_{t=0} = 0$.

$$(2) \frac{d^2 y}{dt^2} + 9y = 30 \operatorname{ch} t, y(0) = 3, \frac{dy}{dt} \Big|_{t=0} = 0.$$

解:对方程进行拉普拉斯变换,有

$$p^2 \bar{y}(p) - 3p + 9 \bar{y}(p) = 30 \frac{p}{p^2-1}, (\text{其中 } \operatorname{ch} t \doteq \frac{p}{p^2-1})$$

$$(p^2 + 9) \bar{y}(p) = \frac{30p + 3p(p^2 - 1)}{p^2 - 1} = \frac{3p^3 + 27p}{p^2 - 1} = \frac{3p(p^2 + 9)}{p^2 - 1}$$

有

$$\bar{y}(p) = \frac{3p}{p^2-1}$$

由拉普拉斯反演,求 $\bar{y}(p)$ 的原函数 $y(t)$, 由于

$$\bar{y}(p) = \frac{3p}{p^2-1} = \frac{3}{2} \left(\frac{1}{p+1} + \frac{1}{p-1} \right)$$

$$\doteq \frac{3}{2} (e^{-t} + e^t) = 3 \operatorname{ch} t$$

其中

$$\frac{1}{p+1} \doteq e^{-t}, \frac{1}{p-1} \doteq e^t$$

所以常微分方程的解为 $y(t) = 3 \operatorname{ch} t$. 且满足 $y(0) = 3, \frac{dy}{dt} \Big|_{t=0} = 0$.

$$(3) \begin{cases} \frac{dy}{dt} + 2y + 2z = 10e^{2t}, \\ \frac{dz}{dt} - 2y + z = 7e^{2t}. \end{cases} \quad \begin{cases} y(0) = 1, \\ z(0) = 3. \end{cases}$$

解:对方程进行拉普拉斯变换后,有

$$\begin{cases} p\bar{y}(p) - 1 + 2\bar{y}(p) + 2\bar{z}(p) = \frac{10}{p-2} \\ p\bar{z}(p) - 3 - 2\bar{y}(p) + \bar{z}(p) = \frac{7}{p-2} \end{cases} \quad \text{其中} \begin{cases} 10e^{2t} \doteq \frac{10}{p-2} \\ 7e^{2t} \doteq \frac{7}{p-2} \end{cases}$$

$$\begin{cases} p\bar{y}(p) + 2\bar{y}(p) + 2\bar{z}(p) = \frac{10}{p-2} + 1 = \frac{p+8}{p-2} \\ p\bar{z}(p) + \bar{z}(p) - 2\bar{y}(p) = \frac{7}{p-2} + 3 = \frac{3p+1}{p-2} \end{cases}$$

$$\begin{cases} (p+2)\bar{y}(p) + 2\bar{z}(p) = \frac{p+8}{p-2} \\ -2\bar{y}(p) + (p+1)\bar{z}(p) = \frac{3p+1}{p-2} \end{cases} \quad (1)$$

$$\begin{cases} (p+2)\bar{y}(p) + 2\bar{z}(p) = \frac{p+8}{p-2} \\ -2\bar{y}(p) + (p+1)\bar{z}(p) = \frac{3p+1}{p-2} \end{cases} \quad (2)$$

解方程,得

$$(p+2)(p+1)\bar{y}(p) + 2(p+1)\bar{z}(p) = \frac{(p+8)(p+1)}{p-2}$$

$$-4\bar{y}(p) + 2(p+1)\bar{z}(p) = \frac{2(3p+1)}{p-2}$$

(1) $\times (p+1)$ 和 (2) $\times 2$ 两式相减

$$[(p+2)(p+1) + 4]\bar{y}(p) = \frac{(p+8)(p+1) - 2(3p+1)}{(p-2)}$$

$$\begin{aligned} \bar{y}(p) &= \frac{(p+8)(p+1) - 2(3p+1)}{(p-2)[(p+2)(p+1) + 4]} = \frac{p^2 + 9p + 8 - 6p - 2}{(p-2)(p^2 + 3p + 2 + 4)} \\ &= \frac{p^2 + 3p + 6}{(p-2)(p^2 + 3p + 6)} = \frac{1}{p-2} \end{aligned}$$

将其代入(1)式

$$\frac{p+2}{p-2} + 2\bar{z}(p) = \frac{p+8}{p-2}$$

$$\bar{z}(p) = \frac{6/2}{p-2} = \frac{3}{p-2}$$

得

$$\begin{cases} \bar{y}(p) = \frac{1}{p-2} \\ \bar{z}(p) = \frac{3}{p-2} \end{cases}$$

由拉普拉斯反演,求 $\bar{y}(p), \bar{z}(p)$ 的原函数 $y(t), z(t)$.

由 $\bar{y}(p) = \frac{1}{p-2}$, 所以 $y(t) = e^{2t}$, 又 $\bar{z}(p) = \frac{3}{p-2}$, 有 $z(t) = 3e^{2t}$

所以常微分方程的解为:

$$\begin{cases} y(t) = e^{2t} \\ z(t) = 3e^{2t} \end{cases} \quad \text{且满足:} \begin{cases} y(0) = 1 \\ z(0) = 3 \end{cases}$$

$$(4) \frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} + y = t^2 e^t, y(0) = \left. \frac{dy}{dt} \right|_{t=0} = 0.$$

解:对方程进行拉普拉斯变换后得

$$p^2 \bar{y}(p) - 2p \bar{y}(p) + \bar{y}(p) = \frac{2}{(p-1)^3}$$

其中由公式 $t^2 f(t) \doteq \frac{d^2}{dp^2} \bar{f}(p)$, 且 $f(t) = e^t, \bar{f}(p) = \frac{1}{p-1}$

$$\text{则 } t^2 e^t \doteq \frac{d^2}{dp^2} \left(\frac{1}{p-1} \right) = (-1) \frac{d}{dp} (p-1)^{-2} = \frac{2}{(p-1)^3}$$

$$\text{所以 } (p^2 - 2p + 1) \bar{y}(p) = \frac{2}{(p-1)^3}, (p-1)^2 \bar{y}(p) = \frac{2}{(p-1)^3}, \bar{y}(p) = \frac{2}{(p-1)^5}$$

拉普拉斯反演求原函数

$$\bar{y}(p) = \frac{2}{(p-1)^5} = \frac{2}{(p-1)^{4+1}}, \text{由 } \frac{1}{(p-1)^{n+1}} \doteq \frac{t^n e^t}{n!}$$

$$\text{得 } y(t) = \frac{2}{4!} t^4 e^t = \frac{1}{12} t^4 e^t, \text{且满足 } y(0) = \left. \frac{dy}{dt} \right|_{t=0} = 0.$$

2. 电压为 E 的直流电源通过电感 L 和电阻 R 对电容 C 充电, 求解充电电流 j 的变化情况.

解: 全电路满足

$$u_C + u_L + u_R = E$$

$$\text{其中 } u_C = \frac{1}{C} \int_0^t j dt \quad (u_C \text{ 电容上的电压})$$

$$u_L = L \frac{dj}{dt} \quad (u_L \text{ 电感上的电压})$$

$$u_R = Rj \quad (u_R \text{ 电阻上的电压})$$

初始条件为 $j(0) = 0$, 即 K 闭合前电路中电流为零. 电路的微分方程是

$$L \frac{dj}{dt} + Rj + \frac{1}{C} \int_0^t j dt = E$$

由拉普拉斯变换后有

$$Lpj(p) + Rj(p) + \frac{1}{Cp} j(p) = \frac{E}{p} \quad \left(\text{其中 } E \doteq \frac{E}{p} \right)$$

$$Lp^2 j(p) + Rpj(p) + \frac{1}{C} j(p) = E$$

$$j(p) = \frac{E}{Lp^2 + Rp + \frac{1}{C}}$$

拉普拉斯反演求原函数

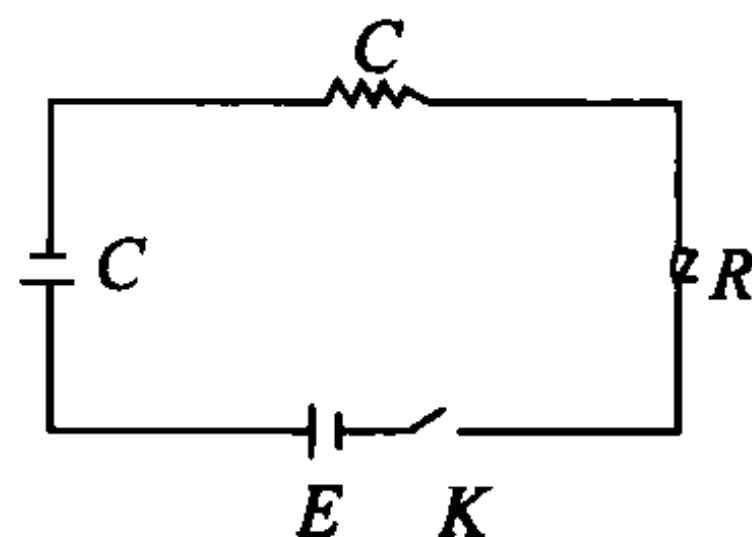
$$j(p) = \frac{1}{Lp^2 + Rp + \frac{1}{C}}$$

$$(a) \text{ 若 } R^2 - 4L/C = 0, \text{ 则 } j(p) = \frac{E/L}{\left(p + \frac{R}{2L}\right)^2}$$

有

$$j(t) = \frac{E}{L} t^2 e^{-\frac{R}{2L}t}$$

(b) 若 $R^2 = 4L/C > 0$,



有
$$j(t) = \frac{E}{L \frac{R^2}{4L^2} - \frac{1}{LC}} e^{-\frac{R}{2L}t} \operatorname{sh} \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} t$$

(c) 若 $R^2 - 4L/C < 0$

有
$$j(t) = \frac{E}{L \frac{1}{LC} - \frac{R^2}{4L^2}} e^{-\frac{R}{2L}t} \sin \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t$$

3. 放射性元素 E_1 蜕变为 E_2 , 元素 E_1 的原子数 N_1 变化规律为 $\frac{dN_1}{dt} = -C_1 N_1$. 元素 E_2 又蜕变为 E_3 , 元素 E_2 的原子数 N_2 变化规律为 $\frac{dN_2}{dt} = C_1 N_1 - C_2 N_2$. 元素 E_3 又蜕变为 E_4 , 元素 E_3 的原子数 N_3 的变化规律为 $\frac{dN_3}{dt} = C_2 N_2 - C_3 N_3$. 元素 E_4 是稳定的, 不再蜕变. 它的原子数 N_4 的变化规律为 $\frac{dN_4}{dt} = C_3 N_3$. 以上 C_1, C_2, C_3 和 C_4 都是常数. 设开始时只有元素 E_1 的 N_0 个原子, 求解 N_4 的变化情况 $N_4(t)$.

解: 由题意有方程, (其中 C_1, C_2, C_3, C_4 为常数).

$$\frac{dN_1}{dt} = -C_1 N_1, \quad \frac{dN_2}{dt} = C_1 N_1 - C_2 N_2$$

$$\frac{dN_3}{dt} = C_2 N_2 - C_3 N_3, \quad \frac{dN_4}{dt} = C_3 N_3$$

开始时 $t = 0$, 有元素 $E_1 = N_0, N_2(0) = N_3(0) = N_4(0) = 0$.

对上述方程进行拉普拉斯变换

$$\frac{dN_1(t)}{dt} \doteq p \bar{N}_1(p) - N_0 \text{ (导数定理)}$$

由 $\frac{dN_1}{dt} + C_1 N_1 = 0$ 得到 $p \bar{N}_1(p) - N_0 + C_1 \bar{N}_1(p) = 0$

$$(p + C_1) \bar{N}_1(p) = N_0, \bar{N}_1(p) = N_0 / (p + C_1)$$

同样可得:

$$p \bar{N}_2(p) + C_2 \bar{N}_2(p) = C_1 \bar{N}_1(p)$$

$$\bar{N}_2(p) = \frac{C_1 \bar{N}_1(p)}{p + C_2} = \frac{C_1 N_0}{(p + C_2)(p + C_1)}$$

$$p \bar{N}_3(p) + C_3 \bar{N}_3(p) = C_2 \bar{N}_2(p)$$

$$\bar{N}_3(p) = \frac{C_2 \bar{N}_2(p)}{p + C_3} = \frac{C_1 C_2 N_0}{(p + C_3)(p + C_2)(p + C_1)}$$

$$\bar{N}_4(p) = \frac{C_3 \bar{N}_3(p)}{p} = \frac{C_1 C_2 C_3 N_0}{p(p + C_1)(p + C_2)(p + C_3)}$$

对 $\bar{N}_4(p)$ 进行拉普拉斯反演求原函数, 设

$$\bar{N}_4(p) = \frac{C_1 C_2 C_3 N_0}{p(p + C_1)(p + C_2)(p + C_3)} = \frac{A_0}{p} + \frac{A_1}{p + C_1} + \frac{A_2}{p + C_2} + \frac{A_3}{p + C_3} \quad (1)$$

$\frac{A_0}{p}$ 的原函数为

$$\frac{A_0}{p} \doteq A_0$$

$\frac{A_1}{p+C_1}$ 的原函数为

$$\frac{A_1}{p+C_1} \doteq A_1 e^{-C_1 t}$$

$\frac{A_2}{p+C_2}$ 的原函数为

$$\frac{A_2}{p+C_2} \doteq A_2 e^{-C_2 t}$$

$\frac{A_3}{p+C_3}$ 的原函数为

$$\frac{A_3}{p+C_3} \doteq A_3 e^{-C_3 t}$$

求解方程(1) 可得

$$A_0 = N_0, \quad A_1 = \frac{C_2 C_3 N_0}{(C_1 - C_2)(C_3 - C_1)}$$

$$A_2 = \frac{C_1 C_3 N_0}{(C_1 - C_2)(C_2 - C_3)}, \quad A_3 = \frac{C_1 C_2 N_0}{(C_3 - C_1)(C_2 - C_3)}$$

$$\overline{N_4}(p) = \frac{A_0}{p} + \frac{A_1}{(p+C_1)} + \frac{A_2}{(p+C_2)} + \frac{A_3}{(p+C_3)}$$

且

$$N_4(t) = A_0 + A_1 e^{-C_1 t} + A_2 e^{-C_2 t} + A_3 e^{-C_3 t}$$

$$\text{得 } N_4(t) = N_0 + \frac{C_2 C_3 N_0 e^{-C_1 t}}{(C_1 - C_2)(C_3 - C_1)} + \frac{C_1 C_3 N_0 e^{-C_2 t}}{(C_1 - C_2)(C_2 - C_3)} + \frac{C_1 C_2 N_0 e^{-C_3 t}}{(C_3 - C_1)(C_2 - C_3)}$$

4. 设地面有一震动, 其速度 $v = H(t)$, 地震仪中的感生电流 j 遵守规律

$$\frac{dj}{dt} + 2cj + c^2 \int_0^t j dt = \lambda \frac{dv}{dt}.$$

这电流通过检流计, 使检流计发生偏转. 偏转 y 遵守规律

$$\frac{d^2 y}{dt^2} + 2c \frac{dy}{dt} + c^2 y = \mu j.$$

求解偏转 y 的变化情况 $y(t)$.

解: 由题意有方程

$$\begin{cases} \frac{dj}{dt} + 2cj + c^2 \int_0^t j dt = \lambda \frac{dv}{dt} & (\text{其中 } v = H(t)) \\ \frac{d^2 y}{dt^2} + 2c \frac{dy}{dt} + c^2 y = \mu j \end{cases}$$

$$j(0) = 0, y(0) = \left. \frac{dy}{dt} \right|_{t=0} = 0$$

$H(t) \doteq \frac{1}{p}$, 有 $dH(t)/dt \doteq p \cdot \frac{1}{p} = 1$, 对方程组进行拉普拉斯变换

$$\begin{cases} p\bar{j}(p) + 2c\bar{j}(p) + \frac{c^2}{p}\bar{j}(p) = \lambda p\bar{H}(p) = \lambda \\ p^2\bar{y}(p) + 2cp\bar{y}(p) + c^2\bar{y}(p) = \mu\bar{j}(p) \end{cases}$$

$$\bar{j}(p) = \frac{\lambda}{p + 2c + \frac{c^2}{p}} = \frac{\lambda p}{p^2 + 2cp + c^2} = \frac{\lambda p}{(p+c)^2}$$

$$\bar{y}(p) = \frac{\mu\bar{j}(p)}{p^2 + 2cp + c^2} = \frac{\mu\lambda p}{(p+c)^4}$$

对 $\bar{y}(p)$ 进行拉普拉斯反演求原函数 $y(t)$

$$\bar{y}(p) = \frac{\lambda\mu p}{(p+c)^4} = \lambda\mu \left[\frac{1}{(p+c)^3} - \frac{c}{(p+c)^4} \right]$$

又 $\bar{y}_1(p) = \frac{\lambda\mu}{(p+c)^3}, \bar{y}_2(p) = \frac{-\lambda\mu c}{(p+c)^4}, \bar{y}(p) = \bar{y}_1(p) + \bar{y}_2(p)$

有 $y_1(t) = \frac{\lambda\mu}{2} t^2 e^{-at}, y_2(t) = \frac{-\lambda\mu c}{6} t^3 e^{-at}$

得 $y(t) = y_1(t) + y_2(t) = \frac{\lambda\mu}{2} t^2 e^{-at} - \frac{\lambda\mu c}{6} t^3 e^{-at} = \frac{1}{2} \lambda\mu t^2 \left(1 - \frac{c}{3} t\right) e^{-at}$

5. 求解交流 RC 电路的方程

$$\begin{cases} Rj + \frac{1}{C} \int_0^t j dt = E_0 \sin \omega t, \\ j(0) = 0. \end{cases}$$

解: 对方程施行拉普拉斯变换得

$$Rj(p) + \frac{1}{Cp} j(p) = E_0 \frac{\omega}{p^2 + \omega^2} \quad (\text{其中 } E_0 \sin \omega t \doteq E_0 \frac{\omega}{p^2 + \omega^2})$$

$$j(p) = \frac{E_0 \omega p}{(p^2 + \omega^2)(Rp + 1/C)}$$

由拉普拉斯反演求得原函数

取 $j(p) = \frac{E_0 \omega p}{(p^2 + \omega^2)(Rp + 1/C)} = \frac{A_1 p}{p^2 + \omega^2} + \frac{A_2}{p^2 + \omega^2} + \frac{A_3}{p + \frac{1}{RC}}$

解得 $A_1 = \frac{E_0}{R^2 \omega C + \frac{1}{C\omega}}, A_2 = \frac{E_0}{R} \left[\frac{\omega}{1 + \frac{1}{R^2 C^2 \omega^2}} \right], A_3 = \frac{-E_0}{R^2 C\omega + \frac{1}{C\omega}}$

$$j(t) = A_1 \cos \omega t + \frac{A_2}{\omega} \sin \omega t + A_3 e^{-\frac{t}{RC}}$$

所以

$$= \frac{E_0}{R^2 + \frac{1}{C^2 \omega^2}} \left[\frac{1}{C\omega} \cos \omega t + R \sin \omega t \right] - \frac{E_0/C\omega}{R^2 + \frac{1}{C\omega}} e^{-\frac{t}{RC}}$$

6. 求解 $\frac{d^2 T}{dt^2} + \frac{\pi^2 a^2}{l^2} T = A \sin \omega t, T(0) = 0, T'(0) = 0.$

解: 对方程进行拉普拉斯变换得

$$p^2 \bar{T}(p) + \frac{\pi^2 a^2}{l^2} \bar{T}(p) = A \frac{\omega}{p^2 + \omega^2}, \quad (\text{其中 } \sin \omega t \doteq \frac{\omega}{p^2 + \omega^2})$$

$$\bar{T}(p) = \frac{A\omega}{\left(p^2 + \frac{\pi^2 a^2}{l^2}\right)(p^2 + \omega^2)}$$

由拉普拉斯反演求得原函数

取 $\bar{T}(p) = \frac{A\omega}{\left(p^2 + \frac{\pi^2 a^2}{l^2}\right)(p^2 + \omega^2)} = \frac{A_1}{p^2 + \omega^2} + \frac{A_2}{p^2 + \pi^2 a^2/l^2}$

解得
$$A_1 = \frac{A\omega}{\pi^2 a^2 / l^2 - \omega^2}, A_2 = -\frac{A\omega}{\pi^2 a^2 / l^2 - \omega^2}$$

$$\begin{aligned} T(t) &= \left[\frac{A_1 \sin \omega t}{\omega} + \frac{A_2 \sin \frac{\pi a}{l} t}{\pi a / l} \right] = \left[\frac{A}{\pi^2 a^2 / l^2 - \omega^2} \sin \omega t - \frac{A\omega l}{(\pi^2 a^2 / l^2 - \omega^2)(\pi a)} \sin \frac{\pi a}{l} t \right] \\ &= \frac{A}{\pi^2 a^2 / l^2 - \omega^2} \left[\sin \omega t - \frac{\omega l}{\pi a} \sin \frac{\pi a}{l} t \right] \\ &= \frac{Al}{\pi a} \frac{1}{\omega^2 - \pi^2 a^2 / l^2} \left(\omega \sin \frac{\pi a}{l} t - \frac{\pi a}{l} \sin \omega t \right) \end{aligned}$$

7. 求解 $\frac{d^2 T}{dt^2} + \omega^2 a^2 T = g(t)$, $T(0) = 0$, $T'(0) = 0$, $g(t)$ 是某个已知函数.

解: 对方程进行拉普拉斯变换得

$$p^2 \bar{T}(p) + \omega^2 a^2 \bar{T}(p) = \bar{g}(p)$$

$$\bar{T}(p) = \frac{\bar{g}(p)}{p^2 + \omega^2 a^2}$$

用拉普拉斯反演求原函数

$$\bar{T}(p) = \frac{\bar{g}(p)}{p^2 + \omega^2 a^2} = \bar{f}(p) \bar{g}(p)$$

其中 $\bar{f}(p) = \frac{1}{p^2 + \omega^2 a^2}$, $f(t) = \frac{1}{\omega a} \sin \omega a t = \frac{1}{\omega a} \frac{1}{2i} (e^{i\omega a t} - e^{-i\omega a t})$, $g(t) \doteq \bar{g}(p)$

由卷积定理

$$f(t) * g(x) \doteq \bar{f}(p) \bar{g}(p)$$

$$\bar{f}(p) \cdot \bar{g}(p) \doteq T(t)$$

$$\begin{aligned} \bar{T}(p) &= \bar{f}(p) \cdot \bar{g}(p) \doteq f(t) * g(t) = \int_0^t g(\tau) f(t-\tau) d\tau \\ &= \frac{1}{\omega a 2i} \int_0^t g(\tau) [e^{i\omega a(t-\tau)} - e^{-i\omega a(t-\tau)}] d\tau = T(t) \end{aligned}$$

8. 求解 $\frac{dT}{dt} + \omega^2 a^2 T = g(t)$, $T(0) = 0$, $g(t)$ 是某个已知函数.

解: 对方程进行拉普拉斯变换得

$$p \bar{T}(p) + \omega^2 a^2 \bar{T}(p) = \bar{g}(p)$$

$$\bar{T}(p) = \frac{\bar{g}(p)}{p + \omega^2 a^2}, \quad \bar{T}(p) = \bar{f}(p) \bar{g}(p)$$

用拉普拉斯反演求原函数

$$f(t) = e^{-\omega^2 a^2 t}, \quad g(t) \doteq \bar{g}(p)$$

$$\bar{T}(p) = \bar{f}(p) \cdot \bar{g}(p) \doteq \int_0^t g(\tau) e^{-\omega^2 a^2(t-\tau)} d\tau = T(t)$$

得

$$T(t) = \int_0^t g(\tau) e^{-\omega^2 a^2(t-\tau)} d\tau$$

9. 埃尔米特方程 $\frac{d^2 y}{dt^2} - 2t \frac{dy}{dt} + \lambda y = 0$ 里的 λ 应取怎样的数值才有可能使方程的解为多项式?

解:对该方程进行拉普拉斯变换

由导数定理

$$f''(t) \doteq p^2 \bar{f}(p) - pf(0) - f'(0)$$

$$f'(t) \doteq p\bar{f}(p) - f(0)$$

$$\frac{d^2 y}{dt^2} - 2t \frac{dy}{dt} + \lambda y = 0$$

有

$$p^2 \bar{y}(p) - py(0) - y'(0) + 2 \frac{d}{dp} [p\bar{y}(p) - y(0)] + \lambda \bar{y}(p) = 0$$

$$2p \frac{d\bar{y}(p)}{dp} + p^2 \bar{y}(p) + \lambda \bar{y}(p) + 2\bar{y}(p) = py(0) + y'(0)$$

$$\frac{d\bar{y}(p)}{dp} + \frac{p^2 + 2 + \lambda}{2p} \bar{y}(p) = \frac{1}{2}y(0) + \frac{1}{2p}y'(0)$$

解得

$$\bar{y}(p) = e^{-p^2/4} p^{-(\frac{1}{2})+1} \int e^{-p^2/4} p^{-(\frac{1}{2}+1)} \left(C_1 + \frac{C_2}{p} \right) dp$$

其中 $C_1 = y(0)/2, C_2 = y'(0)/2$

λ 取何值方能使原函数 $y(t)$ 为多项式? 当 λ 为偶数时, 即可使原函数为多项式, λ 不为偶数时,

$\bar{y}(p)$ 不能表示为 $\frac{1}{p}$ 的形式, 故原函数不为多项式.

10. 拉盖尔方程 $t \frac{d^2 y}{dt^2} + (1-t) \frac{dy}{dt} + \lambda y = 0$ 的 λ 应取怎样的数值才有可能使方程的解为多项式?

解: 对该方程进行拉普拉斯变换后

$$p(p-1) \frac{d\bar{y}(p)}{dp} + (p-\lambda-1)\bar{y}(p) = 0$$

$$\frac{d\bar{y}(p)}{dp} + \frac{p-\lambda-1}{p(p-1)} \bar{y}(p) = 0$$

$$\frac{d\bar{y}(p)}{\bar{y}(p)} = \frac{\lambda+1-p}{p(p-1)} dp$$

$$\ln \bar{y}(p) = \int \frac{-p+\lambda+1}{p(p-1)} dp = - \int \frac{1}{p} dp + \int \frac{\lambda dp}{p(p-1)}$$

$$= -\ln p + \lambda \int \left(\frac{1}{p-1} - \frac{1}{p} \right) dp = -\ln p + \lambda \ln(p-1) - \lambda \ln p + \ln C$$

$$= \ln \left(C \frac{(p-1)^\lambda}{p^{\lambda+1}} \right)$$

$$\bar{y}(p) = C \frac{(p-1)^\lambda}{p^{\lambda+1}}$$

若使方程的解为多项式, 则 $\bar{y}(p)$ 的原函数是多项式当 λ 为正整数时,

$$\bar{y}(p) = \frac{C(p-1)^\lambda}{p^{\lambda+1}}$$

$$= C \left[\frac{1}{p^{\lambda+1}} (p^\lambda - \lambda p^{(\lambda-1)} + \frac{\lambda(\lambda-1)}{2} p^{(\lambda-2)} \dots + (-1)^k \frac{\lambda(\lambda-1) \dots (\lambda-k+1)}{k!} p^{(\lambda-k)} + \dots + (-1)^\lambda \right]$$

$$= C \left[\frac{1}{p} - \frac{\lambda}{p^2} + \dots + \frac{\lambda(\lambda-1) \dots (\lambda-k+1)}{k!} \frac{(-1)^k}{p^{k+1}} + \dots + \frac{(-1)^\lambda}{p^{\lambda+1}} \right]$$

$\bar{y}(p)$ 为 $\frac{1}{p}$ 的多项式, 相应的原函数 $y(t)$ 也为多项式.

11. 有一种船舶减震器利用的是耦合振动原理. 在水面上颠簸的船体不妨看作是一个阻尼振子, 其质量为 M , 劲度系数为 K , 阻尼系数为 R . 减震器则是附着在船体上的振子, 其质量为 m , 劲度系数为 k . 因此, 船体的位移 $X(t)$ 和减震器的位移 $x(t)$ 的运动方程是

$$\begin{cases} M \frac{d^2 X}{dt^2} = F_0 \sin \omega t - KX - R \frac{dX}{dt} - k(X - x), \\ m \frac{d^2 x}{dt^2} = -k(x - X), \end{cases}$$

其中 $F_0 \sin \omega t$ 是使船体颠簸的外力. 在什么条件下, 船体的运动不含有稳定振荡而只含有指数式衰减或衰减振荡?

解: 对 $M \frac{d^2 x}{dt^2} = -k(x - X)$ 进行拉普拉斯变换

$$M[p^2 \bar{x}(p) - px(0) - x(0)] = -k[\bar{x}(p) - \bar{X}(p)]$$

$$\bar{x}(p) = \frac{Mpx(0) + mx(0) + k\bar{X}(p)}{Mp^2 + k}$$

对 $M \frac{d^2 X}{dt^2} = F_0 \sin \omega t - kX - R \frac{dX}{dt} - k(X - x)$ 进行拉普拉斯变换,

$$M[p^2 \bar{X}(p) - pX(0) - X'(0)] = \frac{F_0 \omega}{p^2 + \omega^2} - k\bar{X}(p) - R[p\bar{X}(p) - X(0)] - k[\bar{X}(p) - \bar{x}(p)]$$

若 $t = 0$ 时有 $X(0) = \dot{X}(0) = x(0) = \dot{x}(0) = 0$ 则

$$\bar{x}(p) = \frac{k\bar{X}(p)}{Mp^2 + k}$$

$$Mp^2 \bar{X}(p) = \frac{F_0 \omega}{p^2 + \omega^2} - k\bar{X}(p) - Rp\bar{X}(p) - k[\bar{X}(p) - \bar{x}(p)]$$

$$\bar{X}(p)(Mp^2 + k + Rp + k) - \frac{F_0 \omega}{p^2 + \omega^2} = k\bar{x}(p)$$

得
$$\bar{X}(p) \left[(Mp^2 + k + kp + k) - \frac{k^2}{Mp^2 + k} \right] = \frac{F_0 \omega}{p^2 + \omega^2}$$

$$\bar{X}(p) = \frac{F_0 \omega (mp^2 + k)}{(p^2 + \omega^2) [(mp^2 + k + Rp + k)(mp^2 + k) - k^2]} = \frac{F_0 \omega (mp^2 + k)}{(p^2 + \omega^2) D(p)}$$

其中 $D(p) = (mp^2 + k + Rp + k)(mp^2 + k) - k^2$

$\bar{X}(p) = F_0 \frac{\omega}{p^2 + \omega^2} \cdot \frac{mp^2 + k}{D(p)}$, 船体的运动不含有稳定振荡只含有指数式衰减或衰减振荡的条件是

$$p^2 + k/m = p^2 + \omega^2, \text{ 分母仅为 } D(p) \text{ 形式, 即}$$

$$\omega^2 = k/m$$

当 $\omega^2 = k/m$ 时, 原函数 $z(t)$ 不含有稳定振荡部分而只含有指数式衰减或衰减振荡部分.

12. 用拉普拉斯变换方法求出下列积分.

$$(1) I(t) = \int_0^\infty \frac{\cos tx}{x^2 + a^2} dx$$

解:对 $I(t)$ 进行拉普拉斯变换

$$\begin{aligned}\bar{I}(p) &= \int_0^{\infty} \frac{1}{x^2+a^2} \frac{p}{p^2+x^2} dx \quad (\text{其中 } \cos tx \doteq \frac{p}{p^2+x^2}) \\ &= p \int_0^{\infty} \left(\frac{1}{x^2+a^2} \frac{1}{p^2+x^2} \right) dx = \frac{p}{a^2-p^2} \int_0^{\infty} \left(\frac{1}{x^2+p^2} - \frac{1}{x^2+a^2} \right) dx \\ &= \frac{p}{a^2-p^2} \left[\frac{1}{p} \arctan \frac{x}{p} - \frac{1}{a} \arctan \frac{x}{a} \right] \Big|_0^{\infty} = \frac{\pi}{2} \frac{p}{a^2-p^2} \left[\frac{1}{p} - \frac{1}{a} \right] = \frac{\pi}{2a} \left[\frac{1}{a+p} \right]\end{aligned}$$

对 $\bar{I}(p)$ 进行拉普拉斯反演求出原函数 $I(t)$,

由 $\bar{I}(p) = \frac{\pi}{2a} \left(\frac{1}{a+p} \right)$, (查教材附录二, 拉普拉斯变换函数表(4))

得

$$I(t) = \frac{\pi}{2a} e^{-at}$$

$$(2) I(t) = \int_0^{\infty} \frac{\sin tx}{x} dx$$

解:对 $I(t)$ 进行拉普拉斯变换

$$\begin{aligned}\bar{I}(p) &= \int_0^{\infty} \left(\frac{1}{x} \frac{x}{p^2+x^2} \right) dx, (\text{其中 } \sin tx \doteq \frac{x}{p^2+x^2}) \\ &= \int_0^{\infty} \frac{dx}{p^2+x^2} = \frac{1}{p} \int_0^{\infty} \frac{dx/p}{1+\frac{x^2}{p^2}} = \frac{1}{p} \arctan \left(\frac{x}{p} \right) \Big|_0^{\infty} = \frac{\pi}{2p}\end{aligned}$$

对 $\bar{I}(p)$ 进行拉普拉斯反演, 求出原函数 $I(t)$

由 $\bar{I}(p) = \frac{\pi}{2p}$ (查教材附录二, 拉普拉斯变换函数表(1))

得

$$I(t) = \frac{\pi}{2}$$

$$(3) I(t) = \int_0^{\infty} \frac{\sin tx}{x(x^2+1)} dx$$

解:对 $I(t)$ 进行拉普拉斯变换

$$\begin{aligned}\bar{I}(p) &= \int_0^{\infty} \frac{1}{x(x^2+1)} \frac{x}{(p^2+x^2)} dx = \int_0^{\infty} \frac{1}{(x^2+1)} \frac{1}{(p^2+x^2)} dx \\ &= \frac{1}{p^2-1} \int_0^{\infty} \left(\frac{1}{x^2+1} - \frac{1}{p^2+x^2} \right) dx = \frac{\pi}{2} \frac{1}{p^2-1} \left[1 - \frac{1}{p} \right] \\ &= \frac{\pi}{2p(p+1)}\end{aligned}$$

对 $\bar{I}(p)$ 进行拉普拉斯反演求出原函数 $I(t)$

由 $\bar{I}(p) = \frac{\pi}{2p(p+1)} = \frac{\pi}{2} \left(\frac{1}{p} - \frac{1}{p+1} \right)$ (查教材附录二, 拉普拉斯变换函数表(1)、(4) 项)

得

$$I(t) = \frac{\pi}{2} (1 - e^{-t})$$

$$(4) I(t) = \int_0^{\infty} \frac{\sin^2 tx}{x^2} dx$$

解:对 $I(t)$ 进行拉普拉斯变换

$$\begin{aligned}\bar{I}(p) &= \int_0^{\infty} \frac{1}{2x^2} \left[\frac{1}{p} - \frac{p}{p^2 + (2x)^2} \right] dx = \int_0^{\infty} \frac{1}{2x^2} \left(\frac{p^2 + (2x)^2 - p^2}{p[p^2 + (2x)^2]} \right) dx \\ &= \frac{2}{p} \int_0^{\infty} \frac{dx}{(p^2 + 4x^2)} = \frac{1}{p^2} \int_0^{\infty} \frac{d\frac{2x}{p}}{1 + \left(\frac{2x}{p}\right)^2} = \frac{1}{p^2} \arctan \left(\frac{2x}{p} \right) \Big|_0^{\infty} = \frac{\pi}{2p^2}\end{aligned}$$

对 $\bar{I}(p)$ 进行拉普拉斯反演求出原函数 $I(t)$

由 $\bar{I}(p) = \frac{\pi}{2p^2}$, (查教材附录二, 拉普拉斯变换函数表(2) 项, $n=1$)

得

$$I(t) = \frac{\pi}{2} t$$

第二篇

数学物理方程

数学物理方程可以为学习物理课程提供基础的数学处理工具。数学物理方程涉及偏微分方程,积分方程,保角变换等。本篇介绍物理学中常见的三类偏微分方程及有关定解问题和这些问题的几种常用解法。

第七章 数学物理定解问题

本章知识要点

数学物理方程的导出

(一) 数学物理方程的导出步骤

1. 确定研究的物理量 u .
2. 在所研究的系统中划出一个小部分, 根据物理规律分析邻近部分和这个小部分的作用, 确定这个相互作用在一个短时间内怎样影响物理量 u .
3. 将相互作用的影响用公式表达出来, 经化简整理即得到数学物理方程.

(二) 具体问题的数学物理方程

1. 均匀弦的微小横振动

弦振动方程

$$\rho u_{tt} - Tu_{xx} = 0$$

受迫振动方程

$$\rho u_{tt} - Tu_{xx} = F(x, t)$$

其中, ρ 为弦的线密度, T 为研究区域邻段拉力, F 为单位长度弦所受横向力.

2. 均匀杆的纵振动

杆的纵振动方程

$$\rho u_{tt} - Yu_{xx} = 0$$

杆的受迫纵振动方程

$$\rho u_{tt} - Yu_{xx} = F(x, t)$$

ρ 为杆的密度, Y 是杆材料的杨氏模量, F 是单位长度杆每单位横截面积所受的纵向外力.

3. 传输线方程(电报方程)

对理想导线

传输线方程

$$j_t - a^2 j_{xx} = 0; v_t - a^2 v_{xx} = 0$$

其中, $a^2 = 1/LC$ L 为电感大小, C 为电容大小

4. 均匀薄膜的微小横振动

振动方程

$$\rho u_{tt} - T\Delta u = 0$$

受迫振动方程

$$\rho u_{tt} - T\Delta u = F(x, y, t)$$

其中 ρ 为薄膜单位面积质量, T 为研究区域邻近部分的张力, F 为薄膜单位面积上受到的横向外力.

5. 流体力学与声学方程

声波方程

$$s_{tt} - a^2 \Delta s = 0$$

其中, s 为声波中的空气密度相对变化量, $a^2 = \frac{\gamma P_0}{\rho_0}$, P_0, ρ_0 分别为平衡时空气的压强和密度, γ 为空气定压比热容与定容比热容的比值.

6. 真空中电磁波方程

$$\mathbf{E}_{tt} - a^2 \Delta \mathbf{E} = 0$$

$$\mathbf{H}_{tt} - a^2 \Delta \mathbf{H} = 0$$

其中 $a^2 = 1/\mu_0 \epsilon_0 = c^2$, ϵ_0, μ_0 分别为真空介电常数和磁导率, c 为光速.

7. 扩散方程

扩散定律

$$\mathbf{q} = -D \nabla u$$

三维扩散方程

$$u_t - a^2 \Delta u = 0$$

其中 $a^2 = D$, D 为扩散系数, 假设在空间中均匀.

8. 热传导方程

热传导定律

$$\mathbf{q} = -k \nabla u$$

均匀物体热传导方程

$$u_t - a^2 u_{xx} = f(x, t)$$

$$u_t - a^2 \Delta u = f(x, y, z, t)$$

其中, k 为热传导系数, $a^2 = \frac{k}{c\rho}$, c 为比热容, ρ 为密度, $f = \frac{F}{c\rho}$, F 为热源强度, 即单位时间在单位体积中产生的热量.

9. 稳定浓度分布

有扩散源

$$D \Delta u = -F$$

无扩散源

$$\Delta u = 0$$

其中, F 是扩散源强度.

10. 稳定温度分布

有热源

$$k \Delta u = -F$$

无热源

$$\Delta u = 0$$

其中, F 是热源强度.

11. 静电场

有电荷

$$\Delta V = -\frac{1}{\epsilon_0} \rho$$

无电荷

$$\Delta V = 0$$

其中, ρ 为电荷密度

12. 稳恒电流场

对均匀导电介质

$$\Delta \varphi = 0$$

含电源

$$\Delta \varphi = -f$$

其中, f 是电流源强度的分布

13. 不可压缩流体的无旋稳恒流动

含源或汇

$$\Delta\varphi = -f$$

不含源或汇

$$\Delta\varphi = 0$$

其中, f 为源或汇的强度

14. 杆的微小横振动

$$u_{tt} - a^2 u_{xxxx} = 0$$

其中 $a^2 = YI/\rho$, I 为转动惯量

15. 量子力学的薛定谔方程

薛定谔方程

$$i\hbar u_t = -\frac{\hbar^2}{2m}\Delta u + Vu$$

定态薛定谔方程

$$-\frac{\hbar^2}{2m}\Delta u + Vu = Eu$$

(三) 数学物理方程的分类

按照数学物理方程的形式可分为波动方程、输运方程和稳定场方程等, 上述问题中 1 ~ 6, 14 是波动方程, 7, 8 为输运方程, 9 ~ 13 为稳定场方程.

定解条件

(一) 初始条件

1. 初始条件

对于随时间变化发展的问题, 必须考虑到研究对象的特定“历史”, 追溯到早先某个所谓“初始”时刻的状态, 即初始条件.

2. 初始条件的要求

初始条件应当给出整个系统的初始状态, 而不仅是系统中个别地点的初始状态.

3. 没有初始条件的问题

当一类问题的运动情况经长时间后, 其运动状态仅由外源或外力而引起, 这时处理这类问题完全可以忽略初始条件的影响, 该类问题称为没有初始条件的问题, 例如外力作用下很多个周期后的阻尼振动、稳定场问题等.

(二) 边界条件

1. 边界条件

研究具体的物理系统, 必须考虑研究对象所处的特定“环境”, 而周围环境的影响常体现为边界上的物理状况, 即边界条件.

2. 边界条件的类型

常见的线性边界条件, 在数学上分为第一、二、三类边界条件.

(1) 第一类边界条件: 直接规定了所研究的物理量在边界上的数值, 即

$$u(x, y, z, t)|_{\text{边界 } x_0, y_0, z_0} = f(x_0, y_0, z_0, t)$$

(2) 第二类边界条件: 规定了所研究的物理量在边界外法线方向上方向导数的数值, 即

$$\left. \frac{\partial u}{\partial n} \right|_{\text{边界 } x_0, y_0, z_0} = f(x_0, y_0, z_0, t)$$

(3) 第三类边界条件: 规定了所研究的物理量及其外法向导数的线性组合在边界上的数值, 即

$$(u + Hu_n) \Big|_{\text{边界 } x_0, y_0, z_0} = f(x_0, y_0, z_0, t)$$

其中, f 为已知函数, H 为常数系数.

3. 线性(非线性) 边界条件

若边界条件表示为物理量或物理量各阶导数的线性组合, 则称之为线性边界条件, 否则为非线性边界条件. 其中 $f = 0$ 的边界条件又称之为齐次的.

4. 没有边界条件的问题

物理系统是有限, 必然有边界, 但如果要研究的部分很小, 则可以认为整个物理系统对于要研究的区域是无限大的; 或者说在要研究的区域内发生的物理过程时间很短, 还来不及传到整个系统, 这类问题也就无需求边界条件, 该类问题称为没有边界条件的问题, 例如研究弦的一端问题, 半导体扩散问题等.

(三) 衔接条件

衔接条件: 有时, 由于一些原因, 在所研究的区域里出现跃变点, 泛定方程在跃变点失去意义, 这时需要考虑物理量在跃变点的左右极限或物理量在跃变点两侧导数极限的连续性, 这种条件称为衔接条件.

数学物理方程的分类

(一) 线性二阶偏微分方程

1. 线性方程

若自变数为 $x_1, x_2, \dots, x_n$ 的二阶偏微分方程可表示为

$$\sum_{j=1}^n \sum_{i=1}^n a_{ij} u_{x_i x_j} + \sum_{i=1}^n b_i u_{x_i} + cu + f = 0$$
 其中 a_{ij}, b_i, c, f 只是 $x_1, x_2, \dots, x_n$ 的函数, 就称该方程为线性方程, 当 $f = 0$, 则称之为齐次的, 反之为非齐次的.

2. 叠加定理

若泛定方程和定解条件都是线性的, 那么可以把定解问题的解看作几个部分的线性叠加, 这些部分各自所满足的泛定方程和定解条件的相应的线性叠加必须是原来的泛定方程和定解条件, 该定理即叠加定理.

(二) 两个自变数的方程分类

1. 两个自变量 x, y 的二阶线性偏微分方程的一般形式

$$a_{11} u_{xx} + 2a_{12} u_{xy} + a_{22} u_{yy} + b_1 u_x + b_2 u_y + cu + f = 0$$

其中, $a_{11}, a_{12}, a_{22}, b_1, b_2, c, f$ 只是 x 和 y 的函数.

2. 二阶线性偏微分方程的特征方程和特征线

特征方程
$$a_{11} \left(\frac{dy}{dx} \right)^2 - 2a_{12} \frac{dy}{dx} + a_{22} = 0$$

特征线 特征方程的一般积分, $\xi(x, y) = \text{常数}$ 和 $\eta(x, y) = \text{常数}$

特征方程可分为两个

$$\frac{dy}{dx} = \frac{a_{12} + \sqrt{a_{12}^2 - a_{11}a_{22}}}{a_{11}}$$

$$\frac{dy}{dx} = \frac{a_{12} - \sqrt{a_{12}^2 - a_{11}a_{22}}}{a_{11}}$$

3. 二阶性偏微分方程的类型和划分标准

$$\begin{cases} a_{12}^2 - a_{11}a_{22} > 0 & \text{双曲型} \\ a_{12}^2 - a_{11}a_{22} = 0 & \text{抛物型} \\ a_{12}^2 - a_{11}a_{22} < 0 & \text{椭圆型} \end{cases}$$

4. 三种类型方程的标准形式

(1) 双曲型
$$u_{\alpha\alpha} - u_{\beta\beta} = -\frac{1}{A_{12}}[(B_1 + B_2)u_\alpha + (B_1 - B_2)u_\beta + 2C_f + 2F]$$

(2) 抛物型
$$u_{\eta\eta} = -\frac{1}{A_{22}}[B_1u_\xi + B_2u_\eta + Cu + F]$$

(3) 椭圆型
$$u_{\alpha\alpha} + u_{\beta\beta} = -\frac{1}{A_{12}}[(B_1 + B_2)u_\alpha + (B_2 - B_1)u_\beta + 2C_u + F]$$

(三) 多自变数的方程分类

1. 多自变数的二阶线性偏微分方程的一般形式

$$\sum_{j=1}^n \sum_{i=1}^n a_{ij} u_{x_i x_j} + \sum_{i=1}^n b_i u_{x_i} + cu + f = 0 \quad \text{其中, } a_{ij}, b_i, c, f \text{ 是 } x_1, x_2, \dots, x_n \text{ 的函数.}$$

2. 多自变数的二阶性偏微分方程的分类及其标准形式

(1) 椭圆型: 所有 n 个 $A_{ii} \neq 0$, 且全为同号.

标准形式:
$$\sum_{i=1}^n u_{x_i x_i} + \sum_{i=1}^n B_i u_{x_i} + C_u + F = 0$$

(2) 抛物型: 有某些 $A_{ii} = 0$

标准形式:
$$\sum_{i=1}^{n-m} u_{x_i x_i} + \sum_{i=1}^n B_i u_{x_i} + C_u + F = 0$$

(3) 双曲面: 所有 n 个 $A_{ii} \neq 0$, 其中 $n-1$ 个同号, 另一个反号.

标准形式:
$$u_{x_1 x_1} - \sum_{i=2}^n u_{x_i x_i} + \sum_{i=1}^n B_i u_{x_i} + C_u + F$$

(4) 超双曲型: 所有 n 个 $A_{ii} \neq 0$, 两种符号都不止一个,

标准形式:
$$\sum_{i=1}^m u_{x_i x_i} - \sum_{i=m+1}^n u_{x_i x_i} + \sum_{i=1}^n B_i u_{x_i} + C_u + F = 0$$

(四) 常系数线性方程

处理方法: 当线性方程的系数都是常数时, 除了依据以上方法化简为一般形式外, 还可进一步利用不同的变换化简为更简便的形式.

达朗贝尔公式 定解问题

(一) 达朗贝尔公式

1. 达朗贝尔公式 $u(x, t) = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi$

2. 达朗贝尔公式应用范围

达朗贝尔公式解决的问题为具有 $u_{tt} - a^2 u_{xx} = 0$ 形式的波动方程, 并且不存在边界条件, 其中, 初始条件为 $u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x) (-\infty < x < \infty)$.

(二) 端点的反射

1. 在研究半无限长的自由振动时, 采用“延拓”对问题进行改造, 由于端点处振动 u 为 0, 所以延拓到全空间时初始条件应为奇函数. 延拓之后, 便可在整个空间运用达朗贝尔公式对问题进行求解.

2. 在研究半无限长杆的自由振动时, 采用“延拓”对问题进行改造, 由于端点处伸长 u_x 为 0, 所以延拓到全空间时初始条件应为偶函数, 延拓之后, 便可在整个空间运用达朗贝尔公式对问题进行求解.

(三) 定解问题的整体性和适定性

1. 定解问题除极少例子以外, 都必须同时考虑偏微分方程和定解条件以进行求解, 这就是定解问题的整体性.

2. 定解问题应当满足三个条件: (1) 有解; (2) 其解是唯一的; (3) 解是稳定的. 若定解问题满足上述三个条件, 就称之为适定的.

习题全解

§ 7.1 数学物理方程的导出

1. 拿图 7-1-1 的 B 段弦作为代表, 推导弦振动方程.

证明: 选取 B 段弦为研究对象, 进行受力分析.

在 x 方向, $T_2 \cos \alpha_2 - T_1 \cos \alpha_1 = 0$

在 y 方向, $T_1 \sin \alpha_1 - T_2 \sin \alpha_2 = (\rho ds) u_{tt}$

由于所研究的问题为小振动, 所以可作如下近似

$$\cos \alpha_1 \approx 1, \cos \alpha_2 \approx 1$$

$$\sin \alpha_1 \approx \tan \alpha_1 = -u_x|_x$$

$$\sin \alpha_2 \approx \tan \alpha_2 = -u_x|_{x+dx}$$

则运动方程可化简为

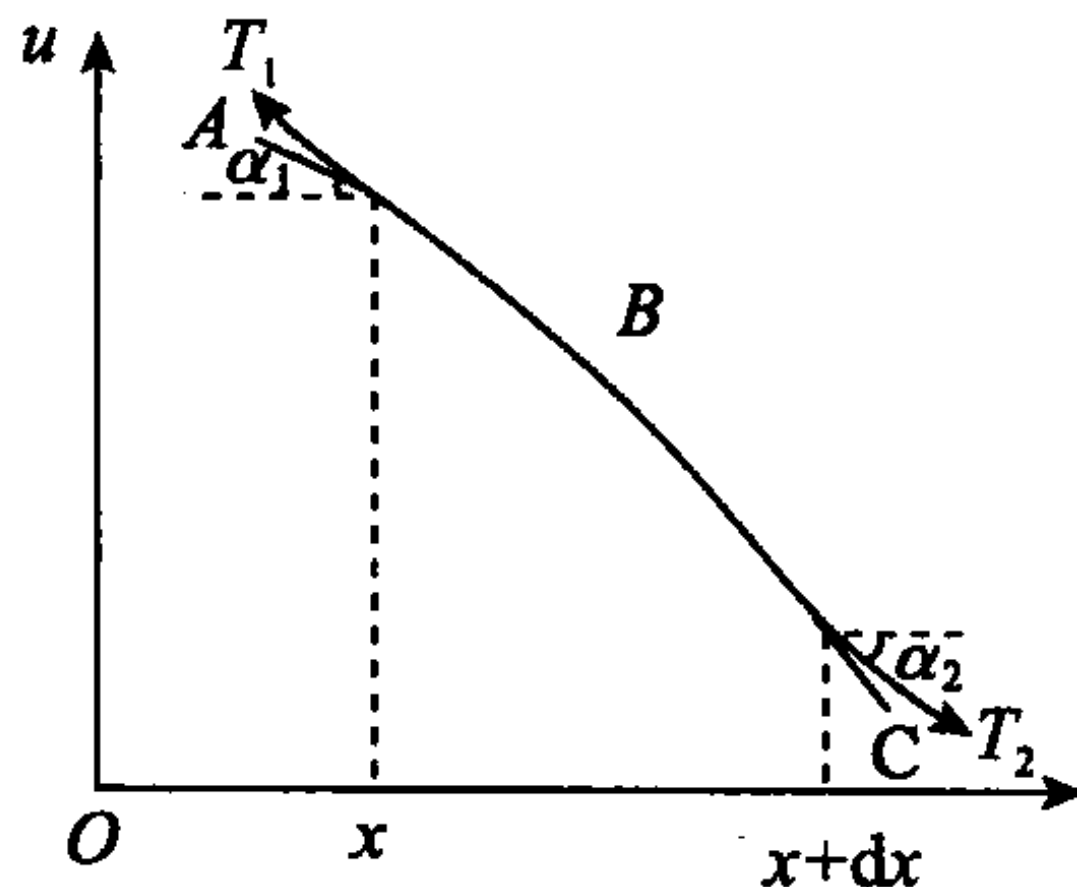


图 7-1-1

$$\begin{cases} T_2 - T_1 = 0 \\ T_2 u_x|_{x+dx} - T_1 u_x|_x = (\rho ds) u_x \end{cases}$$

令

$$T_2 = T_1 = T.$$

又

$$ds = \sqrt{(dx)^2 + (du)^2} = dx \sqrt{1 + (u_x)^2}$$

由上述近似, $u_x \ll 1$,

$$ds \approx dx$$

$$T u_x|_{x+dx} - T u_x|_x = u_x \rho dx$$

$$T \frac{u_x|_{x+dx} - u_x|_x}{dx} = \rho u_x$$

所以

$$\frac{T}{\rho} u_{xx} = u_x$$

令 $a^2 = \frac{T}{\rho}$, 方程化为标准形式

$$u_{xx} - a^2 u_{xx} = 0$$

2. 用均质材料制做细圆锥杆, 试推导它的纵振动方程.

证明: 选取圆锥体 $x \sim x + dx$ 部分为研究对象, 进行受力分析.

在 x 方向, 由胡克定律

$$YS(x+dx)u_x|_{x+dx} - YS(x)u_x|_x = \rho S(x)dx u_{xx}$$

其中, $S(x) = \pi(x \tan \theta)^2$, 代入上式得

$$\rho x^2 dx u_{xx} = Y[(x+dx)^2 u_x|_{x+dx} - x^2 u_x|_x]$$

展开 $(x+dx)^2$ 略去二阶小量 $(dx)^2$, 得 $\frac{\rho}{Y} u_{xx} = \frac{1}{x^2} \left(\frac{x^2(u_x|_{x+dx} - u_x|_x)}{dx} + 2x u_x|_{x+dx} \right)$

近似认为 $u_x|_{x+dx} \approx u_x|_x$ 则

$$\frac{\rho}{Y} u_{xx} = \frac{1}{x^2} \frac{\partial(x^2 u_x)}{\partial x}$$

令 $\frac{\rho}{Y} = \frac{1}{a^2}$, 方程化为标准形式

$$u_{xx} - \frac{a^2}{x^2} \frac{\partial(x^2 u_x)}{\partial x} = 0$$

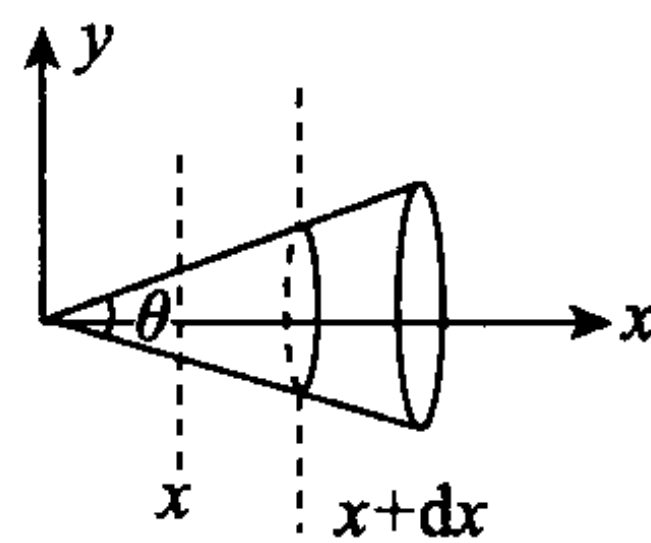


图 7-1-2

3. 弦在阻尼介质中振动, 单位长度的弦所受阻力 $F = -Ru_t$ (比例常数 R 叫作阻力系数), 试推导弦在这阻尼介质中的振动方程.

证明: 选取 B 段弦为研究对象, 进行受力分析. (图同 7-1-1 所示)

在 x 方向

$$T_2 \cos \alpha_2 - T_1 \cos \alpha_1 = 0$$

在 y 方向

$$T_1 \sin \alpha_1 - T_2 \sin \alpha_2 - Ru_{tdx} = (\rho ds) u_{tt}$$

由于所研究的问题为小振动, 所以可作下列近似

$$\cos \alpha_1 \approx \cos \alpha_2 \approx 1$$

$$\sin \alpha_1 \approx \tan \alpha_1 = -u_x|_x$$

$$\sin \alpha_2 \approx \tan \alpha_2 = -u_x|_{x+dx}$$

$$ds = dx \sqrt{1 + (u_x)^2} \approx dx$$

则运动方程可化简为

$$\begin{cases} T_2 - T_1 = 0 \\ T_2 u_x|_{x+dx} - T_1 u_x|_x - Ru_{tx} = \rho dx u_{tt} \end{cases}$$

令 $T_2 = T_1 = T$, 进一步化简得到

$$T \frac{u_x|_{x+dx} - u_x|_x}{dx} - Ru_{tx} = \rho u_{tt}$$

所以

$$\frac{T}{\rho} u_{xx} - \frac{R}{\rho} u_{tx} = u_{tt}$$

令 $\frac{T}{\rho} = a^2$ 则方程可化为

$$u_{tt} - a^2 u_{xx} + \frac{R}{\rho} u_{tx} = 0$$

4. 试推导一维和三维的热传方程(7.1.31) 和(7.1.34)

证明: 以介质中的一小块立方体为研究对象

先考虑 x 方向的热流, 在 x 面流入热量 $q_x|_x dydz$

在 $x+dx$ 面流出热量 $q_x|_{x+dx} dydz$

单位时间内 x 方向净流入热量为

$$-(q_x|_{x+dx} - q_x|_x) dydz = -\frac{\partial q}{\partial x} dx dydz$$

由热传导定律 $q = -k \nabla u$, 即分量形式为

$$q_x = -ku_x, q_y = -ku_y, q_z = -ku_z$$

单位时间内 x 方向净流入热量为

$$\frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) dx dydz$$

同理易得单位时间内 y 方向、 z 方向净流入热量分别为

$$\frac{\partial}{\partial y} \left(k \frac{\partial u}{\partial y} \right) dx dydz, \frac{\partial}{\partial z} \left(k \frac{\partial u}{\partial z} \right) dx dydz$$

由能量守恒定律, 所研究区域单位时间内增加的热量等于该区域单位时间内流入的热量.

$$\text{所以 } \rho c_p \frac{\partial u}{\partial t} dx dydz = \left[\frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial u}{\partial z} \right) \right] dx dydz$$

考虑一维情况, 则 $\rho c_p u_t - \frac{\partial}{\partial x} (ku_x) = 0$

考虑三维情况, 令 $\frac{k}{c_p} = a^2$ 且为均匀物体

则

$$u_t - a^2 \Delta_3 u = 0$$

5. 混凝土浇灌后逐渐放出“水化热”, 放热速率正比于当时尚储存着的水化热密度 Q ,

即 $\frac{dQ}{dt} = -\beta Q$. 试推导浇灌后的混凝土内的热传导方程.

证明: 单位时间内流入研究区域的热量为(图同 7-1-3 所示)

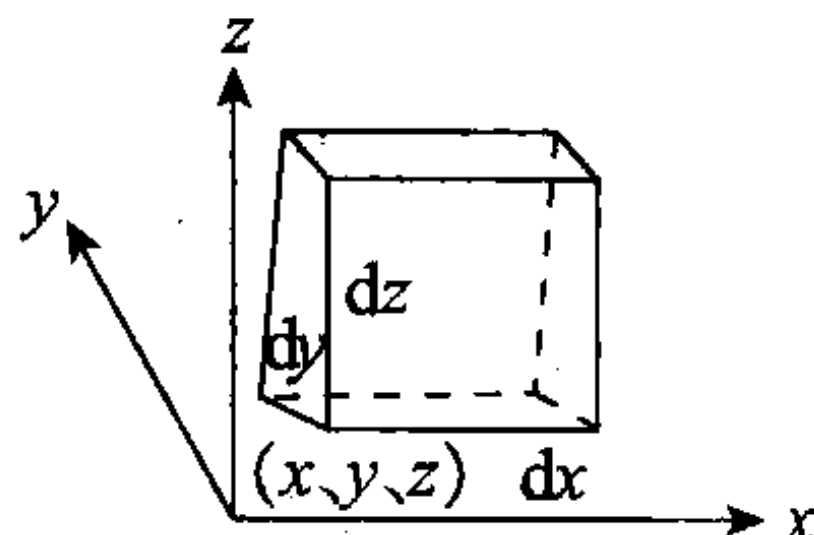


图 7-1-3

$$\left[\frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial u}{\partial z} \right) \right] dx dy dz$$

单位时间内流出该区域的热量为

$$-\beta Q dx dy dz$$

由能量守恒定律得到

$$c\rho u_t dx dy dz = \left[\frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial u}{\partial z} \right) \right] dx dy dz - \beta Q dx dy dz$$

由 $\frac{dQ}{dt} = -\beta Q$ 得

$$Q = Q_0 e^{-\beta t}$$

热传导方程可化为

$$c\rho u_t - \left[\frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial u}{\partial z} \right) \right] = \beta Q_0 e^{-\beta t}$$

6. 均质导线电阻率为 r , 通有均匀分布的直流电, 电流密度为 j . 试推导导线内的热传导方程.

证明: 选取 $x \sim x+dx$ 段导线为研究对象,

由热导定律

$$q_x = -ku_x$$

由焦耳定律

$$p = j^2 r$$

单位时间内, 由能量守恒定律

$$c\rho u_t (Sdx) = \frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) (Sdx) + j^2 r (Sdx),$$

$$c\rho u_t - \frac{\partial}{\partial x} \left(k \frac{\partial u}{\partial x} \right) = j^2 r$$

导线为均匀介质, k 为常数, 则

$$c\rho u_t - ku_{xx} = j^2 r$$

若考虑三维情况, 令 $\frac{k}{c\rho} = a^2$, 则

$$u_t - a^2 \Delta_3 u = j^2 r / c\rho$$

7. 长为 l 柔软均质轻绳, 一端固定在以匀速 ω 转动的竖直轴上. 由于惯性离心力的作用, 这弦的平衡位置应是水平线. 试推导此弦相对于水平线的横振动方程.

证明: 选取轻绳 $x \sim x+dx$ 段为研究对象

分析受力情况, 在 x 方向

$$T(x+dx) \cos \alpha_2 - T(x) \cos \alpha_1 = -(\rho ds) \omega^2$$

在 y 方向

$$T(x+dx) \sin \alpha_2 - T(x) \sin \alpha_1 = (\rho ds) u_{xx}$$

下面推导 $T(x)$ 表达式, 任选某处拉力为 $T(x)$,

则对 $x \sim l$ 段提供向心力

$$T(x) = \int_x^l (\rho dx) \omega^2 x$$

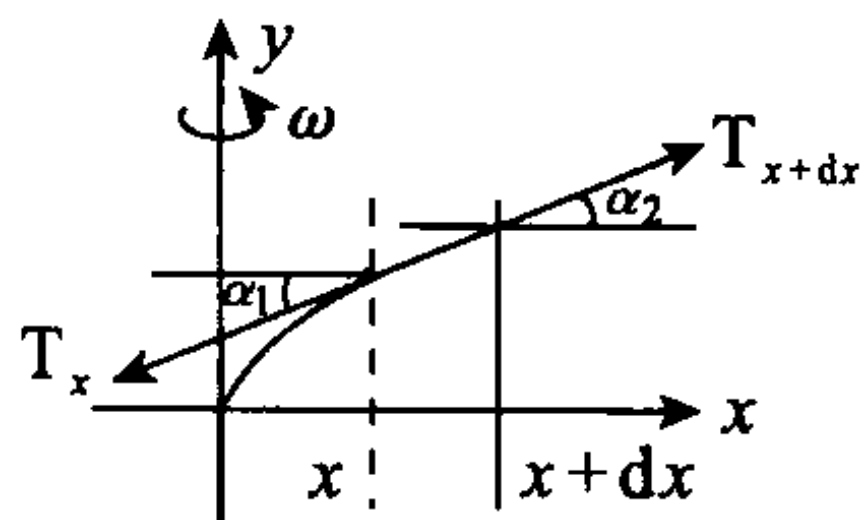


图 7-1-4

即

$$T(x) = \frac{1}{2} \rho \omega^2 (l^2 - x^2)$$

将其代入 y 方向运动方程得到

$$\frac{1}{2} \rho \omega^2 \{ [l^2 - (x+dx)^2] \sin \alpha_2 - [l^2 - x^2] \sin \alpha_1 \} = \rho dx u_{tt}$$

由于所研究问题为小振动, 所以作如下近似

$$\sin \alpha_2 \approx \tan \alpha_2 = u_x |_{x+dx}$$

$$\sin \alpha_1 \approx \tan \alpha_1 = u_x |_x$$

$$ds = dx \sqrt{1 + (u_x)^2} \approx dx$$

则 y 方向运动方程可化为

$$\frac{1}{2} \omega^2 \frac{\{ [l^2 - (x+dx)^2] u_x |_{x+dx} - (l^2 - x^2) u_x |_x \}}{dx} = u_{tt}$$

即

$$\frac{1}{2} \omega^2 \frac{\partial [(l^2 - x^2) u_x]}{\partial x} = u_{tt}$$

$$u_{tt} - \frac{1}{2} \omega^2 \frac{\partial}{\partial x} [(l^2 - x^2) u_x] = 0$$

8. 长为 l 的柔软均质重绳, 上端固定在以匀速 ω 转动的竖直轴上, 由于重力作用, 绳的平衡位置应是竖直线. 试推导此绳相对于竖直线的横振动方程.

选取轻绳 $x \sim x+dx$ 段为研究对象, 分析受力情况

$$\text{在 } x \text{ 方向} \quad T_2 \cos \alpha_2 - T_1 \cos \alpha_1 + \rho g ds = 0$$

$$\text{在 } y \text{ 方向} \quad T_2 \sin \alpha_2 - T_1 \sin \alpha_1 = (\rho ds) u_{tt} - \omega^2 u (\rho ds)$$

由于所研究问题为小振动, 所以作如下近似

$$\cos \alpha_1 \approx \cos \alpha_2 \approx 1$$

$$\sin \alpha_2 \approx \tan \alpha_2 = u_x |_{x+dx}$$

$$\sin \alpha_1 \approx \tan \alpha_1 = u_x |_x$$

$$ds = dx \sqrt{1 + (u_x)^2} \approx dx$$

则 x 方向运动方程可化为

$$T(x+dx) - T(x) + \rho g dx = 0$$

$$\frac{dT}{dx} = -\rho g$$

$$T = -\rho g x + T_0$$

当 $x = l$ 时, $T = 0$, 有 $T_0 = \rho g l$

所以

$$T = \rho g (l - x)$$

则 y 方向运动方程可化为

$$\rho g \{ [l - (x+dx)] u_x |_{x+dx} - (l - x) u_x |_x \} = (\rho dx) u_{tt} - \omega^2 u (\rho dx)$$

即

$$g \frac{\{ [l - (x+dx)] u_x |_{x+dx} - (l - x) u_x |_x \}}{dx} = u_{tt} - \omega^2 u$$

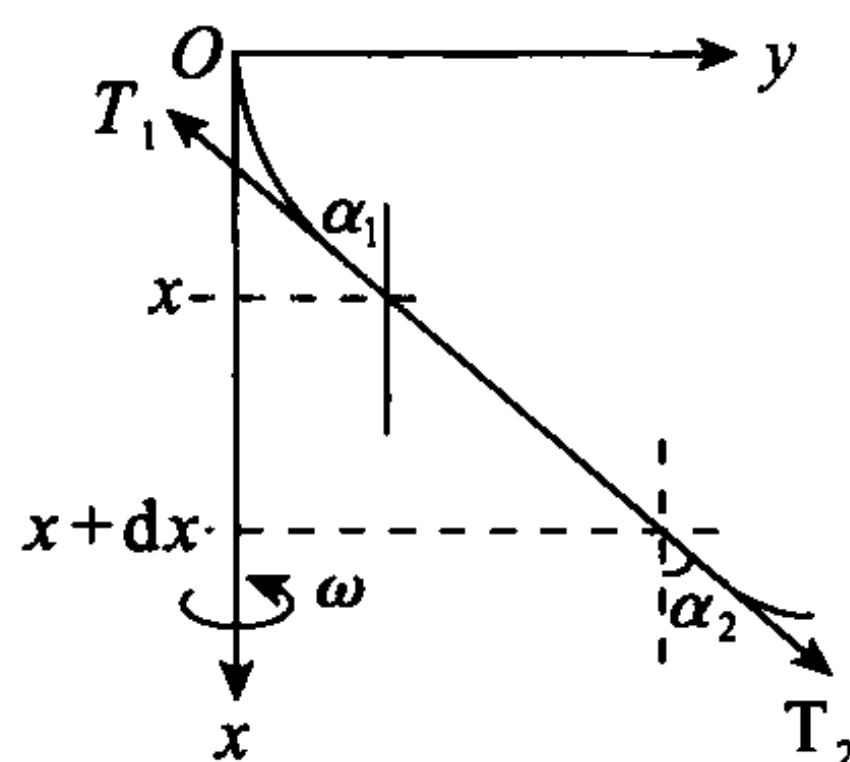


图 7-1-5

$$g \frac{\partial[(l-x)u_x]}{\partial x} = u_{xx} - \omega^2 u$$

$$u_{xx} - g \frac{\partial[(l-x)u_x]}{\partial x} - \omega^2 u = 0$$

9. 推导均匀圆杆的扭转振动方程. 杆半径为 R , 切变模量为 N .

证明: 选取圆柱 $x \sim x+dx$ 部分为研究对象, 分析受力情况.

$$\text{力矩 } L = \int_0^{2\pi} d\varphi \int_0^R r(Nr\theta_x) r dr = 2\pi N\theta_x \int_0^R r^3 dr = \frac{\pi N\theta_x R^4}{2}$$

设单位长度对轴的转动惯量为 I , 则由动量矩定理

$$I dx \frac{\partial^2 \theta}{\partial t^2} = \frac{\pi N R^4}{2} (\theta_x|_{x+dx} - \theta_x|_x)$$

即

$$\frac{\partial^2 \theta}{\partial t^2} = \frac{\pi N R^4}{2I} \frac{\partial^2 \theta}{\partial x^2}$$

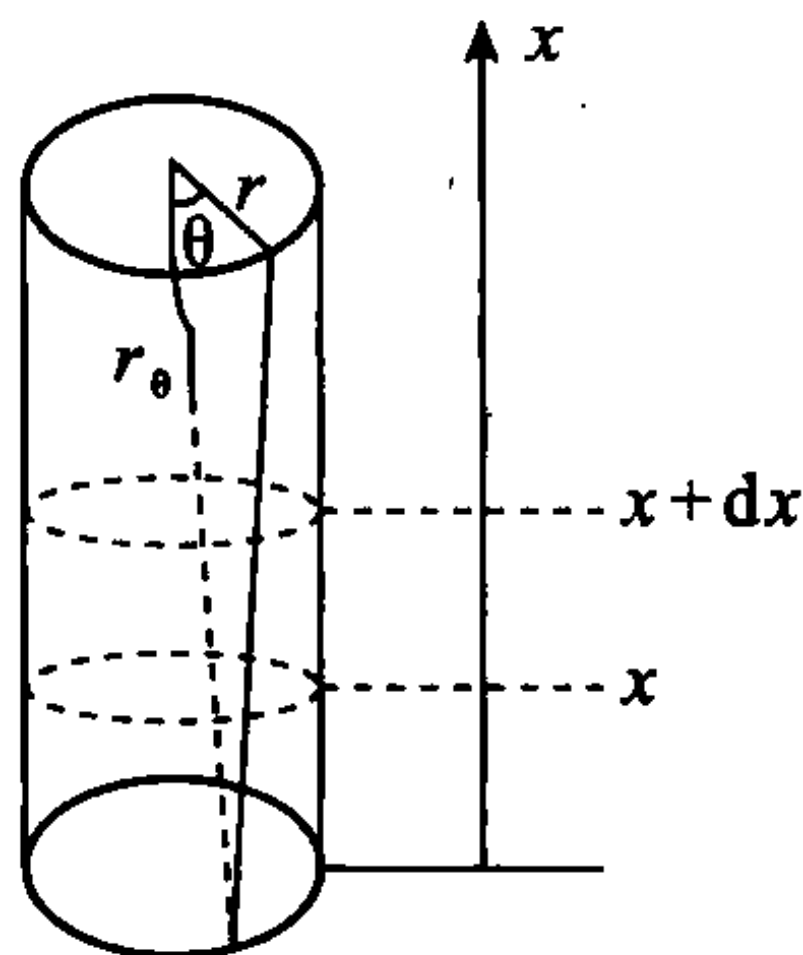


图 7-1-6

10. 推导水槽中的重力波方程. 水槽长 l , 截面为矩形, 两端由刚性平面封闭. 槽中的水在平衡时深度为 h .

证明: 设水槽横截面水平方向为 x , 选取 $x \sim x+dx$ 区域的水为研究对象, 分析受力情况.

在 x 方向, 设 x 处液面高度与平衡时液面高度相差 $H(x)$, 则所研究区域两边压强不同, 得到运动方程

$$[-\rho g H(x+dx) + \rho g H(x)]S = (\rho S dx) u_{xx}$$

化简得

$$u_{xx} = -g H_x$$

对所选区域, 当静止时, 水体积为 $hSdx$.

当运动时, 这部分水高度为 $h+H$, 宽度为 $dx+du$, 所以体积为 $S(h+H)(dx+du)$

由于水的不可压缩性

$$Shdx = S(h+H)(dx+du)$$

略去二阶水量 Hdu , 得

$$Hdx + hdu = 0$$

得

$$H = -hu_x$$

代入水的 x 方向运动方程

$$u_{xx} = -g(-hu_x)_x$$

即

$$u_{xx} = gh u_{xx}$$

§ 7.2 定解条件

1. 长为 l 的均匀弦, 两端 $x=0$ 和 $x=l$ 固定, 弦中张力为 T_0 , 在 $x=h$ 点, 以横向力 F_0 拉弦, 达到稳定后放手任其自由振动. 写出初始条件.

解: 由题目画出弦的受力情况

初始时刻, 设 H 为 $x=h$ 处初始位移

$$\text{当 } 0 \leq x \leq h, u|_{t=0} = \frac{H}{h}x$$

当 $h \leq x \leq l, u|_{t=0} = \frac{H}{l-h}(l-x)$

下面求 H 的表达式,

由弦受力平衡得, 得到

$$\begin{cases} T_2 \cos \alpha_2 - T_1 \cos \alpha_1 = 0 \\ T_2 \sin \alpha_2 + T_1 \sin \alpha_1 = F_0 \end{cases}$$

由于研究问题为小振动, 所以可作如下近似

$$\cos \alpha_2 \approx \cos \alpha_1 \approx 1$$

$$\sin \alpha_2 \approx \tan \alpha_2 = \frac{H}{l-h}$$

$$\sin \alpha_1 \approx \tan \alpha_1 = \frac{H}{h}$$

则运动方程可化为

$$\begin{cases} T_2 - T_1 = 0 \\ \frac{H}{l-h}T_2 + \frac{H}{h}T_1 = F_0 \end{cases}$$

令 $T_2 = T_1 = T$, 则

$$\left(\frac{H}{l-h} + \frac{H}{h}\right)T = F_0$$

$$H = \frac{F_0 h(l-h)}{Tl}$$

将 H 的表达式代回初始位移得到

$$u|_{t=0} = \begin{cases} \frac{F_0(l-h)}{T_0 l}x & (0 \leq x \leq h) \\ \frac{F_0 h}{T_0 l}(l-x) & (h \leq x \leq l) \end{cases}$$

初始速度 $u_t|_{t=0} = 0$

2. 长为 l 的均匀杆两端受拉力 F_0 作用而纵振动. 写出边界条件.

解: 由题目画出均匀杆的受力情况

在 $x=0$ 处, 拉力 F_0 为该端面所受的杨氏弹性力, 则

$$-YS \frac{\partial u}{\partial x} \Big|_{x=0} = -F_0$$

$$YSu_x \Big|_{x=0} = F_0$$

同理, 在 $x=l$ 处

$$YS \frac{\partial u}{\partial x} \Big|_{x=l} = F_0$$

$$YSu_x \Big|_{x=l} = F_0$$

3. 长为 l 的均匀杆, 两端有恒定热流进入, 其强度为 q_0 , 写出这个热传导问题的边界条件.

解: 由题目画出均匀杆的传热情况

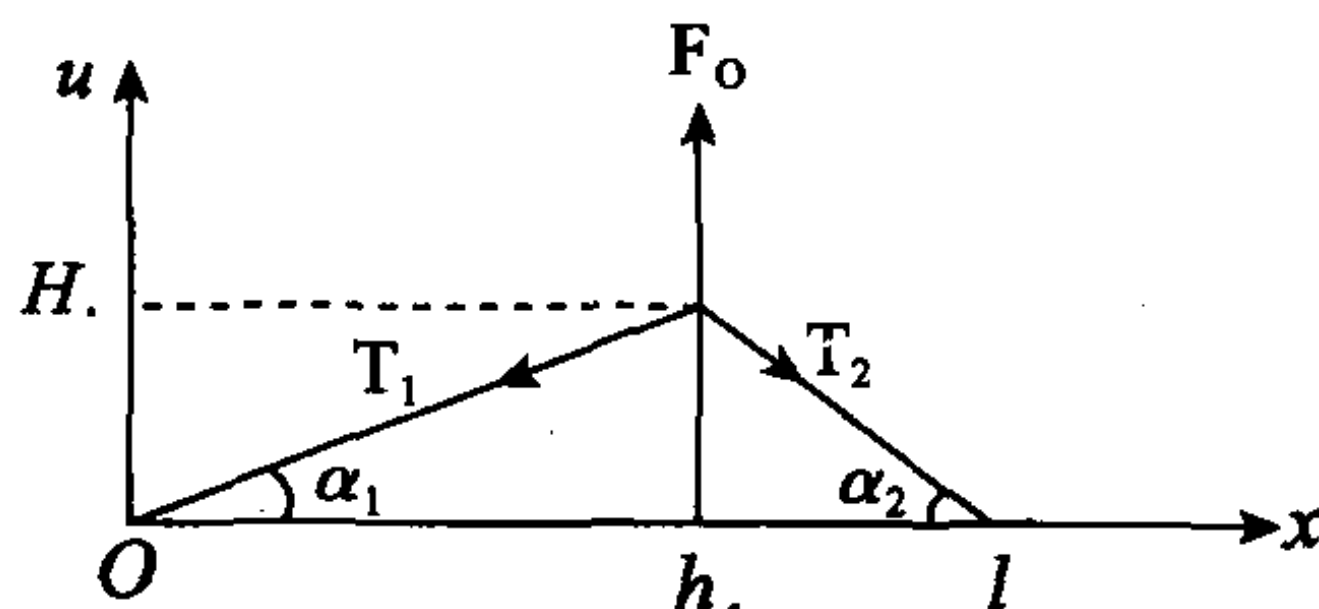


图 7-2-1

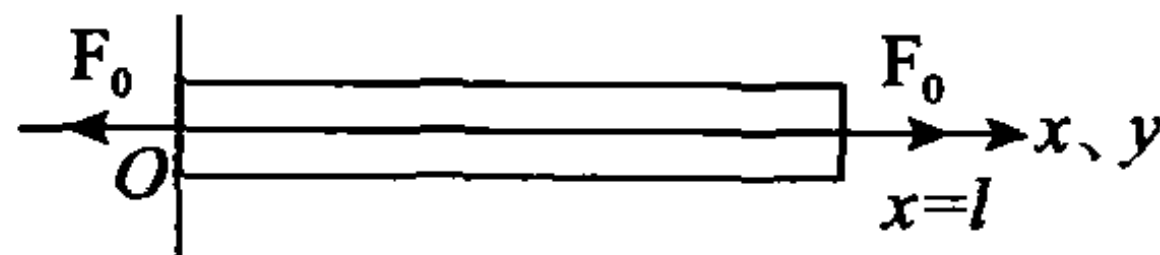


图 7-2-2

由热传导定律 $q_x = -ku_x$

在 $x=0$ 处, $q_x|_{x=0} = -ku_x|_{x=0} = q_0$

在 $x=l$ 处, $q_x|_{x=l} = -ku_x|_{x=l} = -q_0$

整理得, 边界条件为

$$\begin{cases} ku_x|_{x=0} = -q_0 \\ ku_x|_{x=l} = q_0 \end{cases}$$

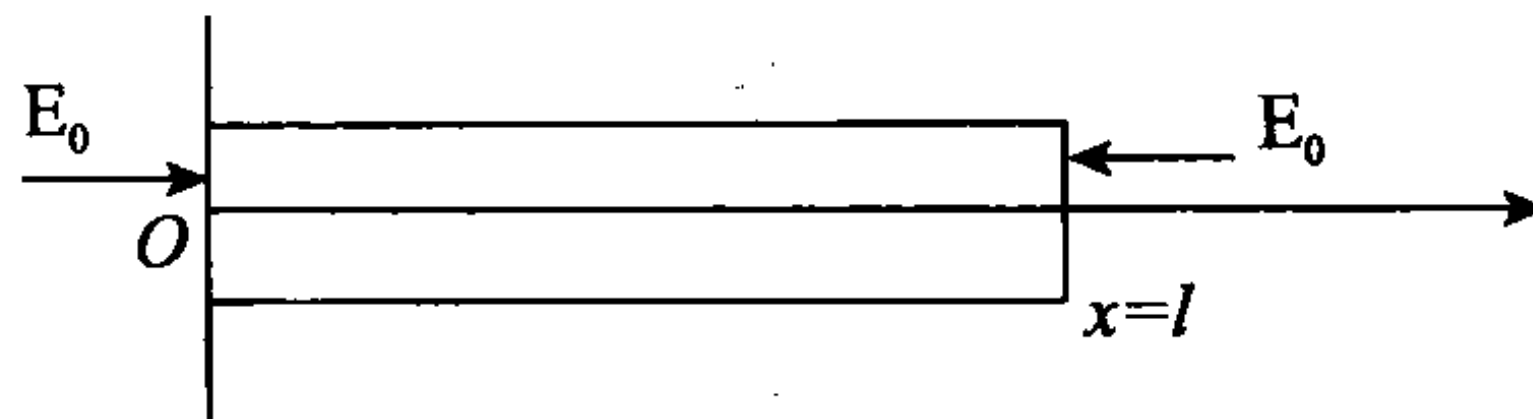


图 7-2-3

4. 半径为 ρ_0 而表面熏黑的金属长圆柱, 受到阳光照射. 阳光方向垂直于柱轴, 热流强度为 q_0 . 周围介质温度为 θ , 写出这个圆柱的热传导问题的边界条件.

解: 由题目画出圆柱横截面散热情况

由于圆柱表面熏黑, 所以其吸收系数为 1, 可以完全吸收阳光热流的法向部分, 即 $q_0 \sin \varphi$.

由自然冷却, 散热热流强度为 $h(u-\theta)$, 建立极坐标, 则

$$-k \frac{\partial u}{\partial \rho} \Big|_{\rho=\rho_0} + q_0 \sin \varphi = h(u-\theta) \Big|_{\rho=\rho_0}$$

$$(ku_\rho + h(u-\theta)) \Big|_{\rho=\rho_0} = \begin{cases} q_0 \sin \varphi, & (0 \leq \varphi < \pi) \\ 0, & (\pi \leq \varphi \leq 2\pi) \end{cases}$$

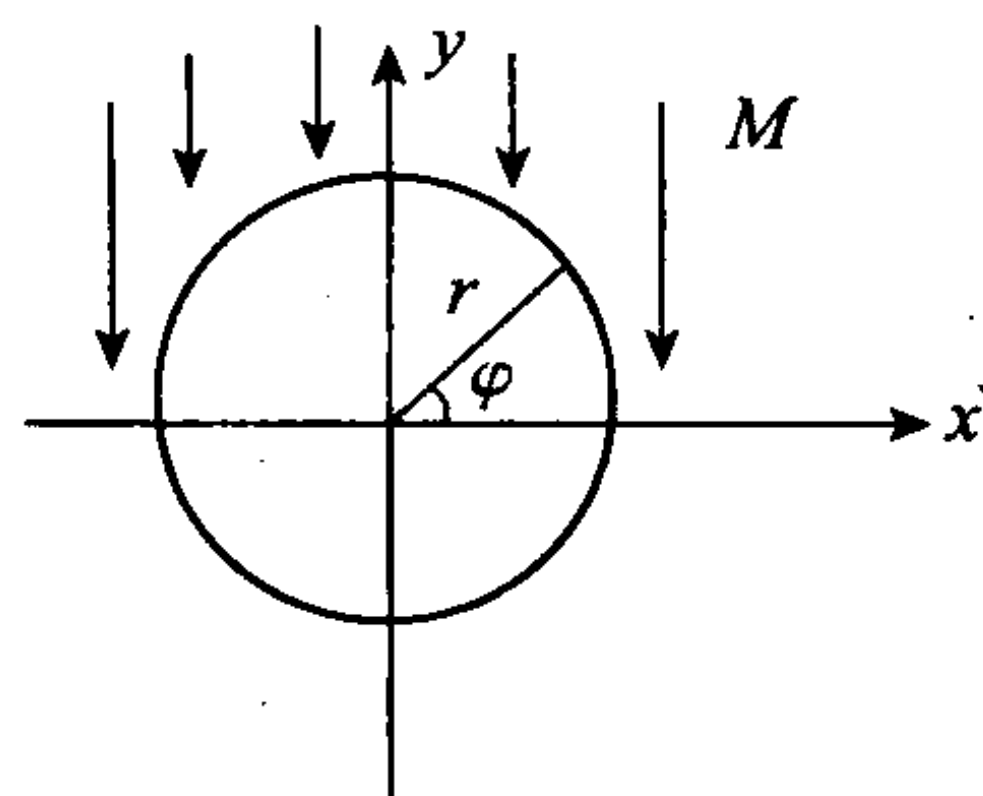


图 7-2-4

5. 习题 1 是否需要衔接条件?

解: 习题 1 不需要衔接条件, 因为当弦开始振动时, F_0 已撤去, 不再影响弦的振动.

6. 一根杆由横截面相同的两段连接而成. 两段的材料不同. 杨氏模量分别是 E^I 和 E^{II} , 密度分别是 ρ^I 和 ρ^{II} . 试写出衔接条件.

解: 由题意画出连接处的受力情况.

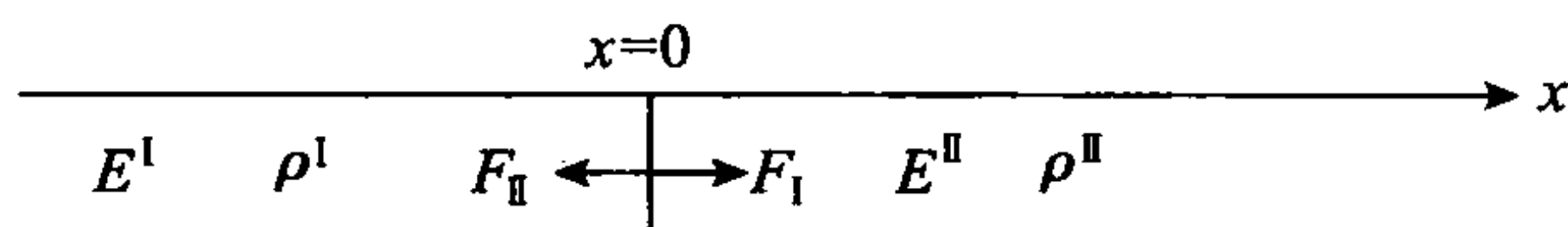


图 7-2-5

设两材料连接处 $x=0$, 则该点位移连续

即

$$u^I|_{x=0^+} = u^{II}|_{x=0^-}$$

在该点的受力情况

$$F_I = E^I S \frac{\partial u^I}{\partial x} \Big|_{x=0}$$

$$F_{II} = -E^{II} S \frac{\partial u^{II}}{\partial x} \Big|_{x=0}$$

两个力为一对反作用力, 大小相等

$$E^I S \frac{\partial u^I}{\partial x} \Big|_{x=0} = E^{II} S \frac{\partial u^{II}}{\partial x} \Big|_{x=0}$$

衔接条件为

$$\begin{cases} u^I|_{x=0} = u^{II}|_{x=0} \\ E^I S u_x^I|_{x=0} = E^{II} S u_x^{II}|_{x=0} \end{cases}$$

7. 写出静电场中电介质表面的衔接条件.

解: 在电介质交界处, 电势连续

$$u_I|_{x=0^-} = u_{II}|_{x=0^+}$$

电位移法向分量连续

$$D_n = \epsilon_1 E_1 = \epsilon_2 E_2$$

即

$$\epsilon_1 \frac{\partial u_I}{\partial x} \Big|_{x=0^-} = \epsilon_2 \frac{\partial u_{II}}{\partial x} \Big|_{x=0^+}$$

衔接条件为

$$\begin{cases} u_I|_{x=0^-} = u_{II}|_{x=0^+} \\ \epsilon_1 u_x^I|_{x=0^-} = \epsilon_2 u_x^{II}|_{x=0^+} \end{cases}$$

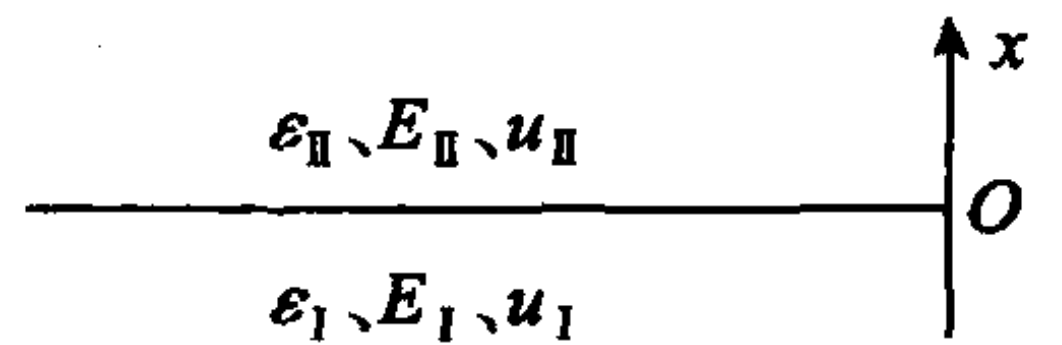


图 7-2-6

8. 一根导热杆由两段构成, 两段热传导系数、比热容、密度分别是 k^I, c^I, ρ^I 和 $k^{II}, c^{II}, \rho^{II}$. 初始温度是 u_0 , 然后保持两端温度为零. 试把这个热传导问题表为定解问题.

解: 由题意画图表示该热导杆

考虑第 I 部分, 由一维热传导方程可知

$$u_t^I - \frac{k^I}{c^I \rho^I} u_{xx}^I = 0$$

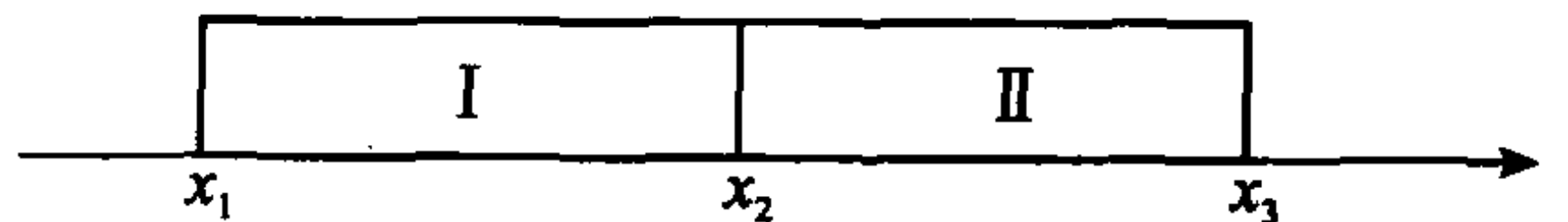


图 7-2-7

由题意, 任意时刻端点温度为 0,

即

$$u^I(x_1, t) = 0$$

初始时刻温度为 u_0 , 即

$$u^I(x, 0) = u_0$$

同理, 考虑第 II 部分, 易得到相似结果

在衔接处, 由于温度分布连续, 所以

$$u^I|_{x=x_2^-} = u^{II}|_{x=x_2^+}$$

又由于从左侧流入 $x = x_2$ 处的热量等于从右侧流出的热量, 所以

$$k^I \frac{\partial u^I}{\partial x} \Big|_{x=x_2^-} = k^{II} \frac{\partial u^{II}}{\partial x} \Big|_{x=x_2^+}$$

综上所述, 该定解问题可表示为

$$\begin{cases} u_t^I - \frac{k^I}{c^I \rho^I} u_{xx}^I = 0 \\ u^I(x_1, t) = 0 \\ u^I(x, 0) = u_0 \end{cases} \quad (x_1 < x < x_2)$$

$$\begin{cases} u_t^{II} - \frac{k^{II}}{c^{II} \rho^{II}} u_{xx}^{II} = 0 \\ u^{II}(x_3, t) = 0 \\ u^{II}(x, 0) = u_0 \end{cases} \quad (x_2 < x < x_3)$$

$$\begin{cases} u^I|_{x=x_2^-} = u^{II}|_{x=x_2^+} \\ k^I \frac{\partial u^I}{\partial x} \Big|_{x=x_2^-} = k^{II} \frac{\partial u^{II}}{\partial x} \Big|_{x=x_2^+} \end{cases}$$

§ 7.3 数学物理方程的分类

1. 把下列方程化为标准形式.

$$(1) au_{xx} + 2au_{xy} + au_{yy} + bu_x + cu_y + u = 0$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = 0$$

$$\text{该方程为抛物型, } \frac{dy}{dx} = \frac{a \pm \sqrt{a^2 - a^2}}{a} = 1$$

令 $\xi = y - x, y = x$, 代入系数计算公式, 得到

$$\begin{cases} A_{22} = a_{11}y_x^2 + 2a_{12}\eta_x y_y + a_{22}\eta_y^2 = a \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = c - b \\ B_2 = a_{11}y_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = b \\ C = c = 1 \\ F = f = 0 \end{cases}$$

$$\text{标准形式为} \quad u_{\eta\eta} = -\frac{1}{a}[(c-b)u_\xi + bu_\eta + u]$$

$$\text{即} \quad u_{\eta\eta} + \frac{c-b}{a}u_\xi + \frac{b}{a}u_\eta + \frac{1}{a}u = 0$$

$$(2) u_{xx} - 2u_{xy} - 3u_{yy} + 2u_x + 6u_y = 0$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = 4 > 0$$

$$\text{该方程为双曲型, } \frac{dy}{dx} = \frac{-1 \pm \sqrt{1+3}}{1} = 1 \text{ 或 } 3,$$

令 $\xi = x - y, \eta = 3x + y$ 代入系数计算公式
得到

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y = 8 \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = -4 \\ B_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = 12 \\ C = 0 \\ F = 0 \end{cases}$$

$$\text{标准形式为} \quad u_{\xi\eta} = -\frac{1}{16}(-4u_\xi + 12u_\eta)$$

$$\text{即} \quad 4u_{\xi\eta} - u_\xi + 3u_\eta = 0$$

$$(3) u_{xx} + 4u_{xy} + 5u_{yy} + u_x + 2u_y = 0$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = -1 < 0$$

$$\text{该方程为椭圆型} \quad \frac{dy}{dx} = \frac{2 \pm \sqrt{-1}}{1} = 2 \pm i$$

$$\text{令} \quad \xi = (2+i)x - y, \eta = (2-i)x - y$$

代入系数计算公式

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y = 2 \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = i \\ B_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = -i \\ C = 0 \\ F = 0 \end{cases}$$

$$u_{\xi\eta} = -\frac{1}{4}(iu_{\xi} - iu_{\eta})$$

令

$$\alpha = \frac{1}{2}(\xi + \eta) = 2x - y$$

$$\beta = \frac{1}{2i}(\xi - \eta) = x$$

代入标准形式

$$u_{\alpha\alpha} + u_{\beta\beta} = -\frac{1}{A_{12}}[(B_1 + B_2)u_{\alpha} + i(B_2 - B_1)u_{\beta} + 2Cu + F]$$

即得

$$u_{\alpha\alpha} + u_{\beta\beta} = -\frac{1}{2}[i(-2i)u_{\beta}]$$

即

$$u_{\alpha\alpha} + u_{\beta\beta} + u_{\beta} = 0$$

将角标变量记号改为 ξ, η , 即得

$$u_{\xi\xi} + u_{\eta\eta} + u_{\eta} = 0$$

$$(4) u_{xx} + y u_{yy} = 0$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = -y$$

① 当 $y < 0$, 则 $a_{12}^2 - a_{11}a_{22} > 0$, 该方程为双曲型, $\frac{dy}{dx} = \frac{0 \pm \sqrt{-y}}{1} = \pm \sqrt{-y}$

令

$$\xi = x + 2\sqrt{y}, y = x - 2\sqrt{-y}$$

代入系数计算公式

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y = 2 \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = -\frac{1}{2\sqrt{-y}} = \frac{2}{y-\xi} \\ B_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = \frac{1}{2\sqrt{-y}} = \frac{2}{\xi-y} \\ C = 0 \\ F = 0 \end{cases}$$

标准形式为

$$u_{\xi\eta} = -\frac{1}{4}\left(\frac{2}{\eta-\xi}u_{\xi} + \frac{2}{\xi-\eta}u_{\eta}\right)$$

即

$$(\xi - \eta)u_{\xi\eta} + \frac{1}{2}(u_{\xi} - u_{\eta}) = 0$$

② 当 $y > 0$, 则 $a_{12}^2 - a_{11}a_{22} < 0$, 该方程为椭圆型

$$\frac{dy}{dx} = \frac{0 \pm \sqrt{yi}}{1} = \pm i\sqrt{y}$$

令

$$\xi = x + 2\sqrt{yi}, y = x - 2\sqrt{yi},$$

进一步又令

$$\alpha = \frac{1}{2}(\xi + \eta) = x, \beta = \frac{1}{2i}(\xi - \eta) = 2\sqrt{y}$$

由系数公式求得

$$A_{12} = 2, B_1 = -\frac{1}{2\sqrt{y}}i, B_2 = \frac{1}{2\sqrt{y}}i$$

$$C = 0, F = 0$$

标准形式为

$$u_{\alpha\alpha} + u_{\beta\beta} = \frac{1}{2} \left(i \cdot i \frac{1}{\sqrt{y}} \right) u_{\beta}$$

即

$$u_{\alpha\alpha} + u_{\beta\beta} - \frac{1}{\beta} u_{\beta} = 0$$

将角标变量记号改为 ξ, η 即得

$$u_{\xi\xi} + u_{\eta\eta} - \frac{1}{\eta} u_{\eta} = 0$$

$$(5) u_{xx} + xu_{yy} = 0$$

$$\text{解: } a_{12}^2 - a_{11}a_{22} = -x$$

① 当 $x < 0$, 则 $a_{12}^2 - a_{11}a_{22} > 0$, 方程为双曲型

$$\frac{dy}{dx} = \frac{0 \pm \sqrt{-x}}{2} = \pm \sqrt{-x}$$

令

$$\xi = \frac{3}{2}y + (\sqrt{-x})^3, \eta = \frac{3}{2}y - (\sqrt{-x})^3$$

代入系数计算公式

$$\begin{cases} A_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y = \frac{9}{2}x \\ B_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y = \frac{3}{4\sqrt{-x}} \\ B_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y = \frac{-3}{4\sqrt{-x}} \\ C = 0 \\ F = 0 \end{cases}$$

标准形式为

$$u_{\xi\eta} = \frac{-1}{9x} \left(\frac{3}{4\sqrt{-x}} u_{\xi} + \frac{-3}{4\sqrt{-x}} u_{\eta} \right)$$

由 $(-\sqrt{x})^3 = \frac{(\xi - \eta)}{2}$, 代入标准形式化简即得

$$u_{\xi\eta} - \frac{1}{6(\xi - \eta)}(u_{\xi} - u_{\eta}) = 0$$

② 当 $x > 0$, 则 $a_{12}^2 - a_{11}a_{22} < 0$, 方程为椭圆型, $\frac{dy}{dx} = \frac{0 \pm \sqrt{x}i}{1} = \pm i\sqrt{x}$

令
$$\xi = \frac{3}{2}y - ix^{3/2} \quad \eta = \frac{3}{2}y + ix^{3/2}$$

进一步又令

$$\alpha = \frac{1}{2}(\xi + \eta) = \frac{3}{2}y, \beta = \frac{1}{2}(\xi - \eta) = -x^{3/2}$$

由系数公式得

$$A_{12} = \frac{9}{2}x, B_1 = -\frac{3i}{4} \cdot \frac{1}{\sqrt{x}}$$

$$B_2 = \frac{3i}{4} \cdot \frac{1}{\sqrt{x}}, C = 0, F = 0$$

标准形式

$$u_{\alpha\alpha} + u_{\beta\beta} = \frac{1}{3x^{3/2}}u_{\beta}$$

将角标记号换为 ξ, η , 且 $x^{3/2} = -\beta$, 即得

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{3\eta}u_{\eta} = 0$$

$$(6) y^2 u_{xx} + x^2 u_{yy} = 0$$

解: $a_{12}^2 - a_{11}a_{22} = -x^2 y^2 < 0$

该方程为椭圆型

$$\frac{dy}{dx} = \frac{0 \pm \sqrt{-x^2 y^2}}{y^2} = \pm i \frac{x}{y}$$

直接令 $\xi = y^2, \eta = x^2$, 则

$$u_x = u_{\eta}\eta_x = 2xu_{\eta}$$

$$u_y = u_{\xi}\xi_y = 2yu_{\xi}$$

$$u_{xx} = 2(u_{\eta} + xu_{\eta\eta}\eta_x) = 4x^2 u_{\eta\eta} + 2u_{\eta}$$

$$u_{yy} = 2(u_{\xi} + yu_{\xi\xi}\xi_y) = 4y^2 u_{\xi\xi} + 2u_{\xi}$$

代回原方程, 整理得

$$4x^2 y^2 u_{\eta\eta} + 2y^2 u_{\eta} + 4x^2 y^2 u_{\xi\xi} + 2x^2 u_{\xi} = 0$$

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{2y^2}u_{\xi} + \frac{1}{2x^2}u_{\eta} = 0$$

$$u_{\xi\xi} + u_{\eta\eta} + \frac{1}{2\xi}u_{\xi} + \frac{1}{2\eta}u_{\eta} = 0$$

$$(7) 4y^2 u_{xx} - e^{2x} u_{yy} - 4y^2 u_x = 0$$

解: $a_{12}^2 - a_{11}a_{22} = 4y^2 e^{2x} > 0$

该方程为双曲型

$$\frac{dy}{dx} = \frac{0 \pm \sqrt{4y^2 e^{2x}}}{4y^2} = \pm \frac{e^x}{2y}$$

令 $\xi = y^2 + e^x, \eta = y^2 - e^x$, 则

$$u_x = u_\xi \xi_x + u_\eta \eta_x = u_\xi e^x - u_\eta e^x = e^x(u_\xi - u_\eta)$$

$$\begin{aligned} u_{xx} &= e^x(u_\xi - u_\eta) + e^x(u_{\xi\xi}\xi_x + u_{\xi\eta}\eta_x - u_{\eta\xi}\xi_x - u_{\eta\eta}\eta_x) \\ &= e^x(u_\xi - u_\eta) + e^{2x}(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}) \end{aligned}$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = 2y(u_\xi + u_\eta)$$

$$\begin{aligned} u_{yy} &= 2(u_\xi + u_\eta) + 2y(u_{\xi\xi}\xi_y + u_{\xi\eta}\eta_y + u_{\eta\xi}\xi_y + u_{\eta\eta}\eta_y) \\ &= 2(u_\xi + u_\eta) + 4y^2(u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}) \end{aligned}$$

代回原方程, 整理得

$$-16y^2 e^{2x} u_{\xi\eta} + (4ye^x - 2e^{2x})u_\xi - (4y^2 e^{2x} + 2e^{2x})u_\eta - 4y^2 e^x(u_\xi - u_\eta) = 0$$

即

$$-16y^{2x} e u_{\xi\eta} - 2e^x u_\xi - 2e^x u_\eta = 0$$

$$8y^2 u_{\xi\eta} + u_\xi + u_\eta = 0$$

由 $y^2 = \frac{1}{2}(\xi + \eta)$, 代入上式即得

$$4(\xi + \eta)u_{\xi\eta} + u_\xi + u_\eta = 0$$

2. 简化下列常系数方程.

$$(1) u_{xx} + u_{yy} + \alpha u_x + \beta u_y + \gamma u = 0$$

解: 尝试作变换 $u(x, y) = e^{\lambda x + \mu y} v(x, y)$, 其中 λ, μ 为待定常数, 则

$$\begin{cases} u_x = e^{\lambda x + \mu y} (v_x + \lambda v) \\ u_{xx} = e^{\lambda x + \mu y} (v_{xx} + 2\lambda v_x + \lambda^2 v) \\ u_y = e^{\lambda x + \mu y} (v_y + \mu v) \\ u_{yy} = e^{\lambda x + \mu y} (v_{yy} + 2\mu v_y + \mu^2 v) \\ u_{xy} = e^{\lambda x + \mu y} (v_{xy} + \lambda v_y + \mu v_x + \lambda \mu v) \end{cases}$$

代回原方程, 化简并整理后得到

$$v_{xx} + v_{yy} + (2\lambda + \alpha)v_x + (2\mu + \beta)v_y + (\lambda^2 + \mu^2 + 2\lambda + \beta\mu + \gamma)v = 0$$

为简便, 取 $\lambda = -\frac{\alpha}{2}, \mu = -\frac{\beta}{2}$, 即作变换

$$u = e^{-\frac{1}{4}(\alpha^2 + \beta^2)x} v$$

代入上式得

$$v_{xx} + v_{yy} + \left(r - \frac{\alpha^2}{4} - \frac{\beta^2}{4}\right)v = 0$$

$$(2) u_{xx} = \frac{1}{a^2} u_y + \alpha u + \beta u_x$$

解: 尝试作变换 $u(x, y) = e^{\lambda x + \mu y} v(x, y)$ 则

$$\begin{cases} u_x = e^{\lambda x + \mu y} (v_x + \lambda v) \\ u_{xx} = e^{\lambda x + \mu y} (v_{xx} + 2\lambda v_x + \lambda^2 v) \\ u_y = e^{\lambda x + \mu y} (v_y + \mu v) \\ u_{yy} = e^{\lambda x + \mu y} (v_{yy} + 2\mu v_y + \mu^2 v) \\ u_{xy} = e^{\lambda x + \mu y} (v_{xy} + \lambda v_y + \mu v_x + \lambda \mu v) \end{cases}$$

代回原方程,化简并整理后得到

$$v_{xx} + (2\lambda - \beta)v_x - \frac{1}{a^2}v_y + \left(\lambda^2 - \frac{\mu}{a^2} - \alpha - \lambda\beta\right)v = 0$$

令零阶导数系数为0,则 $\lambda^2 - \frac{\mu}{a^2} - \alpha - \lambda\beta = 0$,

即 $\mu = -a^2\left(\alpha + \frac{\beta^2}{4}\right)$

作变换 $u = e^{\frac{\beta}{2}x - a^2\left(\alpha + \frac{\beta^2}{4}\right)y}$, v 可使方程化简为

$$v_{xx} - \frac{1}{a^2}v_y = 0$$

$$(3) u_{yy} + \frac{c-b}{a}u_x + \frac{b}{a}u_y + u = 0$$

解:尝试作变换 $u(x, y) = e^{\lambda x + \mu y} \cdot v(x, y)$, 并如(1)、(2)题中过程,代回各阶导数化简后得到

$$v_{yy} + \frac{c-b}{a}v_x + \left(2\mu + \frac{b}{a}\right)v_y + \left(\mu^2 + \lambda\frac{c-b}{a} + \frac{b}{a}\mu + 1\right)v = 0$$

令 v_y 系数为0,则 $\mu = -\frac{b}{2a}$

令 v 系数为0,则 $\mu^2 + \lambda\frac{c-b}{a} + \frac{b}{a}\mu + 1 = 0$, 即

$$\lambda = \frac{4a^2 - b^2}{4a(b-c)}$$

作变换

$$u = \frac{4a^2 - b^2}{e^{4a(b-c)}} - \frac{b}{2a}y$$

可使方程化简为

$$v_{yy} + \frac{c-b}{a}v_x = 0$$

$$(4) u_{xy} + 3u_x + 4u_y + 2u = 0$$

解:尝试作变换 $u(x, y) = e^{\lambda x + \mu y} \cdot v(x, y)$, 并如(1)、(2)题中过程,代回各阶导数化简后得到

$$v_{xy} + (\mu + 3)v_x + (\lambda + 4)v_y + (\lambda\mu + 3\lambda + 4\mu + 2)v = 0$$

令 v_x 系数为0,则 $\mu = -3$. 令 v_y 系数为0,则 $\lambda = -4$

作变换 $u = e^{-4x-3y}$, v 可使方程简化为

$$v_{xy} - 10v = 0$$

$$(5) 2au_{xx} + 2au_{xy} + au_{yy} + 2bu_x + 2cu_y + u = 0$$

解:① 首先将方程化为标准形式

$$a_{12}^2 - a_{11}a_{22} = -a^2 < 0$$

该方程为椭圆型

$$\frac{dy}{dx} = \frac{a \pm \sqrt{-a^2}}{2a} = \frac{1}{2} \pm \frac{i}{2}$$

直接令 $\xi = y - \frac{x}{2}$, $\eta = \frac{x}{2}$, 则

$$\begin{cases} u_x = u_\xi \cdot \xi_x + u_\eta \cdot \eta_x = \frac{1}{2}(u_\eta - u_\xi) \\ u_{xx} = \frac{1}{2}(u_{\eta\xi}\xi_x + u_{\eta\eta}\eta_x - u_{\xi\xi}\xi_x - u_{\xi\eta}\eta_x) \\ \quad = \frac{1}{4}(u_{\xi\xi} + u_{\eta\eta} - 2u_{\xi\eta}) \\ u_y = u_\xi \xi_y + u_\eta \eta_y = u_\xi \\ u_{yy} = u_{\xi\xi}\xi_y + u_{\xi\eta}\eta_y = u_{\xi\xi} \\ u_{xy} = u_{\xi\xi}\xi_x + u_{\xi\eta}\eta_x = \frac{1}{2}(u_{\xi\eta} - u_{\xi\xi}) \end{cases}$$

将各阶导数代回原方程化简, 可得方程标准形式为

$$au_{\xi\xi} + au_{\eta\eta} + (4c - 2b)u_\xi + 2bu_\eta + 2u = 0$$

② 然后尝试作变换 $u = ve^{\lambda\xi + \mu\eta}v$, 则

$$\begin{cases} u_\xi = e^{\lambda\xi + \mu\eta}(v_\xi + \lambda v) \\ u_{\xi\xi} = e^{\lambda\xi + \mu\eta}(v_{\xi\xi} + 2\lambda v_\xi + \lambda^2 v) \\ u_\eta = e^{\lambda\xi + \mu\eta}(v_\eta + \mu v) \\ u_{\eta\eta} = e^{\lambda\xi + \mu\eta}(v_{\eta\eta} + 2\mu v_\eta + \mu^2 v) \\ u_{\xi\eta} = e^{\lambda\xi + \mu\eta}(v_{\xi\eta} + \lambda v_\eta + \mu v_\xi + \lambda\mu v) \end{cases}$$

代回原方程标准形式并化简整理后得到

$$v_{\xi\xi} + v_{\eta\eta} + \left[\frac{2(2c-b)}{a} + 2\lambda \right] v_\xi + \left(\frac{2b}{a} + 2\mu \right) v_\eta + \left[\lambda^2 + \mu^2 + \frac{2(2c-b)}{a}\lambda + \frac{2b}{a}\mu + \frac{2}{a} \right] v = 0$$

令 v_ξ 系数为 0, 则

$$\lambda = -\frac{2c-b}{a}$$

令 v_η 系数为 0, 则

$$\mu = -\frac{b}{a}$$

综上所述, 作变换

$$\begin{cases} \xi = y - \frac{x}{2} \\ \eta = \frac{x}{2} \\ u = e^{\frac{b-2c}{a}\xi - \frac{b}{a}\eta}v \end{cases}$$

可使原方程简化为

$$v_{\xi\xi} + v_{\eta\eta} + \frac{2}{a} \left(\frac{2bc - b^2 - 2c^2}{a} + 1 \right) v = 0$$

§ 7.4 达朗贝尔公式 定解问题

1. 求解无限长弦的自由振动, 设弦的初始位移为 $\varphi(x)$, 初始速度为 $-a\varphi'(x)$.

解: 由题意定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{t=0} = 0 = \varphi(x) \\ u_t|_{t=0} = 0 = -a\varphi'(x) \end{cases}$$

由达朗贝尔公式

$$u(x, t) = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x) dx$$

代入 $\varphi(x) = \varphi(x), \psi(x) = -a\varphi'(x)$, 可得

$$\begin{aligned} u(x, t) &= \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} [-a\varphi'(x)] dx \\ &= \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a}(-a)[\varphi(x+at) - \varphi(x-at)] \\ &= \varphi(x-at) \end{aligned}$$

即无限长弦的自由振动的波只向 x 正方向传播.

2. 求解无限长理想传输线上电压和电流的传播情况, 设初始电压分布为 $A \cos kx$, 初始电流分布为 $\sqrt{G/L} A \cos kx$.

解: (1) 电压传播情况, 定解问题为

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0 & (a^2 = \frac{1}{Lc}) \\ v|_{t=0} = A \cos kx \\ v_t|_{t=0} = -\frac{1}{c} j_x|_{t=0} = \left(-\frac{1}{c}\right) \sqrt{\frac{c}{L}} A k (-\sin kx) \end{cases}$$

由达朗贝尔公式

$$v(x, t) = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x) dx$$

代入 $\varphi(x) = A \cos kx, \psi(x) = \frac{1}{c} \sqrt{\frac{c}{L}} A k \sin kx$ 可得

$$\begin{aligned} v(x, t) &= \frac{1}{2} A [\cos k(x+at) + \cos k(x-at)] + \frac{1}{2} \int_{x-at}^{x+at} A \sin kx dx \\ &= \frac{1}{2} A [\cos k(x+at) + \cos k(x-at)] + \frac{1}{2} (-A) \\ &\quad [\cos k(x+at) - \cos k(x-at)] \\ &= A \cos k(x-at) \end{aligned}$$

(2) 电流传播情况, 定解问题为

$$\begin{cases} j_{tt} - a^2 j_{xx} = 0 & (a^2 = \frac{1}{Lc}) \\ j|_{t=0} = \sqrt{\frac{c}{L}} A \cos kx \\ j_t|_{t=0} = -\frac{1}{L} v_x|_{t=0} = \left(-\frac{1}{L}\right) A k (-\sin kx) \end{cases}$$

由达朗贝尔公式, 并代入

$$\varphi(x) = \sqrt{\frac{c}{L}} A \cos kx, \psi(x) = \frac{Ak}{L} \sin kx$$

可得

$$\begin{aligned} j(x, t) &= \frac{1}{2} \sqrt{\frac{c}{L}} A [\cos k(x+at) + \cos k(x-at)] + \frac{1}{2a} \int_{x-a}^{x+a} \frac{Ak}{L} \sin kx dx \\ &= \frac{1}{2} \sqrt{\frac{c}{L}} A [\cos k(x+at) + \cos k(x-at)] + \frac{1}{2a} + \left(-\frac{A}{L}\right) \\ &\quad [\cos k(x+at) - \cos k(x-at)] \\ &= \sqrt{\frac{c}{L}} A \cos k(x-at) \end{aligned}$$

3. 在 $G/C = R/L$ 条件下求无限长传输上的电报方程的通解.

解: 电报方程为

$$\begin{cases} LCj_u - j_{xx} + (LG + RC)j_t + RGj = 0 \\ LCv_u - v_{xx} + (LG + RC)v_t + RGv = 0 \end{cases}$$

设方程通解为 y

令

$$y = ue^{\lambda x + \mu t}$$

则

$$y_x = e^{\lambda x + \mu t} \cdot (u_x + \lambda u)$$

$$y_t = e^{\lambda x + \mu t} \cdot (u_t + \mu u)$$

$$y_{xx} = e^{\lambda x + \mu t} (u_{xx} + 2\lambda u_x + \lambda^2 u)$$

$$y_u = e^{\lambda x + \mu t} (u_u + 2\mu u_t + \mu^2 u)$$

将其代入电报方程, 并消去 $e^{\lambda x + \mu t}$ 后得

$$LC(u_u + 2\mu u_t + \mu^2 u) - (u_{xx} + 2\lambda u_x + \lambda^2 u) + (LG + RC)(u_t + \mu u) + RG u = 0$$

合并同类项, 得

$$LCu_u - u_{xx} + [2\mu LC + (LG + RC)]u_t - 2\lambda u_x + [LC\mu^2 - \lambda^2 + \mu(LG + RC) + RG]u = 0$$

令 $\lambda = 0, \mu = -\frac{(LG + RC)}{2LC}$, 代入上面方程得

$$LCu_u - u_{xx} + \left[LC\left(-\frac{LG + RC}{2LC}\right)^2 + \left(-\frac{LG + RC}{2LC}\right) \cdot (LG + RC) + RG\right]u = 0$$

由于 $\frac{G}{C} = \frac{R}{L}$, 所以方程可化为

$$LCu_u - u_{xx} = 0, \text{ 即}$$

$$u_u - a^2 u_{xx} = 0 \quad (a^2 = \frac{1}{LC})$$

令初始条件

$$y|_{t=0} = \varphi(x), y_t|_{t=0} = \Psi(x)$$

则

$$u|_{t=0} = e^{-\lambda x - \mu t} y|_{t=0} = \varphi(x)$$

$$u_t|_{t=0} = e^{-\lambda x - \mu t} y_t|_{t=0} = -\mu u|_{t=0} = y_t|_{t=0} - \mu \varphi(x)$$

$$= y_t|_{t=0} - \left(-\frac{G + RC}{2LC}\right) \varphi(x) = \Psi(x) + \frac{R}{L} \varphi(x)$$

令
$$\Psi(x) + \frac{R}{L}\varphi(x) = \psi(x)$$

将 $\varphi(x)$ 、 $\psi(x)$ 代入达朗贝尔公式,得

$$u(x,t) = \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x) dx$$

通解
$$y(x,t) = e^{\mu} u = e^{-\frac{t}{\tau}} \left\{ \frac{1}{2}[\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x) dx \right\}$$

4. 无限长弦在点 $x = x_0$ 受到初始冲击,冲量为 I ,试求解弦的振动.

[提示: $u_t|_{t=0} = (I/\rho)\delta(x-x_0)$]

解:由题意得定解方程

$$\begin{cases} u_{xx} - a^2 u_{tt} = 0 & (a^2 = \frac{T}{\rho}) \\ u|_{t=0} = 0 \\ u_t|_{t=0} = \frac{I}{\rho} \delta(x-x_0) \end{cases}$$

将 $\varphi(x) = 0$, $\psi(x) = a^2 \delta(x-x_0)$ 代入达朗贝尔公式得

$$\begin{aligned} u(x,t) &= \frac{1}{2a} \int_{x-at}^{x+at} a^2 \delta(x-x_0) dx = \frac{I}{2a\rho} \int_{x-at}^{x+at} \delta(x-x_0) dx \\ &= \frac{I}{2a\rho} [H(x+at-x_0) - H(x-at-x_0)] \end{aligned}$$

即
$$u(x,t) = \frac{I}{2a\rho} [H(x-x_0+at) - H(x-x_0-at)]$$

5. 求解细圆锥形均质杆的纵振动. [提示: 泛定方程见 § 7.1 习题 2, 作变换 $u = v/x$.]

解:由题意得细圆锥形均质杆的纵振动方程为

$$u_{xx} - a^2 \frac{1}{x^2} \frac{\partial}{\partial x} (x^2 u_x) = 0 \quad (a^2 = \frac{Y}{\rho})$$

令
$$u(x,t) = \frac{v(x,t)}{x}$$

则
$$u_x = \frac{1}{x} v_x - \frac{v}{x^2}$$

$$\frac{\partial}{\partial x} (x^2 u_x) = \frac{\partial}{\partial x} (x v_x - v) = x v_{xx}$$

$$u_{xx} = \frac{v_{xx}}{x}$$

将以上代入振动方程得

$$\frac{1}{x} v_{xx} - \frac{a^2}{x^2} x v_{xx} = 0$$

即
$$v_{xx} - a^2 v_{xx} = 0$$

$v(x,t)$ 通解为

$$v(x,t) = f_1(x-at) + f_2(x+at)$$

$u(x, t)$ 通解为

$$u(x, t) = \frac{1}{2} [f_1(x - at) + f_2(x + at)]$$

6. 半无限长杆的端点受到纵向力 $F(t) = A \sin \omega t$ 作用, 求解杆的纵振动.

解: 由题意得定解问题为

$$\begin{cases} u_{xx} - a^2 u_{tt} = 0 & (x > 0) (a^2 = \frac{Y}{\rho}) \\ u|_{t=0} = \varphi(x) \\ u_t|_{t=0} = \psi(x) \\ u_x|_{x=0} = \frac{A}{YS} \sin \omega t \end{cases}$$

(1) 当 $x > at$, 认为端点的影响不能传到

则由达朗贝尔公式直接得到

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(x) dx$$

(2) 当 $x < at$, 考虑端点的影响, 对问题进行延拓, 设 $t = 0$ 时刻

$$\Phi(x) = \begin{cases} \varphi(x), & (x \geq 0) \\ \varphi_1(x), & (x < 0) \end{cases} \quad \Psi(x) = \begin{cases} \psi(x), & (x \geq 0) \\ \psi_1(x), & (x < 0) \end{cases}$$

代入达朗贝尔公式可得

$$u(x, t) = \frac{1}{2} [\varphi(x + at) + \varphi_1(x - at)] + \frac{1}{2a} \int_0^{x+at} \psi(x) dx + \frac{1}{2a} \int_{x-at}^0 \psi_1(x) dx$$

$u(x, t)$ 应满足边界条件, 即

$$u_x|_{x=0} = \frac{1}{2} \varphi'(at) + \frac{1}{2} \varphi_1'(-at) + \frac{1}{2a} \psi(at) - \frac{1}{2a} \psi_1(-at)$$

即

$$\varphi_1(x) = \varphi(-x), \psi_1(x) = \psi(-x) + \frac{2Aa}{YS} \sin \frac{\omega}{a} x$$

即可满足边界条件

$$u(x, t) = \frac{1}{2} \left[\varphi(x + at) + \varphi(at - x) + \frac{1}{2a} \int_{x-at}^0 \psi(-x) dx + \frac{1}{2a} \int_0^{x+at} \psi(x) dx + \frac{1}{2a} \int_{x-at}^0 \frac{2Aa}{YS} \sin \frac{\omega}{a} x dx \right]$$

即

$$u = \frac{1}{2} [\varphi(x + at) + \varphi(at - x)] + \frac{1}{2a} \int_0^{x+at} \psi(x) dx + \frac{1}{2} a \int_0^{at-x} \psi(x) dx + \frac{aA}{YS\omega} \cos\left(t - \frac{x}{a}\right) - \frac{aA}{YS\omega}$$

7. 求解半无限长理想传输线上电报方程的解, 端点通过电阻 R 而相接. 初始电压分布 $A \cos kx$, 初始电流分布 $\sqrt{C/L} A \cos kx$. 在什么条件下端点没有反射(这种情况叫作匹配)?

解: 理想传输线的定解问题为

$$\begin{cases} j_{xx} - a^2 j_{tt} = 0 & (x < 0) (a^2 = \frac{1}{LC}) \\ j|_{t=0} = \sqrt{\frac{C}{L}} A \cos kx \\ j_t|_{t=0} = -\frac{1}{L} v_x|_{t=0} = \frac{Ak}{L} \sin kx \\ v_{xx} - a^2 v_{tt} = 0 & (x < 0) \\ v|_{t=0} = A \cos kx \\ v_t|_{t=0} = -\frac{1}{C} j_x|_{t=0} = \frac{Ak}{\sqrt{LC}} \sin kx \end{cases}$$

(1) 当 $|x| > at$, 认为端点的影响不能传到, 则由达朗贝尔公式直接得到

$$\begin{aligned} j(x, t) &= \frac{1}{2} \sqrt{\frac{C}{L}} A [\cos k(x+at) + \cos k(x-at)] + \frac{1}{2a} \cdot \frac{Ak}{L} \int_{x-at}^{x+at} \sin kx dx \\ &= \frac{1}{2} \sqrt{\frac{C}{L}} A [\cos k(x+at) + \cos k(x-at)] + \frac{Ak}{2aL} \times \left(-\frac{1}{k}\right) \\ &\quad [\cos k(x+at) - \cos k(x-at)] \\ &= \sqrt{\frac{C}{L}} A \cos k(x-at) \end{aligned}$$

同理可求, $v(x, t) = A \cos k(x-at)$

(2) 当 $|x| < at$, 考虑端点的影响, 令

$$j(x, t) = \sqrt{\frac{C}{L}} A \cos k(x-at) + j_1(x+at)$$

$$v(x, t) = A \cos k(x-at) + v_1(x+at)$$

由 $j_t = -\frac{1}{L} v_x, v_t = -\frac{1}{C} j_x$ 得

$$\begin{cases} \sqrt{\frac{C}{L}} ka A \sin k(x-at) + a j_1'(x+at) = \frac{A}{L} \sin k(x-at) - v_1'(x+at) \\ Aa \sin k(x-at) + a v_1'(x+at) = \frac{1}{C} \sqrt{\frac{C}{L}} A \sin k(x-at) - j_1'(x+at) \end{cases}$$

解得

$$\sqrt{\frac{L}{C}} j_1'(x+at) = -v_1'(x+at)$$

由 j_1, v_1 相关联, 则可令

$$j(x, t) = \sqrt{\frac{C}{L}} A \cos k(x-at) + j_1(x+at)$$

$$v(x, t) = A \cos k(x-at) - \sqrt{\frac{L}{C}} j_1(x+at)$$

由边界条件 $v|_{x=0} = Rj|_{x=0}$, 则

$$A \cos kat - \sqrt{\frac{L}{C}} j_1(at) = R \sqrt{\frac{C}{L}} A \cos kat + R j_1(at)$$

解得

$$j_1(at) = \frac{\left(1 - R\sqrt{\frac{C}{L}}\right)A}{R + \sqrt{\frac{L}{C}}} \cos kat$$

即

$$j_1(at) = \sqrt{\frac{C}{L}} \cdot \frac{1 - R\sqrt{\frac{C}{L}}}{1 + R\sqrt{\frac{C}{L}}} A \cos kat$$

则

$$v_1(at) = \frac{1 - R\sqrt{\frac{C}{L}}}{1 + R\sqrt{\frac{C}{L}}} A \cos kat$$

$$j(x, t) = \sqrt{\frac{C}{L}} A \cos k(x - at) + \sqrt{\frac{C}{L}} \cdot \frac{1 - R\sqrt{\frac{C}{L}}}{1 + R\sqrt{\frac{C}{L}}} A \cos k(x + at)$$

$$v(x, t) = A \cos k(x - at) + \frac{1 - R\sqrt{\frac{C}{L}}}{1 + R\sqrt{\frac{C}{L}}} A \cos k(x + at)$$

由表达式可知 $(x + at)$ 项为反射波, 要使反射波不存在, 则 $1 - R\sqrt{\frac{C}{L}} = 0$, 即

$$R_0 = \sqrt{\frac{L}{C}}.$$

R_0 称为传输线的特性阻抗.

8. 半无限长弦的初始位移和初始速度都是零, 端点作微小振动 $u|_{x=0} = A \sin \omega t$. 求解弦的振动.

解: 由题意得定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 & (x > 0) \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \\ u|_{x=0} = A \sin \omega t \end{cases}$$

当 $x > at$, 则端点的影响未传到

$$u(x, t) = 0$$

当 $x < at$, 考虑端点的影响, 对问题进行延拓, 设 $t = 0$ 时刻

$$\Phi(x) = \begin{cases} 0, & (x \geq 0) \\ \varphi(x), & (x < 0) \end{cases}, \Psi(x) = \begin{cases} 0, & (x \geq 0) \\ \psi(x), & (x < 0) \end{cases}$$

代入达朗贝尔公式可得

$$u(x, t) = \frac{1}{2} \varphi(x - at) + \frac{1}{2a} \int_{x-at}^0 \psi(x) dx$$

$u(x, t)$ 应满足边界条件, 即

$$\begin{aligned}u|_{x=0} &= \frac{1}{2}\varphi(-at) + \frac{1}{2a}\int_{-a}^0 \psi(x)dx \\&= A\sin \omega t\end{aligned}$$

即 $\varphi(x) = 2A\sin\left(-\frac{\omega}{a}x\right)$, $\psi(x) = 0$ 即可满足边界条件

所以

$$u(x, t) = A\sin \omega\left(t - \frac{x}{a}\right)$$

第八章 分离变数法

本章知识要点

齐次方程的分离变数法

(一) 分离变数法

将方程的解假设为两个互不影响的函数乘积,其中各函数对应各自的自变量互不影响,将这样的形式解代回原方程进行求解的方法,叫做分离变数法.

(二) 分离变数法解题步骤

首先将分离变数形式的试探解代入偏微分方程,从而将其分为若干个常微分方程,同时,代入定解条件把它转化为常微分方程的定解条件,并与相应的常微分方程构成本征值问题.

非齐次振动方程和输运方程

(一) 傅里叶级数法

将方程的解本身展开为傅里叶级数,如 $u(x,t) = \sum_n T_n(t) X_n(x)$,将 $X_n(x)$ 看作基本函数族, T_n 为其傅里叶系数,将待定解代入泛定方程,尝试分离出 $T_n(t)$ 的常微分方程然后求解,这样的方法称为傅里叶级数法.

(二) 冲量定理法

对初始条件为 0 的方程,将外加力看作是一系列瞬时力的叠加,则定解问题的解也是瞬时力所引起的振动的叠加,则 $u(x,t) = \int_0^t v(x,t,\tau) d\tau$,其中 v 满足齐次方程,且 $v|_{t=\tau} = 0, v|_{t=\tau} = f(x,\tau)$,这样的方法称为冲量定理法.

非齐次边界条件的处理

(一) 一般处理方法

寻找一个简单的函数,满足非齐次边界条件定解问题,利用叠加原理可将原问题的解分解为该简单函数和一个齐次边界条件定解问题的解.

(二) 特殊处理方法

根据定解问题的特点,选取特殊的函数分解原问题可将求解过程大大简化.

泊松方程

1. 泊松方程

$$\Delta u = f(x, y, z)$$

2. 特解法

不管边界条件,任取一个泊松方程的特解 v , 然后令 $u = v + w$, 问题便转化为求解 w 的拉普拉斯方程的问题, 这样的方法叫作特解法.

习题全解

§ 8.1 齐次方程的分离变数法

1. 长为 l 的弦, 两端固定, 弦中张力为 T . 在距一端为 x_0 的一点以力 F_0 把弦拉开, 然后突然撤除这力. 求解弦的振动.

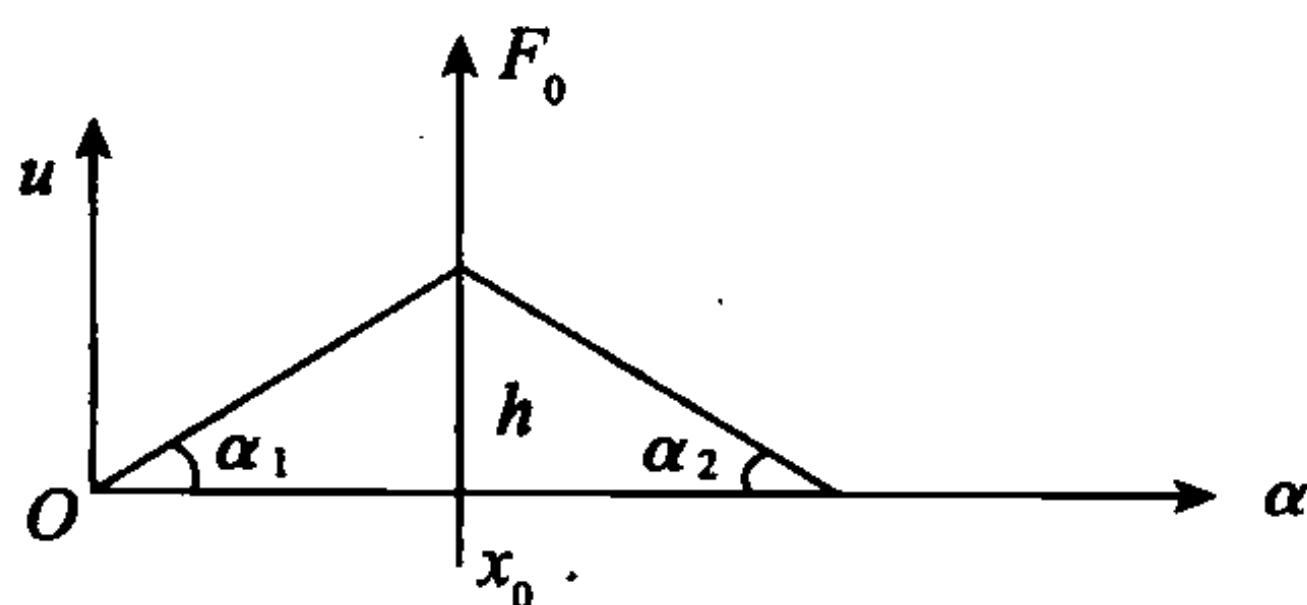


图 8-1-1

解: 由题意得方程

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (0 < x < l) (a^2 = T/p) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = \begin{cases} \frac{F_0}{T} \frac{l-x_0}{l} x, & (0 < x < x_0) \\ \frac{F_0}{T} \frac{x_0}{l} (l-x), & (x_0 < x < l) \end{cases} \\ u_t|_{t=0} = 0 \end{cases}$$

令 $u(x, t) = X(x)T(t)$, 代入泛定方程得

$$\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda^2 \quad (\lambda > 0)$$

进一步可得

$$T'' + a^2 \lambda^2 T = 0$$

$$X'' + \lambda^2 x = 0, X(0) = X(l) = 0$$

对时间变数 t , 解得

$$T(t) = A \cos \lambda at + B \sin \lambda at$$

对空间变数 x , 解得

$$X(x) = C \cos \lambda x + D \sin \lambda x$$

由 $X(0) = 0$ 可知 $C = 0$, 由 $X(l) = 0$ 可知, $\sin \lambda l = 0, \lambda l = n\pi \quad (n = 1, 2, 3 \cdots)$

$$\lambda = \frac{n}{l} \pi \quad (n = 1, 2, 3 \cdots)$$

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x$$

由初始条件 $u_t|_{t=0} = 0$ 知

$$\begin{aligned} u_t|_{t=0} &= \left[\sum_{n=1}^{\infty} \left(-A_n \frac{n\pi a}{l} \sin \frac{n\pi a}{l} t + B_n \frac{n\pi a}{l} \cos \frac{n\pi a}{l} t \right) \sin \frac{n\pi}{l} x \right] \Big|_{t=0} \\ &= \sum_{n=1}^{\infty} B_n \frac{n\pi a}{l} \sin \frac{n\pi}{l} x = 0 \end{aligned}$$

$$B_n = 0$$

则

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{l} t \sin \frac{n\pi}{l} x$$

由初始条件 $u|_{t=0}$ 知

$$u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = \begin{cases} \frac{F_0}{T} \cdot \frac{l-x_0}{l} x, & (0 < x < x_0) \\ \frac{F_0}{T} \cdot \frac{x_0}{l} (l-x), & (x_0 < x < l) \end{cases}$$

A_n 为等号右边傅立叶展开系数

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^{x_0} \frac{F_0}{T} \frac{l-x_0}{l} x \sin \frac{n\pi x}{l} dx + \frac{2}{l} \int_{x_0}^l \frac{F_0}{T} \frac{x_0}{l} (l-x) \sin \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \left\{ \frac{F_0}{T} \frac{l-x_0}{l} \left[\frac{l}{n^2 \pi^2} \sin \frac{n\pi x}{l} - \frac{l^2}{n^2 \pi^2} \frac{n\pi x}{l} \cos \frac{n\pi x}{l} \right] \Big|_0^{x_0} \right. \\ &\quad \left. - \frac{F_0}{T} \frac{x_0}{l} \frac{l^2}{n^2 \pi^2} \left[\sin \frac{n\pi x}{l} - \frac{n\pi x}{l} \cos \frac{n\pi x}{l} \right] \Big|_{x_0}^l \right\} \\ &= \frac{2}{l} \left\{ \frac{F_0}{T} \frac{(l-x_0)l}{n^2 \pi^2} \left[\sin \frac{n\pi x_0}{l} - \frac{n\pi x_0}{l} \cos \frac{n\pi x_0}{l} \right] - \frac{F_0}{T} \frac{x_0}{n\pi} \left[\cos n\pi - \cos \frac{n\pi x_0}{l} \right] \right. \\ &\quad \left. - \frac{F_0}{T} \frac{x_0 l}{n^2 \pi^2} \left[-n\pi \cos n\pi - \sin \frac{n\pi x_0}{l} + \frac{n\pi x_0}{l} \cos \frac{n\pi x_0}{l} \right] \right\} \\ &= \frac{2}{l} \left\{ \left[\frac{F_0 l(l-x_0)}{T n^2 \pi^2} + \frac{F_0 l x_0}{T n^2 \pi^2} \right] \sin \frac{n\pi x_0}{l} + \left[\frac{F_0 l x_0}{T n\pi} - \frac{F_0 x_0^2}{T n\pi} - \frac{F_0 (l-x_0)x_0}{T n\pi} \right] \right. \\ &\quad \left. \cos \frac{n\pi x_0}{l} - \left(\frac{F_0 l x_0}{T n\pi} - \frac{F_0 x_0 l}{T n\pi} \right) \cos n\pi \right\} \\ &= \frac{2}{l} \frac{F_0}{T} \frac{l^2}{n^2 \pi^2} \sin \frac{n\pi x_0}{l} \\ &= \frac{2F_0 l}{n^2 T \pi^2} \sin \frac{n\pi x_0}{l} \end{aligned}$$

所以

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2F_0 l}{T \pi^2} \frac{1}{n^2} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi}{l} x \cos \frac{n\pi a}{l} t$$

2. 求解细杆导热问题, 杆长 l , b 为常数, 两端保持为零度, 初始温度分布

$$u|_{t=0} = bx(l-x)/l^2.$$

解: 由题意得方程

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (0 < x < l) \quad (a^2 = \frac{k}{c\rho}) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = \frac{bx(l-x)}{l^2} \end{cases}$$

令 $u(x, t) = X(x)T(t)$, 代入泛定方程得

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda \quad (\lambda > 0)$$

进一步可得

$$T' + a^2 \lambda T = 0$$

$$X'' + \lambda X = 0, X(0) = X(l) = 0$$

对时间变数 t , 解得

$$T(t) = Ae^{-a^2 \lambda t}$$

对空间变数 x , 解得

$$X(x) = B\cos\sqrt{\lambda}x + C\sin\sqrt{\lambda}x$$

由 $X(0) = 0$ 可知 $B = 0$, 由 $X(l) = 0$ 可知 $\sin\sqrt{\lambda}l = 0, \sqrt{\lambda}l = n\pi (n = 1, 2, 3, \dots)$

$$\sqrt{\lambda} = \frac{n}{l}\pi, \lambda = \frac{n^2\pi^2}{l^2}$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2\pi^2 a^2}{l^2} t} \sin \frac{n\pi x}{l}$$

由初始条件 $u|_{t=0} = \frac{bx(l-x)}{l^2}$ 知,

$$u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = \frac{bx(l-x)}{l^2}$$

A_n 为等号右边傅立叶展开系数

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l \frac{bx(l-x)}{l^2} \sin \frac{n\pi x}{l} dx = \frac{2b}{l^3} \int_0^l x(1-x) \sin \frac{n\pi x}{l} dx \\ &= \frac{2b}{l^3} \left\{ \frac{l^3}{n^2\pi^2} \sin \frac{n\pi x}{l} \Big|_0^l - \frac{l^2}{n\pi} x \cos \frac{n\pi x}{l} \Big|_0^l - \frac{xl^2}{n^2\pi^2} \left(2\sin \frac{n\pi x}{l} - \frac{n\pi x}{l} \cos \frac{n\pi x}{l} \right) \Big|_0^l \right. \\ &\quad \left. + \frac{2l}{n^3\pi^3} \cos \frac{n\pi x}{l} \Big|_0^l \right\} \\ &= \frac{2b}{l^3} \left\{ -\frac{l^3}{n\pi} (-1)^n + \frac{l^3}{n\pi} (-1)^n + \frac{2l^3}{n^3\pi^3} [(-1)^n - 1] \right\} \\ &= \begin{cases} \frac{8b}{\pi^3 n^3} & (n \text{ 为奇数}) \\ 0 & (n \text{ 为偶数}) \end{cases} \end{aligned}$$

所以
$$u(x, t) = \sum_{k=0}^{\infty} \frac{8b}{\pi^3 (2k+1)^3} \exp \left(-\frac{(2k+1)^2 \pi^2 a^2 t}{l^2} \right) \sin \frac{(2k+1)\pi x}{l}$$

3. 两端固定弦, 长为 l . ① 用宽为 2δ 的平面锤敲击弦 $x = x_0$ 点; ② 用宽为 2δ 的余弦式凸面锤敲击弦的 $x = x_0$ 点. 求解弦的振动.

解:(1) 用平面锤敲击,定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = \begin{cases} 0, (0 < x < x_0 - \delta; x_0 + \delta < x < l) \\ v_0, (x_0 - \delta < x < x_0 + \delta) \end{cases} \end{cases}$$

由边界条件为第一类齐次边界条件,可知其本征函数为 $\sin \frac{n\pi}{l}x$,

$$\text{设问题的解 } u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l}t + B_n \sin \frac{n\pi a}{l}t \right) \sin \frac{n\pi}{l}x$$

由初始条件

$$\begin{cases} u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l}x = 0 \\ u_t|_{t=0} = \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi}{l}x = \begin{cases} 0 & (0 < x < x_0 - \delta; x_0 + \delta < x < l) \\ v_0 & (x_0 - \delta < x < x_0 + \delta) \end{cases} \end{cases}$$

则 $A_n = 0$,

$$\begin{aligned} B_n &= \frac{l}{n\pi a} \frac{2}{l} \int_{x_0-\delta}^{x_0+\delta} v_0 \sin \frac{n\pi}{l}x dx \\ &= \frac{2v_0}{n\pi a} \frac{l}{n\pi} \left(-\cos \frac{n\pi}{l}x \right) \Big|_{x_0-\delta}^{x_0+\delta} \\ &= \frac{2v_0 l}{n^2 \pi^2 a} \left[\cos \frac{n\pi}{l}(x_0 - \delta) - \cos \frac{n\pi}{l}(x_0 + \delta) \right] \end{aligned}$$

由三角函数的和差化积公式,可得 $\cos \frac{n\pi}{l}(x_0 - \delta) - \cos \frac{n\pi}{l}(x_0 + \delta) = 2 \sin \frac{n\pi}{l}x_0 \sin \frac{n\pi}{l}\delta$

$$\begin{aligned} B_n &= \frac{4v_0 l}{n^2 \pi^2 a} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \\ u(x, t) &= \frac{4v_0 l}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x_0}{l} \sin \frac{n\pi \delta}{l} \sin \frac{n\pi a}{l}t \sin \frac{n\pi}{l}x \end{aligned}$$

(2) 用余弦锤敲击,定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = \begin{cases} 0, (0 < x < x_0 - \delta, x_0 + \delta < x < l) \\ v_0 \cos \frac{x-x_0}{2\delta} \pi, (x_0 - \delta < x < x_0 + \delta) \end{cases} \end{cases}$$

由边界条件为第一类齐次边界条件,可知其本征函数为 $\sin \frac{n\pi}{l}x$

$$\text{设问题的解 } u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l}t + B_n \sin \frac{n\pi a}{l}t \right) \sin \frac{n\pi}{l}x$$

由初始条件

$$u|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l} x = 0$$

$$u_t|_{t=0} = \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi}{l} x = \begin{cases} 0 & (0 < x < x_0 - \delta < x < l) \\ v_0 \cos \frac{x-x_0}{2\delta} \pi & (x_0 - \delta < x < x_0 + \delta) \end{cases}$$

则 $A_n = 0$,

$$B_n = \frac{l}{n\pi a} \frac{2}{l} \int_{x_0-\delta}^{x_0+\delta} v_0 \cos \frac{x-x_0}{2\delta} \pi \sin \frac{n\pi}{l} x dx$$

由三角函数积化和差公式,可得

$$\begin{aligned} \cos \frac{x-x_0}{2\delta} \pi \sin \frac{n\pi}{l} x &= \frac{\sin\left(\frac{n}{l}x + \frac{x-x_0}{2\delta}\pi\right) + \sin\left(\frac{n}{l}x - \frac{x-x_0}{2\delta}\pi\right)}{2} \\ B_n &= \frac{v_0}{n\pi a} \int_{x_0-\delta}^{x_0+\delta} \left\{ \sin\left[\left(\frac{n}{l} + \frac{1}{2\delta}\right)x + \frac{x_0}{2\delta}\pi\right] + \sin\left[\left(\frac{n}{l} - \frac{1}{2\delta}\right)x + \frac{x_0}{2\delta}\pi\right] \right\} \cdot dx \\ &= \frac{v_0}{n\pi a} \left\{ \frac{1}{\pi} \left(\frac{-2\delta l}{2\delta n + l}\right) \cdot \cos\left[\left(\frac{n}{l} + \frac{1}{2\delta}\right)x - \frac{x_0}{2\delta}\pi\right] \Big|_{x_0-\delta}^{x_0+\delta} \right. \\ &\quad \left. + \frac{1}{\pi} \left(\frac{-2\delta l}{2\delta n - l}\right) \cdot \cos\left[\left(\frac{n}{l} - \frac{1}{2\delta}\right)x + \frac{x_0}{2\delta}\pi\right] \Big|_{x_0-\delta}^{x_0+\delta} \right\} \\ &= \frac{v_0}{n\pi^2 a} 4\delta l \left[\frac{\sin \frac{n}{l}(x_0 + \delta)\pi + \sin \frac{n}{l}(x_0 - \delta)\pi}{2\delta n + l} - \frac{\sin \frac{n}{l}(x_0 + \delta)\pi + \sin \frac{n}{l}(x_0 - \delta)\pi}{2\delta n - l} \right] \\ &= \frac{v_0}{n\pi^2 a} 4\delta l \cdot \frac{\sin \frac{n\pi}{l} x_0 \cos \frac{n\pi\delta}{l}}{l^2 - 4\delta^2 n^2} (2l) \\ &= \frac{8v_0\delta}{n\pi^2 a} \frac{1}{1 - \left(\frac{2\delta n}{l}\right)^2} \sin \frac{n\pi}{l} x_0 \cos \frac{n\pi\delta}{l} \\ u(x, t) &= \frac{8v_0\delta}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n} \frac{1}{1 - \left(\frac{2\delta n}{l}\right)^2} \sin \frac{n\pi x_0}{l} \cos \frac{n\pi\delta}{l} \sin \frac{n\pi a t}{l} \sin \frac{n\pi x}{l} \end{aligned}$$

4. 长为 l 的均匀杆, 两端受压从而长度缩为 $l(1-2\epsilon)$. 放手后自由振动, 求解杆的纵振动.

解: 由题得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 & (0 < x < l) \\ u_x|_{x=0} = u_x|_{x=l} = 0 \\ u|_{t=0} = 2\epsilon\left(\frac{l}{2} - x\right) \\ u_t|_{t=0} = 0 \end{cases}$$

令 $u(x, t) = X(x)T(t)$, 代入泛定方程得

$$\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda^2 \quad (\lambda > 0)$$

进一步可得

$$\begin{cases} T'' + a^2 \lambda^2 T = 0 \\ X'' + \lambda^2 X = 0, X'(0) = X'(l) = 0 \end{cases}$$

对时间变数 t , 解得

$$T(t) = A \cos \lambda a t + B \sin \lambda a t$$

对空间变数 X , 解得 $X(x) = C \cos \lambda x + D \sin \lambda x$, 则 $X'(x) = -C \lambda \sin \lambda x + D \lambda \cos \lambda x$

由 $X'(0) = 0$ 可知, $D = 0$; 由 $X'(l) = 0$ 可知 $\sin \lambda l = n\pi (n = 1, 2, 3, \dots)$

$$\lambda = \frac{n}{l} \pi \quad (n = 1, 2, 3, \dots)$$

$$u(x, t) = \sum_{n=0}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right) \cos \frac{n\pi x}{l}$$

由初始条件

$$u|_{t=0} = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi x}{l} = 2\epsilon \left(\frac{l}{2} - x \right)$$

$$u_t|_{t=0} = \sum_{n=0}^{\infty} \frac{l}{n\pi a} B_n \cos \frac{n\pi x}{l} = 0$$

则 $B_n = 0$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l 2\epsilon \left(\frac{l}{2} - x \right) \cos \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \int_0^l \epsilon l \cos \frac{n\pi x}{l} dx - \frac{2}{l} \int_0^l 2\epsilon x \cos \frac{n\pi x}{l} dx \\ &= \epsilon l \left(\frac{l}{n\pi} \right) \sin \frac{n\pi x}{l} \Big|_0^l = 0 \end{aligned}$$

$$\begin{aligned} \int_0^l 2\epsilon x \cos \frac{n\pi x}{l} dx &= 2\epsilon x \left(\frac{l}{n\pi} \right) \sin \frac{n\pi x}{l} \Big|_0^l - \int_0^l 2\epsilon \left(\frac{l}{n\pi} \right) \sin \frac{n\pi x}{l} dx \\ &= -\frac{2\epsilon l}{n\pi} \left(\frac{-l}{n\pi} \right) \cos \frac{n\pi x}{l} \Big|_0^l = \frac{2\epsilon l^2}{n^2 \pi^2} [(-1)^n - 1] \end{aligned}$$

$$A_n = -\frac{4\epsilon l}{n^2 \pi^2} [(-1)^n - 1] = \begin{cases} \frac{8\epsilon l}{\pi^2 (2k+1)^2} & \text{当 } n = 2k+1 (k = 0, 1, 2, 3, \dots), \\ 0 & \text{当 } n = 2k \end{cases}$$

$$u(x, t) = \frac{8\epsilon l}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \frac{(2k+1)\pi a t}{l} \cos \frac{(2k+1)\pi x}{l}$$

5. 长为 l 的杆, 一端固定, 另一端受力 F_0 而伸长. 求解杆在放手后的振动.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l) \\ u|_{x=0} = 0, u_x|_{x=l} = 0 \\ u|_{t=0} = \int_0^x u_x dx = \int_0^x \frac{F_0}{YS} dx = \frac{F_0 x}{YS} \\ u_t|_{t=0} = 0 \end{cases}$$

令 $u(x, t) = X(x)T(t)$, 代入泛定方程得

$$\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda^2 \quad (\lambda > 0)$$

进一步可得

$$\begin{cases} T'' + a^2 \lambda^2 T = 0 \\ X'' + \lambda^2 X = 0 \quad X(0) = X'(l) = 0 \end{cases}$$

对时间变数 t , 解得 $T(t) = A \cos \lambda at + B \sin \lambda at$

对空间变数 x , 解得 $X(x) = C \cos \lambda x + D \sin \lambda x$

则 $X'(x) = -C \lambda \sin \lambda x + D \lambda \cos \lambda x$

由 $X(0) = 0$ 可知 $C = 0$; 由 $X'(l) = 0$ 可知, $\cos \lambda l = 0, \lambda l = \frac{2n+1}{2} \pi (n = 0, 1, 2, \dots)$

$$\lambda = \frac{2n+1}{2} \frac{\pi}{l}$$

$$u(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \left(n + \frac{1}{2} \right) \frac{\pi at}{l} + B_n \sin \left(n + \frac{1}{2} \right) \frac{\pi at}{l} \right] \sin \left(n + \frac{1}{2} \right) \frac{\pi x}{l}$$

由初始条件

$$u|_{t=0} = \sum_{n=0}^{\infty} A_n \sin \left(n + \frac{1}{2} \right) \frac{\pi x}{l} = \frac{F_0 x}{YS}$$

$$u_t|_{t=0} = \sum_{n=0}^{\infty} \frac{l}{\left(n + \frac{1}{2} \right) \pi a} B_n \sin \left(n + \frac{1}{2} \right) \frac{\pi x}{l} = 0$$

则 $B_n = 0$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l \frac{F_0 x}{YS} \cdot \sin \left(n + \frac{1}{2} \right) \frac{\pi x}{l} dx \\ &= \frac{2F_0}{lYS} \left[-\frac{lx}{\left(n + \frac{1}{2} \right) \pi} \cos \left(n + \frac{1}{2} \right) \frac{\pi x}{l} \Big|_0^l + \int_0^l \frac{l}{\left(n + \frac{1}{2} \right) \pi} \cos \left(n + \frac{1}{2} \right) \frac{\pi x}{l} dx \right] \\ &= \frac{2F_0}{lYS} \frac{l}{\left(n + \frac{1}{2} \right) \pi} \frac{l}{\left(n + \frac{1}{2} \right) \pi} \sin \left(n + \frac{1}{2} \right) \frac{\pi x}{l} \Big|_0^l \\ &= \frac{2F_0 l}{YS \pi^2} \frac{1}{\left(n + \frac{1}{2} \right)^2} (-1)^n \end{aligned}$$

$$u(x, t) = \frac{8F_0 l}{YS \pi^2} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^2} \cos \frac{\left(n + \frac{1}{2} \right) \pi at}{l} \sin \frac{\left(n + \frac{1}{2} \right) \pi x}{l}$$

6. 长为 l 的理想传输线, 远端开路. 先把传输线充电到电压 v_0 , 然后把近端短路. 求解线上电压 $v(x, t)$.

解: 由题意得定解问题为

$$\begin{cases} v_{xx} - a^2 v_{tt} = 0 & (0 < x < l), \quad (a^2 = \frac{1}{LC}) \\ v|_{x=0} = 0 \\ v_x|_{x=l} = -\left(R + L \frac{\partial}{\partial t}\right) j|_{x=l} = 0 \\ v|_{t=0} = v_0 \\ v_t|_{t=0} = -\frac{1}{C} j_x|_{t=0} = 0 \end{cases}$$

由边界条件为第一类和第二类齐次边界条件,可知其本征函数为 $\sin \frac{n+\frac{1}{2}}{l}\pi x$

$$\text{设问题的解 } v(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{n+\frac{1}{2}}{l} a \pi t + B_n \sin \frac{n+\frac{1}{2}}{l} a \pi t \right] \sin \frac{n+\frac{1}{2}}{l} \pi x$$

由初始条件

$$v|_{t=0} = \sum_{n=0}^{\infty} A_n \sin \frac{n+\frac{1}{2}}{l} \pi x = v_0$$

$$v_t|_{t=0} = \sum_{n=0}^{\infty} \frac{(n+\frac{1}{2})\pi a}{l} B_n \sin \frac{n+\frac{1}{2}}{l} \pi x = 0$$

则 $B_n = 0$

$$A_n = \frac{2}{l} \int_0^l v_0 \sin \frac{n+\frac{1}{2}}{l} \pi x dx = -\frac{2v_0}{l} \frac{l}{(n+\frac{1}{2})\pi} \cos \frac{n+\frac{1}{2}}{l} \pi x \Big|_0^l = \frac{4v_0}{(2n+1)\pi}$$

$$\text{所以 } v(x, t) = \frac{4v_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{(2n+1)} \sin \frac{n+\frac{1}{2}}{l} \pi x \cos \frac{(n+\frac{1}{2})\pi a t}{l}$$

7. 长为 l 的杆,上端固定在电梯天花板,杆身竖直,下端自由. 电梯下降,当速度为 v_0 时突然停止. 求解杆的振动.

解:由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, (0 < x < l) \\ u|_{x=0} = u_x|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = v_0 \end{cases}$$

由边界条件为第一类和第二类齐次边界条件,可知其本征函数为 $\sin \frac{n+\frac{1}{2}}{l}\pi x$

$$\text{设问题的解 } u(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{n+\frac{1}{2}}{l} a \pi t + B_n \sin \frac{n+\frac{1}{2}}{l} a \pi t \right] \sin \frac{n+\frac{1}{2}}{l} \pi x$$

由初始条件

$$u|_{t=0} = \sum_{n=0}^{\infty} A_n \sin \frac{n+\frac{1}{2}}{l} \pi x = 0$$

$$u_t|_{t=0} = \sum_{n=0}^{\infty} \frac{(n+\frac{1}{2})\pi a}{l} B_n \sin \frac{n+\frac{1}{2}}{l} \pi x = v_0$$

则 $A_n = 0$,

$$B_n = \frac{1}{\left(n + \frac{1}{2}\right)\pi a} \frac{2}{l} \int_0^l v_0 \sin \frac{n + \frac{1}{2}}{l} \pi x dx = \frac{2v_0}{\left(n + \frac{1}{2}\right)\pi a} \frac{(-l)}{\left(n + \frac{1}{2}\right)\pi} \cos \frac{n + \frac{1}{2}}{l} \pi x \Big|_0^l$$

$$= \frac{2v_0 l}{\left(n + \frac{1}{2}\right)^2 \pi^2 a}$$

$$u(x, t) = \frac{2v_0 l}{\pi^2 a} \sum_{n=0}^{\infty} \frac{1}{\left(n + \frac{1}{2}\right)^2} \sin \frac{n + \frac{1}{2}}{l} a \pi t \sin \frac{n + \frac{1}{2}}{l} \pi x$$

8. 在铀块中,除了中子的扩散运动以外,还进行着中子的增殖过程,每秒钟在单位体积中产生的中子数正比于该处的中子浓度 u ,从而可表为 βu (β 是表示增殖快慢的常数). 研究厚度为 l 的层状铀块. 求临界厚度. (铀块厚度超过临界厚度,则中子浓度将随着时间而急剧增长以致铀块爆炸,原子弹爆炸就是这个原理.)

解:由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} - \beta u = 0, & (0 \leq x \leq l), \quad (a^2 = D) \\ u|_{x=0} = u|_{x=l} = 0 \end{cases}$$

其中 u 为中子浓度,令 $u(x, t) = X(x)T(t)$,代入泛定方程是

$$\frac{T' - \beta T}{a^2 T} = \frac{X''}{X} = -\lambda^2 \quad (\lambda > 0)$$

进一步可得

$$\begin{cases} T' + (a^2 \lambda^2 - \beta) T = 0 \\ X'' + \lambda^2 X = 0, X(0) = X(l) = 0 \end{cases}$$

对空间变数 x ,解得

$$X(x) = B \cos \lambda x + C \sin \lambda x$$

由边界条件

$$\begin{cases} X(0) = B = 0 \\ X(l) = B \cos \lambda l + C \sin \lambda l = 0 \end{cases}$$

$$B = 0, \sin \lambda l = 0, \lambda l = n\pi \quad (n = 1, 2, 3, \dots)$$

$$\lambda = \frac{n\pi}{l}$$

对时间变数 t ,解得

$$T(t) = A e^{-(a^2 \lambda^2 - \beta)t}$$

综上所述,

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{a^2 n^2 \pi^2}{l^2} - \beta\right)t} \sin \frac{n\pi}{l} x$$

对时间指数项进行讨论: ($n = 1$ 时)

① 当 $\beta - \frac{a^2 \pi^2}{l^2} > 0$, 则浓度 u 随时间呈指数增加,最终爆炸

② 当 $\beta - \frac{a^2 \pi^2}{l^2} < 0$, 则浓度 u 随时间呈指数减少,最终熄灭

③ 当 $\beta - \frac{a^2 \pi^2}{l^2} = 0$, 则浓度 u 不随时间发生变化,此时厚度即临界厚度

即

$$L = \frac{a\pi}{\sqrt{\beta}}$$

9. 求解薄膜的恒定表面浓度扩散问题. 薄膜厚度为 l , 杂质从两面进入薄膜. 由于薄膜周围气氛中含有充分的杂质, 薄膜表面上的杂质浓度得以保持为恒定的 N_0 , 对于较大的 t , 把所得答案简化.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{x=0} = N_0 \\ u|_{x=l} = N_0 \\ u|_{t=0} = 0 \end{cases}$$

令 $u(x, t) = v(x, t) + N_0$, 则定解问题转化为

$$\begin{cases} v_t - a^2 v_{xx} = 0 \\ v|_{x=0} = v|_{x=l} = 0 \\ v|_{t=0} = -N_0 \end{cases}$$

由边界条件为第一类齐次边界条件, 可知本征函数为 $\sin \frac{n\pi}{l}x$,

设问题的解为
$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi}{l}x$$

由初始条件

$$v|_{t=0} = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{l}x = -N_0$$

则 A_n 为 $-N_0$ 的傅里叶系数

$$A_n = \frac{2}{l} \int_0^l (-N_0) \sin \frac{n\pi}{l}x dx = \frac{2N_0}{l} \frac{l}{n\pi} \cos \frac{n\pi}{l}x \Big|_0^l = \frac{2N_0}{n\pi} [(-1)^n - 1]$$

$$= \begin{cases} -\frac{4N_0}{\pi(2k+1)} & \text{当 } n = 2k+1, (k=0, 1, 2, 3\cdots) \\ 0 & \text{当 } n = 2k \end{cases}$$

$$v(x, t) = -\frac{4N_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin \frac{(2k+1)\pi}{l}x e^{-\frac{(2k+1)^2 \pi^2 a^2}{l^2} t}$$

$$u(x, t) = v(x, t) + N_0 = N_0 - \frac{4N_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} e^{-\frac{(2k+1)^2 \pi^2 a^2}{l^2} t} \sin \frac{(2k+1)\pi}{l}x$$

当 t 很大时, 时间指数项因子由于随时间呈指数衰减而变得很小, u 的级数收敛很快, 故可忽略 $k \geq 1$ 各项的影响, 则

$$u(x, t) = N_0 - \frac{4N_0}{\pi} e^{-\frac{\pi^2 a^2}{l^2} t} \sin \frac{\pi x}{l}$$

10. 把上题改为限定源扩散. 这是说, 薄膜两面的表层已含有一定的杂质, 比方说, 每单位表面积下杂质总量 Φ_0 , 但此外不再有杂质进入薄膜.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0, & (a^2 = D)(0 \leq x \leq l) \\ u_x|_{x=0} = u_x|_{x=l} = 0 \\ u|_{t=0} = \begin{cases} \Phi_0 \delta(x^+), & (0 < x < \epsilon) \\ 0, & (\epsilon < x < l - \epsilon) \\ \Phi_0 \delta[(x-l)^-], & (l - \epsilon < x < l) \end{cases} \end{cases}$$

其中, ϵ 为小量, 由边界条件为第二类齐次边界条件, 可知其本征函数为 $\cos \frac{n\pi}{l}x$,

设问题的解

$$u(x, t) = \sum_{n=0}^{\infty} A_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \cos \frac{n\pi}{l} x$$

由初始条件

$$u|_{t=0} = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{l} x = \begin{cases} \Phi_0 \delta(x^+), & (0 < x < \epsilon) \\ 0, & (\epsilon < x < l - \epsilon) \\ \Phi_0 \delta[(x-l)^-], & (l - \epsilon < x < l) \end{cases}$$

则 A_n 是等号右边的傅里叶系数

$$\begin{aligned} A_0 &= \frac{1}{l} \int_0^l u|_{t=0} dx = \frac{1}{l} \left[\int_0^\epsilon \Phi_0 \delta(x^+) dx + \int_{l-\epsilon}^l \Phi_0 \delta[(x-l)^-] dx \right] \\ &= \frac{1}{l} [\Phi_0 + \Phi_0] = \frac{2}{l} \Phi_0 \end{aligned}$$

$$\begin{aligned} A_n &= \frac{2}{l} \left[\int_0^\epsilon \Phi_0 \delta(x^+) \cos \frac{k\pi x}{l} dx + \int_{l-\epsilon}^l \Phi_0 \delta[(x-l)^-] \cos \frac{k\pi x}{l} dx \right] \\ &= \frac{2}{l} [\Phi_0 + (-1)^n \Phi_0] \end{aligned}$$

$$= \begin{cases} \frac{4}{l} \Phi_0 & \text{当 } n=2k \quad (k=1, 2, \dots) \\ 0 & \text{当 } n=2k-1 \end{cases}$$

$$u(x, t) = \frac{2\Phi_0}{l} + \frac{4\Phi_0}{l} \sum_{k=1}^{\infty} e^{-\frac{4k^2 \pi^2 a^2}{l^2} t} \cos \frac{2k\pi x}{l}$$

当 t 很大时, u 的级数收敛很快, 忽略 $k \geq 1$ 各项的影响, 则

$$u(x, t) = \frac{2\Phi_0}{l} + \frac{4\Phi_0}{l} e^{-\frac{4\pi^2 a^2}{l^2} t} \cos \frac{2\pi x}{l}$$

11. 在矩形区域 $0 < x < a, 0 < y < b$ 上求解拉普拉斯方程 $\Delta u = 0$, 使满足如下边界条件, 其中 A, B 为常数.

$$u|_{x=0} = Ay(b-y), \quad u|_{x=a} = 0, \quad u|_{y=0} = B \sin \frac{\pi x}{a}, \quad u|_{y=b} = 0.$$

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0, & (0 < x < a, 0 < y < b) \\ u|_{x=0} = Ay(b-y) \\ u|_{x=a} = 0 \\ u|_{y=0} = B \sin \frac{\pi x}{a} \\ u|_{y=b} = 0 \end{cases}$$

令 $u(x, y) = v(x, y) + w(x, y)$, 将问题转化为 v 和 w 的定解问题, 即

$$\begin{cases} \Delta v = 0, (0 < x < a, 0 < y < b) \\ v|_{x=0} = 0 \\ v|_{x=a} = 0 \\ v|_{y=0} = B \sin \frac{\pi x}{a} \\ v|_{y=b} = 0 \end{cases} \quad \begin{cases} \Delta w = 0, (0 < x < a, 0 < y < b) \\ w|_{x=0} = Ay(b-y) \\ w|_{x=a} = 0 \\ w|_{y=0} = 0 \\ w|_{y=b} = 0 \end{cases}$$

(1) 求解 $v(x, y)$, 令 $v(x, y) = X(x)Y(y)$, 代入 $\Delta v = 0$ 得

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2 \quad (\lambda > 0)$$

进一步可得

$$\begin{cases} X'' + \lambda^2 X = 0 \\ Y'' - \lambda^2 Y = 0 \end{cases}$$

对变数 x , 解得

$$X(x) = M \cos \lambda x + N \sin \lambda x$$

对变数 y , 解得

$$Y(y) = C e^{\lambda y} + D e^{-\lambda y}$$

由 x 边界条件得

$$\begin{cases} X|_{x=0} = M = 0 \\ X|_{x=a} = M \cos \lambda a + N \sin \lambda a = 0 \end{cases}$$

解得

$$M = 0, \sin \lambda a = 0,$$

$$\lambda a = n\pi, \lambda = \frac{n\pi}{a}, (n = 1, 2, 3, \dots)$$

$$v(x, y) = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi}{a}y} + D_n e^{-\frac{n\pi}{a}y}) \sin \frac{n\pi x}{a}$$

由 y 边界条件得

$$\begin{cases} v|_{y=0} = \sum_{n=1}^{\infty} (C_n + D_n) \sin \frac{n\pi x}{a} = B \sin \frac{\pi x}{a} \\ v|_{y=b} = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi}{a}b} + D_n e^{-\frac{n\pi}{a}b}) \sin \frac{n\pi x}{a} = 0 \end{cases}$$

比较系数可得

$$\begin{cases} C_n + D_n = 0 & (n \geq 2) \\ C_1 + D_1 = B \\ C_n e^{\frac{n\pi b}{a}} + D_n e^{-\frac{n\pi b}{a}} = 0 \end{cases}$$

解得

$$\begin{cases} C_1 = \frac{B}{1 - e^{\frac{2\pi b}{a}}} \\ D_1 = \frac{B}{1 - e^{-\frac{2\pi b}{a}}} \\ C_n = D_n = 0 (n \geq 2) \end{cases}$$

所以
$$v(x, y) = B \left(\frac{1}{1 - e^{\frac{2\pi b}{a}}} e^{\frac{n\pi}{a}y} + \frac{1}{1 - e^{-\frac{2\pi b}{a}}} e^{-\frac{n\pi}{a}y} \right) \sin \frac{n\pi x}{a}$$

将指数函数化为双曲正弦函数, 即得

$$v(x, y) = \frac{B \sinh \frac{(b-y)}{a}}{\sinh \frac{\pi b}{a}} \sin \frac{\pi x}{a}$$

(2) 求解 $\omega(x, y)$

由 y 的边界条件为第一类齐次边界条件, 通过类似讨论, 可设问题的解为

$$\omega(x, y) = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}}) \sin \frac{n\pi y}{b}$$

由 x 的边界条件可得

$$\begin{cases} \omega|_{x=0} = \sum_{n=1}^{\infty} (C_n + D_n) \sin \frac{n\pi y}{b} = Ay(b-y) \\ \omega|_{x=a} = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}}) \sin \frac{n\pi y}{b} = 0 \end{cases}$$

比较系数可知

$$\begin{aligned} C_n + D_n &= \frac{2}{b} \int_0^b Ay(b-y) \sin \frac{n\pi y}{b} dy = \frac{2A}{b} \int_0^b y(b-y) \sin \frac{n\pi y}{b} dy \\ \int_0^b y \sin \frac{n\pi y}{b} dy &= \left. -\frac{b}{n\pi} y \cos \frac{n\pi y}{b} \right|_0^b + \int_0^b \frac{b}{n\pi} \cos \frac{n\pi y}{b} dy = -\frac{b^2}{n\pi} (-1)^n \\ \int_0^b y^2 \sin \frac{n\pi y}{b} dy &= \left. -\frac{b}{n\pi} y^2 \cos \frac{n\pi y}{b} \right|_0^b + 2 \int_0^b \frac{b}{n\pi} y \cos \frac{n\pi y}{b} dy \\ &= -\frac{b^3}{n\pi} (-1)^n + 2 \left(\frac{b}{n\pi} \right)^2 \left[y \sin \frac{n\pi y}{b} \right]_0^b - \int_0^b \sin \frac{n\pi y}{b} dy \\ &= \frac{b^3}{n\pi} (-1)^n - 2 \left(\frac{b}{n\pi} \right)^3 \cos \frac{n\pi y}{b} \Big|_0^b \\ &= \frac{2b^3}{(n\pi)^3} - b^3 \left[\frac{1}{n\pi} + \frac{2}{(n\pi)^3} \right] (-1)^n \\ C_n + D_n &= \frac{2A}{b} \left\{ -\frac{b^3}{n\pi} (-1)^n - \frac{2b^3}{n^3 \pi^3} + b^3 \cdot \frac{n^2 \pi^2 + 2}{n^3 \pi^3} (-1)^n \right\} \\ &= \frac{4Ab^2}{n^3 \pi^3} [1 - (-1)^n] \end{aligned}$$

当 $n = 2k$ 时, ($k = 1, 2, 3 \dots$)

$$\begin{cases} C_n + D_n = 0 \\ C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} = 0 \end{cases}$$

解得

$$C_n = 0$$

$$D_n = 0$$

当 $n = 2k + 1$ 时

$$\begin{cases} C_n + D_n = \frac{8Ab^2}{n^3\pi^3} \\ C_n e^{\frac{n\pi y}{b}} + D_n e^{-\frac{n\pi y}{b}} = 0 \end{cases}$$

解得

$$\begin{cases} C_n = \frac{-4Ab^2 e^{-\frac{n\pi y}{b}}}{n^3\pi^3 \sinh \frac{n\pi a}{b}} \\ D_n = \frac{4Ab^2 e^{\frac{n\pi y}{b}}}{n^3\pi^3 \sinh \frac{n\pi a}{b}} \end{cases}$$

$$w(x, y) = \sum_{k=0}^{\infty} \frac{8Ab^2 \sinh \frac{(2k+1)\pi(a-x)}{b}}{(2k+1)^3 \pi^3 \sinh \frac{(2k+1)\pi a}{b}} \sin \frac{(2k+1)\pi y}{b}$$

$$u(x, y) = v(x, y) + w(x, y)$$

$$= \frac{B \sinh \frac{\pi(b-y)}{a}}{\sinh \frac{\pi b}{a}} \sin \frac{\pi x}{a} + \sum_{k=0}^{\infty} \frac{8Ab^2 \sinh \frac{(2k+1)\pi(a-x)}{b}}{(2k+1)^3 \pi^3 \sinh \frac{(2k+1)\pi a}{b}} \sin \frac{(2k+1)\pi y}{b}$$

12. 均匀的薄板占据区域 $0 < x < a, 0 < y < \infty$. 边界上的温度

$$u|_{x=0} = 0, \quad u|_{x=a} = 0, \quad u|_{y=0} = u_0, \quad \lim_{y \rightarrow \infty} u = 0.$$

求解板的稳定温度分布.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0 & (0 < x < a, 0 < y < \infty) \\ u|_{x=0} = 0 \\ u|_{x=a} = 0 \\ u|_{y=0} = u_0 \\ u|_{y=\infty} = 0 \end{cases}$$

由 x 边界条件为第一类边界条件, 类似于上题对 $v(x, y)$ 的讨论, 可将问题的解设为

$$u(x, y) = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}}) \sin \frac{n\pi x}{a}$$

由 y 的边界条件可得

$$\begin{cases} u|_{y=0} = \sum_{n=1}^{\infty} (C_n + D_n) \sin \frac{n\pi x}{a} = u_0 \\ u|_{y=\infty} = \sum_{n=1}^{\infty} C_n e^{\frac{n\pi y}{a}} \sin \frac{n\pi x}{a} \Big|_{y \rightarrow \infty} = 0 \end{cases}$$

比较系数可知 $C_n = 0$,

$$D_n = \frac{2}{a} \int_0^a u_0 \sin \frac{n\pi x}{a} dx = \frac{2u_0}{a} \cdot \frac{-a}{n\pi} \cos \frac{n\pi x}{a} \Big|_0^a = \frac{2u_0}{n\pi} [1 - (-1)^n]$$

$$= \begin{cases} \frac{4u_0}{(2k+1)\pi} & \text{当 } n = 2k+1 \quad (k = 0, 1, 2, \dots) \\ 0 & \text{当 } n = 2k \end{cases}$$

所以
$$u(x, y) = \frac{4u_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)} e^{-\frac{(2k+1)\pi y}{a}} \sin \frac{(2k+1)\pi x}{a}$$

13. A 为常数, 在带状区域 $0 < x < a, 0 < y < \infty$ 上求解 $\Delta u = 0$, 并满足边界条件:

$$u \Big|_{x=0} = 0, \quad u \Big|_{x=a} = 0, \quad u \Big|_{y=0} = A \left(1 - \frac{x}{a}\right), \quad \lim_{y \rightarrow \infty} u = 0.$$

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0, (0 < x < a, 0 < y < \infty) \\ u \Big|_{x=0} = u \Big|_{x=a} = 0 \\ u \Big|_{y=0} = A \left(1 - \frac{x}{a}\right) \\ u \Big|_{y \rightarrow \infty} = 0 \end{cases}$$

类似上题讨论, 可设
$$u(x, y) = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}}) \sin \frac{n\pi x}{a}$$

由 y 的边界条件, $y \rightarrow \infty$ 时 $u = 0$ 可知 $e^{\frac{n\pi y}{a}}$ 项系数为 0, 即 $C_n = 0$.

$$u \Big|_{y=0} = \sum_{n=1}^{\infty} D_n \sin \frac{n\pi x}{a} = A \left(1 - \frac{x}{a}\right)$$

$$D_n = \frac{2}{a} \int_0^a A \left(1 - \frac{x}{a}\right) \sin \frac{n\pi x}{a} dx = \frac{2A}{n\pi}$$

$$u(x, y) = \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} e^{-\frac{n\pi y}{a}} \sin \frac{n\pi x}{a}$$

14. 矩形膜, 边长为 l_1 和 l_2 , 边缘固定. 求它的本征振动模式.

解: 由题意得定解问题为

$$\begin{cases} u_{xx} - a^2 \Delta u = 0 \quad (0 < x < l_1, 0 < y < l_2) \\ u \Big|_{x=0} = u \Big|_{x=l_1} = 0 \\ u \Big|_{y=0} = u \Big|_{y=l_2} = 0 \end{cases}$$

令 $u(x, y, t) = X(x)Y(y)T(t)$, 代入泛定方程得

$$\frac{T''}{a^2 T} = \frac{X''Y + XY''}{XY} = -\lambda$$

进一步可得

$$\begin{cases} T'' + \lambda a^2 T = 0 \\ X''Y + XY'' + \lambda XY = 0 \end{cases}$$

对时间变数 t , 解得 $T(t) = A \cos \sqrt{\lambda} at + B \sin \sqrt{\lambda} at$

对空间变数 x, y , 进一步分离, 可得

$$-\frac{Y''}{Y} = \frac{X''}{X} + \lambda$$

令 $\frac{X''}{X} = -\lambda_1$, 则 $X'' + \lambda_1 X = 0$

由边界条件 $X(0) = X(l_1) = 0$, 可知其本征函数为 $\sin \frac{m\pi}{l_1}x; \lambda_1 = \frac{m^2\pi^2}{l_1^2} (m = 1, 2, \dots)$

令 $\frac{Y''}{Y} = -\lambda_2$, 经类似讨论, 可知其本征函数为 $\sin \frac{n\pi}{l_2}y; \lambda_2 = \frac{n^2\pi^2}{l_2^2} (n = 1, 2, \dots)$

$$\lambda = \lambda_1 + \lambda_2 = \frac{m^2\pi^2}{l_1^2} + \frac{n^2\pi^2}{l_2^2} = \omega_{m,n}^2$$

本征振动模式 $u_{m,n}(A_{m,n}\cos\omega_{m,n}at + B_{m,n}\sin\omega_{m,n}at)\sin\frac{m\pi}{l_1}x\sin\frac{n\pi}{l_2}y$

其中 $\omega_{m,n} = \pi\sqrt{\frac{m^2}{l_1^2} + \frac{n^2}{l_2^2}}$

15. 细圆环, 半径为 ρ_0 , 初始温度分布已知为 $f(\varphi)$, φ 是以环心为极点的极角. 环的表面是绝热的. 求解环内温度变化的情况.

解: 由题意得定解问题为

$$\begin{cases} u_t - \frac{a^2}{R^2}u_{\varphi\varphi} = 0 & (0 \leq \varphi < 2\pi) (a^2 = \frac{k}{\rho c}) \\ u_{t=0} = f(\varphi) \\ u(\varphi + 2\pi) = u(\varphi) \end{cases}$$

令 $u(\varphi, t) = \Phi(\varphi)T(t)$, 代入泛定方程, 得

$$-\frac{T'}{\frac{a^2}{R^2}T} = \frac{\Phi''}{\Phi} = -\omega^2$$

进一步可得

$$T' - \omega^2 \frac{a^2}{R^2}T = 0$$

$$\Phi'' + \omega^2\Phi = 0$$

对时间变数 t , 解得

$$T(t) = Ce^{-\frac{\omega^2 a^2}{R^2}t}$$

对角向变数 φ , 解得

$$\Phi(\varphi) = B\cos\omega\varphi + C\sin\omega\varphi$$

由 φ 的自然边界条件可知

$$\Phi(\varphi + 2\pi) = A\cos(\omega\varphi + 2\pi) + B\sin(\omega\varphi + 2\pi) = \Phi(\varphi)$$

$$\omega = n \quad (n = 0, 1, 2, 3, \dots)$$

$$u(\varphi, t) = \sum_{n=0}^{\infty} (A_n\cos n\varphi + B_n\sin n\varphi)e^{-\frac{n^2 a^2}{R^2}t}$$

代入初始条件, 可求得

$$A_0 = \frac{1}{\pi} \int_0^{2\pi} f(\varphi) d\varphi$$

$$A_n = \frac{2}{\pi} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi$$

$$B_n = \frac{2}{\pi} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi$$

16. A, B 为常数, 在圆形域内求解 $\Delta u = 0$, 并分别满足边界条件:

$$(1) u|_{\rho=\rho_0} = A \cos \varphi; \quad (2) u|_{\rho=\rho_0} = A + B \sin \varphi.$$

解: 对极坐标下的拉普拉斯方程, 可表示为

$$u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\varphi\varphi} = 0$$

令 $u(\rho, \varphi) = R(\rho)\Phi(\varphi)$ 代入 $\Delta u = 0$ 可得

$$-\frac{\Phi''}{\Phi} = \frac{1}{R}\left(\rho \frac{dR}{d\rho}\right)' = \lambda \quad (\lambda \geq 0)$$

进一步可得

$$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \rho^2 R'' + \rho R' - \lambda R = 0 \end{cases}$$

对角向变数 φ , 解得 $\Phi(\varphi) = A \cos \sqrt{\lambda}\varphi + B \sin \sqrt{\lambda}\varphi$, 由自然边界条件 $\Phi(\varphi + 2\pi) = \Phi(\varphi)$ 可知

$$\sqrt{\lambda} = m, (m = 0, 1, 2, 3, \dots)$$

$$\Phi(\varphi) = A_m \cos m\varphi + B_m \sin m\varphi$$

对径向变数 ρ , 当 $\sqrt{\lambda} = m = 0$ 时, $R_0 = C_0 + D_0 \ln \rho$

当 $\sqrt{\lambda} = m > 0$ 时, $R_m = C_m \rho^m + D_m \rho^{-m}$

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m})$$

当在圆形域内求解, $\rho = 0$ 处应有限, 故 $D_0 = D_m = 0$.

(1) 当 $u|_{\rho=\rho_0} = A \cos \varphi$, 则 $C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho_0^m = A \cos \varphi$

比较系数可知 $A_1 \rho_0 = A$, 其余系数都为 0,

$$A_1 = \frac{A}{\rho_0}$$

$$u(\rho, \varphi) = \frac{A}{\rho_0} \rho \cos \varphi$$

(2) 当 $u|_{\rho=\rho_0} = A + B \sin \varphi$, 则 $C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho_0^m = A + B \sin \varphi$

比较系数可知 $C_0 = A, B_1 \rho_0 = B$, 其余系数都为 0

所以

$$u(\rho, \varphi) = A + \frac{B}{\rho_0} \rho \sin \varphi$$

17. 半圆形薄板, 半径为 ρ_0 , 板面绝热, 边界直径上温度保持零度, 圆周上保持 u_0 . 求稳定状态下的板上温度分布.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0 & (\rho < \rho_0, 0 < \varphi < \pi) \\ u|_{\varphi=0} = u|_{\varphi=\pi} = 0 \\ u|_{\rho=\rho_0} = u_0 \\ u|_{\rho=0} = \text{有限值} \end{cases}$$

设问题的解

$$u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (C_m \rho^m + D_m \rho^{-m}) (A_m \cos \varphi + B_m \sin m\varphi)$$

由 $\rho = 0$ 处 u 为有限值, 可得 $D_0 = D_m = 0$

$$u(\rho, \varphi) = C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m$$

由 $u|_{\varphi=0} = 0$ 可得 $C_0 + \sum_{m=1}^{\infty} A_m \rho^m = 0$, 则 $A_m = C_0 = 0$

$$u(\rho, \varphi) = \sum_{m=1}^{\infty} B_m \sin m\varphi \cdot \rho^m$$

由 $u|_{\rho=\rho_0} = u_0$ 可得, $\sum_{m=1}^{\infty} B_m \sin m\varphi \cdot \rho_0^m = u_0$

$$\begin{aligned} B_m &= \frac{2}{\pi \rho_0^m} \int_0^\pi u_0 \sin m\varphi d\varphi = \frac{2u_0}{\pi \rho_0^m} \frac{1}{m} \cos m\varphi \Big|_0^\pi = \frac{2u_0}{\rho_0^m m \pi} [1 - (-1)^m] \\ &= \begin{cases} \frac{4u_0}{\rho_0^m} \frac{1}{m\pi}, & m = 2k+1 \quad (k=0, 1, 2, \dots) \\ 0 & m = 2k \end{cases} \end{aligned}$$

所以
$$u(\rho, \varphi) = \frac{4u_0}{\pi} \sum_{k=0}^{\infty} \frac{1}{\rho_0^{2k+1} (2k+1)} \rho^{2k+1} \sin(2k+1)\varphi$$

18. 把教材中例 4 的导体圆柱换为介质圆柱, 介质的“相对”介电常数为 ϵ . 求解柱内外的电场. [提示: 柱内电势必须有限. 在柱面上电势连续, 电位移的法向分量连续.]

解: 由题意得定解问题为

$$\begin{cases} \Delta u_1 = 0 & (0 \leq \rho < a) \\ \Delta u_2 = 0 & (\rho > a) \\ u_1(\varphi + 2\pi) = u_1(\varphi) & u_2(\varphi + 2\pi) = u_2(\varphi) \\ u_1|_{\rho=0} = \text{有限值} & u_2|_{\rho=\infty} = -E_0 \rho \cos \varphi \\ u_1|_{\rho=a} = u_2|_{\rho=a} \\ \epsilon \epsilon_0 \frac{\partial u_1}{\partial \rho} \Big|_{\rho=a} = \epsilon_0 \frac{\partial u_2}{\partial \rho} \Big|_{\rho=a} \end{cases}$$

对球内电势 u_1 , 由于在 $\rho = 0$ 处有限,

设
$$u_1(\rho, \varphi) = C_{01} + \sum_{m=1}^{\infty} (A_{1m} \cos m\varphi + B_{1m} \sin m\varphi) \rho^m$$

对球外电势 u_2 , 设 $u_2(\rho, \varphi) = C_{02} + D_{02} \ln \rho + \sum_{m=1}^{\infty} (A_{2m} \cos m\varphi + B_{2m} \sin m\varphi) (C_{2m} \rho^m + D_{2m} \rho^{-m})$

在 $\rho = \infty$ 处, $u_2|_{\rho=\infty} = -E_0 \rho \cos \varphi$, 比较系数可知

$$C_{02} = 0, D_{02} = 0, C_{2m} = 0 (m \neq 1) \quad A_{21} C_{21} = -E_0$$

代入以上系数得

$$u_2(\rho, \varphi) = -E_0 \rho \cos \varphi + \sum_{m=1}^{\infty} (A_{2m} \cos m\varphi + B_{2m} \sin m\varphi) \rho^{-m}$$

代入衔接条件可得

$$\begin{cases} C_{01} + \sum_{m=1}^{\infty} (A_{1m} \cos m\varphi + B_{1m} \sin m\varphi) a^m = -E_0 \cos \varphi + \sum_{m=1}^{\infty} (A_{2m} \cos m\varphi + B_{2m} \sin m\varphi) a^{-m} \\ \epsilon \sum_{m=1}^{\infty} (A_{1m} \cos m\varphi + B_{1m} \sin m\varphi) m a^{m-1} = -E_0 \cos \varphi \\ + \sum_{m=1}^{\infty} (A_{2m} \cos m\varphi + B_{2m} \sin m\varphi) (-m) a^{-(m+1)} \end{cases}$$

比较 $\cos m\varphi, \sin m\varphi$ 的系数可得

$$\begin{cases} C_{01} = 0 \\ A_{11} a^m = -E_0 a + A_{21} a^{-m} \\ A_{1m} a^m = A_{2m} a^{-m} (m \neq 1) \\ B_{1m} a^m = B_{2m} a^{-m} \end{cases} \quad \text{及} \quad \begin{cases} \epsilon A_{11} = -E_0 - A_{21} a^{-2} \\ \epsilon A_{1m} a^{m-1} = -A_{2m} a^{-(m+1)} m (m \neq 1) \\ \epsilon B_{1m} a^{m-1} \cdot m = -B_{2m} a^{-(m+1)} m \end{cases}$$

解得

$$\begin{cases} C_{01} = 0 \\ A_{11} = \frac{-2E_0}{1+\epsilon} \\ A_{21} = \frac{(\epsilon-1)}{\epsilon+1} a^2 E_0 \\ A_{1m} = A_{2m} = 0, \quad (m \neq 1) \\ B_{1m} = B_{2m} = 0 \end{cases}$$

柱内电势

$$u_1(\rho, \varphi) = -\frac{2E_0}{1+\epsilon} \rho \cos \varphi$$

柱外电势

$$u_2(\rho, \varphi) = -\left(\rho - \frac{\epsilon-1}{\epsilon+1} \frac{a^2}{\rho}\right) E_0 \cos \varphi$$

柱内场强

$$E_1 = -\frac{\partial u_1}{\partial x} = -\frac{\partial}{\partial x} \left(-\frac{2E_0}{1+\epsilon} x\right) = \frac{2E_0}{1+\epsilon}, \text{ 为均强电场}$$

柱内极化强度

$$P_1 = (\epsilon-1)\epsilon_0 E_1 = 2\epsilon_0 \frac{\epsilon-1}{\epsilon+1} E_0$$

柱内束缚电荷面密度 $= P_n = 2\epsilon_0 \frac{\epsilon-1}{\epsilon+1} E_0 \cos \varphi$

19. 半径为 ρ_0 , 表面熏黑了的均匀长圆柱, 在温度为零度的空气中受着阳光照射. 阳光垂直于柱轴, 热流强度为 q_0 . 试求柱内稳定温度分布. [提示: 泛定方程为 $\Delta u = 0$, 边界条件为 $(ku_\rho + hu)]|_{\rho=\rho_0} = f(\varphi)$, $f(\varphi)$ 是热流强度的法向分量. 如取极轴垂直于阳光, 则

$$f(\varphi) = \begin{cases} q_0 \sin \varphi, & (0 < \varphi < \pi), \\ 0, & (\pi < \varphi < 2\pi). \end{cases}$$

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0, & (0 < \rho < a) \\ [ku_\rho + Hu]|_{\rho=a} = \begin{cases} q \sin \varphi, & (0 < \varphi < \pi) \\ 0, & (\pi < \varphi < 2\pi) \end{cases} \end{cases}$$

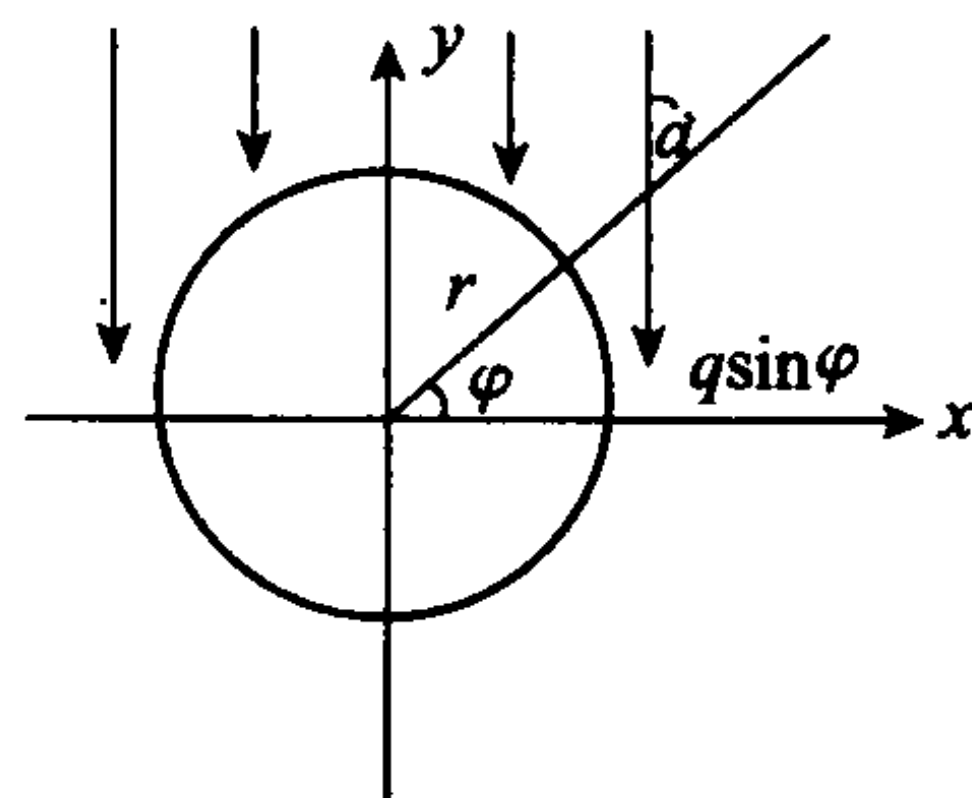


图 8-1-2

设问题的解 $u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi)(C_m \rho^m + D_m \rho^{-m})$

由 $\rho = 0$ 处 u 为有限值, 可知 $D_0 = D_m = 0$, 则

$$u(\rho, \varphi) = C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m$$

代入另一边界条件得

$$\left[HC_0 + H \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m + k \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) m \rho^{m-1} \right] \Big|_{\rho=a} = \begin{cases} q \sin \varphi & (0 < \varphi < \pi) \\ 0 & (\pi < \varphi < 2\pi) \end{cases}$$

$$\text{即 } HC_0 + \sum_{m=1}^{\infty} (Ha^m + kma^{m-1})(A_m \cos m\varphi + B_m \sin m\varphi) = \begin{cases} q \sin \varphi & (0 < \varphi < \pi) \\ 0 & (\pi < \varphi < 2\pi) \end{cases}$$

则通过傅里叶展开, 可求得 C_0, A_m 及 B_m

$$C_0 = \frac{1}{H} \frac{1}{2\pi} \int_0^\pi q \sin \varphi d\varphi = \frac{q}{\pi H}$$

$$\begin{aligned} A_m &= \frac{1}{Ha^m + kma^{m-1}} \frac{1}{\pi} \int_0^\pi q \sin \varphi \cos \varphi d\varphi \\ &= \frac{q}{(Ha^m + kma^{m-1})\pi} \int_0^\pi \frac{\sin(m+1)\varphi + \sin(1-m)\varphi}{2} d\varphi \\ &= \frac{q}{2(Ha^m + kma^{m-1})\pi} \left[\frac{1 - (-1)^{m+1}}{1+m} + \frac{1 - (-1)^{m-1}}{1-m} \right] \\ &= \begin{cases} 0 & m = 2l+1, (l=0, 1, 2, \dots) \\ \frac{q}{2(Ha^{2l} + 2kla^{2l-1})\pi} \frac{2}{1-4l^2} & m = 2l \end{cases} \end{aligned}$$

$$B_m = \frac{1}{(Ha^m + kma^{m-1})} \frac{1}{\pi} \int_0^\pi q \sin \varphi \sin m\varphi d\varphi$$

若 $m \neq 1$, 则 $\sin \varphi$ 与 $\sin m\varphi$ 正交, $B_m = 0$

$$\text{若 } m = 1, \text{ 则 } B_m = \frac{q}{(Ha^m + kma^{m-1})\pi} \int_0^\pi \sin^2 \varphi d\varphi = \frac{q}{2(k+Ha)}$$

$$\text{所以 } u(\rho, \varphi) = \frac{q}{H\pi} + \frac{1}{k+Ha} \frac{q}{2} \rho \sin \varphi + \sum_{l=1}^{\infty} \frac{2q}{\pi a^{2l-1} (2lk+Ha)(1-4l^2)} \rho^{2l} \cos 2l\varphi$$

20. 在以原点为心, 以 ρ_1 和 ρ_2 为半径的两个同心圆围成的环域上求解 $\Delta u = 0$, 使满足边界条件

$$u \Big|_{\rho=\rho_1} = f_1(\varphi), \quad u \Big|_{\rho=\rho_2} = f_2(\varphi).$$

解: 由题意得定解问题为

$$\Delta u = 0 \quad (\rho_1 < \rho < \rho_2).$$

$$u \Big|_{\rho=\rho_1} = f_1(\varphi)$$

$$u \Big|_{\rho=\rho_2} = f_2(\varphi)$$

设问题的解 $u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi)(C_m \rho^m + D_m \rho^{-m})$

由边界条件可得

$$(1) \quad \begin{cases} C_0 + D_0 \ln \rho_1 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho_1^m + D_m \rho_1^{-m}) = f_1(\varphi) \\ C_0 + D_0 \ln \rho_2 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho_2^m + D_m \rho_2^{-m}) = f_2(\varphi) \end{cases}$$

$$\begin{cases} C_0 + D_0 \ln \rho_1 = \frac{1}{2\pi} \int_0^{2\pi} f_1(\varphi) d\varphi \equiv \beta_0^{(1)} \\ C_0 + D_0 \ln \rho_2 = \frac{1}{2\pi} \int_0^{2\pi} f_2(\varphi) d\varphi \equiv \beta_0^{(2)} \end{cases}$$

$$\text{解得 } C_0 = \frac{\beta_0^{(1)} \ln \rho_2 - \beta_0^{(2)} \ln \rho_1}{\ln \rho_2 - \ln \rho_1}, D_0 = \frac{\beta_0^{(1)} - \beta_0^{(2)}}{\ln \rho_1 - \ln \rho_2}$$

$$(2) \quad \begin{cases} (C_m \rho_1^m + D_m \rho_1^{-m}) A_m = \frac{1}{\pi} \int_0^{2\pi} f_1(\varphi) \cos m\varphi d\varphi = \beta_m^{(1)} \\ (C_m \rho_2^m + D_m \rho_2^{-m}) A_m = \frac{1}{\pi} \int_0^{2\pi} f_2(\varphi) \cos m\varphi d\varphi = \beta_m^{(2)} \\ (C_m \rho_1^m + D_m \rho_1^{-m}) B_m = \frac{1}{\pi} \int_0^{2\pi} f_1(\varphi) \sin m\varphi d\varphi = \alpha_m^{(1)} \\ (C_m \rho_2^m + D_m \rho_2^{-m}) B_m = \frac{1}{\pi} \int_0^{2\pi} f_2(\varphi) \sin m\varphi d\varphi = \alpha_m^{(2)} \end{cases}$$

$$\text{解得} \quad \begin{cases} C_m A_m = \frac{1}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} (\rho_2^{-m} \beta_m^{(1)} - \rho_1^{-m} \beta_m^{(2)}) \\ D_m A_m = \frac{-1}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} (\rho_2^{-m} \beta_m^{(1)} - \rho_1^{-m} \beta_m^{(2)}) \\ C_m B_m = \frac{1}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} (\rho_2^{-m} \alpha_m^{(1)} - \rho_1^{-m} \alpha_m^{(2)}) \\ D_m B_m = \frac{-1}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} (\rho_2^{-m} \alpha_m^{(1)} - \rho_1^{-m} \alpha_m^{(2)}) \end{cases}$$

$$\text{则 } u(\rho, \varphi) = \frac{\beta_0^{(1)} \ln \rho_2 - \beta_0^{(2)} \ln \rho_1}{2(\ln \rho_2 - \ln \rho_1)} + \frac{\beta_0^{(2)} - \beta_0^{(1)}}{2(\ln \rho_2 - \ln \rho_1)} \ln \rho + \sum_{m=1}^{\infty} \left[\frac{\beta_m^{(1)} \rho_2^{-m} - \beta_m^{(2)} \rho_1^{-m}}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} \rho^m - \frac{\beta_m^{(1)} \rho_2^m - \beta_m^{(2)} \rho_1^m}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} \rho^{-m} \right] \cos m\varphi + \sum_{m=1}^{\infty} \left[\frac{\alpha_m^{(1)} \rho_2^{-m} - \alpha_m^{(2)} \rho_1^{-m}}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} \rho^m - \frac{\alpha_m^{(1)} \rho_2^m - \alpha_m^{(2)} \rho_1^m}{\rho_1^m \rho_2^{-m} - \rho_1^{-m} \rho_2^m} \rho^{-m} \right] \sin m\varphi$$

21. 求解绕圆柱的水流问题. 在远离圆柱因而未受圆柱干扰处的水流是均匀的, 流速为 v_0 . 圆柱半径为 a .

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0, (\rho > a) \\ u_\rho|_{\rho=a} = 0 \\ u|_{\rho=\infty} \approx v_0 \rho \cos \varphi \end{cases}$$

设问题的解 $u(\rho, \varphi) = C_0 + D_0 \ln \rho + A_0 \varphi + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m \rho^m + D_m \rho^{-m})$

由 $\rho = \infty$ 处边界条件可得 $C_0 = D_0 = 0, C_m = 0 (m \neq 1), A_1 C_1 = V_0, B_1 C_1 = 0$

$$u(\rho, \varphi) = v_0 \rho \cos \varphi + A_0 \varphi \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) \rho^{-m}$$

由 $\rho = a$ 处边界条件可得

$$u_\rho|_{\rho=a} = \left[v_0 \cos \varphi + \sum_{m=1}^{\infty} (a_m \cos m\varphi + b_m \sin m\varphi) (-m) \rho^{-m-1} \right] \Big|_{\rho=a} = 0$$

$$v_0 + a_1(-1)a^{-1-1} = 0 \quad a_m = 0 (m \neq 1), b_m = 0, a_1 = v_0 a^2$$

$$u(\rho, \varphi) = v_0 \rho \cos \varphi + v_0 \frac{a^2}{\rho} \cos \varphi + A_0 \varphi$$

记 $A_0 \frac{\Gamma}{2\pi}$, Γ 称为环流

$$\text{则} \quad u(\rho, \varphi) = v_0 \rho \cos \varphi + v_0 \frac{a^2}{\rho} \cos \varphi + \frac{\Gamma}{2\pi} \varphi$$

22. 长为 l 的理想传输线, 一端接于电动势为 $v_0 \sin \omega t$ 的交流电源, 另一端短路. 求解线上的恒定电振荡并计算输入阻抗.

解: 由题意得定解问题为

$$\begin{cases} v_{xx} - a^2 v_{tt} = 0, (a = \frac{1}{\sqrt{LC}}) \\ v|_{x=0} = v_0 \sin \omega t \\ v|_{x=l} = 0 \end{cases}$$

令 $v(x, t) = X(x)T(t)$ 代入泛定方程得到

$$\frac{T''}{a^2 T} = \frac{X''}{X} = -\lambda \quad (\lambda > 0)$$

进一步可得

$$\begin{cases} T'' + \lambda a^2 T = 0 \\ X'' + \lambda X = 0 \end{cases}$$

对时间变数 t , 解得

$$T(t) = A \cos \sqrt{\lambda} a t + B \sin \sqrt{\lambda} a t$$

对空间变数 x , 解得

$$X(x) = C \cos \sqrt{\lambda} x + D \sin \sqrt{\lambda} x$$

代入边界条件

$$\begin{cases} v|_{x=0} = (A \cos \sqrt{\lambda} a t + B \sin \sqrt{\lambda} a t) C = v_0 \sin \omega t \\ v|_{x=l} = (A \cos \sqrt{\lambda} a t + B \sin \sqrt{\lambda} a t) (C \cos \sqrt{\lambda} l + D \sin \sqrt{\lambda} l) = 0 \end{cases}$$

解得 $A = 0, BC = v_0, \sqrt{\lambda} = \frac{\omega}{a}, C = \sin \sqrt{\lambda} l, D = -\cos \sqrt{\lambda} l$

则

$$v(x, t) = \frac{v_0}{\sin \frac{\omega}{a} l} \sin \omega t \sin \frac{\omega}{a} (l - x)$$

对理想传输线

$$j_i = -\frac{1}{L} v_x = -\frac{1}{L} \left(-\frac{v_0 \omega}{a} a \right) \frac{\sin \omega t \cos \frac{\omega}{a} (l - x)}{\sin \frac{\omega}{a} l}$$

$$= \frac{v_0 \omega}{aL \sin \frac{\omega}{a} l} \sin \omega t \cos \frac{\omega}{a} (l-x)$$

所以

$$j(x, t) = \int \frac{\partial j}{\partial t} dt = \frac{v_0 \omega \cos \frac{\omega}{a} (l-x)}{aL \sin \frac{\omega}{a} l} \int \sin \omega t dt$$

$$= -\frac{v_0 \cos \frac{\omega}{a} (l-x)}{aL \sin \frac{\omega}{a} l} \cos \omega t + f(t)$$

取 $f(t) = 0$, 并代入 $a^2 = \frac{1}{LC}$, 则

$$\begin{cases} v = v_0 \sin[\omega(l-x) \sqrt{LC}] \sin \omega t / \sin(\omega l \sqrt{LC}) \\ j = -v_0 \sqrt{\frac{C}{L}} \cos[\omega(l-x) \sqrt{LC}] \cos \omega t / \sin(\omega l \sqrt{LC}) \end{cases}$$

$$Z_{\text{输入}} = \frac{v_{\text{max}}|_{x=0}}{j_{\text{max}}|_{x=0}} = \frac{v_0 \sin(\omega l \sqrt{LC}) / \sin(\omega l \sqrt{LC})}{V_0 \sqrt{\frac{C}{L}} \cos(\omega l \sqrt{LC}) / \sin(\omega l \sqrt{LC})} = \sqrt{\frac{L}{C}} \tan(\omega l \sqrt{LC})$$

当 $l = \frac{1}{4}\lambda$ (λ 为波长) 时, 输入阻抗无穷大, 即 $Z_{\text{输入}} = \infty$

23. 长为 l 的非理想传输线, 一端接于交流电源 $v_0 \sin \omega t$, 另一端通过阻抗元件 R_0, L_0 和 C_0 而相接. 求解线上的恒定电振荡. 在怎样的条件下不存在反射波(这叫作匹配)?

解: 由题意得定解问题为

$$\begin{cases} LCv_{xx} - v_{xt} + (LG + RC)vt + RGv = 0 \\ v|_{x=0} = v_0 \sin \omega t \\ v|_{x=l} = (jR_0 + L_0 j_t + \frac{1}{C_0} \int_t^{\cdot} j dt)|_{x=l} \end{cases}$$

将电压 $v|_{x=0}$ 写成复数形式 $v_0 e^{i\omega t}$, 约定计算结果取虚部, 则设

$$v = X(x) e^{i\omega t}$$

则 $v_{xx} = X'' e^{i\omega t}$, $v_t = i\omega X e^{i\omega t}$, $v_x = -\omega^2 X(x) e^{i\omega t}$, 代回方程整理后得

$$\frac{X''}{X} = (R + i\omega L)(G + i\omega C)$$

解得 $X(x) = A e^{\sqrt{(R+i\omega L)(G+i\omega C)}x} + B e^{-\sqrt{(R+i\omega L)(G+i\omega C)}x}$

令 $\sqrt{(R+i\omega L)(G+i\omega C)} = \alpha$, 则

$$V = (A e^{\alpha x} + B e^{-\alpha x}) e^{i\omega t}$$

由 $j_x = -Gv - Cv_t$ 得

$$j_x = -GX e^{i\omega t} - C(i\omega)X e^{i\omega t} = -(G + i\omega C)(A e^{\alpha x} + B e^{-\alpha x}) e^{i\omega t}$$

$$j = \int j_x dx = -(G + i\omega C) \left[\int (A e^{\alpha x} + B e^{-\alpha x}) dx \right] e^{i\omega t} + f(t)$$

$$= -\frac{1}{\alpha}(G + i\omega C)[Ae^{\alpha x} - Be^{-\alpha x}]e^{i\omega t} + f(t)$$

将 j 代入 $v_x = -Rj - Ljt$ 得

$$\begin{aligned} v_x &= \frac{R(G + i\omega C)}{\alpha}(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t} + \frac{i\omega L(G + i\omega C)}{\alpha}(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t} - (Rf + Lf') \\ &= \frac{(R + i\omega L)(G + i\omega C)}{\alpha}(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t} - (Rf + Lf') \\ &= \alpha(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t} - (Rf + Lf') \end{aligned}$$

又

$$v_x = X'(x)e^{i\omega t} = \alpha(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t}$$

比较 v_x 两式易得 $Rf(t) + Lf'(t) = 0$, 解得 $f(t) = f_0 e^{-\frac{R}{L}t}$, 整理已得到的结果

$$\begin{cases} v = (Ae^{\alpha x} + Be^{-\alpha x})e^{i\omega t} \\ j = -\sqrt{\frac{G + i\omega C}{R + i\omega L}}(Ae^{\alpha x} - Be^{-\alpha x})e^{i\omega t} + f_0 e^{-\frac{R}{L}t} \end{cases}$$

代入边界条件

$$\begin{cases} v|_{x=0} = v_0 e^{i\omega t} \\ v|_{x=l} = \left(R_0 + i\omega L_0 - \frac{i}{C_0\omega}\right)j|_{x=l} \end{cases}$$

解得

$$\begin{cases} f_0 = 0 \\ A + B = v_0 \\ Ae^{\alpha l} + Be^{-\alpha l} = \left(R_0 + i\omega L_0 - \frac{i}{C_0\omega}\right)\sqrt{\frac{G + i\omega C}{R + i\omega L}}(Ae^{\alpha l} - Be^{-\alpha l}) \end{cases}$$

设

$$M_1 = R_0 + i\omega L_0 - \frac{i}{C_0\omega} + \sqrt{\frac{R + i\omega L}{G + i\omega C}}$$

$$M_2 = R_0 + i\omega L_0 - \frac{i}{C_0\omega} - \sqrt{\frac{R + i\omega L}{G + i\omega C}}$$

则

$$\begin{cases} f_0 = 0 \\ A = \frac{M_2 v_0}{M_1 e^{\alpha l} + M_2 e^{-\alpha l}} \\ B = \frac{M_1 v_0}{M_1 e^{\alpha l} + M_2 e^{-\alpha l} \cdot e^{2\alpha l}} \end{cases}$$

$Ae^{\alpha x + i\omega t}$, 即 $Ae^{\sqrt{(R + i\omega L)(G + i\omega C)} + i\omega t}$ 为反射波, 若使反射波为 0, 则 A 必须为 0,

$M_2 = 0$, 即 $R_0 + i\omega L_0 - \frac{i}{C_0\omega} = \sqrt{\frac{R + i\omega L}{G + i\omega C}}$, 等号右边叫作传输线的特征阻抗.

24. 长为 l 的均匀杆, 一端固定, 另一端在纵向力 $F(t) = F_0 \sin \omega t$ 长期作用下. 求解杆的恒定纵振动.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{x=0} = 0 \\ u_x|_{x=l} = \frac{F_0}{YS} \sin \omega t \end{cases}$$

由于该问题没有初始条件,则振动周期与外力相同,设 $u(x,t) = X(x) \sin \omega t$ 代入泛定方程消去 $\sin \omega t$ 得

$$X'' + \frac{\omega^2}{a^2} X = 0$$

有

$$X = A \cos \frac{\omega}{a} x + B \sin \frac{\omega}{a} x$$

代入边界条件可得

$$\begin{cases} A = 0 \\ -\frac{\omega}{a} A \sin \frac{\omega}{a} l + \frac{\omega}{a} B \cos \frac{\omega}{a} l = \frac{F_0}{YS} \end{cases}$$

解得

$$\begin{cases} A = 0 \\ B = \frac{F_0 a}{YS \omega} \frac{1}{\cos \frac{\omega}{a} l} \end{cases}$$

所以

$$u(x,t) = \frac{a F_0}{YS \omega \cos(\omega l/a)} \sin \frac{\omega x}{a} \sin \omega t$$

25. 长为 l 的均匀弦两端固定,在 x_0 点有一集中的横向力 $F(t) = A \cos \omega t$ 一直作用着,求解弦的恒定横振动.

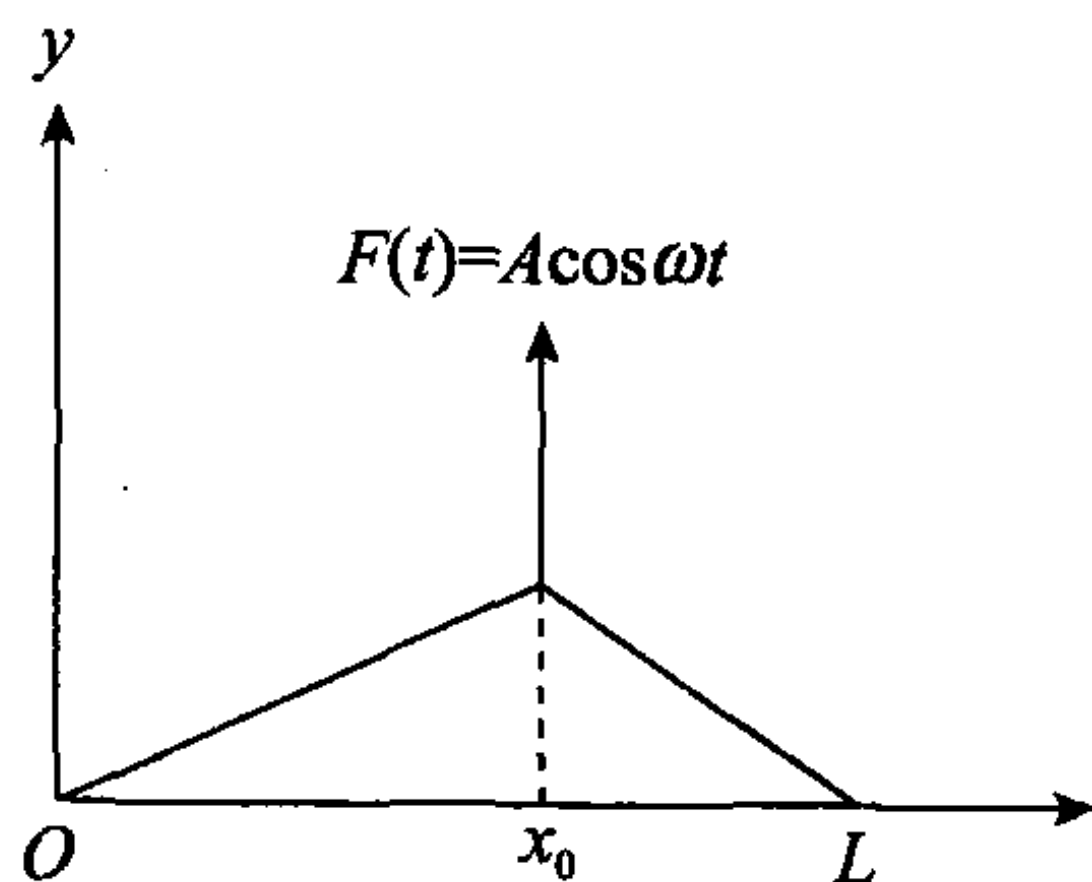
解:由题意可知,该问题应分为 $0 \sim x_0$ 和 $x_0 \sim l$ 两部分讨论.如图

$$\text{对于 } 0 \sim x_0 \quad \begin{cases} U_u^I - a^2 U_{xx}^I = 0 \\ U^I|_{x=0} = 0 \end{cases}$$

$$\text{对于 } x_0 \sim l \quad \begin{cases} U_u^{\text{II}} - a^2 U_{xx}^{\text{II}} = 0 \\ U^{\text{II}}|_{x=l} = 0 \end{cases}$$

有衔接条件

$$\begin{cases} U^I|_{x=x_0} = U^{\text{II}}|_{x=x_0} \\ U_x^I|_{x=x_0} - U_x^{\text{II}}|_{x=x_0} = -\frac{1}{T} F(t) \end{cases}$$



把 U^I, U^{II} 分别进行分离变数 $U^I = x^I(x) T^I(t), U^{\text{II}} = X^{\text{II}}(x) T^{\text{II}}(t)$, 则

$$\begin{cases} X^{I''} + \lambda X^I = 0 & (0 < x < x_0) \\ T^{I''} + a^2 \lambda T^I = 0 & (1) \\ X^I|_{x=0} = 0 \end{cases}$$

$$\begin{cases} X^{\text{II}''} + \lambda X^{\text{II}} = 0 & (x_0 < x < l) \\ T^{\text{II}''} + a^2 \lambda T^{\text{II}} = 0 & (2) \\ X^{\text{II}}|_{x=l} = 0 \end{cases}$$

$$\text{衔接条件:} \quad \begin{cases} X^I|_{x=x_0} \cdot T^I = X^{\text{II}}|_{x=x_0} \cdot T^{\text{II}} & \textcircled{1} \\ X^{I'}|_{x=x_0} \cdot T^I - X^{\text{II}'}|_{x=x_0} \cdot T^{\text{II}} = -\frac{A}{T} \cos \omega t & \textcircled{2} \end{cases}$$

由方程组(1)可得 $X^I = C \sin \sqrt{\lambda} x$ (C 为待定常数)

由方程组(2)可得 $X^{\text{II}} = C' \sin \sqrt{\lambda} x + D' \cos \sqrt{\lambda} x$ (C', D' 为待定常数)

将 $X^{\text{I}}, X^{\text{II}}$ 代入衔接条件,解得

$$C' = -D \frac{\sin \sqrt{\lambda} x_0 \cos \sqrt{\lambda} l}{\sin \sqrt{\lambda} (l - x_0)}$$

$$D' = D \frac{\sin \sqrt{\lambda} x_0 \sin \sqrt{\lambda} l}{\sin \sqrt{\lambda} (l - x_0)}$$

得
$$\begin{cases} X^{\text{I}} = C \sin \sqrt{\lambda} x \\ X^{\text{II}} = C \cdot \frac{\sin \sqrt{\lambda} x_0}{\sin \sqrt{\lambda} (l - x_0)} \sin \sqrt{\lambda} (l - x) \end{cases}$$

对 $T^{\text{I}}, T^{\text{II}}$, 由衔接条件 ① 可知, $T^{\text{I}} = T^{\text{II}}$, 将 $X^{\text{I}}, X^{\text{II}}, T(t)$ 代入衔接条件 ② 得

$$-C \cdot \sqrt{\lambda} \cos \sqrt{\lambda} x_0 T(t) - C \sqrt{\lambda} \frac{\sin \sqrt{\lambda} x_0}{\sin \sqrt{\lambda} (l - x_0)} \cos \sqrt{\lambda} (l - x_0) T(t) = -\frac{A}{T} \cos \omega t$$

$$C \cdot \sqrt{\lambda} \left[\cos \sqrt{\lambda} x_0 + \frac{\sin \sqrt{\lambda} x_0}{\sin \sqrt{\lambda} (l - x_0)} \right] T(t) = (B \cos at + B' \sin \sqrt{\lambda} at) \cdot C \sqrt{\lambda} \cdot \left[\cos \sqrt{\lambda} x_0 + \frac{\sin \sqrt{\lambda} x_0}{\sin \sqrt{\lambda} (l - x_0)} \right] = \frac{A}{T} \cos \omega t$$

比较系数得 $B' = 0, \sqrt{\lambda} a = \omega, BC \cdot \sqrt{\lambda} \left[\cos \sqrt{\lambda} x_0 + \frac{\sin \sqrt{\lambda} x_0}{\sin \sqrt{\lambda} (l - x_0)} \right] = \frac{A}{T}$

$$\text{即 } \sqrt{\lambda} = \frac{\omega}{a}, BC = \frac{Aa}{T\omega} \cdot \frac{\sin \frac{\omega}{a} (l - x_0)}{\sin \frac{\omega}{a} l}$$

将 $\sqrt{\lambda}, BC$ 代回 $U^{\text{I}}, U^{\text{II}}$ 表达式, 即可得

$$u(x, t) = \begin{cases} U^{\text{I}}(l, t) = \frac{Aa}{T\omega} \cdot \frac{\sin \frac{\omega}{a} (l - x_0)}{\sin \frac{\omega}{a} l} \sin \frac{\omega}{a} x \cos \omega t & (0 \leq x \leq x_0) \\ U^{\text{II}}(x, t) = \frac{Aa}{T\omega} \cdot \frac{\sin \frac{\omega}{a} x_0}{\sin \frac{\omega}{a} l} \sin \frac{\omega}{a} (l - x) \cos \omega t & (x_0 \leq x \leq l) \end{cases}$$

26. 一长为 l 的均匀导热细杆, 杆上有热源, 单位长度杆上的热源强度为 $c\rho\beta u$. ($\beta > 0$), $x = 0$ 端绝热, $x = l$ 端保持 0°C , 初始温度分布为 $u_0 x(x - l)$, 试求杆上各处温度如何随时间变化的? 其中 c 为杆的比热容, ρ 为杆的线密度, u_0 为常数, 侧面绝热.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = \beta u \\ u_x|_{x=0} = 0 \quad u|_{x=l} = 0 \\ u|_{t=0} = u_0 x(x - l) \end{cases}$$

令 $u(x, t) = X(x)T(t)$ 代入泛定方程得

$$XT' - a^2 X''T = \beta XT$$

$$\frac{T'}{a^2 T} = \frac{X''}{X} + \frac{\beta}{a^2} = -\lambda \quad (\lambda > 0)$$

解得 $X(x) = A \cos \sqrt{\lambda + \frac{\beta}{a^2}} x + B \sin \sqrt{\lambda + \frac{\beta}{a^2}} x, T(t) = C e^{-\lambda t}$

代入边界条件 $X'(0) = 0, X(l) = 0$, 即得 $B = 0, \sqrt{\lambda + \frac{\beta}{a^2}} l = \frac{2n+1}{2} \pi (n = 0, 1, 2, \dots)$

得 $X(x) = A_n \cos \frac{(2n+1)\pi}{2l} x$

所以 $u(x, t) = X(x)T(t) = \sum_{n=0}^{\infty} A_n e^{-\lambda a^2 t} \cos\left(n + \frac{1}{2}\right) \frac{\pi x}{l}, \lambda = \frac{\left(\frac{n+1}{2}\right)^2 \pi^2}{l^2} - \frac{\beta}{a^2}$

代入初始条件

$$u(x, 0) = \sum_{n=0}^{\infty} A_n \cos\left(n + \frac{1}{2}\right) \frac{\pi x}{l} = u_0 x(x-l)$$

将 $u_0 x(x-l)$ 用 $\cos\left(n + \frac{1}{2}\right) \frac{\pi x}{l}$ 展开

$$A_n = \frac{2}{l} \int_0^l u_0 x(x-l) \cos \frac{\left(n + \frac{1}{2}\right) \pi x}{l} dx$$

经两次分部积分可得

$$A_n = \frac{2u_0 l^2}{\left(n + \frac{1}{2}\right)^3 \pi^3} \left[\left(n + \frac{1}{2}\right) \pi + 2(-1)^{n+1} \right]$$

$$\text{则 } u(x, t) = \sum_{n=0}^{\infty} \frac{2u_0 l^2}{\left(n + \frac{1}{2}\right)^3 \pi^3} \left[\left(n + \frac{1}{2}\right) \pi + 2(-1)^{n+1} \right] e^{-\left[\frac{\left(n + \frac{1}{2}\right)^2 \pi^2}{l^2} - \frac{\beta}{a^2}\right] t} \cos \frac{\left(n + \frac{1}{2}\right) \pi x}{l}$$

27. 一圆环形区域, 内外环半径分别为 ρ_1 和 ρ_2 , 内环上保持温度为 $u_1 \cos^2 \varphi$, 外环上保持温度为 $u_2 \sin \varphi$. 求此圆环区域内的稳定温度分布.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0, (\rho_1 < \rho < \rho_2) \\ u|_{\rho=\rho_1} = u_1 \cos^2 \varphi = \frac{1}{2} u_1 + \frac{1}{2} u_1 \cos 2\varphi \\ u|_{\rho=\rho_2} = u_2 \sin \varphi \end{cases}$$

设一般解 $u(\rho, \varphi) = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin \varphi) \rho^m + \sum_{m=1}^{\infty} (C_m \cos m\varphi + D_m \sin \varphi) \rho^{-m}$

代入边界条件可得

$$\begin{cases} C_0 + D_0 \ln \rho_1 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho_1^m + \sum_{m=1}^{\infty} (C_m \cos m\varphi + D_m \sin m\varphi) \rho_1^{-m} \\ = \frac{1}{2} u_1 + \frac{1}{2} u_1 \cos 2\varphi \\ C_0 + D_0 \ln \rho_2 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho_2^m + \sum_{m=1}^{\infty} (C_m \cos m\varphi + D_m \sin m\varphi) \rho_2^{-m} \\ = u_2 \sin \varphi \end{cases}$$

比较系数可得

$$\begin{cases} C_0 + D_0 \ln \rho_1 = \frac{1}{2} u_1 \\ C_0 + D_0 \ln \rho_2 = 0 \end{cases} \begin{cases} A_1 \rho_1 + \frac{C_1}{\rho_1} = 0 \\ B_1 \rho_1 + \frac{D_1}{\rho_1} = 0 \\ A_2 \rho_1^2 + \frac{C_2}{\rho_1^2} = \frac{1}{2} u_1 \\ B_2 \rho_1^2 + \frac{D_2}{\rho_1^2} = 0 \end{cases} \begin{cases} A_1 \rho_2 + \frac{C_1}{\rho_1} = 0 \\ B_1 \rho_2 + \frac{D_1}{\rho_2} = u_2 \\ A_2 \rho_2^2 + \frac{C_2}{\rho_2^2} = 0 \end{cases}$$

$$A_m = B_m = C_m = D_m = 0 \quad (m \geq 3)$$

解得

$$A_1 = C_1 = 0 \quad B_2 = D_2 = 0$$

$$A_2 = \frac{u_1 \rho_1^2}{2(\rho_1^4 - \rho_2^4)} \quad B_1 = \frac{u_2 \rho_2}{\rho_2^2 - \rho_1^2} \quad C_0 = \frac{u_1 \ln \rho_2}{2 \ln \rho_2 / \rho_1} \quad C_2 = \frac{u_1 \rho_1^2 \rho_2^4}{2(\rho_2^4 - \rho_1^4)} \quad D_0 = \frac{u_1}{2 \ln \rho_1 / \rho_2}$$

$$u(\rho, \varphi) = \frac{u_1}{2} \cdot \frac{\ln \rho_2 - \ln \rho}{\ln \rho_2 - \ln \rho_1} + \frac{u_2 \rho_2}{\ln \rho_2 - \ln \rho_1} \left(\rho - \frac{\rho_1^2}{\rho} \right) \sin \varphi - \frac{u_1 \rho_1^2}{2(\rho_2^4 - \rho_1^4)} \left(\rho^2 - \frac{\rho_2^4}{\rho^2} \right) \cos 2\varphi$$

28. 一长为 l , 截面积相同的均匀细杆, 今将 $x=0$ 端保持为 0°C , $x=l$ 端按牛顿冷却定律向温度为 0°C 的介质散热, 侧面绝热, 原先杆的温度为 u_0 , 求在冷却过程中杆上各处温度的变化.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{x=0} = 0, hu|_{x=l} = -ku_x|_{x=l} \\ u|_{t=0} = u_0 \end{cases}$$

令 $u(x, t) = X(x)T(t)$ 代回泛定方程得

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\omega^2$$

进一步可得

$$\begin{cases} T' + \omega^2 a^2 T = 0 \\ X'' + \omega^2 X = 0, X(0) = 0, hX(l) = -kX'(l) \end{cases}$$

对时间变数 t , 解得 $T(t) = Ce^{-\omega^2 a^2 t}$

对空间变数 x , 解得 $X(x) = A \cos \omega x + B \sin \omega x$

由 $X(0) = 0$ 可知 $A = 0$, 则 $X(x) = B \sin \omega x$

由 $hX(l) = -kX'(l)$ 可知 $hB \sin \omega l = -k\omega B \cos \omega l$, 即 $\omega + \frac{k}{h} \tan \omega l = 0$

令 x_n 为方程 $x + y \tan x = 0$ 的第 n 个根, 其中 $y = \frac{kl}{h}$, 则 $\omega_n = \frac{x_n}{l}$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} D_n e^{-\frac{x_n^2 a^2}{l^2} t} \sin \frac{x_n}{l} x$$

代入初始条件

$$\sum_{n=1}^{\infty} D_n \sin \frac{x_n}{l} x = u_0$$

两边傅里叶展开, 分别代入其正交关系

$$\int_0^l \sin \frac{x_m}{l} x \sin \frac{x_n}{l} x dx + \frac{2l}{y} \cos x_m \cos x_n = 0$$

可求得

$$\frac{D_n l}{2} \left(\frac{\sin x_n \cos x_n}{x_n} - 1 \right) = \frac{u_0 l}{x_n} (1 - \cos x_n)$$

$$D_n = \frac{2u_0(1 - \cos x_n)}{x_n - \sin x_n \cos x_n}$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \frac{2u_0(1 - \cos x_n)}{x_n - \sin x_n \cos x_n} e^{-\frac{x_n^2 a^2}{l^2} t} \sin \frac{x_n}{l} x$$

§ 8.2 非齐次振动方程和输运方程

1. 长为 l 的均匀细杆两端固定, 杆上单位长度受有纵向外力 $f_0 \sin(2\pi x/l) \cos \omega t$, 初始位移为 $[\sin(\pi x/l)]^2$, 初始速度为零, 求解杆的纵振动.

解: 由题意得定解问题为

$$\begin{cases} u_{xx} - a^2 u_{tt} = \frac{f_0}{YS} \sin\left(\frac{2\pi x}{l}\right) \cos \omega t \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = \sin^2\left(\frac{\pi x}{l}\right) \\ u_t|_{t=0} = 0 \end{cases}$$

由第一类齐次边界条件可知泛定方程对应的齐次泛定方程本征函数为 $\sin \frac{n\pi x}{l} (n = 1, 2, 3, \dots)$

$\therefore$ 设问题的解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

将其代入泛定方程得

$$\sum_{n=1}^{\infty} \left(T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n \right) \sin \frac{n\pi x}{l} = f_0 \sin \frac{2\pi x}{l} \cos \omega t$$

比较两边系数可分离出 T_n 的方程, 即

$$\begin{cases} T_2'' + \frac{4\pi^2 a^2}{l^2} T_2 = f_0 \cos \omega t \\ T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n = 0 \quad (n \neq 2) \end{cases}$$

由问题初始条件可得

$$\begin{cases} u|_{t=0} = \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi x}{l} = \sin^2 \frac{\pi x}{l} & (1) \\ u_t|_{t=0} = \sum_{n=1}^{\infty} T'_n(0) \sin \frac{n\pi x}{l} = 0 & (2) \end{cases}$$

由(1)得

$$\begin{aligned} T_n(0) &= \frac{1}{l} \int_0^l \sin^2 \left(\frac{\pi x}{l} \right) \sin \frac{n\pi x}{l} dx = \frac{1}{l} \int_0^l \left[-\frac{1}{2} \cos \left(\frac{2\pi x}{l} \right) + \frac{1}{2} \right] \sin \frac{n\pi x}{l} dx \\ &= \frac{1}{2l} \left[\int_0^l \sin \frac{n\pi x}{l} dx - \int_0^l \cos \frac{2\pi x}{l} \sin \frac{n\pi x}{l} dx \right] \end{aligned}$$

若 $n \neq 2$, 则 $T_n(0) = \frac{1}{2n\pi} [1 - (-1)^n]$

$$= \begin{cases} \frac{1}{n\pi}, & (n \text{ 为奇数}) \\ 0, & (n \text{ 为偶数且 } n \neq 2) \end{cases}$$

若 $n = 2$, 则 $T_n(0) = \frac{1}{2n\pi} [1 - (-1)^n] - \frac{1}{2l} \int_0^l \cos \frac{2\pi x}{l} \sin \frac{2\pi x}{l} dx$

$$= \frac{1}{2n\pi} [1 - (-1)^n]$$

$$T_n(0) = \begin{cases} \frac{1}{n\pi}, & (n = 2k+1), (k = 0, 1, 2, \dots) \\ 0, & (n = 2k) \end{cases}$$

由(2)得

$$T'_n(0) = 0$$

于是构成了完整的 t 的常微分方程定解问题

$$\begin{cases} T''_2 + \frac{4\pi^2 a^2}{l^2} T_2 = f_0 \cos \omega t \\ T_2(0) = 0 \\ T'_2(0) = 0 \end{cases}$$

$$\begin{cases} T_{2k} + \frac{4k^2 \pi^2 a^2}{l^2} T_{2k} = 0 & (k \neq 1) \\ T_{2k}(0) = 0 \\ T'_{2k}(0) = 0 \end{cases}$$

$$\begin{cases} T_{2k+1} + \frac{(2k+1)^2 \pi^2 a^2}{l^2} T_{2k+1} = 0 \\ T_{(2k+1)}(0) = \frac{1}{(2k+1)\pi} \\ T'_{(2k+1)}(0) = 0 \end{cases}$$

解之得

$$\begin{cases} T_2(t) = \frac{f_0}{YS} \frac{1}{\omega^2 - \frac{4\pi^2 a^2}{l^2}} \left(\cos \frac{2\pi a}{l} t - \cos \omega t \right) \\ T_{2k}(t) = 0 \quad (k \neq 1) \\ T_{2k+1}(t) = \frac{-8}{(2k+1)[(2k+1)^2 - 4]\pi} \cos \frac{(2k+1)\pi a t}{l} \end{cases}$$

$$u(x, t) = \frac{f_0}{YS} \frac{1}{\omega^2 - \frac{4\pi^2 a^2}{l^2}} \left(\cos \frac{2\pi a t}{l} - \cos \omega t \right) \sin \frac{2\pi x}{l} + \sum_{k=0}^{\infty} \frac{-8}{(2k+1)[(2k+1)^2 - 4]\pi} \cos \frac{(2k+1)\pi a t}{l} \sin \frac{(2k+1)\pi x}{l}$$

2. 求解热传导问题

$$\begin{cases} u_t - a^2 u_{xx} = A \sin \omega t, \\ u_x \Big|_{x=0} = 0, u \Big|_{x=l} = 0, \\ u \Big|_{t=0} = \varphi(x). \end{cases}$$

解: 由第一类、第二类混合的齐次边界条件可知对应的齐次泛定方程本征函数为

$$\cos\left(n + \frac{1}{2}\right) \frac{\pi}{l} x \quad (n = 0, 1, 2, 3, \dots)$$

设问题的解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) \cos\left(n + \frac{1}{2}\right) \frac{\pi}{l} x$$

将其代入泛定方程得

$$\sum_{n=0}^{\infty} \left[T'_n + \frac{a^2 \left(n + \frac{1}{2}\right)^2 \pi^2}{l^2} T_n \right] \cos\left(n + \frac{1}{2}\right) \frac{\pi}{l} x = A \sin \omega t$$

比较两边系数可得

$$\begin{cases} T'_0 + \frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2}{l^2} T_0 = A \sin \omega t \\ T'_n + \frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2}{l^2} T_n = 0, \quad (n \neq 0) \end{cases}$$

将级数解代入初始条件得

$$u \Big|_{t=0} = \sum_{n=0}^{\infty} T_n(0) \cos\left(n + \frac{1}{2}\right) \frac{\pi}{l} x = \varphi(x)$$

$$\begin{cases} T_n(0) = \frac{1}{l} \int_0^l \varphi(x) dx \equiv \varphi_0 \\ T_n(0) = \frac{2}{l} \int_0^l \varphi(x) \cos\left(n + \frac{1}{2}\right) \frac{\pi}{l} x dx \equiv \varphi_n \end{cases}$$

于是构成了完整的关于力的常微分方程定解问题, 即

$$\begin{cases} T_0' + \frac{\pi^2 a^2}{4l^2} T_0 = A \sin \omega t \\ T_0(0) = \varphi_0 \end{cases}$$

$$\begin{cases} T_n' + \frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2}{l^2} T_n = 0 \\ T_n(0) = \varphi_n \end{cases}$$

解之得

$$\begin{cases} T_0(t) = \frac{A}{\frac{\pi^4 a^4}{16l^4} + \omega^2} \left(\frac{\pi^2 a^2}{4l^2} \sin \omega t - \omega \cos \omega t + \omega \right) \varphi_0 \\ T_n(t) = \varphi_n e^{-\frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} t}, (n \neq 0) \end{cases}$$

$$u(x, t) = \frac{A}{\frac{\pi^4 a^4}{16l^4} + \omega^2} \left(\frac{\pi^2 a^2}{4l^2} \sin \omega t - \omega \cos \omega t + \omega \right) \varphi_0 + \sum_{n=1}^{\infty} \varphi_n e^{-\frac{(n+\frac{1}{2})^2 \pi^2 a^2}{l^2} t}$$

3. 两端固定的弦, 原先静止不动, 单位长度所受横向外力为 $\rho f(x, t) = \rho \Phi(x) \sin \omega t$, 求解弦的振动, 研究共振的可能性, 并求共振时的解.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = \Phi(x) \sin \omega t \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \end{cases}$$

(1) 傅里叶级数法

设问题的解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$$

代入泛定方程得

$$\sum_{n=1}^{\infty} \left(T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n \right) \sin \frac{n\pi x}{l} = \Phi(x) \sin \omega t$$

设

$$f_n = \frac{2}{l} \int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx$$

则比较系数可得

$$T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n = f_n \sin \omega t$$

代入初始条件得

$$\begin{cases} \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi x}{l} = 0 \\ \sum_{n=1}^{\infty} T_n'(0) \sin \frac{n\pi x}{l} = 0 \end{cases}$$

即

$$\begin{cases} T_n(0) = 0 \\ T_n'(0) = 0 \end{cases}$$

解得

$$T_n(t) = \frac{f_n}{n\pi a} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \right]$$

所以
$$u(x, t) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \right] \sin \frac{n\pi x}{l}$$

(2) 冲量定理法, 先将问题转化为求解 $v(x, t, l)$

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0 \\ v|_{x=0} = v_{x=l} = 0 \\ v|_{t=\tau+} = 0, v_t|_{t=\tau+} = \Phi(x) \sin \omega \tau \end{cases}$$

由第一类边界条件, 可知其本征函数为 $\sin \frac{n\pi x}{l}$

设问题的解

$$v(x, t) = \sum_{n=1}^{\infty} \left[A_n(\tau) \cos \frac{n\pi a}{l} (t - \tau) + B_n(\tau) \sin \frac{n\pi a}{l} (t - \tau) \right] \sin \frac{n\pi x}{l}$$

代入初始条件可得

$$\begin{cases} \sum_{n=1}^{\infty} A_n(\tau) \sin \frac{n\pi x}{l} = 0 \\ \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n(\tau) \sin \frac{n\pi x}{l} = \Phi(x) \sin \omega \tau \end{cases}$$

$$A_n(\tau) = 0$$

$$B_n(\tau) = \frac{1}{n\pi a} \frac{2}{l} \int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx \cdot \sin \omega \tau$$

$$v(x, t) = \sum_{n=1}^{\infty} \frac{2 \sin \omega \tau}{n\pi a} \left[\int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx \right] \sin \frac{n\pi a}{l} (t - \tau) \sin \frac{n\pi x}{l}$$

$$u(x, t) = \int_0^t v(x, t, \tau) d\tau = \frac{2}{\pi} a \sum_{n=1}^{\infty} \frac{1}{n} \left[\int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx \right] \sin \frac{n\pi x}{l} \int_0^t \sin \frac{n\pi a}{l} (t - \tau) d\tau$$

$$= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \left[\int_0^l \Phi(x) \sin \frac{n\pi x}{l} dx \right] \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \right] \sin \frac{n\pi x}{l}$$

(3) 讨论共振

当外力频率 ω 等于基音或谐音的频率 $\frac{n\pi a}{l}$ 时

$$\lim_{\omega \rightarrow \frac{n\pi a}{l}} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \right] \xrightarrow{\text{L'Hospital 法则}} \lim_{\omega \rightarrow \frac{n\pi a}{l}} \frac{\left(\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t \right)'}{\left(\omega^2 - \frac{n^2 \pi^2 a^2}{l^2} \right)'}$$

$$= \lim_{\omega \rightarrow \frac{n\pi a}{l}} \frac{\sin \frac{n\pi a}{l} t - \frac{n\pi a t}{l} \cos \omega t}{2\omega} = \frac{1}{2\omega} \sin \frac{n\pi a}{l} t - \frac{1}{2} t \cos \omega t$$

其中第二项振幅为 $\frac{1}{2}t$, 随时间增长, 这就是共振.

4. 两端固定弦在点 x_0 受谐变力 $\rho f(t) = \rho f_0 \sin \omega t$ 作用而振动, 求解振动情况. [提示: 外加力的线密度可表为 $\rho f(x, t) = \rho f_0 \sin \omega t \delta(x - x_0)$.]

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = f_0 \sin \omega t \delta(x - x_0) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \end{cases}$$

(1) 傅里叶级数法

设问题的解 $u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$, 代入泛定方程得

$$\sum_{n=1}^{\infty} \left(T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n \right) \sin \frac{n\pi x}{l} = f_0 \sin \omega t \delta(x - x_0)$$

则比较系数可得

$$T_n'' + \frac{n^2 \pi^2 a^2}{l^2} T_n = \frac{2}{l} \int_0^l \delta(x - x_0) \sin \frac{n\pi x}{l} dx f_0 \sin \omega t = \frac{2f_0}{l} \sin \frac{n\pi x_0}{l} \sin \omega t$$

代入初始条件得

$$\begin{cases} \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi x}{l} = 0 \\ \sum_{n=1}^{\infty} T_n'(0) \sin \frac{n\pi x}{l} = 0 \end{cases}$$

即

$$T_n(0) = 0$$

$$T_n'(0) = 0$$

解得

$$T_n(t) = \frac{f_0}{n\pi a} \sin \frac{n\pi x_0}{l} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 \pi^2 a^2}{l^2}} \right]$$

$$\text{所以 } u(x, t) = \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 a^2 \pi^2}{l^2}} \right] \sin \frac{n\pi x}{l}$$

(2) 冲量定理法

先将问题转化为求解 $v(x, t, \tau)$

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0 \\ v|_{x=0} = v|_{x=l} = 0 \\ v|_{t=\tau^+} = 0 \quad v_t|_{t=\tau^+} = f_0 \delta(x - x_0) \sin \omega \tau \end{cases}$$

由第一类边界条件,可知其本征函数为 $\sin \frac{n\pi x}{l}$.

设问题的解

$$v(x, t) = \sum_{n=1}^{\infty} \left[A_n(\tau) \cos \frac{n\pi a(t-\tau)}{l} + B_n(\tau) \sin \frac{n\pi a(t-\tau)}{l} \right] \sin \frac{n\pi x}{l}$$

代入初始条件可得

$$\begin{cases} \sum_{n=1}^{\infty} A_n(\tau) \sin \frac{n\pi x}{l} = 0 \\ \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n(\tau) \sin \frac{n\pi x}{l} = f_0 \delta(x-x_0) \sin \omega \tau \end{cases}$$

$$A_n(\tau) = 0$$

$$B_n(\tau) = \frac{l}{n\pi a} \frac{2}{l} \int_0^l f_0 \delta(x-x_0) dx \sin \omega \tau$$

$$= \frac{2f_0}{n\pi a} \sin \omega \tau \sin \frac{n\pi x_0}{l}$$

$$v(x, t, \tau) = \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} \cdot \sin \frac{n\pi x}{l}$$

所以

$$\begin{aligned} u(x, t) &= \int_0^t v(x, t, \tau) d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \int_0^t \sin \omega \tau \sin \frac{n\pi a(t-\tau)}{l} d\tau \sin \frac{n\pi x}{l} \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \left(\frac{\sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - \frac{n^2 a^2 \pi^2}{l^2}} \right) \sin \frac{n\pi x}{l} \end{aligned}$$

5. 求解振动问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), \\ u|_{x=0} = 0, u|_{x=l} = 0, \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x). \quad (0 < x < l) \end{cases}$$

解: 首先令 $u = v + w$, 使 v 满足齐次泛定方程, 而 w 满足齐次边界条件及初始条件, 即

$$\begin{cases} v_{tt} - a^2 v_{xx} = 0 \\ v|_{x=0} = v|_{x=l} = 0 \\ v|_{t=0} = \varphi(x) \\ v_t|_{t=0} = \psi(x) \end{cases} \quad \begin{cases} w_{tt} - a^2 w_{xx} = f(x, t) \\ w|_{x=0} = w|_{x=l} = 0 \\ w|_{t=0} = 0 \\ w_t|_{t=0} = 0 \end{cases}$$

(1) 对 $v(x, t)$, 由其具有第一类边界条件, 可知泛定方程的本征函数为 $\sin \frac{n\pi x}{l}$,

设
$$v(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a t}{l} + B_n \sin \frac{n\pi a t}{l} \right) \sin \frac{n\pi x}{l}$$

代入初始条件可得

$$\begin{cases} \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = \varphi(x) \\ \sum_{n=1}^{\infty} \frac{n\pi a}{l} B_n \sin \frac{n\pi x}{l} = \psi(x) \end{cases}$$

将 $\varphi(x)$ 、 $\psi(x)$ 用 $\sin \frac{n\pi x}{l}$ 展开, 比较系数即得

$$A_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx$$

$$B_n = \frac{2}{l} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx$$

(2) 对 $w(x, t)$ 采用冲量定理法

先将问题转化为解 $w_1(x, t, \tau)$

$$\begin{cases} w_{1xx} - a^2 w_{1\tau\tau} = 0 \\ w_1|_{x=0} = w_1|_{x=l} = 0 \\ w_1|_{t=\tau^+} = 0 \\ w_{1t}|_{t=\tau^+} = f(x, \tau) \end{cases}$$

$$\text{设 } w_1(x, t, \tau) = \sum_{n=1}^{\infty} \left[C_n(\tau) \cos \frac{n\pi a}{l}(t-\tau) + D_n(\tau) \sin \frac{n\pi a}{l}(t-\tau) \right] \sin \frac{n\pi x}{l}$$

代入初始条件可得

$$\begin{cases} \sum_{n=1}^{\infty} C_n(\tau) \sin \frac{n\pi x}{l} = 0 \\ \sum_{n=1}^{\infty} \frac{n\pi a}{l} D_n(\tau) \sin \frac{n\pi x}{l} = f(x, \tau) \end{cases}$$

比较系数可得

$$C_n(\tau) = 0$$

$$D_n(\tau) = \frac{l}{n\pi a} \frac{2}{l} \int_0^l f(x, \tau) \sin \frac{n\pi x}{l} dx$$

则

$$w_1(x, t, \tau) = \sum_{n=1}^{\infty} D_n(\tau) \sin \frac{n\pi a}{l}(t-\tau) \sin \frac{n\pi x}{l}$$

得

$$\begin{aligned} w(x, t) &= \int_0^t w_1(x, t, \tau) d\tau \\ &= \sum_{n=1}^{\infty} \int_0^t D_n(\tau) \sin \frac{n\pi a}{l}(t-\tau) d\tau \cdot \sin \frac{n\pi x}{l} \end{aligned}$$

综上所述

$$u(x, t) = v(x, t) + w(x, t)$$

$$= \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{l} t + B_n \sin \frac{n\pi a}{l} t \right) \sin \frac{n\pi x}{l} + \sum_{n=1}^{\infty} \int_0^t D_n(\tau) \sin \frac{n\pi a}{l}(t-\tau) d\tau \cdot \sin \frac{n\pi x}{l}$$

其中

$$A_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx$$

$$B_n = \frac{2}{l} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx$$

$$D_n(\tau) = \frac{2}{n\pi a} \int_0^l f(x, \tau) \sin \frac{n\pi}{l} x dx$$

6. 求解输运问题

$$\begin{cases} u_t - a^2 u_{xx} = -bu_x, & (b \text{ 为常数}), \\ u|_{x=0} = 0, u|_{x=l} = 0 \\ u|_{t=0} = \varphi(x), & (0 < x < l) \end{cases}$$

能否用傅里叶级数法求解?如果不能,要说明原因;如果能,将 $u(x, t)$ 解出来.

解:本题可以用傅里叶级数法求解.

设 $u(x, t) = \sum_{n=1}^{\infty} T_n(t) e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l}$, 本征函数 $e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l}$ 在 $(0, l)$ 上带权重 $e^{-\frac{bx}{2a}}$ 正交

$$\text{则 } u_x(x, t) = \sum_{n=1}^{\infty} T_n(t) \left(\frac{b}{2a^2}\right) e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} + \sum_{n=1}^{\infty} T_n(t) e^{\frac{bx}{2a}} \left(\frac{n\pi}{l}\right) \cos \frac{n\pi x}{l}$$

$$\begin{aligned} u_{xx}(x, t) &= \sum_{n=1}^{\infty} T_n(t) \left(\frac{b}{2a^2}\right)^2 e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} + 2 \sum_{n=1}^{\infty} T_n(t) \left(\frac{b}{2a^2}\right) \left(\frac{n\pi}{l}\right) e^{\frac{bx}{2a}} \cos \frac{n\pi x}{l} \\ &\quad - \sum_{n=1}^{\infty} T_n(t) \left(\frac{n\pi}{l}\right)^2 e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} \end{aligned}$$

$$u_t(x, t) = \sum_{n=1}^{\infty} T'_n(t) e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l}$$

代入泛定方程得

$$\sum_{n=1}^{\infty} \left[T'_n(t) - \left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) T_n(t) \right] e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} = 0$$

可知 $T'_n(t) - \left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2} \right) T_n(t) = 0$ 解得

$$T_n(t) = \varphi_n e^{-\left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2}\right)t}$$

则解可写为

$$u(x, t) = \sum_{n=1}^{\infty} \varphi_n e^{-\left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2}\right)t} e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l}$$

代入初始条件

$$\sum_{n=1}^{\infty} \varphi_n e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} = \varphi(x)$$

两边傅里叶展开, 比较系数可得

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \left(e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l} \right) e^{-\frac{bx}{2a}} dx$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \varphi_n e^{-\left(\frac{b^2}{4a^2} + \frac{n^2 \pi^2 a^2}{l^2}\right)t} e^{\frac{bx}{2a}} \sin \frac{n\pi x}{l}$$

其中 φ_n 由傅里叶展开系数给出.

7. 均匀细导线, 每单位长的电阻为 R , 通以恒定电流 I , 导线表面跟周围温度为 0°C 的介质进行热量交换, 试解线上温度变化, 设初始温度和两端温度都是 0°C .

解: 由题意得定解问题为

方法 1: 傅里叶级数法

$$\begin{cases} u_t - a^2 u_{xx} + \frac{h}{c\rho} u = \frac{1}{c\rho} I^2 R \\ u|_{x=0} = 0 \quad u|_{x=l} = 0 \\ u|_{t=0} = 0 \end{cases}$$

其中 $\frac{h}{c\rho}u$ 为导线表面与周围温度为 0°C 介质进行的能量交换

对应泛定方程的齐次方程 $u_t - a^2 u_{xx} + \frac{h}{c\rho} u = 0$, 应用分离变数量, 令 $u(x, t) = X(x)$

则
$$XT' - a^2 X''T + \frac{h}{c\rho} XT = 0$$

令
$$\frac{T'}{a^2 T} = \frac{X''}{X} - \frac{h}{c\rho} = -\lambda \quad (\lambda > 0)$$

得到
$$\begin{cases} X'' + \left(\lambda - \frac{h}{a^2 c\rho}\right) X = 0 & X(0) = 0, X(l) = 0 \\ T' + \lambda a^2 T = 0 & T(0) = 0 \end{cases}$$

由 X 的方程及边界条件, 可知 $X(x)$ 的本征函数为 $\frac{\sin n\pi x}{l}$, $\sqrt{\lambda - \frac{h}{a^2 c\rho}} \cdot l = n\pi (n = 1, 2, \dots)$

令 $u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l}$, 代入非齐次泛定方程

$$\sum_{n=1}^{\infty} \left[T'_n(t) + \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) T_n(t) \right] \sin \frac{n\pi x}{l} = \frac{1}{c\rho} I^2 R$$

对等式右边 $\frac{I^2 R}{c\rho}$ 用 $\sin \frac{n\pi x}{l}$ 进行傅里叶展开, 比较系数可得

$$T'_n(t) + \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) T_n(t) = \frac{2}{l} \int_0^l \frac{I^2 R}{c\rho} \sin \frac{n\pi x}{l} dx = \frac{2}{n\pi} \cdot \frac{I^2 R}{c\rho} [1 - (-1)^n]$$

当 $n = 2k$ 时, $T'_{2k}(t) + \left[\frac{(2k)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] T_{2k}(t) = 0$, 解得 $T_{2k}(t) = C_{2k} e^{-\left[\frac{(2k)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] t}$

由初始条件, $T_{2k}(0) = 0$, 则 $C_{2k} = 0$

当 $n = 2k+1$ 时, $T'_{2k+1}(t) + \left[\frac{(2k+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] T_{2k+1}(t) = 0$, 另有初始条件 $T_{2k+1}(0) = 0$, 解得:

$$T_{2k+1}(t) = \frac{4I^2 R}{\pi c\rho} \cdot \frac{1}{2k+1} \cdot \frac{1}{(2k+1)^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} \cdot \{1 - e^{-\left[\frac{(2k+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] t}\}$$

将 T_{2k+1} 代回 $u(x, t)$, 并将角标 k 重写成 n , 则

$$u(x, t) = \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} \cdot \frac{1}{(2n+1)^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} \{1 - e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right] t}\} \sin \frac{(2n+1)\pi x}{l}$$

方法 2: 冲量定理法

应用冲量定理法, 先求解

$$\begin{cases} V_t - a^2 V_{xx} + \frac{h}{c\rho} V = 0 \\ V|_{x=0} = 0 \quad V|_{x=l} = 0 \\ V|_{t=\tau} = \frac{I^2 R}{c\rho} \end{cases}$$

类似方法 1 中求齐次泛定方程的解, 用分离变数法解出

$$v(x, t, \tau) = \sum_{n=1}^{\infty} C_n e^{-\left(\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right)(t-\tau)} \sin \frac{n\pi x}{l}$$

其中系数

$$\begin{aligned} C_n &= \frac{2}{l} \int_0^l \frac{I^2 R}{c\rho} \sin \frac{n\pi \xi}{l} d\xi = \frac{2I^2 R}{c\rho} \cdot \frac{1}{n\pi} [1 - (-1)^n] \\ &= \begin{cases} \frac{4I^2 R}{c\rho} \cdot \frac{1}{n\pi} & n \text{ 为奇数} \\ 0 & n \text{ 为偶数} \end{cases} \end{aligned}$$

这样

$$v(x, t, \tau) = \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]t} \sin \frac{(2n+1)\pi x}{l}$$

从而 $u(x, t) = \int_0^t v(x, t, \tau) d\tau$

$$\begin{aligned} &= \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]t} \sin \frac{(2n+1)\pi x}{l} \int_0^t e^{\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]\tau} d\tau \\ &= \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} e^{-\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]t} \sin \frac{(2n+1)\pi x}{l} \cdot \frac{1}{(2n+1)^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} \\ &\quad \cdot \{e^{\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]t} - 1\} \\ &= \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} \cdot \frac{1}{(2n+1)^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} \{1 - e^{\left[\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}\right]t}\} \sin \frac{(2n+1)\pi x}{l} \end{aligned}$$

§ 8.3 非齐次边界条件的处理

1. 求解细杆导热问题. 杆长 l , 初始温度均匀为 u_0 , 两端分别保持温度 u_1 和 u_2 .

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 & (0 < x < l) \\ u|_{x=0} = u_1 \quad u|_{x=l} = u_2 \\ u|_{t=0} = u_0 \end{cases}$$

设 $u(x, t) = v(x, t) + \omega(x, t)$ 且 v, ω 分别满足

$$\begin{cases} v_t - a^2 v_{xx} = 0 \\ v|_{x=0} = u_1 \quad v|_{x=l} = u_2 \end{cases} \quad \begin{cases} \omega_t - a^2 \omega_{xx} = 0 \\ \omega|_{x=0} = \omega|_{x=l} = 0 \end{cases}$$

令 $v = \frac{u_2 - u_1}{l}x + u_1$ 即满足条件, 此时

$$\omega|_{t=0} = u_0 - \frac{u_2 - u_1}{l}x - u_1$$

下面求解关于 w 的定解问题

$$\begin{cases} u_t - a^2 \omega_{xx} = 0 \\ \omega|_{x=0} = \omega|_{x=l} = 0 \\ \omega|_{t=0} = -\frac{u_2 - u_1}{l}x + u_0 - u_1 \end{cases}$$

由第一类齐次边界条件, 可设

$$\omega(x, t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi x}{l} \quad (n = 1, 2, \dots)$$

代入初始条件得

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = -\frac{u_2 - u_1}{l}x + u_0 - u_1$$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l \left(-\frac{u_2 - u_1}{l}x + u_0 - u_1 \right) \sin \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \left[\int_0^l -\frac{u_1 - u_2}{l}x \sin \frac{n\pi x}{l} dx + \int_0^l (u_0 - u_1) \sin \frac{n\pi x}{l} dx \right] \\ &= \frac{2}{l} \cdot \left\{ \frac{(u_2 - u_1)(-1)^n}{n\pi} + \frac{l}{n\pi} (u_0 - u_1) [1 - (-1)^n] \right\} \\ &= \frac{2}{n\pi} \{ (u_2 - u_1)(-1)^n + (u_1 - u_0)[(-1)^n - 1] \} \end{aligned}$$

$$\omega(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{(u_1 - u_0)(-1)^n}{n} + \frac{(u_1 - u_0)[(-1)^n - 1]}{n} \right\} e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi x}{l}$$

$$u(x, t) = v(x, t) + w(x, t)$$

$$= u_1 + \frac{u_2 - u_1}{l}x + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \{ (u_1 - u_0)[(-1)^n - 1] + (u_2 - u_1)(-1)^n \} e^{-\frac{n^2 \pi^2 a^2}{l^2} t} \sin \frac{n\pi x}{l}$$

2. 求解细杆导热问题, 初始温度为零, 一端 $x = l$ 保持零度, 另一端 $x = 0$ 的温度为 At (A 是常数, t 代表时间).

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 & (0 < x < l) \\ u|_{x=l} = 0 & u|_{x=0} = At \\ u|_{t=0} = 0 \end{cases}$$

$$\text{设 } u(x, t) = v(x, t) + w(x, t)$$

$$\text{令 } v(x, t) = At \left(1 - \frac{x}{l} \right) - \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right]$$

则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = 0 \\ v|_{x=0} = At & v|_{x=l} = 0 \\ v|_{t=0} = -\frac{l^2 A}{6a^2} \left[\left(\frac{x}{l} \right)^3 - 3 \left(\frac{x}{l} \right)^2 + 2 \left(\frac{x}{l} \right) \right] \end{cases}$$

则 $\omega(x, t)$ 满足

$$\begin{cases} \omega_t - a^2 \omega_{xx} = 0 \\ \omega|_{x=0} = \omega|_{x=l} = 0 \\ \omega|_{t=0} = \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l}\right)^3 - 3\left(\frac{x}{l}\right)^2 + 2\left(\frac{x}{l}\right) \right] \end{cases}$$

由第一类齐次边界条件, 可设

$$\omega(x, t) = \sum_{n=1}^{\infty} A_n e^{\frac{-n^2 \pi^2 a^2 t}{l^2}} \sin \frac{n\pi x}{l} \quad (n = 1, 2, 3 \cdots)$$

代入初始条件得

$$\begin{aligned} \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} &= \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l}\right)^3 - 3\left(\frac{x}{l}\right)^2 + 2\left(\frac{x}{l}\right) \right] \\ A_n &= \frac{2}{l} \int_0^l \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l}\right)^3 - 3\left(\frac{x}{l}\right)^2 + 2\left(\frac{x}{l}\right) \right] \sin \frac{n\pi x}{l} dx \\ I_1 &= \int_0^l \left(\frac{x}{l}\right) \sin \frac{n\pi x}{l} dx = \frac{l}{n^2 \pi^2} \int_0^l \left(\frac{n\pi x}{l}\right) \sin \frac{n\pi x}{l} d\left(\frac{n\pi x}{l}\right) \\ &= \frac{l}{n^2 \pi^2} \int_0^{n\pi} t \sin t dt = \frac{l}{n^2 \pi^2} (-n\pi)(-1)^n \\ &= -\frac{l}{n\pi} (-1)^n \\ I_2 &= \int_0^l \left(\frac{x}{l}\right)^2 \sin \frac{n\pi x}{l} dx = \frac{l}{n^3 \pi^3} \int_0^l \left(\frac{n\pi x}{l}\right)^2 \sin \frac{n\pi x}{l} d\left(\frac{n\pi x}{l}\right) \\ &= \frac{l}{n^3 \pi^3} \int_0^{n\pi} t^2 \sin t dt \\ &= \frac{l}{n^3 \pi^3} \left(-t^2 \cos t \Big|_0^{n\pi} + 2 \int_0^{n\pi} t \cos t dt \right) \\ &= \frac{l}{n^3 \pi^3} \left[-n^2 \pi^2 (-1)^n + 2t \sin t \Big|_0^{n\pi} - 2 \int_0^{n\pi} \sin t dt \right] \\ &= \frac{l}{n^3 \pi^3} [-n^2 \pi^2 (-1)^n + 2(-1)^n - 2] \\ &= -\frac{l}{n\pi} (-1)^n + \frac{2l}{n^3 \pi^3} [(-1)^n - 1] \\ I_3 &= \int_0^l \left(\frac{x}{l}\right)^3 \sin \frac{n\pi x}{l} dx = \frac{l}{n^4 \pi^4} \int_0^l \left(\frac{n\pi x}{l}\right)^3 \sin \frac{n\pi x}{l} d\left(\frac{n\pi x}{l}\right) \\ &= \frac{l}{n^4 \pi^4} \int_0^{n\pi} t^3 \sin t dt \\ &= \frac{l}{n^4 \pi^4} \left(-t^3 \cos t \Big|_0^{n\pi} + 3 \int_0^{n\pi} t^2 \cos t dt \right) \\ &= \frac{l}{n^4 \pi^4} \left[-n^3 \pi^3 (-1)^n + 3t^2 \sin t \Big|_0^{n\pi} - \int_0^{n\pi} \sin t dt \right] \\ &= \frac{l}{n^4 \pi^4} [-n^3 \pi^3 (-1)^n + 6(-n\pi)(-1)^n] \\ &= -\frac{l}{n\pi} (-1)^n + \frac{6l}{n^3 \pi^3} (-1)^n \end{aligned}$$

$$A_n = \frac{lA}{3a^2} (I_3 - 3I_2 + 2I_1)$$

$$= \frac{LA}{3a^2} \left\{ -\frac{l}{n\pi}(-1)^n - \frac{6l}{n^3\pi^3}(-1)^n + \frac{3l}{n\pi}(-1)^n - \frac{6l}{n^3\pi^3}[(-1)^n - 1] + \frac{2l}{n\pi}(-1)^n \right\}$$

$$= \frac{LA}{3a^2} \frac{6l}{n^3\pi^3} = \frac{2l^2 A}{n^3\pi^3 a^2}$$

$$w(x, t) = \frac{2Al^2}{\pi^3 a^2} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 a^2}{l^2} t}$$

$$u(x, t) = v(x, t) + w(x, t)$$

$$= At \left(1 - \frac{x}{l}\right) - \frac{l^2 A}{6a^2} \left[\left(\frac{x}{l}\right)^3 - 3\left(\frac{x}{l}\right)^2 + 2\left(\frac{x}{l}\right) \right] + \frac{2Al^2}{\pi^3 a^2} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin \frac{n\pi x}{l} e^{-\frac{n^2 \pi^2 a^2}{l^2} t}$$

3. 求解均匀杆的纵振动, 杆长 l , 一端固定, 另一端受纵向力

$$F(t) = F_0 \sin \omega t$$

作用, 初始位移和速度分别是 $\varphi(x)$ 和 $\psi(x)$.

解: 由题意得定解问题为

$$\begin{cases} u_x - a^2 u_{xx} = 0 & (0 < x < l) \\ u|_{x=0} = 0 & u_x|_{x=l} = \frac{F_0}{YS} \sin \omega t \\ u|_{t=0} = \varphi(x) & u_t|_{t=0} = \psi(x) \end{cases}$$

设 $u(x, t) = v(x, t) + w(x, t)$

令

$$v(x, t) = \frac{F_0 a}{YS \omega \cos \frac{\omega l}{a}} \sin \frac{\omega x}{a} \sin \omega t$$

则 $v(x, t)$ 满足

$$\begin{cases} v_x - a^2 v_{xx} = 0 \\ v|_{x=0} = 0, v_x|_{x=l} = \frac{F_0}{YS} \sin \omega t \\ v|_{t=0} = 0 \\ v_t|_{t=0} = \frac{F_0 a}{YS \cos \frac{\omega l}{a}} \sin \frac{\omega x}{a} \end{cases}$$

则 $w(x, t)$ 满足

$$\begin{cases} w_x - a^2 w_{xx} = 0 \\ w|_{x=0} = w_x|_{x=l} = 0 \\ w|_t = \varphi(x), w_t|_{t=0} = \psi(x) - \frac{F_0 a \sin \frac{\omega x}{a}}{YS \cos \frac{\omega l}{a}} \end{cases}$$

由第一类和第二类混合的齐次边界条件可设

$$w(x, t) = \sum_{n=0}^{\infty} \left[A_n \cos \frac{\left(n + \frac{1}{2}\right) \pi a t}{l} + B_n \sin \frac{\left(n + \frac{1}{2}\right) \pi a t}{l} \right] \cdot \sin \frac{\left(n + \frac{1}{2}\right) \pi}{l} x \quad (n = 0, 1, 2, \dots)$$

代入初始条件得

$$\sum_{n=1}^{\infty} A_n \sin \frac{\left(n + \frac{1}{2}\right)\pi}{l} x = \varphi(x)$$

$$\sum_{n=1}^{\infty} \frac{\left(n + \frac{1}{2}\right)\pi a}{l} B_n \sin \frac{\left(n + \frac{1}{2}\right)\pi}{l} x = \psi(x) - \frac{F_0 a \sin \frac{\omega x}{a}}{Y S \cos \frac{\omega l}{a}}$$

将等号右边用傅里叶展开并比较系数可得

$$A_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx$$

$$B_n = \frac{2}{\left(n + \frac{1}{2}\right)\pi a} \int_0^l \left[\psi(x) - \frac{F_0 a \sin \frac{\omega x}{a}}{Y S \cos \frac{\omega l}{a}} \right] \sin \frac{n\pi x}{l} dx$$

$$u(x, t) = v(x, t) + w(x, t)$$

$$= \frac{F_0 a}{Y S \omega \cos \frac{\omega l}{a}} \sin \frac{\omega x}{a} \sin \omega t + \sum_{n=0}^{\infty} \left[A_n \cos \frac{\left(n + \frac{1}{2}\right)\pi a t}{l} + B_n \sin \frac{\left(n + \frac{1}{2}\right)\pi a t}{l} \right] \sin \frac{\left(n + \frac{1}{2}\right)\pi x}{l}$$

其中 A_n, B_n 由上面两积分式确定.

4. 长为 l 的柱形管, 一端封闭, 另一端开放. 管外空气中含有某种气体, 其浓度为 u_0 , 向管内扩散. 求解该气体在管内的浓度 $u(x, t)$.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{x=0} = u_0 \quad u_x|_{x=l} = 0 \\ u|_{t=0} = 0 \end{cases}$$

设 $u(x, t) = u_0 + v(x, t)$, 则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = 0 \\ v|_{x=0} = 0 \quad v_x|_{x=l} = 0 \\ v|_{t=0} = -u_0 \end{cases}$$

设 $u(x, t) = u_0 + v(x, t)$, 则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = 0 \\ v|_{x=0} = 0 \quad v_x|_{x=l} = 0 \\ v|_{t=0} = -u_0 \end{cases}$$

由第一类和第二类混合的齐次边界条件可设

$$v(x, t) = \sum_{n=0}^{\infty} A_n e^{-\frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2}{l^2} t} \sin \frac{\left(n + \frac{1}{2}\right)\pi x}{l}$$

代入初始条件可得

$$\sum_{n=0}^{\infty} A_n \sin \frac{\left(n + \frac{1}{2}\right) \pi x}{l} = -u_0$$

$$\begin{aligned} A_n &= \frac{2}{l} \int_0^l (-u_0) \sin \frac{\left(n + \frac{1}{2}\right) \pi x}{l} dx \\ &= \frac{2}{l} (-u_0) \left[-\frac{l}{\left(n + \frac{1}{2}\right) \pi} \cos \frac{\left(n + \frac{1}{2}\right) \pi x}{l} \right] \Big|_0^l \\ &= -\frac{2u_0}{\left(n + \frac{1}{2}\right) \pi} \end{aligned}$$

所以
$$v(x, t) = -\frac{2u_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{\left(n + \frac{1}{2}\right)} e^{-\frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2 t}{l^2}} \sin \frac{\left(n + \frac{1}{2}\right) \pi x}{l}$$

$$u(x, t) = u_0 - \frac{2u_0}{\pi} \sum_{n=0}^{\infty} \frac{1}{n + \frac{1}{2}} e^{-\frac{\left(n + \frac{1}{2}\right)^2 \pi^2 a^2 t}{l^2}} \sin \frac{\left(n + \frac{1}{2}\right) \pi x}{l}$$

5. 把弹簧上端 $x=0$ 加以固定, 在静止弹簧的下端 $x=l$ 轻轻地挂上质量为 m 的物体, 求解弹簧的纵振动. 弹簧本身的重量可忽略不计.

解: 由题意得定解问题为

$$\begin{cases} u_x - a^2 u_{xx} = 0 & (a^2 = E/\rho) \\ u|_{x=0} = 0 & (ES u_x + m u_x)|_{x=l} = 0 \\ u|_{t=0} = -\frac{mg}{ES} x & u_t|_{t=0} = 0 \end{cases}$$

令 $u(x, t) = x(x)T(t)$, 代入泛定方程得 $XT'' - a^2 X''T = 0$, 即 $\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda (\lambda \geq 0)$

当 $\lambda = 0$, 则 $\begin{cases} X''(x) = 0 \\ T''(t) = 0 \end{cases}$ 代入边界条件解得 $u = -\frac{mg}{ES} x$

当 $\lambda > 0$, 则 $\begin{cases} X'' + \lambda x = 0 \\ x(0) = 0, X'(l) - \frac{\lambda m}{\rho S} X(l) = 0 \end{cases}$, 且 $\begin{cases} T'' + \lambda a^2 T = 0 \\ T'(0) = 0 \end{cases}$

由 X 的泛定方程和边界条件可得

$$X(x) = A_n \sin \sqrt{\lambda_n} x \quad (\text{其中 } \lambda_n \text{ 方程 } \sqrt{\lambda} \tan \sqrt{\lambda} l = \frac{S\rho}{m} \text{ 的第 } n \text{ 个解, } n = 1, 2, 3, \dots)$$

由 T 的泛定方程和边界条件可得

$$T(t) = B_n \cos a \sqrt{\lambda_n} t$$

由

$$u = \sum_{n=1}^{\infty} C_n \sin \sqrt{\lambda_n} x \cos a \sqrt{\lambda_n} t$$

由于本题边界条件不属于第一、二、三类边界条件任意一种,所以关于 x 的本征函数不正交,

$$\text{即} \int_0^l \sin \sqrt{\lambda_n} x \sin \sqrt{\lambda_n} x dx \neq 0, \text{而是} \int_0^l \sin \sqrt{\lambda_n} x \sin \sqrt{\lambda_n} dx + \frac{m}{S\rho} \sin \sqrt{\lambda_n} l \sin \sqrt{\lambda_n} l = 0 (*)$$

当 $t=0, u|_{t=0} = -\frac{mg}{ES}x = \sum_{n=1}^{\infty} C_n \sin \sqrt{\lambda_n} x$, 为求 C_n 需将其代入 $\sin \sqrt{\lambda_n} x$ 所满足的 * 式正交关系

$$\begin{aligned} & \int_0^l \sum_{m=1}^{\infty} C_m \sin \sqrt{\lambda_m} x \sin \sqrt{\lambda_n} x dx + \frac{m}{S\rho} \sum_{m=1}^{\infty} C_m \sin \sqrt{\lambda_m} l \sin \sqrt{\lambda_n} l \\ &= \int_0^l C_n \sin^2 \sqrt{\lambda_n} x dx + \frac{m}{S\rho} C_n \sin^2 \sqrt{\lambda_n} l \\ &= \frac{1}{2} \left(l + \frac{m}{S\rho} \sin^2 \sqrt{\lambda_n} l \right) C_n \end{aligned}$$

$$\text{而 } \lambda_n \text{ 满足方程 } \sqrt{\lambda_n} \tan \sqrt{\lambda_n} l = \frac{S\rho}{m}, \text{ 可求得 } \frac{m}{S\rho} \sin^2 \sqrt{\lambda_n} l = \frac{\rho S m}{\rho^2 S^2 + m^2 \lambda_n}$$

$$\therefore \int_0^l u(x, 0) \sin \sqrt{\lambda_n} x dx + \frac{m}{S\rho} u(l, 0) \sin \sqrt{\lambda_n} l = \frac{1}{2} \left(l + \frac{\rho S m}{\rho^2 S^2 + m^2 \lambda_n} \right) C_n$$

$$\begin{aligned} \text{而} \int_0^l u(x, 0) \sin \sqrt{\lambda_n} x dx + \frac{m}{S\rho} u(l, 0) \sin \sqrt{\lambda_n} l &= \int_0^l \left(-\frac{mg}{ES} x \right) \sin \sqrt{\lambda_n} x dx + \frac{m}{S\rho} \left(-\frac{mg}{ES} l \right) \sin \sqrt{\lambda_n} l \\ &= \frac{mg}{ES\lambda_n} (\sqrt{\lambda_n} \cos \sqrt{\lambda_n} l - \sin \sqrt{\lambda_n} l) - \frac{m^2 g}{\rho ES^2} \sin \sqrt{\lambda_n} l \end{aligned}$$

$$\text{有} \frac{1}{2} \left(l + \frac{\rho S m}{\rho^2 S^2 + m^2 \lambda_n} \right) C_n = \frac{mg}{ES\lambda_n} (\sqrt{\lambda_n} l \cos \sqrt{\lambda_n} l - \sin \sqrt{\lambda_n} l) - \frac{m^2 g}{\rho ES^2} \sin \sqrt{\lambda_n} l$$

$$\text{即 } C_n = \frac{2}{l + \frac{\rho S m}{\rho^2 S^2 + m^2 \lambda_n}} \cdot \left[\frac{mg}{ES\lambda_n} (\sqrt{\lambda_n} l \cos \sqrt{\lambda_n} l - \sin \sqrt{\lambda_n} l) - \frac{m^2 g}{\rho ES^2} \sin \sqrt{\lambda_n} l \right]$$

综合 $\lambda=0$ 和 $\lambda>0$ 情况, 所求解叠加得到

$$\begin{aligned} u(x, t) &= -\frac{mg}{ES}x + \sum_{n=1}^{\infty} \frac{2}{l + \frac{\rho S m}{\rho^2 S^2 + m^2 \lambda_n}} \left[\frac{mg}{ES\lambda_n} (\sqrt{\lambda_n} l \cos \sqrt{\lambda_n} l - \sin \sqrt{\lambda_n} l) - \frac{m^2 g}{\rho ES^2} \sin \sqrt{\lambda_n} l \right] \\ &\quad \sin \sqrt{\lambda_n} x \cos \sqrt{\lambda_n} t \end{aligned}$$

§ 8.4 泊松方程

1. 在圆域 $\rho < \rho_0$ 上求解 $\Delta u = -4$, 边界条件是 $u|_{\rho=\rho_0} = 0$.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = -4, (\rho < a) \\ u|_{\rho=a} = 0 \end{cases}$$

设

$$u(\rho, \varphi) = v(\rho, \varphi) + w(\rho, \varphi)$$

令 $v(\rho, \varphi) = -\rho^2$, 代回泛定方程检验, 得 $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -2 - 2 = -4$, 满足方程

$v(\rho, \varphi) = -\rho^2$ 是问题一个特解, 则对 $w(\rho, \varphi)$, 有

$$\begin{cases} \Delta w = 0 \\ w|_{\rho=a} = 0 - (-a^2) = a^2 \end{cases}$$

显然, $w(\rho, \varphi) = a^2$ 即满足方程及边界条件

$$u(\rho, \varphi) = v + w = -\rho^2 + a^2$$

2. 在圆域 $\rho < \rho_0$ 上求解 $\Delta u = -xy$, 边界条件是 $u|_{\rho=\rho_0} = 0$.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = -xy & (\rho < a) \\ u|_{\rho=a} = 0 \end{cases}$$

设

$$u = v + w$$

令 $v = -\frac{1}{12}(x^3y + xy^3)$ 代回泛定方程检验, 得

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = -\frac{1}{2}xy - \frac{1}{2}xy = -xy$$

满足方程 $v = -\frac{1}{12}(x^3y + xy^3)$ 是问题的一个特解, 即 $v = -\frac{1}{12}\rho^4 \sin \varphi \cos \varphi$,

则对 $w(\rho, \varphi)$, 有

$$\begin{cases} \Delta w = 0 \\ w|_{\rho=a} = 0 - \left(-\frac{1}{12}a^4 \sin \varphi \cos \varphi\right) = \frac{1}{24}a^4 \sin 2\varphi \end{cases}$$

设 $w = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m (C_m \rho^m + D_m \rho^{-m})$

由 $\rho = 0$ 处圆心为有限值, 故舍去 ρ^{-m} 项以及 $\ln \rho$ 项, 则

$$w = C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m$$

由 $\rho = a$ 处边界条件, 可得

$$\begin{aligned} w|_{\rho=a} &= C_0 + \sum_{m=1}^{\infty} (A_m \cos \varphi + B_m \sin m\varphi) a^m \\ &= \frac{1}{24}a^4 \sin 2\varphi \end{aligned}$$

比较系数, 可得

$$\begin{cases} C_0 = 0 \\ A_m = 0 \\ B_2 = \frac{1}{24}a^2 \\ B_m = 0 \quad (m \neq 2) \end{cases}$$

$$w = \frac{1}{24}a^2\rho^2 \sin 2\varphi$$

$$u(\rho, \varphi) = v + w = \frac{1}{24}\rho^2(a^2 - \rho^2)\sin^2\varphi$$

3. 在矩形域 $0 < x < a$, $-b/2 < y < +b/2$ 上求解 $\Delta u = -2$, 且 u 在边界上的值为零.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = -2, (0 < x < a, -\frac{b}{a} < y < \frac{b}{2}) \\ u|_{x=0} = 0, u|_{x=a} = 0 \\ u|_{y=-\frac{b}{2}} = 0, u|_{y=\frac{b}{2}} = 0 \end{cases}$$

取试探解 $u = A + Bx + Cx^2$, 使其满足

$$\begin{cases} \Delta v = -2 \\ v|_{x=0} = v|_{x=a} = 0 \\ 2C = -2 \\ A = 0 \\ Ca^2 + Ba + A = 0 \end{cases}$$

得到

$$\begin{cases} A = 0 \\ B = a \\ C = -1 \end{cases}$$

解得

$$v = ax - x^2$$

令 $u(x, y) = ax - x^2 + w(x, y)$, 则定解问题转化为求解 $w(x, y)$

$$\begin{cases} \Delta w = 0 \\ w|_{x=0} = w|_{x=a} = 0 \\ w|_{y=-\frac{b}{2}} = w|_{y=\frac{b}{2}} = x^2 - ax \end{cases}$$

由 x 的边界条件为第一类边界条件, 可知其本征函数为 $\sin \frac{n\pi}{a}x$, 则可设

$$w(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a}y + B_n \sinh \frac{n\pi}{a}y \right) \sin \frac{n\pi}{a}x$$

代入 y 的边界条件可得

$$\begin{cases} \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi b}{2a} + B_n \sinh \frac{n\pi b}{2a} \right) \sin \frac{n\pi x}{a} = x^2 - ax \\ \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi b}{2a} - B_n \sinh \frac{n\pi b}{2a} \right) \sin \frac{n\pi x}{a} = x^2 - ax \end{cases}$$

$$B_n = 0$$

$$A_n = \frac{1}{\cosh \frac{n\pi b}{2a}} \cdot \frac{2}{a} \int_0^a (x^2 - ax) \sin \frac{n\pi x}{a} dx$$

$$\int_0^a x \sin \frac{n\pi x}{a} dx = \left. -\frac{a}{n\pi} x \cos \frac{n\pi x}{a} \right|_0^a + \int_0^a \left(\frac{a}{n\pi} \right) \cos \frac{n\pi x}{a} dx$$

$$\begin{aligned}
 &= -\frac{a^2}{n\pi}(-1)^n + \frac{a}{n\pi} \frac{a}{n\pi} \sin \frac{n\pi x}{a} \Big|_0^a \\
 &= -\frac{a^2}{n\pi}(-1)^n
 \end{aligned}$$

$$\begin{aligned}
 \int_0^a x^2 \sin \frac{n\pi x}{a} dx &= -\frac{a}{n\pi} x^2 \cos \frac{n\pi x}{a} \Big|_0^a + 2 \int_0^a \frac{a}{n\pi} x \cos \frac{n\pi x}{a} dx \\
 &= -\frac{a^3}{n\pi}(-1)^n + \frac{2a}{n\pi} x \left(\frac{a}{n\pi} \right) \sin \frac{n\pi x}{a} \Big|_0^a - \frac{2a}{n\pi} \left(\frac{a}{n\pi} \right) \int_0^a \sin \frac{n\pi x}{a} dx \\
 &= -\frac{a^3}{n\pi}(-1)^n + \frac{2a^2}{n^2 \pi^2} \cdot \frac{a}{n\pi} \cos \frac{n\pi x}{a} \Big|_0^a \\
 &= -\frac{a^3}{n\pi}(-1)^n + \frac{2a^3}{n^3 \pi^3} [(-1)^n - 1]
 \end{aligned}$$

$$\begin{aligned}
 A_n &= \frac{1}{\cosh \frac{n\pi b}{2a}} \frac{2}{a} \left\{ -\frac{a^3}{n\pi}(-1)^n + \frac{2a^3}{n^3 \pi^3} [(-1)^n - 1] + \frac{a^3}{n\pi}(-1)^n \right\} \\
 &= \frac{4a^2}{\cosh \frac{n\pi b}{2a}} \frac{1}{n^3 \pi^3} [(-1)^n - 1] \\
 &= \begin{cases} \frac{-8a^2}{(2k+1)^3 \pi^3 \cosh \frac{(2k+1)\pi b}{2a}}, & (n=2k+1), (k=0,1,2,\dots) \\ 0, & (n=2k) \end{cases}
 \end{aligned}$$

$$w(x, y) = -\frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3 \cosh \frac{(2k+1)\pi b}{2a}} \cosh \frac{n\pi y}{a} \sin \frac{n\pi x}{a}$$

所以 $u(x, y) = ax - x^2 - \frac{8a^2}{\pi^3} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3 \cosh \frac{(2k+1)\pi b}{2a}} \cosh \frac{n\pi y}{a} \sin \frac{n\pi x}{a}$

4. 在矩形域 $0 < x < a$, $-b/2 < y < +b/2$ 上求解 $\Delta u = -x^2 y$, 且 u 在边界上的值为零.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = -x^2 y \\ u|_{x=0} = 0 \quad u|_{x=a} = 0 \\ u|_{y=-\frac{b}{2}} = 0 \quad u|_{y=\frac{b}{2}} = 0 \end{cases}$$

设 $u(x, y) = v(x, y) + w(x, y)$, 令 $v(x, y) = \frac{xy}{12}(a^3 - x^3)$, 则 $v(x, y)$ 满足

$$\begin{cases} \Delta v = -x^2 y \\ v|_{x=0} = v|_{x=a} = 0 \\ v|_{y=-\frac{b}{2}} = -\frac{xb}{24}(a^3 - b^3) \\ v|_{y=\frac{b}{2}} = \frac{xb}{24}(a^3 - x^3) \end{cases}$$

则 $w(x, t)$ 满足

$$\begin{cases} \Delta w = 0 \\ w|_{x=0} = w|_{x=a} = 0 \\ w|_{y=-\frac{b}{2}} = \frac{xb}{24}(a^3 - x^3) \\ w|_{y=\frac{b}{2}} = -\frac{xb}{24}(a^3 - x^3) \end{cases}$$

由 x 的边界条件为第一类齐次边界条件, 可知其本征函数为 $\sin \frac{n\pi}{a}x$, 则可设

$$w(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a}y + B_n \sinh \frac{n\pi}{a}y \right) \sin \frac{n\pi}{a}x$$

代入 y 的边界条件可得

$$\begin{cases} \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi b}{2a} - B_n \sinh \frac{n\pi b}{2a} \right) \sin \frac{n\pi x}{a} = \frac{xb}{24}(a^3 - x^3) \\ \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi b}{2a} + B_n \sinh \frac{n\pi b}{2a} \right) \sin \frac{n\pi x}{a} = -\frac{xb}{24}(a^3 - x^3) \end{cases}$$

$$A_n = 0$$

$$B_n = \frac{1}{\sinh \frac{n\pi b}{2a}} \cdot \frac{2}{a} \int_0^a \frac{(x^3 - a^3)xb}{24} \sin \frac{n\pi x}{a} dx$$

$$\begin{aligned} \int_0^a x^4 \sin \frac{n\pi x}{a} dx &= \frac{a^5}{n^5 \pi^5} \int_0^{\pi} \left(\frac{n\pi x}{a} \right)^4 d\left(\frac{n\pi y}{a} \right) \\ &= \frac{a^5}{n^5 \pi^5} \int_0^{\pi} t^4 \sin t dt \end{aligned}$$

$$\int_0^{\pi} t^4 \sin t dt = -t^4 \cos t \Big|_0^{\pi} + 4 \int_0^{\pi} t^3 \cos t dt$$

$$= -n^4 \pi^4 (-1)^n + 4t^3 \sin t \Big|_0^{\pi} - 12 \int_0^{\pi} t^2 \sin t dt$$

$$= -n^4 \pi^4 (-1)^n + 12t^2 \cos t \Big|_0^{\pi} - 24 \int_0^{\pi} \cos t dt$$

$$= -n^4 \pi^4 (-1)^n + 12n^4 \pi^2 (-1)^n - 24t \sin t \Big|_0^{\pi} + 24 \int_0^{\pi} \sin t dt$$

$$= -n^4 \pi^4 (-1)^n + 12n^2 \pi^2 (-1)^n + 24[1 - (-1)^n]$$

$$\int_0^a x^4 \sin \frac{n\pi x}{a} dx = \frac{a^5}{n^5 \pi^5} \{ -n^4 \pi^4 (-1)^n + 12n^2 \pi^2 (-1)^n + 24[1 - (-1)^n] \}$$

$$\int_0^a x \sin \frac{n\pi x}{a} dx = -\frac{a}{n\pi} x \cos \frac{n\pi x}{a} \Big|_0^a + \int_0^a \left(\frac{a}{n\pi} \right) \cos \frac{n\pi x}{a} dx = -\frac{a^2}{n\pi} (-1)^n$$

$$B_n = \frac{b}{12a \sinh \frac{n\pi b}{2a}} \left\{ \frac{a^5}{n^5 \pi^5} [-n^4 \pi^4 (-1)^n + 12n^2 \pi^2 (-1)^n + 24 - 24(-1)^n] \right.$$

$$\left. - a^3 \left(-\frac{a^2}{n\pi} \right) (-1)^n \right\} = \frac{b}{\sinh \frac{n\pi b}{2a}} \left\{ \frac{a^4}{n^3 \pi^3} (-1)^n + \frac{2a^4}{n^5 \pi^5} [1 - (-1)^n] \right\}$$

$$w(x, y) = \frac{a^4 b}{\pi^5} \sum_{n=1}^{\infty} \frac{n^2 \pi^2 (-1)^n + 2 - 2(-1)^n}{n^5 \sinh \frac{n\pi b}{2a}} \sinh \frac{n\pi y}{a} \sin \frac{n\pi x}{a}$$

$$u(x, y) = v(x, y) + w(x, y)$$

$$= \frac{xy}{12} (a^3 - x^3) + \frac{a^4 b}{\pi^5} \sum_{n=1}^{\infty} \frac{n^2 \pi^2 (-1)^n + 2 - 2(-1)^n}{n^5 \sinh \frac{n\pi b}{2a}} \sinh \frac{n\pi y}{a} \sin \frac{n\pi x}{a}$$

第九章 二阶常微分方程级数解法 本征值问题

本章知识要点

特殊函数常微分方程

(一) 拉普拉斯方程 $\Delta u = 0$

1. 在球坐标下分离变量, 得到

$$\begin{cases} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - l(l+1)R = 0 & \text{欧拉常微分方程} \\ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} + l(l+1)Y = 0 & \text{球函数方程} \end{cases}$$

球函数 $Y(\theta, \varphi)$ 进一步分解, 得到

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ \sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + [l(l+1) \sin^2 \theta - \lambda] \Theta = 0 \end{cases}$$

经变量替换, $x = \cos \theta$, 可使 θ 方程转化为 l 阶连带勒让德方程, 即

$$(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta = 0$$

若 $m = 0$, 则称之为 l 阶勒让德方程

2. 在柱坐标下分离变量, 得到

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ Z' - \mu Z = 0 \\ \frac{d^2 R}{d\rho^2} + \frac{1}{\rho} \frac{dR}{d\rho} + \left(\mu - \frac{m^2}{\rho^2} \right) R = 0 \end{cases}$$

(1) 当 $\mu > 0$

经变量替换 $x = \sqrt{\mu} \rho$, 可使 ρ 方程转化为 m 阶贝塞尔方程, 即

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + (x^2 - m^2) R = 0$$

(2) 当 $\mu = 0$

ρ 方程可转化为欧拉常微分方程, 即

$$\frac{d^2 R}{d\rho^2} + \frac{1}{\rho} \frac{dR}{d\rho} - \frac{m^2}{\rho^2} R = 0$$

(3) 当 $\mu < 0$

经变量替换, $x = \sqrt{-\mu}$, 可使 ρ 方程转化为虚宗量贝塞尔方程, 即

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} - (x^2 + m^2) R = 0$$

(二) 波动方程 $u_{tt} - a^2 \Delta u = 0$

分离时间变数 t 和空间变数 r , 得到

$$\begin{cases} T'' + k^2 a^2 T = 0 \\ \Delta v + k^2 v = 0 \end{cases}$$

(三) 输运方程 $u_t - a^2 \Delta u = 0$

分离时间变数 t 和空间变数 r , 得到

$$\begin{cases} T' + k^2 a^2 T = 0 \\ \Delta v + k^2 v = 0 \end{cases}$$

(四) 亥姆霍兹方程 $\Delta v + k^2 v = 0$

1. 在球坐标系下分离变量, 得到

$$\begin{cases} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} + l(l+1)Y = 0 & \text{球函数方程} \\ \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + [k^2 r^2 - l(l+1)]R = 0 & l \text{ 阶球贝塞尔方程} \end{cases}$$

球函数方程可进一步分解得到

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ (1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta = 0 \end{cases}$$

$x = \cos \theta$, 连带勒让德方程

2. 在柱坐标下分离变量, 得到

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ Z' + \nu^2 Z = 0 \\ \frac{d^2 R}{d\rho^2} + \frac{1}{\rho} \frac{dR}{d\rho} + \left(k^2 - \nu^2 - \frac{\lambda}{\rho^2} \right) R = 0 \end{cases}$$

令 $\mu = k^2 - \nu^2$, $x = \sqrt{\mu}\rho$, 径向方程转化为 m 阶贝塞尔方程, 即

$$\frac{d^2 R}{dx^2} + \frac{1}{x} \frac{dR}{dx} + \left(1 - \frac{m^2}{x^2} \right) R = 0$$

常点邻域上的级数解法

(一) 级数解法

1. 级数解法:对不能用通常方法解出的二阶常微分方程,可在某个任选点 x_0 的邻域上,把待求的解表示为系数待定的级数,代入方程以逐个确定系数,这种方法称作级数解法.

2. 复变函数 $w(z)$ 的线性二阶常微分方程

$$\frac{d^2 w}{dz^2} + p(z) \frac{dw}{dz} + q(z)w = 0$$

$$w(z_0) = C_0 \quad w'(z_0) = C_1$$

其中 z 为复变数, z_0 为选定的点, C_0, C_1 为复常数.

3. 方程的常点和奇点

对一般的二阶线性常微分方程,若系数函数 $p(z)$ 和 $q(z)$ 在选定的点 z_0 的邻域中解析,则点 z_0 称作方程的常点,若选定的 z_0 是 $p(z)$ 或 $q(z)$ 的奇点,则点 z_0 称作方程的奇点.

4. 常点邻域上的解析解

若 z_0 为二阶线性常微分方程的常点,则在其邻域上存在唯一的解析解,其泰勒级数的形式为

$$w(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$$

其中系数 $a_1, a_2 \cdots a_k$ 待定

(二) 勒让德方程 自然边界条件

1. 勒让德方程的级数解

一般系数递推公式

$$a_{k+2} = \frac{(k-l)(k+l+1)}{(k+2)(k+1)} a_k$$

级数解

$$y(x) = a_0 y_0(x) + a_1 y_1(x)$$

其中, $y_0(x)$ 仅含 x 偶次幂,为偶函数

$y_1(x)$ 仅含 x 奇次幂,为奇函数.

2. 勒让德方程的级数解的收敛性

当 $|x| > 1$,级数解发散,当 $|x| < 1$,级数解收敛,当 $|x| = 1$,级数解发展.

3. 自然边界条件

勒让德方程的自然边界条件是解在 $x = \pm 1$ 保持有限,使级数解退化为多项式.

4. 本征值问题

勒让德方程和自然边界条件构成本征值问题

本征值为 $l(l+1)$,本征函数是 l 阶勒让德多项式.

正则奇点邻域上的级数解法

(一) 奇点邻域上的级数解

定理:若点 z_0 为方程 $w'' + p(z)w' + q(z)w = 0$ 的奇点,则在点 z_0 的邻域 $0 < |z - z_0| < R$ 上,该方程存在两个线性独立解,其形式为

$$w_1(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^{s_1+k} \text{ 和 } w_2(z) = \sum_{k=-\infty}^{\infty} b_k (z - z_0)^{s_2+k}$$

或表示成 $w(z) = Aw_1(z) \ln(z - z_0) + \sum_{k=-\infty}^{\infty} b_k (z - z_0)^{s_2+k}$

(二) 正则奇点邻域上的级数解

1. 正则奇点:若方程奇点 z_0 的邻域 $0 < |z - z_0| < R$ 上,方程的两线性独立解全都具有有限个负幂项,则奇点 z_0 称为方程的正则奇点.

2. 判定方程: $s(s-1) + sp_{-1} + q_{-2} = 0$, 其中 s_1, s_2 即该方程的两个根, p_{-1}, q_{-2} 为系数 $p(z), q(z)$ 按 $(z - z_0)$ 幂展开式的 $-1, -2$ 项系数.

(三) 贝塞尔方程

1. (1) ν 阶贝塞尔方程 $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$ (阶 $\nu \neq$ 整数或奇数) 其中 $x_0 = 0$ 为方程正则奇点

(2) 一般系数递推公式 $a_k = \frac{-1}{(s+k+\nu)(s+k-\nu)} a_{k-2} \quad b_k = \frac{-1}{k(k-2\nu)} b_{k-2}$

(3) 级数解: $y(x) = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$, 其中 $J_{\nu(x)}$ 为 ν 阶贝塞尔函数, 具体形式为

$$J_{\nu(x)} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{k! \Gamma(\nu + k + 1)} \left(\frac{x}{2}\right)^{\nu+2k}$$

(4) ν 阶诺伊曼函数, 当令 ν 阶贝塞尔方程通解 $C_1 = \cot \nu\pi, C_2 = -\csc \nu\pi$ 时, 可得到, 另一线性独立的解, 叫作 ν 阶诺伊曼函数, 即

$$N_\nu(x) = \frac{J_\nu(x) \cos \nu\pi - J_{-\nu}(x)}{\sin \nu\pi}$$

因此 ν 阶贝塞尔方程通解也可取为

$$y(x) = C_1 J_\nu(x) + C_2 N_\nu(x)$$

2. 半奇数 $(l+1/2)$ 阶贝塞尔方程

(1) 半奇数 $(l+1/2)$ 阶贝塞尔方程: $x^2 y'' + xy' + [x^2 - (l + \frac{1}{2})^2]y = 0$ ($l = 0, 1, 2, \dots$) 其中 $x_0 = 0$ 为方程正则奇数.

(2) $(l+1/2)$ 阶贝塞尔方程的通解: $y(x) = C_1 J_{l+1/2}(x) + C_2 J_{-(l+1/2)}(x)$

3. 整数 m 阶贝塞尔方程

(1) 方程: $x^2 y'' + xy' + (x^2 - m^2)y = 0$ (m 为自然数)

其中, $x_0 = 0$ 为方程正则奇点.

(2) 通解:

$$y(x) = C_1 J_m(x) + C_2 N_m(x)$$

4. $x = 0$ 处自然边界条件

当 $x \rightarrow 0$ 时,

$$\begin{cases} J_0(x) \rightarrow 1, J_\nu \rightarrow 0, J_{-\nu}(x) \rightarrow \infty \\ N_\nu(x) \rightarrow \pm \infty, N_m(x) \rightarrow -\infty, (\nu > 0) \end{cases}$$

当研究区域包含 $x = 0$ 时,需排除 $N_0(x)$ 、 $N_m(x)$ 、 $J_{-\nu}(x)$ 、 $N_\nu(x)$,而只剩下 $J_0(x)$ 、 $J_m(x)$ 及 $J_\nu(x)$. 这样即称在 $x = 0$ 处,贝塞尔方程具有自然边界条件.

(四) 虚宗量贝塞尔方程

1. ν 阶虚宗量贝塞尔方程

(1) 方程: $x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} - (x^2 + \nu^2)R = 0$ ($\nu \neq$ 整数或半奇数)

(2) 通解: $y(x) = C_1 I_\nu(x) + C_2 I_{-\nu}(x)$

其中, $I_\nu(x)$ 称为 ν 阶虚宗量贝塞尔函数,具体形式为

$$I_\nu(x) = i^{-\nu} J_\nu(ix) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(\frac{x}{2}\right)^{\nu+2k}$$

2. 整数 m 阶虚宗量贝塞尔方程

(1) 方程: $x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} - (x^2 + m^2)R = 0$ ($m =$ 整数)

(2) 方程的一个解为 $I_m(x)$,另一个线性独立解在 § 11.4 中讨论.

施图姆 - 刘维尔本征值问题

(一) 施图姆 - 刘维尔本征值问题

1. 施图姆 - 刘维尔方程:形式为 $\frac{d}{dx} \left[k(x) \frac{dy}{dx} \right] - q(x)y + \lambda \varphi(x)y = 0$ ($a \leq x \leq b$) 的二阶常微分方程叫作施图姆 - 刘维尔型方程.

2. 施图姆 - 刘维尔方程的转化:将一般的二阶常微分方程 $y'' + a(x)y' + b(x)y + \lambda c(x)y = 0$ 乘上适当的函数 $e^{\int a(x)dx}$,就化成施图姆 - 刘维尔型方程.

3. 施图姆 - 刘维尔本征值问题:当施图姆 - 刘维尔型方程附以齐次的第 一 类、第 二 类或第 三 类边界条件或自然边界条件,就构成了施图姆 - 刘维尔本征值问题.

(二) 施图姆 - 刘维尔本征值问题的共同性质

(1) 如 $k(x)$ 、 $k'(x)$ 、 $q(x)$ 连续或者最多以 $x = a$ 和 $x = b$ 为一阶极点,则存在无限多个本征值,

$$\lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \lambda_4 \leq \dots$$

相应地有无限多个本征函数

$$y_1(x), y_2(x), y_3(x), y_4(x), \dots$$

这些本征函数的排列次序正好使节点个数依次增多.

(2) 所有本征值 $\lambda_n \geq 0$

(3) 相应于不同本征值 λ_m 和 λ_n 的本征函数 $y_m(x)$ 和 $y_n(x)$ 在区间 $[a, b]$ 上带权重 $\rho(x)$ 正交. 即

$$\int_a^b y_m(x) y_n(x) \rho(x) dx = 0 \quad (m \neq n)$$

(4) 本征函数族 $y_1(x), y_2(x), y_3(x), \dots$ 是完备的, 即函数 $f(x)$ 如具有连续一阶导数和分段连续二阶导数, 且满足本征函数族所满足的边界条件, 就可以展开为绝对且一致收敛的级数

$$f(x) = \sum_{n=1}^{\infty} f_n y_n(x)$$

(三) 广义傅里叶级数

1. 对 $f(x) = \sum_{n=1}^{\infty} f_n y_n(x)$ 等号右边的级数叫作广义傅里叶级数, 系数 $f_n (n = 1, 2, \dots)$ 叫作 $f(x)$ 的广义傅里叶级数, 函数族 $y_n(x) (n = 1, 2, \dots)$ 叫作这级数展开的基.

2. 展开基的模的计算公式

$$N_m^2 = \int_a^b [y_m(x)]^2 \rho(x) dx$$

3. 广义傅里叶系数的计算公式

$$f_m = \frac{1}{N_m^2} \int_a^b f(x) y_m(x) \rho(x) dx$$

(四) 复数的本征函数族

若本征函数为实变数的复值函数时, 则

1. 模的定义: $N_m^2 = \int_a^b y_m(x) [y_m(x)]^* \rho(x) dx$

2. 正交关系: $\int_a^b y_m(x) [y_n(x)]^* \rho(x) dx = 0$

3. 广义傅里叶系数公式: $f_m = \frac{1}{N_m^2} \int_a^b f(x) [y_m(x)]^* \rho(x) dx$

(五) 希尔伯特空间

对于一个本征值问题, 它的解构成了某种无限维的空间, 即希尔伯特空间. 它的本征函数族构成了希尔伯特空间中的基, 基底之间带权重正交. 将任一函数 $f(x)$ 在这个空间中展开, 为广义傅里叶级数, 即把 $f(x)$ 表示为各基底的线性组合, 当基底的个数低于空间维数, 则基底是不完备的. 若基底的个数等于空间维数, 则基底是完备的.

习题全解

§ 9.1 特殊函数常微分方程

1. 试用平面极坐标系把二维波动方程分离变数.

解: $u_{tt} - a^2 \Delta_2 u = 0$

令 $u(\rho, \varphi, t) = T(t)v(\rho, \varphi)$ 代入二维波动方程, 得到

$$\frac{T''}{a^2 T} = \frac{\Delta_2 v}{v}$$

左边为时间 t 的函数, 右边是空间变量 r 的函数, 对任意条件等式成立, 则等式两边恒为常数, 记作 $-k^2$.

则分解为两个方程

$$\begin{cases} T'' + k^2 a^2 T = 0 & (1) \\ \Delta_2 v + k^2 v = 0 & (2) \end{cases}$$

对时间变量 t 由(1) 易求得

$$T(t) = A \cos kat + B \sin kat$$

(A, B 为常数)

对空间变量 r , 进一步进行分离变量

(2) 式在极坐标下表达式为

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + k^2 u = 0$$

令 $v(\rho, \varphi) = R(\rho)\Phi(\varphi)$ 代入(2) 式用 $\rho^2/R\Phi$ 遍乘各项并适当移项, 得

$$\frac{\rho^2 R''}{R} + \frac{\rho R'}{R} + R^2 \rho^2 = -\frac{\Phi''}{\Phi}$$

左边为径向 ρ 的函数, 右边是角向 φ 的函数, 对任意条件等式成立, 只能使等式两边恒为常数, 记作 m^2 .

则可分解为两个方程

$$\begin{cases} \Phi'' + m^2 \Phi = 0 \\ \rho^2 R'' + \rho R' + (k^2 \rho^2 - m^2) R = 0 \end{cases}$$

对角向变量 φ , 存在周期性变界条件 $\Phi(\varphi) = \Phi(\varphi + 2\pi)$, 解之易得

$$\Phi(\varphi) = C \sin m\varphi + D \cos m\varphi (m = 0, 1, 2, \dots) (C, D \text{ 为常数})$$

对径向变量 ρ , 令 $x = k\rho$, 代入径向方程得到贝塞尔方程, 即

$$\frac{d^2 R}{dx^2} + \frac{1}{x} \frac{dR}{dx} + \left(1 - \frac{m^2}{x^2}\right) R = 0$$

2. 试用平面极坐标系把二维输运方程分离变数.

解: $u_t - a^2 \Delta_2 u = 0$

令 $u(\rho, \varphi, t) = T(t)v(\rho, \varphi)$ 代入二维输运方程, 得到

$$\frac{T'}{a^2 T} = \frac{\Delta_2 v}{v}$$

左边为时间 t 的函数, 右边是空间变量 r 的函数, 对任意条件等式成立, 则等式两边应恒为常数, 记作 $-k^2$, 则可分解为两个方程

$$\begin{cases} T' + k^2 a^2 T = 0 & (1) \\ \Delta_2 v + k^2 v = 0 & (2) \end{cases}$$

对时间变量 t , 由 ① 式易求得

$$T(t) = Ae^{-k^2 a^2 t}$$

(A 为常数)

对空间变量 r , 讨论与第 1 题完全相同

故直接给出接果

$$\Phi(\varphi) = C\sin m\varphi + D\cos m\varphi (m = 0, 1, 2, \dots)$$

$$\frac{d^2 R}{dx^2} + \frac{1}{x} \frac{dR}{dx} + \left(1 - \frac{m^2}{x^2}\right) R = 0$$

3. 氢原子定态问题的量子力学薛定谔方程是

$$-\frac{h^2}{8\pi^2\mu}\Delta u - \frac{Ze^2}{r}u = Eu,$$

其中, h, μ, Z, e, E 都是常数. 试用球坐标系把这个方程分离变量.

解: $-\frac{h^2}{8\pi^2\mu}\Delta u - \frac{Ze^2}{r}u = Eu$ 在球坐标下的表达式为

$$-\frac{h^2}{8\pi^2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial\varphi^2}\right] - \frac{Ze^2}{r}u = Eu$$

令

$$\frac{h^2}{8\pi^2\mu} = A, Ze^2 = B$$

$u(r, \theta, \varphi) = R(r)Y(\theta, \varphi)$ 代入上式, 用 r^2/ARY 遍乘各项并适当移项, 得

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{r^2}{A}\left(\frac{B}{r} + E\right) = -\frac{1}{Y\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) - \frac{1}{Y\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}$$

左边为径向变量 r 的函数, 右边是角向变量 θ, φ 的函数, 对任意条件等式成立, 则等式两边应恒为常数, 记作 $l(l+1)$, 则可分解为两个方程

$$\begin{cases} \frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{B}{A}r + \frac{E}{A}r^2 - l(l+1)\right]R = 0 \\ \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2} + l(l+1)Y = 0 \end{cases}$$

对径向变量 r , 代入 A, B 具体形式, 可得到径向方程

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{8\pi^2\mu}{h^2}\left(E + \frac{Ze^2}{r}\right) - \frac{l(l+1)}{r^2}\right]R = 0$$

对球函数方程, 令

$$Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$$

代入后化简,过程与拉普拉斯方程在球坐标下化简球函数方程完全相同,故直接给出结果对 φ :

$$\Phi(\varphi) = A \cos m\varphi + B \sin m\varphi (m = 0, 1, 2, \dots)$$

对 θ :

$$(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta = 0$$

$x = \cos \theta$, 连带勒让德方程.

§ 9.2 常点邻域上的级数解法

1. 在 $x_0 = 0$ 的邻域上求解常微分方程 $y'' + \omega^2 y = 0$ (ω 是常数).

解: 设方程的级数解为

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

则

$$y''(x) = \sum_{k=2}^{\infty} a_k k(k-1) x^{k-2}$$

并代入原方程得

$$\sum_{k=2}^{\infty} a_k k(k-1) x^{k-2} + \omega^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

即

$$\sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) x^k + \omega^2 \sum_{k=0}^{\infty} a_k x^k = 0$$

令各次幂的系数为 0, 得到系数递推公式

$$a_{k+2} = \frac{\omega^2}{(k+2)(k+1)} a_k$$

则对偶次项

$$\begin{aligned} a_2 &= -\frac{\omega^2}{2!} a_0, a_4 = -\frac{\omega^2}{4 \cdot 3} a_2 = \frac{\omega^4}{4!} a_0 \\ a_6 &= -\frac{\omega^2}{6 \cdot 5} a_4 = \frac{\omega^4}{6 \cdot 5 \cdot 4 \cdot 3} a_2 = -\frac{\omega^6}{6!} a_0 \\ &\vdots \end{aligned}$$

则对奇次项

$$\begin{aligned} a_3 &= -\frac{\omega^2}{3 \cdot 2} a_1, a_5 = -\frac{\omega^2}{5 \cdot 4} a_3 = \frac{\omega^4}{5!} a_1 \\ a_7 &= -\frac{\omega^2}{7 \cdot 6} a_5 = \frac{\omega^4}{7 \cdot 6 \cdot 5 \cdot 4} a_3 = -\frac{\omega^6}{7!} a_1 \\ &\vdots \end{aligned}$$

故

$$y(x) = a_0 y_0(x) + a_1 y_1(x)$$

其中 $y_0(x) = 1 - \frac{(\omega x)^2}{2!} + \frac{(\omega x)^4}{4!} - \frac{(\omega x)^6}{6!} + \dots + (-1)^k \frac{(\omega x)^{2k}}{(2k)!} + \dots = \cos \omega x$

$$y_1(x) = (\omega x) - \frac{1}{3!} (\omega x)^3 + \frac{1}{5!} (\omega x)^5 + \dots + (-1)^k \frac{(\omega x)^{2k+1}}{(2k+1)!} + \dots = \sin \omega x$$

$$y = a_0 y_0(x) + a_1 y_1(x) = a_0 \cos \omega x + a_1 \sin \omega x.$$

2. 在 $x_0 = 0$ 的邻域上求解 $y'' - xy = 0$.

解: 设方程的级数解

$$y(x) = \sum_{k=0}^{\infty} a_k x^k$$

则

$$y''(x) = \sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}$$

将 $y(x)$ 、 $y''(x)$ 代回原方程得

$$\sum_{k=2}^{\infty} (k-1)ka_k x^{k-2} - \sum_{k=0}^{\infty} a_k x^{k+1} = 0, \text{ 即 } \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k - \sum_{k=1}^{\infty} a_{k-1} x^k = 0$$

令各次幂的系数为 0, 得到系数递推公式

$$a_{k+2} = \frac{1}{(k-1)k} a_{k-1}$$

则

$$a_2 = a_{-1} = 0, a_5 = a_8 = \cdots = a_{3k+2} = 0$$

$$a_3 = \frac{1}{2 \cdot 3} a_0 \quad a_6 = \frac{1}{5 \cdot 6} a_3 = \frac{1 \cdot 4}{6!} a_0$$

$$a_{3k} = \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} a_0 \quad a_4 = \frac{1}{3 \cdot 4} a_1 \quad a_7 = \frac{1}{6 \cdot 7} a_4 = \frac{2 \cdot 5}{7!} a_1$$

$$a_{3k+1} = \frac{2 \cdot 5 \cdots (3k-1)}{(3k+1)!} a_1$$

故

$$y(x) = a_0 y_0(x) + a_1 y_1(x)$$

其中

$$y_0(x) = 1 + \frac{1}{2 \cdot 3} x^3 + \cdots + \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} x^{3k} + \cdots$$

$$y_1(x) = x + \frac{1}{3 \cdot 4} x^4 + \cdots + \frac{2 \cdot 5 \cdots (3k-1)}{(3k+1)!} x^{3k+1} + \cdots$$

级数解收敛半径

$$R_0 = \lim_{k \rightarrow \infty} \left| \frac{a_{3k}}{a_{3(k+1)}} \right| = \lim_{k \rightarrow \infty} \left| \frac{1 \cdot 4 \cdots (3k-2)}{(3k)!} \cdot \frac{[3(k+1)]!}{1 \cdot 4 \cdots [3(k+1)-2]} \right|$$

$$= \lim_{k \rightarrow \infty} | (3k+2)(3k+3) | = \infty$$

$$R_1 = \lim_{k \rightarrow \infty} \left| \frac{a_{3k}}{a_{3(k+1)+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{2 \cdot 5 \cdots (3k-1)}{(3k)!} \cdot \frac{[3(k+1)+1]!}{2 \cdot 5 \cdots [3(k+1)-1]} \right|$$

$$= \lim_{k \rightarrow \infty} | (3k+3)(3k+4) | = \infty$$

该级数解收敛半径为 ∞ .

3. 在 $x_0 = 0$ 的邻域上求解埃尔米特方程

$$y'' - 2xy' + (\lambda - 1)y = 0.$$

λ 取什么数值可使级数解退化为多项式? 这些多项式乘以适当常数使最高幂项成为 $(2x)^n$ 形式, 就叫作埃尔米特多项式, 记作 $H_n(x)$. 写出前几个 $H_n(x)$.

解: 设方程的级数解 $y(x) = \sum_{k=0}^{\infty} a_k x^k$,

$$\text{则 } y'(x) = \sum_{k=1}^{\infty} a_k k x^{k-1}, y''(x) = \sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}$$

将 $y(x)$ 、 $y'(x)$ 、 $y''(x)$ 代回原方程得

$$\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2} - \sum_{k=1}^{\infty} 2a_k \cdot kx^k + \sum_{k=0}^{\infty} (\lambda-1)a_k x^k = 0$$

即
$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k - \sum_{k=1}^{\infty} 2ka_k x^k + \sum_{k=0}^{\infty} (\lambda-1)a_k x^k = 0$$

令各次幂的系数为 0, 得到系数递推公式

$$(k+2)(k+1)a_{k+2} - 2ka_k + (\lambda-1)a_k = 0$$

即

$$a_{k+2} = \frac{2k+1-\lambda}{(k+2)(k+1)} a_k$$

则对偶次项

$$a_2 = \frac{1-\lambda}{1 \cdot 2} a_0$$

$$a_4 = \frac{5-\lambda}{3 \cdot 4} a_2 = \frac{(1-\lambda)(5-\lambda)}{4!} a_0$$

$$a_6 = \frac{9-\lambda}{5 \cdot 6} a_4 = \frac{(9-\lambda)(5-\lambda)}{6 \cdot 5 \cdot 4 \cdot 3} a_2 = \frac{(9-\lambda)(5-\lambda)(1-\lambda)}{6!} a_0$$

$\vdots$

$\vdots$

对奇次项

$$a_3 = \frac{3-\lambda}{2 \cdot 3} a_1$$

$$a_5 = \frac{7-\lambda}{3 \cdot 4} a_3 = \frac{(7-\lambda)(3-\lambda)}{4!} a_1$$

$$a_7 = \frac{11-\lambda}{6 \cdot 7} a_5 = \frac{(11-\lambda)(7-\lambda)}{7 \cdot 6 \cdot 5 \cdot 4} a_3 = \frac{(11-\lambda)(7-\lambda)(3-\lambda)}{7!} a_1$$

$\vdots$

$\vdots$

故

$$y(x) = a_0 y_0(x) + a_1 y_1(x)$$

其中 $y_0(x) = 1 + \frac{1-\lambda}{2!} x^2 + \frac{(1-\lambda)(5-\lambda)}{4!} x^4 + \dots + \frac{(1-\lambda)(5-\lambda)\dots(4k-3-\lambda)}{(2k)!} x^{2k} + \dots$

$$y_1(x) = x + \frac{3-\lambda}{3!} x^3 + \frac{(3-\lambda)(7-\lambda)}{5!} x^5 + \dots + \frac{(3-\lambda)(7-\lambda)\dots(4k-1-\lambda)}{(2k+1)!} x^{2k+1} + \dots$$

收敛半径为无穷大

当 $\lambda = 4k-3$ 时, $y_0(x)$ 退化为多项式

当 $\lambda = 4k-1$ 时, $y_1(x)$ 退化为多项式

其中 $k = 1, 2, 3, \dots$

埃米尔特多项式, 取 $k = 1$, 则

当 $\lambda = 4k-3 = 1$, $y_0(x) = 1$, 记为 $H_0(x) = 1$

当 $\lambda = 4k-1 = 3$, $y_1(x) = x$, 记为 $H_1(x) = 2y_1(x) = 2x$

取 $k = 2$, 则

当 $\lambda = 4k-3 = 5$, $y_0(x) = 1 - 2x^2$, 记为 $H_2(x) = -2y_0(x) = (2x^2)^2 - 2$

当 $\lambda = 4k-1 = 7$, $y_1(x) = x - \frac{2}{3}x^3$ 记为 $H_3(x) = -12y_1(x) = (2x)^3 - 6 \cdot (2x)$

4. 在 $x_0 = 0$ 的邻域上求解 $(1-x^2)y'' - 6xy' + 6y = 0$ 即

$$(1-x^2)y'' - 2(2+1)xy' + [3(3+1) - 2(2+1)]y = 0.$$

在勒让德方程的级数解(9.2.7)和(9.2.8)之中以 $l=3$ 代入,并求它的二阶导数,然后跟本题的答案比较一下.

解:设方程的级数解 $y(x) = \sum_{k=0}^{\infty} a_k x^k$, 则 $y'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$, $y''(x) = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$

将 $y(x)$ 、 $y'(x)$ 、 $y''(x)$ 代回原方程得到

$$(1-x^2) \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} - 6 \sum_{k=1}^{\infty} k a_k x^{k-1} + 6 \sum_{k=0}^{\infty} a_k x^k = 0$$

令各次幂的系数为 0, 得到系数的递推公式

$$a_{k+2} = \frac{(k+6)(k-1)}{(k+1)(k+2)} a_k$$

则对偶次项

$$a_2 = \frac{6 \cdot (-1)}{1 \cdot 2} a_0$$

$$a_4 = \frac{8 \cdot 1}{3 \cdot 4} a_2 = \frac{-48}{4!} a_0$$

$$a_6 = \frac{10 \cdot 3}{5 \cdot 6} a_4 = \frac{10 \cdot 8 \cdot 3 \cdot 1}{5 \cdot 6 \cdot 3 \cdot 4} a_2 = \frac{(10 \cdot 8 \cdot 6) \cdot [3 \cdot 1 \cdot (-1)]}{6!} a_0$$

$\vdots$

$\vdots$

对奇次项

$$a_3 = \frac{(-5) \cdot 0}{2 \cdot 3} a_1 = 0$$

$$a_5 = a_7 = a_9 = \cdots = 0$$

故 $y(x) = a_0 y_0(x) + a_1 y_1(x)$

$$\text{其中, } y_0(x) = 1 + \frac{(-1) \cdot 6}{2!} x^2 + \frac{(-1)(+1) \cdot 6 \cdot 8}{4!} x^4 + \cdots +$$

$$\frac{(2k-3)(2k-5)\cdots(-1) \cdot 6 \cdot 8 \cdots (2k+4)}{(2k)!} x^{2k} + \cdots$$

$$y_1(x) = x.$$

当 $l=3$, 代入勒让德方程的解, 得到

$$Y_0(x) = 1 + \frac{(-3) \cdot 4}{2!} x^2 + \cdots + \frac{(2n-5)(2n-7)\cdots(-3) \cdot 4 \cdot 6 \cdots (2n+2)}{(2n)!} x^{2n} + \cdots$$

$$Y_1(x) = x + \frac{(-2) \cdot 5}{3!} x^3$$

对 $Y_0(x)$ 、 $Y_1(x)$ 分别求二阶导, 得到

$$\begin{aligned} Y_0''(x) &= (-3) \cdot 4 + \frac{(-1)(-3) \cdot 4 \cdot 6 \cdot 3}{3!} x^2 + \cdots \\ &\quad + \frac{(2n-5)\cdots(-3) \cdot 4 \cdot 6 \cdots (2n+2)(2n-1)}{(2n-1)!} x^{2n-2} + \cdots \\ &= (-3) \cdot 4 \cdot y_0(x) \end{aligned}$$

$$Y''_1(x) = (-2) \cdot 5 \quad x = (-2) \cdot 5y_1(x)$$

故本题的解是三阶勒让德方程的解的二阶导数.

5. 在 $x_0 = 0$ 的邻域上求解雅可俾方程

$$(1-x^2)y'' + [\beta - \alpha - (\alpha + \beta + 2)x]y' + \lambda(\alpha + \beta + \lambda + 1)y = 0$$

解: 设方程的级数解为 $y(x) = \sum_{k=0}^{\infty} a_k x^k$

$$\text{则 } y'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}, y''(x) = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$$

将 $y(x)$ 、 $y'(x)$ 、 $y''(x)$ 代入原方程得到

$$(1-x)^2 \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} + [\beta - \alpha - (\alpha + \beta + 2)x] \sum_{k=1}^{\infty} k a_k x^{k-1} + \lambda(\alpha + \beta + \lambda + 1) \sum_{k=0}^{\infty} a_k x^k = 0$$

令各次幂的系数为 0, 得到系数的递推公式

$$a_{k+2} = \frac{(k-\lambda)(k+\alpha+\beta+\lambda+1)}{(k+2)(k+1)} a_k + \frac{\alpha-\beta}{k+2} a_{k+1}$$

可以写出前面几个系数, 但难以写出一般的系数公式.

§ 9.3 正则奇点邻域上的级数解法

1. 在 $x_0 = 0$ 的邻域上求解 $x^2 y'' + xy' - m^2 y = 0$ (m^2 是常数). [这个方程即(8.1.67).]

解: $x^2 y'' + xy' - m^2 y = 0$ (m^2 是常数)

$$y'' + \frac{1}{x} y' - \frac{m^2}{x^2} y = 0$$

$x_0 = 0$ 是 $P(x) = \frac{1}{x}$ 的一阶极点, $q(x) = \frac{m^2}{x^2}$ 的二阶极点, 所以是方程的正则奇点.

设方程的解为 $y = \sum_{k=s}^{\infty} a_k x^k$ 代入方程, 交各幂次及系数列于下表

	x^s	x^{s+1}	...	x^{s+k}	...
$x^2 y''$	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k)(s+k-1)a_{s+k}$	...
xy'	$s a_s$	$(s+1)a_{s+1}$	...	$(s+k)a_{s+k}$	...
$-m^2 y$	$-m^2 a_s$	$-m^2 a_{s+1}$	...	$-m^2 a_{s+k}$	...

令最低幂项系数为 0, 得判定方程

$$[s(s-1) + s - m^2] a_s = 0$$

由于 $a_s \neq 0$, 有 $s^2 = m^2$, 所以 $s_1 = m, s_2 = -m$.

令 x^{s+k} 项系数为 0, 得系数递推公式

$$[(s+k)(s+k-1) + (s+k) - m^2] a_{s+k} = 0$$

(1) 当 $s_1 = m$ 时

$$[(m+k)(m+k-1) + (m+k) - m^2] a_{s+k} = k(2m+k) a_{s+k} = 0$$

若 $k \neq 0, m$ 为常数, 则 $a_{s_1+k} = 0$

若 $k = 0$, 则 $a_{s_1} \neq 0$, 得一特解 $y_0(x) = Cx^m$ (C 为常数)

(2) 当 $s_2 = -m$ 时,

$$[(m+k)(-m+k-1) + (-m+k) - m^2]a_{s_1+k} = k(k-2m)a_{s_1+k} = 0$$

若 $k \neq 0, m$ 为常数, 则 $a_{s_1+k} = 0$

若 $k = 0$, 则 $a_{s_1} \neq 0$, 得另一特解 $y_1(x) = Dx^{-m}$ (D 为常数)

综上所述 $y(x) = Cx^m + D \frac{1}{x^m}$

2. 在 $x_0 = 0$ 的邻域上求解 $x^2 y'' + 2xy' - l(l+1)y = 0$. [这个方程即(9.1.2).]

解: $x^2 y'' + 2xy' - l(l+1)y = 0$

$$y'' + \frac{2}{x}y' - \frac{l(l+1)}{x^2}y = 0$$

$x = 0$ 是 $p(x) = \frac{2}{x}$ 的一阶极点, $q(x) = -\frac{l(l+1)}{x^2}$ 的二阶极点, 所以是方程的正则奇点.

设方程的解为 $y = \sum_k a_k x^k$ 代入方程, 将各幂次及系数列于下表

	x^s	x^{s+1}	...	x^{s+k}	...
$x^2 y''$	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k)(s+k-1)a_{s+k}$	...
$2xy'$	$2s a_s$	$2(s+1)a_{s+1}$	...	$2(s+k)a_{s+k}$	...
$-l(l+1)y$	$-l(l+1)a_s$	$-l(l+1)a_{s+1}$	...	$-l(l+1)a_{s+k}$	...

令最低次幂项系数为 0, 得判定方程

$$[s(s-1) + 2s - l(l+1)]a_s = 0$$

由于 $a_s \neq 0$, 有 $s^2 + s - l(l+1) = 0$, 所以 $s_1 = l, s_2 = -(l+1)$

令 x^{s+k} 项系数为 0, 得系数递推公式

$$[(s+k)(s+k-1) + 2(s+k) - l(l+1)]a_{s+k} = 0$$

(1) 当 $s_1 = l$ 时

$$[(l+k)(l+k-1) + 2(l+k) - l(l+1)]a_{s+k} = k(k+2l+1)a_{s+k} = 0$$

若 $k \neq 0, l$ 为常数, 则 $a_{s_1+k} = 0$

若 $k = 0$, 则 $a_{s_1} \neq 0, \therefore$ 得一特解 $y_0(x) = a_0 x^l$

(2) 当 $s_2 = -(l+1)$ 时

$$[(-1+k)(-l-1+k-1) + 2(-l-1+k) - l(l+1)]a_{s_2+k} = k_1 k - 2(-1)a_{s_2+k} = 0$$

若 $k \neq 0, l$ 为常数, 则 $a_{s_2+k} = 0$

若 $k = 0$, 则 $a_{s_2} \neq 0, \therefore$ 得另一特解 $y_1(x) = a_1 x^{-(l+1)}$

综上所述 $y(x) = a_0 x^l + a_1 \frac{1}{x^{l+1}}$

3. 在 $x_0 = 0$ 的邻域上求拉盖尔方程 $xy'' + (1-x)y' + \lambda y = 0$ 的有限解. λ 取什么数值可使级数退化为多项式? 这些多项式乘以适当常数使最高幂项成为 $(-x)^n$ 形式就叫作拉盖尔多项式, 记

作 $L_n(x)$. 写出前几个 $L_n(x)$.

解: $xy'' + (1-x)y' + \lambda y = 0$

$$y'' + \frac{1-x}{x}y' + \frac{\lambda}{x}y = 0$$

由于 $x_0 = 0$ 是 $p(x) = \frac{1-x}{x}$ 和 $q(x) = \frac{\lambda}{x}$ 的一阶极点, 所以是方程的正则极点

设方程的解为 $y = \sum_{k=s}^{\infty} a_k x^k$ 代入方程, 将各幂次及系数列于下表

	x^{s-1}	x^s	...	x^{s+k}	...
xy''	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
y'	$s a_s$	$(s+1)a_{s+1}$	...	$(s+k+1)a_{s+k+1}$	...
$-xy'$	0	$-s a_s$	...	$-(s+k)a_{s+k}$	...
xy	0	λa_s	...	λa_{s+k}	...

令最低次幂项系数为 0, 得系数递推公式

$$[s(s-1) + s]a_s = 0$$

由 $a_s \neq 0$, 有 $s^2 = 0$, 所以 $s_1 = s_2 = 0$

令 x^{s+k} 项系数为 0, 得系数递推公式

$$[(s+k+1)(s+k) + (s+k+1)]a_{s+k+1} + [-(s+k) + \lambda]a_{s+k} = 0$$

当 $s_1 = s_2 = 0$ 时

$$[(k+1) \cdot k + (k+1)]a_{s+k+1} + [\lambda - k]a_{s+k} = 0$$

$$a_{k+1} = \frac{k-\lambda}{(k+1)}a_k$$

则由系数递推公式可得

$$a_0 = a_0; a_1 = -\lambda a_0; a_2 = \frac{1-\lambda}{2^2}a_1 = \frac{(-\lambda)(1-\lambda)}{2^2}a_0;$$

$$a_3 = \frac{2-\lambda}{3^2}a_2 = \frac{(-\lambda)(1-\lambda)(2-\lambda)}{3^2 \cdot 2^2}a_0$$

$$a_k = \frac{(-\lambda)(1-\lambda)\cdots(k-1-\lambda)}{(k!)^2}a_0$$

另一特解形式为 $\ln x$, 在 $x_0 = 0$ 处发散, 故舍去.

综上所述,

$$y(x) = a_0 \left[1 + \frac{-\lambda}{(1!)^2}x + \frac{(-\lambda)(1-\lambda)}{(2!)^2}x^2 + \cdots + \frac{(-\lambda)(1-\lambda)\cdots(k-1-\lambda)}{(k!)^2}x^k + \cdots \right]$$

当 λ 取整数时, $y(x)$ 退化为 λ 次多项式

为保证最高次项成为 $(-x)^n$ 形式, 则常数 a_0 可任取

当 $\lambda = 0$, $y_0(x) = 1$ 有 $L_0(x) = 1$

当 $\lambda = 1$, $y_1(x) = 1-x$ 有 $L_1(x) = 1-x$

当 $\lambda = 2$, $y_2(x) = 1-2x+\frac{1}{2}x^2$ 有 $L_2(x) = (-x)^2-4x+2$

当 $\lambda = 3, y_3(x) = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3$ 有 $L_3(x) = (-x)^3 + 9x^2 - 18x + 6$

当 $\lambda = 4, y_4(x) = 1 - 4x + 3x^2 - \frac{2}{3}x^3 + \frac{1}{24}x^4$ 有 $L_4(x) = (-x)^4 - 16x^3 + 72x^2 - 96x + 24$

4. 在 $x_0 = 0$ 的邻域上求 $y'' - 2\lambda y' + \left[\frac{2z}{x} - \frac{l(l+1)}{x^2}\right]y = 0$ 的有限解. λ 取什么数值可使级数退化为多项式?

解: $y'' - 2\lambda y' + \left[\frac{2z}{x} - \frac{l(l+1)}{x^2}\right]y = 0, x^2 y'' - 2\lambda x^2 y' + [2zx - l(l+1)]y = 0$

由 $x_0 = 0$ 是 $p(x) = -2\lambda$ 的常点, $q(x) = \frac{2z}{x} - \frac{l(l+1)}{x^2}$ 的二阶极点, 所以是方程的正则奇点.

设方程的解为 $y = \sum_{k=s}^{\infty} a_k x^k$ 代入方程, 将各幂次及系数列于下表

	x^s	x^{s+1}	...	x^{s+k}	...
$-2\lambda x^2 y$	0	$-2\lambda s a_s$	...	$-2\lambda(s+k-1)a_{s+k-1}$	...
$2zx y$	0	$2z a_s$	...	$2z a_{s+k-1}$	...
$-l(l+1)y$	$\lambda - l(l+1)a_s$	$-l(l+1)a_{s+1}$	...	$-l(l+1)a_{s+k}$	...

令最低次幂项系数为 0, 得判定方程

$$[s(s-1) - l(l+1)]a_s = 0$$

由 $a_s \neq 0$, 有 $s^2 - s - l(l+1) = 0$ 所以 $s_1 = l+1, s_2 = -l$.

令 x^{s+k} 项系数为 0, 得系数递推公式

$$[(s+k)(s+k-1) - l(l+1)]a_{s+k} + [-2\lambda(s+k-1) + 2z]a_{s+k-1} = 0$$

当 $s_1 = l+1$ 时

$$[(l+1+k)(l+1+k-1) - l(l+1)]a_{s_1+k} + [-2\lambda(l+1+k-1) + 2z]a_{s_1+k-1} = 0$$

$$a_{s_1+k} = \frac{2\lambda(l+k) - 2z}{k(k+2l+1)} a_{s_1+k-1}$$

则由系数递推公式可得

$$\begin{aligned} a_{l+1} &= a_0 \quad a_{l+2} = \frac{2[\lambda + (l+1) - z]}{1(2+2l)} a_{l+1} = \frac{2[\lambda(l+1) - z]}{1(2+2l)} a_0 \\ a_{l+3} &= \frac{2[\lambda(l+2) - z]}{2(3+2l)} a_{l+2} = \frac{2^2[\lambda(l+2) - z][\lambda(l+1) - z]}{2(3+2l)(2+2l)} a_0 \\ a_{l+k} &= \frac{2^k[\lambda(l+k) - z][\lambda(l+k-1) - z] \cdots [\lambda(l+1) - z]}{k!(k+2l+1)(k+2l) \cdots (2+2l)} a_0 \end{aligned}$$

$$\text{得一解 } y_{0(x)} = a_0 x^{l+1} \left\{ 1 + \frac{\lambda(l+1) - z}{2+2l} x + \cdots + \frac{[2^k(\lambda(l+k) - z) \cdots [\lambda(l+1) - z]]}{k!(k+2l+1) \cdots (2+2l)} x^k + \cdots \right\}$$

当 $s_2 = -l$ 时, 解中含 x^{-l} 负幂次项, 在 $x_0 = 0$ 处发散, 故舍去.

综上所述,

$$y(x) = a_0 x^{l+1} \left[1 + \frac{(l+1 - z/\lambda)}{1!(2l+2)} (2\lambda x) + \frac{(l+2 - z/\lambda)(1+2 - z/\lambda)}{2!(2l+2)(2l+3)} (2\lambda x)^2 + \cdots \right]$$

当 z/λ 为整数时, $y(x)$ 退化为 $(z/\lambda - l - 1)$ 次多项式

5. 在 $x_0 = 1$ 的邻域上, 求勒让德方程 $(1-x^2)y'' - 2xy' + l(l+1)y = 0$ 的有限解.

解: $(1-x^2)y'' - 2xy' + l(l+1)y = 0$ 由于 $x_0 = 1$ 处不方便计算, 可先进行换元.

$$\text{令 } t = \frac{1}{2}(1-x), \text{ 则 } \begin{cases} \frac{\partial y}{\partial x} = \frac{\partial y}{\partial t} \frac{\partial t}{\partial x} = -\frac{1}{2} \frac{\partial y}{\partial t} \\ \frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x} \left(-\frac{1}{2} \frac{\partial y}{\partial t} \right) = -\frac{1}{2} \frac{\partial^2 y}{\partial t^2} \cdot \frac{\partial t}{\partial x} = \frac{1}{4} \frac{\partial^2 y}{\partial t^2} \end{cases}$$

代入原方程得到

$$(t-t^2)y'' + (1-2t)y' + l(l+1)y = 0$$

$$y'' + \frac{1-2t}{t(1-t)}y' + \frac{l(l+1)}{t(1-t)}y = 0$$

由 $t_0 = 0$ 是 $p(t) = \frac{1-2t}{t(1-t)}$ 和 $q(t) = \frac{l(l+1)}{t(1-t)}$ 的一阶极点, 所以 $t_0 = 0$ 是方程的正则奇点

设方程的解为 $y(t) = \sum_{k=s}^{\infty} a_{k+t}$ 代入方程, 将各幂次及系数列于下表

	t^{s-1}	t^s	...	t^{s+k}	
$-t^2 y''$	0	$-s(s-1)a_s$	...	$-(s+k)(s+k-1)a_{s+k}$	
ty''	$s(s-1)a_s$	$(s+1)a_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
y'	sa_s	$(s+1)a_{s+1}$	...	$(s+k+1)a_{s+k+1}$	...
$-2ty'$	0	$-2sa_s$	...	$-2(s+k)a_{s+k}$	...
$l(l+1)y$	0	$l(l+1)a_s$	...	$l(l+1)a_{s+k}$	...

令最低次幂系数为 0, 得到判定方程

$$[s(s-1) + s]a_s = 0$$

由 $a_s \neq 0$ 有 $s^2 - s + s = 0$ 所以 $s_1 = s_2 = 0$

令 t^{s+k} 项系数为 0, 得系数递推公式

$$[-(s+k)(s+k-1) - 2(s+k) + l(l+1)]a_{s+k} + [(s+k+1)(s+k) + (s+k+1)]a_{s+k+1} = 0$$

当 $s_1 = s_2 = 0$

$$[-k(k-1) - 2k + l(l+1)]a_{s+k} + [(k+1)k + (k+1)]a_{s+k+1} = 0$$

$$a_{k+1} = \frac{(k-l)(l+k+1)}{(k+1)^2} a_k$$

由系数递推公式可知

$$a_0 = a_0 \quad a_1 = \frac{-l(l+1)}{12} a_0 = (1-l) \frac{\Gamma(l+2)}{\Gamma(l)} \cdot \frac{1}{12} a_0$$

$$a_2 = \frac{(1-l)(l+2)}{2^2} a_1 = (1-l) \frac{(l-1)(l+2)}{2^2} \cdot (-1) \frac{\Gamma(l+2)}{\Gamma(l)} a_0 = (-1)^2 \frac{\Gamma(l+3)}{\Gamma(l-1)} \frac{a_0}{(2!)^2}$$

$$a_k = (-1)^k \frac{\Gamma(l+k+1)}{\Gamma(l-k+1)} \cdot \frac{1}{(k!)^2} a_0 \quad \therefore \text{得到一特解.}$$

另一特解形式为 $\ln t$, 在 $t_0 = 0$ 处发散, 故舍去.

综上所述,将 $t = \frac{1-x}{2}$ 代回,即得

$$y(x) = a_0 \left[1 - \frac{\Gamma(l+2)}{\Gamma(l)} \frac{1}{(1!)^2} \left(\frac{1-x}{2} \right) + \frac{\Gamma(l+3)}{\Gamma(l-1)} \frac{1}{(2!)^2} \left(\frac{1-x}{2} \right)^2 + \dots + (-1)^k \frac{\Gamma(l+k+1)}{\Gamma(l-k+1)} \frac{1}{(k!)^2} \left(\frac{1-x}{2} \right)^k \right]$$

6. 在 $x_0 = 0$ 的邻域上求解 $xy'' - xy' + y = 0$.

解: $xy'' - xy' + y = 0$

$$y'' - y' + \frac{1}{x}y = 0$$

由 $x_0 = 0$ 是 $p(x) = -1$ 的常点, $q(x) = \frac{1}{x}$ 的一阶极点, 所以是方程的正则奇点

设方程的解为 $\sum_{k=s}^{\infty} a_k x^k$, 代入方程, 将各幂次及系数列于下表

	x^{s-1}	x^s	...	x^{s+k}	...
xy''	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$-xy'$	0	$-s a_s$	...	$-(s+k)a_{s+k}$	...
y	0	a_s	...	a_{s+k}	...

令最低次幂系数为 0, 得判定方程

$$s(s-1)a_s = 0$$

由 $a_s \neq 0$ 有 $s^2 - s = 0$ 所以 $s_1 = 1, s_2 = 0$

令 x^{s+k} 项系数为 0, 得系数递推公式

$$(s+k+1)(s+k)a_{s+k+1} + [-(s+k)+1]a_{s+k} = 0$$

当 $s_1 = 1$

$$[(k+2)(k+1)]a_{s+k+1} + [-(1+k)+1]a_{s+k} = 0$$

$$a_{k+2} = \frac{ka_{k+1}}{(k+2)(k+1)}$$

若 $k = 0$, 则 $a_2 = 0 \cdot a_1 = 0$, 所以 $a_k = 0$ ($k \neq 1$)

$\therefore$ 得到一特解

$$y_1 = a_1 x$$

$\because s_1 - s_2 = 1$ 为整数, 对应 $s_2 = 0$ 的特解形式应为

$$y_0(x) = Ax \ln x + \sum_{k=0}^{\infty} b_k x^k.$$

则

$$\begin{cases} y_0'(x) = A + A \ln x + \sum_{k=1}^{\infty} k b_k x^{k-1} \\ y_0''(x) = \frac{A}{x} + \sum_{k=2}^{\infty} k(k-1) b_k x^{k-2} \end{cases}$$

代入原方程, 将各幂次及系数列于下表

	x^0	$x \ln x$	x	$\dots$	x^k	$\dots$
xy''_2	A	0	$2b^2$	$\dots$	$(k+1)kb_{k+1}$	$\dots$
$-xy'_2$	0	$-A$	$-A-b_1$	$\dots$	$-k b k$	$\dots$
y_2	b_0	A	b_1	$\dots$	b_k	$\dots$

令各幂次项系数为 0, 得

$$\begin{cases} A + b_0 = 0 \\ -a + A = 0 \\ 2b^2 - A - b_1 + b_1 = 0 \\ \vdots \\ (k+1)kb_{k+1} + (-k+1)b_k = 0 \end{cases}$$

A 为任意值; $b_0 = -A$; b_1 为任意值; $b_2 = \frac{1}{2}A$, $b_3 = \frac{2-1}{2 \cdot 3}b_2 = \frac{1}{2!} \cdot \frac{1}{3!}A$

$$b_4 = \frac{2}{3 \cdot 4}b_2 = \frac{2!}{3!4!}A$$

$$b_k = \frac{(k-2)!}{(k-1)!k!}A$$

$$y_0(x) = Ax \ln x + \left[-A + b_1 x + 1 \frac{0!}{1!2!}Ax^2 + \dots + \frac{(k-2)!}{(k-1)!k!}Ax^k + \dots \right]$$

不妨令 $b_1 = 0$

$$\text{则 } y(x) = y_0(x) + y_1(x) = a_1 x + A \left[x \ln x - 1 + \frac{0!}{1!2!}x^2 + \frac{1!}{2!3!}x^3 + \dots + \frac{(k-2)!}{(k-1)!k!}x^k + \dots \right]$$

改变常数项表示, 即是

$$y(x) = a_0 x + a_1 \left[x \ln x - 1 + \frac{0!}{1!2!}x^2 + \frac{1!}{2!3!}x^3 + \dots + \frac{(k-2)!}{(k-1)!k!}x^k + \dots \right]$$

7. 在 $x_0 = 0$ 的邻域上求解 $xy'' + y = 0$.

解: $xy'' + y = 0$.

$$y'' + \frac{1}{x}y = 0$$

$x_0 = 0$ 是 $p(x) = \frac{1}{x}$ 的一阶极点, $q(x) = 0$ 的常点, 所以是方程的正则奇点

设方程的解为 $y = \sum_{k=s}^{\infty} a_k x^k$, 代入方程, 将各幂次及系数列于下表

	x^{s-1}	x^s	$\dots$	x^{s+k}
xy''	$s(s-1)a_s$	$(s+1)s_{s+1}$	$\dots$	$(s+k+1)(s+k)a_{s+k+1}$
y	0	a_s	$\dots$	a_{s+k}

令最低次幂系数为 0, 得判定方程

$$s(s-1)a_s = 0$$

由 $a_s \neq 0$ 有 $s^2 - s = 0$ 所以 $s_1 = 1, s_2 = 0$.

令 x^{s+k} 项系数为 0, 得系数递推公式

$$(s+k+1)(s+k)a_{s+k+1} + a_{s+k} = 0$$

当 $s_1 = 1$ 时

$$(k+2)(k+1)a_{s+k+1} + a_{s+k} = 0$$

$$a_{k+2} = -\frac{1}{(k+2)(k+1)}a_{k+1}$$

由系数递推公式得

$$a_1 = a_1, \quad a_2 = \frac{(-1)}{1 \cdot 2}a_1, \quad a_3 = \frac{(-1)}{2 \cdot 3}a_2 = \frac{(-1)^2}{2!3!}a_1, \quad a_k = \frac{(-1)^{k-1}}{(k-1)!k!}a_1$$

$$\text{得一特解 } y_1(x) = a_1 \left[x + \frac{(-1)}{1 \cdot 2}x^2 + \frac{(-1)^2}{2!3!}x^3 + \cdots + \frac{(-1)^{k-1}}{(k-1)!k!}x^k + \cdots \right]$$

由 $s_1 - s_0 = 1$ 为整数, 对应 $s_2 = 0$ 的特解形式应为

$$y_0(x) = Ay'(x)\ln x + \sum_{k=0}^{\infty} b_k x^k$$

$$\text{则 } \begin{cases} y_2'(x) = \frac{A}{x}y_1(x) + Ay_1'(x)\ln x + \sum_{k=1}^{\infty} kb_k x^{k-1} \\ y_2''(x) = -\frac{A}{x^2}y_1(x) + \frac{2A}{x}y_1'(x) + Ay_1''(x)\ln x + \sum_{k=1}^{\infty} k(k-1)b_k x^{k-2} \end{cases}$$

代入方程得

$$-\frac{A}{x}y_1(x) + 2Ay_1'(x) + Axy_1''(x)\ln x + \sum_{k=2}^{\infty} k(k-1)b_k x^{k-1} + Ay_1(x)\ln x + \sum_{k=0}^{\infty} b_k x^k = 0.$$

$$Axy_1''(x)\ln x + Ay_1(x)\ln x = A\ln x[xy_1''(x) + y_1(x)] = A\ln x \cdot 0 = 0$$

方程化简为

$$2Ay_1'(x) - \frac{A}{x}y_1(x) + \sum_{k=2}^{\infty} [k(k-1)b_k + b_{k-1}]x^{k-1} + b_0 = 0$$

将 $y_1(x), y_1'(x)$ 具体形式代入, 将各幂次及系数列于下表

	x^0	x^1	$\cdots x^{k-1}$	$\cdots$	
$2Ay_1'$	$2Aa_1$	$-2Aa_1$	$\cdots$	$2Aa_1 \frac{k(-1)^{k-1}}{(k-1)!k!}$	$\cdots$
$-\frac{A}{x}y_1$	$-Aa_1$	$2Aa_1$	$\cdots$	$-Aa_1 \cdot \frac{(-1)^{k-1}}{(k-1)!k!}$	$\cdots$
b_0	b_0	0	$\cdots$	0	$\cdots$
$\sum_{k=2}^{\infty} [k(k-1)b_k + b_{k+1}]x^{k-1}$	0	$2b_2 - b_1$	$\cdots$	$k(k-1)b_k + b_{k-1}$	$\cdots$

令各幂次项系数为 0, 得

$$\begin{cases} 2Aa_1 - Aa_1 + b_0 = 0 \\ -2Aa_1 + 2Aa_1 + 2b_2 - b_1 = 0 \\ \vdots \\ (2Aa_1 k - Aa_1) \frac{(-1)^{k-1}}{(k-1)!k!} + k(k-1)b_k + b_{k-1} = 0 \end{cases}$$

解得

$$b_0 = -Aa_1; b_1 \text{ 为任意值}; b_k = \left[\frac{Aa_1(2k-1)(-1)^k}{(k-1)!k!} - b_{k-1} \right] \frac{1}{k(k-1)}$$

不妨设 $b_1 = 0$

$$\begin{aligned} \text{则 } b_2 &= \frac{3}{1!2!}, Aa_1 \cdot \frac{1}{2} \quad b_3 = \left(\frac{-5}{2!3!} Aa_1 - b_2 \right) \cdot \frac{1}{3 \cdot 2} = \left(\frac{-5A!}{2!3!} - \frac{3Aa_{11}}{1!2!} \cdot \frac{1}{2} \right) \cdot \frac{1}{2 \cdot 3} \\ &= \frac{-Aa_1}{2!3!} x \left(\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} \right) \end{aligned}$$

$$\text{所以 } b_k = \frac{(-1)^k Aa_1}{(k-1)!k!} \left[\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots + \frac{(2k-1)}{(k-1)k} \right]$$

得另一特解

$$y_0(x) = Aa_1 y_1(x) \ln x + Aa_1 \left\{ -1 + \frac{1}{1!2!} \left(\frac{3}{1 \cdot 2} \right) x^2 + \cdots + \frac{(-1)^k}{(k-1)!k!} \left[\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots + \frac{2k-1}{(k-1)k} \right] x^k + \cdots \right\}$$

综上所述,改变常数即特解的表示符号,即得

$$\begin{cases} y_0(x) = a_0 \left[x - \frac{1}{1!2!} x^2 + \frac{1}{2!3!} x^3 + \cdots + \frac{(-1)^{k-1}}{(k-1)!k!} x^k + \cdots \right] \\ y_1(x) = a_1 y_0(x) \ln x + a_1 \left\{ -1 + \frac{1}{1!2!} \left(\frac{3}{1 \cdot 2} \right) x^2 + \cdots + \frac{(-1)^k}{(k-1)!k!} \left[\frac{3}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \cdots + \frac{2k-1}{(k-1)k} \right] x^k + \cdots \right\} \\ y_{(x)} = y_0(x) + y_1(x) \end{cases}$$

8. 在 $x_0 = 0$ 的邻域上求解高斯方程(超几何级数微分方程)

$$x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' + \alpha\beta y = 0. \alpha, \beta, \gamma \text{ 为常数}$$

$$\text{解: } x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' + \alpha\beta y = 0$$

$$y'' + \frac{(1+\alpha+\beta)x - \gamma}{x(x-1)} y' + \frac{\alpha\beta}{x(x-1)} y = 0$$

$$x_0 = 0 \text{ 是 } p(x) = \frac{(1+\alpha+\beta)x - \gamma}{x(x-1)} \text{ 和 } q(x) = \frac{\alpha\beta}{x(x-1)} \text{ 的一阶极点}$$

所以是方程的正则奇点.

设方程的解为 $y_{(x)} = \sum_{k=s}^{\infty} a_k x^k$ 代入方程,将各次幂及系数列于下表

	x^{s-1}	x^s	$\cdots x^{s+k}$	$\cdots$	
$x^2 y''$	0	$s(s-1)a_s$	$\cdots$	$(s+k)(s+k-1)a_{s+k}$	$\cdots$
$-xy''$	$-s(s-1)a_s$	$-(s+1)s a_{s+1}$	$\cdots$	$-(s+k+1)(s+k)a_{s+k+1}$	$\cdots$
$(1+\alpha+\beta)xy'$	0	$(1+\alpha+\beta)s a_s$	$\cdots$	$(1+\alpha+\beta)(s+k)a_{s+k}$	$\cdots$
$-\gamma y'$	$-\gamma s a_s$	$-\gamma(s+1)a_{s+1}$	$\cdots$	$-\gamma(s+k+1)a_{s+k+1}$	$\cdots$
$\alpha\beta y$	0	$\alpha\beta a_s$	$\cdots$	$\alpha\beta a_{s+k}$	$\cdots$

令最低次幂系数为0,得判定方程

$$[-s(s-1)-\gamma s]a_s=0$$

由 $a_s \neq 0$ 得 $-s^2+s-\gamma s=0$ 所以 $s_1=0; s_2=1-\gamma$

令 x^{s+k} 项系数为0,得系数递推公式

$$[(s+k)(s+k-1)+(1+\alpha+\beta)(s+k)+\alpha\beta]a_{s+k}+[-(s+k+1)(s+k)-\gamma(s+k+1)]a_{s+k+1}=0$$

当 $s_1=0$

$$[k(k-1)+(1+\alpha+\beta)k+\alpha\beta]a_{s_1+k}+[-(k+1)k-\gamma(k+1)]a_{s_1+k+1}=0$$

$$a_{k+1}=\frac{(k+\alpha)(k+\beta)}{(k+\gamma)(k+1)}a_k \quad (1)$$

由系数递推公式可得

$$a_0=a_0, a_1=\frac{\alpha\beta}{\gamma}a_0, a_2=\frac{(\alpha+1)(\beta+1)}{(\gamma+1)2}a_1=\frac{(\alpha+1)\alpha\beta(\beta+1)}{(\gamma+1)\gamma\cdot 2!}a_0$$

$$a_k=\frac{(\alpha+k-1)(\alpha+k-2)\cdots\alpha\beta(\beta+1)\cdots(\beta+k-1)}{(\gamma+k-1)(\gamma+k-2)\cdots\gamma\cdot k!}a_0$$

得一特解

$$y_0(x)=a_0\left[1+\frac{\alpha\beta}{1!\gamma}x+\frac{(\alpha+1)\alpha\beta(\beta+1)}{(\gamma+1)\gamma\cdot 2!}x^2+\cdots+\frac{(\alpha+k-1)\cdots\alpha\beta\cdots(\beta+k-1)}{(\gamma+k-1)\cdots\gamma\cdot k!}x^k+\cdots\right]$$

$$\equiv a_0F(\alpha,\beta,\gamma,x)$$

当 $s_2=1-\gamma$ (γ 不为整数)

$$[(1-\gamma+k)(k-\gamma)+(1+\alpha+\beta)(1-\gamma+k)+\alpha\beta]a_{s_2+k}+[-(2-\gamma+k)(1-\gamma+k)-\gamma(2-\gamma+k)]a_{s_2+k+1}=0$$

$$a_{k+1}=\frac{(k+\alpha+1-\gamma)(k+\beta+1-\gamma)}{(k+1)(k+2-\gamma)}a_k \quad (2)$$

由递推公式对比可知(2)式是(1)式中 α, β, γ 被 $(\alpha+1-\gamma)(\beta+1-\gamma)(2-\gamma)$ 所替换,所以另一特解应为

$$y_1(x)=a_1x^{1-\gamma}F(\alpha+1-\gamma,\beta+1-\gamma,2-\gamma,x)$$

综上所述, $y(x)=y_0(x)+y_1(x)=a_0F(\alpha,\beta,\gamma,x)+a_1x^{1-\gamma}F(\alpha+1-\gamma,\beta+1-\gamma,2-\gamma,x)$

9. 在 $x_0=0$ 的邻域上求解汇合超几何级数微分方程

$$xy''+(\gamma-x)y'-\alpha y=0, \alpha, \gamma \text{ 为常数}$$

解: $xy''+(\gamma-x)y'-\alpha y=0$

$$y''-\frac{\gamma-x}{x}y'-\frac{\alpha}{x}y=0$$

$x_0=0$ 是 $p(x)=\frac{\gamma-x}{x}$ 和 $q(x)=-\frac{\alpha}{x}$ 的一阶极点

所以是方程的正则极点

设方程的解为 $y(x)=\sum_{k=s}^{\infty}a_kx^k$ 代入方程,将各次幂及系数列于下表

	x^{s-1}	x^s	...	x^{s+k}	...
xy''	$s(s-1)a_s$	$(s+1)s a_{s+1}$	...	$(s+k+1)(s+k)a_{s+k+1}$	...
$-xy'$	0	$-s a_s$	...	$-(s+k)a_{s+k}$	...
$-xy$	0	$-\alpha a_s$	...	$-\alpha a_{s+k}$	...

令最低幂次系数为0,得判定方程

$$[s(s-1) + \gamma s]a_s = 0$$

由 $a_s \neq 0$ 有 $s^2 - s + \gamma s = 0$ 所以 $s_1 = 0, s_2 = 1 - \gamma$

令 x^{s+k} 项系数为0,得系数递推公式

$$[(s+k+1)(s+k) + \gamma(s+k+1)]a_{s+k+1} + [-(s+k) - \alpha]a_{s+k} = 0$$

当 $s_1 = 0$,

$$[(k+1)k + \gamma(k+1)]a_{s+k+1} + [-k - \alpha]a_{s+k} = 0$$

$$a_{k+1} = \frac{k + \alpha}{(k+1)(k + \gamma)} a_k$$

与上题(1)式比较,可知上式不含 $(k + \beta)$ 项,故解为

$$y_0(x) = a_0 \left[1 + \frac{\alpha}{1!\gamma} + \frac{\alpha(\alpha+1)}{2!\gamma(\gamma+1)} + \dots + \frac{\alpha(\alpha+1)\dots(\alpha+k-1)}{2!\gamma(\gamma+1)\dots(\gamma+k-1)} + \dots \right] \equiv a_0 F(\alpha, \gamma, x)$$

当 $s_1 = 1 - \gamma$

$$[(2+k-\gamma)(1+k-\gamma) + \gamma(2+k-\gamma)]a_{s+k+1} + [-(1+k-\gamma) - \alpha]a_{s+k} = 0$$

$$a_{k+1} = \frac{(k + \alpha + 1 - \gamma)}{(k+1)(k + 2 - \gamma)} a_k$$

与上题(1)式比较,可知上式不含 β 项及 $(\alpha + 1 - \gamma)$ 替代 $\alpha, 2 - \gamma$ 替代 γ 故解为

$$y_1(x) = a_1 F(\alpha + 1 - \gamma, 2 - \gamma, x)$$

综上所述, $y(x) = y_0(x) + y_1(x) = a_0 F(\alpha, \gamma, x) + a_1 F(\alpha + 1 - \gamma, 2 - \gamma, x)$

§ 9.4 施图姆 - 刘维尔本征值问题

1. 把下列二阶线性常微分方程化成施图姆 - 刘维尔型方程的形式

(1) 高斯方程(超几何级数微方程) $x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' + \alpha\beta y = 0$; (2) 汇合超几何级数微分方程 $xy + (\gamma - x)y' - \alpha y = 0$;

解:(1) $x(x-1)y'' + [(1+\alpha+\beta)x - \gamma]y' + \alpha\beta y = 0$

首先将方程化为一般形式

$$y'' + \frac{[(1+\alpha+\beta)x - \gamma]}{x(x-1)}y' + \alpha\beta y = 0$$

则 $a(x) = \frac{[(1+\alpha+\beta)x - \gamma]}{x(x-1)}, b(x) = \alpha\beta.$

$\gamma(x) = 0$

$$\int a(x)dx = \int \left[\frac{1+\alpha+\beta}{x-1} - \frac{\gamma}{x(x-1)} \right] dx = (1+\alpha+\beta)\ln(x-1) + \gamma \left(\frac{1}{x} - \frac{1}{x-1} \right) dx$$

$$= (1 + \alpha + \beta - \gamma) \ln(x-1) + \gamma \ln x + C$$

$$e^{\int a(x) dx} = (x-1)^{(1+\alpha+\beta-\gamma)} x^\gamma e^c$$

为简便不妨取 $c = 0$, 则将 $e^{\int a(x) dx}$ 乘原方程即将其转化为施图姆-刘维尔型方程. 即

$$\frac{d}{dx} \left[x^\gamma (x-1)^{1+\alpha+\beta-\gamma} \frac{dy}{dx} + \alpha \beta x^{\gamma-1} (x-1)^{\alpha+\beta-\gamma} y \right] = 0$$

$$(2) xy'' + (\gamma - x)y' - \delta y = 0$$

首先将方程化为一般形式

$$y'' + \frac{\gamma - x}{x} y' - \frac{\delta}{x} y = 0$$

则

$$a(x) = \frac{\gamma - x}{x}, b(x) = -\frac{\delta}{x}, \lambda c(x) = 0$$

$$\int a(x) dx = \int \frac{\gamma - x}{x} dx = \gamma \ln x - x + C$$

取 $C = 0$, 则

$$e^{\int a(x) dx} = x^\gamma e^{-x}$$

将 $e^{\int a(x) dx}$ 乘原方程即将其转化为施图姆-刘维尔型方程, 即

$$\frac{d}{dx} \left[x^\gamma e^{-x} \frac{dy}{dx} \right] - \alpha x^{\gamma-1} e^{-x} y = 0$$

2. 长为 l 的柔软均质重绳, 上端固定在以匀速 ω 转动的竖直轴上, 由于重力作用, 绳的平衡位置应是竖直线. 此绳相对于竖直线的横振动方程为 $u_{xx} - g \frac{\partial}{\partial x} \left[(l-x) \frac{\partial u}{\partial x} \right] - \omega^2 u = 0$ (取 x 轴向下, 原点在固定端). 试写出该绳的边界条件.

解: 由题意得绳的示意图

[解法一] 由绳的顶端固定, 可知该点不会有振动产生位移. $\therefore u|_{x=0} = 0$

在绳的末端. 由物理规律可知末端振动的位移不可能无限.

$\therefore u|_{x=l} = \text{有限值}$.

即该定解问题的边界条件为

$$\begin{cases} u|_{x=0} = 0 \\ u|_{x=l} = \text{有限值} \end{cases}$$

[解法二] 由该绳的振动方程

$$u_{xx} - g \frac{\partial}{\partial x} \left[(l-x) \frac{\partial u}{\partial x} \right] - \omega^2 u = 0.$$

先将其进行分离变数, 令 $u(x, t) = X(x)T(t)$.

则代入原方程得到

$$\frac{T''}{T} = \frac{g \frac{\partial}{\partial x} [(l-x)x']}{x} + \omega^2 = -\lambda^2. \quad (\lambda > 0)$$

对 x 变数, 化为一般形式的二阶常微分方程.

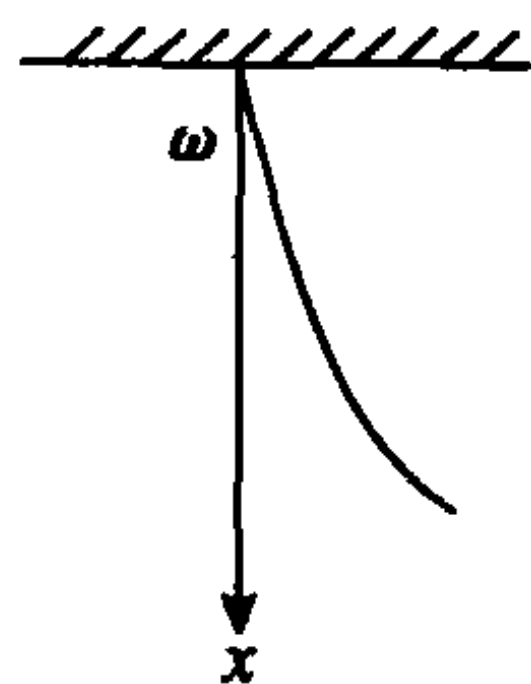


图 9-4-1

即

$$x'' + \frac{-1}{l-x}x' + \frac{\omega^2 + \lambda^2}{g(l-x)}x = 0.$$

$$\text{则 } p(x) = -\frac{1}{l-x}, q(x) = \frac{\omega^2 + \lambda^2}{g(l-x)}$$

$$p_{-1} = [p(x)(x-l)]|_{x=l} = 1$$

$$q_{-2} = [q(x)(x-l)^2]|_{x=l} = 0$$

代入判定方程 $s(s-1) + sp_{-1} + q_{-2} = 0$.

解得 $s_1 = s_2 = 0$

可知 u 中有一个解 $|n(x-l)|$ 项在 $x=l$ 处发散. 由方程的自然边界条件, 该解应排除. 则 $X|_{x=l}$ 应为有限值. 即 $u|_{x=l}$ 为有限值. 再加上 $x=0$ 端固定.

即得边界条件 $\begin{cases} u|_{x=0} = 0 \\ u|_{x=l} = \text{有限值} \end{cases}$

3. 求解下列本征值问题, 证明各题中不同的本征函数互相正交, 并求出模的平方.

$$(1) \begin{cases} X'' + \lambda X = 0 \\ X(a) = 0, X(b) = 0; \end{cases} \quad (2) \begin{cases} X'' + \lambda X = 0, \\ X'(0) = 0, X(l) + hX'(l) = 0. \end{cases}$$

$$\text{证明: (1)} \begin{cases} X'' + \lambda X = 0 & (\lambda > 0) \\ X(a) = 0 & X(b) = 0 \end{cases}$$

由 $x'' + \lambda x = 0$ 可解得

$$X(x) = A\cos\sqrt{\lambda}x + B\sin\sqrt{\lambda}x$$

则代入边界条件

$$\begin{cases} X(a) = A\cos\sqrt{\lambda}a + B\sin\sqrt{\lambda}a = 0 \\ X(b) = A\cos\sqrt{\lambda}b + B\sin\sqrt{\lambda}b = 0 \end{cases}$$

解得

$$\begin{cases} A = -\sin\sqrt{\lambda}a \\ B = \cos\sqrt{\lambda}a \\ \sqrt{\lambda} = \frac{n\pi}{b-a} \quad (n = 1, 2, 3, \dots) \end{cases}$$

则本征函数

$$X_n = -\sin\left(\frac{n\pi}{b-a}a\right)\cos\left(\frac{n\pi}{b-a}x\right) + \cos\left(\frac{n\pi}{b-a}a\right)\sin\frac{n\pi}{b-a}x$$

即

$$x_n = \sin\left[\frac{n\pi}{b-a}(x-a)\right] \quad (n = 1, 2, 3, \dots)$$

则

$$\int_a^b X_m X_n dx = \int_a^b \sin\frac{m\pi}{b-a}(x-a) \cdot \sin\frac{n\pi}{b-a}(x-b) dx$$

由三角函数积化和差得

$$\sin\left[\frac{m\pi}{b-a}(x-a)\right]\sin\left[\frac{n\pi}{b-a}(x-b)\right] = \frac{1}{2} \left\{ \cos\left[\frac{(m-n)\pi}{b-a}(x-a)\right] - \cos\left[\frac{(m+n)\pi}{b-a}(x-a)\right] \right\}$$

则 $\int_a^b x_m x_n dx$ 结果为两正弦值之和.

即 $\sin\left[\frac{(n-m)\pi}{b-a}(x-a)\right]\Big|_a^b, \sin\left[\frac{(n+m)\pi}{b-a}(x-a)\right]\Big|_a^b,$

显然两项为 0, $\therefore \int_a^b X_m X_n dx = 0$ 即证明其正交性

$$\begin{aligned} \text{模 } N_n: \quad N_n^2 &= \int_a^b \left[\sin \frac{n\pi}{b-a}(x-a) \right]^2 dx \\ N_n^2 &= \frac{1}{2} \left[1 - \cos \frac{2n\pi}{b-a}(x-a) \right] dx. \end{aligned}$$

易知 $\int_a^b \cos \left[\frac{2n\pi}{b-a}(x-a) \right] dx = 0$. 则 $N_n^2 = \frac{1}{2} \int_a^b 1 dx = \frac{1}{2}(b-a)$

$$(2) \quad \begin{cases} X'' + \lambda X = 0 & (\lambda > 0) \\ X'(0) = 0, X(l) + hX'(l) = 0 \end{cases}$$

由 $X'' + \lambda X = 0$ 可解得

$$X(x) = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$$

则 $X'(x) = -A\sqrt{\lambda} \sin \sqrt{\lambda} x + B\sqrt{\lambda} \cos \sqrt{\lambda} x$, 代入边界条件 $x = 0$, 则

$$X'(0) = B\sqrt{\lambda} = 0. \quad \text{有 } B = 0. \text{ 有 } x(x) = A \cos \sqrt{\lambda} x.$$

代入边界条件 $x = l$, 则

$$A \cos \sqrt{\lambda} l - hA \sqrt{\lambda} \sin \sqrt{\lambda} l = 0$$

$$\text{即 } \cos \sqrt{\lambda} l - h \sqrt{\lambda} \sin \sqrt{\lambda} l = 0$$

令 x_n 为 $x - \frac{l}{h} \cot x = 0$ 的根, 则 $\sqrt{\lambda} l = x_n, \sqrt{\lambda} = \frac{x_n}{l} (n = 1, 2, 3 \dots)$

则本征函数为 $X_n = \cos \frac{x_n}{l} x \quad (n = 1, 2, 3 \dots)$

则 $\int_0^l X_m X_n dx = \int_0^l \cos \frac{x_m}{l} x \cos \frac{x_n}{l} x dx$ 由三角函数积化和差得

$$\cos \frac{x_m}{l} x \cos \frac{x_n}{l} x = \frac{1}{2} \left[\cos \frac{x_m + x_n}{l} x + \cos \frac{x_m - x_n}{l} x \right]$$

$$\begin{aligned} \int_0^l X_m X_n dx &= \frac{1}{2} \int_0^l \left(\cos \frac{x_m + x_n}{l} x + \cos \frac{x_m - x_n}{l} x \right) dx \\ &= \frac{1}{2} \left(\frac{l}{x_m + x_n} \sin \frac{x_m + x_n}{l} x + \frac{l}{x_m - x_n} \sin \frac{x_m - x_n}{l} x \right) \Big|_0^l \end{aligned}$$

则

$$\begin{aligned} &= l \frac{x_m \sin x_m \cos x_n - x_n \sin x_n \cos x_m}{x_m^2 - x_n^2} \\ &= l \sin x_m \sin x_n \frac{x_m \cot x_n - x_n \cot x_m}{x_m^2 - x_n^2} \end{aligned}$$

x_m, x_n 是方程 $x - \frac{l}{h} \cot x = 0$ 的根, $\cot x_m = \frac{h}{l} x_m, \cot x_n = \frac{h}{l} x_n$

$$x_m \cot x_n - x_n \cot x_m = \frac{h}{l} x_m x_n - \frac{h}{l} x_n x_m = 0$$

所以 $\int_0^l x_m x_n dx = 0$, 即证明其正交性.

$$\text{模 } N_n : N_n^2 = \int_0^l \left(\cos \frac{x_n}{l} x \right)^2 dx$$

$$N_n^2 = \frac{1}{2} \int_0^l \left(\cos \frac{2x_n}{l} x + 1 \right) dx = \frac{1}{2} \left(\frac{l}{2x_n} \sin \frac{2x_n}{l} x + x \right) \Big|_0^l = \frac{l}{2} \left[1 + \frac{1}{2x_n} \sin 2x_n \right]$$

4. 定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 & (a^2 = E/\rho), \\ u \Big|_{x=0} = 0, & (ESu_x + mu_t) \Big|_{x=l} = 0, \\ u \Big|_{t=0} = -\frac{mg}{ES} x, & u_t \Big|_{t=0} = 0. \end{cases}$$

用分离变数法求解过程中, 出现本征值问题

$$\begin{cases} X'' + \lambda X = 0, \\ X(0) = 0, X'(l) - \frac{\lambda m}{\rho s} X(l) = 0. \end{cases}$$

本征函数 $X(x) = C \sin \sqrt{\lambda} x$, 本征值 λ 是超越方程 $\sqrt{\lambda} \cos \sqrt{\lambda} l - \frac{\lambda m}{\rho s} \sin \sqrt{\lambda} l = 0$ 的根.

不难验证: 对于 $\lambda_m \neq \lambda_n$, $\int_0^l \sin \sqrt{\lambda} x, \sin \sqrt{\lambda} x dx \neq 0$.

为什么对应于不同本征值的本征函数不正交?

解: 尽管 $X(x)$ 的二阶常微分方程满足施图姆—刘维尔型方程的形式. 但是 $X(x)$ 满足的边界条件不属于第一类、第二类或第三类边界条件. 也不属于自然边界条件. 所以无法构成施图姆—刘维尔本征值问题, 所以其不同本征值的本征函数不正交.

第十章 球 函 数

本章知识要点

轴对称球函数

(一) 球函数

1. 球函数

球函数方程 $\frac{1}{\sin \theta} \frac{\partial Y}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} + l(l+1)Y = 0$ 的解 $Y(\theta, \varphi)$ 称为球函数. 经过分离变数可得到 $Y(\theta, \varphi)$ 形式为 $Y(\theta, \varphi) = (A \cos m\varphi + B \sin m\varphi) \Theta(\theta)$ 其中 $\Theta(\theta)$ 可以由连带勒让德方程 $(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta = 0$ 解出, 式中 $x = \cos \theta$

2. 轴对称球函数

当 $m=0$ 时, 满足周期条件的解是 $\Phi(\varphi) = \text{常数}$. 这样球函数便以球坐标极轴为对称轴. 同时 $\Theta(\theta)$ 遵从的连带勒让德方程简化为勒让德方程.

(二) 勒让德多项式

1. 勒让德多项式的表达式

$$P_l(x) = \sum_{k=0}^{[l/2]} (-1)^k \frac{(2l-2k)!}{2^l k! (l-k)! (l-2k)!} x^{l-2k} \quad (x = \cos \theta)$$

需记住前几项勒让德多项式

$$P_0(x) = 1 \quad P_1(x) = x = \cos \theta \quad P_2(x) = \frac{1}{2}(3x^2 - 1) = \frac{1}{2}(2\cos 2\theta + 1)$$

2. 勒让德多项式的微分表示

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^2} (x^2 - x^1)^l$$

这叫做罗德里格斯公式

3. 勒让德多项式的积分表示

$$P_l(x) = \frac{1}{2\pi!} \frac{1}{2^l} \oint_C \frac{(z^2 - 1)^l}{(z - x)^{l+1}} dz$$

其中 C 为 z 平面上围绕 $z = x$ 点的任一闭合回路, 这叫做施列夫利积分.

(三) 第二类勒让德函数

从勒让德方程第一个解 $P_l(x)$ 得出具有统一形式的第二个线性独立解称为第二类勒让德函数

$$Q_l(x) = P_l(x) \int \frac{e^{\int \frac{2x}{1-x^2} dx}}{[P_l(x)]^2} dx = P_l(x) \int \frac{1}{(1-x^2)[P_l(x)]^2} dx$$

(四) 勒让德多项式的正交关系

不同阶的勒让德多项式在区间 $(-1, 1)$ 上正交

$$\int_{-1}^{+1} P_k(x) P_l(x) dx = 0 \quad (k \neq l)$$

x 变回到原来的变数 θ 则为

$$\int_0^\pi P_k(\cos \theta) P_l(\cos \theta) \sin \theta d\theta = 0 \quad (k \neq l)$$

(五) 勒让德多项式的模

$$N_l = \sqrt{\int_{-1}^{+1} [P_l(x)]^2 dx} = \sqrt{\frac{2}{2l+1}} \quad (l = 0, 1, 2, \dots)$$

(六) 广义傅里叶级数

把定义在 x 区间 $[-1, 1]$ 上的函数 $f(x)$ 或定义在 θ 区间 $[0, \pi]$ 上的函数, $f(\theta)$ 展开为广义傅

里叶级数

$$\begin{cases} f(x) = \sum_{l=0}^{\infty} f_l P_l(x) \\ \text{系数 } f_l = \frac{2l+1}{2} \int_{-1}^{+1} f(x) P_l(x) dx \end{cases}$$

即

$$\begin{cases} f(\theta) = \sum_{l=0}^{\infty} f_l P_l(\cos \theta) \\ \text{系数 } f_l = \frac{2l+1}{2} \int_0^\pi f(\theta) P_l(\cos \theta) \sin \theta d\theta \end{cases}$$

(七) 拉普拉斯方程的轴对称定解问题

如果拉普拉斯方程的定解问题具有对称轴, 那么 $m = 0$. 问题与 φ 无关可直接将问题的解设

为 $u(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$. 再由边界条件确定未知系数.

(八) 母函数及递推公式

1. 函数 $\frac{1}{\sqrt{1-2r\cos\theta+r^2}}$ 叫做勒让德多项式的母函数(或生成函数)

2. 勒让德多项式的递推公式

勒让德多项式的递推公式有多种. 下面是几个常见递推公式

$$(k+1)P_{k+1}(x) - (2k+1)xP_k(x) + kP_{k-1}(x) = 0$$

$$P_k(x) = P'_{k+1}(x) - 2xP'_k(x) + P'_{k-1}(x) \quad (k \geq 1)$$

$$(2k+1)P_k(x) = P'_{k+1}(x) - P'_{k-1}(x)$$

连带勒让德函数

(一) 连带勒让德函数

1. 连带勒让德函数的表达式通常记作 $P_l^m(x)$.

$$P_l^m(x) = (1-x^2)^{\frac{m}{2}} P_l^{[m]}(x)$$

下面给出 $l=1, 2$ 的连带勒让德函数具体形式

$$P_1^1(x) = (1-x^2)^{1/2} = \sin \theta$$

$$P_2^1(x) = (1-x^2)^{1/2}(3-x) = \frac{3}{2} \sin 2\theta$$

$$P_2^2(x) = 3(1-x^2) = 3\sin^2 \theta = \frac{3}{2}(1-\cos 2\theta)$$

2. 连带勒让德函数的微分及积分表示

$$(1) P_l^m(x) = \frac{(1-x^2)^{\frac{m}{2}}}{2^l l!} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l \quad \text{罗德里格斯公式}$$

$$(2) P_l^m(x) = \frac{(1-x^2)^{\frac{m}{2}}}{2^l} \frac{1}{2\pi i} \frac{(l+m)!}{l!} \oint_C \frac{(z^2-1)^l}{(z-x)^{l+m+1}} dz. \quad \text{其中 } C \text{ 为 } z \text{ 平面上围绕 } z=x \text{ 的任意}$$

闭合回路, 也叫作施列夫利积分.

$$(3) P_l^m(x) = \frac{i^m}{2\pi} \frac{(l+m)!}{l!} \int_{-\pi}^{\pi} e^{-im\psi} [\cos \theta + i \sin \theta \cos \psi]^l d\psi \quad \text{拉普拉斯积分}$$

(二) 连带勒让德函数的正交关系

同一 m 不同阶 l 的连带勒让德函数在区间 $[-1, 1]$ 上正交

$$\int_{-1}^{+1} P_k^m(x) P_l^m(x) dx = 0 \quad (k \neq l)$$

如果从 x 回到原来的变数 θ , 则

$$\int_0^{\pi} P_k^m(\cos \theta) P_l^m(\cos \theta) \sin \theta d\theta = 0 \quad (k \neq l)$$

(三) 连带勒让德函数的模

$$N_l^m = \sqrt{\frac{(l+m)! 2}{(l-m)! (2l+1)}}$$

(四) 广义傅里叶级数

定义在 x 区间 $[-1, 1]$ 上的函数 $f(x)$, 或定义在 θ 区间 $[0, \pi]$ 上的函数 $f(\theta)$ 展开为广义傅里

叶级数

$$\begin{cases} f(x) = \sum_{l=0}^{\infty} f_l P_l^m(x) \\ \text{系数 } f_l = \frac{2l+1}{2} \frac{(l-m)!}{(l+m)!} \int_{-1}^{+1} f(x) P_l^m(x) dx \end{cases}$$

即

$$\begin{cases} f(\theta) = \sum_{l=0}^{\infty} f_l P_l^m(\cos \theta) \\ \text{系数 } f_l = \frac{2l+1}{2} \frac{(l-m)!}{(l+m)!} \int_0^\pi f(\theta) P_l^m(\cos \theta) \sin \theta d\theta \end{cases}$$

(五) 连带勒让德函数的递推公式

下面列出常见递推公式

$$\begin{aligned} (2k+1)xP_k^m(x) &= (k+m)P_{k-1}^m(x) + (k-m+1)P_{k+1}^m(x) \\ (2k+1)(1-x^2)^{1/2}P_k^m(x) &= P_{k+1}^{m+1}(x) - P_{k-1}^{m+1}(x) \\ (x^2-1)P_k'(x) &= kxP_k(x) - kP_{k-1}(x) \quad (k \geq 1) \end{aligned}$$

一般的球函数

(一) 球函数的表达式

球函数方程的解称为球函数, 一般情况下可表示为

$$Y_l^m(\theta, \varphi) = P_l^m(\cos \theta) \begin{cases} \sin m\varphi \\ \cos m\varphi \end{cases} \begin{pmatrix} m=0,1,2,\dots,l \\ l=0,1,2,\dots \end{pmatrix}$$

l 叫做球函数的阶

(二) 球函数的正交关系

球函数中任意两个在球面 S 上 ($0 \leq \theta \leq \pi, 0 \leq 2\pi$) 正交. 即

$$\iint_S Y_l^m(\theta, \varphi) Y_k^n(\theta, \varphi) \sin \theta d\theta d\varphi = 0 \quad (m \neq n, l \neq k)$$

(三) 球函数的模

$$N_l^m = \sqrt{\frac{2\pi\delta_m}{2l+1} \frac{(l+m)!}{(l-m)!}}$$

(四) 球面上的函数的广义傅里叶级数

定义在球面 S 上的函数 $f(\theta, \varphi)$ 可用球函数展开为二重广义傅里叶级数

$$\begin{aligned} f(\theta, \varphi) &= \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} [A_l^m \cos m\varphi + B_l^m \sin m\varphi] P_l^m(\cos \theta) \\ \text{其中} \begin{cases} A_l^m = \frac{2l+1}{2\pi\delta_m} \cdot \frac{(l-m)!}{(l+m)!} \int_0^\pi \int_0^{2\pi} f(\theta, \varphi) P_l^m(\cos \theta) \cos m\varphi \sin \theta d\theta d\varphi \\ B_l^m = \frac{2l+1}{2\pi} \frac{(l-m)!}{(l+m)!} \int_0^\pi \int_0^{2\pi} f(\theta, \varphi) P_l^m(\cos \theta) \sin m\varphi \sin \theta d\theta d\varphi \end{cases} \end{aligned}$$

(五) 拉普拉斯方程的非定解问题

如果拉普拉斯方程的定解问题不具有对称轴, 那么 $m \neq 0$ 需要用一般的球函数求解. 可将问题的解设为

$$u(r, \theta, \varphi) = \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} r^l [A_l^m \cos m\varphi + B_l^m \sin m\varphi] P_l^m(\cos \theta) + \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} \frac{1}{r^{l+1}} [C_l^m \cos m\varphi + D_l^m \sin m\varphi] P_l^m(\cos \theta)$$

$m\varphi]P_l^m(\cos\theta)$

(六) 正交归一化的球函数

定义: $Y_m(\theta, \varphi) = \frac{1}{(N_l^m)} Y_l^m(\theta, \varphi) = \sqrt{\frac{D(l+1)}{4\pi} \cdot \frac{(l-|m|)!}{(l+|m|)!}} P_l^{|m|}(\cos\theta) e^{im\varphi}$

该函数即正交归一化的球函数

(七) 加法公式

勒让德多项式用一般球函数展开可得

$$P_l(\cos\theta) = \sum_{m=0}^l \frac{2}{f_m} \frac{(l-m)!}{(l+m)!} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi) P_l^m(\cos\theta_0) P_l^m(\cos\theta)$$

该式称为球函数的加法公式

(八) 复数形式的球函数及其性质

1. 复数形式的球函数

根据亥拉公式将球函数方程的分离变数解重新组合为复数形式. 即

$$Y_l^m(\theta, \varphi) = P_l^{|m|}(\cos\theta) e^{im\varphi} \begin{cases} m = -l, -l+1, \dots, 0, 1, \dots, l \\ l = 0, 1, 2, 3, \dots \end{cases}$$

2. 正交关系 $\iint Y_l^m(\theta, \varphi) Y_k^n(\theta, \varphi) \sin\theta d\theta d\varphi = 0 \quad (m \neq n \text{ 或 } l \neq k)$

3. 模 $N_l^m = \sqrt{\frac{4\pi}{2l+1} \cdot \frac{(l+|m|)!}{(l-|m|)!}}$

4. 广义傅里叶级数

$$f(\theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l C_l^m P_l^{|m|}(\cos\theta) e^{im\varphi}$$

其中 $C_l^m = \frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!} \int_0^\pi \int_0^{2\pi} f(\theta, \varphi) P_l^{|m|}(\cos\theta) [e^{im\varphi}]^* \sin\theta d\theta d\varphi$

5. 正交归一化

$$Y_m = \sqrt{\frac{(2l+1)}{4\pi} \cdot \frac{(l-|m|)!}{(l+|m|)!}} P_l^{|m|}(\cos\theta) e^{im\varphi}$$

6. 加法公式

$$P_l(\cos\theta) = \sum_{m=-l}^l \frac{(l-m)!}{(l+m)!} P_l^m(\cos\theta_0) P_l^m(\cos\theta) e^{im(\varphi-\varphi_0)}$$

(九) 四极子、多极子

两个偶极子、偶极矩相等而方向相反, 两者相距很近, 构成四极子. 依此类推下去还有八极子、十六极子等所谓多极子, 一系列多极子可以代替区域 T 里的电荷对远场的作用.

习题全解

§ 10.1 轴对称球函数

1. 以勒让德多项式为基, 在区间 $[-1, 1]$ 上把下列函数展开为广义傅里叶级数.

(1) $f(x) = x^4 + 2x^3$

解法一: 比较系数法

由于 $f(x)$ 最高为四次方, 所以勒让德多项式展开级数只包含 $l = 0 \sim 4$ 的项且 $P_4(x)$ 系数为 $\frac{8}{35}$, $P_3(x)$ 系数为 $\frac{4}{5}$. 由于 x^2 项系数为 0, 故 $P_2(x)$ 系数

$$f_2 = - \left[\frac{8}{35} \left(-\frac{30}{8} \right) \right] \frac{2}{3} = \frac{4}{7}$$

同理, 由于 x 项和常数项系数都为 0. 可得

$$P_1 \text{ 系数 } f_1 = \frac{6}{5}, P_0 \text{ 系数 } f_0 = \frac{1}{5}$$

$$\text{所以 } f(x) = \frac{1}{5}P_0(x) + \frac{6}{5}P_1(x) + \frac{4}{7}P_2(x) + \frac{4}{5}P_3(x) + \frac{8}{35}P_4(x)$$

解法二: 公式法

$$f_0 = \frac{1}{2} \int_{-1}^1 (x^4 + 2x^3) P_0(x) dx = \frac{1}{2} \int_{-1}^1 (x^4 + 2x^3) dx = \frac{1}{5}$$

$$f_1 = \frac{3}{2} \int_{-1}^1 (x^4 + 2x^3) P_1(x) dx = \frac{3}{2} \int_{-1}^1 (x^5 + 2x^4) dx = \frac{6}{5}$$

$$f_2 = \frac{5}{2} \int_{-1}^1 (x^4 + 2x^3) P_2(x) dx = \frac{5}{2} \int_{-1}^1 \frac{1}{2} (3x^6 + 6x^5 - x^4 - 2x^3) dx = \frac{4}{7}$$

$$f_3 = \frac{7}{2} \int_{-1}^1 (x^4 + 2x^3) P_3(x) dx = \frac{7}{2} \int_{-1}^1 \frac{1}{2} (x^4 + 2x^3) (5x^3 - 3x) dx = \frac{4}{5}$$

$$f_4 = \frac{9}{2} \int_{-1}^1 (x^4 + 2x^3) P_4(x) dx = \frac{9}{2} \int_{-1}^1 (x^4 + 2x^3) \frac{1}{8} (35x^4 - 30x^2 + 3) dx = \frac{8}{35}$$

当 $l > 4$, 系数积分为 0

$$\text{所以 } f(x) = \frac{1}{5}P_0(x) + \frac{6}{5}P_1(x) + \frac{4}{7}P_2(x) + \frac{4}{5}P_3(x) + \frac{8}{35}P_4(x)$$

$$(2) f(x) = \begin{cases} x^2, & (0 \leq x \leq 1) \\ 0, & (-1 \leq x < 0) \end{cases}$$

直接运用公式法

$$\begin{aligned} f_l &= \frac{2l+1}{2} \int_{-1}^1 f(x) P_l(x) dx = \frac{2l+1}{2} \int_0^1 x^2 P_l(x) dx = \frac{2l+1}{2^{l+1} l!} \int_0^1 x^2 \frac{d^l}{dx^l} (x^2 - 1)^l dx \\ &= \frac{2l+1}{2^{l+1} l!} \left\{ \left[x^2 \frac{d^{l-1}}{dx^{l-1}} (x^2 - 1)^l \right] \Big|_0^1 - 2 \int_0^1 x \frac{d^{l-1}}{dx^{l-1}} (x^2 - 1)^l dx \right\} \end{aligned}$$

$$= -\frac{(2l+1)}{2^{l+1}l!} 2 \left\{ \left[x \frac{d^{l-2}}{dx^{l-2}} (x^2-1)^l \right] \Big|_0^1 - \int_0^1 \frac{d^{l-2}}{dx^{l-2}} (x^2-1)^l dx \right\} = \frac{(2l+1)}{2^l l!} \frac{d^{l-3}}{dx^{l-3}} (x^2-1)^l \Big|_0^1$$

$$= -\frac{(2l+1)}{2^l l!} \left[\frac{d^{l-3}}{dx^{l-3}} \sum_{k=0}^l C_l^k x^{2k} (x-1)^{l-k} \right] \Big|_{x=0}$$

当 $l > 3$ 且 l 为偶数, 则 $f_l = 0$. 当 $l > 3$ 且 l 为奇数, 则设 $l = 2n+1$.

$$f_{2n+1} = \frac{(-1)^{n+1} (4n+3)(2n-3)!!}{(2n+4)!!} (n=2,3,4,\dots)$$

对 $l=0,1,2,3$, 有

$$f_0 = \frac{1}{2} \int_{-1}^1 f(x) P_0(x) dx = \frac{1}{2} \int_0^1 x^2 dx = \frac{1}{6}$$

$$f_1 = \frac{3}{2} \int_{-1}^1 f(x) P_1(x) dx = \frac{3}{2} \int_0^1 x^3 dx = \frac{3}{8}$$

$$f_2 = \frac{5}{2} \int_{-1}^1 f(x) P_2(x) dx = \frac{5}{2} \int_0^1 \frac{1}{2} (3x^4 - x^2) dx = \frac{1}{3}$$

$$f_3 = \frac{7}{3} \int_{-1}^1 f(x) P_3(x) dx = \frac{7}{2} \int_0^1 \frac{1}{2} (5x^5 - 3x^3) dx = \frac{7}{48}$$

$$\text{所以 } f(x) = \frac{1}{6} P_0(x) + \frac{3}{8} P_1(x) + \frac{1}{3} P_2(x) + \frac{7}{48} P_3(x) + \sum_{n=2}^{\infty} \frac{(-1)^{n+1} (4n+4)(2n-3)!!}{(2n+4)!!} P_{2n+1}(x)$$

(3) $f(x) = x^n$. (n 为正整数)

由系数公式计算

$$f_l = \frac{2l+1}{2} \int_{-1}^1 x^n P_l(x) dx = \frac{2l+1}{2} \frac{1}{2^l l!} \int_{-1}^1 \frac{d^l (x^2-1)^l}{dx^l} x^n dx$$

$$= \frac{2l+1}{2} \frac{1}{2^l l!} \left\{ \left[x^n \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \right] \Big|_{-1}^{+1} - n \int_{-1}^1 x^{n-1} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx \right\}$$

$$\frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \Big|_{-1}^{+1} = 0$$

$$\text{所以 } f_l = -\frac{2l+1}{2} \frac{1}{2^l l!} n \int_{-1}^{+1} x^{n-1} \frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l dx$$

分部积分 n 次, 可得

$$f_l = (-1)^n \frac{2l+1}{2} \frac{n!}{2^l l!} \int_{-1}^1 1 \frac{d^{l-n}}{dx^{l-n}} (x^2-1)^l dx$$

$$\text{当 } l > n, \text{ 则 } f_l = (-1)^n \frac{2l+1}{2} \cdot \frac{n!}{2^l l!} \left[\frac{d^{l-n-1}}{dx^{l-n-1}} (x^2-1)^l \right] \Big|_{-1}^{+1} = 0$$

$$\text{当 } l = n, \text{ 则 } f_l = (-1)^n \frac{2l+1}{2} \frac{n!}{2^l l!} \int_{-1}^1 (x^2-1)^0 dx = (-1)^n (2l+1) \frac{n!}{2^l l!}$$

当 $l < n$, 则分部积分 l 次

$$f_l = (-1)^l \frac{(2l+1)}{2} \cdot \frac{n(n-1)\cdots(n-l+1)}{2^l l!} \int_{-1}^1 x^{n-l} \frac{d^{n-l}}{dx^{n-l}} (x^2-1)^l dx$$

2. 用一层不导电的物质把半径为 r_0 的导体球壳分隔为两个半球壳, 使半球各充电到电势为 v_1 和 v_2 . 试计算电场中的电势分布.

解:由题意得方程

(1) 对球壳内部 $r < r_0$ 有

$$\begin{cases} \Delta u_1 = 0 \\ u_1|_{r=0} = \text{有限值} \\ u_1|_{r=r_0} = \begin{cases} v_1 & (0 \leq \theta < \frac{\pi}{2}) \\ v_2 & (\frac{\pi}{2} < \theta \leq \pi) \end{cases} \end{cases}$$

设问题的解 $u_1 = \sum_{l=0}^{\infty} (A_l r^l + B_l \frac{1}{r^{l+1}} P_l(\cos \theta))$ 由 $r=0$ 处 u_1 有限知 $B_l = 0$, 则

$$u_1 = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$$

由 $r=r_0$ 处边界条件知 $\sum_{l=0}^{\infty} A_l r_0^l P_l(\cos \theta) = \begin{cases} v_1 & (0 \leq \theta < \frac{\pi}{2}) \\ v_2 & (\frac{\pi}{2} < \theta \leq \pi) \end{cases}$ $A_l r_0^l$ 即为等式右边勒让德多项式展开的系数, 则

$$\begin{aligned} A_l &= \frac{2l+1}{2r_0^l} \left[\int_{-1}^0 v_2 P_l(x) dx + \int_0^1 v_1 P_l(x) dx \right] \\ &= \frac{(2l+1)}{2r_0^l} v_2 \int_{-1}^0 P_l(x) dx + \frac{2(+1)v_1}{2r_0^l} \int_0^1 P_l(x) dx \end{aligned}$$

$$\int_{-1}^0 P_l(x) dx = \int_1^0 P_l(-x)(-dx) = \int_0^1 P_l(-1) dx = (-1)^l \int_0^1 P_l(x) dx$$

$$\text{又} \int_0^1 P_l(x) dx = \frac{1}{2^l l!} \int_0^1 \frac{d^l}{dx^l} (x^2 - 1)^l dx = \frac{1}{2^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} (x^2 - 1)^l \right] \Big|_0^1$$

$$= \frac{-1}{2^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} \sum_{k=0}^l C_l^k x^{2k} (-1)^{l-k} \right] \Big|_{x=0} = \begin{cases} 0 & (l \text{ 为偶数且 } l \neq 0) \\ \frac{(-1)^l (2l)!}{2^{2l+1} (l+1)! l!} & (l \text{ 为奇数}) \end{cases}$$

$$A_l = \frac{v_1 + (-1)^l v_2}{r_0^l} \frac{2l+1}{2} \int_0^1 P_l(x) dx$$

$$= \begin{cases} 0 & (l = 2k \text{ 且 } k \neq 0) \\ \frac{v_1 - v_2}{2} (-1)^k \frac{(4k+3)(2k+1)!}{(2k+2)!!(2k+1)!!} \frac{1}{r_0^{2k+1}} & (l = 2k+1) \end{cases}$$

$$A_0 = \frac{v_1 + v_2}{2} \int_0^1 P_0(x) dx = \frac{v_1 + v_2}{2}$$

$$\text{球壳内电势 } u_1 = \frac{v_1 + v_2}{2} + \frac{v_1 - v_2}{2} \sum_{k=0}^{\infty} (-1)^k \times \frac{(4k+3)(2k+1)!!}{(2k+1)(2k+2)!!} \left(\frac{r}{r_0}\right)^{2k+1} \cdot P_{2k+1}(\cos \theta)$$

(2) 对球壳外部 $r > r_0$, 有

$$\begin{cases} \Delta u_2 = 0 \\ u_2|_{r \rightarrow \infty} = 0 \\ u_2|_{r=r_0} = \begin{cases} v_1 & (0 \leq \theta < \frac{\pi}{2}) \\ v_2 & (\frac{\pi}{2} < \theta \leq \pi) \end{cases} \end{cases}$$

设问题的解 $u_2 = \sum_{l=0}^{\infty} (C_l r^l + D_l \frac{1}{r^{l+1}}) P_l \cos^2 \theta$

由 $r = \infty$ 处 u_2 为 0 知 $C_l = 0$, 则 $u_2 = \sum_{l=0}^{\infty} \frac{D_l}{r^{l+1}} P_l(\cos^2 \theta)$

由 $r = r_0$ 处边界条件可知, $\frac{D_l}{r_0^{l+1}}$ 即为其勒让德多项式的展开系数, 则类似于球内部的计算, 可得

$$D_0 = \frac{r_0}{2} (v_1 + v_2) \int_0^1 P_0(x) dx = \frac{r_0}{2} (v_1 + v_2)$$

$$D_{2k+1} = \frac{v_1 - v_2}{2} (-1)^k (4k+3) \frac{(2k)!}{(2k+2)!!(2k)!!}$$

$$D_{2k} = 0 \quad (k \neq 0)$$

$$\therefore \text{球壳外电势 } u_2 = \frac{v_1 + v_2}{2} \cdot \frac{r_0}{r} + \frac{v_1 - v_2}{2} \sum_{k=0}^{\infty} (-1)^k \times \frac{(4k+3)(2k)!}{(2k+2)!!(2k)!!} \left(\frac{r_0}{r}\right)^{2k+2} P_{2k+1}(\cos^2 \theta)$$

3. 一空心圆球区域, 内半径为 r_1 , 外半径为 r_2 , 内球面上有恒定电势 u_0 , 外球面上电势保持为 $u_1 \cos^2 \theta$, u_0, u_1 均为常数, 试求内外球面之间空心圆球区域中的电势分布.

解: 由题意得方程

$$\begin{cases} \Delta u = 0 & (r_1 \leq r \leq r_2) \\ u|_{r=r_1} = u_0 \\ u|_{r=r_2} = u_1 \cos^2 \theta \end{cases}$$

设问题的解 $u = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$

由边界条件可知 u 中含 $l=0, l=2$ 阶勒让德多项式, 则

$$\begin{aligned} u &= A_0 + \frac{B_0}{r} + \left(A_2 r^2 + \frac{B_2}{r^3} \right) P_2(\cos \theta) \\ &= A_0 + \frac{B_0}{r} + \left(A_2 r^2 + \frac{B_2}{r^3} \right) \cdot \frac{1}{2} (3 \cos^2 \theta - 1) \end{aligned}$$

当 $r = r_1$

$$u|_{r=r_1} = A_0 + \frac{B_0}{r_1} + \left(A_2 r_1^2 + \frac{B_2}{r_1^3} \right) \frac{1}{2} (3 \cos^2 \theta - 1) = u_0$$

当 $r = r_2$

$$u|_{r=r_2} = A_0 + \frac{B_0}{r_2} + \left(A_2 r_2^2 + \frac{B_2}{r_2^3} \right) \cdot \frac{1}{2} (3 \cos^2 \theta - 1) = u_1 \cos^2 \theta$$

比较系数可得

$$\begin{cases} A_0 + \frac{B_0}{r_1} - \frac{1}{2}A_2r_1^2 + \frac{B_2}{r_1^3} = u_0 \\ A_2r_1^2 + \frac{B_2}{r_1^3} = 0 \\ A_0 + \frac{B_0}{r_2} - \frac{1}{2}A_2r_2^2 + \frac{B_2}{r_2^3} = 0 \\ \frac{3}{2}\left(A_2r_2^2 + \frac{B_2}{r_2^3}\right) = u_1 \end{cases}$$

解得

$$\begin{cases} A_0 = \left(\frac{u_1}{3}r_2 - u_0r_1\right)\frac{1}{r_2 - r_1} \\ B_0 = \left(u_0 - \frac{u_1}{3}\right)\frac{r_1r_2}{r_2 - r_1} \\ A_2 = \frac{2}{3}u_1\frac{r_2^3}{r_2^5 - r_1^5} \\ B_2 = \frac{2}{3}u_1\frac{r_2^3r_1^5}{r_2^5 - r_1^5} \end{cases}$$

空心区域电势

$$u = \frac{\frac{1}{3}u_1r_2 - u_0r_1}{r_2 - r_1} + \frac{\left(u_0 - \frac{1}{3}u_1\right)r_1r_2}{r_2 - r_1} \frac{1}{r} + \left[\frac{\frac{2}{3}u_1r_2^3}{r_2^5 - r_1^5}r^2 - \frac{\frac{2}{3}u_1r_1^5r_2^3}{r_2^5 - r_1^5} \frac{1}{r^3} \right] P_2(\cos \theta)$$

4. 半径为 r_0 的半球的球面保持一定温度 u_0 , 半球底面(1) 保持 0°C , (2) 绝热, 试求这个半球里的稳定温度分布.

解: 由题意得方程

$$\begin{cases} \Delta u = 0, & (0 \leq \theta < \frac{\pi}{2}) \\ u|_{r=r_0} = u_0 \\ u|_{r=0} = \text{有限值} \\ u|_{\theta=\frac{\pi}{2}} = 0 \end{cases}$$

将 θ 延拓到 $(\frac{\pi}{2}, \pi]$ 上

设问题的解 $u = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$

由 $r=0$ 处为有限值可知应舍去 $\frac{1}{r^{l+1}}$ 项.

则 $u = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$

由 $r=r_0$ 处边界条件可知 $A_l r_0^l$ 即为 u_0 的勒让德展开式的系数. 又由于 θ 为 0 到 $\frac{\pi}{2}$

$$A_l = \frac{2l+1}{2r_0^l} \int_0^1 u_0 P_l(x) dx$$

$$\begin{aligned}
&= \frac{(2l+1)u_0}{2r_0^l} \int_0^1 P_l(x) dx = \frac{2(l+1)u_0}{2r_0^l} \cdot \frac{1}{2^l l!} \int_0^1 \frac{d^l}{dx^l} (x^2-1)^l dx \\
&= \frac{(2l+1)u_0}{2^{l+1} r_0^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} (x^2-1)^l \right] \Big|_0^1 = \frac{-(2l+1)u_0}{2^{l+1} r_0^l l!} \left[\frac{d^{l-1}}{dx^{l-1}} \sum_{k=0}^l C_l^k x^{2k} (-1)^{l-k} \right] \Big|_{x=0} \\
&= \begin{cases} 0 & (l \text{ 为偶数且 } l \neq 0) \\ l & \\ \frac{(-1)^l (2l)! (2l+1) u_0}{2^{2l+2} (l+1)! l! r_0^l} & (l \text{ 为奇数}) \end{cases}
\end{aligned}$$

当 $l=0$, 则边界条件 $u|_{\theta=\frac{\pi}{2}}=0$ 可知, u 中 A_0 应为 0.

$$\begin{cases} A_0 = 0 \\ A_l = \frac{(-1)^l (4l+3)(2l-1)!! u_0}{(2l+2)!! \cdot r_0^{2l+1}} \end{cases}$$

所以半球的电势为

$$\text{当 } 0 \leq \theta < \frac{\pi}{2}, u = u_0 \sum_{k=1}^{\infty} (-1)^k (4k+3) \frac{(2k-1)!!}{(2k+2)!!} \left(\frac{r}{r_0}\right)^{2k+1} P_{2k+1}(\cos \theta)$$

$$\text{当 } \theta = \frac{\pi}{2}, u = u_0$$

5. 在本来是匀强的静电场 E_0 中放置导体球, 球的半径为 r_0 . 试求解球外的静电场. [提示: 参考教材 §8.1 中例 4]

解: 令导体球电势为 0, 由题意得方程

$$\begin{cases} \Delta u = 0 & (\gamma > r_0) \\ u|_{\gamma=\infty} = -E_0 \gamma \cos \theta \\ u|_{\gamma=r_0} = 0 \end{cases}$$

设问题的解为

$$u = \sum_{l=0}^{\infty} \left(A_l \gamma^l + \frac{B_l}{\gamma^{l+1}} \right) P_l(\cos \theta)$$

当 $\gamma = \infty$ 时,

$$u|_{\gamma=\infty} = \sum_{l=0}^{\infty} A_l \gamma^l P_l(\cos \theta) = -E_0 \gamma \cos \theta$$

当 $\gamma = r_0$ 时

$$u|_{\gamma=r_0} = \sum_{l=0}^{\infty} \left(A_l r_0^l + \frac{B_l}{r_0^{l+1}} \right) P_l(\cos \theta) = 0$$

比较系数可得

$$\begin{cases} A_1 = -E_0 \\ A_l = 0 & (l \neq 1) \\ A_1 r_0 + \frac{B_1}{r_0^2} = 0 \\ B_l = 0 & (l \neq 1) \end{cases}$$

解得

$$\begin{cases} A_1 = -E_0 \\ A_l = 0 \quad (l \neq 1) \\ B_1 = E_0 \gamma_0^3 \\ B_l = 0 \quad (l \neq 1) \end{cases}$$

球外电势为

$$u = -E_0 \gamma \cos \theta + E_0 \frac{\gamma_0^3}{\gamma^2} \cos \theta$$

6. 均匀介质球, 半径为 r_0 , 介电常数为 ϵ . 把介质球放在点电荷 $4\pi\epsilon_0 q$ 的电场中, 球心跟点电荷相距 $d (d > r_0)$, 求解这个静电场中的电势.

解: 取球心为球坐标系的极点. 极轴通过点电荷. 则极轴是对称轴, 问题与 φ 无关.

由题意得到球内外电势 u_1, u_2 所满足的条件为

$$\begin{cases} \Delta u_1 = 0 \quad (r < r_0) \\ u_1|_{r=0} = \text{有限值} \end{cases} \quad \begin{cases} \Delta u_2 = q\delta(z-d) \\ u_2|_{r=\infty} = 0 \end{cases}$$

边界条件

$$\begin{cases} u_1|_{r=r_0} = u_2|_{r=r_0} \\ \epsilon_0 \frac{\partial u_1}{\partial r} \Big|_{r=r_0} = \epsilon_0 \frac{\partial u_2}{\partial r} \Big|_{r=r_0} \end{cases}$$

对球内电势 u_1 , 由于在 $r=0$ 处有限. 所以设 $u_1 = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$

对球外电势 u_2 , 考虑没有顺球时, 球外电势本来是 $\frac{q}{\sqrt{d^2 - 2dr \cos \theta + r^2}}$

即为 u_2 泛定方程的一个特解, 又由于 u_2 在无穷处为 0, 所以设

$$u_2 = \frac{q}{\sqrt{d^2 - 2dr \cos \theta + r^2}} + \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta)$$

由勒让德多项式的母函数可知, 在 $r=r_0$ 处

$$\frac{q}{\sqrt{d^2 - 2dr_0 \cos \theta + r_0^2}} = q \sum_{l=0}^{\infty} \frac{r_0^l}{d^{l+1}} P_l(\cos \theta)$$

由边界条件 $u_1|_{r=r_0} = u_2|_{r=r_0}$ 得

$$\epsilon \sum_{l=1}^{\infty} l A_l r_0^{l-1} P_l(\cos \theta) = \sum_{l=1}^{\infty} l q \frac{r_0^{l-1}}{d^{l+1}} P_l(\cos \theta) + \sum_{l=0}^{\infty} -(l+1) \frac{B_l}{r_0^{l+2}} P_l(\cos \theta), \text{ 比较系数可得}$$

$$\begin{cases} A_l r_0^l = q \frac{r_0^l}{d^{l+1}} + \frac{B_l}{r_0^{l+1}} \\ \epsilon l A_l r_0^{l-1} = l q \frac{r_0^{l-1}}{d^{l+1}} - (l+1) \frac{B_l}{r_0^{l+2}} \quad (l \neq 0) \\ B_0 = 0 \end{cases}$$

解得

$$\begin{cases} A_l = \frac{q(2l+1)}{[(\epsilon+1)l+1]d^{l+1}} \\ B_l = \frac{-q(\epsilon-1)lr_0^{2l+1}}{[(\epsilon+1)l+1]d^{l+1}} \end{cases}$$

综上所述

$$\text{球内电势} \quad u_1 = q \sum_{l=0}^{\infty} \frac{2l+1}{[(\epsilon+1)l+1]d^{l+1}} r^l P_l(\cos \theta)$$

$$\text{球外电势} \quad u_2 = \frac{q}{\sqrt{d^2 - 2rd(\cos \theta + r^2)}} - q(\epsilon-1) \sum_{l=0}^{\infty} \frac{lr_0^{2l+1}}{[(\epsilon+1)l+1]d^{l+1}} \frac{1}{r^{l+1}} P_l(\cos \theta)$$

7. 半径为 r_0 , 表面熏黑的均匀球, 在温度为 0°C 的空气中, 受着阳光的照射, 阳光的热流强度为 q_0 . 求解小球里的稳定温度分布.

解: 取太阳光直射方向为极轴, 则温度分布与 ϕ 无关. 仅有上半球受到照射. 由题意得方程及边界条件

$$\begin{cases} \Delta u = 0 & (r < r_0) \\ u|_{r=0} = \text{有限值} \\ \left(\frac{\partial u}{\partial r} + Hu\right)\Big|_{r=r_0} = \begin{cases} q_0 \cos \theta & (0 \leq \theta < \frac{\pi}{2}) \\ 0 & (\frac{\pi}{2} < \theta \leq \pi) \end{cases} \end{cases}$$

其中 $H = \frac{h}{k}$, h 为热交换系数, k 为热传导系数

由于 $r=0$ 处 u 为有限值, 所以设

$$u = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$$

$$\text{则 } H \sum_{l=0}^{\infty} A_l r_0^l P_l(r_0^l P_l(\cos \theta)) + \sum_{l=1}^{\infty} A_l l r^{l-1} P_l(\cos \theta) = \begin{cases} q_0 \cos \theta & (0 \leq \theta < \frac{\pi}{2}) \\ 0 & (\frac{\pi}{2} < \theta \leq \pi) \end{cases}$$

将等号右边用勒让德多项式展开令 $\cos \theta = x$

$$f_0 = \frac{1}{2} \int_0^1 q_0 x dx = \frac{1}{4} q_0$$

$$f_1 = \frac{1}{3} \int_0^1 q_0 x x dx = \frac{1}{9} q_0$$

$$\begin{aligned} f_l &= \frac{1}{2l+1} \int_0^1 q_0 x P_l(x) dx = \frac{q_0}{2l+1} \int_0^1 x \frac{d^l(x^2-1)^l}{dx^l} dx \\ &= \frac{q_0}{2l+1} \left\{ \left[x \frac{d^{l-1}(x^2-1)^l}{dx^{l-1}} \right] \Big|_0^1 - \int_0^1 \frac{d^{l-1}(x^2-1)^l}{dx^{l-1}} dx \right\} \\ &= \frac{q_0}{2l+1} \left[- \frac{d^{l-2}(x^2-1)^l}{dx^{l-2}} \right] \Big|_0^1 = \frac{-q_0}{2l+1} \left[\frac{d^{l-2}}{dx^{l-2}} \sum_{k=0}^l C_l^k x^{2k} (-1)^{l-k} \right] \Big|_{x=0} \\ &= \begin{cases} 0 & (l \text{ 为奇数}) \\ \frac{(-1)^{n+1} (4n+1) (2n-2)!}{2(2n-2)!! (2n+2)!!} & (l=2n \text{ 且 } n \neq 0) \end{cases} \end{aligned}$$

比较系数可得

$$A_0 = \frac{q_0}{4H}, A_1 = \frac{1}{Hr_0+1} \frac{q_0}{2}, A_{2n+1} = 0$$

$$A_{2n} = \frac{(-1)^{n+1} q_0}{Hr_0 - 2n} \frac{4n+1}{2} \frac{(2n-2)!}{(2n-2)!!(2k+2)!!} \frac{1}{r_0^{2n-1}}$$

综上所述,温度分布

$$u = \frac{q_0}{4H} + \frac{1}{Hr_0 + 1} \frac{q_0 r}{2} P_1(\cos \theta) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} q_0}{(Hr_0 + 2k)r_0^{2k-1}} \frac{(3k+1)(2k-2)!}{2(2k-2)!!(2k+2)!!} r^{2k} P_{2k}(\cos \theta)$$

8. 求解 § 7.1 习题 7 的轻绳的振动,初始位移 $\varphi(x)$, 初始速度 $\psi(x)$.

解:由题意得定解问题

$$\begin{cases} u_{xx} - a^2 \frac{\partial}{\partial x} \left[(l^2 - x^2) \frac{\partial u}{\partial x} \right] = 0 & (a^2 = \frac{\omega^2}{2}) \\ u|_{x=0} = 0 \\ u|_{x=l} = \text{有限值} \\ u|_{t=0} = \varphi(x) \\ ut|_{t=0} = \psi(x) \end{cases}$$

令 $u(x, t) = X(x)T(t)$ 代入泛定方程得到

$$\frac{(l^2 - x^2)x'' - 2xX'}{X} = \frac{T''}{a^2 T} = -k(k+1)$$

进一步得到

$$\begin{cases} (l^2 - x^2)X'' - 2xX' + k(k+1)X = 0, X(0) = 0. & (1) \\ T'' + a^2 k(k+1)T = 0. & (X(l) = \text{有限值}) \end{cases} \quad (2)$$

对空间变量 x , 令 $\xi = \frac{x}{l}$, 代入方程(1), 将其转化为 k 阶勒让德方程, 即

$$\begin{cases} (1 - \xi^2) \frac{d^2 X}{d\xi^2} - 2\xi \frac{dX}{d\xi} + k(k+1)X = 0 \\ X|_{\xi=0} = 0 \\ X|_{\xi=1} = \text{有限值} \end{cases}$$

解得 $X_k(\xi) = P_k(\xi) = P_k\left(\frac{x}{l}\right)$

由 $X|_{x=l} = \text{有限值}$, 可知 $k = 0, 1, 2, \dots$, 由 $X|_{x=0} = 0$, 可知 $P_k(0) = 0$

则 k 为奇数, 设为 $k = 2n+1 (n = 0, 1, 2, 3, \dots)$

方程(1) 解为 $P_{2n+1}\left(\frac{x}{l}\right)$, 对时间变量 t , 解方程(2) 得

$$T = A \cos[\sqrt{k(k+1)}at] + B \sin[\sqrt{k(k+1)}at]$$

即 $T_n = A_n \cos \sqrt{(2n+1)(2n+2)}at + B_n \sin \sqrt{(2n+1)(2n+2)}at$

综上所述, 振动 u 应为

$$u(x, t) = \sum_{n=0}^{\infty} [A_n \cos \sqrt{(2n+1)(2n+2)}at + B_n \sin \sqrt{(2n+1)(2n+2)}at] P_{2n+1}\left(\frac{x}{l}\right)$$

其中, 当 $t = 0$ 时,

$$u \Big|_{z=0} = \sum_{n=0}^{\infty} A_n P_{2n+1} \left(\frac{x}{l} \right) = \varphi(x)$$

$$u_z \Big|_{z=0} = \sum_{n=0}^{\infty} B_n \sqrt{(2n+1)(2n+2)} a P_{2n+1} \left(\frac{x}{l} \right) = \psi(x)$$

用勒让德多项式展开 $\varphi(x), \psi(x)$, 并比较系数, 即得 A_n, B_n 表达式

$$\begin{cases} A_{2n+1} = \frac{4n+3}{l} \int_0^l \varphi(x) P_{2n+1} \left(\frac{x}{l} \right) dx \\ B_{2n+1} = \frac{4n+3}{\sqrt{(2n+1)(2n+2)} a l} \int_0^l \psi(x) P_{2n+1} \left(\frac{x}{l} \right) dx \end{cases}$$

9. 细导线首尾相接而构成圆环, 环的半径为 r_0 , 环上带电 $4\pi\epsilon_0 q$ 单位. 求圆环周围电场中的静电势. [在初等电学课程中已经知道, 圆环轴上距环心 r 处的电势为 $q/\sqrt{r_0^2+r^2}$. 这可用来作为电势在 $\theta=0$ 和 $\theta=\pi$ 方向的值.]

解: 将垂直于环面且过环心的轴为极轴, 以环心为极点.

(1) 对环内, 有

$$\begin{cases} \Delta u_1 = 0 \\ u_1|_{r=0} = \text{有限值} \\ u_1|_{\theta=0} = u_1|_{\theta=\pi} = \frac{q}{\sqrt{r_0^2+r^2}} \end{cases}$$

由 $u_1|_{r=0}$ 为有限值, 所以设

$$u_1 = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$$

由 θ 方向边界条件可得

$$\sum_{l=0}^{\infty} A_l r^l P_l(+1) = \frac{q}{\sqrt{r_0^2+r^2}}$$

$$\text{即 } \sum_{l=0}^{\infty} A_l r^l = \frac{q}{r_0} \frac{1}{\sqrt{1+\left(\frac{r}{r_0}\right)^2}}$$

$$= \frac{q}{r_0} \left\{ 1 - \frac{1}{2} \left(\frac{r}{r_0} \right)^2 + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{r}{r_0} \right)^4 - \dots + (-1)^k \frac{(2k-1)!!}{(2k)!!} \left(\frac{r}{r_0} \right)^{2k} + \dots \right\}$$

$$A_0 = q/r_0, A_{2k+1} = 0$$

$$A_{2k} = (-1)^k \frac{(2k-1)!!}{(2k)!!} \frac{q}{r_0^{2k+1}}$$

所以

$$u_1 = \frac{q}{r_0} + \frac{q}{r_0} \sum_{k=1}^{\infty} (-1)^k \frac{(2k-1)!!}{(2k)!!} \left(\frac{r}{r_0} \right)^{2k} P_{2k}(\cos \theta)$$

(2) 对环外, 有

$$\begin{cases} \Delta u_2 = 0 \\ u_2|_{r=\infty} = \text{有限值} \\ u_2|_{\theta=0} = u_2|_{\theta=\pi} = \frac{q}{\sqrt{r_0^2+r^2}} \end{cases}$$

由 $u_2|_{r=\infty}$ 为有限值, 所以设

$$u_2 = \sum_{l=0}^{\infty} B_l \frac{1}{r^{l+1}} P_l(\cos \theta)$$

由 θ 方向边界条件可得

$$\begin{aligned} \sum_{l=0}^{\infty} B_l \frac{1}{r^{l+1}} P_l(\cos \theta) &= \frac{q}{\sqrt{r_0^2 + r^2}} = \frac{q}{r} \cdot \frac{1}{\sqrt{1 + \left(\frac{r_0}{r}\right)^2}} \\ &= \frac{q}{r} \left[1 - \frac{1}{2} \left(\frac{r_0}{r}\right)^2 + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{r_0}{r}\right)^4 + \cdots + (-1)^k \frac{(2k-1)!!}{(2k)!!!} \left(\frac{r_0}{r}\right)^{2k} + \cdots \right] \end{aligned}$$

$$B_0 = q, \quad B_{2k+1} = 0, \quad B_{2k} = (-1)^k \frac{(2k-1)!!}{(2k)!!!} q r_0^{2k}$$

所以
$$u_2 = \frac{q}{r} + \frac{q}{r_0} \sum_{k=1}^{\infty} (-1)^k \frac{(2k-1)!!}{(2k)!!!} \left(\frac{r_0}{r}\right)^{2k+1} P_{2k}(\cos \theta)$$

10. 证明递推公式(10.1.67)、(10.1.68)、(10.1.69).

解: 由母函数

$$\frac{1}{\sqrt{1-2rx+r^2}} = \sum_{l=0}^{\infty} r^l P_l(x)$$

两边对 x 求导, 得

$$\frac{r}{(1-2rx+r^2)^{3/2}} = \sum_{l=0}^{\infty} r^l P'_l(x) \quad (1)$$

由递推公式

$$(k+1)P_{k+1}(x) - 2(k+1)xP_k(x) + kP_{k-1}(x) = 0$$

两边求导, 得

$$(k+1)P'_{k+1}(x) - (2k+1)P_k(x) - (2k+1)xP'_k(x) + kP'_{k-1}(x) = 0 \quad (2)$$

由(教材公式 10.1.64) 知

$$P_k(x) = P'_{k+1}(x) - 2xP'_k(x) + P'_{k-1}(x) \quad (3)$$

由(2) + $k \times (3)$ 可得(教材公式 10.1.67), 即

$$P'_{k+1}(x) = (k+1)P_k(x) + xP'_k(x) \quad (4)$$

由(2) + $(k+1) \times (3)$ 可得(教材公式 10.1.68), 即

$$kP_k(x) = xP'_k(x) - P'_{k-1}(x) \quad (5)$$

由(4) 将序列 k 变为 $k-1$, 可得

$$P'_k(x) = kP_{k-1}(x) + xP'_{k-1}(x) \quad (6)$$

由(5) $\times x$ - (6) 并移项, 可得(教材公式 10.1.69), 即

$$(x^2 - 1)P'_k(x) = kxP_k(x) - kP_{k-1}(x)$$

11. 利用勒让德多项式的递推公式, 计算定积分

$$I = \int_0^1 P_l(x) dx.$$

解:
$$I = \int_0^1 P_l(x) dx$$

由递推公式(10,1,66)可得

$$(2l+1)P_l(x) = P'_{l+1}(x) - P'_{l-1}(x)$$

$$\begin{aligned} \text{则 } I &= \int_0^1 \frac{1}{2l+1} [P'_{l+1}(x) - P'_{l-1}(x)] dx = \frac{1}{2l+1} [P_{l+1}(x) - P_{l-1}(x)] \Big|_0^1 \\ &= \frac{1}{2l+1} [P_{l-1}(0) - P_{l+1}(0)] \end{aligned}$$

其中 $l = 1, 2, 3, \dots$,

当 $l = 2n, P_{l-1}(0) = P_{l+1}(0) = 0$, 则 $I = 0$

当 $l = 2n+1$

$$P_{l-1}(0) = (-1)^n \frac{(2n-1)!!}{(2n)!!}$$

$$P_{l+1}(0) = (-1)^{n+1} \frac{(2n+1)!!}{(2n+2)!!}$$

$$\text{则 } I = (-1)^n \frac{(2n-1)!!}{(2n+2)!!}$$

其中 $n = 1, 2, 3, \dots$

$$\text{当 } l = 0 \text{ 时, } I = \int_0^1 P_0(x) dx = 1$$

$$\text{当 } l = 1 \text{ 时, } I = \int_0^1 P_1(x) dx = \frac{1}{2}$$

§ 10.3 一般的球函数

1. 用球函数把下列函数展开.

$$(1) (1 + 3\cos \theta) \sin \theta \cos \varphi$$

$$\text{解: } (1 + 3\cos \theta) \sin \theta \cos \varphi = \sin \theta \cos \varphi + 3\sin \theta \cos \theta \cos \varphi$$

$$= \sin \theta \cos \varphi + \frac{3}{2} \sin 2\theta \cos \varphi = P_1^1(\cos \theta) \cos \varphi + P_2^1(\cos \theta) \cos \varphi$$

$$(2) (1 - |\cos \theta|)(1 + \cos 2\varphi)$$

$$\text{解: } (1 - |\cos \theta|)(1 + \cos 2\varphi) = (1 - |\cos \theta|) + (1 - |\cos \theta|) \cos 2\varphi$$

对 $1 - |\cos \theta|$, 写作 $1 - |x|$ ($1 - sx \leq 1$)

用 $P_l(x)$ 展形, 由于 $1 - |x|$ 为偶函数, 当 l 为奇数时, 被积函数为奇函数.

故奇次项系数为 0

$$f_0 = \frac{1}{2} \int_{-1}^1 (1 - |x|) dx = \frac{1}{2} \left[\int_{-1}^0 (1 + x) dx + \int_0^1 (1 - x) dx \right]$$

$$= \frac{1}{2} \left[\left(x + \frac{1}{2} x^2 \right) \Big|_{-1}^0 + \left(x - \frac{1}{2} x^2 \right) \Big|_0^1 \right] = \frac{1}{2}$$

$$f_2 = \frac{5}{2} \int_{-1}^1 (1 - |x|) \frac{1}{2} (3x^2 - 1) dx$$

$$= \frac{5}{4} \left[\int_{-1}^1 (3x^2 - 1) dx + \int_{-1}^0 x(3x^2 - 1) dx + \int_0^1 x(3x^2 - 1) dx \right]$$

$$= \frac{5}{4} \left[(x^3 - x) \Big|_{-1}^1 + \left(\frac{3}{4}x^4 - \frac{1}{2}x^2 \right) \Big|_{-1}^0 - \left(\frac{3}{4}x^4 - \frac{1}{2}x^2 \right) \Big|_0^1 \right] = -\frac{5}{8}$$

$$\begin{aligned} f_{2n} &= \frac{4n+1}{2} \int_{-1}^1 (1-|x|) P_{2n}(x) dx = \frac{4n+1}{2} \int_{-1}^1 (1-|x|) \frac{1}{2^{2n}(2n)!} \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} dx \\ &= \frac{4n+1}{2^{2n}(2n)!} \int_0^1 (1-x) \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} dx \\ &= \frac{4n+1}{2^{2n}(2n)!} \left\{ \left[(1-x) \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n} \right] \Big|_0^1 - \int_0^1 \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n} d(1-x) \right\} \end{aligned}$$

由于 $\frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n}$ 只有奇次幂故 $x=0$ 代入其必为 0.

$$f_{2n} = \frac{4n+1}{2^{2n}(2n)!} \int_0^1 \frac{d^{2n-1}}{dx^{2n-1}} (x^2-1)^{2n} dx = \frac{4n+1}{2^{2n}(2n)!} \left[\frac{d^{2n-2}}{dx^{2n-2}} (x^2-1)^{2n} \right] \Big|_0^1$$

运用二项式定理

$$(x^2-1)^{2n} = \sum_{k=0}^{2n} \frac{(2n)!}{(2n-k)!k!} (x^2)^{2n-k} (-1)^k$$

取 $k=n+1$ 的项

$$\begin{aligned} f_{2n} &= \left(\frac{4n+1}{2^{2n}(2n)!} \right) \frac{d^{2n-2}}{dx^{2n-2}} \left[\frac{(2n)!}{(n-1)!(n+1)!} x^{2n-2} (-1)^{n+1} \right] = (-1)^n \frac{(4n+1)(2n-2)!}{[2^{n-1}(n-1)!][2^{n+1}(n+1)!]} \\ &= (-1)^n \frac{(4n+1)(2n-2)!}{(2n-2)!!(2n+2)!!} = (-1)^n \frac{(4n+1)(2n-3)!!}{(2n+2)!!}, (n=2,3,4,\dots) \end{aligned}$$

对 $(1-|\cos \theta|)$, 写作 $1-|x|$. 用 $P_l^2(x)$ 展开, 由于 $1-|x|$ 为偶函数, 当 l 为奇数时, 被积函数为奇函数, 故奇次项系数为 0.

$$\begin{aligned} f_{2n} &= \frac{4n+1}{2} \cdot \frac{(2n-2)!}{(2n+2)!} \int_{-1}^1 (1-|x|) P_{2n}^2(x) dx \\ &= \frac{(4n+1)(2n-2)!}{(2n+2)!2^{2n}(2n)!} \int_0^1 (1-x)(1-x^2) \frac{d^{2n+2}}{dx^{2n+2}} (x^2-1)^{2n} dx \\ &= \frac{(4n+1)(2n+1)!}{(2n+2)!2^{2n}(2n)!} \left\{ \left[(1-x)(1-x^2) \frac{d^{2n+1}}{dx^{2n+1}} (x^2-1)^{2n} \right] \Big|_0^1 - \int_0^1 \frac{d^{2n+1}}{dx^{2n+1}} (x^2-1)^{2n} d[(1-x)(1-x)^2] \right\} \end{aligned}$$

由于 $\frac{d^{2n+1}}{dx^{2n+1}} (x^2-1)^{2n}$ 只有奇次幂, 故 $x=0$ 代入其必为 0

$$\begin{aligned} f_{2n} &= -\frac{(4n+1)(2n-2)!}{(2n+2)!2^{2n}(2n)!} \int_0^1 (3x+1)(x-1) \frac{d^{2n+1}}{dx^{2n+1}} (x^2-1)^{2n} dx \\ &= -\frac{(4n+1)(2n-2)!}{(2n+2)!2^{2n}(2n)!} \left\{ \left[(3x+1)(x-1) \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} \right] \Big|_0^1 - \int_0^1 \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} d[(3x+1)(x-1)] \right\} \\ &\quad \left[(3x+1)(x-1) \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} \right] \Big|_0^1 = -\frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} \Big|_{x=0} \end{aligned}$$

由前面二项式定理展开的 $(x^2-1)^{2n}$ 进行求导 $(2n)$ 次, 得到

$$\left[(3x+1)(x-1) \frac{d^{2n}}{dx^{2n}} (x^2-1)^{2n} \right] \Big|_0^1 = (-1)^n 2^n (2n)! \frac{(2n-1)!}{(2n)!!}$$

利用类似的讨论,分部积分后利用二项式展开求导,易得到积分项

$$\int_0^1 \frac{d^{2n}}{dx^{2n}}(x^2-1)^{2n} d[(3x+1)(x-1)] = \int_0^1 \left[\frac{d^{2n}}{dx^{2n}}(x^2-1)^{2n} \right] \cdot 2(3x-1) dx$$

$$= 6 \frac{(2n-2)!(2n)!(-1)^n}{(n+1)!(n-1)!}$$

$$f_{2n} = -(-1)^n \frac{(4n+1)(2n-2)!(2n-1)!!}{(2n+2)!(2n)!!} \times \left[1 + \frac{6}{(2n-1)(2n+2)} \right]$$

综上所述, $(1-|\cos \theta|)(1+\cos 2\varphi)$ 可展开为

$$\frac{1}{2} - \frac{5}{8} P_2(\cos \theta) + \sum_{n=2}^{\infty} (-1)^n (4n+1) \frac{(2n-3)!!}{(2n+2)!!} P_{2n}(\cos \theta) - \sum_{n=1}^{\infty} (-1)^n (4n+1)$$

$$\left[1 + \frac{6}{(2n-1)(2n+2)} \right] \times \frac{(2n-2)!(2n-1)!!}{(2n+2)!(2n)!!} P_{2n}^2(\cos \theta) \cos^2 \varphi$$

2. 在半径为 r_0 的球形区域, 在(1) 球的内部和(2) 球的外部, 分别求解定解问题

$$\begin{cases} \Delta u = 0, \\ u|_{r=r_0} = 4 \sin^2 \theta \left(\cos \varphi \sin \varphi + \frac{1}{2} \right). \end{cases}$$

解: 设方程的解为

$$u(r, \theta, \varphi) = \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} r^l [A_l^m \cos m\varphi + B_l^m \sin m\varphi] P_l^m(\cos \theta) + \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} \frac{1}{r^{l+1}} [C_l^m \cos m\varphi + D_l^m \sin m\varphi] P_l^m(\cos \theta)$$

由边界条件中 $\cos \varphi \sin \varphi$ 一项及常数项可知 $m = 0, 2$

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \left[r^l A_l P_l^0(\cos \theta) + \frac{1}{r^{l+1}} C_l P_l^0(\cos \theta) \right] + \sum_{l=2}^{\infty} \left[r^l B_l^2 \sin 2\varphi P_l^2(\cos \theta) + \frac{D_l^2}{r^{l+1}} \sin^2 \varphi P_l^2(\cos \theta) + \frac{D_l^2}{r^{l+1}} \sin 2\varphi P_0^2(\cos \theta) \right]$$

(1) 在球内部, 球心处有界, 舍去 $\frac{1}{r^{l+1}}$ 项, 得

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} r^l A_l P_l^0(\cos \theta) + \sum_{l=2}^{\infty} r^l B_l^2 P_l^2(\cos \theta) \sin 2\varphi$$

(2) 在球外部, 问题在无穷远处有界, 舍去 r^l 项, 得

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \frac{C_l}{r^{l+1}} P_l^0(\cos \theta) + \sum_{l=2}^{\infty} \frac{D_l}{r^{l+1}} P_l^2(\cos \theta) \sin 2\varphi$$

将边界条件用球函数展开, 得到

$$u|_{r=r_0} = 2 \sin^2 \theta + 2 \sin^2 \theta \sin 2\varphi$$

$$= 2(1 - \cos^2 \theta) + 2(1 - \cos^2 \theta) \sin 2\varphi$$

$$= \frac{4}{3} P_0^0(\cos \theta) - \frac{4}{3} P_2^0(\cos \theta) + \frac{2}{3} P_2^2(\cos \theta) \sin 2\varphi$$

比较(1)、(2) 两解与边界条件的球函数展开式, 即可得到最终解.

$$(1) u = \frac{4}{3} P_0^0(\cos \theta) - \frac{4}{3} \left(\frac{r}{r_0} \right)^2 P_2^0(\cos \theta) + \frac{2}{3} \left(\frac{r}{r_0} \right)^2 P_2^2(\cos \theta) \sin 2\varphi$$

$$(2) u = \frac{4}{3} \frac{r_0}{r} P_0^0(\cos \theta) - \frac{4}{3} \left(\frac{r_0}{r}\right)^3 P_2^0(\cos \theta) + \frac{2}{3} \left(\frac{r_0}{r}\right)^2 P_2^2(\cos \theta) \sin 2\varphi$$

3. 在半径为 r_0 的球形区域, 在(1) 球的内部和(2) 球的外部, 分别求解定解问题

$$\begin{cases} \Delta u = 0, \\ \left. \frac{\partial u}{\partial r} \right|_{r=r_0} = \cos^2 \theta \cos^2 \varphi - \cos^2 \varphi + \frac{1}{3}. \end{cases}$$

解: 由题意得方程

$$\begin{cases} \Delta u = 0 \\ u|_{r=r_0} = \cos^2 \theta \cos^2 \varphi - \cos^2 \varphi + \frac{1}{3} \end{cases}$$

将边界条件用球函数展开, 得到

$$u_r|_{r=r_0} = (\cos^2 \theta - 1) \cdot \frac{1}{2} \cos^2 \varphi + \frac{1}{2} (\cos^2 \theta - 1) + \frac{1}{3} = -\frac{1}{6} P_2^2(\cos \theta) \cos^2 \varphi + \frac{1}{3} P_2^0(\cos \theta)$$

比较解与边界条件, 知 $m = 0, 2$

(1) 在球内部, 则球心处有界, 设问题的解为

$$u(r, \theta, \varphi) = \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} r^l (A_l^m \cos m\varphi + B_l^m \sin m\varphi) P_l^m(\cos \theta)$$

则

$$u = \sum_{l=0}^{\infty} r^l A_l P_l^0(\cos \theta) + \sum_{l=2}^{\infty} r^l (A_l^2 \cos 2\varphi + B_l^2 \sin 2\varphi) P_l^2(\cos \theta)$$

$$u_r|_{r=r_0} = \sum_{l=1}^{\infty} r_0^{l-1} A_l P_l^0(\cos \theta) + \sum_{l=2}^{\infty} r_0^{l-1} (A_l^2 \cos 2\varphi + B_l^2 \sin 2\varphi) P_l^2(\cos \theta)$$

对比边界条件可得

$$u_r|_{r=r_0} = 2r_0 A_2 P_2^0(\cos \theta) + 2r_0 (A_2^2 \cos 2\varphi + B_2^2 \sin 2\varphi) P_2^2(\cos \theta)$$

$$\begin{cases} 2r_0 A_2^0 = \frac{1}{3} \\ B_2^2 = 0 \\ 2r_0 A_2^2 = -\frac{1}{6} \end{cases} \quad \begin{cases} A_2^0 = \frac{1}{6r_0} \\ B_2^2 = 0 \\ A_2^2 = -\frac{1}{12r_0} \end{cases}$$

$$\text{所以 } u = \frac{1}{6} \frac{r^2}{r_0} P_2^0(\cos \theta) - \frac{1}{12} \frac{r^1}{r_0} P_2^2(\cos \theta) \cos 2\varphi$$

(2) 在球外部, 则问题在无穷处有界

设问题的解为

$$u(r, \theta, \varphi) = \sum_{l=m}^{\infty} \sum_{m=0}^{\infty} \frac{1}{r^{l+1}} (C_l^m \cos m\varphi + D_l^m \sin m\varphi) P_l^m(\cos \theta)$$

$$\text{则 } u = \sum_{l=0}^{\infty} \frac{1}{r^{l+1}} C_l^0 P_l^0(\cos \theta) + \sum_{l=2}^{\infty} \frac{1}{r^{l+1}} (C_l^2 \cos 2\varphi + D_l^2 \sin 2\varphi) P_l^2(\cos \theta)$$

$$u_r|_{r=r_0} = \sum_{l=0}^{\infty} -(l+1) \frac{1}{r_0^{l+2}} C_l^0 P_l^0(\cos \theta) + \sum_{l=2}^{\infty} (-1)(l+1) \frac{1}{r_0^{l+2}} (C_l^2 \cos 2\varphi + D_l^2 \sin 2\varphi) P_l^2(\cos \theta)$$

对比边界条件可得

$$u_r|_{r=r_0} = \frac{-3}{r_0^4} C_2^0 P_2^0(\cos \theta) + \frac{-3}{r_0^4} (C_2^2 \cos 2\varphi + D_2^2 \sin^2 \varphi) P_2^2(\cos \theta)$$

$$\begin{cases} \frac{-3}{r_0^4} C_2^0 = \frac{1}{3} \\ D_2^2 = 0 \\ \frac{-3}{r_0^4} C_2^2 = -\frac{1}{6} \end{cases} \quad \begin{cases} C_2^0 = -\frac{1}{9} r_0^4 \\ D_2^2 = 0 \\ C_2^2 = \frac{1}{18} r_0^4 \end{cases}$$

所以
$$u = -\frac{1}{9} \frac{r_0^4}{r^3} P_2^0(\cos \theta) + \frac{1}{18} \frac{r_0^4}{r^3} P_2^2(\cos \theta) \cos^2 \varphi$$

4. 在半径为 r_0 的球的内部区域($r < r_0$), 求解定解问题

$$\begin{cases} \Delta u = 0, \\ \left(u + H \frac{\partial u}{\partial r} \right) \Big|_{r=r_0} = u_0 \sin \theta (\sin \theta + \cos \theta \sin \varphi). \quad (H, u_0 \text{ 为常数}) \end{cases}$$

解: 由问题在球内部, 则球心处有界, 设问题的解为

$$u(r, \theta, \varphi) = \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} r^l (A_l^m \cos m\varphi + B_l^m \sin m\varphi) P_l^m(\cos \theta)$$

将边界条件用球函数展开得到

$$\begin{aligned} (u + Hu_r) \Big|_{r=r_0} = r_0 &= u_0 (1 - \cos^2 \theta) + \frac{1}{2} u_0 \sin 2\theta \sin \varphi \\ &= u_0 \left[\frac{2}{3} P_0^0(\cos \theta) - \frac{2}{3} P_2^0(\cos \theta) + \frac{1}{3} P_2^1(\cos \theta) \sin \theta \right] \end{aligned}$$

比较解与边界条件, 知 $m = 0, 1$

$$\text{则 } u = \sum_{l=0}^{\infty} r^l A_l P_l^0(\cos \theta) + \sum_{l=1}^{\infty} r^l (A_l^1 \cos \varphi + B_l^1 \sin \varphi) P_l^1(\cos \theta)$$

$$u_r = \sum_{l=1}^{\infty} l r^{l-1} A_l P_l^0(\cos \theta) + \sum_{l=1}^{\infty} l r^{l-1} (A_l^1 \cos \theta + B_l^1 \sin \varphi) P_l^1(\cos \theta)$$

$$\begin{aligned} \text{则 } (u + Hu_r) \Big|_{r=r_0} &= A_0^0 P_0^0(\cos \theta) + \sum_{l=1}^{\infty} (r_0^l + H l r_0^{l-1}) A_l^0 P_l^0(\cos \theta) + \sum_{l=1}^{\infty} (r_0^l + H l r_0^{l-1}) (A_l^1 \cos \theta \\ &+ B_l^1 \sin \varphi) P_l^1(\cos \theta) \end{aligned}$$

对比边界条件可得

$$(u + Hu_r) \Big|_{r=r_0} = A_0^0 P_0^0(\cos \theta) + (r_0^2 + 2Hr_0) A_2^0 P_2^0(\cos \theta) + (r_0^2 + 2Hr_0) (A_2^1 \cos \theta + B_2^1 \sin \varphi) P_2^1(\cos \theta)$$

$$\begin{cases} A_0^0 = \frac{2}{3} u_0 \\ (r_0^2 + 2Hr_0) A_2^0 = -\frac{2}{3} u_0 \\ A_2^1 = 0 \\ (r_0^2 + 2Hr_0) B_2^1 = \frac{1}{3} u_0 \end{cases}$$

$$\therefore \begin{cases} A_0^0 = \frac{2}{3}u_0 \\ A_2^0 = -\frac{2}{3} \cdot \frac{u_0}{r_0(r_0+2H)} \\ A_2^1 = 0 \\ B_2^1 = \frac{1}{3} \frac{u_0}{r_0(r_0+2H)} \\ B_2^1 = \frac{1}{3} \frac{u_0}{r_0(r_0+2H)} \end{cases}$$

$$u(r, \theta, \varphi) = \frac{2}{3}u_0 P_0^0(\cos \theta) - \frac{2u_0}{3r_0(r_0+2H)} r^2 P_2^0(\cos \theta) + \frac{u_0}{3r_0(r_0+2H)} r^2 P_2^0(\cos \theta)$$

5. 在内半径为 r_1 、外半径为 r_2 的空心球区域($r_1 < r < r_2$), 求解定解问题

$$\begin{cases} \Delta u = 0, \\ u|_{r=r_1} = u_1 \cos \theta, \\ u|_{r=r_2} = u_2 \sin \theta \cos \theta \sin \varphi \quad (u_1, u_2 \text{ 为常数}) \end{cases}$$

解: 设方程的解为

$$u(r, \theta, \varphi) = \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} r^l [A_l^m \cos m\varphi + B_l^m \sin m\varphi] P_l^m(\cos \theta) + \sum_{m=0}^{\infty} \sum_{l=m}^{\infty} \frac{1}{r^{l+1}} [C_l^m \cos m\varphi + D_l^m \sin m\varphi] P_l^m(\cos \theta)$$

将边界条件用球函数展开, 得到

$$u|_{r=r_1} = u_1 \cos \theta = u_1 P_1^0(\cos \theta)$$

$$u|_{r=r_2} = u_2 \sin \theta \cos \theta \sin \varphi = \frac{u_2}{3} P_2^1(\cos \theta) \sin \varphi$$

比较解和边界条件可得 $m = 0, 1$, 则

$$u = \sum_{l=0}^{\infty} \left[r^l A_l^0 P_l^0(\cos \theta) + \frac{1}{r^{l+1}} C_l^0 P_l^0(\cos \theta) \right] = \sum_{l=1}^{\infty} [A_l^1 \cos \varphi + B_l^1 \sin \varphi] P_l^1(\cos \theta) + \frac{1}{r^{l+1}} (C_l^1 \cos \varphi + D_l^1 \sin \varphi) P_l^1(\cos \theta)$$

对比边界条件可得

$$u = r A_1^0 P_1^0(\cos \theta) + \frac{1}{r^2} C_1^0 P_1^0(\cos \theta) + r^2 (A_2^1 \cos 2\varphi + B_2^1 \sin 2\varphi) P_2^1(\cos \theta) + \frac{1}{r^3} (C_2^1 \cos 2\varphi + D_2^1 \sin 2\varphi) P_2^1(\cos \theta)$$

$$\begin{cases} r_1 A_1^0 + \frac{1}{r_1^2} C_1^0 = u_1 \\ r_1^2 B_2^1 + \frac{1}{r_2^2} D_2^1 = \frac{1}{3} u_2 \\ r_2 A_1^0 + \frac{1}{r_2^2} C_1^0 = 0 \\ r_2 B_2^1 + \frac{1}{r_2^3} D_2^1 = 0 \end{cases}$$

解得

$$\begin{cases} A_1^0 = \frac{u_1 r_1^2}{r_1^3 - r_2^3} \\ B_2^1 = \frac{u_2 r_2^3}{r_2^5 - r_1^5} \\ C_1^0 = \frac{u_1 r_1^2}{r_1^3 - r_2^3} (-r_2^3) \\ D_2^1 = \frac{u_2 r_2^3}{r_2^5 - r_1^5} (-r_1^5) \end{cases}$$

$$u = \frac{u_1 r_1^2}{r_2^3 - r_1^3} \left(-r + r_2^3 \cdot \frac{1}{r^2} \right) P_1^0(\cos \theta) + \frac{u_2 r_2^3}{3(r_2^5 - r_1^5)} \left(r^2 - r_1^5 \cdot \frac{1}{r^3} \right) P_2^1(\cos \theta) \sin \varphi$$

6. 在半径为 r_0 的球的内部区域($r < r_0$), 求解泊松方程问题

$$\begin{cases} \Delta u = A r \cos \theta, & (A \text{ 为常数}) \\ u|_{r=r_0} = 0. \end{cases}$$

解: 由题意得方程

$$\begin{cases} \Delta u = A r \cos \theta \\ u|_{r=r_0} = 0 \end{cases}$$

首先寻找方程的特解

令 $v = \frac{A}{10} r^3 \cos \theta$, 代回方程验证满足方程, 则在 $r = r_0$ 处

$$v|_{r=r_0} = \frac{A}{10} r_0^3 \cos \theta$$

令 $u = v + \omega$,

$$\begin{cases} \Delta \omega = 0 \\ \omega|_{r=r_0} = -\frac{A}{10} r_0^3 \cos \theta \end{cases}$$

由于问题在球内部有界

$$\text{设 } \omega = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$$

比较解和边界条件可知 $l = 1$, 即得 $\omega = A_1 r \cos \theta$

$$A_1 r_0 = -\frac{A}{10} r_0^3, \text{ 有 } A_1 = -\frac{A}{10} r_0^2, \omega = -\frac{A}{10} r_0^2 r \cos \theta.$$

$$\text{所以 } u = v + \omega = \frac{A}{10} (r^2 - r_0^2) r \cos \theta$$

第十一章 柱函数

本章知识要点

三类柱函数

(一) 贝塞尔方程

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + (x^2 - m^2)R = 0$$

虚宗量贝塞尔方程

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} - (x^2 + m^2)R = 0$$

贝塞尔方程的通解为 $y = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$ 或 $y = C_1 J_\nu(x) + C_2 N_\nu(x)$

前者对应 ν 为非整数的情形, 后者无论 ν 是否为整数都适用.

$J_\nu(x)$ 称为 ν 阶贝塞尔函数, $N_\nu(x)$ 称为 ν 阶诺伊曼函数.

又取线性独立的 $H_\nu^{(1)}(x) = J_\nu(x) + iN_\nu(x)$ 和 $H_\nu^{(2)}(x) = J_\nu(x) - iN_\nu(x)$, 称之为第一种汉克尔函数与第二种汉克尔函数.

从而贝塞尔方程的通解又可写成 $y = C_1 H_\nu^{(1)}(x) + C_2 H_\nu^{(2)}(x)$

$J_\nu(x)$ 称为第一类柱函数 $N_\nu(x)$ 称为第二类柱函数, $H_\nu(x)$ 称为第三类柱函数.

(二) 渐近行为

$$\begin{aligned} \text{当 } x \rightarrow 0 \text{ 时} \quad J_\nu(x) &\rightarrow 0 & J_{-\nu}(x) &\rightarrow \infty & J_0(x) &\rightarrow 1 \\ N_0(x) &\rightarrow -\infty & N_\nu(x) &\rightarrow \pm \infty \end{aligned}$$

$$\text{当 } x \rightarrow \infty \text{ 时} \quad H_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{i(x - \nu\pi/2 - \frac{\pi}{4})}$$

$$H_\nu^{(2)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{i(x - \nu\pi/2 - \frac{\pi}{4})}$$

$$J_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} \cos(x - \nu\pi/2 - \frac{\pi}{4})$$

$$N_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} \sin(x - \nu\pi/2 - \frac{\pi}{4})$$

递推公式

$$\frac{d}{dx} \left[\frac{J_\nu(x)}{x^\nu} \right] = -\frac{J_{\nu+1}(x)}{x^\nu} \quad \frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x)$$

用 $Z_\nu(x)$ 代表 ν 阶的第一、二、三类柱函数, 则有

$$\frac{d}{dx} \left[\frac{Z_\nu(x)}{x^\nu} \right] = -\frac{Z_{\nu+1}(x)}{x^\nu}, \quad \frac{d}{dx} [x^\nu Z_\nu(x)] = x^\nu Z_{\nu-1}(x)$$

$$\text{进一步有} \quad \begin{cases} Z_\nu^1 - \nu Z_\nu/x = -Z_{\nu+1} \\ Z_\nu^1 + \nu Z_\nu/x = Z_{\nu-1} \end{cases} \Rightarrow \begin{cases} Z_{\nu-1} - Z_{\nu+1} = 2Z_\nu^1 \\ Z_{\nu+1} - 2\nu Z_\nu/x + Z_{\nu-1} = 0 \end{cases}$$

贝塞尔方程

(一) 贝塞尔函数与本征值问题

在 $\mu > 0$ 的情况下, $R(\rho)$ 是 m 阶贝塞尔方程的解, 对应圆柱内部的问题

$$R(\rho) = J_m(x) = J_m(\sqrt{\mu}\rho)$$

(1) 第一类齐次边界条件 $R(\rho_0) = 0$, ρ_0 为圆柱的半径, 即柱侧齐次边界条件
本征值为 $\mu_n^{(m)} = (x_n^{(m)}/\rho_0)^2$ $x_n^{(m)}$ 是 $J_m(x)$ 的第 n 个零点.

(2) 第二类齐次边界条件 $R'(\rho_0) = 0$, 即 $[dJ_m(\sqrt{\mu}\rho)/d\rho]|_{\rho=\rho_0} = \sqrt{\mu}J'_m(\sqrt{\mu}\rho_0) = 0$
本征值为 $\mu_n^{(m)} = (x_n^{(m)}/\rho_0)^2$ $x_n^{(m)}$ 是 $J'_m(x)$ 的第 n 个零点.

(3) 第三类齐次边界条件 $R(\rho_0) + HR'(\rho_0) = 0$, 即 $J_m(\sqrt{\mu}\rho_0) + H\sqrt{\mu}J'_m(\sqrt{\mu}\rho_0) = 0$
 $x_0 = \sqrt{\mu}\rho_0$, $h = \rho_0/H$, 由 Z_ν 的递推公式 $Z'_\nu - \nu Z_\nu/x = -Z_{\nu+1}$ 可得

$$J_m(x_0) = \frac{x_0}{h+m} J_{m+1}(x_0) \quad (*) \text{ 本征值是 } \mu_n^{(m)} = (x_n^{(m)}/\rho_0)^2 \text{ 其中 } x_n^{(m)} \text{ 是 } (*) \text{ 式的第 } n \text{ 个根.}$$

(二) 贝塞尔函数的正交关系

不同本征值的同防贝塞尔函数在区间 $[0, \rho_0]$ 上带权重 ρ 正交

$$\int_0^{\rho_0} J_m(\sqrt{\mu_n}\rho) J_m(\sqrt{\mu_l}\rho) \rho d\rho = 0$$

(三) 贝塞尔函数的模

贝塞尔函数 $J_m(\sqrt{\mu_n^{(m)}}\rho)$ 的模 $N_n^{(m)}$ 定义为 $[N_n^{(m)}]^2 = \int_0^{\rho_0} [J_m(\sqrt{\mu_n^{(m)}}\rho)]^2 \rho d\rho$

对第一类边界齐次条件 $J_m(\sqrt{\mu_n^{(m)}}\rho_0) = 0$: $[N_n^{(m)}]^2 = \frac{1}{2} \rho_0^2 [J'_m(\sqrt{\mu_n^{(m)}}\rho_0)]^2$

对第二类齐次边界条件 $J'_m(\sqrt{\mu_n^{(m)}}\rho_0) = 0$: $[N_n^{(m)}]^2 = \frac{1}{2} \left(\rho_0^2 - \frac{m^2}{\mu_n^{(m)}} \right) [J_m(\sqrt{\mu_n^{(m)}}\rho_0)]^2$

对第三类齐次边界条件 $J'_m = -J_m/\sqrt{\mu_n^{(m)}}H$: $[N_n^{(m)}]^2 = \frac{1}{2} \left(\rho_0^2 - \frac{m^2}{\mu_n^{(m)}} + \frac{\rho_0^2}{\mu_n^{(m)}} H \right) [J_m(\sqrt{\mu_n^{(m)}}\rho_0)]^2$

(四) 傅里叶 - 贝塞尔级数

$$\begin{cases} f(\rho) = \sum_{n=1}^{\infty} f_n J_m(\sqrt{\mu_n^{(m)}}\rho) \\ \text{系数 } f_n = \frac{1}{[N_n^{(m)}]^2} \int_0^{\rho_0} f(\rho) J_m(\sqrt{\mu_n^{(m)}}\rho) \rho d\rho \end{cases}$$

(五) 诺伊曼函数

研究圆柱内部问题, 圆柱轴上的自然边界条件排除了诺伊曼函数

但若考虑空心圆柱之类的区域并不涉及 $\rho = 0$ 的自然边界条件, 就不能排除诺伊曼函数

虚宗量贝塞尔方程

如果圆柱上下底面具有齐次边界条件, 应该考虑 $\mu \leq 0$ 的分离变量解

在 $\mu + \nu^2 < 0$ 的情况下, $R(\rho)$ 应是虚宗量贝塞尔方程 $x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} - (x^2 + m^2)R = 0$

($x = \nu\rho$) 该方程的一个解是 m 阶虚宗量贝塞尔函数 $I_m(x) = \sum_{k=0}^{\infty} \frac{1}{k!(m+k)!} \left(\frac{x}{2}\right)^{m+2k}$

寻找线性独立的另一解, $K_\nu(x) = \frac{\pi}{2} \frac{I_\nu(x) - I_{-\nu}(x)}{\sin \nu\pi}$

令 $\nu \rightarrow$ 整数 m 得 m 阶虚宗量汉克尔函数 $K_m(x) = \lim_{\nu \rightarrow m} \frac{\pi}{2} \frac{I_\nu - I_{-\nu}}{\sin \nu\pi}$

当 $x \rightarrow 0$ 时, $K_m(x) \rightarrow \infty$

当 $x \rightarrow \infty$ 时, $I_m(x) \rightarrow \infty, K_m(x) \rightarrow 0$

球贝塞尔方程

用球坐标系对 Helmholtz 方程分离变数得的球贝塞尔方程

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + [k^2 r^2 - (l+1)]R = 0.$$

令 $x = kr$ $R(r) \rightarrow y(x)$ 则 $R(r) = \sqrt{\frac{\pi}{2x}} y(x)$

则方程度为 $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left[x^2 - \left(l + \frac{1}{2}\right)^2\right]y = 0$

(一) 线性独立解

$$J_{l+\frac{1}{2}}(x) \quad J_{-(l+\frac{1}{2})}(x) \quad N_{l+\frac{1}{2}}(x) \quad H_{l+\frac{1}{2}}^{(1)}(x) \quad H_{l+\frac{1}{2}}^{(2)}(x)$$

$$\text{球贝塞尔函数 } j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}}(x) \quad j_{-l}(x) = \sqrt{\frac{\pi}{2x}} J_{-l+\frac{1}{2}}(x)$$

$$\text{球诺伊曼函数 } n_l(x) = \sqrt{\frac{\pi}{2x}} N_{l+\frac{1}{2}}(x)$$

$$\text{球汉克尔函数 } h_l^{(1)}(x) = \sqrt{\frac{\pi}{2x}} H_{l+\frac{1}{2}}^{(1)}(x) \quad h_l^{(2)}(x) = \sqrt{\frac{\pi}{2x}} H_{l+\frac{1}{2}}^{(2)}(x)$$

(二) 递推公式

$$Z_l(x) = \sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}}(x)$$

有 $Z_{l+\frac{1}{2}}(x) - (2l+1)Z_{l+\frac{1}{2}}(x)/x + Z_{l-\frac{1}{2}}(x) = 0$ 将 Z 用 z 表示

$$Z_{l+1}(x) = \frac{2l+1}{x} z_l - z_{l-1}$$

(三) 初等函数表示式

$$j_0(x) = \frac{\sin x}{x}$$

$$n_0(x) = -\frac{\cos x}{x}$$

$$j_1(x) = \frac{1}{x^2}(\sin x - x \cos x)$$

$$n_1(x) = -\frac{1}{x^2}(\cos x + x \sin x)$$

$$j_2(x) = \frac{1}{x^3}[3(\sin x - x \cos x) - x^2 \sin x] \quad n_2(x) = -\frac{1}{x^3}[3(\cos x + x \sin x) - x^2 \cos x]$$

$$\vdots$$

$$\vdots$$

由球汉克尔函数的定义, $h_l^{(1)} = j_l(x) + in_l(x)$ $h_l^{(2)} = j_l(x) - in_l(x)$

$$\text{有 } h_0^{(1)}(x) = -\frac{i}{x} e^{ix}$$

$$h_0^{(2)}(x) = \frac{i}{x} e^{-ix}$$

$$h_1^{(1)}(x) = \left(-\frac{i}{x^2} - \frac{1}{x}\right) e^{ix}$$

$$h_1^{(2)}(x) = \left(\frac{i}{x^2} - \frac{1}{x}\right) e^{-ix}$$

$$h_2^{(1)}(x) = \left(-\frac{3i}{x^3} - \frac{3}{x^2} + \frac{i}{x}\right) e^{ix}$$

$$h_2^{(2)}(x) = \left(\frac{3i}{x^3} - \frac{3}{x^2} - \frac{i}{x}\right) e^{-ix}$$

$$\vdots$$

$$\vdots$$
(四) $x \rightarrow 0$ 和 $x \rightarrow \infty$ 时的行为

当 $x \rightarrow 0$ 时, $n_l(x) \rightarrow \infty$

当 $x \rightarrow \infty$ 时, $j_l(x) \sim \frac{1}{x} \cos\left(x - \frac{l+1}{2}\pi\right)$

$$n_l(x) \sim \frac{1}{x} \sin\left(x - \frac{l+1}{2}\pi\right) \quad h_l^{(1)}(x) \sim \frac{1}{x} e^{ix} (-i)^{l+1}$$

$$h_l^{(2)}(x) \sim \frac{1}{x} e^{-ix} i^{l+1}$$

(五) 球形区域内的本征值问题

球贝塞尔方程写成 Sturm-Liouville 型即是 $\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) - l(l+1)R + k^2 r^2 R = 0$

在 $r=0$ 有自然边界条件, 应取 $j_l(kr)$, 舍去 $n_l(kr)$ 且 $j_l(kr)$ 满足球面 $r=r_0$ 上的第一, 第二或第三类齐次边界条件.

习题全解

§ 11.2 贝塞尔方程

1. 计算不定积分 $\int x^4 J_1(x) dx$. [提示: 反复运用(教材 P₂₆₅ 的公式 11.1.7) 和(教材 P₂₆₇ 的公式 11.2.5).]

解法一: 计算积分 $\int x^4 J_1(x) dx$

由公式(11.1.7) $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n-1}(x)$ 得

$$\int x^4 J_1(x) dx = \int x^2 x^2 J_1(x) dx = x^4 J_2(x) - 2 \int x^3 J_2(x) dx = x^4 J_2(x) - 2x J_3(x) + C$$

解法二:由公式(11.2.5) $J'_0(x) = -J_1(x)$. 原式有:

$$\begin{aligned} \int x^4 J_1(x) dx &= - \int x^4 J'_0(x) dx = - \int x^4 dJ_0(x), (\text{分部积分后有}) \\ &= -x^4 J_0(x) + 4 \int x^3 J_0(x) dx. \end{aligned}$$

再由公式(11.1.7) $\frac{d}{dx}[x^n J_n(x)] = x^n \cdot J_{n-1}(x)$, 得

$$\begin{aligned} \int x^3 J_0(x) dx &= \int x^2 [x J_0(x)] dx = \int x^2 \frac{d}{dx}[x J_1(x)] dx \\ &= \int x^2 d[x J_1(x)] = x^2 [x J_1(x)] - 2 \int x^2 J_1(x) dx, \\ &= x^3 J_1(x) - 2 \int x^2 J_1(x) dx. \end{aligned}$$

再分别由公式(11.1.7) 和公式(11.2.5) 得:

$$\begin{aligned} \int x^2 J_1(x) dx &= - \int x^2 J'_0(x) dx = - \int x^2 dJ_0(x) \\ &= -x^2 J_0(x) + \int J_0(x) dx^2 = -x^2 J_0(x) + 2 \int x J_0(x) dx \end{aligned}$$

由公式(11.1.7)

$$\int x J_0(x) dx = \int \frac{d}{dx}[x J_1(x)] dx = x J_1(x) + C$$

所以

$$\begin{aligned} \int x^4 J_1(x) dx &= -x^4 J_0(x) + 4 \int x^3 J_0(x) dx \\ &= -x^4 J_0(x) + 4x^3 J_1(x) - 8 \int x^2 J_1(x) dx \\ &= -x^4 J_0(x) + 4x^3 J_1(x) + 8x^2 J_0(x) - 16 \int x J_0(x) dx \\ &= -x^4 J_0(x) + 4x^3 J_1(x) + 8x^2 J_0(x) - 16x J_1(x) + C \\ &= x^2(8 - x^2) J_0(x) + 4x(x^2 - 4) J_1(x) + C \end{aligned}$$

2. 计算不定积分 $\int J_3(x) dx$.

解: 计算积分 $\int J_3(x) dx$

由公式 $\frac{d}{dx} \left(\frac{J_m(x)}{x^m} \right) = -\frac{J_{m+1}(x)}{x^m}$ 得

$$\int J_3(x) dx = \int x^2 \left(\frac{J_3}{x^2} \right) dx = \int x^2 \left(\frac{J_2}{x^2} \right)' dx = -J_2 + 2 \int x^{-1} J_2 dx = -J_2 - 2 \frac{J_1}{x} + C$$

再根据递推公式 $J_{n+1} - \frac{2nJ_n}{x} + J_{n-1} = 0$ 得 $J_2 = \frac{2J_1}{x} - J_0$

代入上式可得 $\int J_3(x) dx = J_0 - \frac{2J_1}{x} - 2 \frac{J_1}{x} + c = J_0 - \frac{4J_1}{x} + C$

3. 在区间 $[0, 1]$ 上, 第一类齐次边界条件下, 用零阶贝塞尔函数把 $f(x) = 1$ 展开为傅里叶-贝塞尔级数.

解: 展开公式如下 $f(\rho) = \sum_{n=1}^{\infty} f_n J_m\left(\frac{x_n^{(m)}}{\rho_0} \rho\right)$

其中系数 $f_n = \frac{1}{[N_n^{(m)}]^2} \int_0^{\rho_0} f(\rho) J_m\left(\frac{x_n^{(m)}}{\rho_0} \rho\right) \rho d\rho$

设定 $x = \rho$, 有 $f(x) = f(\rho) = 1$ $\rho_0 = 1$ 根据第一类边界条件,

$$[N_n^{(0)}]^2 = \frac{1}{2} \cdot 1 \cdot [J_1(x_n^{(0)})]^2$$

$$\begin{aligned} \text{从而得到 } f_n &= \frac{2}{[J_2(x_n^{(0)})]^2} \int_0^1 J_0(x_n^{(0)} \rho) \rho d\rho = \frac{2}{[J_1(x_n^{(0)})]^2} \frac{1}{(x_n^{(0)})^2} \int_0^1 (x_n^{(0)}(\rho) J_0(x_n^{(0)} \rho) d(x_n^{(0)} \rho) \\ &= \frac{2}{[x_n^{(0)} J_1(x_n^{(0)})]^2} (x_n^{(0)} \rho) J_1(x_n^{(0)} \rho) \Big|_0^1 = \frac{2}{x_n^{(0)} J_1(x_n^{(0)})} \end{aligned}$$

$$\text{因此 } 1 = 2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} \rho)}{x_n^{(0)} J_1(x_n^{(0)})} = 2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)})}{x_n^{(0)} J_1(x_n^{(0)})} \quad (x_n^{(0)} \text{ 为 } J_0(x) \text{ 的第 } n \text{ 个零点})$$

4. 半径为 ρ_0 、高为 L 的圆柱体, 下底和侧面保持零度, 上底温度分布为 $f(\rho) = \rho^2$, 求柱体内各点的稳恒温度分布.

解: 写出定解问题满足的方程及边界条件

$$\begin{cases} \Delta u = 0 & \text{在柱坐标系中考虑问题 } z = 0 \text{ 的面为下底面} \\ u|_{z=0} = 0 & \text{柱轴为 } z \text{ 轴} \\ u|_{z=L} = \rho^2 & \text{由侧面是第一类边界条件从而本绝值入由 } J_0(\lambda R) = 0 \text{ 决定} \\ u|_{\rho=R} = 0 & \text{即 } \lambda_n^{(0)} = \frac{x_n^{(0)}}{R} (n = 1, 2, \dots) \text{ 在题中 } R = \rho_0 \end{cases}$$

$$\text{在柱内 } u = \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{R} z} + B_n e^{-\frac{x_n^{(0)}}{R} z}) J_0(\lambda_n^{(0)} \rho)$$

$$\text{由边界条件得 } \begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = 0 \\ \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{R} L} + B_n e^{-\frac{x_n^{(0)}}{R} L}) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = \rho^2 \end{cases}$$

$$\Rightarrow A_n + B_n = 0 \quad A_n = \frac{1}{R^2 [J_1(x_n^{(0)})]^2 \operatorname{sh}\left(\frac{x_n^{(0)}}{R} L\right) \int_0^R \rho^2 J_0\left(\frac{x_n^{(0)}}{R} \rho\right) \rho d\rho}$$

$$\text{求解积分 } \int_0^R \rho^3 J_0\left(\frac{x_n^{(0)}}{R} \rho\right) d\rho = \frac{R^4}{(x_n^{(0)})^4} [x^3 J_1(x) + 2x^2 J_0(x) - 4x J_1(x)]_0^{\frac{x_n^{(0)}}{R}}$$

$$= \frac{R^4}{x_n^{(0)}} J_1(x_n^{(0)}) - \frac{4R^4}{(x_n^{(0)})^3} J_1(x_n^{(0)})$$

$$\text{有 } A_n = \frac{R^2}{x_n^{(0)} J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L/R)} \left(1 - \frac{4}{(x_n^{(0)})^2}\right)$$

将 A_n 表达式代入, 并由 $A_n = -B_n$ 可得:

$$u = 2R^2 \sum_{n=1}^{\infty} \frac{J_0(x_n^{(0)} \rho/R) \operatorname{sh}(x_n^{(0)} Z/R)}{x_n^{(0)} J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L/R)} \cdot \left[1 - \frac{4}{(x_n^{(0)})^2} \right], (R = \rho_0)$$

5. 半径为 ρ_0 而高为 L 的圆柱体, 下底温度分布为 $u_0 \rho^2$, 上底温度保持为 u_1 , 侧面绝热, 求柱体内的稳恒温度分布.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0 \\ u|_{\rho=\rho_0} = 0, u|_{\rho=0} = \text{有限值} \\ u|_{z=0} = u_0 \rho^2, u|_{z=L} = u_1 \end{cases}$$

因柱侧是齐次第二类边界条件, 对应 $\mu \geq 0$ 部分, 解的形式为

$$J_m(\sqrt{\mu}\rho) \begin{cases} e^{\sqrt{\mu}z} \\ e^{-\sqrt{\mu}z} \end{cases} \begin{cases} \cos m\varphi \\ \sin m\varphi \end{cases}$$

由边界条件与 φ 无关, 得 $m = 0$, 则解可简化为

$$J_0(\sqrt{\mu}\rho) \begin{cases} e^{\sqrt{\mu}z} \\ e^{-\sqrt{\mu}z} \end{cases}$$

由 $u_\rho|_{\rho=\rho_0} = 0$ 可得本征值 $\mu_n^{(1)} = (x_n^{(1)}/\rho_0)^2$. 其中 $x_n^{(1)}$ 是 $T_1(x)$ 的第 n 个正的零点, 考虑 $\mu = 0$ 时, 由 $u|_{\rho=0} = \text{有限值}$, 可知应舍去 $\ln \rho, \rho^{-m}$ 项, $z(z)$ 应为 $(\frac{1}{z})$, 综上所述, 解的形式为

$$u(\rho, z) = A_0 + B_0 z + \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(1)} z}{\rho_0}} + B_n e^{-\frac{x_n^{(1)} z}{\rho_0}}) J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right)$$

将上式代入柱顶, 柱底边界条件

$$\begin{cases} A_0 + \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) = u_0 \rho^2 \\ A_0 + B_0 L + \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(1)} L}{\rho_0}} + B_n e^{-\frac{x_n^{(1)} L}{\rho_0}}) J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) = u_1 \end{cases}$$

把右边的 $u_0 \rho^2, u_1$ 分别展开为傅里叶—贝塞尔级数, 比较左边系数, 即得

$$\begin{cases} A_0 = \frac{2}{\rho_0^2} \int_0^{\rho_0} u_0 \rho^2 \cdot \rho d\rho \\ A_0 + B_0 L = \frac{2}{\rho_0^2} \int_0^{\rho_0} u_1 \rho d\rho \\ \begin{cases} A_n + B_n = \frac{2}{\rho_0^2 [J_0(x_n^{(1)})]^2} \int_0^{\rho_0} \frac{1}{2} u_0 \rho^2 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho \\ A_n e^{\frac{x_n^{(1)} L}{\rho_0}} + B_n e^{-\frac{x_n^{(1)} L}{\rho_0}} = \frac{2}{\rho_0^2 [J_0(x_n^{(1)})]^2} \int_0^{\rho_0} u_1 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho \end{cases} \end{cases}$$

由上式可求得 A_0, B_0, A_n, B_n 代回 $u(\rho, z)$, 取得到最终解

$$u(\rho, z) = \frac{u_0}{2} \rho_0^2 + \frac{u_1 - (u_0 \rho_0^2/2)}{L} z + \sum_{n=1}^{\infty} \frac{4u_0 \rho_0^2}{(x_n^{(1)})^2 J_0(x_n^{(1)}) \operatorname{sh}(x_n^{(1)} L/\rho_0)} \times \operatorname{sh} \frac{x_n^{(1)}}{\rho_0} (L - z) J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right)$$

6. 圆柱体半径为 ρ_0 而高为 L , 上底有均匀分布的强度为 q_0 的热流进入, 下底则有同样的热流流出, 柱侧保持温度为零度, 求柱体内的稳恒温度分布.

解:写出定解问题

$$\begin{cases} \Delta u = 0 \\ \frac{\partial u}{\partial z} \Big|_{z=0} = \frac{q_0}{k_0} \\ \frac{\partial u}{\partial z} \Big|_{z=L} = \frac{q_0}{k} \\ u \Big|_{\rho=\rho_0} = 0 \end{cases}$$

侧面是第一类边界条件,解与 φ 值无关

有 $J_0(\omega R) = 0, \omega = \frac{x_n^{(0)}}{R} (n = 1, 2, \dots), R = \rho_0$

$$u = \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{\rho_0} z} + B_n e^{-\frac{x_n^{(0)}}{\rho_0} z}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right), (*)$$

由边界条件定出 A_n, B_n

$$\frac{\partial u}{\partial z} \Big|_{z=0} = \sum_{n=1}^{\infty} (A_n - B_n) \frac{x_n^{(0)}}{\rho_0} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = \frac{q_0}{k}$$

$$A_n - B_n = \frac{2q_0}{\rho^2 \frac{x_n^{(0)}}{\rho_0} \left[J_1\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \right]^2} \int_0^R J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho \quad (1)$$

解得

$$A_n - B_n = \frac{2q_0}{k\rho \frac{(x_n^{(0)})^2}{\rho_0^2} J_1\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)}$$

同理由

$$\frac{\partial u}{\partial z} \Big|_{z=L} = \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{\rho_0} L} - B_n e^{-\frac{x_n^{(0)}}{\rho_0} L}) \frac{x_n^{(0)}}{\rho_0} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = \frac{q_0}{k}$$

得

$$A_n e^{\frac{x_n^{(0)}}{\rho_0} L} - B_n e^{-\frac{x_n^{(0)}}{\rho_0} L} = \frac{2q_0}{k\rho \left(\frac{x_n^{(0)}}{\rho_0}\right)^2 J_1\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)} \quad (2)$$

由此可解出 A_n, B_n , 代入(*)式可得 u

$$u(\rho, z) = \sum_{n=1}^{\infty} \frac{2q_0 \rho_0}{k(x_n^{(0)})^2 J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} \left[\operatorname{ch} \frac{x_n^{(0)}}{\rho_0} z - \operatorname{ch} \frac{x_n^{(0)}}{\rho_0} (L - z) \right] J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)$$

7. 半径为 ρ_0 的圆形膜, 边缘固定, 初始形状是旋转抛物面 $u|_{t=0} = (1 - \rho^2 / \rho_0^2) u_0$, 初始速度为零, 求解膜的振动情况.

解: 取以圆形膜中心为原点的极坐标系, 写出定解问题

$$u_{tt} - a^2 \left(u_{\rho\rho} + \frac{1}{\rho} u_{\rho} \right) = 0$$

$$\text{边界条件: } u|_{\rho=0} \text{ 有限} \quad u|_{\rho=\rho_0} = 0 \quad \text{初始条件} \begin{cases} u|_{t=0} = \left(\frac{1 - \rho^2}{\rho_0^2} \right) u_0 \\ u_t|_{t=0} = 0 \end{cases}$$

$$\text{设 } u = u(\rho) T(t) \text{ 分离变量得} \begin{cases} T'' + \lambda^2 a^2 T = 0 \\ u'' + \frac{1}{\rho} u' + \lambda^2 u = 0 \end{cases} \quad (1) \quad (2)$$

(2) 的有限解为 $u = J_0(\lambda \rho)$ 令 $x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点 ($n = 1, 2, \dots$)

则

$$\lambda_n^{(0)} = \frac{x_n^{(0)}}{\rho_0}, \nu_n = J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)$$

$$T_n = A_n \cos \frac{x_n^{(0)}}{\rho_0} t + B_n \sin \frac{x_n^{(0)}}{\rho_0} t.$$

从而
$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{x_n^{(0)}}{\rho_0} at + B_n \sin \frac{x_n^{(0)}}{\rho_0} at \right) J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right)$$

由初始条件得
$$A_n = \frac{2u_0}{\rho_0^2 [J_1(x_n^{(0)})]^2} \int_0^R \left(\frac{1-\rho^2}{\rho_0^2} \right) J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \rho d\rho$$

计算积分得
$$A_n = \frac{2u_0}{\rho_0^2 [J_1(x_n^{(0)})]^2} \left[\frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) - \frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) + \frac{4\rho_0^2}{[x_n^{(0)}]^3} J_1(x_n^{(0)}) \right]$$

$$= \frac{8u_0}{[x_n^{(0)}]^3 J_1(x_n^{(0)})}$$

于是
$$u = 8u_0 \sum_{n=1}^{\infty} \frac{1}{[x_n^{(0)}]^3 J_1(x_n^{(0)})} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \cos \frac{x_n^{(0)}}{\rho_0} at$$

8. 半径为 ρ_0 的半圆形膜, 边缘固定, 求其本征频率和本征振动.

解: 用极坐标, 定解问题为 $u_{tt} - a^2 \Delta u = 0$ ($0 \leq \varphi \leq \pi, 0 \leq \rho \leq \rho_0$)

$u|_{\rho=0}$ 有限 $u|_{\rho=\rho_0} = 0$ $u|_{\varphi=0} = u|_{\varphi=\pi} = 0$

设 $u = \nu(\rho)\phi(\varphi)T(t)$ 代入得

$$\begin{cases} \phi'' + m^2 \phi = 0 \\ \phi|_{\varphi=0} = \phi|_{\varphi=\pi} = 0 \end{cases} \quad (1)$$

$$(1) \quad \begin{cases} \nu'' + \frac{1}{\rho} \nu' + \left(\lambda^2 - \frac{m^2}{\rho^2} \right) \nu = 0 \\ \nu|_{\rho=0} \text{ 有限}, \nu|_{\rho=\rho_0} = 0 \end{cases} \quad (2)$$

$$T'' + \lambda^2 a^2 T = 0 \quad (3)$$

(1) 的解: $\phi = A_1 \cos m\varphi + A_2 \sin m\varphi$ 由边界条件知 $\phi_m = A_m \sin m\varphi$

(2) 的解 $\nu = B_1 J_m(\lambda\rho), J_m(\lambda\rho_0) = 0$ 得 $\nu_n = B_n J_m \left(\frac{x_n^{(m)}}{\rho_0} \rho \right)$

$x_n^{(m)}$ 为 $J_m(\lambda\rho)$ 的第 n 个零点

(3) 的解:
$$T_{n,m} = C \cos \frac{x_n^{(m)}}{\rho_0} at + D \sin \frac{x_n^{(m)}}{\rho_0} at$$

以上代入即得
$$u = \left[A_{n,m} \cos \frac{x_n^{(m)}}{\rho_0} at + B_{n,m} \sin \frac{x_n^{(m)}}{\rho_0} at \right] J_m \left(\frac{x_n^{(m)}}{\rho_0} \rho \right) \sin m\varphi$$

本征频率 $\omega_{n,m} = \frac{a}{\rho_0} x_n^{(m)}$

9. 求长圆柱形轴块的临界半径. [“临界”一词的意义见 § 8.1 习题 8.]

解: 取柱坐标系, 扩散方程为 $u_t - a^2 \Delta u = \beta u$ a^2 为扩散系数, β 为增殖常数 圆柱轴的侧面强予防护, 设半径为 R , 有 $u|_{\rho=R} = 0$

故定解问题为
$$\begin{cases} u_t - a^2 \left(u_{\varphi\varphi} + \frac{1}{\rho} u_{\rho\rho} \right) = \beta u \\ u|_{\rho=R} = 0 \end{cases}$$

分离变量得
$$T = Ae^{\lambda t} \quad \nu'' + \frac{1}{\rho} \nu' + \frac{(\beta - \lambda)}{a^2} \nu = 0 \quad \nu|_{\rho=R} = 0$$

$$\nu(\rho) = B J_0 \left(\frac{\sqrt{\beta - \lambda}}{a} \rho \right) \quad \text{由边界条件 } J_0 \left(\frac{\sqrt{\beta - \lambda}}{a} \rho \right) = 0 \Big|_{\rho=R}$$

$$\text{由 } x_n^{(0)} = \frac{\sqrt{\beta - \lambda_n^{(0)}}}{a} R \Rightarrow \lambda_n^{(0)} = \beta - \left(\frac{ax_n^{(0)}}{R} \right)^2$$

当 β 一定时, $\beta - \left(\frac{ax_n^{(0)}}{R} \right)^2 = 0$ 得 $R = \frac{ax_n^{(0)}}{\sqrt{\beta}}$ 称为临界尺寸

$$x_n^{(0)} \text{ 中最小的一个零点为 } x_1^{(0)} = 2.4048 \text{ 代入可得 } R = \frac{2.4048a}{\sqrt{\beta}}$$

10. 把温度为室温 u_0 的样品放入温度为 U_0 的烘炉内保温, 但是, 样品里的温度不可能立刻就转变为 U_0 , 它与 U_0 之差随着时间而衰减, 今约定把这差值降到初始值的 $1/e$ 之时作为保温开始时间, 试计算圆柱形样品放入烘炉多长时间才可计算保温时间.

解: 定解问题 $u_t - a^2 \Delta u = 0$

$$\begin{cases} u|_{\rho=R} = u_0 & \text{取 } u_0 \text{ 作为零点 } v = u - u_0 \\ u|_{z=0} = u_0 \\ u|_{z=H} = u_0 \\ u|_{t=0} = u_0 \end{cases} \quad \text{左边的方程组可改写成} \begin{cases} v_t - a^2 \Delta v = 0 & \rho < R, 0 \leq z \leq H \\ v|_{t=0} = u_0 - u_0 \\ v|_{\rho=R} = v|_{z=0} = v|_{z=H} = 0 \end{cases}$$

可通过分离变量法求解, 得

$$e^{-k^2 a^2 t} \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} \cos hz \\ \sin hz \end{Bmatrix} J_m(\sqrt{k^2 - h^2} \rho)$$

与 φ 无关, 有 $m = 0$, 由上下底边界条件, 取 $\sin hz, h = \frac{m\pi}{H} (m = 1, 2, \dots)$

$$\text{由侧面边界条件 } \sqrt{k^2 - h^2} = \frac{x_n^{(0)}}{R}$$

$$\text{从而 } v = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} C_{m,n} e^{-\left[\left(\frac{x_n^{(0)}}{R}\right)^2 + \left(\frac{m\pi}{H}\right)^2\right] a^2 t} \sin \frac{m\pi}{H} z J_0\left(\frac{x_n^{(0)}}{R} \rho\right)$$

其中 $n = 1, m = 1$ 的项随 t 减小最慢可把该项降为 $t = 0$ 时值的 $\frac{1}{e}$ 作为保温开始的时间即

$$e^{-\left[\left(\frac{x_1^{(0)}}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2\right] a^2 t} = e^{-1}$$

$$\text{则 } t = \frac{1}{a^2 \left[\left(\frac{2.4048}{R} \right)^2 + \left(\frac{\pi}{H} \right)^2 \right]}$$

11. 匀质圆柱, 半径 ρ_0 、高 L , 上下底面固定, 侧面自由, 初始位移为零, 初始速度为 $u_0 \rho^2 z$, 求柱内各处的振动情况.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta u = 0 & (0 < \rho < \rho_0) \\ u|_{\rho=0} = \text{有限值} & u_{\rho}|_{\rho=\rho_0} = 0 \\ u|_{z=0} = 0 & u|_{z=L} = 0 \\ u|_{t=0} = 0 & u_t|_{t=0} = u_0 \rho^2 z \end{cases}$$

$$\text{在柱坐标下求解该问题, 可知其解的形式为 } u = \begin{Bmatrix} J_m(\sqrt{\mu} \rho) \\ N_m(\sqrt{\mu} \rho) \end{Bmatrix} \begin{Bmatrix} \cos \mu z \\ \sin \mu z \end{Bmatrix} \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} \cos kat \\ \sin kat \end{Bmatrix}$$

($\mu' > 0$, 若 $\mu' = 0, \nu = 0, k = 0$. 不符合题意)

由 $\rho = 0$ 时, u 取有限值, 可知解中不包含 $N_m(\sqrt{\mu'}\rho)$ 项, 问题不包含 φ , 可知 $m = 0$

则
$$u = J_0(\sqrt{\mu'}\rho) \begin{Bmatrix} \cos \nu z \\ \sin \nu z \end{Bmatrix} \begin{Bmatrix} \cos kat \\ \sin kat \end{Bmatrix}$$

代入边界条件 $u|_{z=0} = 0, u|_{z=L} = 0$ 可知, 解中不包含 $\cos \nu^2$ 项, 且 $\nu = \frac{\rho\pi}{L} (\rho = 1, 2, 3, \dots)$

代入初始条件 $u|_{t=0} = 0$, 可知解中不包含 $\cos kat$ 项.

代入边界条件 $u|_{\rho=\rho_0} = 0$. 即 $\sqrt{\mu'} J'_0(\sqrt{\mu'}\rho_0) = 0$, 由于 $J'_0(x) = -J_1(x)$.

有 $-\sqrt{\mu'} J_1(\sqrt{\mu'}\rho_0) = 0$. 则 $\sqrt{\mu'} = \frac{x_n^{(1)}}{\rho_0}$ ($x_n^{(1)}$ 为 $J_1(x)$ 的第 n 个零点, $n = 0, 1, 2, \dots$)

所以
$$u = \sum_{n=0}^{\infty} \sum_{\rho=1}^{\infty} A_{np} J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \sin \frac{\rho\pi z}{L} \sin \left(\sqrt{\left(\frac{x_n^{(1)}}{\rho_0}\right)^2 + \frac{\rho^2 \pi^2}{L^2}} at \right)$$

代入初始条件 $u_t|_{t=0} = u_0 \rho^2 z$, 得到

$$\sum_{n=0}^{\infty} \sum_{\rho=1}^{\infty} A_{np} J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \sin \frac{\rho\pi z}{L} \sqrt{\left(\frac{x_n^{(1)}}{\rho_0}\right)^2 + \frac{\rho^2 \pi^2}{L^2}} a = u_0 \rho^2 z$$

对右边进行广义傅里叶展开, 得到

$$A_{np} \sqrt{\left(\frac{x_n^{(1)}}{\rho_0}\right)^2 + \frac{\rho^2 \pi^2}{L^2}} a = \frac{u_0}{[N_n^{(0)}]^2} \int_0^{\rho_0} \rho^2 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho \cdot \frac{2}{L} \int_0^L z \sin \frac{\rho\pi z}{L} dz$$

其中, $J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right)$ 满足第二类齐次边界条件

$$[N_n^{(0)}] = \frac{1}{2} \rho_0^2 [J_0(x_n^{(1)})]^2$$

令 $x = \frac{x_n^{(1)}}{\rho_0} \rho$

$$\int_0^{\rho_0} \rho^2 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho = \left(\frac{\rho_0}{x_n^{(1)}}\right)^4 \int_0^{x_n^{(1)}} x^3 J_0(x) dx. \text{ 由贝塞尔函数递推公式}$$

$\frac{d}{dx}[x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x)$. 可知 $x J_0(x) = \frac{d}{dx}[x J_1(x)]$. 代入

$$\begin{aligned} \int_0^{x_n^{(1)}} x^3 J_0(x) dx &= \int_0^{x_n^{(1)}} x^2 \frac{d[x J_1(x)]}{dx} dx = \int_0^{x_n^{(1)}} x^3 J_1'(x) dx + \int_0^{x_n^{(1)}} x^2 J_1(x) dx \\ &= x^3 J_1(x) \Big|_0^{x_n^{(1)}} - 3 \int_0^{x_n^{(1)}} x^2 J_1(x) dx + \int_0^{x_n^{(1)}} x^2 J_1(x) dx \\ &= -2 \int_0^{x_n^{(1)}} x^2 J_1(x) dx = -2[x^2 J_2(x)] \Big|_0^{x_n^{(1)}} = -2(x_n^{(1)})^2 J_2(x_n^{(1)}) \end{aligned}$$

由递推关系 $J_{\rho+1}(x) - \frac{2\nu J_\nu(x)}{x} + J_{\nu-1}(x) = 0$ 得

$$J_2(x) - \frac{2J_1(x)}{x} + J_0(x) = 0.$$

当 $x = x_n^{(1)}$, 得 $J_2(x_n^{(1)}) = -J_0(x_n^{(1)})$ 有 $\int_0^{x_n^{(1)}} x^3 J_0(x) dx = 2(x_n^{(1)})^2 J_0(x_n^{(1)})$

所以
$$\int_0^{\rho_0} \rho^2 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho = \left(\frac{\rho_0}{x_n^{(1)}}\right)^4 2(x_n^{(1)})^2 J_0(x_n^{(1)}) = \frac{2\rho_0^4 J_0(x_n^{(1)})}{(x_n^{(1)})^2}$$

$$\frac{2}{L} \int_0^L z \sin \frac{\rho \pi z}{L} dz = \frac{2}{L} \left(-\frac{L}{\rho \pi} z \cos \frac{\rho \pi z}{L} \Big|_0^L + \frac{L}{\rho \pi} \int_0^L \cos \frac{\rho \pi z}{L} dz \right) = \frac{2L}{\rho \pi} (-1)^{\rho+1}$$

$$A_{\rho} \sqrt{\left(\frac{x_n^{(1)}}{\rho_0}\right)^2 + \frac{\rho^2 \pi^2}{L^2} a} = \frac{u_0}{\frac{1}{2} \rho_0^2 [J_0 x_n^{(1)}]^2} \cdot \frac{2 \rho_0^4 J_0(x_n^{(1)})}{(x_n^{(1)})^2} \cdot \frac{2L}{\rho \pi} (-1)^{\rho+1}$$

得到
$$A_{\rho} = \frac{(-1)^{\rho+1} 8 u_0 L \rho_0^2}{a \rho \pi (x_n^{(1)})^2 J_0(x_n^{(1)})} \cdot \frac{1}{\sqrt{(x_n^{(1)}/\rho_0)^2 + (\rho \pi/L)^2}}$$

$$u(\rho, z, t) = \sum_{n=0}^{\infty} \sum_{\rho=1}^{\infty} \frac{(-1)^{\rho+1} 8 u_0 L \rho_0^2}{a \rho \pi (x_n^{(1)})^2 J_0(x_n^{(1)})} \cdot \frac{1}{\sqrt{(x_n^{(1)}/\rho_0)^2 + (\rho \pi/L)^2}} \sin \sqrt{(x_n^{(1)}/\rho_0)^2 + (\rho \pi/L)^2} at \cdot \sin \frac{\rho \pi z}{L} \cdot J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right)$$

12. 半径为 ρ_0 的圆形膜, 边缘固定、初始位移、初始速度为零, 每单位质量上的作用力为 $f = A \sin \omega t$. 求解膜的振动情况.

解: $u_{tt} - a^2 u = f$ 取极坐标系, 振动与 φ 无关. 从而方程变为 $u_{tt} - a^2 \left(u_{\rho\rho} + \frac{1}{\rho} u_{\rho} \right) = f$

当 $f = A \sin \omega t$ 时, 令 $u = v(\rho) \sin \omega t$ 则
$$\begin{cases} v'' + \frac{1}{\rho} v' + \frac{\omega^2}{a^2} v = -\frac{A}{a^2} \\ v|_{\rho=R} = 0 \end{cases}$$

令 $w = v + \frac{A}{a^2}$ $x = \frac{w}{a} \rho$ 定解问题改写成
$$\begin{cases} w'' + \frac{1}{x} w' + w = 0 \\ w|_{x=\frac{w}{a} R} = \frac{A}{a^2} \end{cases}$$

$w = C J_0(x)$ 代入边界条件: $w = \frac{A}{a^2} \frac{1}{J_0\left(\frac{w}{a} R\right)} J_0(x)$

即 $v = \frac{A}{a^2} \left[\frac{J_0\left(\frac{\omega}{a}\right)}{J_0\left(\frac{\omega}{a} R\right)} - 1 \right] \therefore u = \frac{A}{a^2} \left[\frac{J_0\left(\frac{\omega}{a}\right)}{J_0\left(\frac{\omega}{a} R\right)} - 1 \right] \sin \omega t$

13. 匀质圆柱, 半径 ρ_0 、高 L , 试求解圆柱内部的热传导问题.

$$\begin{cases} u_t - a^2 \Delta u = 0, \\ u|_{\rho=\rho_0} = u_1 z^2, u|_{\rho=0} = \text{有限值}, \\ u|_{z=0} = 0, \frac{\partial u}{\partial z} \Big|_{z=L} = 2L u_1, \\ u|_{t=0} = u_0 \end{cases}$$

其中 u_0, u_1 为常数.

解: 令 $u(\rho, z, t) = v(\rho, z, t) + u_1 z^2$, 则 v 满足的定解问题为

$$\begin{cases} v_t - a^2 \Delta v = 2u_1 & (0 < \rho < \rho_0) \\ v|_{\rho=\rho_0} = 0 & v|_{\rho=0} = \text{有限值} \\ v|_{z=0} = 0 & v_z|_{z=L} = 0 \\ v|_{t=0} = u_0 - u_1 z^2 \end{cases}$$

在柱坐标下求解该问题, 可知其坐标部分解的形成成为

$$\begin{cases} J_m(\sqrt{\mu'}\rho) \\ N_m(\sqrt{\mu'}\rho) \end{cases} \begin{cases} (\cos \nu z) \\ (\sin \nu z) \end{cases} \begin{cases} \cos m\varphi \\ \sin m\varphi \end{cases}$$

($\mu' > 0$, 若 $\mu' = 0, \nu = 0, k = 0$ 不符合题意)

由 $\rho = 0$ 时, u 取有限值, 可知解中不包含 $N_m(\sqrt{\mu'}\rho)$ 项, 由问题不包含 φ 可知 $m = 0$. 则坐标

部分解化为 $J_0(\sqrt{\mu'}\rho) \begin{cases} \cos \nu z \\ \sin \nu z \end{cases}$

代入边界条件 $v|_{z=0} = 0, v_z|_{z=L} = 0$ 可知解中不包含 $\cos \nu z$ 项, 且 $\nu = \frac{(p + \frac{1}{2})\pi}{L}$ ($p = 0, 1, 2, \dots$)

代入边界条件 $v|_{\rho=\rho_0} = 0$, 即 $J_0(\sqrt{\mu'}\rho_0) = 0$, 则 $\sqrt{\mu'} = \frac{x_n^{(0)}}{\rho_0}$ ($x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点, $n = 1, 2, 3, \dots$)

设解的时间部分为 $T_p(t)$. 则

$$v(\rho, z, t) = \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} T_p(t) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L}$$

将 $v(\rho, z, t)$ 代回泛定方程得

$$\sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \left\{ T_p(t) + \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 T_p(t) \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L} = 2u_1$$

对右边进行广义傅里叶展开, 得到

$$T_p(t) + \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 T_p(t) = \frac{2u_1}{[N_n^{(0)}]^2} \int_0^{\rho_0} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho \cdot \frac{2}{L} \int_0^L \sin \frac{(p + \frac{1}{2})\pi z}{L} dz$$

$T_p(t)$ 满足一阶线性非齐次常微分方程, 其解由齐次通解 $T_p(t)$ 和非齐次特解 $T_p^*(t)$ 组成.

$$(1) T_p(t) = C_p \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} \quad (C_p \text{ 为待定常数})$$

(2) 对 $y' + Ay = B$ 形式的方程. 其特解为 $y^* = \frac{B}{A}(1 - e^{-At})$. 在本题中

$$A = \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2$$

$$\begin{aligned}
B &= \frac{2u_1}{[N_n^{(0)}]^2} \int_0^A J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho \cdot \frac{2}{L} \int_0^L \sin \frac{(p + \frac{1}{2})\pi z}{L} dz \\
&= \frac{2u_1}{\frac{1}{2}\rho_0^2 [J_1(x_n^{(0)})]^2} \cdot \frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) \cdot \frac{2}{(p + \frac{1}{2})\pi} \\
&= \frac{8u_1}{(p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \\
T_p^*(t) &= \frac{8u_1}{a^2 (p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \cdot \frac{1}{\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2}} \cdot \\
&\quad \left[1 - \exp \cdot \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} \right]
\end{aligned}$$

(3) 将 $v(\rho, z, t)$ 代入初始条件得

$$\sum_{n=1}^{\infty} \sum_{p=0}^{\infty} T_p(0) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L} = u_0 - u_1 z^2$$

$$\begin{aligned}
\text{则 } T_p(0) &= \frac{1}{[N_n^{(0)}]^2} \int_0^A J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho \cdot \frac{2}{L} \int_0^L (u_0 - u_1 z^2) \sin \frac{(p + \frac{1}{2})\pi z}{L} dz \\
&= \frac{1}{\frac{1}{2}\rho_0^2 [J_1(x_n^{(0)})]^2} \cdot \frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) \cdot \frac{2}{(p + \frac{1}{2})\pi} \cdot \left[u_0 + (-1)^{p+1} \frac{2u_1 L}{(p + \frac{1}{2})\pi} + \frac{2u_1 L}{(p + \frac{1}{2})^2 \pi^2} \right] \\
&= \frac{4}{(p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L}{(p + \frac{1}{2})\pi} + \frac{2u_1 L}{(p + \frac{1}{2})^2 \pi^2} \right]
\end{aligned}$$

而 $T_p(0) = T_{pc}(0) + T_p^*(0) = C_p$

$$\text{所以 } C_p = \frac{4}{(p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L}{(p + \frac{1}{2})\pi} + \frac{2u_1 L}{(p + \frac{1}{2})^2 \pi^2} \right]$$

$$\begin{aligned}
\therefore v(\rho, z, t) &= \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} T_p(t) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L} \\
&= \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} T_{pc}(t) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L} + \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} T_p^*(t) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2})\pi z}{L} \\
&= \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{4}{(p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L}{(p + \frac{1}{2})\pi} + \frac{2u_1 L}{(p + \frac{1}{2})^2 \pi^2} \right] e^{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t}
\end{aligned}$$

$$\begin{aligned}
& + \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{8u_1}{a^2(p+\frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \cdot \frac{1}{\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}} \left\{ 1 - e^{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}\right] a^2 t} \right\} \\
& J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \cdot \sin \frac{(p+\frac{1}{2})}{L} \pi z. \\
& \text{所以 } u(\rho, z, t) = u_1 z^2 + \\
& \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{4}{(p+\frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L}{(p+\frac{1}{2})\pi} + \frac{2u_1 L}{(p+\frac{1}{2})^2 \pi^2} \right] e^{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}\right] a^2 t} \cdot \\
& J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p+\frac{1}{2})\pi z}{L} + \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{8u_1}{a^2(p+\frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \cdot \frac{1}{\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}} \left\{ 1 - \right. \\
& \left. e^{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}\right] a^2 t} \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p+\frac{1}{2})\pi z}{L}
\end{aligned}$$

14. 环形膜, 内、外半径分别是 ρ_1 和 ρ_2 , 内外边缘都保持固定, 试求它的本征振动和本征频率.

解: 写出定解问题
$$\begin{cases} u_{rr} - a^2 \Delta u = 0 & (0 \leq \varphi \leq 2\pi, \rho_1 < \rho < \rho_2) \\ u|_{\rho=\rho_1} = u|_{\rho=\rho_2} = 0 \end{cases}$$

采用分离变量法 设 $u = \nu(\rho)\phi(\varphi)T(t)$ 代入得

$$\phi'' + m^2 \phi = 0 \quad \nu'' + \frac{1}{\rho} \nu' + \left(\lambda^2 - \frac{m^2}{\rho^2}\right) \nu = 0 \quad T'' + \lambda^2 a^2 T = 0 \quad u|_{\rho=\rho_1} = \nu|_{\rho=\rho_1} = 0$$

从而有 $\phi = A_1 \cos m\varphi + A_2 \sin m\varphi$

$$\nu = B_1 J_m(\lambda\rho) \quad J_m(\lambda\rho_1) = 0 \quad J_m(\lambda\rho_2) = 0$$

$$\therefore \nu_n = B_n J_m\left(\frac{x_n^{(m)}}{\rho_2} \rho\right) \quad x_n^{(m)} \text{ 为 } J_m(\lambda\rho) \text{ 的第 } n \text{ 个零点}$$

$$\text{关于 } T, T_{n,m} = C \cos \frac{x_n^{(m)}}{\rho_2} at + D \sin \frac{x_n^{(m)}}{\rho_2} at$$

在此基础上去掉 $\rho = \rho_1$ 那部分的振动情况

$$\begin{aligned}
u = & \left[A_{n,m} \cos \frac{x_n^{(m)}}{\rho_2} at + B_{n,m} \sin \frac{x_n^{(m)}}{\rho_2} at \right] J_m\left(\frac{x_n^{(m)}}{\rho_2} \rho\right) [\cos m\varphi + \sin m\varphi] \\
& - \left[A_{n,m}^1 \cos \frac{x_n^{(m)}}{\rho_1} at + B_{n,m}^1 \sin \frac{x_n^{(m)}}{\rho_1} at \right] J_m\left(\frac{x_n^{(m)}}{\rho_1} \rho\right) [\cos m\varphi + \sin m\varphi]
\end{aligned}$$

15. 半径为 ρ_0 的长圆柱面, 其径向速度分布为

$$v = v_0 \cos \varphi \cos \omega t.$$

试求解这个长圆柱面在空气中辐射出去的声场中的速度势. 设 ρ_0 远小于声波波长.

解: 定解问题
$$\begin{cases} u_{rr} - a^2 \Delta u = 0 & (1) \\ u_r|_{\rho=\rho_0} = v_0 \cos \varphi e^{-i\omega t} & (2) \end{cases}$$

由(2)式得 $m = 1$ $\therefore u = A_1 \cos \varphi H_1^{(1)}\left(\frac{\omega}{a}\rho\right)e^{-i\omega t} + A_0 H_0^{(1)}\left(\frac{\omega}{a}\rho\right)e^{-i\omega t}$ 代入边界条件有

$$A_1 \cos \varphi \frac{\partial}{\partial \rho} \left[H_1^{(1)}\left(\frac{\omega}{a}\rho\right) \right] = \nu_0 \cos \varphi \quad \text{则得到} \quad A_1 \frac{\partial}{\partial \rho} \left[H_0^{(1)}\left(\frac{\omega}{a}\rho\right) \right] = \nu_0 \quad A_0 = 0$$

$$\text{由于 } \rho_0 < \lambda = \frac{2\pi}{k} = 2\pi \frac{a}{\omega}$$

$$\text{即 } A \left[\frac{\partial}{\partial \rho} \left(1 + i \frac{2}{\pi} \ln \frac{\omega \rho}{2a} + ic \right) \right]_{\rho=\rho_0} = \nu_0 \quad \text{即 } iA \frac{2}{\pi \rho_0} = \nu_0$$

$$\text{得 } A = -\frac{i}{2} \pi \nu_0 \rho_0 \quad u = -i \frac{\pi \nu_0 \rho_0}{2} \cos \varphi H_1^{(1)}\left(\frac{\omega}{a}\rho\right)e^{-i\omega t} \quad \text{结果取 } u \text{ 的实部}$$

16. 半径为 ρ_0 的长圆柱面上一条母线作谐振动, 即柱面径向速度为 $v = v_0 \delta(\varphi - \varphi_0) \cos \omega t$. 试求解这个长圆柱在空气中辐射出去的声场中的速度势, 设 $\rho_0 \ll \lambda$ (声波波长).

$$\text{解: 由于是长圆柱} \quad \therefore \text{本题与 } z \text{ 无关, 度势的定解问题是} \begin{cases} u_{xx} - a^2 \Delta_2 u = 0 \\ \frac{\partial u}{\partial \rho} \Big|_{\rho=\rho_0} = v_0 \delta(\varphi - \varphi_0) \cos \omega t \\ \phantom{\frac{\partial u}{\partial \rho} \Big|_{\rho=\rho_0}} = v_0 \delta(\varphi - \varphi_0) e^{-i\omega t} \end{cases}$$

$$\text{由分离变量得解的形式为 } u_{(r, \varphi, t)} = \sum_{m=0}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \cdot H_m^{(1)}(k\rho) e^{-i\omega t}$$

$$\text{代入边界条件得} \quad A_0 = -i \frac{v_0 \rho_0}{4} \quad B_0 = 0 \quad A_m = -i \frac{v_0 \rho_0^{m+1} k^m}{2^m m!} \cos m\varphi_0$$

$$B_m = -i \frac{v_0 \rho_0^{m+1} k^m}{2^m m!} \sin m\varphi_0 \quad (m \neq 0)$$

将上述参数代入 u 的表达式最终取实部, 即得答案为

$$\text{Re} \left\{ \left[\frac{v_0 \rho_0}{4} H_0^{(1)}(k\rho) + \sum_{m=1}^{\infty} H_m^{(1)}(k\rho) \times \frac{v_0 k^m \rho_0^{m+1}}{2^m m!} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi) \right] e^{-i(\omega t + \frac{\pi}{4})} \right\}$$

§ 11.4 虚宗量贝塞尔方程

1. 试证明虚宗量贝塞尔函数 $I_\nu(x)$ 下列递推关系

$$(1) \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] = \frac{I_{\nu+1}(x)}{x^\nu}; \quad \frac{d}{dx} [x^\nu I_\nu(x)] = x^\nu I_{\nu-1}(x).$$

$$\text{解: } I_\nu(x) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(\frac{x}{2} \right)^{\nu+2k}$$

$$(1) \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] = \frac{d}{dx} \left[\sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{2k} \right] = \sum_{k=1}^{\infty} \frac{2k}{k! \Gamma(\nu + k + 1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{2k-1}$$

令 $k = l+1$, 则

$$\begin{aligned} \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] &= \sum_{l=0}^{\infty} \frac{2(l+1)}{(l+1)! \Gamma(\nu + l + 1 + 1)} \left(\frac{1}{2} \right)^{\nu+2l+2} x^{2l+1} \\ &= \frac{1}{x^\nu} \sum_{l=0}^{\infty} \frac{1}{l! \Gamma(\nu + 1 + l + 1)} \left(\frac{1}{2} \right)^{\nu+1+2l} x^{\nu+1+2l} \\ &= \frac{I_{\nu+1}(x)}{x^\nu} \end{aligned}$$

$$\frac{d}{dx}[x^\nu I_\nu(x)] = \frac{d}{dx}\left[\sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2}\right)^{\nu+2k} x^{2k+\nu}\right] = \sum_{k=0}^{\infty} \frac{2(k+\nu)}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2}\right)^{\nu+2k} x^{2k+2\nu-1}$$

由 Γ 函数性质 $\Gamma(x) = \frac{1}{x} \Gamma(x+1)$, 则

$$\begin{aligned} \frac{d}{dx}[x^\nu I_\nu(x)] &= \sum_{k=0}^{\infty} \frac{2}{k! [\Gamma(\nu+k+1)/(k+\nu)]} \left(\frac{1}{2}\right)^{\nu+2k} x^{2k+2\nu-1} \\ &= x^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu-1+k+1)} \left(\frac{1}{2}\right)^{\nu-1+2k} x^{\nu-1+2k} \\ &= x^\nu I_{\nu-1}(x) \end{aligned}$$

$$(2) I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu I_\nu(x)}{x}; I_{\nu-1}(x) + I_{\nu+1}(x) = 2I'_\nu(x)$$

解: 将上两式左边展开, 即得

$$I'_\nu - \frac{\nu I_\nu(x)}{x} = I_{\nu+1}(x) \quad (1)$$

$$I'_\nu - \frac{\nu I_\nu(x)}{x} = I_{\nu-1}(x) \quad (2)$$

则 ② - ① 得

$$I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu I_\nu(x)}{x}$$

② + ① 得

$$I_{\nu-1}(x) + I_{\nu+1}(x) = 2I'_\nu(x)$$

2. 试证明虚宗量汉克函数 $K_\nu(x)$ 下列递推关系

$$(1) \frac{d}{dx}\left[\frac{K_\nu(x)}{x^\nu}\right] = -\frac{K_{\nu+1}(x)}{x^\nu}; \frac{d}{dx}[x^\nu K_\nu(x)] = x^\nu K_{\nu-1}(x)$$

$$\text{解: } K_\nu(x) = \frac{\pi}{2} \cdot \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu \pi}$$

$$(1) \text{ 同第 1 题推导易得 } \frac{d}{dx}\left[\frac{I_\nu(x)}{x^\nu}\right] = \frac{I_{\nu-1}(x)}{x^\nu}$$

$$\begin{aligned} \frac{d}{dx}\left[\frac{K_\nu(x)}{x^\nu}\right] &= \frac{d}{dx}\left[\frac{\pi}{2} \cdot \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu \pi \cdot x^\nu}\right] = \frac{\pi}{2 \sin \nu \pi} \frac{d}{dx}\left[\frac{I_{-\nu}(x) - I_\nu(x)}{x^\nu}\right] \\ &= \frac{\pi}{2 \sin \nu \pi} \left[\frac{I_{-\nu-1}(x)}{x^\nu} - \frac{I_{\nu+1}(x)}{x^\nu}\right] \\ &= \frac{1}{x^\nu} \left[\frac{\pi}{2} \cdot \frac{I_{-\nu-1}(x) - I_{\nu+1}(x)}{\sin(\nu+1)\pi}\right] \end{aligned}$$

$$\text{由于 } \sin(\nu+1)\pi = \sin \nu \pi \cos \pi + \sin \pi \cos \nu \pi = -\sin \nu \pi$$

$$\text{得 } \frac{d}{dx}\left[\frac{K_\nu(x)}{x^\nu}\right] = -\frac{1}{x^\nu} \left[\frac{\pi}{2} \cdot \frac{I_{-\nu-1}(x) - I_{\nu+1}(x)}{\sin(\nu+1)\pi}\right] = -\frac{K_{\nu+1}(x)}{x^\nu}$$

$$\text{同第 1 题推导易得 } \frac{d}{dx}[x^\nu I_\nu(x)] = x^\nu I_{\nu+1}(x)$$

$$\frac{d}{dx}[x^\nu K_\nu(x)] = \frac{d}{dx}\left[\frac{\pi}{2} \cdot \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu \pi} \cdot x^\nu\right]$$

$$\begin{aligned}
&= \frac{\pi}{2\sin\nu\pi} \frac{d}{dx} \{ [L_\nu(x) - L_{\nu-1}(x)] x^\nu \} \\
&= \frac{\pi}{2\sin\nu\pi} [x^\nu L_{\nu+1}(x) - x^\nu L_{\nu-1}(x)] \\
&= x^\nu \left[\frac{\pi}{2} \cdot \frac{L_{\nu+1}(x) - L_{\nu-1}(x)}{\sin\nu\pi} \right] \\
&= -x^\nu \left[\frac{\pi}{2} \cdot \frac{L_{\nu+1}(x) - L_{\nu-1}(x)}{\sin(\nu+1)\pi} \right] \\
&= -x^\nu K_{\nu-1}(x)
\end{aligned}$$

$$(2) K_{\nu-1}(x) - K_{\nu+1}(x) = -\frac{2\nu K_\nu(x)}{x}; K_{\nu-1}(x) + K_{\nu+1}(x) = -2K'_\nu(x)$$

解:将上两式左边展开,即得

$$K_{\nu-1}(x) + \frac{\nu K_\nu(x)}{x} = -\nu K'_\nu(x) \quad (1)$$

$$K_{\nu+1}(x) - \frac{\nu K_\nu(x)}{x} = -\nu K'_\nu(x) \quad (2)$$

则 ① - ② 得

$$K_{\nu-1}(x) - K_{\nu+1}(x) = \frac{-2\nu K_\nu(x)}{x}$$

① + ② 得

$$K_{\nu-1}(x) + K_{\nu+1}(x) = -2\nu K'_\nu(x)$$

3. 匀质圆柱半径为 ρ_0 、高为 L . 下底保持温度 u_1 , 上底保持温度 u_2 . 侧面温度分布为 $f(z) = (2u_2/L^2)(z - L/2) + (u_1/L)(L - z)$. 求解柱体内各点的稳恒温度.

解:写出定解问题

$$\begin{cases} \Delta u = 0 \\ u|_{\rho=\rho_0} = \frac{2u_2}{L^2} \left(z - \frac{1}{2}L \right) + \frac{u_1}{L} (L - z) \\ u|_{\rho=0} \text{ 有限} \\ u|_{z=0} = u_1, u|_{z=L} = u_2 \end{cases}$$

把 u 写成 $u = v + \omega$, 可以将定解问题改写

$$(1) \quad \begin{cases} \Delta v = 0 \\ v|_{\rho=\rho_0} = \frac{2u_2}{L^2} \left(z - \frac{1}{2}L \right) + \frac{u_1}{L} (L - z) \\ v|_{\rho=0} \text{ 有限} \\ v|_{z=0} = 0 \quad v|_{z=L} = 0 \end{cases}$$

$$(2) \quad \begin{cases} \Delta \omega = 0 \\ \omega|_{\rho=\rho_0} = 0 \\ \omega|_{\rho=0} \text{ 有限} \\ \omega|_{z=0} = u_1, \omega|_{z=L} = u_2 \end{cases}$$

其中(1)的解可以参照课本 P₃₅₈ 进行, (2)可参照教材 § 11.2 中的例 4.

先解(2),因柱测是齐次边界条件,查表得结果与 m 无关,即 $m = 0$

可以得到解的形式为: $J_0(\sqrt{\mu}\rho) \begin{Bmatrix} e^{\sqrt{\mu}z} \\ e^{-\sqrt{\mu}z} \end{Bmatrix}$

柱侧为第一类齐次边界条件,满足 $R(\rho_0) = 0$ 即 $J_0(\sqrt{\mu}\rho_0) = 0$

因此我们得到本征值 $\mu_n^{(0)} = (x_n^{(0)}/\rho_0)^2$ 其中 $x_n^{(0)}$ 是 $J_0(x)$ 的第 n 个零点.

以上为 $\mu > 0$ 的情况.

参数 $\mu = 0$,在 $m = 0$ 时,查得 $Z(z) = \begin{Bmatrix} 1 \\ Z \end{Bmatrix}$

把解叠加可以得到 $\omega(\rho, z) = A_0 + B_0 z + \sum_{n=1}^{\infty} (A_n e^{zx_n^{(0)}/\rho_0} + B_n e^{-zx_n^{(0)}/\rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)$

由边界条件 $\omega|_{z=0} = u_1$ 和 $\omega|_{z=L} = u_2$ 可确定系数

$$\text{有} \begin{cases} A_0 + \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = u_1 \\ A_0 + B_0 L + \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} L/\rho_0} + B_n e^{-x_n^{(0)} L/\rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = u_2 \end{cases}$$

然后再求解(1),可使用特解法,得出 v 后,与 ω 的解组合可得到 u 的解.

4. 匀质圆柱半径为 ρ_0 、高为 L . 上底有均匀分布的强度为 q_0 的热流进入,下底保持温度 u_0 ,侧面温度分布为 $f(z)$,求解柱体内各点的稳恒温度.

$$\text{解:写出定解问题} \quad \begin{cases} \Delta u = 0 \\ u_z|_{z=L} = q_0 \\ u|_{z=0} = u_0 \\ u|_{\rho=\rho_0} = f(z) \quad u|_{\rho=0} = \text{有限值} \end{cases}$$

$$\text{将温标统一,规定 } u_0 \text{ 时为 } 0, \text{ 即有} \quad \begin{cases} \Delta u = 0 \\ u_z|_{z=L} = q_0 \\ u|_{z=0} = 0 \\ u|_{\rho=\rho_0} = f(z) - u_0 \end{cases}$$

将 u 分成两部分 $u = v + \omega$ 则问题转化为两个方程组

$$(1) \quad \begin{cases} \Delta v = 0 \\ v_z|_{z=L} = q_0 \\ v|_{z=0} = 0 \\ v|_{\rho=\rho_0} = f(z) \quad v|_{\rho=0} = \text{有限值} \end{cases}$$

$$(2) \quad \begin{cases} \Delta \omega = 0 \\ \omega_z|_{z=L} = 0 \\ \omega|_{\rho=0} = \text{有限值} \\ \omega|_{z=0} = 0 \\ \omega|_{\rho=\rho_0} = 0 \end{cases}$$

(1) 对于 $\mu < 0$, 问题与 φ 无关, 即 $m = 0$

解的形式如下: $I_0(\nu\rho) \begin{cases} \cos \nu z \\ \sin \nu z \end{cases}$

对于 $\mu = 0$, $R(\rho) = 1$ $Z(z) = \begin{cases} 1 \\ Z \end{cases}$

将两者组成 v 的解 $v = A_0 + B_0 z + \sum_{p=1}^{\infty} [A_p I_0(\nu\rho) \sin \nu z + B_p I_0(\nu\rho) \cos \nu z]$

由边界条件 $v|_{z=0} = 0$ 得 $A_0 + \sum_{p=1}^{\infty} B_p I_0(\nu\rho) = 0$

有 $B_p = 0, A_0 = 0$

由 $v_z|_{z=L} = 0$ 得 $B_0 + \sum A_p I_0(\nu\rho) \nu \cos \nu z \Big|_{z=L} = 0$ $B_0 = 0$

得 $\cos \nu L = 0$ 即 $\nu = \frac{(2p+1)\pi}{2L}$, 得 $v = \sum_{p=1}^{\infty} A_p I_0 \left[\frac{(2p+1)\pi}{2L} \rho \right] \sin \frac{2p+1}{2L} \pi z$

再利用 $v|_{\rho=\rho_0} = f(z)$ 即可确定 A_p

(2) 参见教材 § 11.2 中例 4 的解法, 对于 $\mu > 0$ 查得解为 $J_m(\sqrt{\mu}\rho) \begin{cases} e^{\sqrt{\mu}z} \\ e^{-\sqrt{\mu}z} \end{cases} \begin{cases} \cos m\varphi \\ \sin m\varphi \end{cases}$

边界条件均与 φ 无关, 有 $m = 0$, 于是解可写成 $J_0(\sqrt{\mu}\rho) \begin{cases} e^{\sqrt{\mu}z} \\ e^{-\sqrt{\mu}z} \end{cases}$

所以 $\omega = \sum J_0(\sqrt{\mu}\rho) (Ae^{\sqrt{\mu}z} + Be^{-\sqrt{\mu}z}) + A_0 + B_0 z$

由 $\rho = \rho_0$ 时, $\omega = 0$ 得 $\mu_n^{(0)} = \left(\frac{x_n^{(0)}}{\rho_0} \right)^2$ $x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点.

得 $\omega = \sum_{n=1}^{\infty} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) (A_n e^{\frac{x_n^{(0)}}{\rho_0} z} + B_n e^{-\frac{x_n^{(0)}}{\rho_0} z}) + A_0 + B_0 z$

由 $\omega|_{z=0} = 0$ 得 $A_0 + \sum_{n=1}^{\infty} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) (A_n + B_n) = 0$

由 $\omega_z|_{z=L} = 0$ 得 $B_0 + \sum_{n=1}^{\infty} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \left(A_n \frac{x_n^{(0)}}{\rho_0} e^{\frac{x_n^{(0)}}{\rho_0} L} - B_n \frac{x_n^{(0)}}{\rho_0} e^{-\frac{x_n^{(0)}}{\rho_0} L} \right) = 0$

由此可以确定各项系数.

将 v, ω 加起来得到 u 的表达式.

5. 如图 11-4-1 电子光学透镜的某一部件由两个中空圆柱筒组成, 其电势分别为 $+v_0$ 和 $-v_0$. 在圆柱中间隙缝的边缘处电势可近似表为 $v = v_0 \sin \frac{\pi z}{2\delta}$. 求圆柱筒内的电势分布. 圆柱两

端边界条件可近似表为 $v|_{z=\pm l} = \pm v_0$. 圆柱筒的半径为 ρ_0 .

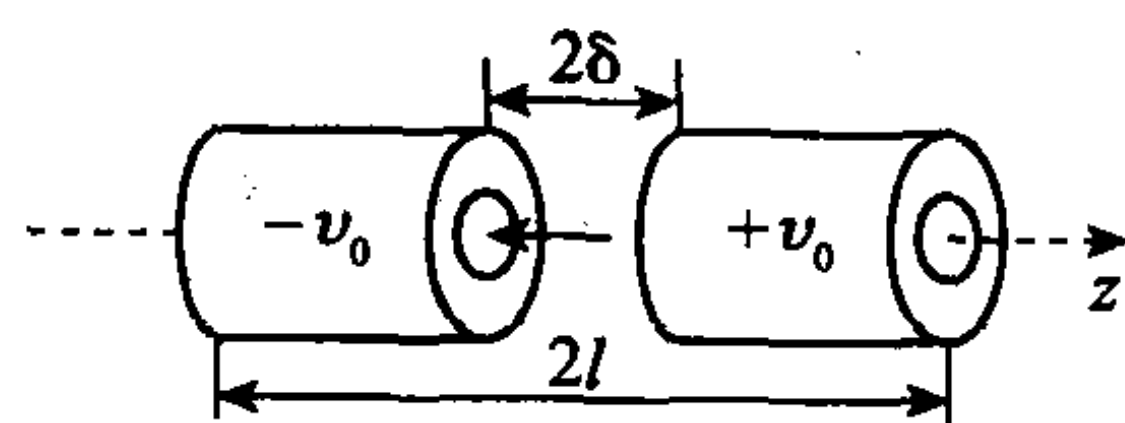


图 11-4-1

解:定理问题为

$$\begin{cases} \Delta u = 0 \\ u|_{z=0} = v_0 \sin \frac{\pi z}{2\delta} \Big|_{z=0} = 0 \\ u|_{z=L} = v_0 \\ u|_{\rho=R} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta} & (0 \leq z \leq \delta) \\ v_0 & (\delta \leq z \leq l) \end{cases} \end{cases}$$

将 u 写成 $u = v + \omega$, 有

$$(1) \quad \begin{cases} \Delta v = 0 \\ v|_{z=0} = 0 \\ v|_{z=l} = 0 \\ v|_{\rho=R} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta} & (0 \leq z \leq \delta) \\ v_0 & (\delta \leq z \leq l) \end{cases} \end{cases}$$

$$(2) \quad \begin{cases} \Delta \omega = 0 \\ \omega|_{z=0} = 0 \\ \omega|_{z=L} = v_0 \\ \omega|_{\rho=R} = 0 \end{cases}$$

解(1):问题与 φ 无关. 由第一类边界条件得 $Z_n = C_n \sin \frac{n\pi}{l} z$

$$\text{由 } v|_{\rho=R} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta} & (0 \leq z \leq \delta) \\ v_0 & (\delta \leq z \leq l) \end{cases} \text{ 得 } \sum_{n=1}^{\infty} C_n I_0\left(\frac{n\pi R}{l}\right) \sin \frac{n\pi}{l} z = \begin{cases} v_0 \sin \frac{\pi z}{2\delta} \\ v_0 \end{cases}$$

$$\text{由此可确定 } C_n \text{ 的值 } v = \frac{2v_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left[(-1)^{n+1} + \frac{l^2}{l^2 - 4\delta^2 n^2} \cos \frac{n\pi\delta}{l} \right] \frac{I_0\left(\frac{n\pi}{l}\rho\right)}{I_0\left(\frac{n\pi}{l}R\right)} \sin \frac{n\pi}{l} z$$

解(2):柱侧为齐次边界条件,与 φ 无关,有 $m=0$.

由表知本征值为 $\lambda_n^{(0)} = \frac{x_n^{(0)}}{k} (n=1,2,\dots)$ $x_n^{(0)}$ 是 $J_0(x)$ 的第 n 个零点

$$\text{解为 } \omega = \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{R} z} + B_n e^{-\frac{x_n^{(0)}}{R} z}) J_0\left(\frac{x_n^{(0)}}{R} \rho\right)$$

$$\text{由 } \omega|_{z=0} = 0 \text{ 得 } \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = 0 \Rightarrow A_n = -B_n$$

$$\text{由 } \omega|_{z=L} = v_0 \text{ 得 } \sum_{n=1}^{\infty} (A_n e^{\frac{x_n^{(0)}}{R} L} + B_n e^{-\frac{x_n^{(0)}}{R} L}) J_0\left(\frac{x_n^{(0)}}{R} \rho\right) = v_0$$

由此可求得 A_n, B_n .

将 v, ω 相加,即可写成 u 的表达式,直接给出结果

$$\omega = 2\nu_0 \sum_{n=1}^{\infty} \frac{J_0\left(\frac{x_n^{(0)}}{R}\rho\right)}{x_n^{(0)} J_1(x_n^{(0)})} \frac{\operatorname{sh}\left(\frac{x_n^{(0)}}{R}z\right)}{\operatorname{sh}\left(\frac{x_n^{(0)}}{R}l\right)}$$

$u = v + \omega$ 可以求得结果

6. 利用虚宗量贝塞尔函数重解 § 11.2 的习题 6.

解: 由题意, 得定解问题为

$$\begin{cases} \Delta u = 0 & (0 \leq \rho \leq \rho_0) \\ u|_{\rho=\rho_0} = 0 & u|_{\rho=0} = \text{有限值} \\ ku_z|_{z=0} = q_0 & ku_z|_{z=L} = q_0 \end{cases}$$

将边界条件化为齐次边界条件, 令 $u = v + \frac{q_0}{k}z$, 则

$$\begin{cases} \Delta v = 0 \\ v|_{\rho=\rho_0} = -\frac{q_0}{k}z & v|_{\rho=0} \text{ 有限值} \\ kv_z|_{z=0} = 0 & kv_z|_{z=L} = 0 \end{cases}$$

由于问题研究的是圆柱区域内, 所以对应 $\mu \leq 0$, 当 $\mu = 0$ 时, 对应解为 $\frac{q_0}{R}z$

$$\text{当 } \mu < 0 \text{ 时, } v(\rho, \varphi, z) = \begin{Bmatrix} I_m(\nu\rho) \\ K_m(\nu\rho) \end{Bmatrix} \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} \cos \nu z \\ \sin \nu z \end{Bmatrix}$$

由问题的对称性可知 $m = 0$. 由 $v|_{\rho=0}$ 为有限值可知应舍去 $K_m(\nu\rho)$, 则

$$v(\rho, \varphi, z) = I_0(\nu\rho) \begin{Bmatrix} \cos \nu z \\ \sin \nu z \end{Bmatrix}$$

由 z 的边界条件可知 $\nu = \frac{n\pi}{L}$ 且 $\sin \nu z$ 系数为 0, 则

$$v(\rho, \varphi, z) = \sum_{n=0}^{\infty} C_n I_0\left(\frac{n\pi\rho}{L}\right) \cos \frac{n\pi z}{L}$$

将上式代入 $\rho = \rho_0$ 边界条件, 则

$$\sum_{n=0}^{\infty} C_n I_0\left(\frac{n\pi\rho_0}{L}\right) \cos \frac{n\pi z}{L} = -\frac{q_0}{k}z$$

将等号右边用 $\cos \frac{n\pi z}{L}$ 展开, 比较两边系数可知

$$\begin{aligned} C_0 &= \frac{1}{L} \int_0^L \left(-\frac{q_0}{k}z\right) dz = -\frac{q_0 L}{2k} \\ C_n &= \frac{\frac{2}{L} \int_0^L \left(-\frac{q_0}{k}z\right) \cos \frac{n\pi z}{L} dz}{I_0\left(\frac{n\pi\rho_0}{L}\right)} = -\frac{2q_0 L}{kn^2 \pi^2 I_0(n\pi\rho_0/L)} [(-1)^n - 1] \end{aligned}$$

当 n 为偶数时, $C_n = 0$. 将 $v(\rho, \varphi, z)$ 可写为

$$v(\rho, \varphi, z) = -\frac{q_0 L}{2k} + \sum_{n=1}^{\infty} \frac{4q_0 L}{k(2n-1)^2 \pi^2 I_0[(2n-1)\pi\rho_0/L]} I_0\left[\frac{(2n-1)\pi\rho}{L}\right] \cos \frac{(2n-1)\pi z}{L}$$

$$\therefore u(\rho, \varphi, z) = v(\rho, \varphi, z) + \frac{q_0}{k}z = \frac{q_0}{k}z - \frac{q_0 L}{2k} + \sum_{n=1}^{\infty} \frac{4q_0 L}{k(2n-1)^2 \pi^2 I_0[(2n-1)\pi\rho_0/L]} I_0\left[\frac{(2n-1)\pi}{L}\rho\right] \cos \frac{(2n-1)\pi z}{L}$$

7. 匀质圆柱, 半径为 ρ_0 , 高为 L . 下底保持温度 u_1 , 上底温度分布为 $u_2\rho^2$, 侧面温度分布为 $u_0 z$. 求解柱体内各点的稳恒温度.

解: 由题意得定解问题为

$$\begin{cases} \Delta u = 0 & (0 \leq \rho \leq \rho_0) \\ u|_{\rho=\rho_0} = u_0 z & u|_{\rho=0} = \text{有限值} \\ u|_{z=L} = u_2 \rho^2 & u|_{z=0} = u_1 \end{cases}$$

令 $u = v + w$ 且 v, w 满足如下条件

$$\begin{cases} \Delta v = 0 \\ v|_{\rho=\rho_0} = u_0 z & v|_{\rho=0} = \text{有限值} \\ v|_{z=L} = 0 & v|_{z=0} = 0 \end{cases} \quad \begin{cases} \Delta w = 0 \\ w|_{\rho=\rho_0} = 0 & w|_{\rho=0} = \text{有限值} \\ w|_{z=L} = u_2 \rho^2 & w|_{z=0} = u_1 \end{cases}$$

对 v 定解问题, 上下底为齐次边界条件, 对应于 $\mu \leq 0$ 部分, 由柱轴上的自然边界条件, 且问题与 φ 无关即 $m = 0$, 得到解为

$$I(\gamma^2 \rho) \begin{Bmatrix} \cos \gamma^2 z \\ \sin \gamma^2 z \end{Bmatrix}$$

由上下底齐次边界条件, 将 $\cos \gamma^2 z$ 舍去, 本征值 $\gamma^2 = \frac{p^2 \pi^2}{L^2}$ (p 为自然数), 则

$$v = \sum_{p=1}^{\infty} A_p I_0\left(\frac{p\pi}{L}\rho\right) \sin \frac{p\pi z}{L}$$

为确定系数 A_p 将上式代入柱侧边界条件

$$\sum_{p=1}^{\infty} A_p I_0\left(\frac{p\pi}{L}\rho\right) \sin \frac{p\pi z}{L} = u_0 z$$

将上式右端进行傅里叶展开, 然后比较两边系数得

$$A_p = \frac{1}{I_0\left(\frac{p\pi}{L}\rho_0\right)} \frac{z}{L} \int_0^L u_0 z \sin \frac{p\pi z}{L} dz$$

积分后可得

$$A_p = \frac{(-1)^{p+1} 2u_0 L}{p\pi I_0(p\pi\rho_0/L)}$$

$$\therefore v = \sum_{p=1}^{\infty} \frac{(-1)^{p+1} 2u_0 L}{p\pi I_0(p\pi\rho_0/L)} I_0\left(\frac{p\pi}{L}\rho\right) \sin \frac{p\pi z}{L}$$

对 w 的定解问题, 柱侧为齐次的第二类边界条件, 对应于 $\mu \geq 0$ 部分. 由柱轴上的自然边界条件, 且问题与 φ 无关. 即 $m = 0$, 得到解为

$$J(\sqrt{\mu}\rho) \begin{Bmatrix} e^{\sqrt{\mu}z} \\ \rho^{-\sqrt{\mu}z} \end{Bmatrix}$$

由柱侧齐次边界条件 $J_0(\sqrt{\mu}\rho_0) = 0$ 得到本征值 $\mu^{(n)} = (x_n^{(0)}/\rho_0)^2$, 其中 $x_n^{(0)}$ 是 $J_0(x)$ 的第 n 个

正根,则

$$\begin{cases} (A_n + B_n)J_0\left(\frac{x_n^{(0)}}{\rho_0}e\right) = u_1 \\ (A_n e^{\frac{x_n^{(0)}}{\rho_0}L} + B_n e^{-\frac{x_n^{(0)}}{\rho_0}L})J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) = u_2\rho^2 \end{cases}$$

解得

$$\begin{cases} A_n = \frac{Ae^{-\frac{x_n^{(0)}}{\rho_0}L} - B}{e^{-\frac{x_n^{(0)}}{\rho_0}L}/\rho_0 - e^{\frac{x_n^{(0)}}{\rho_0}L}/\rho_0} \\ B_n = \frac{Ae^{\frac{x_n^{(0)}}{\rho_0}L} - B}{e^{\frac{x_n^{(0)}}{\rho_0}L}/\rho_0 - e^{-\frac{x_n^{(0)}}{\rho_0}L}/\rho_0} \end{cases}$$

其中

$$A = \frac{2u_1}{x_n^{(0)}J_1(x_n^{(0)})}, B = \frac{2u_2\rho_0}{x_n^{(0)}J_1(x_n^{(0)})}\left(1 - \frac{4}{[x_n^{(0)}]^2}\right)$$

代入,将 $e^{\frac{x_n^{(0)}}{\rho_0}z/\rho_0}, e^{-\frac{x_n^{(0)}}{\rho_0}z/\rho_0}$ 化为双曲正弦函数得到

$$w = \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)}J_1(x_n^{(0)})\operatorname{sh}\frac{x_n^{(0)}}{\rho_0}L} \cdot \left\{ u_1 \operatorname{sh}\frac{x_n^{(0)}}{\rho_0}(L-z) + u_2\rho_0^2\left[1 - \frac{4}{(x_n^{(0)})^2}\right] \times \operatorname{sh}\frac{x_n^{(0)}}{\rho_0}z \right\} \cdot J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)$$

由 $u = v + w$ 得

$$\begin{aligned} u(\rho, z) = & \sum_{p=1}^{\infty} \frac{(-1)^{p+1}2u_0L}{p\pi I_0(p\pi\rho_0/L)} I_0\left(\frac{p\pi\rho}{L}\right) \sin\frac{p\pi z}{L} + \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)}J_1(x_n^{(0)})\operatorname{sh}\frac{x_n^{(0)}}{\rho_0}L} \\ & \times \left\{ u_1 \operatorname{sh}\frac{x_n^{(0)}}{\rho_0}(L-z) + u_2\rho_0^2\left[1 - \frac{4}{(x_n^{(0)})^2}\right] \operatorname{sh}\frac{x_n^{(0)}}{\rho_0}z \right\} \cdot J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \end{aligned}$$

8. 半径为 ρ_0 、高为 L 的圆柱,上底绝热,下底保持温度 u_0 ,侧面有均匀分布的强度为 q_0 的热流进入,求柱外匀质介质中各点的稳恒温度.

解:由题意得定解问题为

$$\begin{cases} \Delta u = 0 & (\rho \geq \rho_0) \\ ku_\rho|_{\rho=\rho_0} = q_0 & u|_{\rho \rightarrow \infty} = \text{有限值} \\ u_z|_{z=L} = 0 & u|_{z=0} = u_0 \end{cases}$$

将边界条件化为齐次的,令 $u = v + u_0$,则

$$\begin{cases} \Delta v = 0 & (\rho \geq \rho_0) \\ kv_\rho|_{\rho=\rho_0} = q_0 & v|_{\rho \rightarrow \infty} = \text{有限值} \\ v_z|_{z=L} = 0 & v|_{z=0} = 0 \end{cases}$$

由于研究区域为圆柱体,所以对应 $\mu \leq 0$,当 $\mu = 0$,对应解为 u_0 .当 $\mu < 0$,对应解为

$$v(\rho, \varphi, z) = \begin{Bmatrix} I_m(\nu\rho) \\ K_m(\nu\rho) \end{Bmatrix} \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} \cosh \nu z \\ \sinh \nu z \end{Bmatrix}$$

由问题的轴对称性可知 $m = 0$,由 $v|_{\rho \rightarrow \infty}$ 为有限值可知应舍去 $I_m(\nu\rho)$,则

$$v(\rho, \varphi, z) = K_0(\nu\rho) \begin{Bmatrix} \cosh \nu z \\ \sinh \nu z \end{Bmatrix}$$

由 z 的边界条件可知 $\nu = \frac{(n + \frac{1}{2})\pi}{L}$ 且 $\cos \nu z$ 系数为 0, 则

$$v(\rho, \varphi, z) = \sum_{n=0}^{\infty} C_n K_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho \right] \sin \frac{(n + \frac{1}{2})\pi z}{L}$$

将上式对 ρ 求导后代入 $u_\rho|_{\rho=\rho_0} = q_0/k$, 则

$$\sum_{n=0}^{\infty} C_n \cdot \frac{(n + \frac{1}{2})\pi}{L} \cdot K'_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho_0 \right] \sin \frac{(n + \frac{1}{2})\pi z}{L} = q_0/k$$

将等号右边用 $\sin \frac{(n + \frac{1}{2})\pi z}{L}$ 展开, 比较两边系数可知

$$C_n = \frac{\frac{2}{L} \int_0^L \frac{q_0}{k} \sin \frac{(n + \frac{1}{2})\pi z}{L} dz}{\frac{(n + \frac{1}{2})\pi}{L} K'_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho_0 \right]} = \frac{2q_0 L}{k(n + \frac{1}{2})^2 \pi^2 K'_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho_0 \right]}$$

$$\text{则 } v(\rho, \varphi, z) = \sum_{n=0}^{\infty} \frac{2q_0 L}{k(n + \frac{1}{2})^2 \pi^2 K'_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho_0 \right]} K_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho \right] \sin \frac{(n + \frac{1}{2})\pi z}{L}$$

$$\therefore u(\rho, \varphi, z) = v(\rho, \varphi, z) + u_0$$

$$= u_0 + \sum_{n=0}^{\infty} \frac{2q_0 L}{k(n + \frac{1}{2})^2 \pi^2 K'_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho_0 \right]} K_0 \left[\frac{(n + \frac{1}{2})\pi}{L} \rho \right] \sin \frac{(n + \frac{1}{2})\pi z}{L}$$

§ 11.5 球贝塞尔方程

1. 以 $\psi_l(x)$ 代表 $j_l(x), n_l(x), h_l^{(1)}(x), h_l^{(2)}(x)$ 中任一种, 证明递推关系

$$\textcircled{1} \psi_{l-1} + \psi_{l+1} = \frac{2l+1}{x} \psi_l$$

解: 用 $\psi_l(x)$ 代表 $j_l(x), n_l(x), h_l^{(1)}(x), h_l^{(2)}(x)$. 用 $Z_\nu(x)$ 代表 $J_\nu(x), N_\nu(x), H_\nu^{(1)}(x), H_\nu^{(2)}(x)$

(1) 由公式(11.1.13) 即 $Z_{\nu+1}(x) - 2\nu Z_\nu(x)/x + Z_{\nu-1}(x) = 0$

等式两边同乘以 $\sqrt{\frac{\pi}{2x}}$, 且令 $\nu = l + \frac{1}{2}$, 则

$$\sqrt{\frac{\pi}{2x}} Z_{(l+1)+\frac{1}{2}}(x) + \sqrt{\frac{\pi}{2x}} Z_{(l-1)+\frac{1}{2}}(x) = \frac{2(l + \frac{1}{2}) \sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}}(x)}{x}$$

由 $\sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}}(x) = \psi_l(x)$ 得

$$\psi_{l-1}(x) + \psi_{l+1}(x) = \frac{(2l+1)\psi_l(x)}{x}$$

$$\textcircled{2} l\psi_{l-1} - (l+1)\psi_{l+1} = (2l+1)\psi'_l.$$

$$\begin{aligned} \text{(2) 解: } (2l+1)\psi'_l &= (2l+1) \left[\sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}}(x) \right]' \\ &= (2l+1) \left[\left(-\frac{1}{2}\right) \sqrt{\frac{\pi}{2x^3}} Z_{l+\frac{1}{2}}(x) + \sqrt{\frac{\pi}{2x}} Z'_{l+\frac{1}{2}}(x) \right] \\ &= \sqrt{\frac{\pi}{2x}} \left[-\left(l+\frac{1}{2}\right) \frac{Z_{l+\frac{1}{2}}(x)}{x} + 2\left(l+\frac{1}{2}\right) Z'_{l+\frac{1}{2}}(x) \right] \end{aligned}$$

由公式(11.1.12)(11.1.13)可知

$$\frac{\left(l+\frac{1}{2}\right)Z_{l+\frac{1}{2}}(x)}{x} = \frac{1}{2} [Z_{l+1+\frac{1}{2}}(x) + Z_{l-1+\frac{1}{2}}(x)]$$

$$2Z'_{l+\frac{1}{2}}(x) = Z_{l-1+\frac{1}{2}}(x) - Z_{l+1+\frac{1}{2}}(x)$$

$$\begin{aligned} (2l+1)\psi'_l &= \sqrt{\frac{\pi}{2x}} \left[-\frac{1}{2} Z_{l+1+\frac{1}{2}}(x) - \frac{1}{2} Z_{l-1+\frac{1}{2}}(x) + \left(l+\frac{1}{2}\right) Z_{l-1+\frac{1}{2}}(x) - \left(l+\frac{1}{2}\right) Z_{l+1+\frac{1}{2}}(x) \right] \\ &= \sqrt{\frac{\pi}{2x}} [- (l+1) Z_{l+1+\frac{1}{2}}(x) + l Z_{l-1+\frac{1}{2}}(x)] \\ &= - (l+1) \sqrt{\frac{\pi}{2x}} Z_{l+1+\frac{1}{2}}(x) + l \sqrt{\frac{\pi}{2x}} Z_{l-1+\frac{1}{2}}(x) \\ &= l\psi_{l-1}(x) - (l+1)\psi_{l+1}(x) \end{aligned}$$

2. 证明

$$\textcircled{1} [x^3 j_{-1}(x) j_1(x)]' = x^3 j_0(x) [j_{-1}(x) - j_1(x)] = x^2 [x j_0^2(x)]';$$

$$\textcircled{2} \int x^2 j_0^2(x) dx = \frac{1}{2} x^3 [j_0^2(x) - j_{-1}(x) j_1(x)] + C.$$

$$\text{证明: } \textcircled{1} \text{ 已知 } j_0(x) = \frac{\sin x}{x} \quad j_{-1} = \frac{\cos x}{x} \quad j_1 = \frac{1}{x^2} (\sin x - x \cos x)$$

$$x^3 j_{-1}(x) j_1(x) = x^3 \frac{\cos x}{x} \frac{1}{x^2} (\sin x - x \cos x) = \sin x \cos x - x \cos^2 x$$

$$\text{由此 } [x^3 j_{-1}(x) j_1(x)]' = (\sin x \cos x - x \cos^2 x)' = \cos^2 x + \sin^2 x - \cos^2 x + 2x \cos x \sin x = -\sin^2 x + 2x \sin x \cos x$$

$$\text{而 } x^3 j_0(x) [j_{-1} - j_1(x)] = x^3 \frac{\sin x}{x} \left[\frac{\cos x}{x} - \frac{1}{x^2} (\sin x - x \cos x) \right]$$

$$= x \cos x \sin x - \sin^2 x + x \cos x \sin x = 2x \cos x \sin x - \sin^2 x$$

$$x^2 [x j_0^2(x)]' = x^2 \left[x \frac{\sin^2 x}{x^2} \right]' = x^2 \left(\frac{\sin^2 x}{x} \right)' = x^2 \frac{2 \sin x \cos x - \sin^2 x}{x^2} = 2x \sin x \cos x - \sin^2 x$$

$$\text{综上 } [x^3 j_{-1}(x) j_1(x)]' = x^3 j_0(x) [j_{-1}(x) - j_1(x)] = x^2 [x j_0^2(x)]'$$

$$\textcircled{2} \text{ 同理, 由 } j_0(x) = \frac{\sin x}{x} \quad j_{-1} = \frac{\cos x}{x} \quad j_1 = \frac{1}{x^2} (\sin x - x \cos x)$$

$$x^2 j_0^2(x) = \sin^2 x \quad \int x^2 j_0^2(x) dx = \int \sin^2 x dx = \int \frac{1 - \cos 2x}{2} dx$$

$$= \frac{1}{2}x - \frac{1}{4}\sin 2x + C = \frac{1}{2}x - \frac{1}{2}\sin x \cos x + C$$

$$\text{又 } \frac{1}{2}x^3 [j_0^2(x) - j_{-1}(x)j_1(x)] + C = \frac{x^3}{2} \left[\frac{\sin^2 x}{x^2} - \frac{\cos x}{x^3}(\sin x - x \cos x) \right] + C$$

$$= \frac{1}{2}(x \sin^2 x - \cos x \sin x + x \cos^2 x) + C = \frac{1}{2}(x - \cos x \sin x) + C \text{ 证毕}$$

3. 证明

$$\textcircled{1} [x^3 j_{l-1}(x) j_{l+1}(x)]' = x^3 j_l(x) [j_{l-1}(x) - j_{l+1}(x)] = x^2 [x j_l^2(x)]';$$

$$\textcircled{2} \int x^2 j_l^2(x) dx = \frac{1}{2} x^3 [j_l^2(x) - j_{l-1}(x) j_{l+1}(x)] + C.$$

[提示: 将(11.5.4)代入(11.1.8)、(11.1.9)得出跟 $j_l(x)$ 有关的求导公式, 利用之]

$$\text{证明: 由(11.5.4)式 } Z_l = \sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}} \quad \text{得 } Z_{l+1} = \frac{2l+1}{x} Z_l - Z_{l-1}$$

$$(11.1.8): [Z_l(x)/x^l]' = -Z_{l+1}/x^l$$

$$(11.1.9) [x^l Z_l]' = x^l Z_{l-1}$$

$$\text{由(11.1.8)得 } \left[\frac{Z_l}{x^{l+\frac{1}{2}}} \right]' = -\frac{Z_{l+1}}{x^{l+\frac{1}{2}}} \quad \text{即 } \left[\frac{\sqrt{\frac{2x}{\pi}} Z_l}{x^{l+\frac{1}{2}}} \right]' = -\frac{\sqrt{\frac{2x}{\pi}} Z_{l+1}}{x^{l+\frac{1}{2}}}$$

$$\text{将(11.5.4)代入(11.1.9)} \quad \left[\sqrt{\frac{2x}{\pi}} Z_l x^{l+\frac{1}{2}} \right]' = x^{l+\frac{1}{2}} \sqrt{\frac{2x}{\pi}} Z_{l-1}$$

4. 确定球形铀块的临界半径. [“临界”一词参看 §8.1 习题8]

$$\text{解: 采用分离变量法, 定解问题为 } \begin{cases} u_t - a^2 \Delta u = \beta u \\ u|_{r=R} = 0 \end{cases}$$

其中 a^2 为扩散系数, β 为增殖系数, R 为球半径.

$$u_{(r,t)} = v(r) T(t) \quad \text{代入方程有 } T' - KT = 0 \quad (1)$$

$$r^2 v'' + 2rv' + \frac{r^2}{a^2}(\beta - k)v = 0 \quad \text{为零阶球贝塞尔方程}$$

$$\text{解为 } v = j_0\left(\frac{\sqrt{\beta-k}r}{a}\right) \quad \text{代入边界条件 } j_0\left(\frac{\sqrt{\beta-k}R}{a}\right) = \frac{\sin \frac{\sqrt{\beta-k}R}{a}}{\frac{\sqrt{\beta-k}R}{a}} = 0$$

$$\text{由此得: } \sqrt{\beta-k} \cdot \frac{R}{a} = n\pi \quad (n=1, 2, \dots)$$

$$\text{即 } k_n = \beta - \frac{n^2 \pi^2}{R^2} a^2 \quad \text{由临界半径定义(取 } n=1 \text{) 得}$$

$$\text{临界半径 } R_k = \frac{\pi a}{\sqrt{\beta}}$$

5. 匀质球, 半径为 r_0 , 初始温度分布为 $f(r)$. 把球面温度保持为零度而使它冷却. 求解球内各处温度变化情况.

解:定解问题可写成

$$\begin{cases} u_t - a^2 \Delta u = 0 \\ u|_{r=r_0} = 0 \\ u|_{t=0} = f(r) \end{cases}$$

由题意得解的形式与 φ 和 θ 均无关. 形式为 $j_0(kr)e^{-k^2 a^2 t}$

由 $j_0(kr) = \frac{\sin kr}{kr}$ 代入得 $u(r, t) = \sum_{n=1}^{\infty} C_n \frac{\sin kr}{kr} e^{-k^2 a^2 t}$

由边界条件 $u|_{r=r_0} = 0$ 得 $\frac{\sin kr_0}{kr_0} = 0$ 从而有 $\sin kr_0 = 0$ 即 $k_n = \frac{n\pi}{r_0}$ ($n = 1, 2, \dots$)

得

$$\sum_{n=1}^{\infty} C_n \frac{\sin\left(\frac{n\pi}{r_0}r\right)}{\frac{n\pi}{r_0}r} e^{-\frac{a^2 n^2 \pi^2}{r_0^2}t}$$

再由 $u|_{t=0} = f(r)$ 得 $\sum_{n=1}^{\infty} C_n j_0\left(\frac{n\pi}{r_0}r\right) = f(r)$

C_n 可以被确定 有 $C_n = \frac{\int_0^{r_0} f(r) j_0\left(\frac{n\pi}{r_0}r\right) r^2 dr}{\int_0^{r_0} \left[j_0\left(\frac{n\pi}{r_0}r\right)\right]^2 r^2 dr}$

$$\int_0^{r_0} f(r) r^2 \frac{\sin\left(\frac{n\pi}{r_0}r\right)}{\frac{n\pi}{r_0}r} dr$$

将 j_0 表达式代入 $C_n = \frac{\int_0^{r_0} f(r) r^2 \frac{\sin\left(\frac{n\pi}{r_0}r\right)}{\frac{n\pi}{r_0}r} dr}{\int_0^{r_0} \frac{\sin^2\left(\frac{n\pi}{r_0}r\right)}{\left(\frac{n\pi}{r_0}r\right)^2 r^2 dr}$

得到 $C_n = \frac{2\pi n}{r_0^2} \int_0^{r_0} f(r) r \sin\left(\frac{n\pi r}{r_0}\right) dr$

代入表达式得 $u = \sum_{n=1}^{\infty} \frac{2\pi n}{r_0^2} j_0\left(\frac{n\pi r}{r_0}\right) e^{-\frac{a^2 n^2 \pi^2}{r_0^2}t} \int_0^{r_0} f(r) r \sin\left(\frac{n\pi r}{r_0}\right) dr$

6. 匀质球, 半径为 r_0 , 初始温度分布为 $f(r) \cos \theta$. 把球面温度保持为零度而使它冷却. 求解球内各处温度变化情况.

解: 写出定解问题

$$\begin{cases} u_t - a^2 \Delta u = 0 \\ u|_{t=0} = f(r) \cos \theta \\ u|_{r=r_0} = 0 \end{cases}$$

解的形式与 φ 无关. $\therefore m = 0$. 由 $u|_{t=0} = f(r) \cos \theta$ 得 $l = 1$

因此解的形式为 $P_1(\cos \theta) j_1(kr) e^{-a^2 k^2 t}$

由 $u|_{r=r_0} = 0$ 可得 $j_1(kr)|_{r=r_0} = \frac{\sin kr_0 - kr_0 \cos kr_0}{kr_0}$

即 $\tan kr_0 = kr_0$ 为超越方程. 假设第 n 个根为 $x_n = k_n r_0$

有 $k_n = \frac{x_n}{r_0}$ 代入解的表达式.

得 $u = \sum_{n=1}^{\infty} C_{1,2} P_1(\cos \theta) j_1(k_n r) e^{-a^2 k_n^2 t}$

由初始条件 $u|_{t=0} = f(r) \cos \theta$ 推知 $\sum_{n=1}^{\infty} C_{1,n} P_1(\cos \theta) j_1(k_n r) = f(r) \cos \theta$ 从而可确定 $C_{1,n}$

即
$$C_{1,n} = \frac{\int_0^{r_0} f(r) j_1(k_n r) r^2 dr}{\int_0^{r_0} j_1^2(k_n r) r^2 dr}$$

$$\begin{aligned} \text{计算 } \int_0^{r_0} [j_1(k_n r)]^2 r^2 dr &= \frac{\pi}{2k_n} \int_0^{r_0} [J_+(k_n r)]^2 r dr = \frac{\pi}{2k_n} \frac{1}{2} r_0^2 [J_+(k_n r_0)]^2 \\ &= \frac{\pi}{2k_n} \frac{r_0}{2} \left[J_+(k_n r_0) - \frac{3}{2} \frac{J_+(k_n r_0)}{k_n r_0} \right]^2 = \frac{\pi}{2k_n} \frac{r_0^2}{2} [J_2(k_n r_0)]^2 \\ &= \frac{1}{2} r_0^3 \left[\sqrt{\frac{\pi}{2k_0 r_0}} J_+(k_n r_0) \right]^2 = \frac{r_0^3}{2} [j_0(k_n r_0)]^2 \end{aligned}$$

解可完全确定为 $u = \frac{2}{r_0^3} \sum_{n=1}^{\infty} \frac{1}{[j_0(k_n r_0)]^2} j_1(k_n r) P_1(\cos \theta) e^{-a^2 k_n^2 t} \int_0^{r_0} f(r) r^1 \times j_1(k_n r) dr$

7. 半径为 $2r_0$ 的匀质球, 初始温度 $= \begin{cases} u_0 (0 < r < r_0) \\ 0 (r_0 < r < 2r_0) \end{cases}$. 把球面保持为零度而使它冷却. 求解

球内温度变化情况.

解: 由题意写出定解问题
$$\begin{cases} u_t - a^2 \Delta u = 0 \\ u|_{r=2r_0} = 0 \\ u|_{t=0} = \begin{cases} u_0 & (0 < r < r_0) \\ 0 & (r_0 < r < 2r_0) \end{cases} \end{cases}$$

由条件知: 解的形式与 θ, φ 无关 $l=0, m=0$.

解的形式可写作 $j_0(kr) e^{-a^2 k^2 t}$

根据 $u|_{r=2r_0} = 0$ 得 $j_0(2kr_0) = \frac{\sin 2kr_0}{2kr_0} = 0$

$2kr_0 = n\pi \quad (n=1, 2, \dots)$ 其中 $k_n = \frac{n\pi}{2r_0} \quad (n=1, 2, \dots)$

所以 $u = \sum_{n=1}^{\infty} C_n j_0\left(\frac{n\pi}{2r_0} r\right) e^{-\left(\frac{n\pi}{2r_0}\right)^2 t}$

再根据 $u|_{t=0} = \begin{cases} u_0 & (0 < r < r_0) \\ 0 & (r_0 < r < 2r_0) \end{cases}$ 可以定出系数 C_n

$$\begin{aligned} C_n &= \frac{\int_0^{r_0} u_0 j_0(k_n r) r^2 dr}{\int_0^{2r_0} [j_0(k_n r)]^2 r^2 dr} = \int_0^{r_0} u_0 \frac{\sin(k_n r)}{k_n r} r^2 dr / \int_0^{2r_0} \frac{\sin^2(k_n r)}{(k_n r)^2} r^2 dr \\ &= \frac{\frac{u_0}{k_n} \int_0^{r_0} r_0 \sin(k_n r) dr}{\frac{1}{k_n^2} \int_0^{2r_0} \sin^2(k_n r) dr} \end{aligned}$$

$$\text{先计算分母: } \frac{1}{k_n^2} \int_0^{2r_0} \sin^2(k_n r) dr = \frac{1}{k_n^2} \frac{1}{k_n} \left[\frac{1}{2} k_n r - \frac{1}{4} \sin(2k_n r) \right] \Big|_0^{2r_0}$$

$$= \frac{1}{k_n^3} \left[\frac{1}{2} k_n (2r_0) - \frac{1}{4} \sin(4k_n r_0) \right] = \frac{r_0}{k_0^2}$$

再计算分子:

$$\begin{aligned} \frac{u_0}{k_n} \int_0^{r_0} r \sin(k_n r) dr &= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin(k_n r) - \frac{1}{k_n} r \cos(k_n r) \right] \Big|_0^{r_0} = \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin(k_n r_0) - \frac{r_0}{k_n} \cos(k_n r_0) \right] \\ &= \frac{u_0}{k_0} \left[\frac{1}{k_n^2} \sin\left(\frac{n\pi}{2}\right) - \frac{r_0}{k_n} \cos\left(\frac{n\pi}{2}\right) \right] = \frac{u_0}{k_n^2} \left[\frac{1}{k_0} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right] \end{aligned}$$

$$\text{得 } C_n = \frac{\frac{u_0}{k_n^2} \left[\frac{1}{k_n} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right]}{\frac{r_0}{k_0^2}} = \frac{u_0}{r_0} \left[\frac{1}{k_n} \sin\left(\frac{n\pi}{2}\right) - r_0 \cos\left(\frac{n\pi}{2}\right) \right]$$

$$= \frac{u_0}{r_0 k_n} \sin\left(\frac{n\pi}{2}\right) - u_0 \cos\left(\frac{n\pi}{2}\right)$$

由此将 C_n 代入: 可得解的表达式

$$u = \sum_{n=1}^{\infty} \left(\frac{u_0}{r_0 k_n} \sin \frac{n\pi}{2} - u_0 \cos \frac{n\pi}{2} \right) \frac{\sin k_n r}{k_n r} e^{-k_n^2 a^2 t} = \sum_{n=1}^{\infty} \left(\frac{4r_0 u_0}{n^2 \pi^2} \sin \frac{n\pi}{2} - \frac{2r_0 u_0}{n\pi} \cos \frac{n\pi}{2} \right) \frac{1}{r} \sin(k_n r) e^{-k_n^2 a^2 t}$$

$$\cos \frac{n\pi}{2} = \begin{cases} 0 & (n = 2k+1) \\ (-1)^k & (n = 2k) \end{cases} \quad \sin \frac{n\pi}{2} = \begin{cases} 0 & (n = 2k) \\ (-1)^k & (n = 2k+1) \end{cases}$$

$$\text{所以} \quad u = \sum_{n=1}^{\infty} A_n \frac{1}{r} \sin(k_n r) e^{-a^2 k_n^2 t}$$

$$\text{其中 } A_{2k} = (-1)^{k+1} \frac{u_0 r_0}{k\pi} \quad A_{2k+1} = (-1)^k \frac{4u_0 r_0}{(2k+1)^2 \pi^2}$$

8. 匀质球, 半径为 r_0 , 初始温度为 U_0 . 放在温度为 u_0 的空气中自由冷却(按照牛顿冷却定律跟空气交换热量), 求解球内各处温度变化情况.

$$\text{解: 定解问题写成} \begin{cases} u_t - a^2 \Delta u = 0 \\ (u + H u_r) \Big|_{r=r_0} = u_0 \quad (\text{其中 } H = \frac{k}{h}) \\ u \Big|_{t=0} = U_0 \end{cases}$$

考虑将边界条件化成齐次改变温标

$$\text{得} \quad \begin{cases} u_t - a^2 \Delta u = 0 \\ (u + H u_r) \Big|_{r=r_0} = 0 \\ u \Big|_{t=0} = U_0 - u_0 \end{cases} \quad \text{求解之后加上 } u_0 \text{ 即可.}$$

定解条件与 θ, φ 无关 即有 $l = 0, m = 0$

从而球内的形式解为 $j_0(kr) e^{-a^2 k^2 t}$

$$\text{根据 } u + H u_r \Big|_{r=r_0} = 0 \text{ 可求本征值 } k, j_0(kr) = \frac{\sin kr}{kr} e^{-a^2 k^2 t}$$

$$\therefore j_0'(kr) = \frac{\cos kr \cdot kr - \sin kr}{(kr)^2} k \cdot e^{-a^2 k^2 t}$$

由 $u + Hu_r|_{r=r_0} = 0$ 得 $\frac{\sin kr_0}{kr_0} + Hk \frac{kr_0 \cos kr_0 - \sin kr_0}{k^2 r_0^2} = 0$

即 $r_0 \sin kr_0 + Hkr_0 \cos kr_0 - H \sin kr_0 = 0$ 进一步有

$$r_0 \tan kr_0 - H \tan kr_0 = -Hkr_0 \Rightarrow \tan kr_0 = \frac{Hkr_0}{H - r_0}$$

将 $\tan kr_0 = \frac{H}{H - r_0} kr_0$ 的第 n 个根记作 k_n

所以
$$u = \sum_{n=1}^{\infty} C_n \frac{\sin(k_n r)}{k_n r} e^{-a^2 k_n^2 t}$$

再根据 $u|_{t=0} = U_0 - u_0$ 得: $\sum_{n=1}^{\infty} C_n \frac{\sin(k_n r)}{k_n r} = U_0 - u_0$ 可确定 C_n

$$\text{从而 } C_n = \frac{(U_0 - u_0) \int_0^{r_0} \frac{\sin(k_n r)}{k_n r} r^2 dr}{\int_0^{r_0} \frac{\sin^2(k_n r)}{k_n^2 r^2} r^2 dr}$$

得

$$C_n = \frac{k_n (U_0 - u_0) \int_0^{r_0} r \sin(k_n r) dr}{\int_0^{r_0} \sin^2(k_n r) dr}$$

$$\begin{aligned} \text{先计算分母: } \int_0^{r_0} \sin^2(k_n r) dr &= \frac{1}{k_n} \left[\frac{k_n r}{2} - \frac{1}{4} \sin(2k_n r) \right] \Big|_0^{r_0} = \frac{1}{k_n} \left[\frac{k_n r_0}{2} - \frac{1}{4} \sin(2k_n r_0) \right] \\ &= \frac{r_0}{2} - \frac{1}{k_n^2} \sin(k_n r_0) \cos(k_n r_0) \end{aligned}$$

再计算分子:

$$\begin{aligned} k_n (U_0 - u_0) \int_0^{r_0} r \sin(k_n r) dr &= (U_0 - u_0) k_n \left[\frac{1}{k_n^2} \sin(k_n r) - \frac{1}{k_n} r \cos(k_n r) \right] \Big|_0^{r_0} \\ &= (U_0 - u_0) \left[\frac{\sin(k_n r_0)}{k_n} - r_0 \cos(k_n r_0) \right] \end{aligned}$$

$$\begin{aligned} \text{由此 } C_n &= \frac{(U_0 - u_0) \left[\frac{\sin(k_n r_0)}{k_n} - r_0 \cos(k_n r_0) \right]}{\frac{1}{2k_n} [r_0 k_n - \sin(k_n r_0) \cos(k_n r_0)]} \\ &= 2(U_0 - u_0) \frac{\sin(k_n r_0) - r_0 k_n \cos(k_n r_0)}{r_0 k_n - \sin(k_n r_0) \cos(k_n r_0)} \\ &= \frac{2r_0 (U_0 - u_0)}{H} \frac{\sin(k_n r_0)}{k_n r_0 - \sin(k_n r_0) \times \cos(k_n r_0)} \end{aligned}$$

$$\text{代入解的表达式得 } u = u_0 + \sum_{n=1}^{\infty} \frac{2(U_0 - u_0)r_0}{H} \cdot \frac{\sin k_n r_0}{k_n r_0 - \sin(k_n r_0) \cdot \cos(k_n r_0)} \frac{\sin k_n r}{k_n r} e^{-k_n^2 a^2 t}$$

9. 半径为 r_0 的球面径向速度分布为 $v = v_0 \frac{1}{4} (3 \cos 2\theta + 1) \cos \omega t$. 试求解这个球在空气中辐射出去的声场中的速度势, 设 $r_0 \ll$ 波长 λ (声波波长). 本题径向速度对空间中的方向的依赖性由因子 $\frac{1}{4} (3 \cos \theta + 1)$ 即 $P_2(\cos \theta)$ 描写, 因而是轴对称四极声源.

解:由题意,速度势满足三维波动方程 $u_{xx} - a^2 \Delta v = 0$

$a^2 = \frac{\rho_0 \gamma}{\rho_0}$ p_0 为起始压强 ρ_0 为起始密度 γ 为定压比热与定容比热之比

声波传播的速度与速度势关系 $\mathbf{v} = \nabla v$

$$\left. \frac{\partial v}{\partial r} \right|_{r=r_0} = v_0 \frac{1}{4} (3 \cos 2\theta + 1) \cos \omega t = v_0 P_2(\cos \theta) e^{-i\omega t}$$

最后结果取实部

$$\text{形式解为 } \begin{Bmatrix} h_l^{(1)}(kr) \\ h_l^{(2)}(kr) \end{Bmatrix} P_l^m(\cos \theta) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} e^{i\omega t} \\ e^{-i\omega t} \end{Bmatrix}$$

$$\text{由 } \left. \frac{\partial v}{\partial r} \right|_{r=r_0} = v_0 P_2(\cos \theta) e^{-i\omega t} \text{ 得: } l=2, m=0 \quad \omega = ka$$

另外, $h_l^{(1)}(kr)$ 含 e^{ikr} 与 e^{-ikr} 得出 $e^{i(kr-ka)}$; 满足辐射波条件排除 $h_l^{(2)}$ 项
解的形式为 $u = A h_2^{(1)}(kr) P_2(\cos \theta) e^{-i\omega t}$

$$\text{所以} \quad u = A \left[-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right] e^{ikr} P_2(\cos \theta) e^{-i\omega t}$$

$$\text{代入 } \left. \frac{\partial u}{\partial r} \right|_{r=r_0} = v_0 P_2(\cos \theta) e^{-i\omega t} \text{ 得:}$$

$$A \left[\frac{qi}{k^4 r_0^4} + \frac{9}{k^3 r_0^3} - \frac{2i}{(kr_0)^2} + \frac{1}{kr_0} \right] k e^{ikr_0} = v_0$$

根据条件 $r_0 \ll \lambda \quad kr_0 \ll 2\pi \quad \therefore e^{ikr_0} \neq 1$

$$\frac{qi}{(kr_0)^4} A_k \approx v_0 \quad \text{即得 } A = -i \frac{v_0 k^3 r_0^4}{9}$$

$$u = -i \frac{u_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2(\cos \theta) e^{-i\omega t}$$

$$\text{取实部} \quad u = \text{Re} \left[-i \frac{u_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2(\cos \theta) e^{-i\omega t} \right]$$

$$\text{在远场即 } r \gg 0 \quad u = -i \frac{u_0 k^3 r_0^4}{9} \left[-\frac{3i}{k^3 r^4} - \frac{3}{k^2 r^2} + \frac{i}{kr} \right] e^{ikr} P_2(\cos \theta) e^{-i\omega t}$$

$$\text{只取第三项} \quad u = \frac{u_0 k^2 r_0^4}{9r} e^{i(kr-a)} P_2(\cos \theta)$$

$$\text{取实部,} \quad u = \frac{u_0 k^2 r_0^4}{9r} P_2(\cos \theta) \cos k(r-at)$$

10. 半径为 r_0 的球面径向速度分布为 $v = v_0 \frac{3}{2} (1 - \cos 2\theta) \sin 2\varphi \cos \omega t$. 试求解这个球在空气中辐射出去的声场中的速度势, 设 $r_0 \ll \lambda$ (声波波长). 本题是非轴对称的四极声源.

解:速度势满足三维波动方程 $u_{xx} - a^2 \Delta u = 0$

$$\begin{aligned} \left. \frac{\partial u}{\partial r} \right|_{r=r_0} &= u_0 \frac{3}{2} (1 - \cos 2\theta) \sin 2\varphi \cos \omega t \\ &= u_0 P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t} \end{aligned}$$

最后结果取实部

由于解的形式为 $\begin{Bmatrix} h_l^{(1)}(kr) \\ h_l^{(2)}(kr) \end{Bmatrix} P_l^m(\cos \theta) \begin{Bmatrix} \cos m\varphi \\ \sin m\varphi \end{Bmatrix} \begin{Bmatrix} e^{i\omega t} \\ e^{-i\omega t} \end{Bmatrix}$

且由 $\frac{\partial u}{\partial r}|_{r=r_0} = \nu_0 \frac{3}{2}(1 - \cos 2\theta) \sin 2\varphi \cos \omega t$ 知 $l = 2, m = 2$.

由于为辐射法,故舍去 $h_l^{(2)}(kr)$ 项,有

$$u = A h_2^{(1)}(kr) P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t}$$

$$A \frac{d}{dr} [h_2^{(1)}(kr)] \Big|_{r=r_0} = u_0 \text{ 在 } r_0 \ll \lambda \text{ 时, } kr_0 \ll 2\pi, e^{ikr_0} \approx 1$$

$$h_2^{(1)}(kr) = \left[-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right] \approx -i \frac{3}{(kr)^3}$$

$$A \frac{d}{dr} \left[-i \frac{3}{(kr)^3} \right] \Big|_{r=r_0} = \nu_0$$

$$A = -i \frac{\nu_0 r_0^4 k^3}{9}$$

$$\text{所以 } u = \operatorname{Re} \left[-i \frac{u_0 r_0^4 k^3}{9} h_2^{(1)}(kr) P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t} \right]$$

在远场时 $r \gg 0, h_2^{(1)}(kr) \approx i \frac{1}{kr} e^{ikr}$

有

$$u = \operatorname{Re} \left[-i \frac{u_0 r_0^4 k^3}{9} i \frac{1}{kr} e^{ikr} P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t} \right]$$

有

$$u = \frac{u_0 r_0^4 k^2}{9r} P_2^2(\cos \theta) \sin 2\varphi \cos k(r - at)$$

第十二章 格林函数法

本章知识要点

泊松方程的格林函数法

设 $u(\mathbf{r})$ 和 $v(\mathbf{r})$ 在区域 T 及其边界 Σ 上有连续一阶导数. 而在 T 中具有连续二阶导数, 则

$$\iint_{\Sigma} u \nabla v \cdot d\mathbf{s} = \iiint_T \nabla \cdot (u \nabla v) dV = \iiint_T u \Delta v dV + \iiint_T \nabla u \cdot \nabla v dV \quad (1)$$

上式称为第一格林公式

同理又有

$$\iint_{\Sigma} v \nabla u \cdot d\mathbf{s} = \iiint_T v \Delta u dV + \iiint_T \nabla u \cdot \nabla v dV \quad (2)$$

(1), (2) 两式相减得

$$\iint_{\Sigma} (u \nabla v - v \nabla u) \cdot d\mathbf{s} = \iiint_T (u \Delta v - v \Delta u) dV$$

即

$$\iint_{\Sigma} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) dS = \iiint_T (u \Delta v - v \Delta u) dV \quad (3)$$

$\frac{\partial}{\partial n}$ 表示沿边界 Σ 的外法向求导数. (3) 式称为第二格林公式.

泊松方程与第一类边界条件构成的定解问题称为第一边值问题(狄里希利问题), 与第二类边界条件构成的定解问题称为第二边值问题, 即诺依曼问题, 与第三类边界条件构成的定解问题称为第三边值问题.

泊松方程的基本积分方式

$$u(\mathbf{r}_0) = \iiint_T v(\mathbf{r}, \mathbf{r}_0) f(\mathbf{r}) dV - \iint_{\Sigma} \left[v(\mathbf{r}, \mathbf{r}_0) \frac{\partial u(\mathbf{r})}{\partial n} - u(\mathbf{r}) \frac{\partial v(\mathbf{r}, \mathbf{r}_0)}{\partial n} \right] dS$$

泊松方程第一边值问题的格林函数 $G(\mathbf{r}, \mathbf{r}_0)$, 解为

$$u(\mathbf{r}_0) = \iiint_T G(\mathbf{r}, \mathbf{r}_0) f(\mathbf{r}) dV + \iint_{\Sigma} \varphi(\mathbf{r}) \frac{\partial G(\mathbf{r}, \mathbf{r}_0)}{\partial n} dS$$

泊松方程第三边值问题的格林函数 $G(\mathbf{r}, \mathbf{r}_0)$, 解为

$$u(\mathbf{r}_0) = \iiint_T G(\mathbf{r}, \mathbf{r}_0) f(\mathbf{r}) dV - \frac{1}{\alpha} \iint_{\Sigma} G(\mathbf{r}, \mathbf{r}_0) \varphi(\mathbf{r}) dS$$

用电像法求柱林函数

(一) 无界空间的格林函数基本解

无界区域的格林函数称为方程的基本解.

一个边值问题的格林函数 G 分成两部分, $G = G_0 + G_1$ (G_0 为基本解)

对三维泊松方程, G_0 满足 $\Delta G_0 = \delta(\mathbf{r} - \mathbf{r}_0)$

G_1 满足 $\Delta G_1 = 0$ 为(拉普拉斯方程). 其中 $\Delta G_0 = \delta(\mathbf{r} - \mathbf{r}_0)$ 描述点源在无界空间产生的稳定场. 静电场情况下, $G_0 = -\frac{1}{4\pi} |\mathbf{r} - \mathbf{r}_0|$

(二) 用电像法求格林函数.

基本思想: 用一个设想的等效点电荷束代替所有的感应电荷. 可得到 G_1 的类似 G_0 形式的解. 这个等效电荷称为电像.

最典型的例子是导体球面, 一个点电荷在球外某处, 球面因感应而产生电荷. 这些感应电荷可以被球内某一点的点电荷代替. 在球外空间产生同样的电势效果.

含时间的格林函数

持续作用的力在一般强迫振动的定解问题中可以看作是前后相继的脉冲力 $f(\mathbf{r}, \tau)\delta(t - \tau)d\tau$ 的叠加.

进一步, 将连分布在三维空间的脉冲力看作是许多点上的力的叠加. 写成

$$f(\mathbf{r}, t) = \iiint_V \int_t f(\mathbf{r}_0, \tau) \delta(\mathbf{r} - \mathbf{r}_0) \delta(t - \tau) d\mathbf{r}_0 d\tau$$

单位脉冲点力引起的振动记为 $G(\mathbf{r}, \mathbf{r}_0, t, t_0)$, 变为波动问题的格林函数. 求出 G , 然后叠加可以算出任意 $f(\mathbf{r}, t)$ 引起的振动.

$$\begin{aligned} G_{tt} - a^2 \Delta G &= \delta(\mathbf{r} - \mathbf{r}_0) \delta(t - t_0) \\ G \text{ 满足 } \left(a \frac{\partial G}{\partial n} + \beta G \right) \Big|_{\Sigma} &= 0 \\ G|_{t=0} = 0 \quad G_t|_{t=0} &= 0 \end{aligned}$$

推广的格林公式及其应用

$$\text{算符 } L \quad L = a_{11} \frac{\partial^2}{\partial x^2} + 2a_{12} \frac{\partial^2}{\partial x \partial y} + a_{22} \frac{\partial^2}{\partial y^2} + b_1 \frac{\partial}{\partial x} + b_2 \frac{\partial}{\partial y} + c$$

$$\text{如果两个算符 } L, M \text{ 满足 } vLu - uMv = \frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y}$$

则 L, M 互称为伴随算符, 当 $L = M$ 时, 则称为自伴算符.

(一) 椭圆形方程

$$Lu = u_{xx} + u_{yy} + b_1 u_x + b_2 u_y + cu = f(x, y)$$

L 的伴随算符由 $Mv = v_{xx} + v_{yy} - (b_1 v)_x - (b_2 v)_y + cv$ 给出.

(二) 抛物型方程

$$Lu = \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial y} = f \quad \text{算符 } L \text{ 的伴随算符 } M = \frac{\partial^2}{\partial x^2} + \frac{\partial}{\partial y}$$

(三) 双曲型方程

$$Lu = u_{xx} - u_{yy} + b_1 u_x + b_2 u_y + cu = f$$

算符 L 的伴随算符 M 由下式给出 $Mv = v_{xx} - v_{yy} - (b_1 v)_x - (b_2 v)_y + cv$

习题全解

§ 12.2 用电像法求格林函数

1. 在圆 $\rho = a$ 内求解拉普拉斯方程的第一边值问题 $\Delta_2 u = 0 (\rho < a), u|_{\rho=a} = f(\varphi)$.

解: 圆的第一边值问题的格林函数

$$G(M, M_0) = \frac{1}{2\pi} \left[1 - \ln \frac{1}{|r - r_0|} + \ln \frac{1}{|r - r_1|} + \ln \frac{a}{\rho_0} \right], \text{ 由电像位置关系}$$

$$\rho_1 = \frac{a^2}{\rho_0}$$

由几何关系得:

$$|r - r_0|^2 = \rho^2 - 2\rho\rho_0 \cos \varphi + \rho_0^2$$

$$|r - r_1|^2 = \rho^2 - 2\rho\rho_1 \cos \varphi + \rho_1^2$$

$$\text{代入可得: } G = \frac{1}{2\pi} \left[\ln \sqrt{\rho^2 - 2\rho\rho_0 \cos \varphi + \rho_0^2} - \ln \sqrt{\rho^2 - 2\rho\rho_1 \cos \varphi + \rho_1^2} + \ln \frac{a}{\rho_0} \right]$$

$$\begin{aligned} \frac{\partial G}{\partial n} \Big|_{\epsilon} &= \frac{\partial G}{\partial \rho} \Big|_{\rho=a} = \frac{1}{2\pi} \left[\frac{\rho - \rho_0 \cos \varphi}{\rho^2 - 2\rho\rho_0 \cos \varphi + \rho_0^2} \right] - \frac{1}{2\pi} \left[\frac{\rho - \rho_1 \cos \varphi}{\rho^2 - 2\rho\rho_1 \cos \varphi + \rho_1^2} \right] \Big|_{\rho=a} \\ &= \frac{1}{2\pi a} \frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0 \cos \varphi + \rho_0^2} \end{aligned}$$

$$G(M, M_0) \Big|_{\epsilon} = 0, ds = a d\varphi$$

$$\text{代入解的积分公式有 } u(\rho_0, \varphi_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - \rho_0^2) f(\varphi)}{a^2 - 2a\rho_0 \cos \varphi + \rho_0^2} d\varphi$$

2. 在半平面 $y > 0$ 内求解拉普拉斯方程的第一边值问题 $\Delta_2 u = 0 (y > 0), u|_{y=0} = f(x)$.

解: 由平面第一边值问题的格林函数满足 $\begin{cases} \Delta G(M, M_0) = \delta(x - x_0) \cdot \delta(y - y_0) \\ G|_{y=0} = 0, \end{cases}$ 的解

$$\text{为 } G(M, M_0) = \frac{1}{2\pi} \left[-\ln \frac{1}{|r - r_0|} + \ln \frac{1}{|r - r_1|} \right]$$

$$\text{其中 } |r - r_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2} \quad |r - r_1| = \sqrt{(x - x_0)^2 + (y + y_0)^2}$$

$$\frac{\partial G}{\partial n} \Big|_{\epsilon} = -\frac{\partial G}{\partial y} \Big|_{y=0} = -\frac{1}{2\pi} \left[\frac{y - y_0}{(x - x_0)^2 + (y - y_0)^2} - \frac{y + y_0}{(x - x_0)^2 + (y + y_0)^2} \right] \Big|_{y=0}$$

$$= \frac{1}{\pi} \frac{y_0}{(x-x_0)^2 + y_0^2}$$

代入积分公式可得 $u(x_0, y_0) = \frac{y_0}{\pi} \int_{-\infty}^{+\infty} \frac{f(x)}{(x-x_0)^2 + y_0^2} dx$

3. 在圆形域 $\rho \leq a$ 上求解 $\Delta u = 0$ 使满足边界条件 ① $u|_{\rho=a} = A \cos \varphi$, ② $u|_{\rho=a} = A + B \sin \varphi$.

解: (1) 定解问题为 $\Delta u = 0$ $u|_{\rho=0} = A \cos \varphi$

由前面的结论得 $u(\rho_0, \varphi_0) = \frac{a^2 - \rho_0^2}{2\pi} \int_0^{2\pi} \frac{A \cos \varphi d\varphi}{a^2 - 2a\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2}$

在圆内求解 有 $a > \rho_0$

$$\text{改写 } u(\rho_0, \varphi_0) = \frac{\left[1 - \left(\frac{\rho_0}{a}\right)^2\right]}{2\pi} A \int_0^{2\pi} \frac{\cos \varphi d\varphi}{1 - 2\left(\frac{\rho_0}{a}\right) \cos(\varphi - \varphi_0) + \left(\frac{\rho_0}{a}\right)^2}$$

由欧拉公式 $\cos \varphi = \frac{1}{2}(e^{i\varphi} + e^{-i\varphi})$ 令 $z = e^{i\varphi}$

$$\cos(\varphi - \varphi_0) = \frac{1}{2}(e^{-i\varphi_0} z + e^{i\varphi_0}), d\varphi = \frac{dz}{iz}$$

$$\int_0^{2\pi} \frac{\cos \varphi d\varphi}{1 - 2\left(\frac{\rho_0}{a}\right) \cos(\varphi - \varphi_0) + \left(\frac{\rho_0}{a}\right)^2} = \frac{-1}{2i} \oint_{|z|=1} \frac{(1+z^2) dz}{z[e^{-i\varphi_0} - (1+\epsilon^2)z + \epsilon e^{i\varphi_0}]} \quad (\epsilon = \frac{\rho_0}{a})$$

在 $|z|=1$ 内有两个单极点 0 与 $\epsilon e^{i\varphi_0}$ 留数为 $\frac{1}{\epsilon} e^{-i\varphi_0}, \frac{\epsilon}{e^{i\varphi_0}} \frac{1+\epsilon^2 e^{i\varphi_0}}{\epsilon^2 - 1}$

$$\text{留数之和为 } \frac{\epsilon^2 + \epsilon^4 e^{i\varphi_0} + e^{-i\varphi_0} e^{i\varphi_0} (\epsilon^2 - 1)}{\epsilon e^{i\varphi_0} (\epsilon^2 - 1)} = \frac{2\epsilon}{\epsilon^2 - 1} \cos \varphi_0$$

由留数定理得

$$u(\rho_0, \varphi_0) = \frac{\left[1 - \left(\frac{\rho_0}{a}\right)^2\right]}{2\pi} A \cdot 2\pi \left(\frac{2\epsilon}{\epsilon^2 - 1}\right) \cos \varphi_0 = \frac{A\rho_0}{a} \cos \varphi_0$$

$$(2) \begin{cases} \Delta u = 0 \\ u|_{\rho=a} = A + B \sin \varphi \end{cases}$$

$$\begin{aligned} u(\rho, \varphi_0) &= \frac{a^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{A + B \sin \varphi}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2} d\varphi \\ &= \frac{1 - \epsilon^2}{2\pi} \int_0^{2\pi} \frac{A + B \sin \varphi}{1 - 2\epsilon \cos(\varphi - \varphi_0) + \epsilon^2} d\varphi \quad (\epsilon = \frac{\rho}{a}) \end{aligned}$$

$$\text{计算 } \int_0^{2\pi} \frac{A + B \sin \varphi}{1 - 2\epsilon \cos(\varphi - \varphi_0) + \epsilon^2} d\varphi \quad \text{令 } z = e^{i\varphi}$$

$$\text{从而得到上述积分为 } \frac{1}{2} \oint_{|z|=1} \frac{2Aiz + Bz^2 - B}{z[\epsilon e^{-i\varphi_0} z^2 - (1+\epsilon^2)z + \epsilon e^{i\varphi_0}]} dz$$

$$\text{留数之和为 } \frac{-B\epsilon^2 + B + B(\epsilon^2 e^{i2\varphi_0} - 1) + 2A\epsilon e^{i\varphi_0}}{\epsilon e^{i\varphi_0} (\epsilon^2 - 1)} \text{ 根据留数定理可得积分}$$

$$\text{值为 } 2\pi i \times \text{留数和} = \frac{2\pi(A + B\epsilon \sin \varphi_0)}{1 - \epsilon^2}$$

代入 $u(\rho, \varphi_0)$ 的表达式可得:

$$u(\rho_0, \varphi_0) = \frac{1 - \epsilon^2}{2\pi} \frac{2\pi(A + B\epsilon \sin \varphi_0)}{1 - \epsilon^2} = A + B\epsilon \sin \varphi_0 = A + B \frac{\rho_0}{a} \sin \varphi_0$$

4. 对于一般的 $f(\varphi)$, 积分公式(12.2.22)里的积分可能不那么容易计算. 试将 $1/[a^2 - 2a\rho\cos(\varphi - \varphi_0) + \rho^2]$ 展开为傅里叶级数, 然后逐项积分.

作为对照, 再用分离变数法求解圆内的第一边值问题.

解: 圆内第一边值问题解的积分公式为

$$u(\rho_0, \varphi_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho_0^2}{a^2 - 2a\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2} f(\varphi) d\varphi = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - \epsilon^2}{1 - 2\epsilon \cos(\varphi - \varphi_0) + \epsilon^2} f(\varphi) d\varphi, \quad (\epsilon = \frac{\rho_0}{a})$$

$$\text{由} \quad \frac{1 - \epsilon^2}{1 - 2\epsilon \cos(\varphi - \varphi_0) + \epsilon^2} = 1 + 2 \sum_{n=1}^{\infty} \epsilon^n \cos n(\varphi - \varphi_0)$$

$$\begin{aligned} u(\rho_0, \varphi_0) &= \frac{1}{2\pi} \int_0^{2\pi} f(\varphi) \left[1 + 2 \sum_{n=1}^{\infty} \epsilon^n \cos n(\varphi - \varphi_0) \right] d\varphi \\ &= \sum_{n=0}^{\infty} \frac{1}{\sigma_n} \left(\frac{\rho_0}{a} \right)^n \left[\cos n\varphi_0 \cdot \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi + \sin n\varphi_0 \cdot \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi \right] \\ &= \sum_{n=0}^{\infty} [A_n \cos n\varphi_0 + B_n \sin n\varphi_0] \rho_0^n \end{aligned}$$

$$A_n \text{ 对应 } \frac{1}{\delta_n \pi a^n} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi \quad B_n \text{ 对应 } \frac{1}{\pi a^n} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi$$

$$\text{分离变量解可以写成 } u(\rho, \varphi) = \sum_{n=0}^{\infty} (A_n \cos n\varphi + B_n \sin n\varphi) \rho^n$$

$$\text{代入给定的边界条件: } \sum (A_n a^n \cos n\varphi + B_n a^n \sin n\varphi) = f(\varphi)$$

$$\text{由 Fourier 分析知 } A_n = \frac{1}{\delta_n \pi a^n} \int_0^{2\pi} f(\varphi) \cos n\varphi d\varphi, B_n = \frac{1}{\pi a^n} \int_0^{2\pi} f(\varphi) \sin n\varphi d\varphi$$

由此可见, 最该结果是一致的

5. 试求层状空间 $0 < z < H$ 第一边值问题的格林函数.

$$\text{解: 层状空间 } 0 < z < H \text{ 第一边值问题} \begin{cases} \Delta G = \delta(\mathbf{r} - \mathbf{r}_0) \\ G|_{z=0} = 0 \\ G|_{z=H} = 0 \end{cases}$$

在层状空间. 由镜像法, 假设 $z = 0, z = H$ 为平面镜, 就会出现无穷多个像对无穷多个像进行叠加处理即可.

$$G = -\frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \frac{1}{[(x-x_0)^2 + (y-y_0)^2 + (z-2nH-z_0)^2]^{\frac{3}{2}}} + \frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \frac{1}{[(x-x_0)^2 + (y-y_0)^2 + (z-2nH+z_0)^2]^{\frac{3}{2}}}$$

§ 12.4 用冲量定理法求格林函数

1. 两端固定的弦在线密度为 $\rho f(x, t) = \rho \Phi(x) \sin \omega t$ 的横向力作用下振动. 求解其振动情况.

研究共振的可能性,并求共振时的解.

解:由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = \Phi(x) \sin \omega t \\ u|_{x=0} = 0 & u|_{x=l} = 0 \\ u|_{t=0} = 0 & u_t|_{t=0} = 0 \end{cases}$$

该问题对应的格林函数满足

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x-\xi)\delta(t-\tau) \\ G|_{x=0} = 0 & G|_{x=l} = 0 \\ G|_{t=0} = 0 & G_t|_{t=0} = 0 \end{cases}$$

由冲量定理,问题可转化为

$$\begin{cases} G_t - a^2 G_{xx} = 0 \\ G|_{x=0} = 0 & G_t|_{x=l} = 0 \\ G|_{t=\tau+0} = 0 & G_t|_{t=\tau+0} = \delta(x-\xi) \end{cases}$$

利用分离变数法,可求得

$$G(x, t, \xi, \tau) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a}{l} (t-\tau) \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}$$

以此得到

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l \Phi(\xi) \sin \omega \tau \cdot \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a}{l} (t-\tau) \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} d\xi d\tau \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \int_0^t \sin \omega \tau \sin \frac{n\pi a}{l} (t-\tau) d\tau \sin \frac{n\pi x}{l} \end{aligned}$$

其中

$$\begin{aligned} \int_0^t \sin \omega \tau \sin \frac{n\pi a}{l} (t-\tau) d\tau &= \int_0^t \frac{1}{2} \left[\cos(\omega \tau - \frac{n\pi a}{l} t + \frac{n\pi a}{l} \tau) - \cos(\omega \tau + \frac{n\pi a}{l} t - \frac{n\pi a}{l} \tau) \right] d\tau \\ &= \frac{1}{2} \left\{ \int_0^t \cos \left[(\omega + \frac{n\pi a}{l}) \tau - \frac{n\pi a}{l} t \right] d\tau - \int_0^t \cos \left[(\omega - \frac{n\pi a}{l}) \tau + \frac{n\pi a}{l} t \right] d\tau \right\} \\ &= \frac{1}{2} \left\{ \frac{1}{\omega + n\pi a/l} \sin \left[(\omega + \frac{n\pi a}{l}) \tau - \frac{n\pi a}{l} t \right] \Big|_0^t - \frac{1}{\omega - n\pi a/l} \sin \left[(\omega - \frac{n\pi a}{l}) \tau + \frac{n\pi a}{l} t \right] \Big|_0^t \right\} \\ &= \frac{1}{2} \left[\frac{1}{\omega + n\pi a/l} (\sin \omega t + \sin \frac{n\pi a}{l} t) - \frac{1}{\omega - n\pi a/l} (\sin \omega t - \sin \frac{n\pi a}{l} t) \right] \\ &= \frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \end{aligned}$$

$$\text{有} \quad u(x, t) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right] \sin \frac{n\pi x}{l}$$

当外力频率等于基音或谐音的频率, 即 $\omega = \frac{n\pi a}{l}$, 则 $[\quad]$ 为 $\frac{0}{0}$ 型. 求其极限可知其为 $(\frac{1}{2\omega} \sin \omega$

$t - \frac{1}{2} t \cos \omega t)$. 其中第二部分的振幅为 $\frac{1}{2} t$. 随时间而增长, 这就是共振.

2. 两端固定的弦在点 x_0 受谐变力 $\rho f(t) = \rho f_0 \sin \omega t$ 作用而振动, 求解振动情况.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = f_0 \sin \omega t \delta(x - x_0) \\ u|_{x=0} = 0 & u|_{x=l} = 0 \\ u|_{t=0} = 0 & u_t|_{t=0} = 0 \end{cases}$$

该问题对应的格林函数满足

$$\begin{cases} G_{tt} - a^2 G_{xx} = \delta(x - \xi) \delta(t - \tau) \\ G|_{x=0} = 0 & G|_{x=l} = 0 \\ G|_{t=0} = 0 & G_t|_{t=0} = 0 \end{cases}$$

由冲量定理, 问题可转化为

$$\begin{cases} G_{tt} - a^2 G_{xx} = 0 \\ G|_{x=0} = 0 & G|_{x=l} = 0 \\ G|_{t=\tau} = 0 & G|_{t=\tau+0} = \delta(x - \xi) \end{cases}$$

利用分离变数法, 可求得

$$G(x, t, \xi, \tau) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a}{l} (t - \tau) \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}$$

以此得到

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f_0 \delta(\xi - x_0) \sin \omega \tau \cdot \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a}{l} (t - \tau) \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} d\xi d\tau \\ &= \frac{2f_0}{\pi a} \cdot \sum_{n=1}^{\infty} \frac{1}{n} \int_0^t \delta(\xi - x_0) \sin \frac{n\pi \delta}{l} d\xi \int_0^t \sin \omega \tau \sin \frac{n\pi a}{l} (t - \tau) d\tau \sin \frac{n\pi x}{l} \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \left[\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right] \sin \frac{n\pi x}{l} \end{aligned}$$

3. 长为 l 的均匀细导线, 每单位长的电阻为 R , 通过恒定的电流 I , 导线表面跟周围温度为零度的介质进行热量交换. 试求解导线上的温度变化. 设初始温度和导线两端温度都是零度.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} + \frac{h}{c\rho} u = \frac{1}{c\rho} I^2 R \\ u|_{x=0} = 0 & u|_{x=l} = 0 \\ u|_{t=0} = 0 \end{cases}$$

其中 $\frac{h}{c\rho} u$ 为导线表面与周围温度为 0°C 介质进行的能量交换.

该问题对应的格林函数满足

$$\begin{cases} G_{tt} - a^2 G_{xx} + \frac{h}{c\rho} G = \delta(x - \xi) \delta(t - \tau) \\ G|_{x=0} = 0 & G|_{x=l} = 0 \\ G|_{t=0} = \delta(x - \xi) \end{cases}$$

由冲量定理, 问题可化为:

$$\begin{cases} G_t - a^2 G_{xx} + \frac{h}{c\rho} G = 0 \\ G|_{x=0} = 0 \quad G|_{x=l} = 0 \\ G|_{t=\tau=0} = \delta(x-\xi) \end{cases}$$

利用分离变数法可求得

$$G(x, t, \xi, \tau) = \frac{2}{l} \sum_{n=1}^{\infty} e^{-(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho})(t-\tau)} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}$$

以此得到

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l \frac{I^2 R}{c\rho} \cdot \frac{2}{l} \sum_{n=1}^{\infty} e^{-(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho})(t-\tau)} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} d\xi d\tau \\ &= \frac{2I^2 R}{lc\rho} \sum_{n=1}^{\infty} \int_0^l \sin \frac{n\pi \xi}{l} d\xi \int_0^t e^{-(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho})(t-\tau)} \\ &= \frac{2I^2 R}{lc\rho} \sum_{n=1}^{\infty} \frac{l}{n\pi} [1 - (-1)^n] \frac{1}{n^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} [1 - e^{-(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho})t}] \sin \frac{n\pi x}{l} \end{aligned}$$

当 n 为偶数, 则对应项为 0, 所以 $u(x, t)$ 又可写成

$$u(x, t) = \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} \cdot \frac{1}{(2n+1)^2 \pi^2 a^2 / l^2 + \frac{h}{c\rho}} [1 - e^{-(\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho})t}] \sin \frac{(2n+1)\pi x}{l}$$

4. 求解一维半无界空间的输运问题 $u_t - a^2 u_{xx} = 0, u|_{x=0} = At, u|_{t=0} = 0$.

解: 令 $u(x, t) = At + v(x, t)$. 则

$$\begin{cases} v_t - a^2 v_{xx} = -A & (x > 0) \\ v|_{x=0} = 0 \\ v|_{t=0} = 0 \end{cases}$$

因为是第一类齐次边界条件, 应将问题奇延拓到 $x < 0$ 的半空间中去, 于是

$$\begin{cases} v_t - a^2 v_{xx} = \begin{cases} -A & (x > 0) \\ A & (x < 0) \end{cases} \\ v|_{t=0} = 0 \end{cases}$$

则此时问题对应的格林函数满足

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x-\xi)\delta(t-\tau) \\ G|_{t=0} = 0 \end{cases}$$

由冲量定理问题又可转达化为

$$\begin{cases} G_t - a^2 G_{xx} = 0 \\ G|_{t=\tau=0} = \delta(x-\xi) \end{cases}$$

利用分离变数法可求得

$$G(x, t, \xi, \tau) = \int_{-\infty}^{\infty} \delta(x_0 - \xi) \left[\frac{1}{2a \sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] dx = \frac{1}{2a \sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)}$$

以此得到

$$v(x, t) = \int_{\tau=0}^t \left\{ \int_{-\infty}^0 A \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi + \int_0^{\infty} (-A) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(\xi-x)^2}{4a^2(t-\tau)}} \right] d\xi \right\} d\tau$$

在右边第一个积分中令 $z = (x - \xi)/2a\sqrt{t - \tau}$, $dz = -d\xi/2a\sqrt{t - \tau}$

在右边第二个积分中令 $z = (\xi - x)/2a\sqrt{t - \tau}$, $dz = d\xi/2a\sqrt{t - \tau}$

$$\begin{aligned} \text{于是 } v(x, t) &= \int_{\tau=0}^t \left(-\frac{A}{\sqrt{\pi}} \int_{\infty}^{x/2a\sqrt{t-\tau}} e^{-z^2} dz - \frac{A}{\sqrt{\pi}} \int_{-x/2a\sqrt{t-\tau}}^{\infty} e^{-z^2} dz \right) d\tau \\ &= -A \int_{\tau=0}^t \frac{2}{\sqrt{\pi}} \int_0^{x/2a\sqrt{t-\tau}} e^{-z^2} dz d\tau \\ &= -A \int_{\tau=0}^t \operatorname{erf}(x/2a\sqrt{t-\tau}) d\tau \end{aligned}$$

$$\therefore u(x, t) = At + v(x, t) = At - A \int_0^t \operatorname{erf}(x/2a\sqrt{t-\tau}) d\tau = A \int_0^t \operatorname{erfc}(x/2a\sqrt{t-\tau}) d\tau, (x > 0)$$

5. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0$, $u|_{x=0} = f(t)$, $u|_{t=0} = \varphi(x)$.

解: 令 $u(x, t) = v(x, t) + f(t)$, 则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = -f'(t) & (x > 0) \\ v|_{x=0} = 0 \\ v|_{t=0} = \varphi(x) - f(0) \end{cases}$$

因为是第一类齐次边界条件, 所以进行奇延拓到 $x < 0$ 的半空间中, 则

$$\begin{cases} v_t - a^2 v_{xx} = \begin{cases} -f'(t) & (x > 0) \\ f'(t) & (x < 0) \end{cases} \\ v|_{t=0} = \begin{cases} \varphi(x) - f(0) & (x > 0) \\ -\varphi(-x) + f(0) & (x < 0) \end{cases} \end{cases}$$

该问题对应的格林函数为

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x - \xi)\delta(t - \tau) \\ G|_{t=0} = 0 \end{cases}$$

由冲量定理得到

$$\begin{cases} G_t - a^2 G_{xx} = 0 \\ G|_{t=\tau=0} = \delta(x - \xi) \end{cases}$$

解得

$$G(x, t; \xi, \tau) = \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)}$$

在求 $v(x, t)$ 时, 注意到本题初始条件非零, 所以应保留 $\iiint_T (uG)|_{t=t_0} dv_0$ 项. 则

$$\begin{aligned} v(x, t) &= \int_{\tau=0}^t \int_{\xi=-\infty}^0 f'(\tau) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)} d\xi d\tau + \int_{\tau=0}^t \int_{\xi=0}^{\infty} (-f'(\tau)) \\ &\quad \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)} d\xi d\tau + \int_{\xi=-\infty}^0 [-\varphi(-\xi) + f(0)] \frac{1}{2a\sqrt{\pi-t}} e^{-(x-\xi)^2/4a^2 t} d\xi d\tau + \int_{\xi=0}^{\infty} [\varphi(\xi) - f(0)] \\ &\quad \frac{1}{2a\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2 t} d\xi d\tau \end{aligned}$$

右边第一项积分中, 令 $z = \frac{x-\xi}{2a\sqrt{t-\tau}}$, $dz = -\frac{d\xi}{2a\sqrt{t-\tau}}$, 则第一项积分可化为

$$-\int_0^t \int_{-\infty}^{\frac{x}{2a\sqrt{t-\tau}}} f'(\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

右边第二项积分中, 令 $z = \frac{\xi-x}{2a\sqrt{t-\tau}}$, $dz = \frac{d\xi}{2a\sqrt{t-\tau}}$, 则第二项积分可化为

$$-\int_0^t \int_{-\frac{x}{2a\sqrt{t-\tau}}}^{\infty} f'(\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

所以前两项积分合并为 $-\int_0^t f'(\tau) \int_{-\frac{x}{2a\sqrt{t-\tau}}}^{\frac{x}{2a\sqrt{t-\tau}}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau = -\int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau$

右边第三项积分中, 令 $z = \frac{x-\xi}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$

则第三项积分可化为

$$-\int_{-\infty}^{\frac{x}{2a\sqrt{t}}} [-\varphi(-x+2a\sqrt{t}z) + f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

右边第四项积分中, 令 $z = \frac{\xi-x}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$, 则第四项积分可化为

$$\int_{-\frac{x}{2a\sqrt{t}}}^{\infty} [\varphi(x+2a\sqrt{t}z) - f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

所以后两项积分合并为

$$-f(0) \left[\int_{-\infty}^{\frac{x}{2a\sqrt{t}}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz + \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \frac{1}{\sqrt{\pi}} e^{-z^2} dz \right] + \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

进一步合并得 $-f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) + \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$

$\therefore v(x, t) = -\int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) + \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz -$

$$\int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

$\therefore u(x, t) = f(t) - \int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) + \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(x+2a\sqrt{t}\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$

$$- \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(-x+2a\sqrt{t}\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

其中, 可将 $f(t)$ 写成 $\int_0^t f'(\tau) d\tau + f(0)$, 则

$$f(t) - \int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$= \int_0^t f'(\tau) \left[1 - \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) \right] d\tau + f(0) \left[1 - \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) \right]$$

$$= \int_0^t f'(\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau + f(0) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$\therefore u(x, t) = f(0) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right) + \int_0^t f'(\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{-\frac{x}{2a\sqrt{t}}}^{\infty} \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz \quad (x > 0)$$

6. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0, u_x \Big|_{x=0} = q(t), u \Big|_{t=0} = 0$.

解: 令 $u(x, t) = v(x, t) - q(t)e^{-x}$, 则

$$\begin{cases} v_t - a^2 v_{xx} = q'(t)e^{-x} - a^2 q(t)e^{-x} & (x > 0) \\ v_x \Big|_{x=0} = 0 \\ v \Big|_{t=0} = q(0)e^{-x} \end{cases}$$

因为是第二类齐次边界条件, 所以进行偶延拓到 $x < 0$ 的半空间中去, 得到

$$\begin{cases} v_t - a^2 v_{xx} = \begin{cases} [q'(t) - a^2 q(t)]e^{-x} & (x > 0) \\ [q'(t) - a^2 q(t)]e^x & (x < 0) \end{cases} \\ v \Big|_{t=0} = \begin{cases} q(0)e^{-x} & (x > 0) \\ q(0)e^x & (x < 0) \end{cases} \end{cases}$$

问题对应的格林函数 G 满足

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x - \xi)\delta(t - \tau) \\ G \Big|_{t=0} = 0 \end{cases}$$

由冲量定理得到

$$\begin{cases} G_t - a^2 G_{xx} = 0 \\ G \Big|_{t=\tau+0} = \delta(x - \xi) \end{cases}$$

解得

$$G(x, t; \xi, \tau) = \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)}$$

求 $v(x, t)$ 时, 注意到本题初始条件非零, 所以应保留 $\int_0^t \int_{-\infty}^{\infty} (uG) \Big|_{t=0} dv_0$ 项, 则

$$\begin{aligned} v(x, t) &= \int_{\tau=0}^t \int_{-\infty}^{\infty} [q'(\tau) - a^2 q(\tau)] e^{\xi} \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi d\tau + \\ &\int_{\tau=0}^t \int_{-\infty}^{\infty} [q'(\tau) - a^2 q(\tau)] e^{-\xi} \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x+\xi)^2}{4a^2(t-\tau)}} d\xi d\tau + \\ &\int_{-\infty}^0 q(0) e^{\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} q(0) e^{-\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x+\xi)^2}{4a^2 t}} d\xi \end{aligned}$$

对前两项积分, 对 e 指数进行配方, 得

$$\begin{aligned} &\int_{\tau=0}^t \int_{-\infty}^{\infty} [q'(\tau) - a^2 q(\tau)] \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{[x+2a^2(t-\tau)-\xi]^2}{4a^2(t-\tau)}} e^{a^2(t-\tau)} e^x d\xi d\tau + \\ &\int_{\tau=0}^t \int_0^{\infty} [q'(\tau) - a^2 q(\tau)] \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{[x-2a^2(t-\tau)-\xi]^2}{4a^2(t-\tau)}} e^{a^2(t-\tau)} e^{-x} d\xi d\tau \end{aligned}$$

在第一项积分中, 令 $z = \frac{x+2a^2(t-\tau)-\xi}{2a\sqrt{t-\tau}}$,

$dz = -\frac{d\xi}{2a\sqrt{t-\tau}}$, 则第一项积分可化为

$$\begin{aligned} & \int_0^t \int_{-\infty}^{\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}}} -[q'(\tau) - a^2 q(\tau)] \frac{1}{\sqrt{\pi}} e^{-z^2} e^{a^2(t-\tau)} e^x dz d\tau \\ &= \int_0^t -\frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \left[\int_0^{\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_{-\infty}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] d\tau \\ &= \int_0^t -\frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \left\{ \operatorname{erf} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] - 1 \right\} d\tau \\ &= \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \operatorname{erfc} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau \end{aligned}$$

在第二项积分中, 令 $z = \frac{\xi - [x - 2a^2(t-\tau)]}{2a\sqrt{t-\tau}}$, $dz = \frac{d\xi}{2a\sqrt{t-\tau}}$, 则第二项积分可化为

$$\begin{aligned} & \int_0^t \int_{-\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}}}^{\infty} [q'(\tau) - a^2 q(\tau)] \frac{1}{\sqrt{\pi}} e^{-z^2} e^{a^2(t-\tau)} e^{-x} dz d\tau \\ &= \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^{-x} \left[\int_{-\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}}}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\infty} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] d\tau \\ &= \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^{-x} \left\{ -\operatorname{erf} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] + 1 \right\} d\tau \\ &= \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^{-x} \operatorname{erfc} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau \end{aligned}$$

前两项积分可合并为

$$\int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] + e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau$$

在第三、四项积分中, 分别令 $z = \frac{x+2a^2t-\xi}{2a\sqrt{t}}$ 和 $z = \frac{\xi - [x - 2a^2t]}{2a\sqrt{t}}$, 并进行与第一、二项积分中类似的变换可得到第三、四项积分合并为

$$\frac{1}{2} q(0) e^{a^2 t} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2 t + x}{2a\sqrt{t}} \right] + e^{-x} \operatorname{erfc} \left[\frac{2a^2 t - x}{2a\sqrt{t}} \right] \right\}$$

$$\therefore u(x, t) = -q(t) e^{-x} + v(x, t)$$

$$= -q(t) e^{-x} + \frac{1}{2} q(0) e^{a^2 t} \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) + e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right] +$$

$$\int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] + e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau \quad (x > 0)$$

§ 12.5 推广的格林公式及其应用

1. 求解 $u_t - a^2 u_{xx} = f(x, t)$, $u|_{t=0} = \varphi(x)$, $u_t|_{t=0} = \psi(x)$.

解:
$$\begin{cases} u_x - a^2 u_{xx} = f(x, t) & \text{将方程化为标准型} & \text{根据方程的特征方程为 } x \pm at = c \\ u|_{t=0} = \varphi(x) & \text{令 } y = at & \text{可将定解问题改写成 } u_{xx} - u_{yy} = -\frac{1}{a^2} f\left(x, \frac{y}{a}\right) \\ u|_{t=0} = \psi(x) & u|_{y=0} = \varphi(x) & u_y|_{y=0} = \psi(x) \end{cases}$$

求 Riemann 函数 R , 方程 $u_{xx} - u_{yy} - \frac{1}{a^2} f\left(x, \frac{y}{a}\right)$ 的系数

$$a_{11} = 1, a_{22} = -1, a_{12} = b_1 = b_2 = c = 0$$

算符 $L = M = \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$ 是自律的, R 是 $Mu = 0$ 的解

$$u = e^{\int_0^s \frac{b_1+b_2}{2} ds} = 1 \text{ (在 } M_2 M_0 \text{ 上)} \quad u = e^{\int_0^s \frac{b_1-b_2}{2} ds} = 1 \text{ (在 } M_0 M_1 \text{ 上)}$$

取 $R(M, M_0) = 1$ 可以满足上述条件.

定解问题的解为

$$u(x, y)R(x, y) = \frac{1}{2} [u(x_1, y_1)R(x_1, y_1) + u(x_2, y_2)R(x_2, y_2)] + \frac{1}{2} \int_{M_1 M_2} [A \cos(n, x) + B \cos(n, y)] dl - \frac{1}{2} \iint_{M_0 M_1 M_2} R f, ds$$

其中 $A = -(uR)_x + (2u_x - b_1 u)R, B = (uR)_y - (2u_y - b_2 u)R$

$M_1(x_1, y_1)$ 坐标 $x_1 = x - y = x - at, y_1 = 0$

$M_2(x_2, y_2)$ 坐标 $x_2 = x + y = x + at, y_2 = 0$

$$R(M_1, M_0) = R(M_2, M_0) = 1.$$

即 $u|_{y_1=0} = \varphi(x_1) = \varphi(x - at), u|_{y_2=0} = \varphi(x_2) = \varphi(x + at)$

由 A, B 表达式得 $A = u_x, B = -u_y$

对于 $M_1 M_2$, 有 $y = 0, \cos(n, x) = 0, \cos(n, y) = -1$

在区域 $M_0 M_1 M_2$ 内, $R = 1 \quad dx = adt$

$$u(x, y) = \frac{1}{2} [\varphi(x - at) + \varphi(x + at)] + \frac{1}{2a} \int_{x-at}^{x+at} \varphi(\xi) d\xi + \frac{1}{2a} \int_0^a \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau$$

2. 求解 $x^2 u_x - t^2 u_{xx} = 0, u|_{t=1} = \varphi(x), u_t|_{t=1} = \psi(x).$

解: 将方程化成标准型 $x^2 - t^2 \left(\frac{dx}{dt}\right)^2 = 0$ 特征线为 $\ln x \pm \ln t = c$

不妨设 $y = \ln x, z = \ln t$ 定解问题为 $u_{yy} - u_{zz} - u_y + u_z = 0, u|_{z=0} = \varphi(e^y)$

$$u_z|_{z=0} = u_t \frac{dt}{dz} \Big|_{t=1} = \psi(e^y)$$

求 Ricmann 函数 R 系数为 $a_{11} = 1 \quad a_{22} = -1 \quad b_1 = -1 \quad b_2 = 1 \quad a_{12} = 0 \quad c = 0$

$$M = \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} + \frac{\partial}{\partial y} - \frac{\partial}{\partial z} \quad MR = 0 \Rightarrow R = e^{\int_0^s \frac{b_1+b_2}{2} ds} = 1 \text{ (在 } M_2 M_0 \text{ 上)}$$

$$R = e^{\int_0^s \frac{b_1-b_2}{2} ds} = e^{-(z-z_0)} \text{ (在 } M_0 M_1 \text{ 上)}$$

取 $R = e^{-\frac{1}{2}(y-y_0+z-z_0)}$ $M_1(x_1, y_1)$ 的坐标为 $y_1 = y_0 - z_0, z_1 = 0.$

$M_2(x_2, y_2)$ 的坐标为 $y_2 = y_0 + z_0 \quad z_2 = 0$

$$R(y_1, z_1) = e^{-\frac{1}{2}(y_1 - z_1 - y_0 + 0 - z_0)} = e^{y_0} = t_0.$$

$$R(y_2, z_2) = e^{-\frac{1}{2}(y_2 + z_2 - z_0 + 0 - y_0)} = 1$$

$$u(y_1, z_1)|_{z_1=0} = 0 = \varphi(e^{y_1}) = \varphi\left(\frac{x_0}{t_0}\right) \quad u(y_2, z_2)|_{z_2=0} = \varphi(e^{y_2}) = \varphi(x_0 t_0)$$

解的形式为 $\frac{1}{2}[u(x_1, y_1)R(x_1, y_1) + u(x_2, y_2)R(x_2, y_2)] + \frac{1}{2} \int_{M_1 M_2} [A \cos(n, x) + B \cos(n, y)] dl - \frac{1}{2} \iint_{M_1 M_2 M_2} R f_1 ds$

在 $M_1 M_2$ 上, $y = 0, \cos(n, y) = 0, \cos(n, z) = -1$

$$B = uR_z - u_z R + b_2 uR = \frac{1}{2} uR - Ru_z \quad (R_z = -\frac{1}{2} R)$$

$$B|_{z=0} = \frac{1}{2} \varphi(e^y) e^{-\frac{1}{2}y} e^{\frac{1}{2}(y_0 + z_0)} - \varphi(e^y) e^{-\frac{1}{2}y} e^{\frac{1}{2}(y_0 + z_0)}$$

由 $e^y = x$, 得 $e^{-\frac{1}{2}y} = x^{-\frac{1}{2}}, e^{\frac{1}{2}(y_0 + z_0)} = (x_0 t_0)^{\frac{1}{2}}$

$$B|_{z=0} = \frac{1}{2} (x_0 t_0)^{\frac{1}{2}} x^{-\frac{1}{2}} \varphi(x) - (x_0 t_0)^{\frac{1}{2}} x^{-\frac{1}{2}} \varphi(x), dy = \frac{dx}{x}$$

$$\frac{1}{2} \int_{M_1 M_2} (-B) dl = -\frac{\sqrt{x_0 t_0}}{4} \int_{\frac{x_0}{t_0}}^{x_0 t_0} \varphi(x) x^{-\frac{3}{2}} dx + \frac{\sqrt{x_0 t_0}}{2} \int_{\frac{x_0}{t_0}}^{x_0 t_0} \varphi(x) x^{-\frac{3}{2}} dx$$

由 $f = 0$ 得,

$$u_{(x_0, t_0)} = \frac{1}{2} t_0 \varphi\left(\frac{x_0}{t_0}\right) + \frac{1}{2} \varphi(x_0, t_0) - \frac{\sqrt{x_0 t_0}}{4} \int_{x_0/t_0}^{x_0 t_0} \varphi(x) x^{-\frac{3}{2}} dx + \frac{\sqrt{x_0 t_0}}{2} \int_{x_0/t_0}^{x_0 t_0} \varphi(x) x^{-\frac{3}{2}} dx$$

第十三章 积分变换法

本章知识要点

积分变换

积分变换,即将函数 $f(x)$ 经过某种可选的积分手续 $F(p) = \int k(x, p) f(x) dx$ 变成另一类函数 $F(p)$. $F(p)$ 称为 $f(x)$ 的像函数. $f(x)$ 称为原函数. $k(x, p)$ 是 x, p 的已知函数,称为积分变换核.

Fourier变换与Laplace变换是两种常用于求解数理方程的积分变换法.前者适用于求解无界空间的定解问题.后者适用于微分方程的初值问题等.

傅里叶变换

1. 定义:设函数在 $(-\infty, +\infty)$ 连续,分段光滑且绝对可积.

则 $G(\omega) = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx$ 为 $f(x)$ 的 Fourier 变换.

$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} G(\omega) e^{i\omega x} d\omega$ 为 $G(\omega)$ 的 Fourier 逆变换.

2. 性质:① 线性: $F(\alpha f_1 + \beta f_2) = \alpha F(f_1) + \beta F(f_2)$

② 延迟: $F[e^{i\omega_0 x} f(x)] = G(\omega - \omega_0)$

③ 位移: $F[f(x - x_0)] = e^{-i\omega x_0} F[f(x)]$

④ 相似: $F[f(ax)] = \frac{1}{|a|} G\left(\frac{\omega}{a}\right)$

⑤ 微分:若当 $|x| \rightarrow \infty$ 时, $f^{(n-1)} x \rightarrow 0$, 则 $F[f^{(n)}(x)] = (i\omega)^n F[f(x)]$

⑥ 积分: $F\left[\int_{x_0}^x f(\epsilon) d\epsilon\right] = \frac{1}{i\omega} F[f(x)]$

⑦ 卷积: $F[f_1(x) * f_2(x)] = F[f_1(x)] \cdot F[f_2(x)]$

$F[f_1(x) \cdot f_2(x)] = \frac{1}{2\pi} F[f_1(x)] * F[f_2(x)]$

卷积: $f_1(x) * f_2(x) = \int_{-\infty}^{+\infty} f_1(\epsilon) f_2(x - \epsilon) d\epsilon$

Laplace 变换

1. 定义: 设 $f(t)$ 满足 (1) $t < 0$ 时, $f(t) = 0$ (2) $t \geq 0$ 时, $f(t)$ 及 $f'(t)$ 除有限个一类间断点外, 处处连续.

(3) 当 $t \rightarrow \infty$ 时, 存在常数 M 及 $\beta_0 \geq 0$ 使 $|f(t)| \leq Me^{\beta_0 t}$, $0 < t < \infty$ 则称 $F(p) = \int_0^\infty f(t)e^{-pt} dt$ 为函数 $f(t)$ 的 Laplace 变换.

记作 $L[f(t)] = F(p)$

称 $f(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} F(p)e^{pt} dp$ 为 $F(p)$ 的 Laplace 变换.

$F(p)$, $f(t)$ 互为像函数与原函数.

2. 性质: ① 线性: $L(\alpha f_1 + \beta f_2) = \alpha L(f_1) + \beta L(f_2)$

② 延迟: $L[e^{p_0 t} f(t)] = F(p - p_0)$, $R_c(p - p_0) > \beta_0$

③ 位移: $L[f(t - \tau)] = e^{-p\tau} F(p)$

④ 相似: $L[f(at)] = \frac{1}{a} F\left(\frac{p}{a}\right)$

⑤ 微分: $L[f^{(n)}(t)] = p^n L[f(t)] - p^{n-1} f(0) - p^{n-2} f'(0) - \dots - f^{(n-1)}(0)$

⑥ 积分: $\left[\int_0^t f(\tau) d\tau \right] = \frac{1}{p} L[f(t)]$

⑦ 卷积: $L[f_1(t) * f_2(t)] = L[f_1(t)] \cdot L[f_2(t)]$

习题全解

§ 13.1 傅里叶变换法

1. 求解无限长传输线上的电振荡传播问题. $G:C = R:L$ 的情形跟 $G:C \neq R:L$ 的情形有什么不同?

解: 无限长传输线上的电振荡传播问题方程为 $LCu_x - u_{xx} + (LG + RC)u_t + RG u$ 对方程变数 x 进行傅里叶变换得

$$v'' + \frac{LG + RC}{LC} v' + \frac{RG + k^2}{LC} v = 0$$

解常微分方程得

$$u(t, k) = A \exp \left[-\frac{R}{2L} t - \frac{G}{2C} t + \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t \right] +$$

$$B \exp \left[-\frac{R}{2L} t - \frac{G}{2C} t - \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C} \right)^2 - \frac{4k^2}{LC}} t \right]$$

考虑到 $t \rightarrow \infty$ 时 u 不可能无限, 所以 $A = 0$. 则对 $v(t, k)$ 进行逆变换得

$$u(x, t) = B' \int e^{ikx} \cdot \exp \left[-\frac{R}{2L}t - \frac{G}{2C}t - \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}}t \right] dk$$

其中 B' 为待定常数, $u(x, k) = e^{ikx} T(t, k) = \exp \left[-\frac{R}{2L}t - \frac{G}{2C}t - \frac{1}{2} \sqrt{\left(\frac{R}{L} - \frac{G}{C}\right)^2 - \frac{4k^2}{LC}}t \right]$

当 $|k| < \frac{\sqrt{LC}}{2} \left| \frac{R}{L} - \frac{G}{C} \right|$ 则 $T(t, k)$ 只衰减不振荡, 传输线相当于高通滤波器.

随 k 不断变化, 称为色散现象. 当 $G:C = R:L$, 则传播速度与频率 k 无关即无色散.

2. 研究半无限长细杆导热问题. 杆端 $x=0$ 温度保持为零. 初始温度分布为 $K(e^{-\lambda x} - 1)$.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 & (x \geq 0) \\ u|_{x=0} = 0 \\ u|_{t=0} = K(e^{-\lambda x} - 1) \end{cases}$$

因为是第一类边界条件, 所以对问题进行奇延拓到 $x < 0$ 的半空间中.

$$\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{t=0} = \varphi(x) = \begin{cases} K(e^{-\lambda x} - 1) & (x > 0) \\ -K(e^{\lambda x} - 1) & (x < 0) \end{cases} \end{cases}$$

对定解问题进行傅里叶变换, 得

$$\begin{cases} u' + k^2 a^2 u = 0 \\ u|_{t=0} = \Phi(k) \end{cases}$$

解泛定方程式并代入初始条件可知

$$u(t, k) = \Phi(k) e^{-k^2 a^2 t}$$

再进行傅里叶逆变换

$$u(x, t) = F^{-1}[u(t, k)] = \int_{-\infty}^{\infty} \Phi(k) e^{-k^2 a^2 t} e^{ikx} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(\xi) e^{-ik\xi} d\xi \right] e^{-k^2 a^2 t} e^{ikx} dk$$

交换积分次序

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\xi) \left[\int_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{ik(x-\xi)} dk \right] d\xi$$

引用积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{\beta k} dk = (\sqrt{\pi}/\alpha) e^{\beta^2/4\alpha^2}$, 置 $\alpha = a\sqrt{t}$, $\beta = i(x-\xi)$ 以利用此积分公式, 即得

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} \varphi(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi \\ &= \int_{-\infty}^0 [-K(e^{\lambda \xi} - 1)] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} [K(e^{-\lambda \xi} - 1)] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \\ &= \int_{-\infty}^0 K \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} (-K) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \\ &\quad \int_{-\infty}^0 (-Ke^{\lambda \xi}) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} Ke^{-\lambda \xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \end{aligned}$$

对右边前两项积分, 第一项中令 $z = \frac{x-\xi}{2a\sqrt{t}}$, $dz = -\frac{d\xi}{2a\sqrt{t}}$, 则第一项积分可化为

$$\int_{-\frac{x}{2a\sqrt{t}}}^{\frac{x}{2a\sqrt{t}}} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

第二项中令 $z = \frac{\xi - x}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$, 则第二项积分可化为 $\int_{-\frac{x}{2a\sqrt{t}}}^{\infty} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$

所以前两项积分可合并为 $\int_{-\frac{x}{2a\sqrt{t}}}^{\frac{x}{2a\sqrt{t}}} (-K) \frac{1}{\sqrt{\pi}} e^{-z^2} dz = (-K) \int_0^{\frac{x}{2a\sqrt{t}}} \frac{L}{\sqrt{\pi}} e^{-z^2} dz = -\operatorname{kerf}\left(\frac{x}{2a\sqrt{t}}\right)$

对右边后两项积分, 对 e 指数进行配方, 得

$$\int_{-\infty}^0 (-K) \frac{1}{2a\sqrt{\pi t}} e^{\frac{[\xi - (x + 2a^2\lambda t)]^2}{4a^2 t}} \cdot e^{a^2\lambda^2 t} e^{\lambda x} d\xi + \int_0^{\infty} K \frac{1}{2a\sqrt{\pi t}} e^{\frac{[\xi - (x - 2a^2\lambda t)]^2}{4a^2 t}} \cdot e^{a^2\lambda^2 t} e^{-\lambda x} d\xi$$

第三项积分中, 令 $z = -\frac{\xi - (x + 2a^2\lambda t)}{2a\sqrt{t}}$, $dz = -\frac{d\xi}{2a\sqrt{t}}$, 则第三项积分可化为

$$\begin{aligned} \int_{-\infty}^{\frac{x+2a^2\lambda t}{2a\sqrt{t}}} \frac{K}{\sqrt{\pi}} e^{-z^2} e^{a^2\lambda^2 t} e^{\lambda x} dz &= \frac{K}{2} e^{a^2\lambda^2 t} e^{\lambda x} \left[\int_{-\infty}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\frac{x+2a^2\lambda t}{2a\sqrt{t}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{K}{2} e^{a^2\lambda^2 t} e^{\lambda x} \left[-1 + \operatorname{erf}\left(\frac{x+2a^2\lambda t}{2a\sqrt{t}}\right) \right] \\ &= -\frac{K}{2} e^{a^2\lambda^2 t} e^{\lambda x} \operatorname{erfc}\left(\frac{x+2a^2\lambda t}{2a\sqrt{t}}\right) \end{aligned}$$

第四项积分中, 令 $z = \frac{\xi - (x - 2a^2\lambda t)}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$, 则第四项积分可化为

$$\begin{aligned} \int_{-\frac{x-2a^2\lambda t}{2a\sqrt{t}}}^{\infty} \frac{K}{\sqrt{\pi}} e^{-z^2} e^{a^2\lambda^2 t} e^{-\lambda x} dz &= \frac{K}{2} e^{a^2\lambda^2 t} e^{-\lambda x} \left[\int_{-\frac{x-2a^2\lambda t}{2a\sqrt{t}}}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\infty} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{K}{2} e^{a^2\lambda^2 t} e^{-\lambda x} \left[-\operatorname{erf}\left(\frac{-x+2a^2\lambda t}{2a\sqrt{t}}\right) + 1 \right] \\ &= \frac{K}{2} e^{a^2\lambda^2 t} e^{-\lambda x} \operatorname{erfc}\left(\frac{-x+2a^2\lambda t}{2a\sqrt{t}}\right) \end{aligned}$$

$$\therefore u(x, t) = \frac{K}{2} e^{a^2\lambda^2 t} \left[e^{-\lambda x} \operatorname{erfc}\left(\frac{-x+2a^2\lambda t}{2a\sqrt{t}}\right) - e^{\lambda x} \operatorname{erfc}\left(\frac{x+2a^2\lambda t}{2a\sqrt{t}}\right) \right] - \operatorname{Kerf}\left(\frac{x}{2a\sqrt{t}}\right)$$

3. 半无界杆, 杆端 $x = 0$ 有谐变热流 $B \sin \omega t$ 进入, 求长时间以后的杆上温度分布 $u(x, t)$.

解: 由题意得定解问题为

$$\begin{cases} u_t - a^2 u_{xx} = 0 & (x > 0) \\ u_x|_{x=0} = \frac{B}{K} \sin \omega t & (K \text{ 为热传导系数}) \end{cases}$$

令 $v(x, t) = u(x, t) + \frac{B}{K} e^{-x} \sin \omega t$ 则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = \frac{a^2 B}{K} e^{-x} \sin \omega t - \frac{\omega B}{K} e^{-x} \cos \omega t & (x > 0) \\ v_x|_{x=0} = 0 \end{cases}$$

因为是第二类齐次边界条件, 所以对问题进行偶延拓到 $x < 0$ 的半空间中去, 则

$$\begin{cases} v_t - a^2 v_{xx} = \frac{a^2 B}{K} e^{-|x|} \sin \omega t - \frac{\omega B}{K} e^{-|x|} \cos \omega t \\ v_x|_{x=0} = 0 \end{cases}$$

对方程两边进行傅里叶变换, 利用公式 $F(e^{-\lambda|x|}) = \frac{1}{\pi\alpha} \cdot \frac{a^2}{a^2 + k^2}$ 得

$$V' + k^2 a^2 V = \frac{a^2 B}{K} \cdot \frac{1}{\pi} \cdot \frac{1}{1+k^2} \sin \omega t - \frac{\omega B}{K} \cdot \frac{1}{\pi} \cdot \frac{1}{1+k^2} \cos \omega t$$

由阶段性常微分方程的解法可知 $V(t, k)$ 的形式应为 $[C(k) \cos \omega t + D(k) \sin \omega t]$ 将其代回方程得

$$[k^2 a^2 D(k) - \omega C(k)] \sin \omega t + [\omega D(k) + k^2 a^2 C(k)] \cos \omega t = \frac{a^2 B}{\pi K} \cdot \frac{1}{1+k^2} \sin \omega t - \frac{\omega B}{\pi K} \cdot \frac{1}{1+k^2} \cos \omega t$$

比较两边系数可知

$$\begin{cases} k^2 a^2 D(k) - \omega C(k) = \frac{a^2 B}{\pi K} \cdot \frac{1}{1+k^2} \\ \omega D(k) + k^2 a^2 C(k) = -\frac{\omega B}{\pi K} \cdot \frac{1}{1+k^2} \end{cases}$$

解得

$$\begin{cases} C(k) = \frac{a^2 B}{\pi K} \cdot \frac{(-\omega)}{k^4 a^4 + \omega^2} \\ D(k) = \frac{B}{\pi K} \cdot \frac{1}{1+k^2} \cdot \frac{k^2 a^4 - \omega^2}{k^4 a^4 + \omega^2} \end{cases}$$

$$\text{则 } V(t, k) = -\frac{a^2 B}{\pi K} \cdot \frac{(-\omega)}{k^4 a^4 + \omega^2} \cos \omega t + \frac{B}{\pi k} \cdot \frac{1}{1+k^2} \cdot \frac{k^2 a^4 - \omega^2}{k^4 a^4 + \omega^2} \sin \omega t$$

对方程两边进行逆傅里叶变换得

$$v(x, t) = F^{-1}[C(k)] \cos \omega t + F^{-1}[D(k)] \sin \omega t.$$

其中

$$C(k) = \frac{-B}{K\pi} \cdot \frac{\omega/a^2}{\left[\left(k + \sqrt{\frac{\omega}{2a^2}}\right)^2 + \frac{\omega}{2a^2}\right] \left[\left(k - \sqrt{\frac{\omega}{2a^2}}\right)^2 + \frac{\omega}{2a^2}\right]}$$

由傅里叶变换公式 $F[\exp(i\beta x - \alpha|x|)] = \frac{1}{\pi x} \cdot \frac{a^2}{a^2 + (k - \beta)^2}$ 可知

$$F[e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}] = \frac{1}{\pi\sqrt{\frac{\omega}{2a^2}}} \cdot \frac{\frac{\omega}{2a^2}}{\frac{\omega}{2a^2} + (k + \sqrt{\frac{\omega}{2a^2}})^2}$$

$$F[e^{i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}] = \frac{1}{\pi\sqrt{\frac{\omega}{2a^2}}} \cdot \frac{\frac{\omega}{2a^2}}{\frac{\omega}{2a^2} + (k - \sqrt{\frac{\omega}{2a^2}})^2}$$

$$\therefore C(k) = -\frac{2\pi B}{K} F[e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}] F[e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}]$$

由傅里叶积分卷积定理可知 $F[f_1(x) * f_2(x)] = 2\pi F_1(\omega) F_2(\omega)$, 则

$$C(x) = F^{-1}[C(k)] = -\frac{B}{K} [(e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}) * (e^{i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|})]$$

$$\begin{aligned} D(k) &= \frac{B}{K\pi} \cdot \frac{1}{1+k^2} \cdot \frac{k^2 a^4 (k^2+1) - (k^4 a^4 + \omega^2)}{k^4 a^4 + \omega^2} = \frac{B}{K\pi} \cdot \left[\frac{k^2 a^4}{k^4 a^4 + \omega^2} - \frac{1}{1+k^2} \right] \\ &= \frac{B}{K\pi} \left[\frac{k^2}{\left[\left(k + \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right]} - \frac{k^2}{\left[\left(k - \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right]} - \frac{1}{1+k^2} \right] \\ &= \frac{B}{K} \left[\frac{1}{2\pi \left[\left(k + \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right]} + \frac{1}{2\pi \left[\left(k - \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right]} - \frac{1}{\pi \left[\left(k + \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right] \left[\left(k - \sqrt{\frac{\omega}{2a^2}} \right)^2 + \frac{\omega}{2a^2} \right]} - \frac{1}{1+k^2} \right] \end{aligned}$$

用上述傅里叶变换公式及卷积定理,类似求 $F^{-1}[C(k)]$ 过程可得

$$D(x) = F^{-1}[D(k)]$$

$$\begin{aligned} &= \frac{B}{K} \left[\frac{\sqrt{2}a}{2\sqrt{\omega}} e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{\sqrt{\frac{\omega}{2a^2}}|x|} + \frac{\sqrt{2}a}{2\sqrt{\omega}} e^{i\sqrt{\frac{\omega}{2a^2}}x} e^{\sqrt{\frac{\omega}{2a^2}}|x|} - [(e^{-i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}) \right. \\ &\quad \left. * (e^{i\sqrt{\frac{\omega}{2a^2}}x} e^{-\sqrt{\frac{\omega}{2a^2}}|x|}) - e^{-x}] \right] \end{aligned}$$

因为问题为求解长时间后的分布,所以分布的时间变化频率与外界输入变化频率相同,则 $u(x,t)$ 解的形式应为 $A(x)\sin[\omega t + \varphi(x)]$. 结合已求得的

$v(x,t) = C(x)\cos\omega t + D(x)\sin\omega t$, 可求得

$$v(x,t) = \frac{aB}{K\sqrt{\omega}} e^{-x\sqrt{\frac{\omega}{2a^2}}} \sin\left(\omega t - x\sqrt{\frac{\omega}{2a^2}} - \frac{\pi}{4}\right) + \frac{B}{K} e^{-x} \sin\omega t$$

$$\therefore u(x,t) = v(x,t) = \frac{B}{K} e^{-x} \sin\omega t = \frac{aB}{K\sqrt{\omega}} e^{-x\sqrt{\frac{\omega}{2a^2}}} \sin\left(\omega t - x\sqrt{\frac{\omega}{2a^2}} - \frac{\pi}{4}\right)$$

4. 应用泊松公式计算下述定解问题求解. $u_{xx} - a^2 \Delta u = 0$, 初始速度为零, 初始位移在某个单位球内为 1, 在球外为零.

解: 由题意得定解问题为

$$\begin{cases} u_{xx} - a^2 \Delta u = 0 \\ u|_{t=0} = \begin{cases} 1 & (\text{单位球内}) \\ 0 & (\text{单位球外}) \end{cases} \\ u_t|_{t=0} = 0 \end{cases}$$

将坐标系建立在单位球心, 已知泊松公式为 $u(x,t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S'_a} \frac{\varphi(\vec{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S'_a} \frac{\varphi(\vec{r}')}{at} dS$ 则

由定解条件知 $\varphi(\vec{r}')$ 始终为 0, 当在球内

(1) 当 S'_a 完全在单位球内时, $r+at < 1$, 即当 $t < \frac{1-r}{a}$ 时, $\varphi(\vec{r}') = 1$

$$\text{则 } u(x, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{1}{at} dS' = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{4\pi} \frac{1}{at} \cdot (at)^2 d\Omega = \frac{1}{4\pi a} \cdot 4\pi a = 1$$

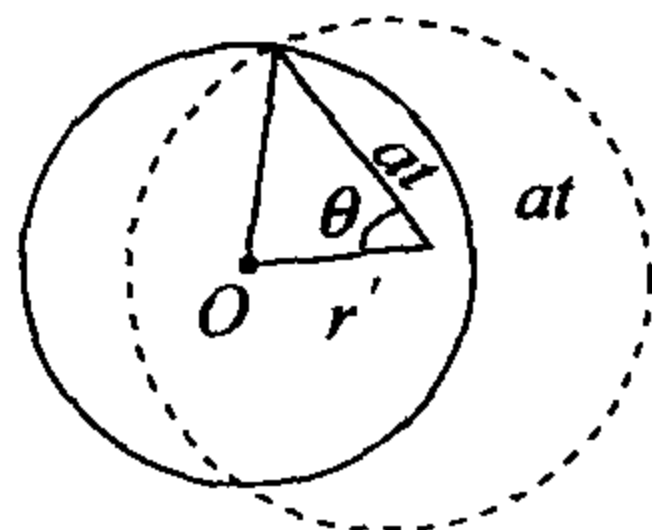
(2) 当 S_a 部分在单位球内时, $r + at > 1, at - r < 1$. 即当 $\frac{1-r}{a} < t < \frac{1+r}{a}, \varphi(\vec{r'})$

$$= \begin{cases} 1 & (\text{球内}) \\ 0 & (\text{球外}) \end{cases}$$

$$\text{则 } u(x, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{1}{at} dS' = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^\theta \frac{1}{at} \cdot 2\pi(at)^2 \sin \theta' d\theta' = \frac{1}{2} \frac{\partial}{\partial t} (t - t \cos \theta)$$

其中, 由两球关系可知 $\cos \theta = \frac{a^2 t^2 + r^2 - 1}{2rat}$

$$\text{则 } u(x, t) = \frac{1}{2} \frac{\partial}{\partial t} \left(t - \frac{a^2 t^2 + r^2 - 1}{2ra} \right) = \frac{1}{2} \left(1 - \frac{at}{r} \right) = \frac{1}{2r} (r - at)$$



(3) 当 S_a 完全包括单位球时, $at - r > 1$. 即当 $t > \frac{r+1}{a}$ 时, $\varphi(\vec{r'}) = 0$, 则

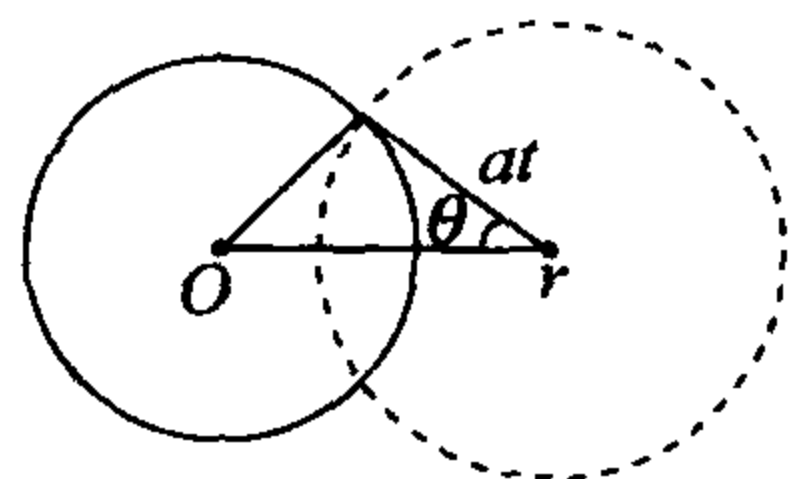
$$u(x, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{0}{at} dS' = 0$$

当在球外

(1) 当 S_a 完全在单位球外时, $r - at > 1$. 即当 $t < \frac{r-1}{a}$ 时, $\varphi(\vec{r'}) = 0$ 则 $u(x, t) = 0$.

(2) 由 S_a 部分在单位球内时 $r = at < 1$, 即当 $\frac{r-1}{a} < t < \frac{r+1}{a}$ 时,

$$\varphi(\vec{r'}) = \begin{cases} 1 & (\text{球内}) \\ 0 & (\text{球外}) \end{cases} \text{ 则}$$



$$u(x, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{1}{at} dS' = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^\theta \frac{1}{at} \cdot 2\pi(at)^2 \sin \theta' d\theta' = \frac{1}{2} \frac{\partial}{\partial t} (t - t \cos \theta)$$

其中, 由两球关系可知 $\cos \theta = \frac{a^2 t^2 + r^2 - 1}{2rat}$

$$\text{则 } u(x, t) = \frac{1}{2} \frac{\partial}{\partial t} \left(t - \frac{a^2 t^2 + r^2 - 1}{2ra} \right) = \frac{1}{2} \left(1 - \frac{at}{r} \right) = \frac{1}{2r} (r - at)$$

(3) 当 S_a 完全包括单位球时, $r - at < -1$, 即当 $t > \frac{r+1}{a}$ 时, $\varphi(\vec{r'}) = 0$. 则 $u(x, t) = 0$

5. 应用泊松公式计算下述定解问题的解. $u_x - a^2 \Delta u = 0$, 初始速度为零, 初始位移在球 $r = r_0$ 以内为 $A \cos(\pi r / 2r_0)$, 在球外为零.

解: 由题意得定解问题为

$$\begin{cases} u_x - a^2 \Delta u = 0 \\ u|_{t=0} = \begin{cases} A \cos(\pi r / 2r_0) & (\text{球内}) \\ 0 & (\text{球外}) \end{cases} \\ u_t|_{t=0} = 0 \end{cases}$$

已知泊松公式为 $u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{\varphi(\vec{r'})}{at} dS' + \frac{1}{4\pi a} \iint_{S_a} \frac{\varphi(\vec{r'})}{at} dS'$. 由定解条件知 $\varphi(\vec{r'}) = 0$, 则

$$u(\vec{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{\varphi(\vec{r}')}{at} dS'$$

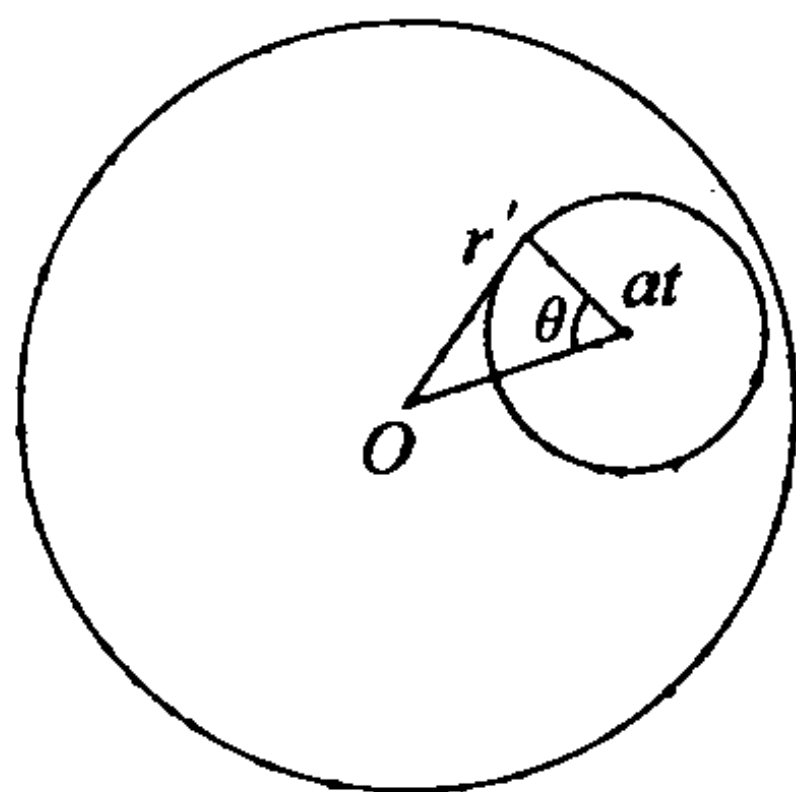
(I) 当 $r < r_0$ 时

(1) 当 $r + at < r_0$, 即 $t < \frac{r_0 - r}{a}$ 时, S_a 完全在半径为 r_0 球内, 则

$$\begin{aligned} u(r, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{A \cos(\frac{\pi r'}{2r_0})}{at} dS' \\ &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^\pi \frac{A \cos(\frac{\pi r'}{2r_0})}{at} \cdot 2\pi(at)^2 \sin \theta d\theta \end{aligned}$$

由两球几何关系可知, $r'^2 = r^2 + a^2 t^2 - 2rat \cos \theta$, 则 $2r' dr' = 2rat \sin \theta d\theta$

$$\begin{aligned} u(r, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{r-at}^{r+at} \frac{A \cos(\frac{\pi r'}{2r_0})}{at} \cdot 2\pi(at)^2 \cdot \frac{r'}{rat} dr' \\ &= \frac{A}{2ra} \frac{\partial}{\partial t} \int_{r-at}^{r+at} r' \cos(\frac{\pi r'}{2r_0}) dr' \\ &= \frac{A}{2r} \left[(r+at) \cos \frac{\pi(r+at)}{2r_0} + (r-at) \cos \frac{\pi(r-at)}{2r_0} \right] \end{aligned}$$



(2) 当 $r + at > r_0, r - at > -r_0$, 即 $\frac{r_0 - r}{a} < t < \frac{r_0 + r}{a}$ 时, S_a 部分在半径

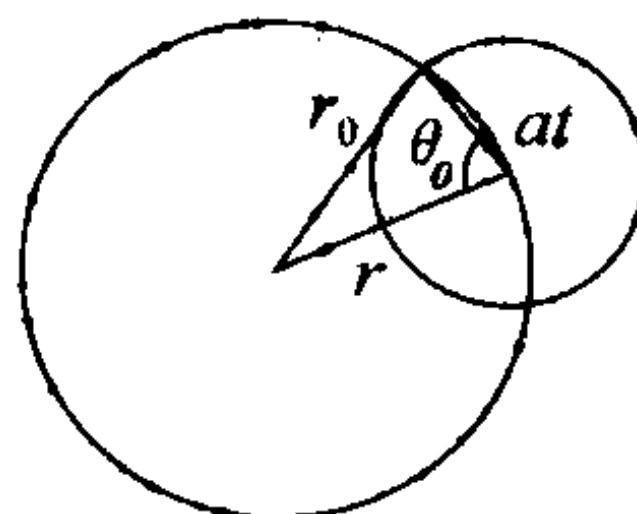
为 r_0 球内, 则

$$u(r, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{\theta_0} \frac{A \cos(\frac{\pi r'}{2r_0})}{at} \cdot 2\pi(at)^2 \sin \theta d\theta$$

由两球几何关系可知 $r_0^2 = r^2 + a^2 t^2 - 2rat \cos \theta_0$

所以

$$\begin{aligned} u(r, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{r-at}^{r_0} \frac{A \cos(\frac{\pi r'}{2r_0})}{at} \cdot 2\pi(at)^2 \frac{r'}{2at} dr' \\ &= \frac{A}{2ra} \frac{\partial}{\partial t} \int_{r-at}^{r_0} r' \cos(\frac{\pi r'}{2r_0}) dr' \\ &= \frac{A}{2r} (r-at) \cos \frac{\pi(r-at)}{2r_0} \end{aligned}$$



(3) 当 $r - at < -r_0$, 即 $t > \frac{r_0 + r}{a}$ 时, S_a 包围了半径为 r_0 的球, 此时 $\varphi(\vec{r}') = 0$, 则

$$u(r, t) = 0$$

(II) 当 $r > r_0$ 时

(1) 当 $r - at > r_0$ 时, 即 $t < \frac{r - r_0}{a}$ 时, S_a 完全在半径为 r_0 球外, 则 $\varphi(\vec{r}') = 0, \therefore u(r, t) = 0$.

(2) 当 $r - at < r_0, r - at > -r_0$, 即 $\frac{r - r_0}{a} < t < \frac{r + r_0}{a}$ 时, S_a 部分在半径为 r_0 球内, 则

$$u(r, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_0^{2\pi} \frac{A \cos(\frac{\pi r_1}{2r_0})}{at} \cdot 2\pi(at)^2 \sin \theta d\theta. \text{ 情况与(I)(2) 完全相同}$$

$$\therefore u(r, t) = \frac{A}{2r} (r - at) \cos \frac{\pi(r - at)}{2r_0}$$

(3) 当 $r - at < r_0$, 即 $t > \frac{r + r_0}{a}$ 时, S_a 包围了半径为 r_0 的球, 此时 $\varphi(r') = 0$. 则

$$u(r, t) = 0$$

6. 二维波动, 初始速度为零, 初始位移在圆 $\rho = 1$ 以内为 1, 在圆外为零. 试求 $u|_{\rho=0}$.

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 \Delta_2 u = 0 \\ u|_{t=0} = \begin{cases} 1 & (\text{圆内}) \\ 0 & (\text{圆外}) \end{cases} \\ u_t|_{t=0} = 0 \end{cases}$$

已知二维问题的泊松公式为 $u(x, y, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_t} \frac{\varphi(x', y')}{\sqrt{a^2 t^2 - (x - x')^2 - (y - y')^2}} dx' dy' +$

$$\frac{1}{2\pi a} \frac{\varphi(x', y')}{\sqrt{a^2 t^2 - (x - x')^2 - (y - y')^2}} dx' dy'$$

由定解条件知 $\varphi(x', y') = 0$, 则

$$u|_{\rho=0} = u(0, 0, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_t^{(0,0)}} \frac{\varphi(x', y')}{\sqrt{a^2 t^2 - x'^2 - y'^2}} dx' dy'$$

(1) 当 $at < 1$, 即 $t < \frac{1}{a}$ 时, $\Sigma_t^{(0,0)}$ 完全在单位圆内, 则

$$\begin{aligned} u|_{\rho=0} &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_t^{(0,0)}} \frac{1}{\sqrt{a^2 t^2 - x'^2 - y'^2}} dx' dy' = \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^{at} \int_0^{2\pi} \frac{1}{\sqrt{a^2 t^2 - \rho'^2}} \rho' d\theta d\rho' \\ &= \frac{1}{2\pi a} \cdot 2\pi \cdot \frac{\partial}{\partial t} \left[\frac{1}{2} \cdot (-2 \sqrt{a^2 t^2 - \rho'^2}) \Big|_0^{at} \right] = \frac{1}{a} \frac{\partial}{\partial t} (at) = 1 \end{aligned}$$

(2) 当 $at > 1$, 即 $t > \frac{1}{a}$ 时, $\Sigma_t^{(0,0)}$ 完全将单位圆包围, 则

$$\begin{aligned} u|_{\rho=0} &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \iint_{\Sigma_t^{(0,0)}} \frac{1}{\sqrt{a^2 t^2 - x'^2 - y'^2}} dx' dy' = \frac{1}{2\pi a} \frac{\partial}{\partial t} \int_0^1 \int_0^{2\pi} \frac{1}{\sqrt{a^2 t^2 - \rho'^2}} \rho' d\theta d\rho' \\ &= \frac{1}{2\pi a} \cdot 2\pi \cdot \frac{\partial}{\partial t} \left[\frac{1}{2} (-2 \sqrt{a^2 t^2 - \rho'^2}) \Big|_0^1 \right] = \frac{1}{a} \frac{\partial}{\partial t} (at - \sqrt{a^2 t^2 - 1}) = 1 - \frac{at}{\sqrt{a^2 t^2 - 1}} \end{aligned}$$

7. 求解三维无界空间中的输运问题 $u_t - a^2 \Delta u = 0, u|_{t=0} = \varphi(x, y, z)$.

解: 由题意可知定解问题为

$$\begin{cases} u_t - a^2 \Delta u = 0 \\ u|_{t=0} = \varphi(x, y, z) \end{cases}$$

作傅里叶变换问题转化为常微分方程的初始值问题

$$\begin{cases} U' + k^2 a^2 U = 0 \\ U|_{t=0} = \Phi(k_1, k_2, k_3) \end{cases}$$

这个问题的解为

$$U(t, \bar{k}) = \Phi(k_1, k_2, k_3) e^{-k^2 a^2 t}$$

再进行傅里叶逆变换

$$\begin{aligned} U(\bar{r}, t) &= \iiint_{-\infty}^{\infty} [\Phi(k_1, k_2, k_3) e^{-k^2 a^2 t}] e^{i\bar{k} \cdot \bar{r}} dk_1 dk_2 dk_3 \\ &= \frac{1}{8\pi^3} \iiint_{-\infty}^{\infty} \left[\iiint_{-\infty}^{\infty} \varphi(x', y', z') e^{-i\bar{k} \cdot \bar{r}'} dx' dy' dz' \right] e^{-k^2 a^2 t} e^{i\bar{k} \cdot \bar{r}} dk_1 dk_2 dk_3 \end{aligned}$$

交换积分次序

$$u(\bar{r}, t) = \frac{1}{8\pi^3} \iiint_{-\infty}^{\infty} \varphi(x', y', z') \left[\iiint_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\bar{k}(\bar{r} - \bar{r}')} dk_1 dk_2 dk_3 \right] dx' dy' dz'$$

引用积分公式

$$\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{i\beta k} dk = (\sqrt{\pi}/\alpha) e^{-\beta^2/4\alpha^2}$$

$$\text{则 } \iiint_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\bar{k}(\bar{r} - \bar{r}')} dk_1 dk_2 dk_3 = \left[\int_{-\infty}^{\infty} e^{-k_1^2 a^2 t} e^{i k_1 (x - x')} dk_1 \right] \left[\int_{-\infty}^{\infty} e^{-k_2^2 a^2 t} e^{i k_2 (y - y')} dk_2 \right] \left[\int_{-\infty}^{\infty} e^{-k_3^2 a^2 t} e^{i k_3 (z - z')} dk_3 \right]$$

对三个括号中, 分别令 $\alpha = a\sqrt{t}$, $\beta = i(x - x')$, $i(y - y')$, $i(z - z')$ 利用积分公式可得

$$\iiint_{-\infty}^{\infty} e^{-k^2 a^2 t} e^{i\bar{k}(\bar{r} - \bar{r}')} dk_1 dk_2 dk_3 = \left(\frac{\sqrt{\pi}}{a\sqrt{t}} \right)^3 e^{\frac{-(x-x')^2}{4a^2 t}} e^{\frac{-(y-y')^2}{4a^2 t}} e^{\frac{-(z-z')^2}{4a^2 t}} = \left(\frac{\sqrt{\pi}}{a\sqrt{t}} \right)^3 e^{\frac{-(\bar{r}-\bar{r}')^2}{4a^2 t}}$$

$$\text{则 } u(\bar{r}, t) = \frac{1}{8\pi^3} \iiint_{-\infty}^{\infty} \varphi(x', y', z') \cdot \left(\frac{\sqrt{\pi}}{a\sqrt{t}} \right)^3 e^{\frac{-(\bar{r}-\bar{r}')^2}{4a^2 t}}$$

$$\text{即 } u(\bar{r}, t) = \iiint_{-\infty}^{\infty} \varphi(x', y', z') \frac{1}{(2a\sqrt{\pi t})^3} e^{\frac{-(\bar{r}-\bar{r}')^2}{4a^2 t}} dx' dy' dz'$$

8. 教材中例 6 研究三维无界空间中的自由振动是从初始($t = 0$) 状态推算以后($t > 0$) 的状态. 试重新求解例 6, 从初始状态反推以前($t < 0$) 的状态.

解: 由例 6 可知, 无论 $t > 0$ 还是 $t < 0$, 都有

$$\begin{aligned} u(\bar{r}, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \varphi(\bar{r}') \left[\iiint_{-\infty}^{\infty} \frac{1}{4\pi^2} \cdot \frac{1}{ik} (e^{ik a t} - e^{-ik a t}) e^{i\bar{k} \cdot (\bar{r} - \bar{r}')} dk_1 dk_2 dk_3 \right] dV' \\ &\quad + \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \varphi(\bar{r}') \left[\iiint_{-\infty}^{\infty} \frac{1}{4\pi^2} \cdot \frac{1}{ik} (e^{ik a t} - e^{-ik a t}) e^{i\bar{k} \cdot (\bar{r} - \bar{r}')} dk_1 dk_2 dk_3 \right] dV' \end{aligned}$$

此时, 由于 $t < 0$, 不能直接引用 $F\left[\frac{1}{r}\delta(r-c)\right] = \frac{1}{(2\pi)^2} \frac{1}{ik} (e^{ikc} - e^{-ikc})$ ($c > 0$)

令 $-t = t'$, 则 $t' > 0$. 于是利用上式得到

$$u(\mathbf{r}, t) = \frac{(-1)}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \varphi(\mathbf{r}') \iiint_{-\infty}^{\infty} F\left[\frac{1}{r} \delta(r - at') e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} dk_1 dk_2 dk_3\right] dV' + \frac{(-1)}{4\pi a} \iiint_{-\infty}^{\infty} \varphi(\mathbf{r}') \\ \iiint_{-\infty}^{\infty} F\left[\frac{1}{r} \delta(r - at') e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} dk_1 dk_2 dk_3\right] dV'$$

应用延迟定理

$$u(\mathbf{r}, t) = \frac{(-1)}{4\pi a} \frac{\partial}{\partial t} \iiint_{-\infty}^{\infty} \frac{\varphi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \delta(|\mathbf{r} - \mathbf{r}'| - at') dV' + \frac{(-1)}{4\pi a} \iiint_{-\infty}^{\infty} \frac{\varphi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \delta(|\mathbf{r} - \mathbf{r}'| - at') dV'$$

由于被积式中出现 $\delta(|\mathbf{r} - \mathbf{r}'| - at')$, 所以仅当 $|\mathbf{r} - \mathbf{r}'| = at'$ 时积分不为 0, 即对 $\mathbf{r}'$ 的积分只需在球面 $S_{a'}$ 上进行, 则

$$u(\mathbf{r}, t) = \frac{(-1)}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{a'}} \frac{\varphi(\mathbf{r}')}{at'} dS' + \frac{(-1)}{4\pi a} \iint_{S_{a'}} \frac{\varphi(\mathbf{r}')}{at'} dS'$$

$$\because t' = |t| = -t$$

$$\therefore u(\mathbf{r}, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{|t|}} \frac{\varphi(\mathbf{r}')}{at} dS' + \frac{1}{4\pi a} \iint_{S_{|t|}} \frac{\varphi(\mathbf{r}')}{at} dS'$$

9. 求解一维半无界空间的输运问题 $u_t - a^2 u_{xx} = 0, u|_{x=0} = At, u|_{t=0} = 0$.

解: 令 $u(x, t) = At + v(x, t)$. 则

$$\begin{cases} v_t - a^2 v_{xx} = -A & (x > 0) \\ v|_{x=0} = 0 \\ v|_{t=0} = 0 \end{cases}$$

因为是第一类齐次边界条件, 应将问题奇延拓到 $x < 0$ 的半空间中去. 于是

$$\begin{cases} v_t - a^2 v_{xx} = \begin{cases} -A & (x > 0) \\ A & (x < 0) \end{cases} \\ v|_{t=0} = 0 \end{cases}$$

代入式(13.1.3)可知

$$v(x, t) = \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) \left[\frac{1}{2a\sqrt{t(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ = \int_0^t \left[\int_{-\infty}^0 A \cdot \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(\xi-x)^2}{4a^2(t-\tau)}} d\xi + \int_0^{\infty} (-A) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{x^2+\xi^2}{4a^2(t-\tau)}} d\xi \right] d\tau$$

在右边[]内第一个积分中令 $z = (x - \xi)/2a\sqrt{t}$, $dz = -d\xi/2a\sqrt{t}$

在右边[]内第二个积分中令 $z = (\xi - x)/2a\sqrt{t}$, $dz = d\xi/2a\sqrt{t}$ 于是

$$v(x, t) = \int_0^t \left(-\frac{A}{\sqrt{\pi}} \int_{-\infty}^{x/2a\sqrt{t-\tau}} e^{-z^2} dz - \frac{A}{\sqrt{\pi}} \int_{-x/2a\sqrt{t-\tau}}^{\infty} e^{-z^2} dz \right) d\tau \\ = -A \int_{\tau=0}^t \frac{2}{\sqrt{\pi}} \int_0^{x/2a\sqrt{t-\tau}} e^{-z^2} dz d\tau \\ = -A \int_{\tau=0}^t \operatorname{erf}(x/2a\sqrt{t-\tau}) d\tau$$

$$\therefore u(x, t) = At + v(x, t) = At - A \int_0^t \operatorname{erf}(x/2a \sqrt{t-\tau}) d\tau = A \int_0^t \operatorname{erfc}(x/2a \sqrt{t-\tau}) d\tau$$

10. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0, u|_{x=0} = f(t), u|_{t=0} = \varphi(x)$.

解: 令 $u(x, t) = v(x, t) + f(t)$, 则 $v(x, t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = -f'(t) & (x > 0) \\ v_{x=0} = 0 \\ v|_{t=0} = \varphi(x) - \varphi(0) \end{cases}$$

因为是第一类齐次边界条件, 所以进行奇延拓到 $x < 0$ 的半空间中, 则

$$\begin{cases} v_t - a^2 v_{xx} = f_0(t) = \begin{cases} -f'(t) & (x > 0) \\ f'(t) & (x < 0) \end{cases} \\ v|_{t=0} = \varphi_0(x) = \begin{cases} \varphi(x) - f(0) & (x > 0) \\ -\varphi(-x) + f(0) & (x < 0) \end{cases} \end{cases}$$

对定解问题进行傅里叶变换, 问题转化为

$$\begin{cases} v' + k^2 a^2 v = F_0(t, k) \\ v|_{t=0} = \Phi_0(k) \end{cases}$$

用 $e^{k^2 a^2 t}$ 乘通方程各项得

$$\frac{d}{dt}[v(t, k)e^{k^2 a^2 t}] = F_0(t, k)e^{k^2 a^2 t}$$

方程两边同时对 t 从 0 到 t 积分, 并代入初始条件得

$$V(t, k)e^{k^2 a^2 t} - \Phi(k) = \int_0^t F_0(\tau, k)e^{k^2 a^2 \tau} d\tau$$

即

$$\begin{aligned} V(t, k) &= \Phi_0(k)e^{-k^2 a^2 t} + e^{-k^2 a^2 t} \int_0^t F_0(\tau, k)e^{k^2 a^2 \tau} d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \varphi_0(\xi) e^{-k\xi} d\xi e^{-k^2 a^2 t} + \frac{1}{2\pi} \int_0^t \int_{-\infty}^{+\infty} f_0(\xi, \tau) e^{-k\xi} d\xi e^{-k^2 a^2 (t-\tau)} d\tau \end{aligned}$$

方程两边进行逆傅叶变换

$$v(x, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} \varphi_0(\xi) e^{-k\xi} e^{-k^2 a^2 t} d\xi \right] e^{kx} dk + \frac{1}{2\pi} \int_0^t \left[\int_{-\infty}^{+\infty} f_0(\xi, \tau) e^{-k\xi} e^{-k^2 a^2 (t-\tau)} d\xi d\tau \right] e^{kx}$$

对右边两个积分分别变换积分次序得

$$v(x, t) = \int_{-\infty}^{+\infty} \varphi_0(\xi) \left[\int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-k^2 a^2 t} e^{k(x-\xi)} dk \right] d\xi + \int_0^t \int_{-\infty}^{+\infty} f_0(\xi, \tau) \left[\int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-k^2 a^2 (t-\tau)} e^{k(x-\xi)} dk \right] d\xi d\tau$$

引用积分分式 $\int_{-\infty}^{+\infty} e^{-a^2 k^2} e^{k\beta} dk = (\sqrt{\pi}/a) e^{\beta^2/4a^2}$. 置 $\alpha = a\sqrt{t}$ (或 $a\sqrt{t-\tau}$), $\beta = i(x-\xi)$, 利用比积分公式得到

$$v(x, t) = \int_{-\infty}^{+\infty} \varphi_0(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^t \int_{-\infty}^{+\infty} f_0(\xi, \tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2 (t-\tau)}} \right] d\xi d\tau$$

代入 $\varphi_0(x), f_0(t)$ 表达式, 即得

$$v(x, t) = \int_0^t \int_{-\infty}^0 [-\varphi(-\xi) + f(0)] \frac{1}{2a\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2 t} d\xi d\tau + \int_0^t \int_0^{\infty} [\varphi(\xi) - f(0)] \frac{1}{2a\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2 t}$$

$$d\xi d\tau + \int_0^t \int_{-\infty}^0 f'(\tau) = \frac{1}{2a\sqrt{\pi t-\tau}} e^{-(x-\xi)^2/4a^2(t-\tau)} d\xi d\tau + \int_0^t \int_0^\infty (-f'(\tau)) \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)} d\xi d\tau$$

右边第一项积分中, 令 $z = \frac{x-\xi}{2a\sqrt{t}}$, $dz = -\frac{d\xi}{2a\sqrt{t}}$. 则第一项积分可化为

$$-\int_{-\infty}^{x/2a\sqrt{t}} [-\varphi(-x+2a\sqrt{t}\tau) + f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

右边第二项积分中, 令 $z = \frac{\xi-x}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$, 则第二项积分可化为

$$\int_{-x/2a\sqrt{t}}^\infty [\varphi(x+2a\sqrt{t}z) - f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

所以前两项积分合并为

$$-f(0) \left[\int_{-\infty}^{x/2a\sqrt{t}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz + \int_{-x/2a\sqrt{t}}^\infty \frac{1}{\sqrt{\pi}} e^{-z^2} dz \right] + \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

进一步合并得 $-f(0)\operatorname{erf}(x/2a\sqrt{t}) + \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$

右边第三项积分中, 令 $z = \frac{x-\xi}{2a\sqrt{t-\tau}}$, $dz = \frac{-d\xi}{2a\sqrt{t-\tau}}$, 则第三项积分可化为 $-\int_0^t \frac{x}{2a\sqrt{t-\tau}} f'(\tau)$

$$\frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

右边第四项积分中, 令 $z = \frac{\xi-x}{2a\sqrt{t-\tau}}$, $dz = \frac{d\xi}{2a\sqrt{t-\tau}}$, 则第四项积分可化为

$$-\int_0^t \int_{-\frac{x}{2a\sqrt{t-\tau}}}^\infty f'(\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

所以后两项积分合并为

$$-\int_0^t f'(\tau) \int_{-\frac{x}{2a\sqrt{t-\tau}}}^\infty \frac{1}{\sqrt{\pi}} e^{-z^2} dz = -\int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau$$

$$v(x,t) = -\int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) + \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz -$$

$$\int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

$$u(x,t) = f(t) - \int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right) + \int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz -$$

$$\int_{-\frac{x}{2a\sqrt{t}}}^\infty \varphi(-x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

其中, 可将 $f(t)$ 写成 $\int_0^t f'(\tau) d\tau + f(0)$, 则

$$f(t) - \int_0^t f'(\tau) \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau - f(0) \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$= \int_0^t f'(\tau) \left[1 - \operatorname{erf}\left(\frac{x}{2a\sqrt{t-\tau}}\right)\right] d\tau + f(0) \left[1 - \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)\right]$$

$$= \int_0^t f'(\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau + f(0) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$u(x,t) = f(0) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right) + \int_0^t f'(\tau) \operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right) d\tau + \int_{-x/2a\sqrt{t}}^{\infty} \varphi(x+2a\sqrt{t}z) \frac{1}{\sqrt{\pi}} e^{-z^2} dz - \int_{x/2a\sqrt{t}}^{\infty} \varphi(-x+2a\sqrt{t}\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

11. 在一维半无界空间中求解 $u_t - a^2 u_{xx} = 0, u_x \Big|_{x=0} = q(t), u \Big|_{t=0} = 0$.

解: 令 $u(x,t) = v(x,t) - q(t)e^{-x}$, 则 $v(x,t)$ 满足

$$\begin{cases} v_t - a^2 v_{xx} = q'(t)e^{-x} - a^2 q(t)e^{-x} & (x > 0) \\ v_x \Big|_{x=0} = 0 \\ v \Big|_{t=0} = q(0)e^{-x} \end{cases}$$

因为是第二类齐次边界条件, 所以进行偶延拓到 $x < 0$ 的半空间中去, 得到

$$\begin{cases} v_t - a^2 v_{xx} = f(x,t) = \begin{cases} [q'(t) - a^2 q(t)]e^{-x} & (x > 0) \\ [q'(t) - a^2 q(t)]e^x & (x < 0) \end{cases} \\ v \Big|_{t=0} = \varphi(x) = \begin{cases} q(0)e^{-x} \\ q(0)e^x \end{cases} & (x < 0) \end{cases}$$

对定解问题进行傅里叶变换, 问题转化为

$$\begin{cases} V' + k^2 a^2 V = F(t,k) \\ V \Big|_{t=0} = \Phi(k) \end{cases}$$

用 $e^{k^2 a^2 t}$ 乘遍方程各项得

$$\frac{d}{dt}[V(t,k)e^{k^2 a^2 t}] = F(t,k)e^{k^2 a^2 t}$$

方程两边同时对 t 从 0 到 t 积分, 并代入初始条件得

$$V(t,k)e^{k^2 a^2 t} - \Phi(k) = \int_0^t F(\tau,k)e^{k^2 a^2 \tau} d\tau$$

$$\text{即 } V(t,k) = \Phi(k)e^{-k^2 a^2 t} + e^{-k^2 a^2 t} \int_0^t F(\tau,k)e^{k^2 a^2 \tau} d\tau$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\xi) e^{-k\xi} d\xi \cdot e^{-k^2 a^2 t} + \frac{1}{2\pi} \int_0^t \int_{-\infty}^{\infty} f(\xi,\tau) e^{-k\xi} d\xi e^{-k^2 a^2 (t-\tau)} d\tau$$

方程两边进行逆傅里叶变换

$$v(x,t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \varphi(\xi) e^{-k\xi} e^{-k^2 a^2 t} d\xi \right] e^{kx} dk + \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_0^t \int_{-\infty}^{\infty} f(\xi,\tau) e^{-k\xi} e^{-k^2 a^2 (t-\tau)} d\xi d\tau \right] e^{kx} dk$$

对右边两个积分分别变换积分次序得

$$v(x,t) = \int_{-\infty}^{\infty} \varphi(\xi) \left[\int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-k^2 a^2 t} e^{k(x-\xi)} dk \right] d\xi + \int_0^t \int_{-\infty}^{\infty} f(\xi,\tau) \left[\int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-k^2 a^2 (t-\tau)} e^{k(x-\xi)} dk \right] d\xi d\tau$$

利用积分公式 $\int_{-\infty}^{\infty} e^{-\alpha^2 k^2} e^{ikx} dk = (\sqrt{\pi}/\alpha) e^{x^2/4\alpha^2}$, 置 $\alpha = a\sqrt{t}$ (或 $a\sqrt{t-\tau}$), $\beta = i(x-\xi)$, 利用此积分公式得到

$$v(x, t) = \int_{-\infty}^{\infty} \varphi(\xi) \left[\frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} \right] d\xi + \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau$$

代入 $\varphi(x)$, $f(x, t)$ 表达式, 即得

$$\begin{aligned} v(x, t) = & \int_{-\infty}^0 q(0) e^{\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \int_0^{\infty} q(0) e^{-\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi + \\ & \int_0^t \int_{-\infty}^0 [q'(\tau) - a^2 q(\tau)] e^{\xi} \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi d\tau + \\ & \int_0^t \int_0^{\infty} [q'(\tau) - a^2 q(\tau)] e^{-\xi} \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} d\xi d\tau \end{aligned}$$

对前两项积分, 对 e 指数进行配方得

$$\int_{-\infty}^0 q(0) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x+2a^2 t)-\xi)^2}{4a^2 t}} e^{a^2 t} e^x d\xi d\tau + \int_0^{\infty} q(0) \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-2a^2 t)-\xi)^2}{4a^2 t}} e^{a^2 t} e^{-x} d\xi d\tau$$

在第一项积分中, 令 $z = \frac{(x+2a^2 t)-\xi}{2a\sqrt{t}}$, $dz = -\frac{d\xi}{2a\sqrt{t}}$, 则第一项积分可化为

$$\begin{aligned} \int_{-\infty}^{\frac{x+2a^2 t}{2a\sqrt{t}}} -q(0) \frac{1}{\sqrt{\pi}} e^{-z^2} e^{a^2 t} e^x dz d\tau &= -\frac{1}{2} q(0) e^{a^2 t} e^x \left[\int_0^{\frac{x+2a^2 t}{2a\sqrt{t}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_{\infty}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{1}{2} q(0) e^{a^2 t} e^x \operatorname{erfc} \left[\frac{x+2a^2 t}{2a\sqrt{t}} \right] \end{aligned}$$

在第二项积分中, 令 $z = \frac{\xi-(x-2a^2 t)}{2a\sqrt{t}}$, $dz = \frac{d\xi}{2a\sqrt{t}}$, 则第二项积分可化为

$$\begin{aligned} \int_{-\frac{x-2a^2 t}{2a\sqrt{t}}}^{\infty} q(0) \frac{1}{\sqrt{\pi}} e^{-z^2} e^{a^2 t} e^{-x} dz d\tau &= \frac{1}{2} q(0) e^{a^2 t} e^{-x} \left[\int_{-\frac{x-2a^2 t}{2a\sqrt{t}}}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\infty} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] \\ &= \frac{1}{2} q(0) e^{a^2 t} e^{-x} \operatorname{erfc} \left[\frac{x-2a^2 t}{2a\sqrt{t}} \right] \end{aligned}$$

第三、四项积分中, 分别令 $z = \frac{[x+2a^2(t-\tau)]-\xi}{2a\sqrt{t-\tau}}$ 和 $z = \frac{\xi-[x-2a^2(t-\tau)]}{2a\sqrt{t-\tau}}$, 并进行与第一、二项积分中类似的变换, 可得

$$\text{第三项积分为 } \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \operatorname{erfc} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau$$

$$\text{第四项积分为 } \int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^{-x} \operatorname{erfc} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau$$

$$u(x, t) = -q(t) e^{-x} + v(x, t)$$

$$= -q(t) e^{-x} + \frac{1}{2} q(0) e^{a^2 t} \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) + e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right] +$$

$$\int_0^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{2a^2(t-\tau) + x}{2a\sqrt{t-\tau}} \right] + e^{-x} \operatorname{erfc} \left[\frac{2a^2(t-\tau) - x}{2a\sqrt{t-\tau}} \right] \right\} d\tau$$

12. 教材中例 7 研究三维无界空间中的受迫振动是从初始($t=0$) 状态推算以后($t>0$) 的状态. 试重新求解例 7, 从初始($t=0$) 状态反推以前($t<0$) 的状态.

解: 由例 7 可知, 无论 $t>0$ 还是 $t<0$, 都有

$$u(\mathbf{r}, t) = \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^t f(\mathbf{r}', \tau) \frac{1}{(2\pi)^3} \iiint_{-\infty}^{\infty} \frac{2\pi}{ik} [e^{ika(t-\tau)} - e^{-ika(t-\tau)}] e^{ik(\mathbf{r}-\mathbf{r}')} dk_1 dk_2 dk_3 d\tau dV'$$

此时, 由于 $t<0$, 不能直接引用 $\mathcal{F}\left[\frac{1}{r}\delta(r-c)\right] = \frac{1}{(2\pi)^2} \frac{1}{ik} [e^{ikc} - e^{-ikc}]$ ($c>0$)

令 $-t = t'$, 则 $t'>0$. 于是利用上式得到

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^{t'} f(\mathbf{r}', \tau) \iiint_{-\infty}^{\infty} \frac{1}{(2\pi)^2} \cdot \frac{1}{ik} [e^{-ika(t'+\tau)} - e^{ika(t'+\tau)}] e^{ik(\mathbf{r}-\mathbf{r}')} dk_1 dk_2 dk_3 d\tau dV' = \\ &= \frac{(-1)}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^{t'} f(\mathbf{r}', \tau) \iiint_{-\infty}^{\infty} \mathcal{F}\left\{\frac{1}{r}\delta[r-a(t'+\tau)]\right\} e^{ik(\mathbf{r}-\mathbf{r}')} dk_1 dk_2 dk_3 d\tau dV' \end{aligned}$$

应用延迟定理

$$u(\mathbf{r}, t) = \frac{(-1)}{4\pi a} \iiint_{-\infty}^{\infty} \int_0^{t'} f(\mathbf{r}', \tau) \frac{1}{|\mathbf{r}-\mathbf{r}'|} \delta[|\mathbf{r}-\mathbf{r}'|-a(t'+\tau)] d\tau dV'$$

利用 $\delta(-x) = \delta(x)$ 以及 $\delta(ax) = \frac{\delta(x)}{|a|}$ 可得

$$u(\mathbf{r}, t) = \frac{(-1)}{4\pi a^2} \iiint_{-\infty}^{\infty} \int_0^{t'} f(\mathbf{r}', \tau) \frac{1}{|\mathbf{r}-\mathbf{r}'|} \delta(t'+\tau - \frac{|\mathbf{r}-\mathbf{r}'|}{a}) d\tau dV'$$

由于 $t<0$. 所以对 τ 的积分上限小于下限, 则

$$\int_0^{t'} f(\mathbf{r}', \tau) \frac{1}{|\mathbf{r}-\mathbf{r}'|} \delta(t'+\tau - \frac{|\mathbf{r}-\mathbf{r}'|}{a}) d\tau = - \frac{f(\mathbf{r}, -t' + \frac{|\mathbf{r}-\mathbf{r}'|}{a})}{|\mathbf{r}-\mathbf{r}'|}$$

$$\therefore u(\mathbf{r}, t) = \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\mathbf{r}, -t' + \frac{|\mathbf{r}-\mathbf{r}'|}{a})}{|\mathbf{r}-\mathbf{r}'|} dV'$$

$$\because t' = |t| = -t$$

$$\therefore u(\mathbf{r}, t) = \frac{1}{4\pi a^2} \iiint_{-\infty}^{\infty} \frac{f(\mathbf{r}, t + \frac{|\mathbf{r}-\mathbf{r}'|}{a})}{|\mathbf{r}-\mathbf{r}'|} dV'$$

本问题的 $f(\mathbf{r}, t)$ 中的 $t \leq 0$, 所以上面的积分不必在全空间进行, 只需在条件 $t + \frac{|\mathbf{r}-\mathbf{r}'|}{a} \leq 0$

下积分. 即对 $\mathbf{r}'$ 的积分只需在球体 $T_{a|t|}^{\mathbf{r}}$ 中进行.

$$\therefore u(\mathbf{r}, t) = \frac{1}{4\pi a^2} \iiint_{T_{a|t|}^{\mathbf{r}}} \frac{f(\mathbf{r}, t + \frac{|\mathbf{r}-\mathbf{r}'|}{a})}{|\mathbf{r}-\mathbf{r}'|} dV'$$

该式称为超前势.

13. 试用降维法由泊松公式(13.1.4) 推出一维波动的达朗贝尔公式(7.4.7).

解: 泊松公式为

$$u(r, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_a} \frac{\varphi(r')}{at} dS' + \frac{1}{4\pi a} \iint_{S_a} \frac{\varphi(r')}{at} dS'$$

对于一维问题, 球面 S_a 上的积分应代之以 x 轴的 L_a^x 上的积分, 其中 L_a^x 对应为球面 S_a 上过圆心的圆周, 则 dS' 对应于 $dx' \cdot 2\pi at$, 则一维情况下 $u(x, t)$ 为

$$\begin{aligned} u(x, t) &= \frac{1}{4\pi a} \frac{\partial}{\partial t} \int_{x-a}^{x+a} \frac{\varphi(x')}{at} \cdot (2\pi at) dx' + \frac{1}{4\pi a} \int_{x-a}^{x+a} \frac{\varphi(x')}{at} \cdot (2\pi at) dx' \\ &= \frac{1}{2a} \frac{\partial}{\partial t} \int_{x-a}^{x+a} \varphi(x') dx' + \frac{1}{2a} \int_{x-a}^{x+a} \varphi(x') dx' \\ &= \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-a}^{x+a} \varphi(x') dx' \end{aligned}$$

该式即为达朗贝尔公式.

§ 13.2 拉普拉斯变换法

1. 求解一维无界空间中的扩散问题 $u_t - a^2 u_{xx} = 0, u|_{t=0} = \varphi(x)$.

解: 一维无界空间的扩散问题 $\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{t=0} = \varphi(x) \end{cases}$

对定解方程组进行 L 变换

$$\text{得 } p\bar{u} - \varphi(x) - a^2 \bar{u}_{xx} = 0 \quad \text{求解 } a^2 y'' - py = -\varphi(x) \Rightarrow y'' - \frac{p}{a^2} y = -\frac{1}{a^2} \varphi(x)$$

对应的齐次方程解为 $y = Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x}$

下面求方程的特解, 使用参数变易法.

$$y' = \frac{\sqrt{p}}{a} Ae^{\frac{\sqrt{p}}{a}x} - \frac{\sqrt{p}}{a} Be^{-\frac{\sqrt{p}}{a}x} + A'e^{\frac{\sqrt{p}}{a}x} + B'e^{-\frac{\sqrt{p}}{a}x}$$

$$A'e^{\frac{\sqrt{p}}{a}x} + B'e^{-\frac{\sqrt{p}}{a}x} = 0$$

$$\begin{aligned} \text{则有 } y' &= \frac{\sqrt{p}}{a} Ae^{\frac{\sqrt{p}}{a}x} - \frac{\sqrt{p}}{a} Be^{-\frac{\sqrt{p}}{a}x}, y'' = \frac{p}{a^2} (Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x}) + \frac{\sqrt{p}}{a} (A'e^{\frac{\sqrt{p}}{a}x} - B'e^{-\frac{\sqrt{p}}{a}x}) \\ &= \frac{p}{a^2} (Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x}) \end{aligned}$$

$$\text{代入原方程得 } \frac{\sqrt{p}}{a} (A'e^{\frac{\sqrt{p}}{a}x} - B'e^{-\frac{\sqrt{p}}{a}x}) = -\frac{1}{a^2} \varphi(x) \quad (*)$$

$$\text{与 } A'e^{\frac{\sqrt{p}}{a}x} + B'e^{-\frac{\sqrt{p}}{a}x} = 0 \text{ 相加得 } A' = \frac{-1}{2a\sqrt{p}} e^{-\frac{\sqrt{p}}{a}x} \varphi(x)$$

$$\text{积分求得 } A' = -\frac{1}{2a\sqrt{p}} \int e^{-\frac{\sqrt{p}}{a}\epsilon} \varphi(\epsilon) d\epsilon$$

$$\text{同理, } (*) \text{ 与 } A'e^{\frac{\sqrt{p}}{a}x} + B'e^{-\frac{\sqrt{p}}{a}x} = 0 \text{ 相减得 } B = \frac{1}{2a\sqrt{p}} e^{\frac{\sqrt{p}}{a}x} \varphi(x)$$

$$B' = \frac{1}{2a\sqrt{p}} \int e^{\frac{\sqrt{p}}{a}\epsilon} \varphi(\epsilon) d\epsilon$$

将 A, B 代入解的形式得特解为 $-\frac{1}{2a\sqrt{p}}e^{\frac{\sqrt{p}}{a}x}\int e^{-\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon + \frac{1}{2a\sqrt{p}}e^{-\frac{\sqrt{p}}{a}x}\int e^{\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon$

$$\begin{aligned}\text{通解为 } y &= Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a\sqrt{p}}e^{\frac{\sqrt{p}}{a}x}\int e^{-\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon + \frac{1}{2a\sqrt{p}}e^{-\frac{\sqrt{p}}{a}x}\int e^{\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon \\ &= Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a}e^{\frac{\sqrt{p}}{a}x}\int \frac{1}{\sqrt{p}}e^{-\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon + \frac{1}{2a}e^{-\frac{\sqrt{p}}{a}x}\int \frac{1}{\sqrt{p}}e^{\frac{\sqrt{p}}{a}\epsilon}\varphi(\epsilon)d\epsilon\end{aligned}$$

解在 $x \rightarrow \infty$ 有限, $A = B = 0$

$$\bar{u}(p) = y = \frac{1}{2a}\int_x^{+\infty} \frac{1}{\sqrt{p}}e^{-\frac{\sqrt{p}}{a}(\epsilon-x)}\varphi(\epsilon)d\epsilon + \frac{1}{2a}\int_{-\infty}^x \frac{1}{\sqrt{p}}e^{\frac{\sqrt{p}}{a}(\epsilon-x)}\varphi(\epsilon)d\epsilon$$

$$\begin{aligned}\text{反演之后得 } u &= \frac{1}{2a}\int_x^{+\infty} \frac{\varphi(\epsilon)}{\sqrt{\pi t}}e^{-\frac{(x-\epsilon)^2}{4a^2t}}d\epsilon + \frac{1}{2a}\int_{-\infty}^x \frac{\varphi(\epsilon)}{\sqrt{\pi t}}e^{-\frac{(x-\epsilon)^2}{4a^2t}}d\epsilon \\ &= \int_{-\infty}^{+\infty} \frac{\varphi(\epsilon)}{2a\sqrt{\pi t}}e^{-\frac{(x-\epsilon)^2}{4a^2t}}d\epsilon\end{aligned}$$

2. 求解硅片的限定源扩散问题, 将硅片的厚度当作无限大, 这是半无界空间的定解问题 $u_t -$

$$a^2 u_{xx} = 0, u_x \Big|_{x=0} = 0, u \Big|_{t=0} = \Phi_0 \delta(x-0). \quad [\text{本题即 § 13.1 例 4, 可对照.}]$$

$$\text{解: 定解问题} \begin{cases} u_t - a^2 u_{xx} = 0 & (x > 0) \\ u_x \Big|_{x=0} = 0 \\ u \Big|_{t=0} = \varphi_0 \delta(x-0) \end{cases}$$

$$\text{对其进行 } L \text{ 变换} \begin{cases} p\bar{u} - \varphi_0 \delta(x-0) - a^2 \bar{u}_{xx} = 0 & (x > 0) \\ \bar{u}_x \Big|_{x=0} = 0 \end{cases}$$

$$\begin{aligned}\text{根据第 1 题的结论, 可知通解为 } \bar{u}(x, p) &= Ae^{\frac{\sqrt{p}}{a}x} + Be^{-\frac{\sqrt{p}}{a}x} - \frac{1}{2a}\int_x^{(+\infty)} \frac{e^{\sqrt{p}(\frac{\epsilon-x}{a})}}{\sqrt{p}}\varphi_0 \delta(\epsilon-0)d\epsilon + \frac{1}{2a}\int_{(-\infty)}^x \\ &\frac{e^{\sqrt{p}(\frac{x-\epsilon}{a})}}{\sqrt{p}}\varphi_0 \delta(\epsilon-0)d\epsilon\end{aligned}$$

当 $x \rightarrow +\infty$ 时, $\bar{u}$ 有界 $\therefore A = 0$

$$\bar{u}_x = -\frac{\sqrt{p}}{a}Be^{\frac{\sqrt{p}}{a}x} - \frac{1}{2a}\frac{e^0}{\sqrt{p}}\varphi_0 \delta(x-0) + \frac{1}{2a}\frac{e^0}{\sqrt{p}}\varphi_0 \delta(x-0) = -\frac{\sqrt{p}}{a}Be^{\frac{\sqrt{p}}{a}x}$$

$B = 0$ 即有

$$\bar{u} = -\frac{\varphi_0}{2a}\int_{-\infty}^x \frac{\delta(\epsilon-0)}{\sqrt{p}}e^{\sqrt{p}\frac{x-\epsilon}{a}}d\epsilon + \frac{\varphi_0}{2a}\int_{-\infty}^x \frac{\delta(\epsilon-0)}{\sqrt{p}}e^{\sqrt{p}\frac{\epsilon-x}{a}}d\epsilon = \frac{\varphi_0}{2a}\frac{1}{\sqrt{p}}e^{\sqrt{p}\frac{x}{a}} + \frac{\varphi_0}{2a}\frac{1}{\sqrt{p}}e^{-\sqrt{p}\frac{x}{a}}$$

$$\begin{aligned}\text{及演得 } u_{(x,t)} &= \frac{\varphi_0}{2a}\frac{1}{\sqrt{\pi t}}e^{-\frac{x^2}{4a^2t}} + \frac{\varphi_0}{2a}\frac{1}{\sqrt{\pi t}}e^{-\frac{x^2}{4a^2t}} \\ &= \frac{\varphi_0}{a}\frac{1}{\sqrt{\pi t}}e^{-\frac{x^2}{4a^2t}}\end{aligned}$$

3. 求解一维无界空间的有源输运问题 $u_t - a^2 u_{xx} = f(x, t), u \Big|_{t=0} = 0$. [本题即 § 13.1 例 3, 可对照.]

解: 方程为 $u_t - a^2 u_{xx} = f(x, t)$ $u|_{t=0} = 0$

对方程进行 L 变换得 $p\bar{u} - a^2 \bar{u}_{xx} = \bar{f}(xp)$

由之前第一题的讨论知 $\bar{u} = Ae^{\frac{\sqrt{x}}{a}} + Be^{-\frac{\sqrt{x}}{a}} + \frac{1}{2a} \int_x^\infty \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{x}}{a}(\xi-x)} \bar{f}(\xi p) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{\sqrt{p}} e^{\frac{\sqrt{x}}{a}(\xi-x)} \bar{f}(\xi p) d\xi$

同样, 根据 $x \rightarrow \pm\infty$ 时, $\bar{u}$ 有限 $\therefore A = B = 0$

即 $\bar{u} = \frac{1}{2a} \int_x^\infty \frac{1}{\sqrt{p}} e^{-\frac{\sqrt{x}}{a}(\xi-x)} \bar{f}(\xi p) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{1}{\sqrt{p}} e^{\frac{\sqrt{x}}{a}(\xi-x)} \bar{f}(\xi p) d\xi$

反演 $\bar{f}(\xi, p) \Rightarrow f(x, t)$ 且 $\frac{1}{\sqrt{p}} e^{-\frac{\sqrt{x}}{a}(\xi-x)} \Rightarrow \frac{1}{\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}}$

运用卷积定理得 $u = \int_{-\infty}^{+\infty} \int_0^t \frac{1}{2a \sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} f(\xi, \tau) d\tau d\xi$

4. 求解一维半无界空间的输运问题 $u_t - a^2 u_{xx} = 0, u|_{t=0} = 0$, 边界条件是 $u|_{x=0} = f(t)$. [本题为 § 13.1 习题 10 的一部分, 可对照.]

解: $\begin{cases} u_t - a^2 u_{xx} = 0 \\ u|_{t=0} = 0 \\ u|_{x=0} = f(x) \end{cases}$ 对该定解问题进行拉氏变换 $\begin{cases} p\bar{u} - a^2 \bar{u}_{xx} = 0 \\ \bar{u}|_{x=0} = \bar{f}(p) \end{cases}$

第一个方程的通解在之前已经解出, 可写成 $\bar{u} = Ae^{\frac{\sqrt{x}}{a}} + Be^{-\frac{\sqrt{x}}{a}}$

在一维半界空间 $x \rightarrow \infty$ 时, $\bar{u}$ 有限 $\therefore A = 0$

即 $\bar{u} = Be^{-\frac{\sqrt{x}}{a}}$

由 $\bar{u}|_{x=0} = \bar{f}(p)$ 得 $B = \bar{f}(p) \Rightarrow \bar{u} = \bar{f}(p) e^{-\frac{\sqrt{x}}{a}} = p \left[\bar{f}(p) - \frac{1}{p} e^{-\frac{\sqrt{x}}{a}} \right]$

反演 $\bar{f}(p) \Rightarrow f(t)$ $\frac{1}{p} e^{-\frac{\sqrt{x}}{a}} \Rightarrow \operatorname{erfc} \frac{x}{2a\sqrt{t}}$

由卷积定理 $\bar{f}(p) \frac{1}{p} e^{-\frac{\sqrt{x}}{a}} \Rightarrow \int_0^t f(t-\tau) \operatorname{erfc} \frac{x}{2a\sqrt{\tau}} d\tau$

$$u(x, t) = \frac{d}{dt} \int_0^t f(t-\tau) \operatorname{erfc} \left(\frac{x}{2a\sqrt{\tau}} \right) d\tau$$

5. 求解无界弦的受迫振动 $u_{tt} - a^2 u_{xx} = f(x, t), u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x)$

解: 由题意得定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t) & (-\infty < x < +\infty) \\ u|_{t=0} = \varphi(x) \\ u_t|_{t=0} = \psi(x) \end{cases}$$

对泛定方程施行拉普拉斯变换得

$$p^2 \bar{u} - p u_t|_{t=0} - u|_{t=0} - a^2 \bar{u}_{xx} = \bar{f}(x, p) = 0$$

代入初始条件得

$$p^2 \bar{u} - p\varphi - \psi - \bar{f} - a^2 \bar{u}_{xx} = 0$$

该非齐次常微分方程的通解是

$$\bar{u}(x, p) = Ae^{px/a} + Be^{-px/a} - \frac{1}{2a}e^{px/a} \int^{(x)} \frac{e^{-p\xi/a}}{p} [\varphi(\xi) + \bar{f}(\xi, p) + p\varphi(\xi)] d\xi + \frac{1}{2a}e^{-px/a} \cdot \int^{(x)} \frac{e^{p\xi/a}}{p} [\varphi(\xi) + \bar{f}(\xi, p) + p\varphi(\xi)] d\xi$$

考虑到当 $x \rightarrow \infty$ 和 $x \rightarrow -\infty$ 时 $\bar{u}$ 不应为无穷大, 所以 $A = 0, B = 0$, 另为保证积分收敛, 第一个积分下限取为 ∞ , 第二个积分下限取为 $-\infty$.

$$\begin{aligned} \therefore \bar{u}(x, p) &= \frac{1}{2a} \int_{\infty}^x \frac{e^{-p(\xi-x)/a}}{p} [\varphi(\xi) + \bar{f}(\xi, p) + p\varphi(\xi)] d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} [\varphi(\xi) + \bar{f}(\xi, p) + p\varphi(\xi)] d\xi \\ &= \left[\frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} \varphi(\xi) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} \varphi(\xi) d\xi \right] + \\ &\quad \left[\frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} \bar{f}(\xi, p) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} \bar{f}(\xi, p) d\xi \right] + \\ &\quad \left[\frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} p\varphi(\xi) d\xi + \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} p\varphi(\xi) d\xi \right] \end{aligned}$$

对第一个[], 运用延迟定理, $\frac{1}{p} \Rightarrow H(t)$, 则 $\frac{e^{-p(\xi-x)/a}}{p} \Rightarrow H\left(t - \frac{\xi-x}{a}\right) = \begin{cases} 1 & (\xi < x+at) \\ 0 & (\xi > x+at) \end{cases}$

$$\therefore \frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} \varphi(\xi) d\xi \Rightarrow \frac{1}{2a} \int_x^{x+at} \varphi(\xi) d\xi$$

$$\text{同理, } \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} \varphi(\xi) d\xi \Rightarrow \frac{1}{2a} \int_{x-at}^x \varphi(\xi) d\xi$$

对第三个[], 可比较第一个[], $\varphi(\xi)$ 代替了 $\varphi(\xi)$ 且多一个因子 p , 则对第一个[] 的原函数将 ψ 换为 φ 并对 t 求导即得到第三个[] 的原函数.

$$\begin{aligned} \frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} p\varphi(\xi) d\xi &\Rightarrow \frac{1}{2a} \varphi(x+at) \\ \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} p\varphi(\xi) d\xi &\Rightarrow \frac{1}{2} \varphi(x-at) \end{aligned}$$

对第二个[], 运用卷积定理

$$\bar{f}_1(p) \bar{f}_2(p) \Rightarrow \int_0^t f_1(\tau) f_2(t-\tau) d\tau$$

$$\frac{1}{2a} \int_x^{\infty} \frac{e^{-p(\xi-x)/a}}{p} \bar{f}(\xi, p) d\xi = \frac{1}{2a} \int_0^t \int_x^{\infty} f(\xi, \tau) H\left(t-\tau-\frac{\xi-x}{a}\right) d\xi d\tau = \frac{1}{2a} \int_0^t \int_x^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau$$

$$\text{同理 } \frac{1}{2a} \int_{-\infty}^x \frac{e^{-p(x-\xi)/a}}{p} \bar{f}(\xi, p) d\xi = \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^x f(\xi, \tau) d\xi d\tau$$

$$\text{得到 } u(x, t) = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \varphi(\xi) d\xi + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau$$

第十四章 保角变换法

本章知识要点

保角变换的基本性质

用适当的变换 $\zeta = \zeta(z), z = z(\zeta)$

即

$$\begin{cases} \xi = \xi(x, y) \\ \eta = \eta(x, y) \end{cases} \quad \begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}$$

可以将复杂的边界条件变成简单的边界,经过该变换, *Laplace* 方程 $u_{xx} + u_{yy} = 0$ 就转成如下形式:

$$(\xi_x^2 + \xi_y^2)u_{\xi\xi} + 2(\xi_x\eta_x + \xi_y\eta_y)u_{\xi\eta} + (\eta_x^2 + \eta_y^2)u_{\eta\eta} + (\xi_{xx} + \xi_{yy})u_{\xi} + (\eta_{xx} + \eta_{yy})u_{\eta} = 0$$

若 ζ 在研究区域上是 z 的解析函数,则

$$\xi_x^2 + \xi_y^2 = |\zeta'(z)|^2, \eta_x^2 + \eta_y^2 = |\zeta'(z)|^2, \xi_x\eta_x + \xi_y\eta_y = 0, \xi_{xx} + \xi_{yy} = 0, \eta_{xx} + \eta_{yy} = 0$$

Laplace 方程化为 $|\zeta'(z)|^2(u_{\xi\xi} + u_{\eta\eta}) = 0$

$\zeta = \zeta(z)$ 所表征的代换称为保角变换或保角映象.

某些常用的保角变换

(一) 线性变换

线性函数 $\zeta(z) = az + b$ (a, b 为复常数) 的导数, $\zeta'(z) = a$ 是常数,图形各部分按同比例放大而形状不变.

(二) 幂函数和根式变换

幂函数 $\zeta(z) = z^n$ 的导数 $\zeta'(z) = nz^{n-1}$. 在原点, 导数为 0, 交角并不保持不变, 在原点的交角放大为 n 倍, 原点以外任一有限远点, 变再保持不变.

根式 $\zeta(z) = \sqrt[n]{z}$ 是逆变换, 在原点的交角缩小为 $\frac{1}{n}$ 倍.

(三) 指数函数和对数函数变换

$$\text{指数函数 } \zeta(z) = e^z = e^x e^{iy} \quad |\zeta| = e^x \quad \text{Arg}\zeta = y$$

那 z 平面上平行于实轴的直线 $y = \text{常数}$ 变为 ζ 平面上 $\text{Arg}\zeta = \text{常数}$ (通过原点的射线), z 平

面上平行于虚轴的直线 $x = \text{常数}$ 变为 ζ 平面上 $|\zeta| = \text{常数}$ (原点作圆心的圆)

(四) 反演变换

变换 $\zeta = \frac{R^2}{z}$ (R 为实常数) 称为反演变换.

采用指数式 $z = \rho e^{i\varphi}$ 则 $\zeta = \frac{R^2}{\rho} e^{-i\varphi}$

反演变换在多平面包括无限深点在内都是保角的.

(五) 分式线性变换

$\zeta = \frac{az+b}{cz+d}$ ($ad-bc \neq 0$) 的分子分母均线性.

反演变换是分式线性变换的特例, 即 $a=0, d=0, b/c$ 为实常数

变换可化为 $\zeta = \frac{a}{c} + \frac{b-ad/c}{cz+d} = \frac{a}{c} + \frac{(bc-ad)/c^2}{z+d/c}$

即可分解成相继的三个变换: $z_1 = z + \frac{d}{c}, z_2 = \frac{(bc-ad)/c^2}{z_1}$

$\zeta = z_2 + \frac{a}{c}$ 分式线性变换是在包括 ∞ 在内的全平面上保角的变换.

分式线性变换的特点是圆保持为圆(直线作为圆的特例)

而且对于圆的对称点保持为对称点.

(六) 儒可夫斯基变换

儒可夫斯基函数 $\zeta(z) = \frac{1}{2}(z + \frac{1}{z})$

其实部与虚部分别为
$$\begin{cases} \xi = \frac{1}{2}(x + \frac{x}{x^2+y^2}) = \frac{1}{2}(\rho + \frac{1}{\rho})\cos\varphi \\ \eta = \frac{1}{2}(y - \frac{y}{x^2+y^2}) = \frac{1}{2}(\rho - \frac{1}{\rho})\sin\varphi \end{cases}$$

z 平面上的同心圆族 $|z| = \rho_0$ 变为 ζ 平面上的 $\xi = \frac{1}{2}(\rho_0 + \frac{1}{\rho_0})\cos\varphi, \eta = \frac{1}{2}(\rho_0 - \frac{1}{\rho_0})\sin\varphi$

消去 φ 得 $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} = 1, a = \frac{1}{2}(\rho_0 + \frac{1}{\rho_0}), b = \frac{1}{2}|\rho_0 - \frac{1}{\rho_0}|$ 表示椭圆族.

因此, 儒可夫斯基函数把圆变为椭圆, 射线变为双曲线, 同心圆族变为共焦点椭圆族, 共点射线族变为共焦双曲线族.

(七) 施瓦兹 - 克利斯多菲变换(多角形变换)

公式: $z = z_0 + A \int (\zeta - b_1)^{-\alpha_1/\pi} (\zeta - b_2)^{-\alpha_2/\pi} \cdots (\zeta - b_n)^{-\alpha_n/\pi} d\zeta$

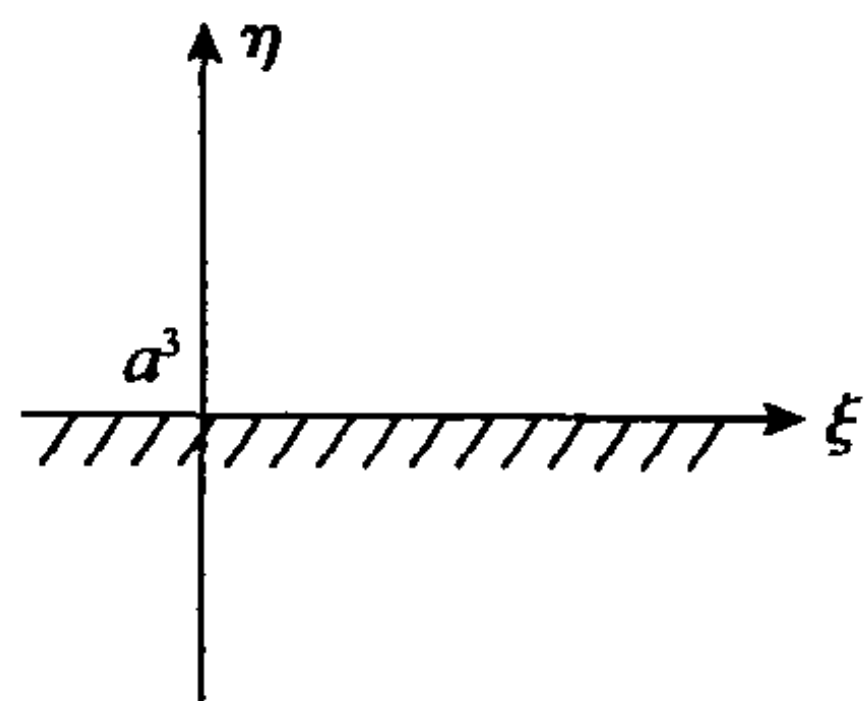
对于三角形, 施瓦兹 - 克利斯多菲变换的 b_1, b_2, b_3 可以任意指定, 但对 $n > 3$ 的多角形不能任意指点 $b_1, b_2, \cdots, b_n$, 多角形任意三个顶点可以连成三角形, 所以 $b_1, b_2, \cdots, b_n$ 中可以任意指定三个数值, 而其系的不可指定, 必须根据顶点的对应关系加以确定.

习题全解

§ 14.2 某些常用的保角变换

1. 教材中例 1 的二面角的二等分面上有一带电细导线, 平行于二面角的顶角线, 相距为 a , 导线每单位长度带电量为 Q . 试求电势分布.

解: 作变换 $\zeta = z^3$ z 平面换到了 ζ 平面, 在 η 轴上距原点 a^3 处有一导线, 下半平面为导体.



由镜像电荷可以求生在 ζ 平面的复势电势为 $-\frac{Q}{2\pi\epsilon_0} \ln \frac{\zeta - a^3 i}{\zeta + a^3 i}$, 由此

在 z 平面的复势为 $-\frac{Q}{2\pi\epsilon_0} \ln \frac{z^3 - a^3 i}{z^3 + a^3 i}$

结果取实部 $\varphi = \operatorname{Re} \left[-\frac{Q}{2\pi\epsilon_0} \ln \frac{z^3 - a^3 i}{z^3 + a^3 i} \right]$

2. 接地甚长空心金属圆柱半径为 a , 柱内有细导线, 平行于柱轴, 相距 b . 导线每单位长度带电量为 Q . 试求圆柱内电势分布.

解: 作分式变换 $\zeta = \frac{z-b}{z-\frac{a^2}{b}}$ 得到 z 平面的 b 变成 ζ 平面的原点

$x=b$ 的对称点 $x=\frac{a^2}{b}$ 转化为无限远的点

$z=a$ 代入 $\zeta = \frac{z-b}{z-\frac{a^2}{b}} = \frac{b}{a}$ 为 ζ 平面上圆半径.

在 ζ 平面, 圆内电势 $u = -\frac{Q}{2\pi\epsilon_0} \ln \frac{\zeta}{\frac{b}{a}} = -\frac{Q}{2\pi\epsilon_0} \ln \frac{a\zeta}{b}$

在 z 平面, 复势为 $-\frac{Q}{2\pi\epsilon_0} \ln \frac{a}{b} \frac{z-b}{z-\frac{a^2}{b}} = -\frac{Q}{2\pi\epsilon_0} \ln \frac{a(z-b)}{bz-a^2}$,

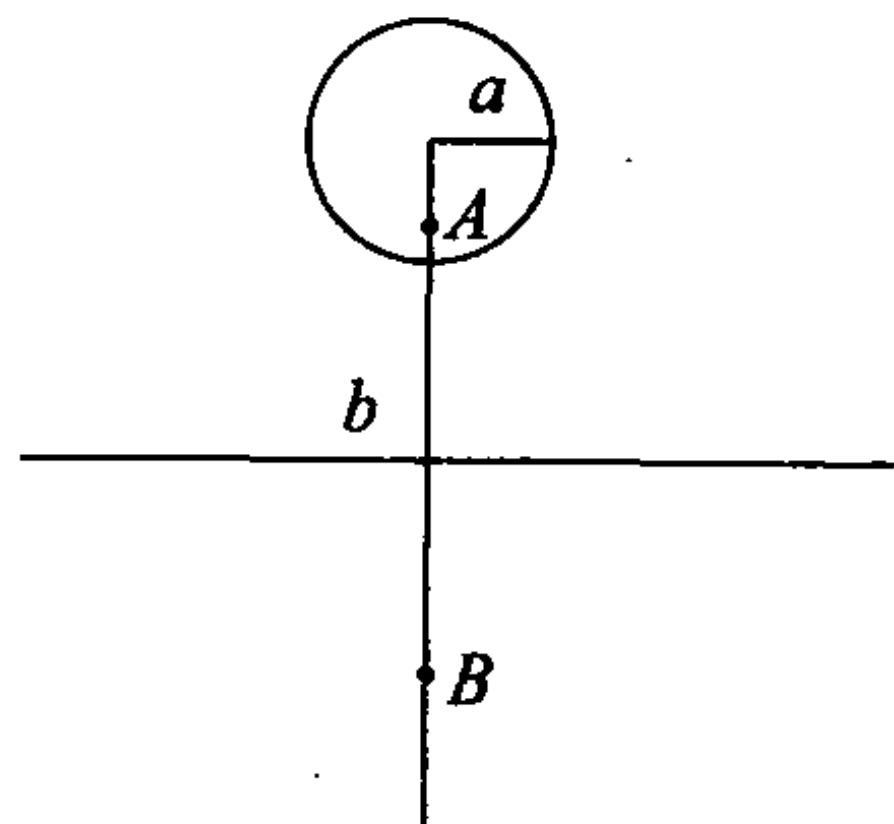
取实部得到电势为 $u = -\frac{Q}{2\pi\epsilon_0} \ln \sqrt{\frac{a^2(x-b)^2 + a^2 y^2}{(a-bx)^2 + b^2 y^2}}$

3. 甚长金属圆柱的轴平行于甚大金属平板, 两者相距 b , 平板接地. 圆柱半径为 a . 试求每单位长度的电容量.

解: 找两点 A, B , 对于圆与实轴对称, 令 A 坐标为 $y_1 i$, B 为 $-y_2 i$.

由对称得 $\begin{cases} (b-y_1)(b+y_2) = a^2 \\ y_1 = y_2 \end{cases} \Rightarrow y_1 = y_2 = \sqrt{b^2 - a^2}$

考虑作分式线性变换 $\zeta = z - \frac{y_1 i}{z + y_2}$, z 平面上的圆与实轴变为 ζ



平面上的同心圆

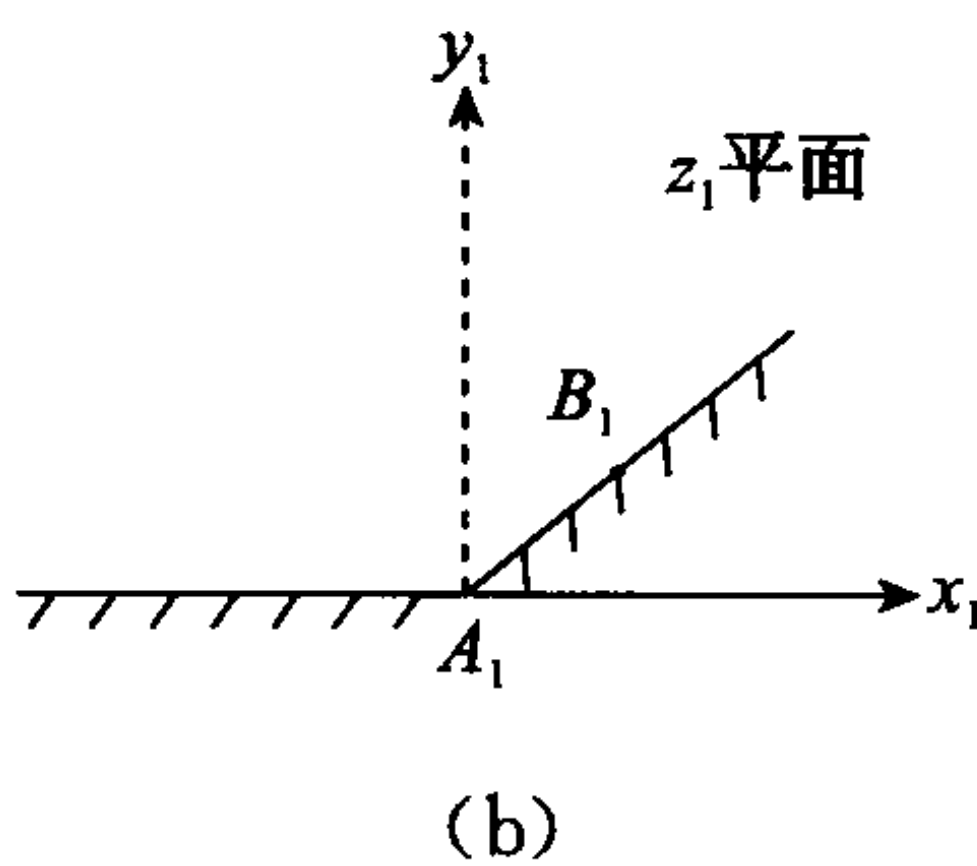
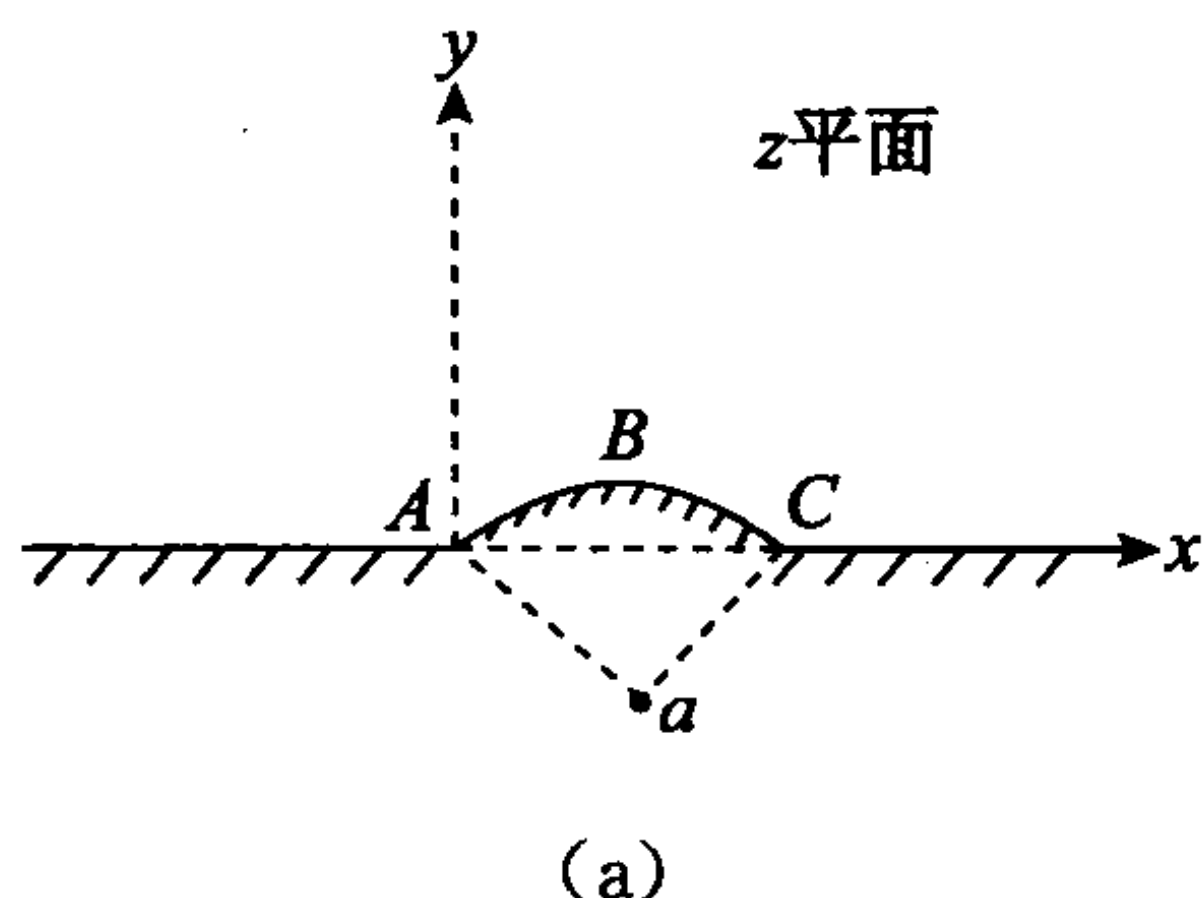
$$R_1 = \left| \frac{(b-a-y_1)i}{(b-a+y_2)i} \right| = \frac{b-a-\sqrt{b^2-a^2}}{b-a+\sqrt{b^2-a^2}} = \frac{a}{b+\sqrt{b^2-a^2}}$$

$$R_2 = \left| \frac{-iy_1}{iy_2} \right| = 1, \text{ 于是电容 } c = \frac{2\pi\epsilon_0}{\ln \frac{R_2}{R_1}} = \frac{2\pi\epsilon_0}{\ln \left(\frac{b+\sqrt{b^2-a^2}}{a} \right)} = \frac{2\pi\epsilon_0}{\ln \left(\frac{b}{a} + \sqrt{\frac{b^2}{a^2}-1} \right)}$$

保角变换对电容无影响, 因此 z 平面上 $c = \frac{2\pi\epsilon_0}{\ln \left(\frac{b}{a} + \sqrt{\frac{b^2}{a^2}-1} \right)}$

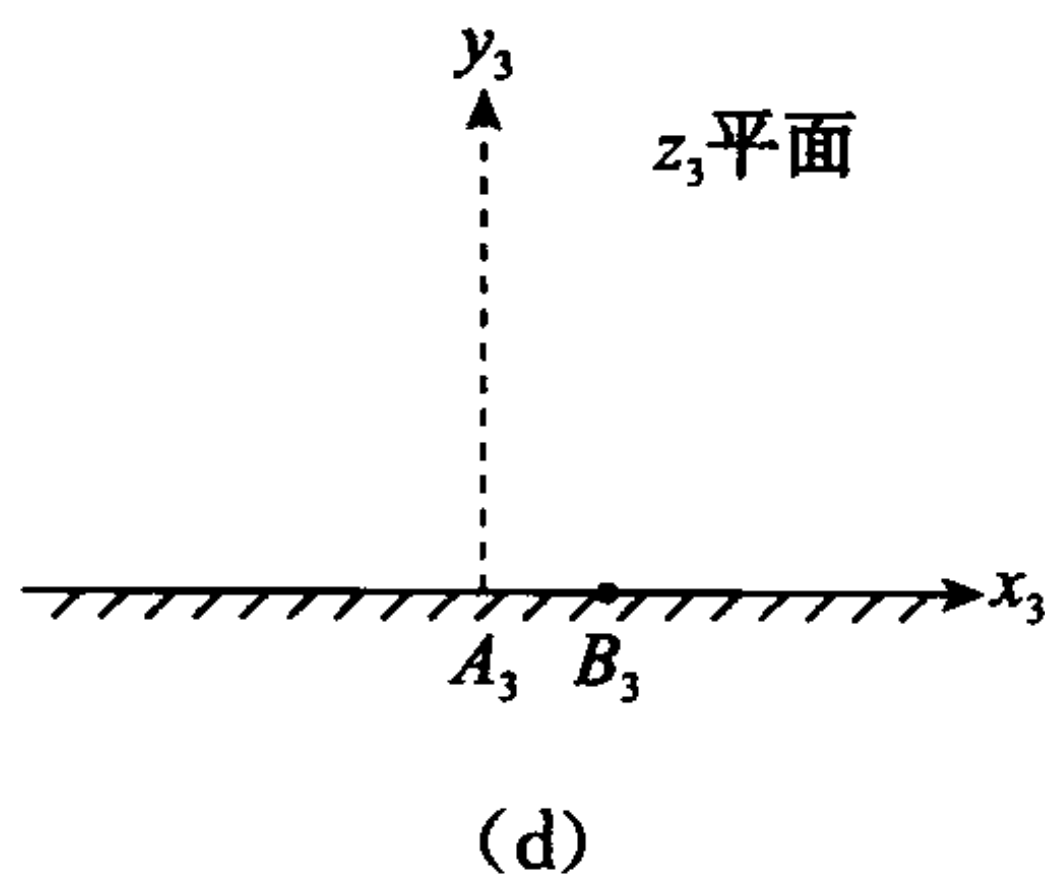
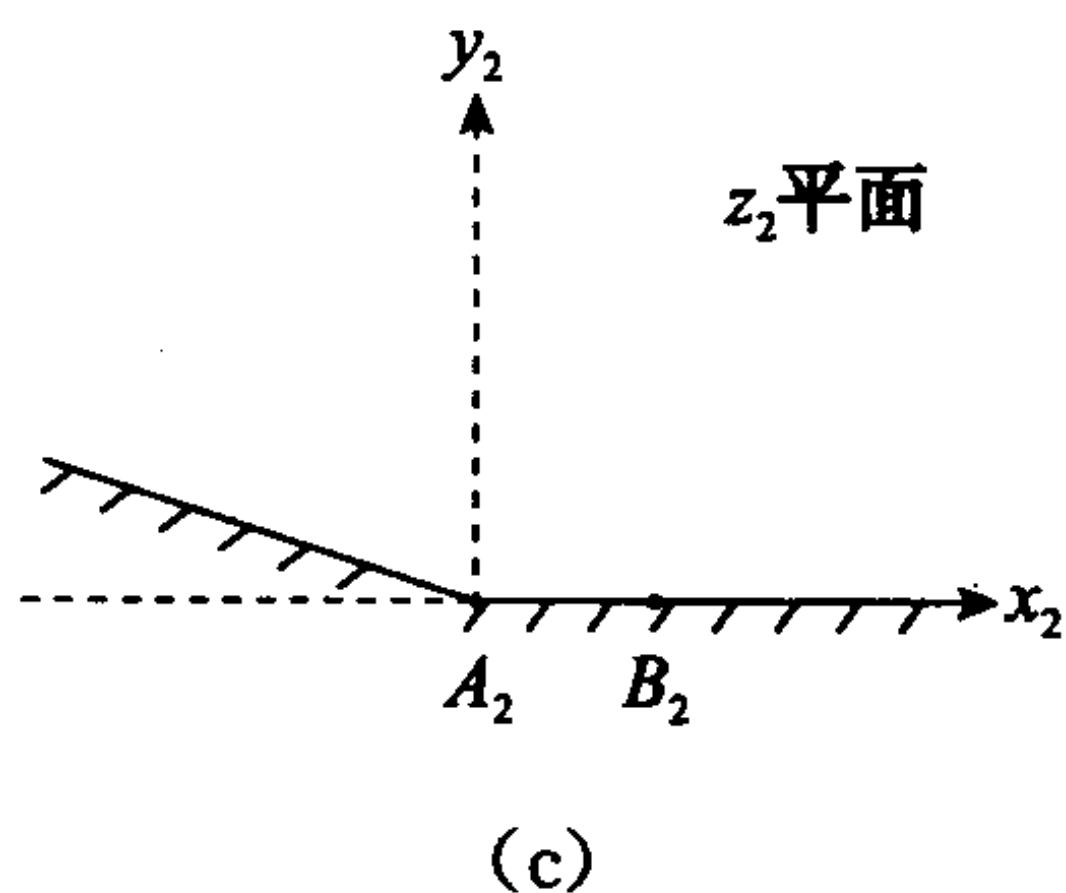
4. 甚大金属平面上有柱形隆起, 其横截面为弓形. 弓形在 0 和 a 之间, 弓形的弧的半径为 a . 求解带电后的静电场.

解: 这个图形可看作两段圆弧组成, 一段是通过 $O(A \text{ 点})$, $\frac{1}{2}a + \left(1 - \frac{\sqrt{3}}{2}\right)ai (B \text{ 点})$ 和 $a(C \text{ 点})$ 的弓形, 另一段是通过 $a(C \text{ 点})$, 无限远点和 $O(A \text{ 点})$ 的圆弧, 如图(a) 所示.



(1) 作分式线性变换 $z_1 = \frac{z}{a-z}$, 将 $z=0(A \text{ 点})$, $z = \frac{1}{2}a + \left(1 - \frac{\sqrt{3}}{2}\right)ai (B \text{ 点})$, 和 $z=a(C \text{ 点})$ 变到 $z_1=0(A \text{ 点})$, $z_1 = \frac{\sqrt{3}}{2} + \frac{1}{2}i (B_1 \text{ 点})$ 和 $z_1=\infty (C_1 \text{ 点})$. 因而将两段圆弧变为 z_1 平面上通过 O 和 ∞ 的圆弧. 即通过原点的射线, 它们之间保持夹角为 $\frac{5\pi}{6}$, 如图(b) 所示.

(2) 作辐角变换 $z_2 = z_1 e^{-i\frac{\pi}{6}}$, 即将两条射线顺时针旋转 $\frac{\pi}{6}$, 使得 A_2 点为 $z_2=0$, C_2 点为 $z_2=\infty$, B_2 点为 $z_2=1$. 如图(c) 所示.



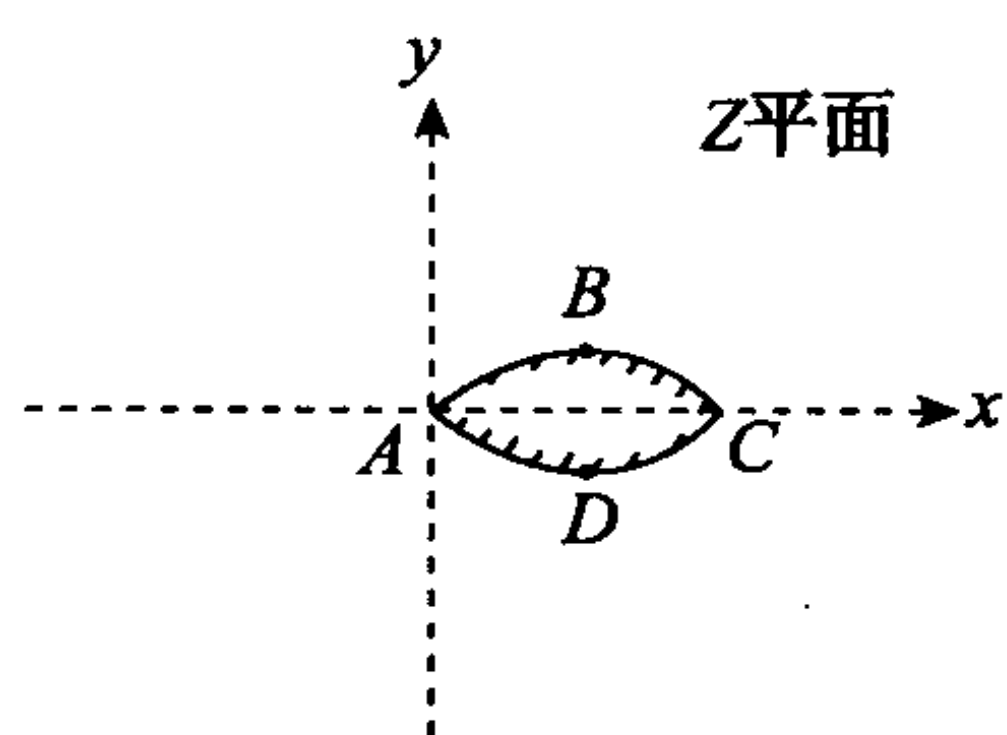
(3) 作根式变换 $z_3 = z_2^{\frac{1}{5}}$, 将两条射线之间的夹角放大到原来的 $\frac{6}{5}$ 倍, 即将其放大为 π , 成为半平角使 A_3 为 $z_3 = 0$, B_3 为 $z_3 = 1$, C_3 为 $z_3 = \infty$.

(4) 作分式线性变换 $\xi = \frac{az_3}{z_3 + 1}$, 将 $z_3 = 0$ (A_3 点), $z_3 = 1$ (B_3 点) 和 $z_3 = \infty$ (C_3 点) 变到 $\xi = 0$ (A_4 点), $\xi = \frac{a}{2}$ (B_4 点) 和 $\xi = a$ (C_4 点), 比较 z 平面内图形, 可知弓形突起已被消除如图(e) 所示.

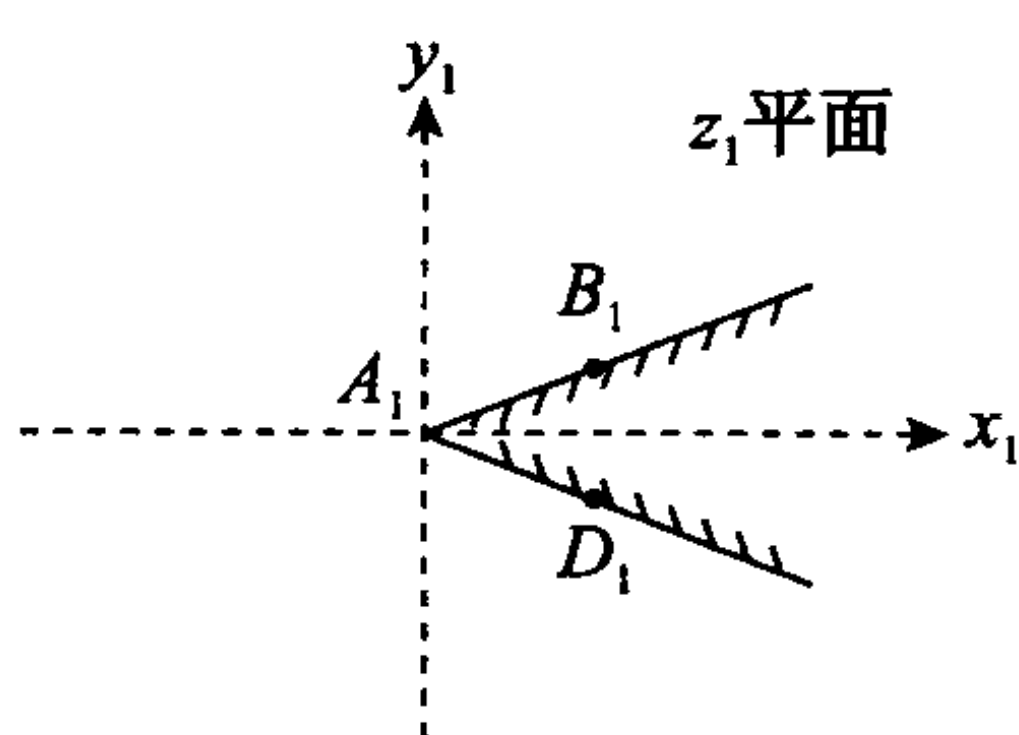
(5) 在 ξ 平面内, 物理问题转化为无限大均匀带电平面外的电势, 易知其电势 $= C_1 \operatorname{Im} \xi + C_2$, 其中常数 C_1 的物理意义为电场强度, 常数 C_2 的物理意义为带电平面的电势.

5. 长金属柱, 其横截面由两段圆弧围成, 这两段圆弧是相等的, 其半径为 a , 交点在 0 和 a . 求解金属柱带电后的周围静电场.

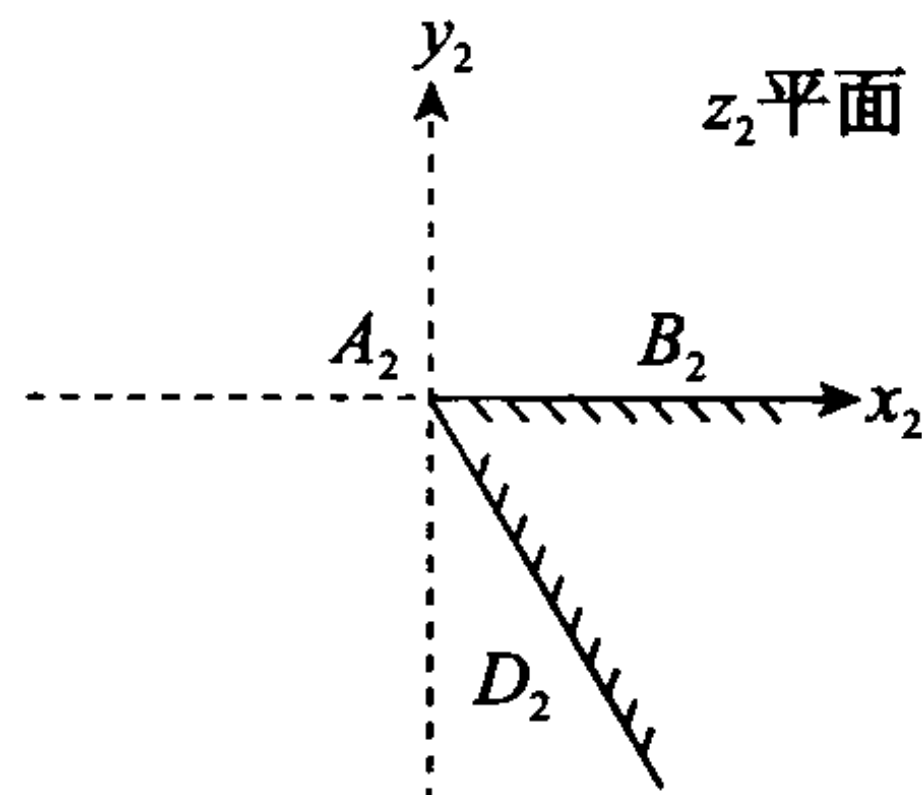
解: 这个图形可看作两段圆弧组成, 一段是通过 O (A 点), $\frac{1}{2}a + \left(1 - \frac{\sqrt{3}}{2}\right)ai$ (B 点), a (C 点) 的弓形, 另一段是通过 a (C 点), $\frac{1}{2}a - \left(1 - \frac{\sqrt{3}}{2}\right)ai$ (D 点) 和 O (A 点) 的圆弧, 如图(a) 所示.



(a)



(b)

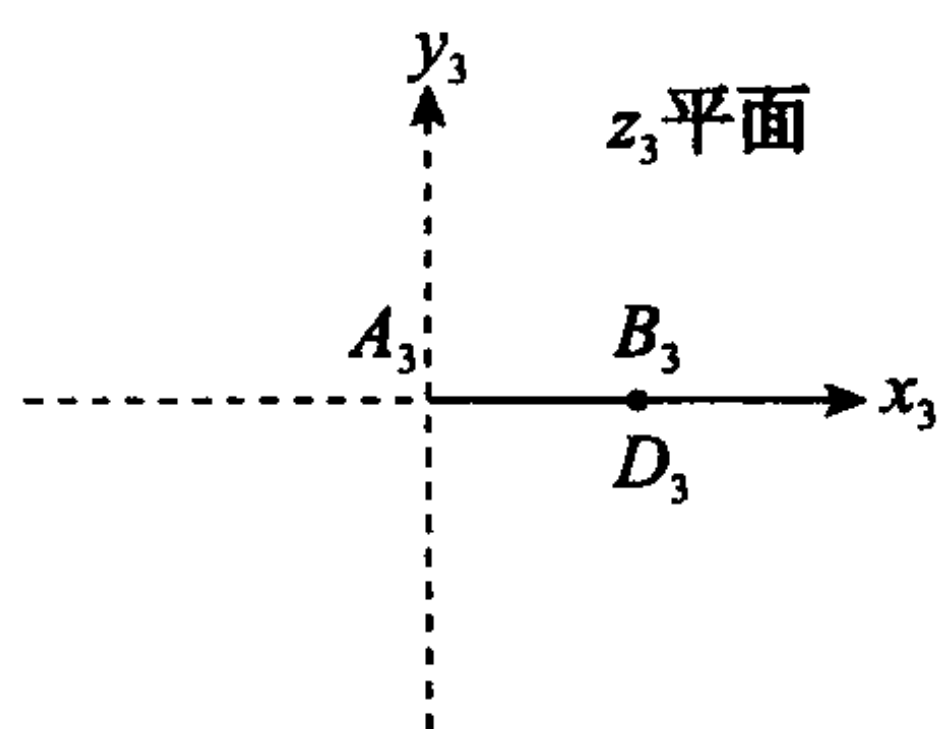


(c)

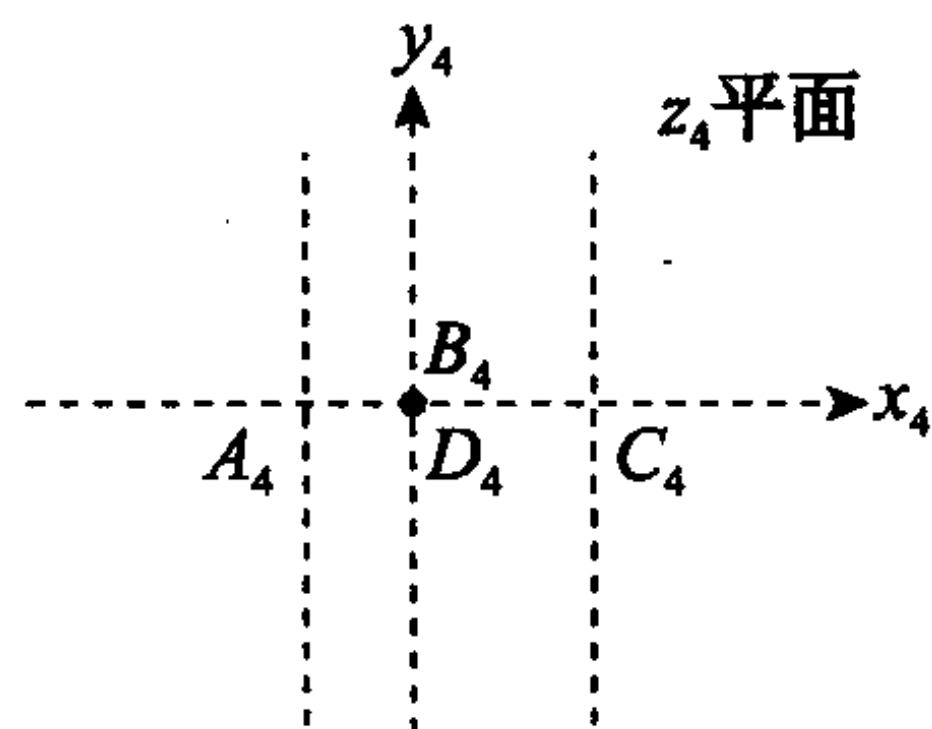
(1) 作分式线性变换 $z_1 = \frac{2}{a-z}$, 将 $z = 0$ (A 点), $z = \frac{1}{2}a + \left(1 - \frac{\sqrt{3}}{2}\right)ai$ (B 点), $z = a$ (C 点) 和 $z = \frac{1}{2}a - \left(1 - \frac{\sqrt{3}}{2}\right)ai$ (D 点) 变到 $z_1 = 0$ (A_1 点), $z_1 = \frac{\sqrt{3}}{2} + \frac{1}{2}i$ (B_1 点), $z_1 = \infty$ (C_1 点) 和 $z_1 = \frac{\sqrt{3}}{2} - \frac{1}{2}i$ (D_1 点). 因而将两段圆弧变为 z_1 平面上通过 O 和 ∞ 的圆弧, 即通过原点的射线, 它们之间的夹角保持为 $\frac{5}{3}\pi$, 如图(b) 所示.

(2) 作辐角变换 $z_2 = z_1 e^{-i\frac{\pi}{6}}$, 即将两条射线顺时针旋转 $\frac{\pi}{6}$, 使得 A_2 点为 $z_2 = 0$, C_2 点为 $z_2 = \infty$, B_2 点为 $z_2 = 1$, D_2 点为 $z_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$, 如图(c) 所示.

(3) 作根式变换 $Z_3 = Z_2^{\frac{1}{5}}$, 将两条射线之间的夹角放大到原来的 $\frac{6}{5}$ 倍, 即将其放大为 2π , 成为全平面, 使 A_3 为 $Z_3 = 0$, B_3 为 $Z_3 = 1$, C_3 为 $Z_3 = \infty$, D_3 为 $Z_3 = 1$, 如图(d) 所示.



(d)

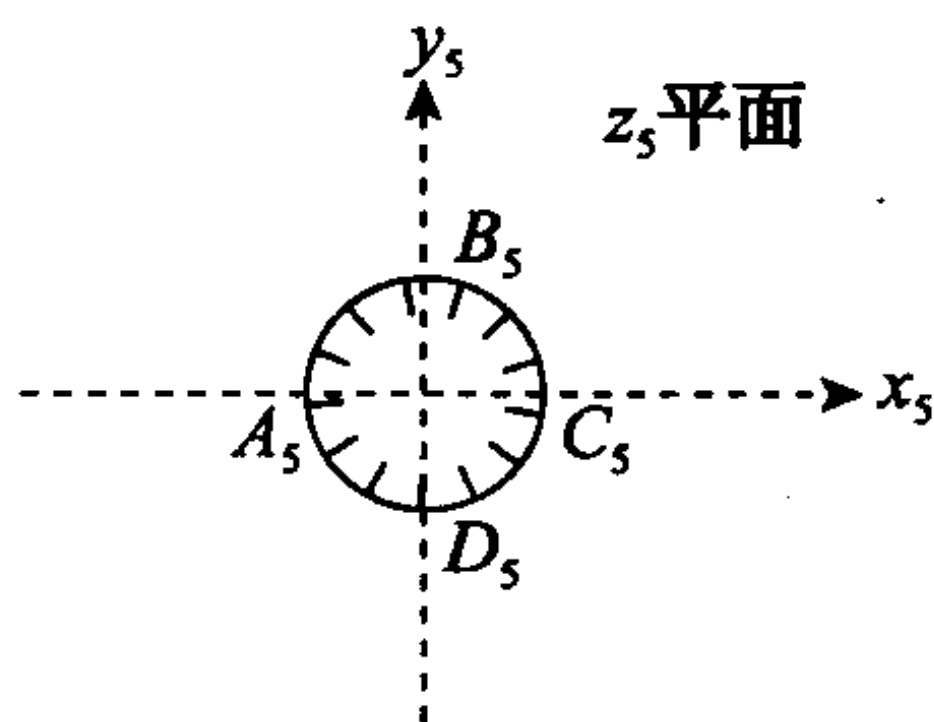


(e)

(4) 作分式线性变换 $Z_4 = \frac{Z_3-1}{Z_3+1}$, 将 $Z_3 = 0$ (A_3 点), $Z_3 = 1$ (B_3 点), $Z_3 = \infty$ (C_3 点) 和 $Z_3 = 1$ (D_3 点) 变到 $Z_4 = -1$ (A_4 点), $Z_4 = 0$ (B_4 点), $Z_4 = 1$ (C_4 点) 和 $Z_4 = 0$ (D_4 点), 即区域变为 Z_4 平面上 $-1 \leq \operatorname{Re} Z_4 \leq 1$ 的带域, 如图(e)所示.

(5) 利用儒阔夫斯基函数, 令 $Z_4 = \frac{1}{2} \left(\xi + \frac{1}{\xi} \right)$, 将椭圆域(实际为直线段)变为圆形域, 即使得 A_5 点为 $\xi = -1$, B_5 点为 $\xi = i$, C_5 点为 $\xi = 1$, D_5 点为 $\xi = -i$, 区域为圆形, 如图(f)所示.

(6) 在 ξ 平面内, 物理问题转化为无限长带电柱体外的电势, 易知其电势 $= C_1 \ln |\xi| + C_2$, 其中, 常数 C_1 与圆柱体带电量有关. 常数 C_2 的物理意义为带电柱体的电势.



(f)

6. 试将下列区域保角变换为圆.

(1) 弓形 $\operatorname{Im} z \geq 1, |z| \leq 2$.

(2) 圆 $|z| = 2$ 外, 除去第一象限.

(3) 两个相切的圆 $|z| \leq 2$ 和 $|z-3| \leq 1$ 以外的区域.

(4) 圆 $|z| \leq 3$ 外, 除去突起 $\begin{cases} \operatorname{Im} z = 0, \\ 3 \leq \operatorname{Re} z \leq 4. \end{cases}$

(5) 圆 $|z| \leq 1$ 内, 带有割线 $\begin{cases} \operatorname{Im} z = 0, \\ 0 \leq \operatorname{Re} z \leq 1. \end{cases}$

(6) 心脏线的内部 $|z| \leq \cos^2 \left[\frac{1}{2} \arg z \right]$.

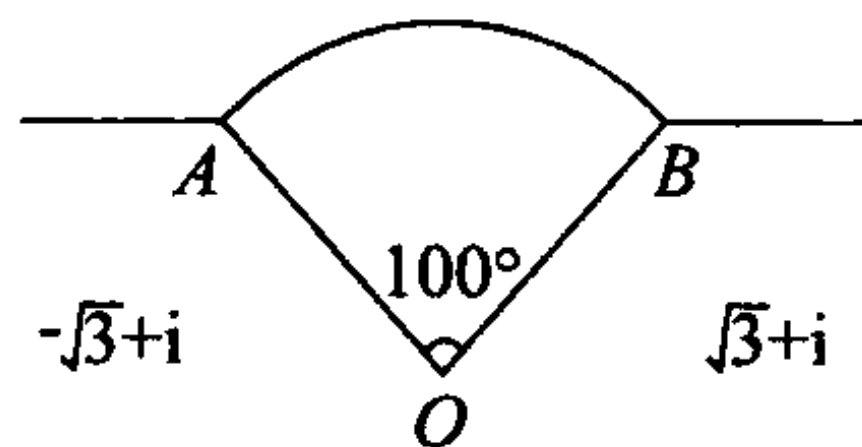
(7) 双纽线一支 $z \leq \sqrt{\cos[2 \arg z]}$.

解: (1) 方形 $\operatorname{Im} z \geq 1, |z| \leq 2$.

以原点为圆心, z 为半径的弧长及经过 $-\sqrt{3}+i$ 与 $\sqrt{3}+i$ 的直线所围的图形.

变换 $z_1 = \frac{z + \sqrt{3} + i}{-z + \sqrt{3} + i}$ 则 A 点变为 z_1 平面上的原点, B 点变成 z_1 平面上的无穷远点.

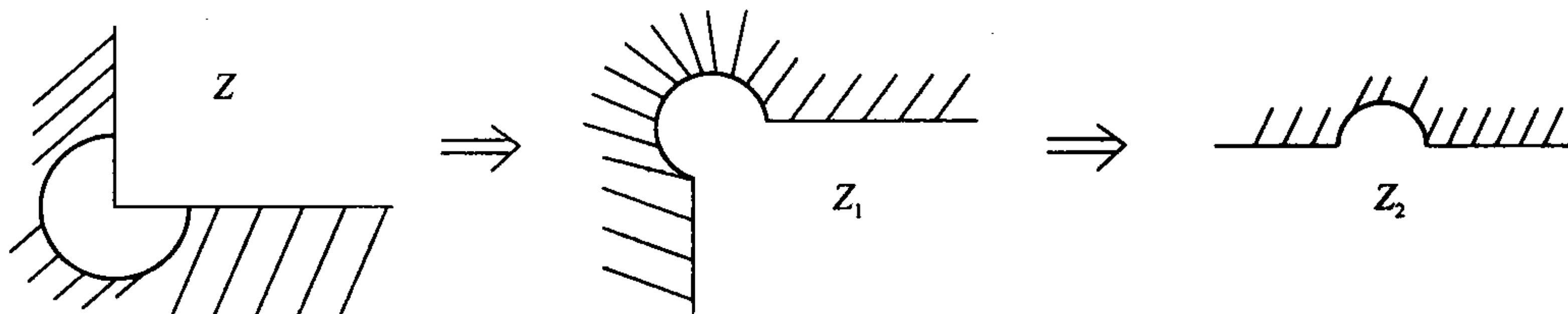
变换 $z_2 = z_1^3$



则 z_1 的 $\frac{\pi}{3}$ 以内部分变成 z_2 的上半平面.

再作变换 $\zeta = \frac{z_2 - i}{z_2 + i}$, 圆的半径 $R = 1$.

(2) 圆 $|z| = z$ 外, 第一象限作变换 $z_1 = e^{-i\frac{\pi}{2}}z$, 再作变换 $z_2 = z_1^{\frac{1}{2}}$

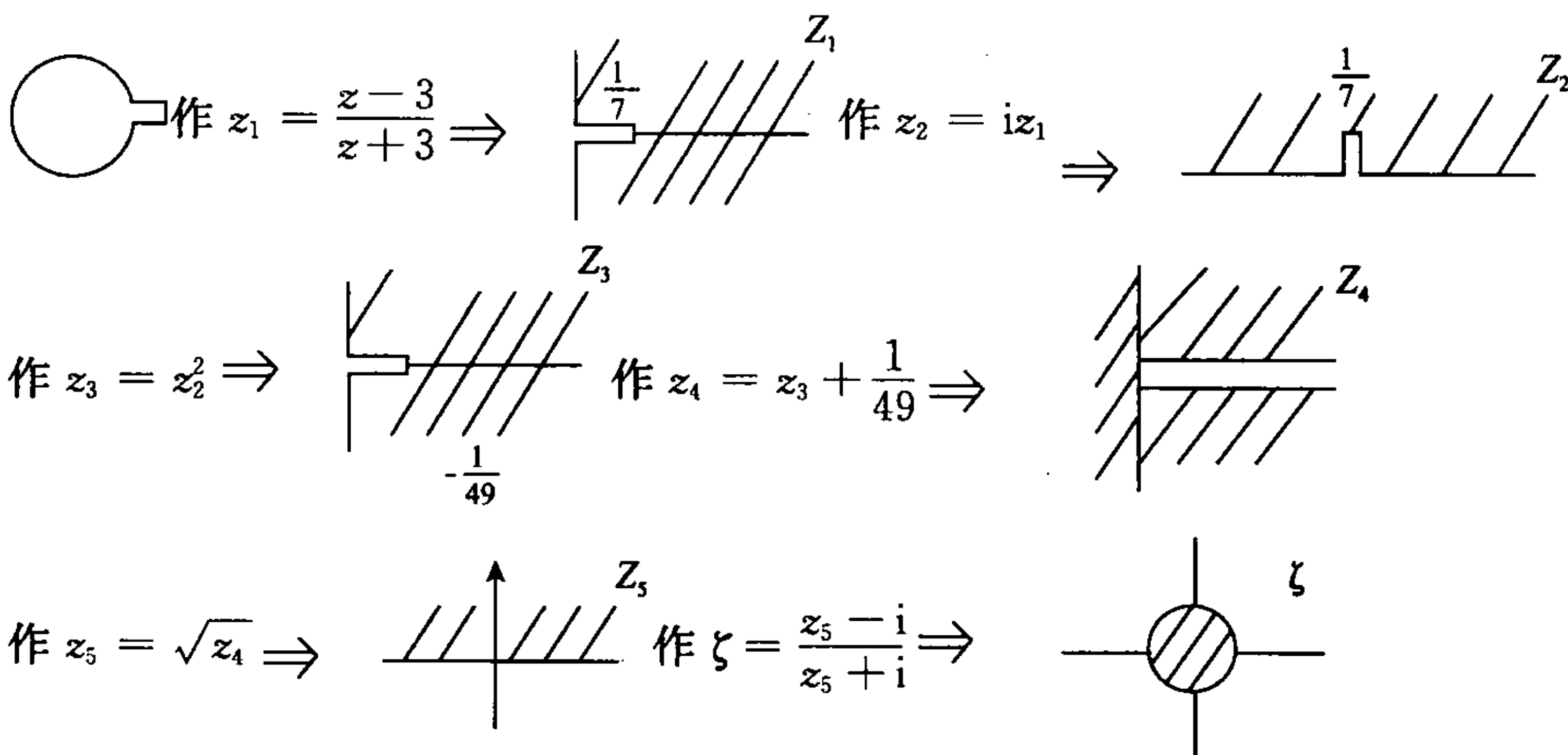


$z_3 = \frac{1}{2} \left(\frac{z_2}{z_1} + \frac{z_1}{z_2} \right)$, 将 z_2 阴影可变为 z_3 的上半平面, 再作 $\zeta = \frac{z_3 - i}{z_3 + i}$ 变成 ζ 平面上的一个圆, 半径 $R = 1$.

(3) 两个相切圆 $|z| \leq 2$ 和 $|z - 3| \leq 1$, 以外的区域, 变换 $z_1 = \frac{1}{z - 2}$, 则 B 变成无穷远, A 变为 $-\frac{1}{4}$, c 变为 $\frac{1}{2}$, AB 上半圆变成 Z_1 的实轴下部, $z_1 = -\frac{1}{2} (z = 0)$, $z_2 = e^{iz_1}$, $z_3 = z_2 e^{\frac{1}{2}}$, $z_4 = z_2^{\frac{1}{2}} \pi$

$\zeta = \frac{z_3 - i}{z_4 + i}$, 则变为圆.

(4) 圆 $|z| \leq 3$ 除去突起 $\begin{cases} \operatorname{Im} z = 0 \\ 3 \leq \operatorname{Re} z \leq 4 \end{cases}$



(5) 在图形区域有割线 $\begin{cases} \operatorname{Im} z = 0 \\ 0 \leq \operatorname{Re} z \leq 1 \end{cases}$, 先用分式变换 $z_1 = \frac{z - 1}{z + 1}$ 变成直线, 然后 $z_2 = -iz_1$,

再作变换 $z_3 = z_2^2$, $z_4 = z_3 + 1$, $z_5 = \sqrt{z_4}$, $\zeta = \frac{z_5 - i}{z_5 + i}$ (图形流程与上题相似, 故略去).

(6) 心脏线 $|z| \leq \cos^2 \left(\frac{1}{2} \arg z \right)$ 内部

令 $\zeta = \sqrt{z}$, $\therefore z = \zeta^2$, 即有 $\zeta^2 \leq \cos^2[\frac{1}{2} \arg \zeta^2]$

$\therefore |\zeta|^2 = \cos^2[\arg \zeta]$, $\therefore |\zeta| = \cos \arg \zeta$

使用极坐标, $\rho = \cos \varphi$, 表示的为圆.

(7) 双细线一支, $|z| \leq \sqrt{\cos[2\arg z]}$, 同样令 $\zeta = z^2$, 即 $z = \sqrt{\zeta}$

$\therefore z$ 平面上的双细线与 ζ 平面上变成 $|\zeta| = 1 \times \cos \arg \zeta$ 正是圆.

7. 研究甚长带电导体薄带周围的静电场, 带宽为 $2a$.

解: 带电导体可看成焦点是 $-a, a$, 半长轴为 a 的椭圆.

于是由儒可夫斯基变换 $\frac{z}{a} = \frac{1}{2} \left(\zeta + \frac{1}{\zeta} \right)$, 可以将带电导体变为 ζ 平面上的半径为 1 的圆柱.

柱外电势 $\varphi = c_1 \ln |\zeta| + C_2$, 将 ζ 变回 z 即可求得 z 平面上的电势.

$$\zeta + \frac{1}{\zeta} = \frac{2z}{a}, \zeta^2 - \frac{2z}{a}\zeta + 1 = 0.$$

$$\zeta = \frac{\frac{2z}{a} \pm \sqrt{\frac{4z^2}{a^2} - 4}}{2} = \frac{2z}{a} \pm \sqrt{\frac{z^2}{a^2} - 1} \text{ 代入 } \varphi = c_1 \ln |\zeta| + C_2 \text{ 即可.}$$

8. 研究甚长带电椭圆导体柱周围的静电场, 椭圆半长轴为 a , 半短轴为 b .

解: 用儒可夫斯基变换 $\frac{z}{\sqrt{a^2 - b^2}} = \frac{1}{2} \left(\zeta + \frac{1}{\zeta} \right)$, 将 z 平面的椭圆变为 ζ 平面上的圆, 半径为

$$\frac{a+b}{\sqrt{a^2 - b^2}}, \text{ 电势 } \varphi = c_1 \ln |\zeta| + c_2$$

9. 两个椭圆柱构成柱形电容器, 横截面是两个共焦点椭圆, 半长轴分别是 a_1 和 a_2 , 半短轴分别是 b_1 和 b_2 . 试求每单位长度的电容量.

解: 两个焦点距的一半 $c = \sqrt{a_1^2 - b_1^2} = \sqrt{a_2^2 - b_2^2}$, 变换 $\frac{z}{c} = \frac{1}{2} \left(\zeta + \frac{1}{\zeta} \right)$, 则共焦点椭圆变成 ζ 平面的同心圆.

$$R_1 = \frac{a_1 + b_1}{c}, R_2 = \frac{a_2 + b_2}{c}, \text{ 则所求电容量为 } c = \frac{2\pi\epsilon_0}{\ln \frac{R_2}{R_1}} = \frac{2\pi\epsilon_0}{\ln \frac{a_2 + b_2}{a_1 + b_1}}$$

10. 求解二维稳恒水流通过宽度为 $2a$ 的闸门的情形.

解: 二维稳恒水流通过宽度为 $2a$ 的闸门, 闸门处有 $\begin{cases} \operatorname{Im} z = 0 \\ -a < \operatorname{Re} z < a \end{cases}$, 考虑到水没有横向流

动, 所以速度势相力, 变换 $\frac{z}{a} = \frac{1}{2} \left(\zeta + \frac{1}{\zeta} \right)$. 闸门变为半径为 1 的圆柱, 根据静电势类推, 得到速度势为 $c_1 \ln |\zeta| + c_2$

11. 图 14-1 是六角“星”, 六个臂彼此相隔 60° , 各自长度为 1. 试将“星”的外部变为 ζ 平面的单位圆外部.

[提示: $z_1 = z^3$, $\zeta_1 = \zeta^3$, 再找 z_1 和 ζ_1 之间的变换.]

解: 变换 $z_1 = z^3$ 和 $\zeta_1 = \zeta^3$ 可以使 z 平面成为 z_1 的三叶交叉平面, 原来六角星度为三叶上沿

实轴的一段割线 $(-1 \sim 1)$.

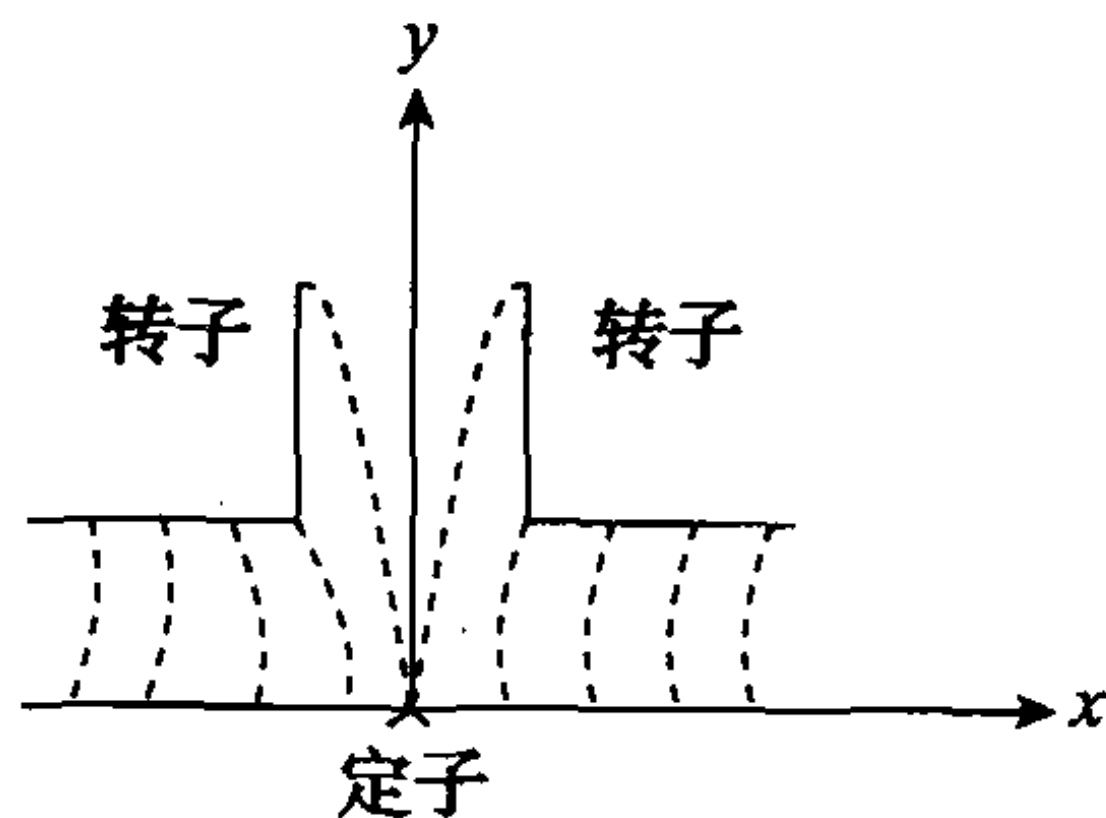
ζ 平面单位圆的外部变成 ζ_1 的三叶叉平面的单位圆外部,作儒可夫斯基变换 $z_1 = \frac{1}{2}(\zeta_1 + \frac{1}{\zeta_1})$,可将 z_1 平面的图形变成 ζ_1 平面上的图形,于是满足条件的变换为 $z^3 = \frac{1}{2}(\zeta^3 + \frac{1}{\zeta^3})$

12. 研究电机的转子和定子之间(图 14-2)的磁场. 求最大磁场和最小磁场之比.

解: 如图取“ x ”处为坐标系原点, x 轴向右, y 轴向上, 作变换

$$z = \frac{2i}{\pi} \left[\operatorname{harctan} \left(\frac{h}{H} \sqrt{\frac{\xi_1}{\xi_1 - a^2}} + \operatorname{Harcch} \sqrt{\frac{\xi_1}{\xi_1 - a^2}} \right) \right],$$

$$(a^2 = 1 + \frac{h^2}{H^2})$$



使得 z 平面内 $(0,0), (\infty, 0 \text{ 或 } h), (H, h), (0 \text{ 或 } H, \infty)$ 变为 ξ_1 平面内 $(0,0), (1,0), (a^2, 0), (\pm\infty, 0)$

再作变换 $\xi = \frac{V}{\pi} \ln \frac{1 + \sqrt{\xi_1}}{1 - \sqrt{\xi_1}}$

使得 ξ_1 平面内 $(0,0), (1,0), (\pm\infty, 0)$ 变为 ξ 平面内 $(0,0), (\infty, 0 \text{ 或 } iv), (0, v)$, 则复势 $= \xi$.

磁场

$$B = H = \left| \frac{d\xi}{dz} \right| = \left| \frac{d\xi}{d\xi_1} \cdot \frac{d\xi_1}{dz} \right| = \left| \frac{d\xi}{d\xi_1} / \frac{dz}{d\xi_1} \right|$$

$$= \left| \frac{V}{\pi \sqrt{\xi_1}(1 - \xi_1)} / \frac{iH \sqrt{\xi_1 - a^2}}{\pi(\xi_1 - 1) \sqrt{\xi_1}} \right| = \left| \frac{V}{H \sqrt{\xi_1 - 1}} \right|$$

在 $\xi_1 = 0$ 处, $B = B_{\min}$, 在 $\xi_1 = 1$ 处, $B = B_{\max}$

$$\therefore B_{\max} : B_{\min} = \frac{a}{\sqrt{a^2 - 1}} = \frac{\sqrt{H^2 + h^2}}{h}$$

13. 求 z 平面的半无界长条 $0 < \operatorname{Re} z < a, \operatorname{Im} z > 0$ 上的调和函数, 边界条件是 $u|_{x=0} = 0$, $u|_{x=a} = 0, u|_{y=0} = u_0$.

解: 作变换 $\xi = -\cos \frac{\pi z}{a}$, 使得 z 平面内 $(0,0), (a,0), (0 \text{ 或 } a, \infty)$ 变为 ξ 平面内 $(-1,0), (1,0), (\pm\infty, 0)$, 问题转化为半平面内求解拉普拉斯方程的第一边值问题.

利用 § 12.2 小节例 4 结论可得. 在 ξ 平面内

$$u = \frac{u_0}{\pi} \arctan \frac{2\eta}{\xi^2 + \eta^2 - 1}$$

还原到自变数

$$u = \frac{u_0}{\pi} \arctan \frac{2\operatorname{sh}(\pi y/a) \sin(\pi x/a)}{\operatorname{sh}^2(\pi y/a) - \sin^2(\pi x/a)}$$

14. 将可变电容器中的静电场所占空间(图 14-3)变为上半平面.

解: 如图取动片右端点为坐标系原点, x 轴向右, y 轴向上.

作变换 $z = \frac{h_2}{\pi} \ln(1 - \xi) + \frac{h_1}{\pi} \ln\left(1 + \frac{h_2}{h_1} \xi\right)$

使得 z 平面内 $(-\infty, 0$ 或 $-h_2)$, $(\infty, -h_2$ 或 $h_1)$, $(-\infty, h_1$ 或 $0)$, $(0, 0)$ 变为 ξ 平面内 $(1, 0)$, $(\pm\infty, 0)$, $(-h_1/h_2, 0)$, $(0, 0)$. 设定片和动片电势分别为 v_0 和 0 . 问题转化为半平面内求解拉普拉斯方程的第一边值问题, 利用 §12.2 小节例 4 结论可得, 在 ξ 平面内

$$u = V_0 - \frac{V_0}{\pi} \arctan \frac{y(1 + h_1/h_2)}{\xi^2 + y^2 + \left(\frac{h_1}{h_2} - 1\right)\xi - h_1/h_2}$$

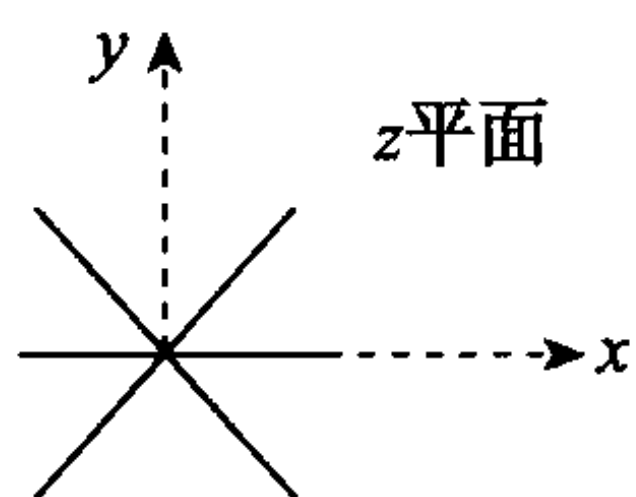
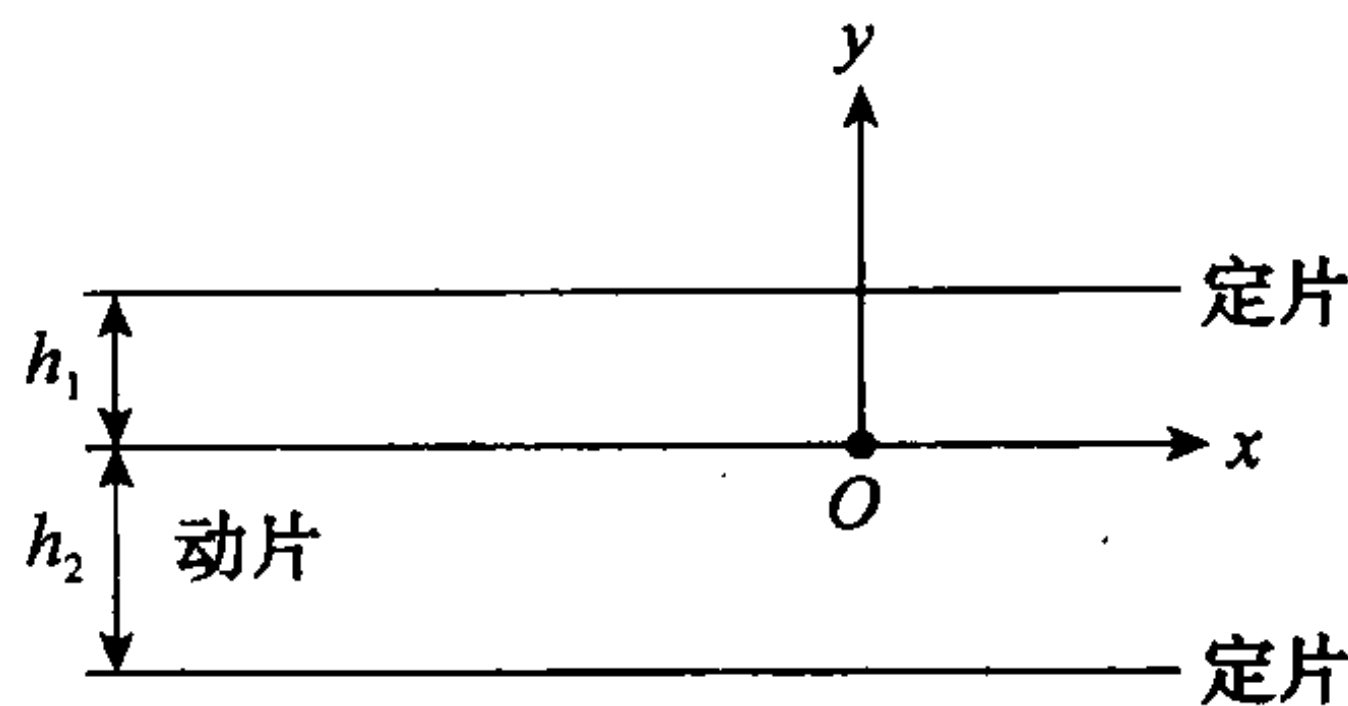


图 14-1

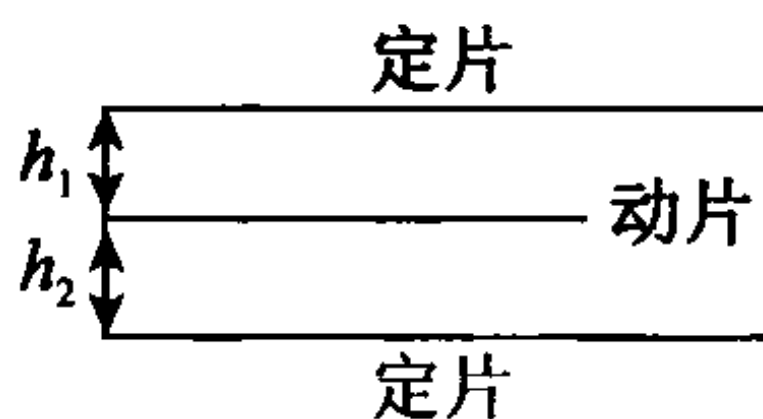


图 14-3

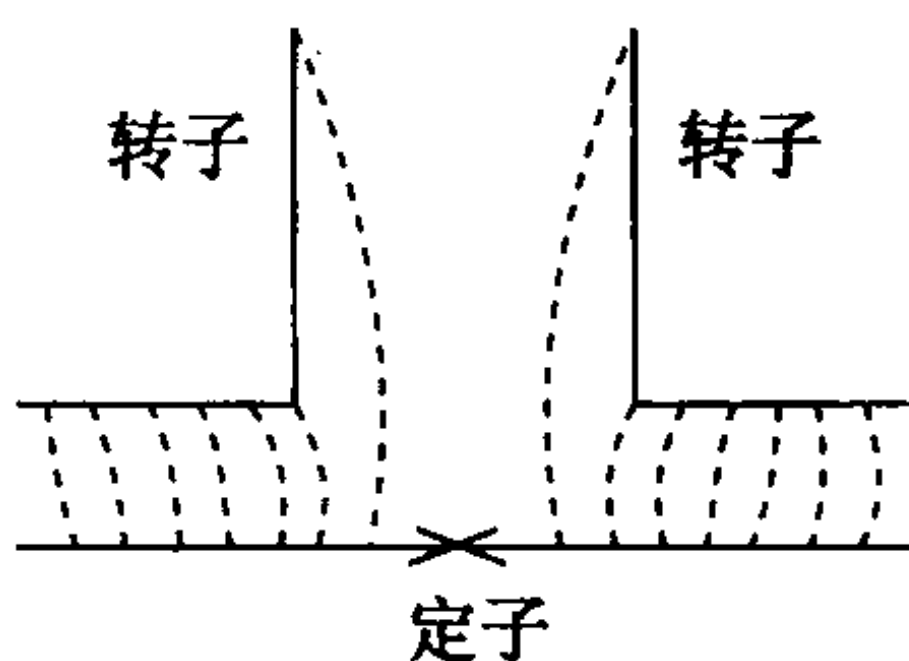


图 14-2

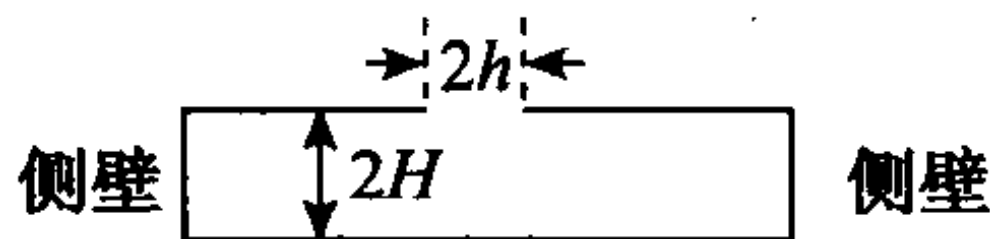


图 14-4

解: 如图取 D 形盒的中心点为原点, x 轴向右, y 轴向上.

作变换 $z = iH + \frac{4Ha}{\pi(a^2 - 1)^2} \left[\xi_1 + \frac{a^2}{\xi_1} \right] - \frac{2H}{\pi} \ln \frac{\xi_1 - a}{\xi_1 + a}$

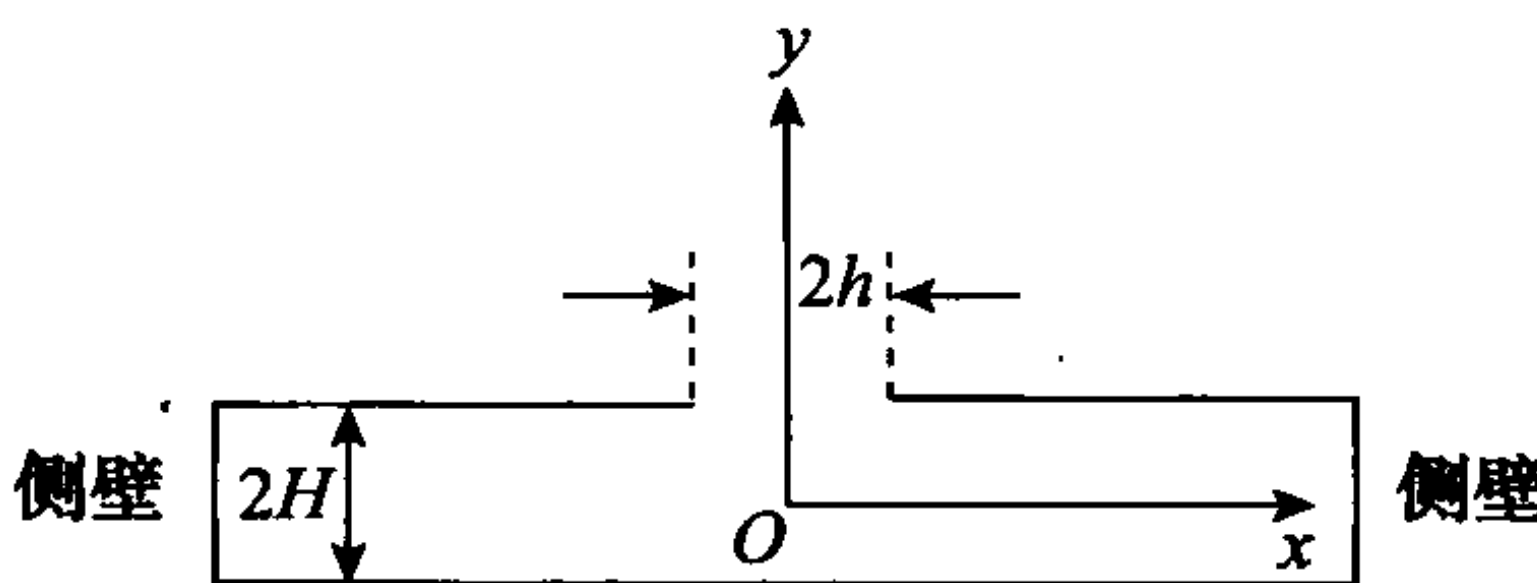
其中 a 满足 $\frac{4Ha}{\pi(a^2 - 1)^2} (a^2 + 1) - \frac{2H}{\pi} \ln \frac{a - 1}{a + 1} = h$

z 平面内 $(\pm\infty, H)$, $(-h, H)$, $(-\infty, \pm h)$, $(-h, -H)$, $(\pm\infty, -H)$, $(h, -H)$, $(\infty, \pm H)$, (h, H) 变为 $(\pm\infty, 0)$, $(-a^2, 0)$, $(-a, 0)$, $(-1, 0)$, $(0, 0)$, $(1, 0)$, $(a, 0)$, $(a^2, 0)$

再作变换 $\xi = \ln \xi_1$, 使研究区域变为带形区域, 则电势 $v = v_1 + \frac{v_2 - v_1}{\pi} \eta$, 即 $v = v_1$

$+ \frac{v_2 - v_1}{\pi} \arg \xi_1$.

再还原到自变数即得到 z 平面下的解.



第十五章 非线性数学物理问题简介

本章知识要点

孤立子

1. 孤立子: 是一种稳定的、局域的孤立波, 彼此相互作用(碰撞)后, 各自仍保持原来的形状、速度, 最多只有相位及位置的变化.

2. 描述孤立子的一些重要方程

(1) KdV 方程 $u_t + \alpha u u_x + u_{xxx} = 0$

(2) Sine-Gordon 方程 $u_x - u_{xx} + \sin u = 0$

(3) 非线性薛定谔方程 $i u_t + u_{xx} + \beta |u|^2 u = 0$

(4) Toda 点阵方程
$$\begin{cases} \frac{dq_n}{dt} = \frac{p_n}{m} \\ \frac{dp_n}{dt} = \exp[-(q_n - q_{n-1})] - \exp[-(q_{n+1} - q_n)] \end{cases}$$

3. 非线性傅里叶变换法: 也称逆散射方法, 借助于线性方程的本征值问题, 散射变换, 散射数据演化及逆散射变换, 将非线性问题化为几个线性问题来求解.

混沌

1. 混沌: 来自用确定论方法描述的系统中的无规运动, 也称内在随机性.

2. 分岔: 对于含参数的系统, 当参数变化到某些临界值时, 系统的动力学性态发生定性变化, 这种现象称为分岔, 发生分岔的参数值称为分岔点.

倍周期分岔: 在某个参数值左右, 系统动力学状态变化周期加倍的现象.

切分岔: 在混沌区的窗口中, 随着参数减小向混沌运动状态过渡的现象.

3. 吸引子: 对每一个固定的参数值, 从任何初始值出发进行迭代时, 经过一段暂态过程, 最后趋向于一定的终态, 称为吸引子.

同期状态对应的吸引子称为周期吸引子, 或平庸吸引子.

混沌状态对应的吸引子称为奇怪吸引子.

4. 混沌的表征方法: 庞加莱截面法、功率谱分析法、分形维数、李雅普诺夫指数.

5. 通向混沌的方式: 倍周期分岔通向混沌、阵发性通向混沌、经准周期通向混沌.

[G e n e r a l I n f o r m a t i o n]

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内容提要 = 本书根据《数学物理方法（第四版）全程导学及习题全解》教学大纲的要求及教学计划的编写，将教材中出现的难点、知识点、重点归纳，并将在教学中遇到的问题给予解答，且将课后习题全解帮助学生更好更快的掌握所学知识。